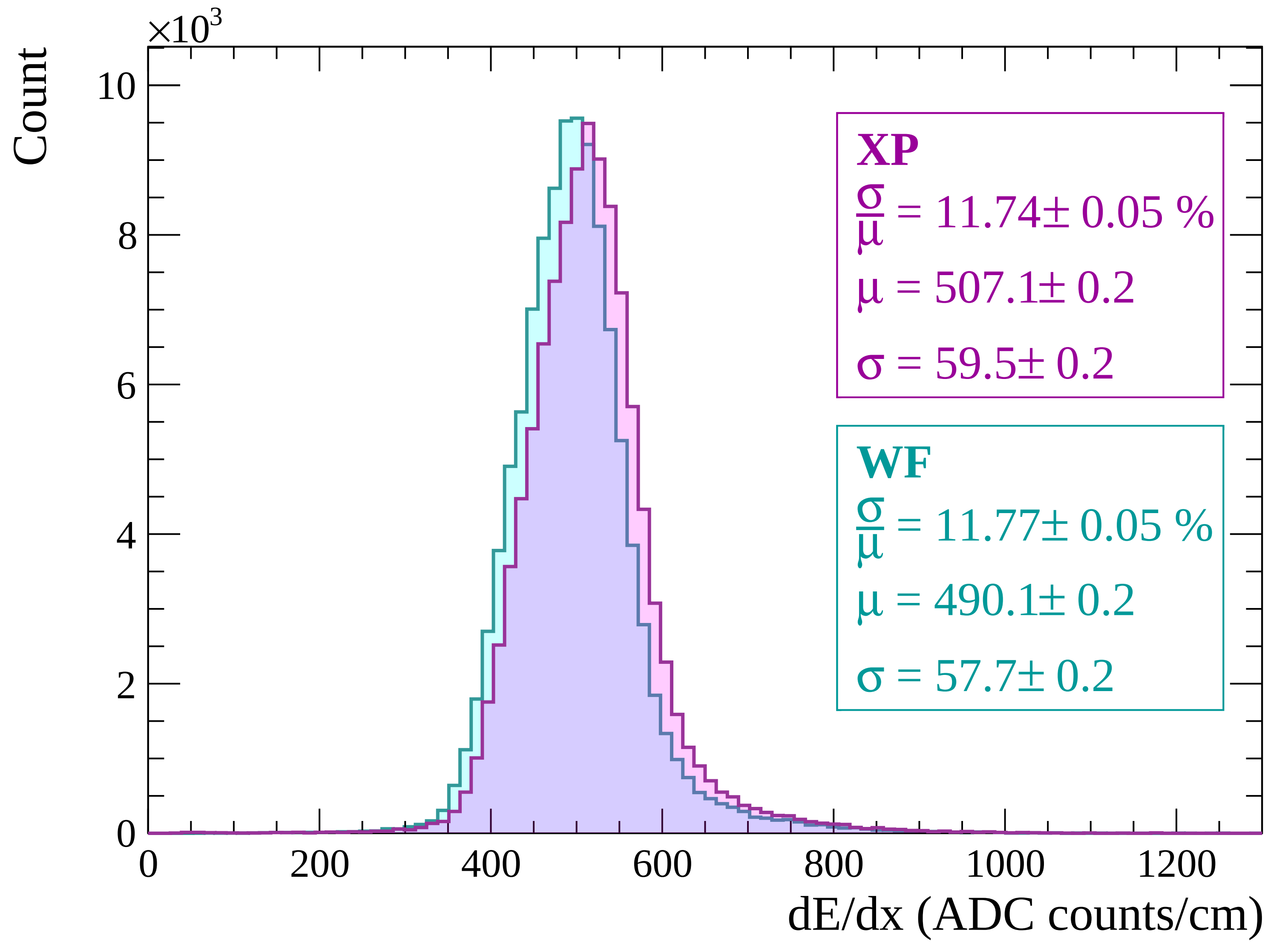
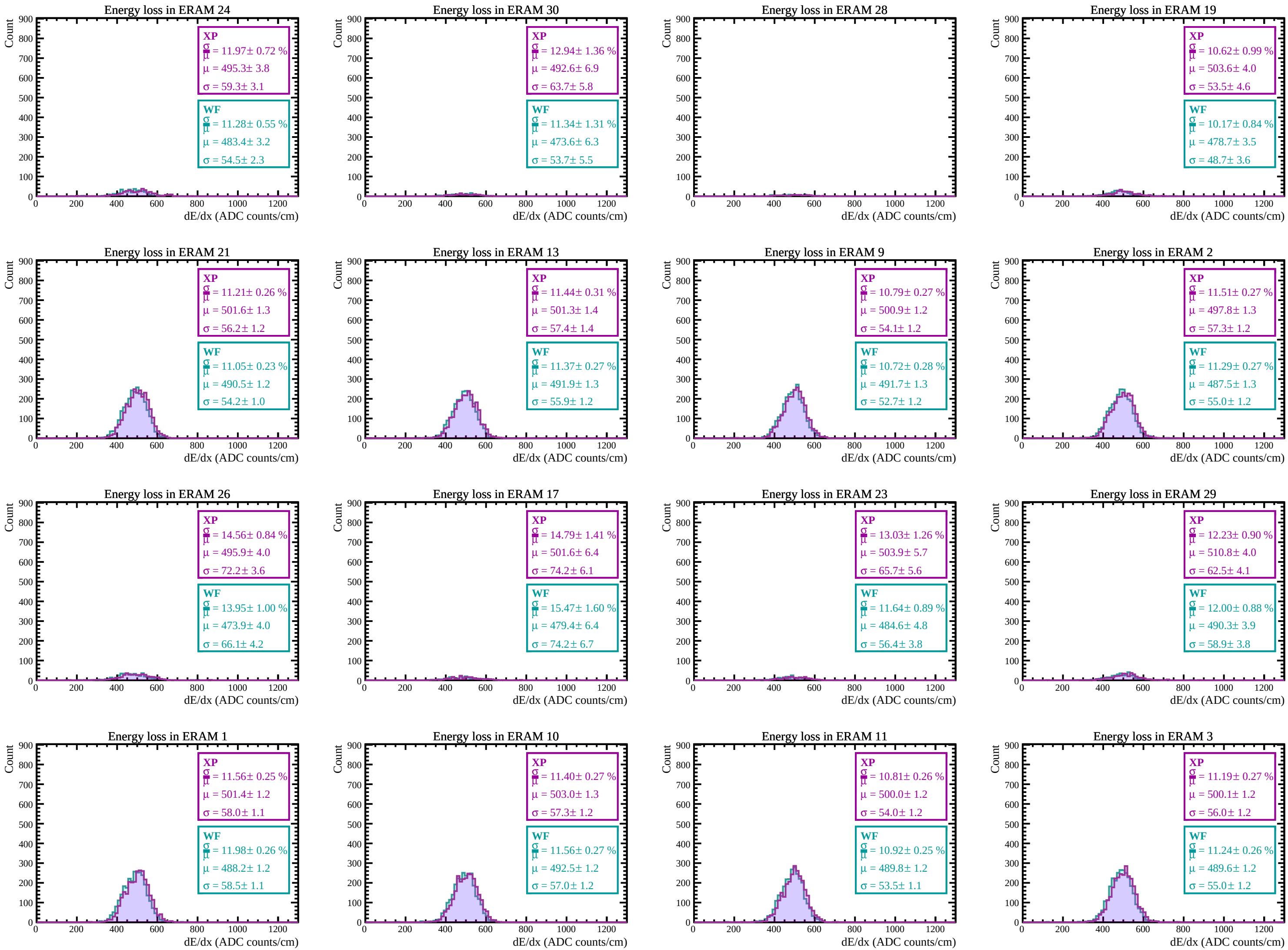
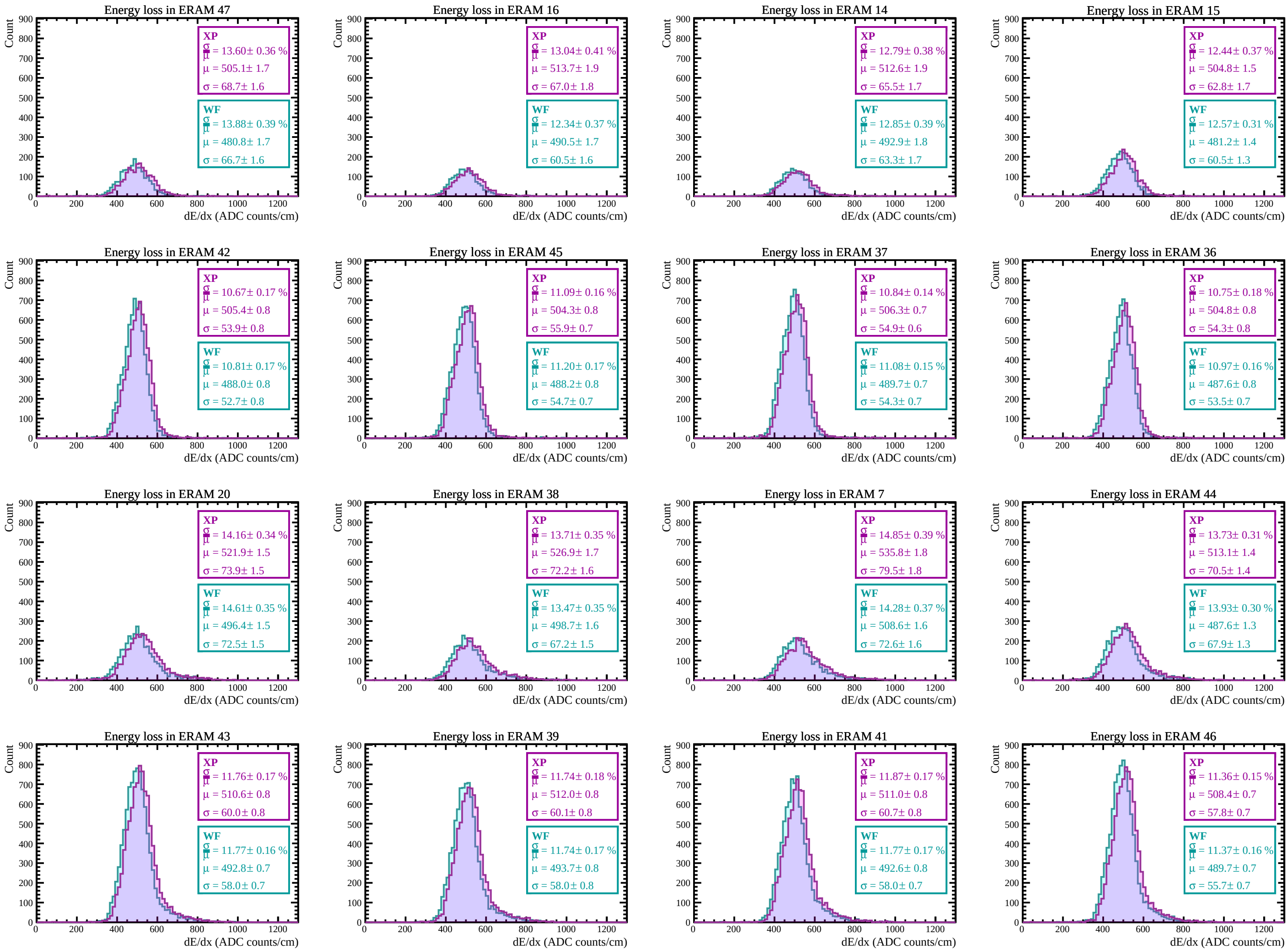
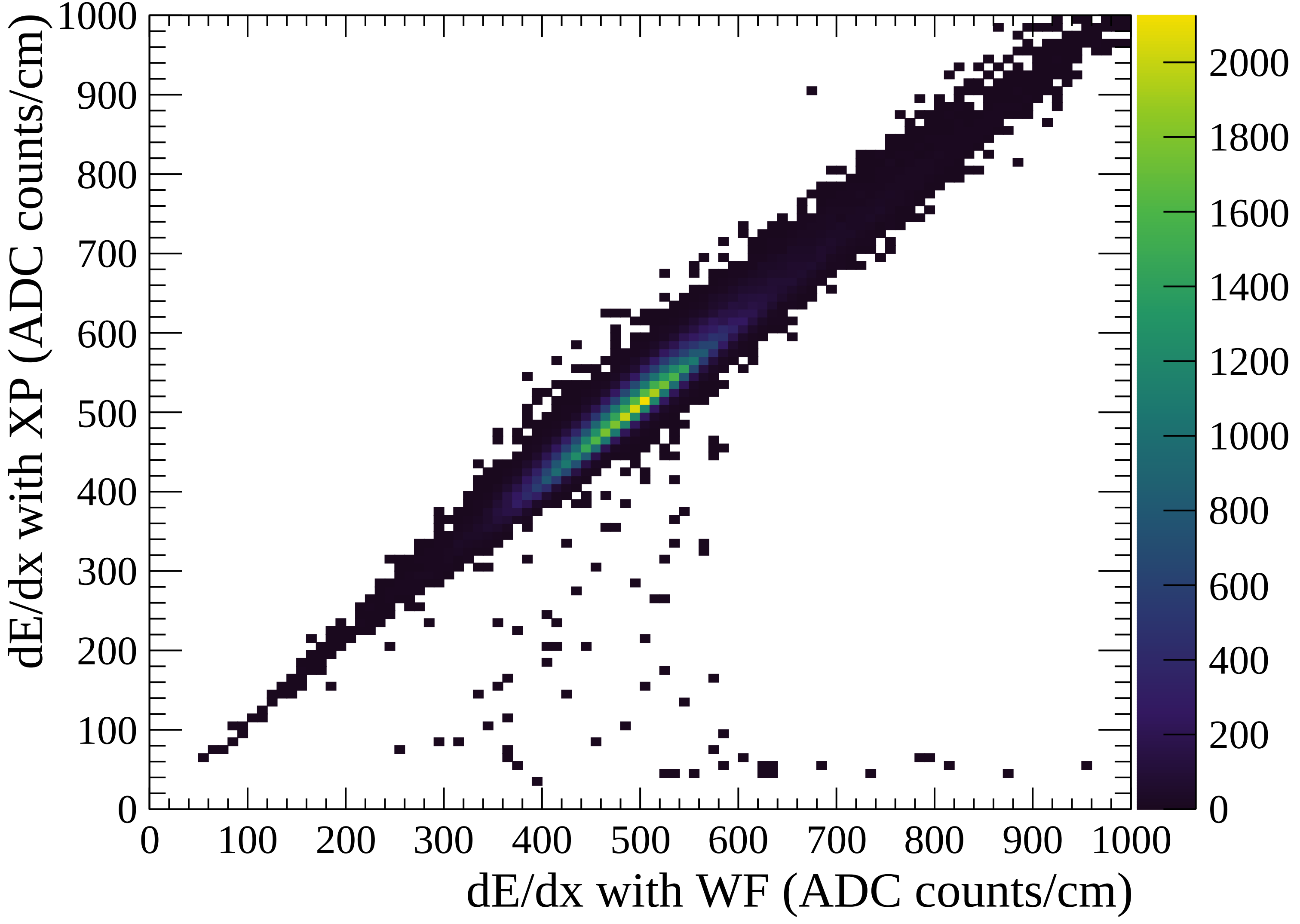
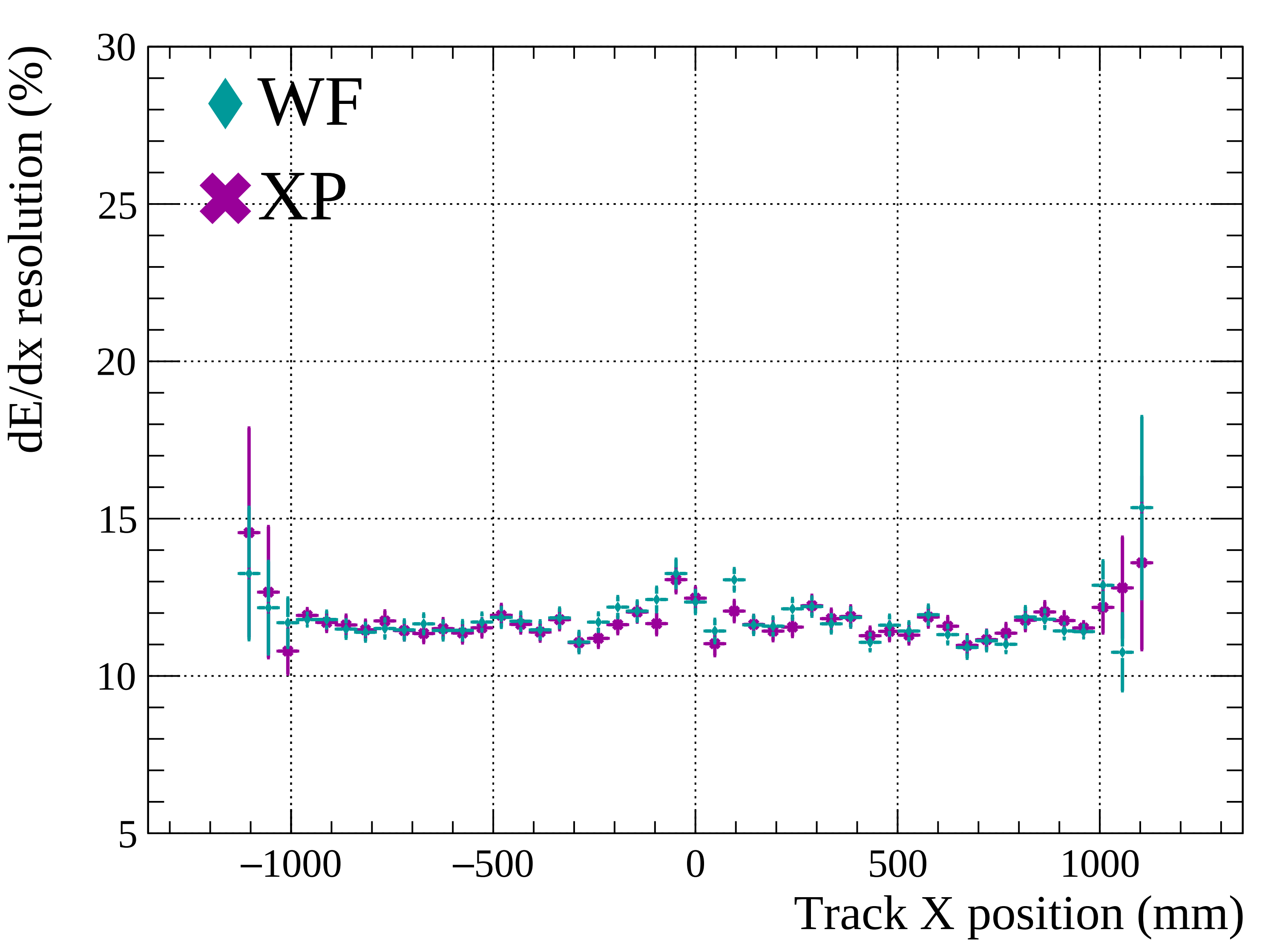


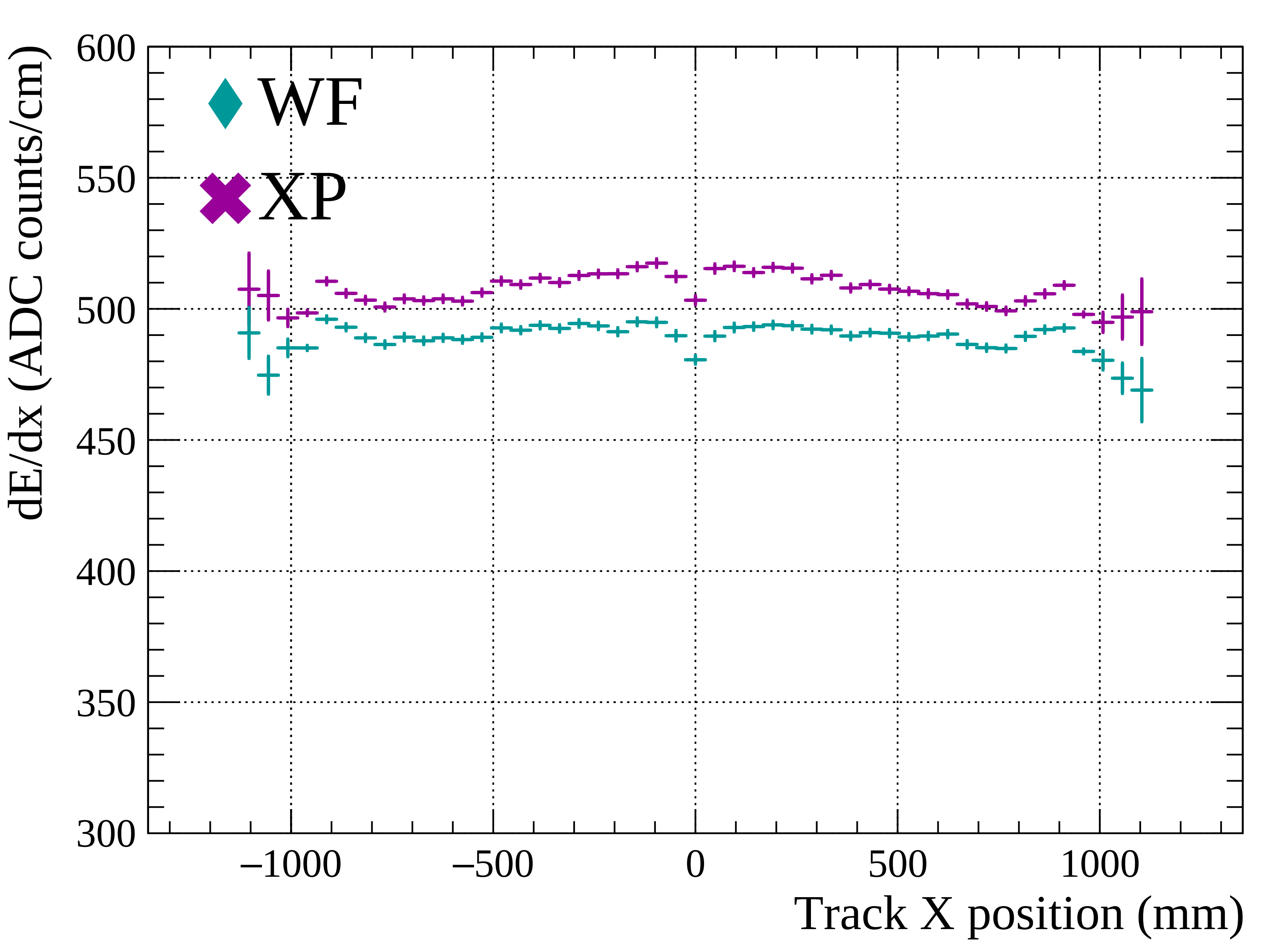
Count
×10³
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.74± 0.05 %
μ = 507.1± 0.2
σ = 59.5± 0.2
WF
σ/μ = 11.77± 0.05 %
μ = 490.1± 0.2
σ = 57.7± 0.2

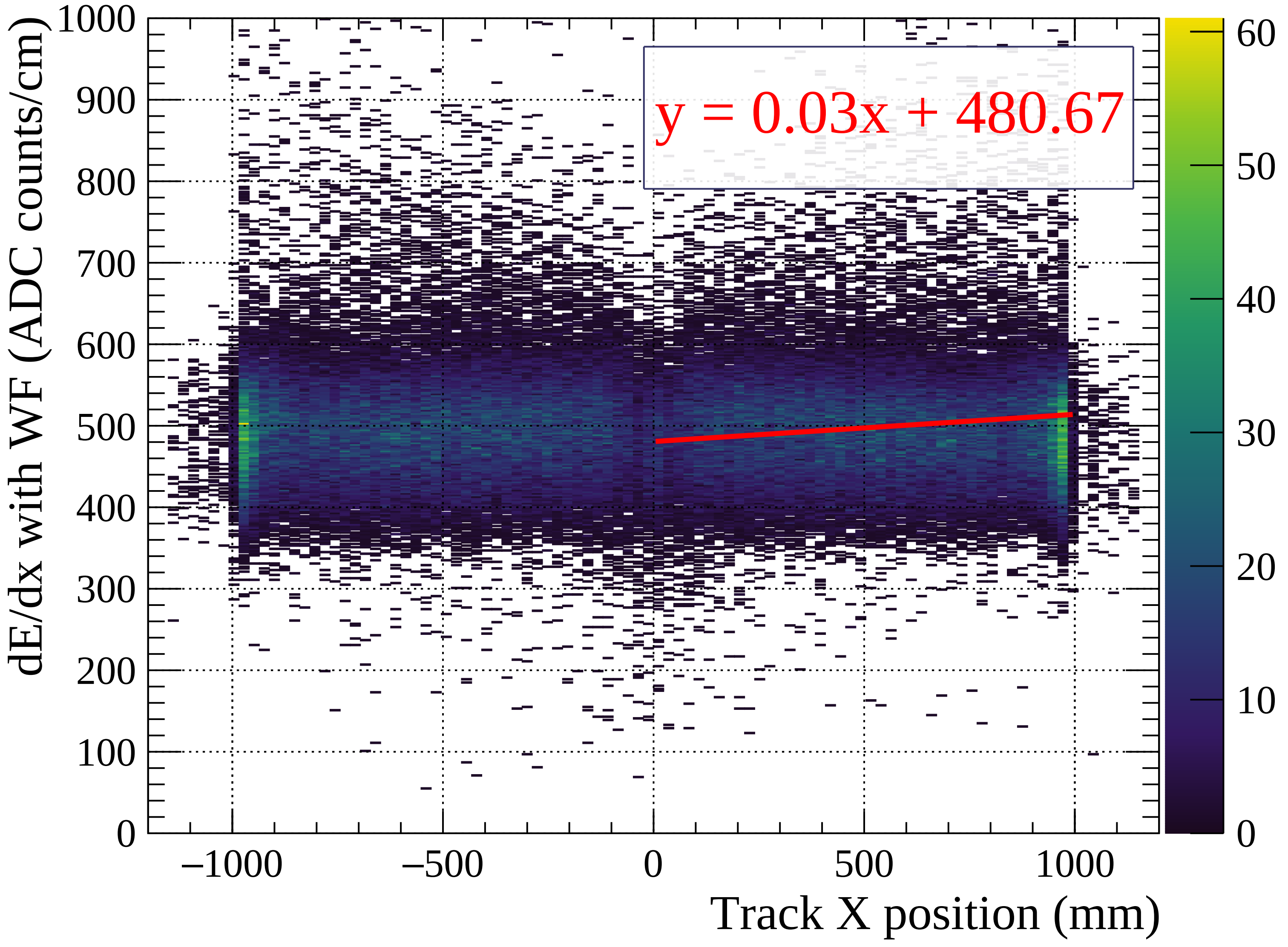

Energy loss in ERAM 24
XP
σ/μ = 11.97± 0.72 %
μ = 495.3± 3.8
σ = 59.3± 3.1
WF
σ/μ = 11.28± 0.55 %
μ = 483.4± 3.2
σ = 54.5± 2.3
Energy loss in ERAM 30
XP
σ/μ = 12.94± 1.36 %
μ = 492.6± 6.9
σ = 63.7± 5.8
WF
σ/μ = 11.34± 1.31 %
μ = 473.6± 6.3
σ = 53.7± 5.5
Energy loss in ERAM 28
Energy loss in ERAM 19
XP
σ/μ = 10.62± 0.99 %
μ = 503.6± 4.0
σ = 53.5± 4.6
WF
σ/μ = 10.17± 0.84 %
μ = 478.7± 3.5
σ = 48.7± 3.6
Energy loss in ERAM 21
XP
σ/μ = 11.21± 0.26 %
μ = 501.6± 1.3
σ = 56.2± 1.2
WF
σ/μ = 11.05± 0.23 %
μ = 490.5± 1.2
σ = 54.2± 1.0
Energy loss in ERAM 13
XP
σ/μ = 11.44± 0.31 %
μ = 501.3± 1.4
σ = 57.4± 1.4
WF
σ/μ = 11.37± 0.27 %
μ = 491.9± 1.3
σ = 55.9± 1.2
Energy loss in ERAM 9
XP
σ/μ = 10.79± 0.27 %
μ = 500.9± 1.2
σ = 54.1± 1.2
WF
σ/μ = 10.72± 0.28 %
μ = 491.7± 1.3
σ = 52.7± 1.2
Energy loss in ERAM 2
XP
σ/μ = 11.51± 0.27 %
μ = 497.8± 1.3
σ = 57.3± 1.2
WF
σ/μ = 11.29± 0.27 %
μ = 487.5± 1.3
σ = 55.0± 1.2
Energy loss in ERAM 26
XP
σ/μ = 14.56± 0.84 %
μ = 495.9± 4.0
σ = 72.2± 3.6
WF
σ/μ = 13.95± 1.00 %
μ = 473.9± 4.0
σ = 66.1± 4.2
Energy loss in ERAM 17
XP
σ/μ = 14.79± 1.41 %
μ = 501.6± 6.4
σ = 74.2± 6.1
WF
σ/μ = 15.47± 1.60 %
μ = 479.4± 6.4
σ = 74.2± 6.7
Energy loss in ERAM 23
XP
σ/μ = 13.03± 1.26 %
μ = 503.9± 5.7
σ = 65.7± 5.6
WF
σ/μ = 11.64± 0.89 %
μ = 484.6± 4.8
σ = 56.4± 3.8
Energy loss in ERAM 29
XP
σ/μ = 12.23± 0.90 %
μ = 510.8± 4.0
σ = 62.5± 4.1
WF
σ/μ = 12.00± 0.88 %
μ = 490.3± 3.9
σ = 58.9± 3.8
Energy loss in ERAM 1
XP
σ/μ = 11.56± 0.25 %
μ = 501.4± 1.2
σ = 58.0± 1.1
WF
σ/μ = 11.98± 0.26 %
μ = 488.2± 1.2
σ = 58.5± 1.1
Energy loss in ERAM 10
XP
σ/μ = 11.40± 0.27 %
μ = 503.0± 1.3
σ = 57.3± 1.2
WF
σ/μ = 11.56± 0.27 %
μ = 492.5± 1.2
σ = 57.0± 1.2
Energy loss in ERAM 11
XP
σ/μ = 10.81± 0.26 %
μ = 500.0± 1.2
σ = 54.0± 1.2
WF
σ/μ = 10.92± 0.25 %
μ = 489.8± 1.2
σ = 53.5± 1.1
Energy loss in ERAM 3
XP
σ/μ = 11.19± 0.27 %
μ = 500.1± 1.2
σ = 56.0± 1.2
WF
σ/μ = 11.24± 0.26 %
μ = 489.6± 1.2
σ = 55.0± 1.2
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
XP
σ/μ = 13.60± 0.36 %
μ = 505.1± 1.7
σ = 68.7± 1.6
WF
σ/μ = 13.88± 0.39 %
μ = 480.8± 1.7
σ = 66.7± 1.6
Energy loss in ERAM 16
XP
σ/μ = 13.04± 0.41 %
μ = 513.7± 1.9
σ = 67.0± 1.8
WF
σ/μ = 12.34± 0.37 %
μ = 490.5± 1.7
σ = 60.5± 1.6
Energy loss in ERAM 14
XP
σ/μ = 12.79± 0.38 %
μ = 512.6± 1.9
σ = 65.5± 1.7
WF
σ/μ = 12.85± 0.39 %
μ = 492.9± 1.8
σ = 63.3± 1.7
Energy loss in ERAM 15
XP
σ/μ = 12.44± 0.37 %
μ = 504.8± 1.5
σ = 62.8± 1.7
WF
σ/μ = 12.57± 0.31 %
μ = 481.2± 1.4
σ = 60.5± 1.3
Energy loss in ERAM 42
XP
σ/μ = 10.67± 0.17 %
μ = 505.4± 0.8
σ = 53.9± 0.8
WF
σ/μ = 10.81± 0.17 %
μ = 488.0± 0.8
σ = 52.7± 0.8
Energy loss in ERAM 45
XP
σ/μ = 11.09± 0.16 %
μ = 504.3± 0.8
σ = 55.9± 0.7
WF
σ/μ = 11.20± 0.17 %
μ = 488.2± 0.8
σ = 54.7± 0.7
Energy loss in ERAM 37
XP
σ/μ = 10.84± 0.14 %
μ = 506.3± 0.7
σ = 54.9± 0.6
WF
σ/μ = 11.08± 0.15 %
μ = 489.7± 0.7
σ = 54.3± 0.7
Energy loss in ERAM 36
XP
σ/μ = 10.75± 0.18 %
μ = 504.8± 0.8
σ = 54.3± 0.8
WF
σ/μ = 10.97± 0.16 %
μ = 487.6± 0.8
σ = 53.5± 0.7
Energy loss in ERAM 20
XP
σ/μ = 14.16± 0.34 %
μ = 521.9± 1.5
σ = 73.9± 1.5
WF
σ/μ = 14.61± 0.35 %
μ = 496.4± 1.5
σ = 72.5± 1.5
Energy loss in ERAM 38
XP
σ/μ = 13.71± 0.35 %
μ = 526.9± 1.7
σ = 72.2± 1.6
WF
σ/μ = 13.47± 0.35 %
μ = 498.7± 1.6
σ = 67.2± 1.5
Energy loss in ERAM 7
XP
σ/μ = 14.85± 0.39 %
μ = 535.8± 1.8
σ = 79.5± 1.8
WF
σ/μ = 14.28± 0.37 %
μ = 508.6± 1.6
σ = 72.6± 1.6
Energy loss in ERAM 44
XP
σ/μ = 13.73± 0.31 %
μ = 513.1± 1.4
σ = 70.5± 1.4
WF
σ/μ = 13.93± 0.30 %
μ = 487.6± 1.3
σ = 67.9± 1.3
Energy loss in ERAM 43
XP
σ/μ = 11.76± 0.17 %
μ = 510.6± 0.8
σ = 60.0± 0.8
WF
σ/μ = 11.77± 0.16 %
μ = 492.8± 0.7
σ = 58.0± 0.7
Energy loss in ERAM 39
XP
σ/μ = 11.74± 0.18 %
μ = 512.0± 0.8
σ = 60.1± 0.8
WF
σ/μ = 11.74± 0.17 %
μ = 493.7± 0.8
σ = 58.0± 0.8
Energy loss in ERAM 41
XP
σ/μ = 11.87± 0.17 %
μ = 511.0± 0.8
σ = 60.7± 0.8
WF
σ/μ = 11.77± 0.17 %
μ = 492.6± 0.8
σ = 58.0± 0.7
Energy loss in ERAM 46
XP
σ/μ = 11.36± 0.15 %
μ = 508.4± 0.7
σ = 57.8± 0.7
WF
σ/μ = 11.37± 0.16 %
μ = 489.7± 0.7
σ = 55.7± 0.7
Count
dE/dx (ADC counts/cm)

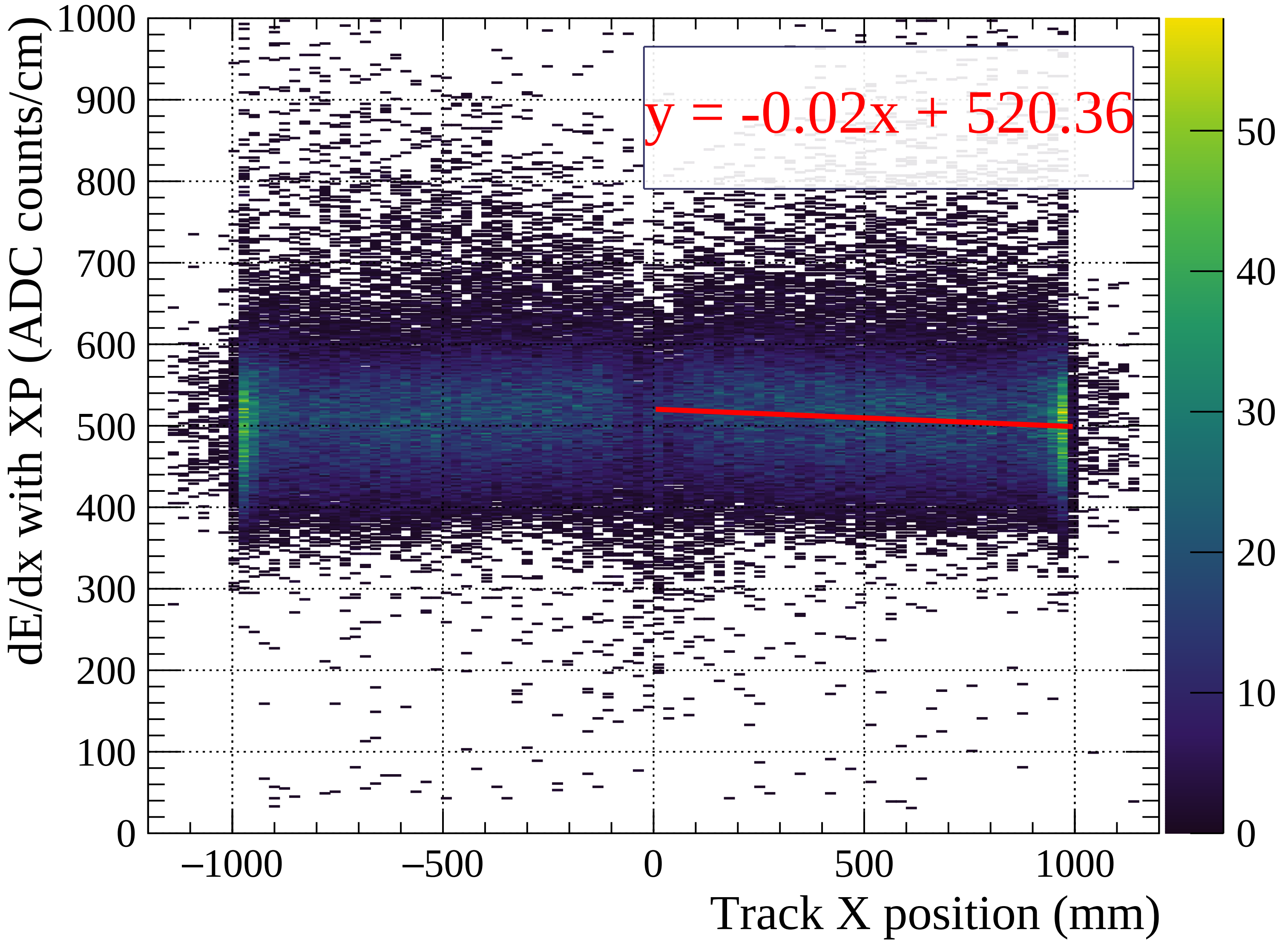

dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
2000
1800
1600
1400
1200
1000
800
600
400
200
0

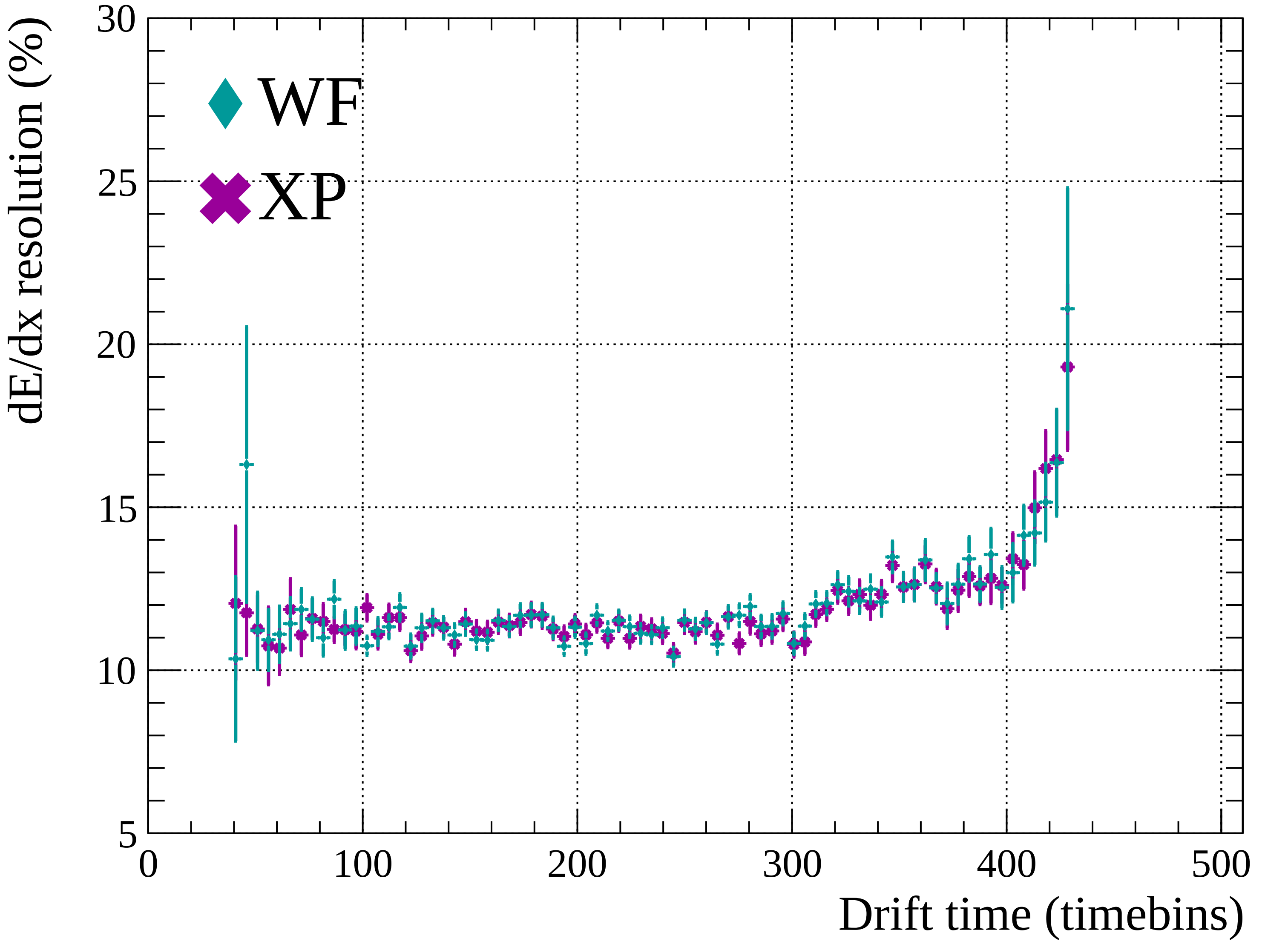

WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000


WF
XP
dE/dx (ADC counts/cm)
Track X position (mm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000

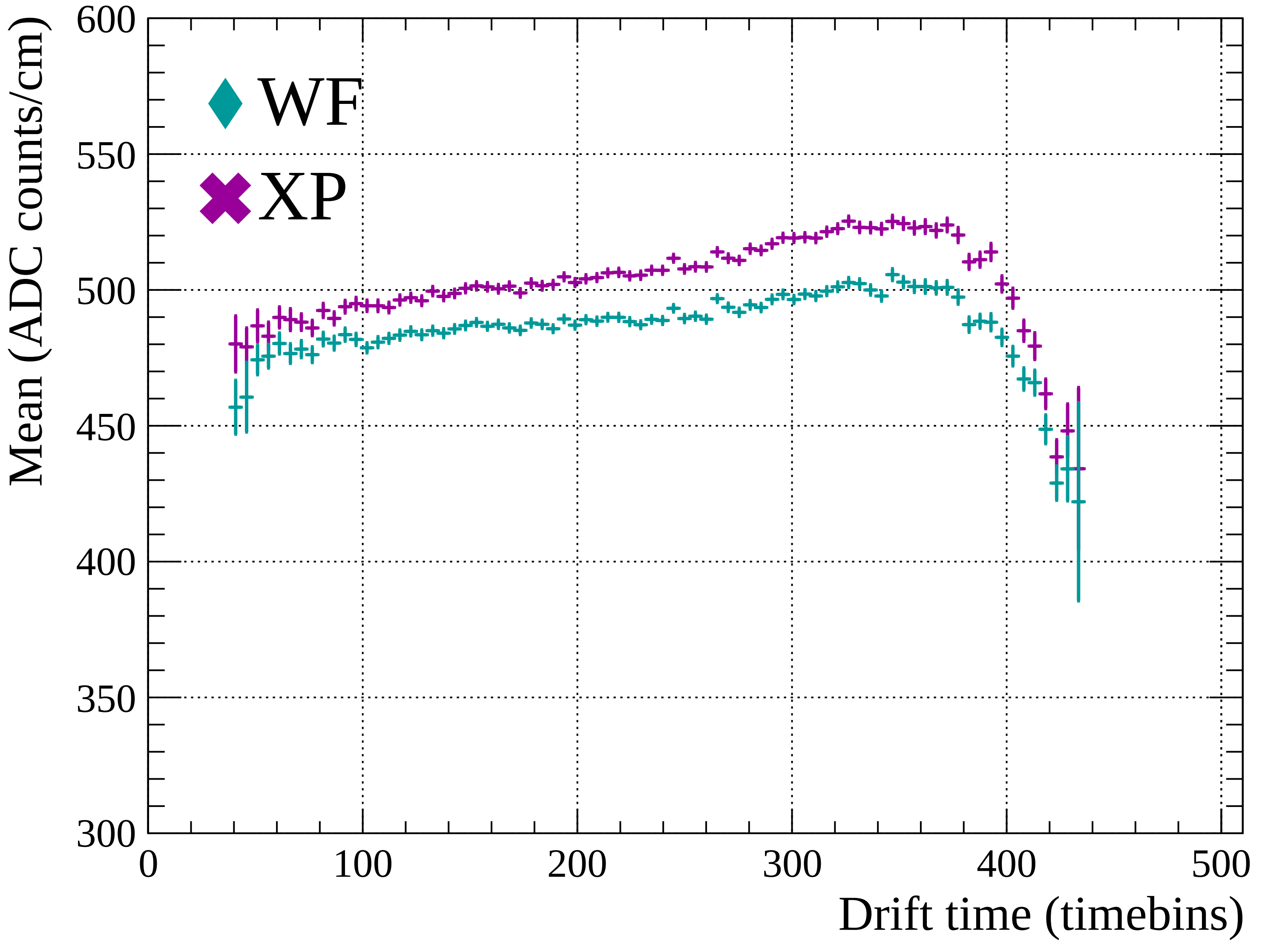


y = 0.03x + 480.67
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50
60

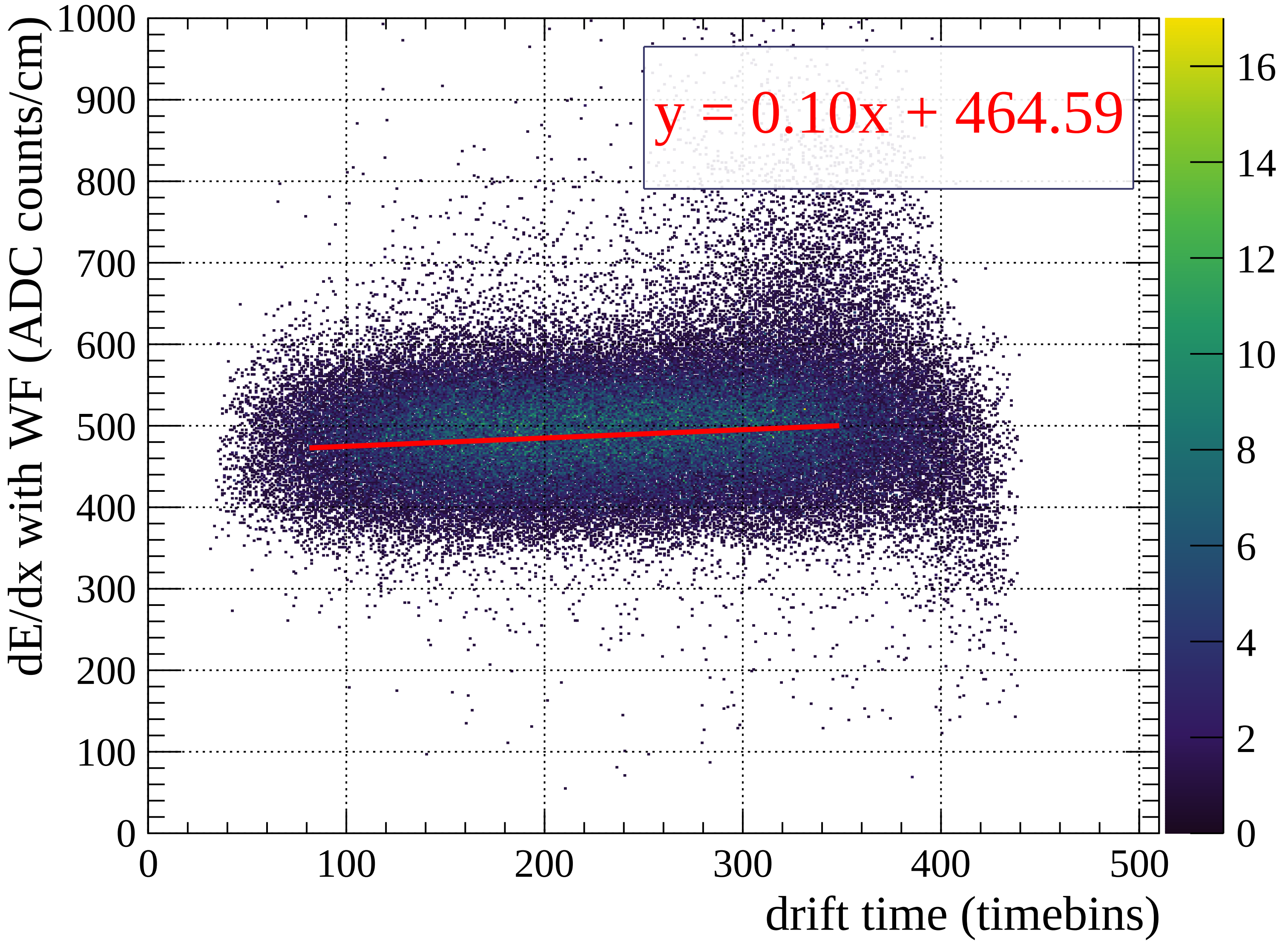

y = -0.02x + 520.36
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
10
20
30
40
50



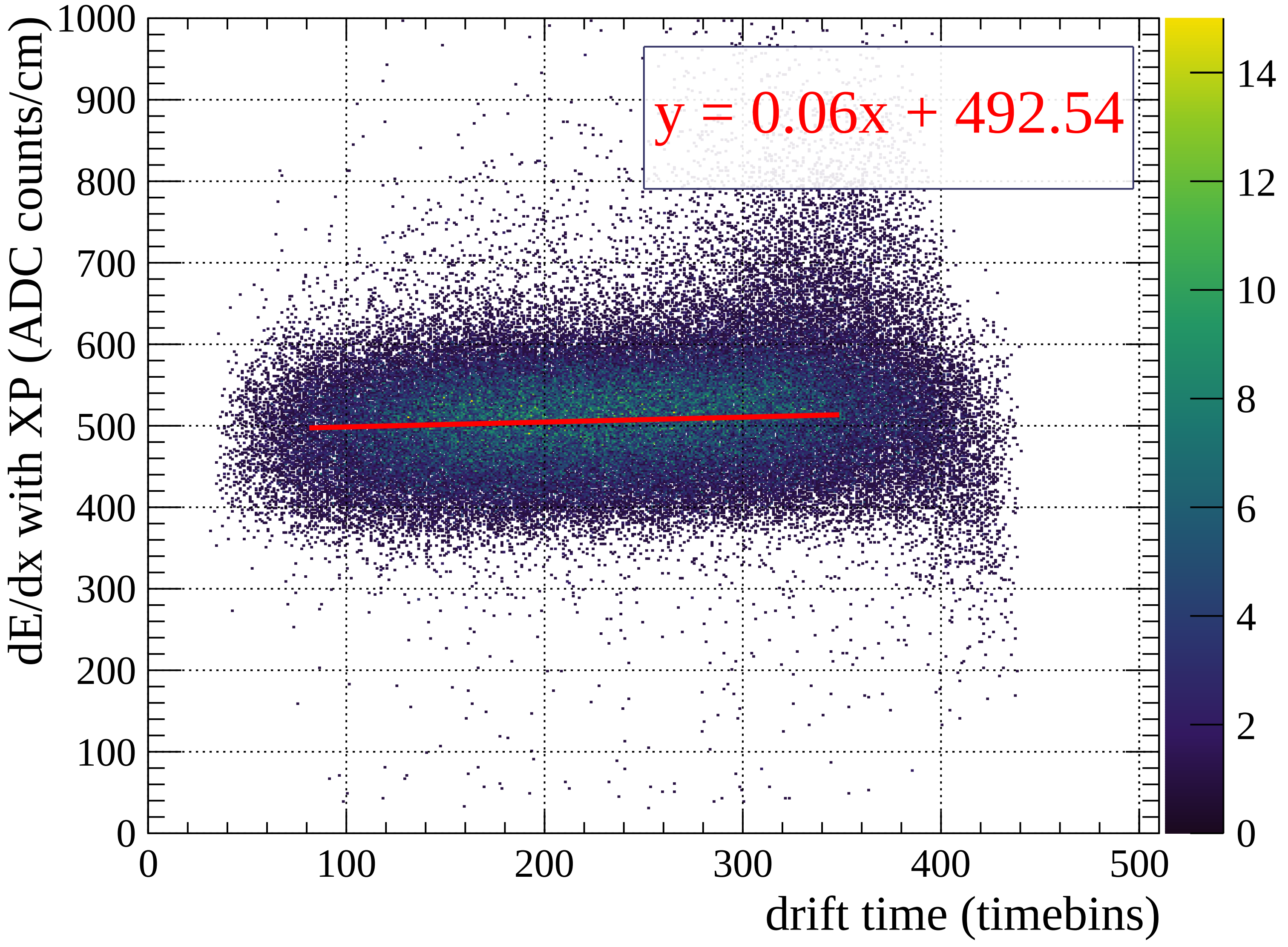
dE/dx resolution (%)
Drift time (timebins)
WF
XP
5
10
15
20
25
30
0
100
200
300
400
500

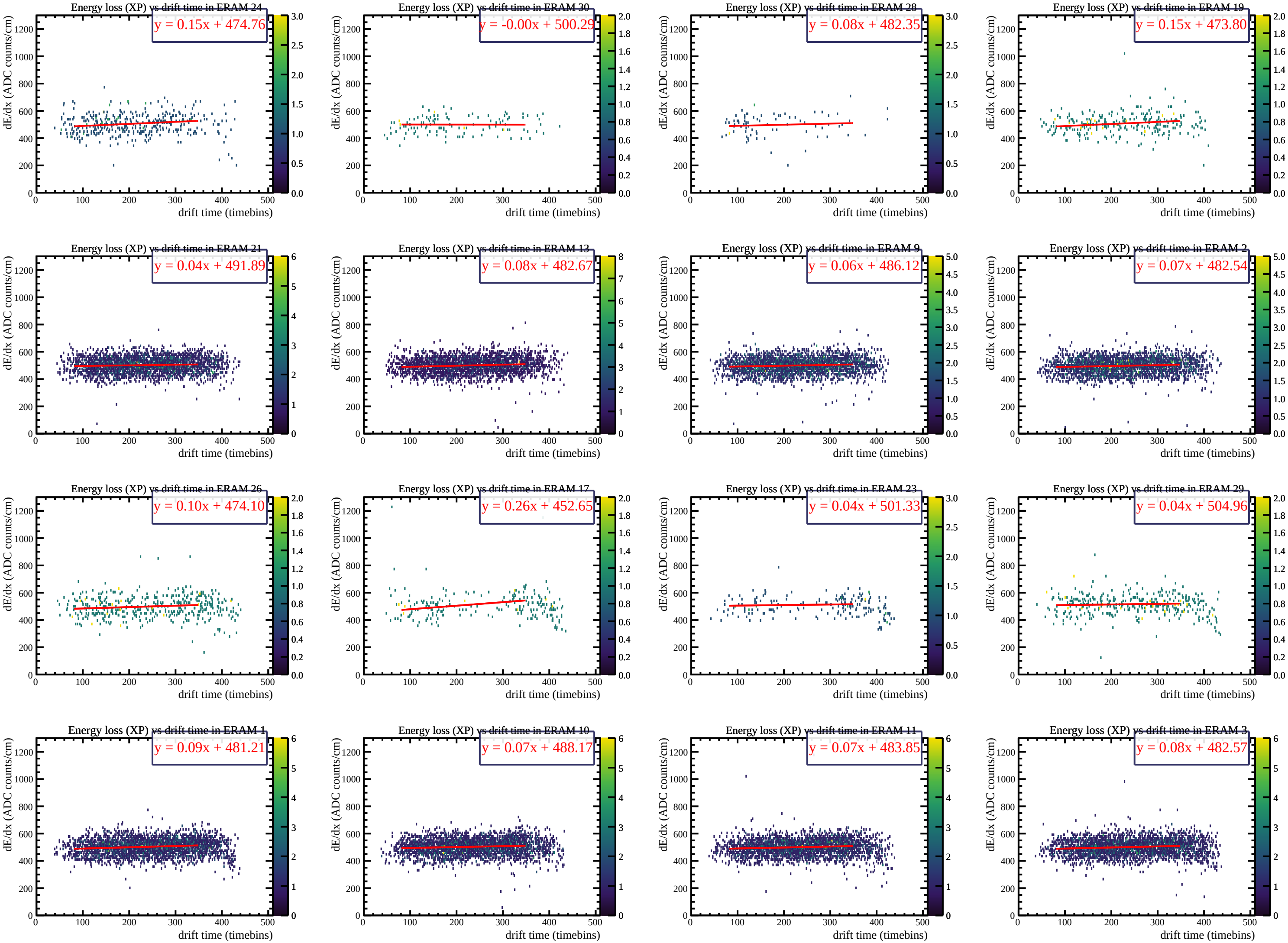


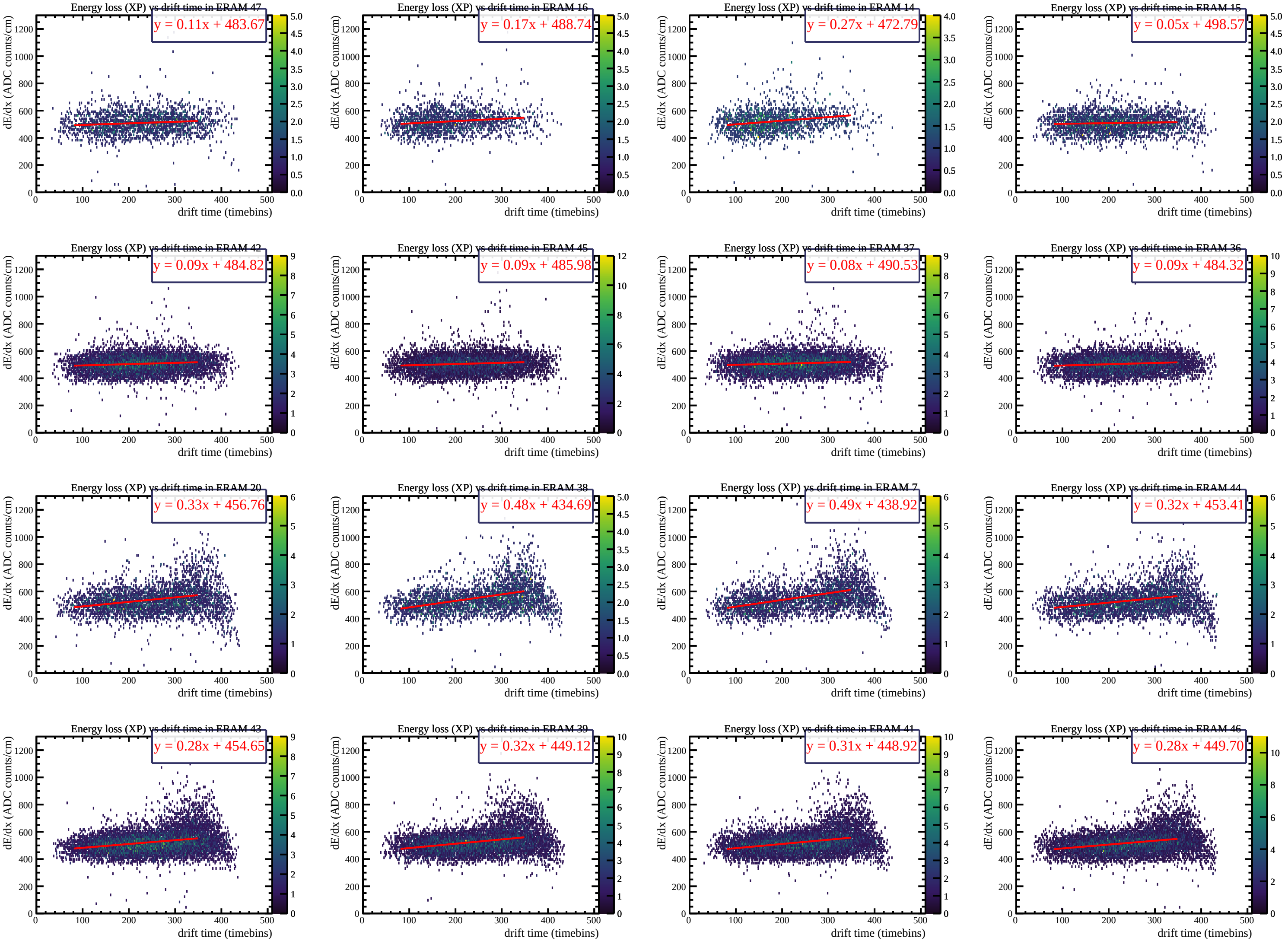
WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

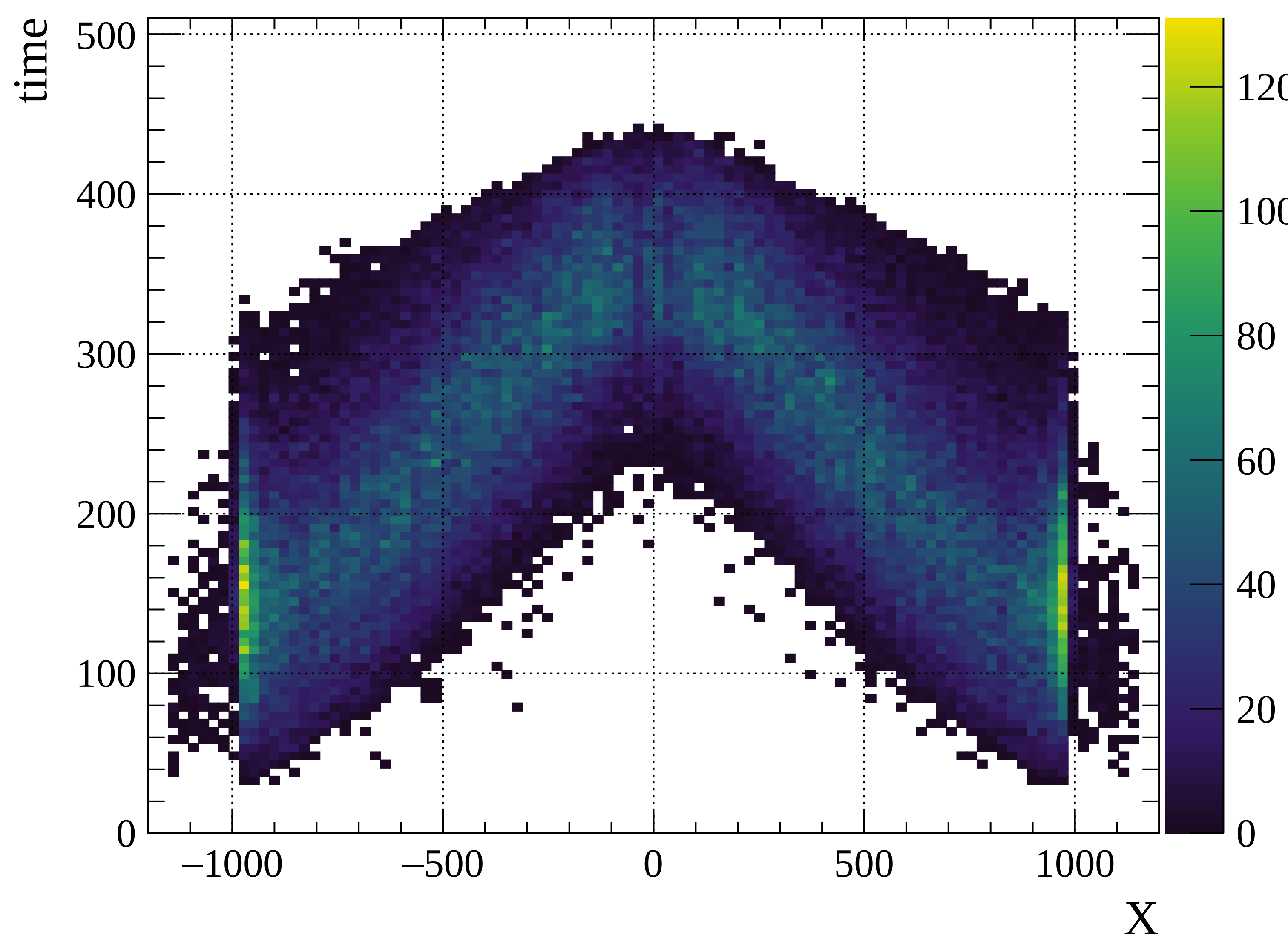


y = 0.10x + 464.59
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16

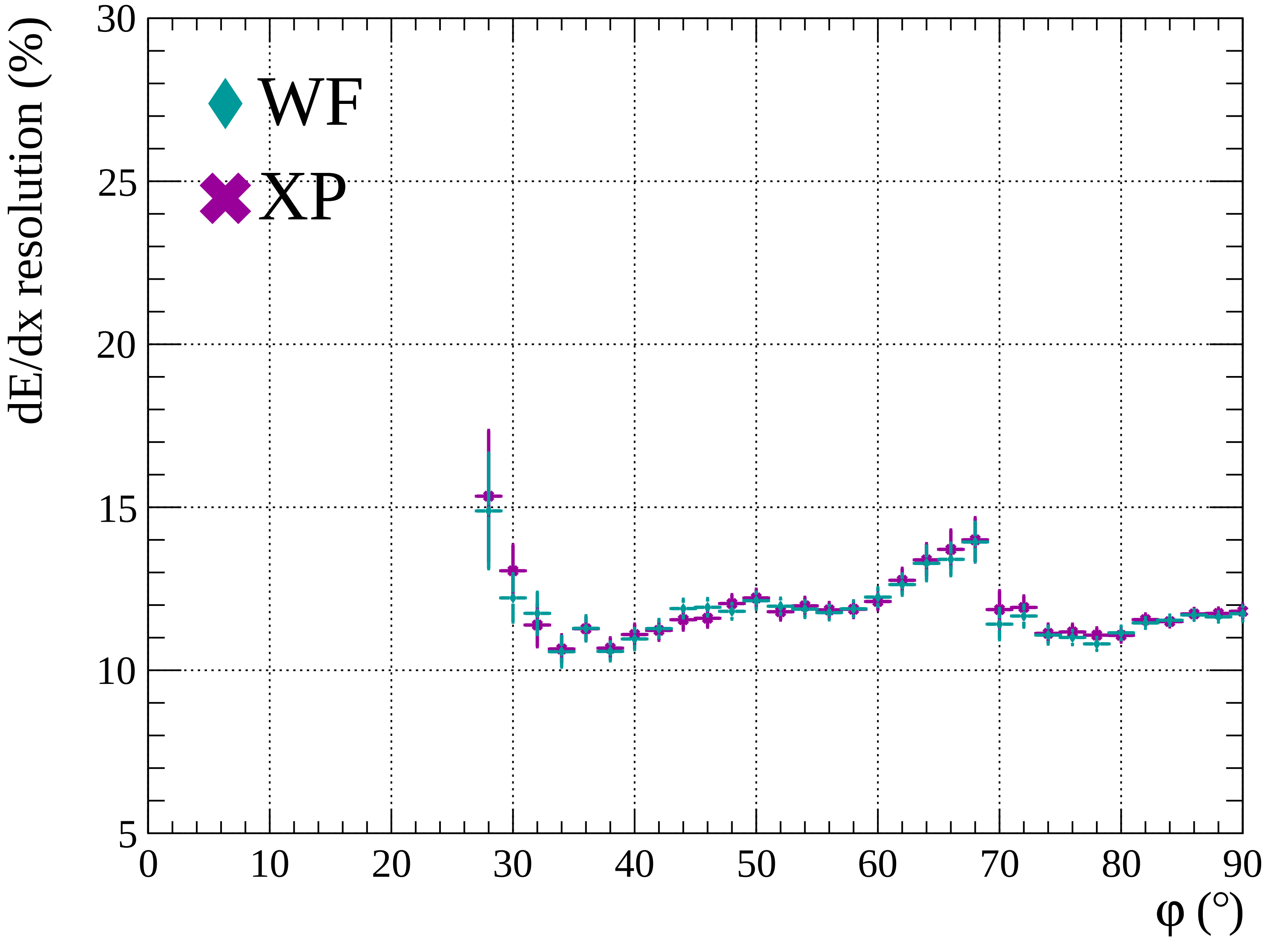

y = 0.06x + 492.54
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14

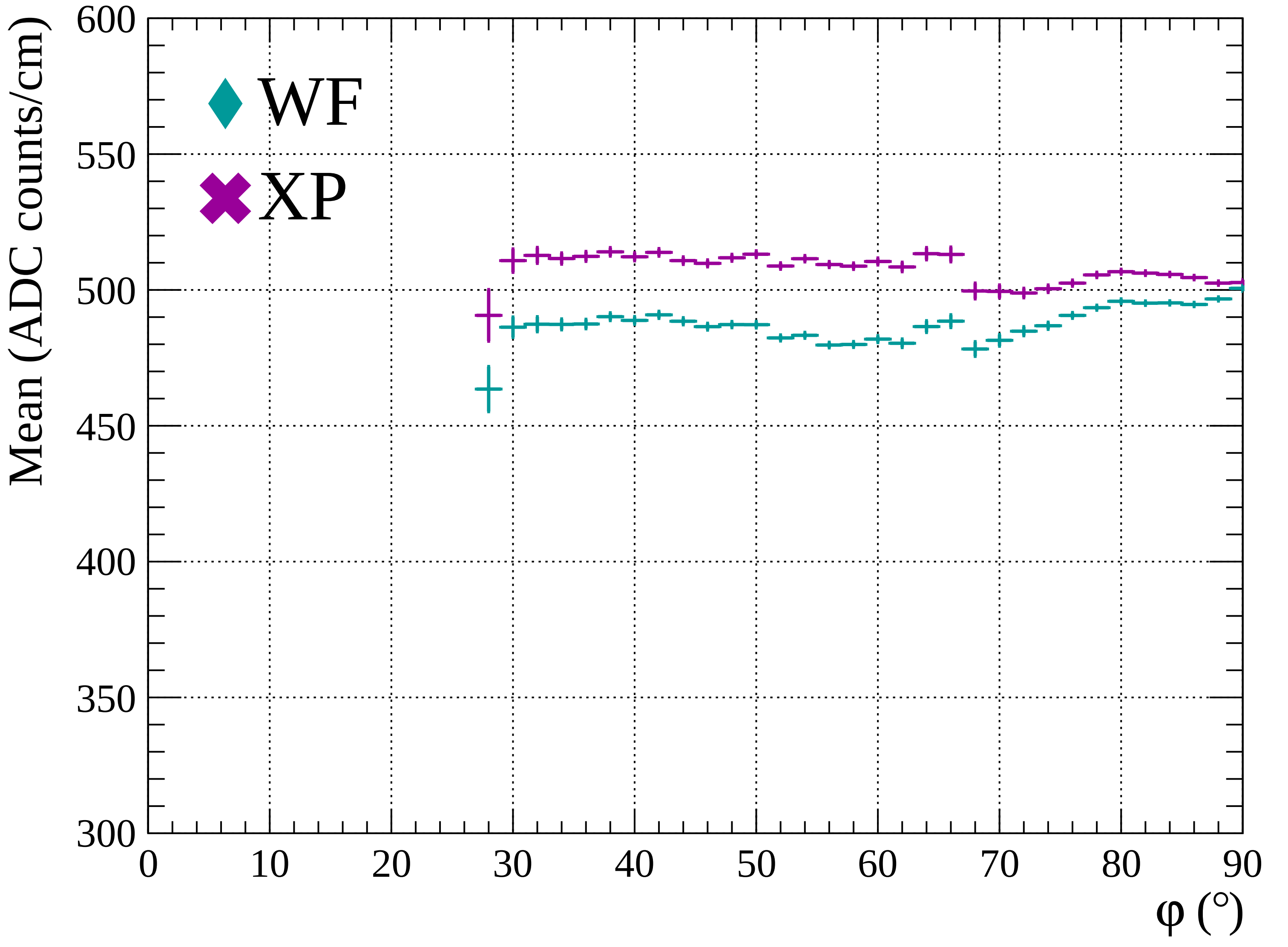

Energy loss (XP) vs drift time in ERAM 24
y = 0.15x + 474.76
Energy loss (XP) vs drift time in ERAM 30
y = -0.00x + 500.29
Energy loss (XP) vs drift time in ERAM 28
y = 0.08x + 482.35
Energy loss (XP) vs drift time in ERAM 19
y = 0.15x + 473.80
Energy loss (XP) vs drift time in ERAM 21
y = 0.04x + 491.89
Energy loss (XP) vs drift time in ERAM 13
y = 0.08x + 482.67
Energy loss (XP) vs drift time in ERAM 9
y = 0.06x + 486.12
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 482.54
Energy loss (XP) vs drift time in ERAM 26
y = 0.10x + 474.10
Energy loss (XP) vs drift time in ERAM 17
y = 0.26x + 452.65
Energy loss (XP) vs drift time in ERAM 23
y = 0.04x + 501.33
Energy loss (XP) vs drift time in ERAM 29
y = 0.04x + 504.96
Energy loss (XP) vs drift time in ERAM 1
y = 0.09x + 481.21
Energy loss (XP) vs drift time in ERAM 10
y = 0.07x + 488.17
Energy loss (XP) vs drift time in ERAM 11
y = 0.07x + 483.85
Energy loss (XP) vs drift time in ERAM 3
y = 0.08x + 482.57
dE/dx (ADC counts/cm)
drift time (timebins)

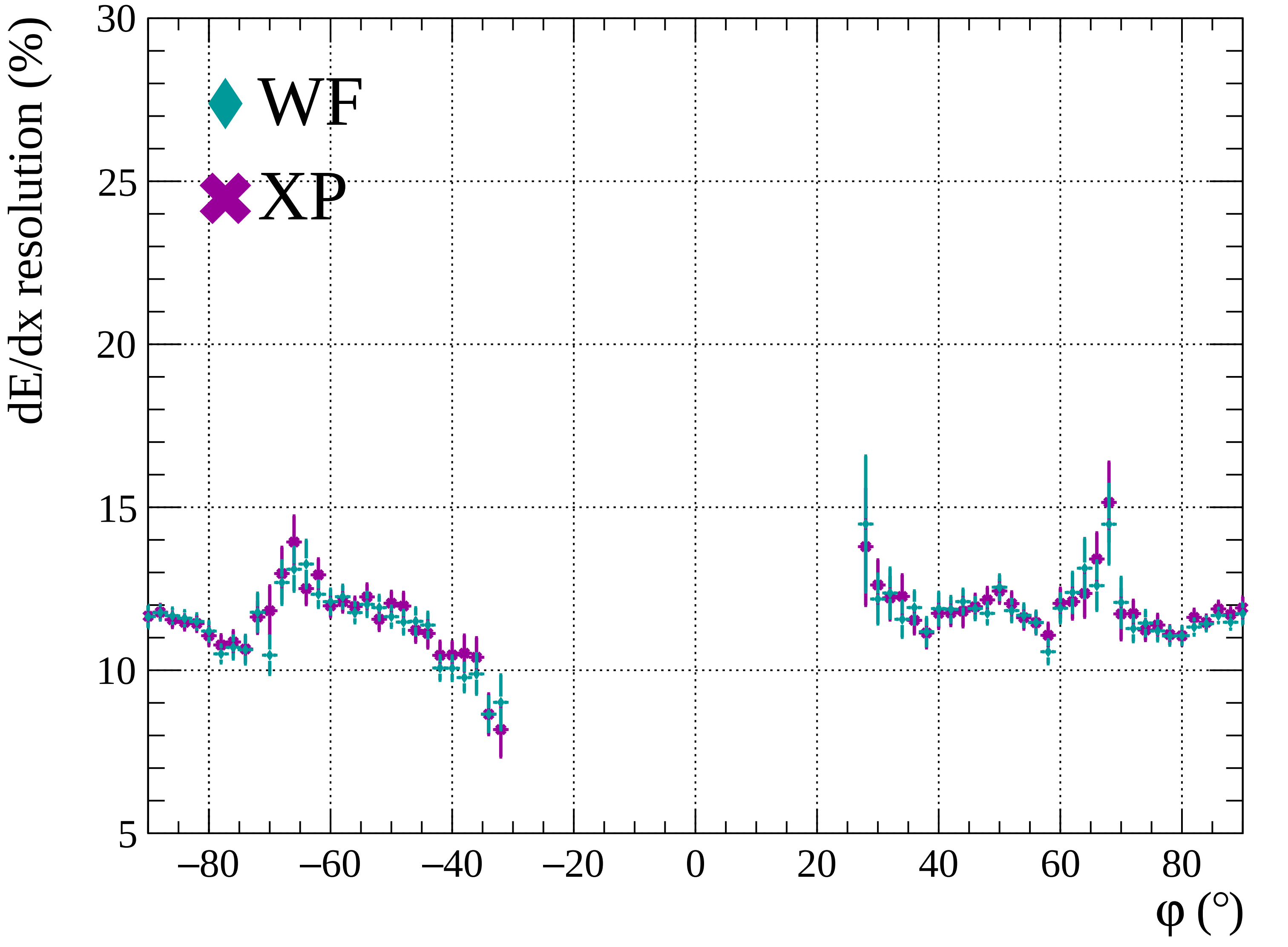

Energy loss (XP) vs drift time in ERAM 47
y = 0.11x + 483.67
Energy loss (XP) vs drift time in ERAM 16
y = 0.17x + 488.74
Energy loss (XP) vs drift time in ERAM 14
y = 0.27x + 472.79
Energy loss (XP) vs drift time in ERAM 15
y = 0.05x + 498.57
Energy loss (XP) vs drift time in ERAM 42
y = 0.09x + 484.82
Energy loss (XP) vs drift time in ERAM 45
y = 0.09x + 485.98
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 490.53
Energy loss (XP) vs drift time in ERAM 36
y = 0.09x + 484.32
Energy loss (XP) vs drift time in ERAM 20
y = 0.33x + 456.76
Energy loss (XP) vs drift time in ERAM 38
y = 0.48x + 434.69
Energy loss (XP) vs drift time in ERAM 7
y = 0.49x + 438.92
Energy loss (XP) vs drift time in ERAM 44
y = 0.32x + 453.41
Energy loss (XP) vs drift time in ERAM 43
y = 0.28x + 454.65
Energy loss (XP) vs drift time in ERAM 39
y = 0.32x + 449.12
Energy loss (XP) vs drift time in ERAM 41
y = 0.31x + 448.92
Energy loss (XP) vs drift time in ERAM 46
y = 0.28x + 449.70
dE/dx (ADC counts/cm)
drift time (timebins)



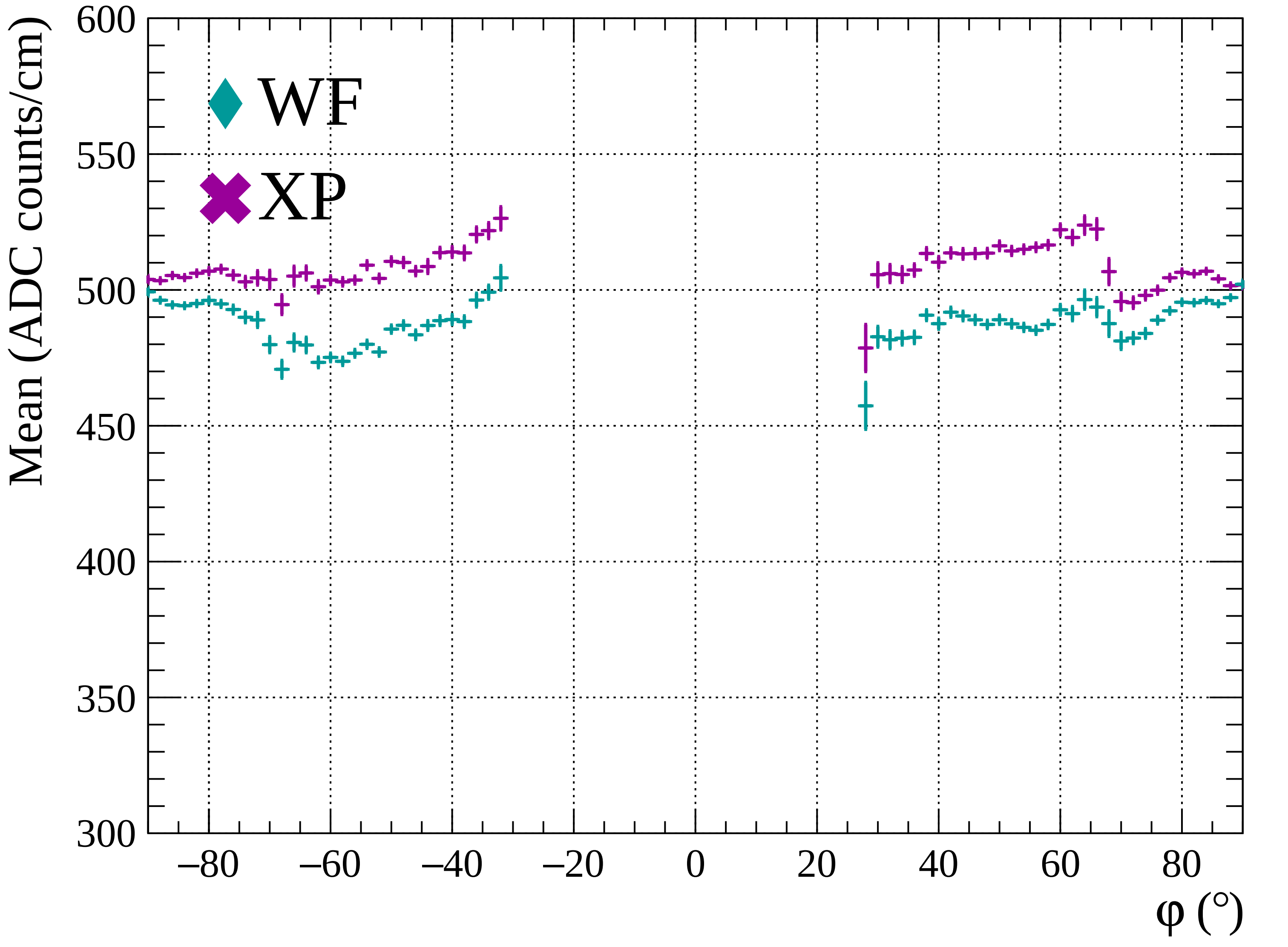
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
120
100
80
60
40
20
0


dE/dx resolution (%)
φ (°)
WF
XP
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90

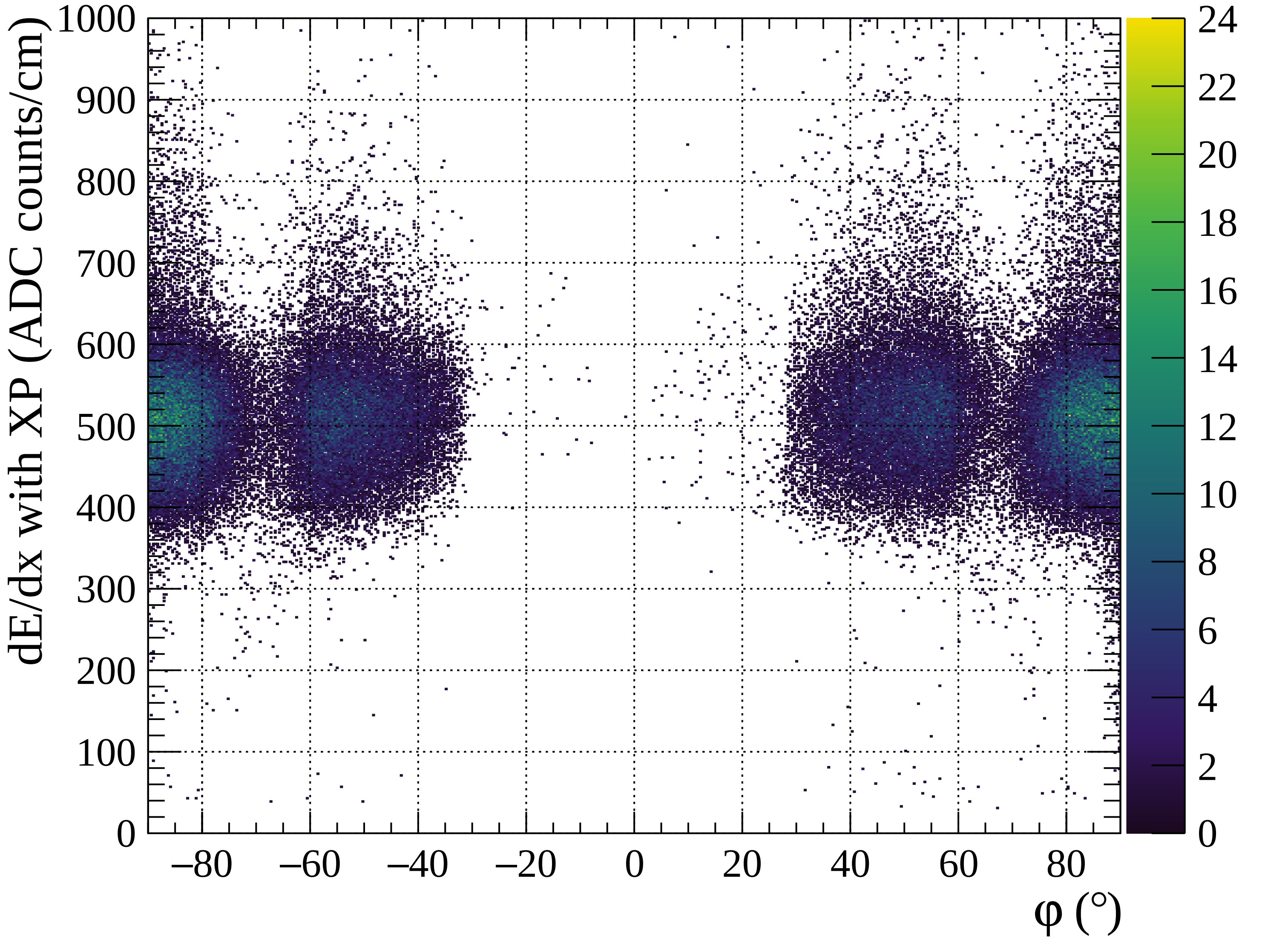

WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90



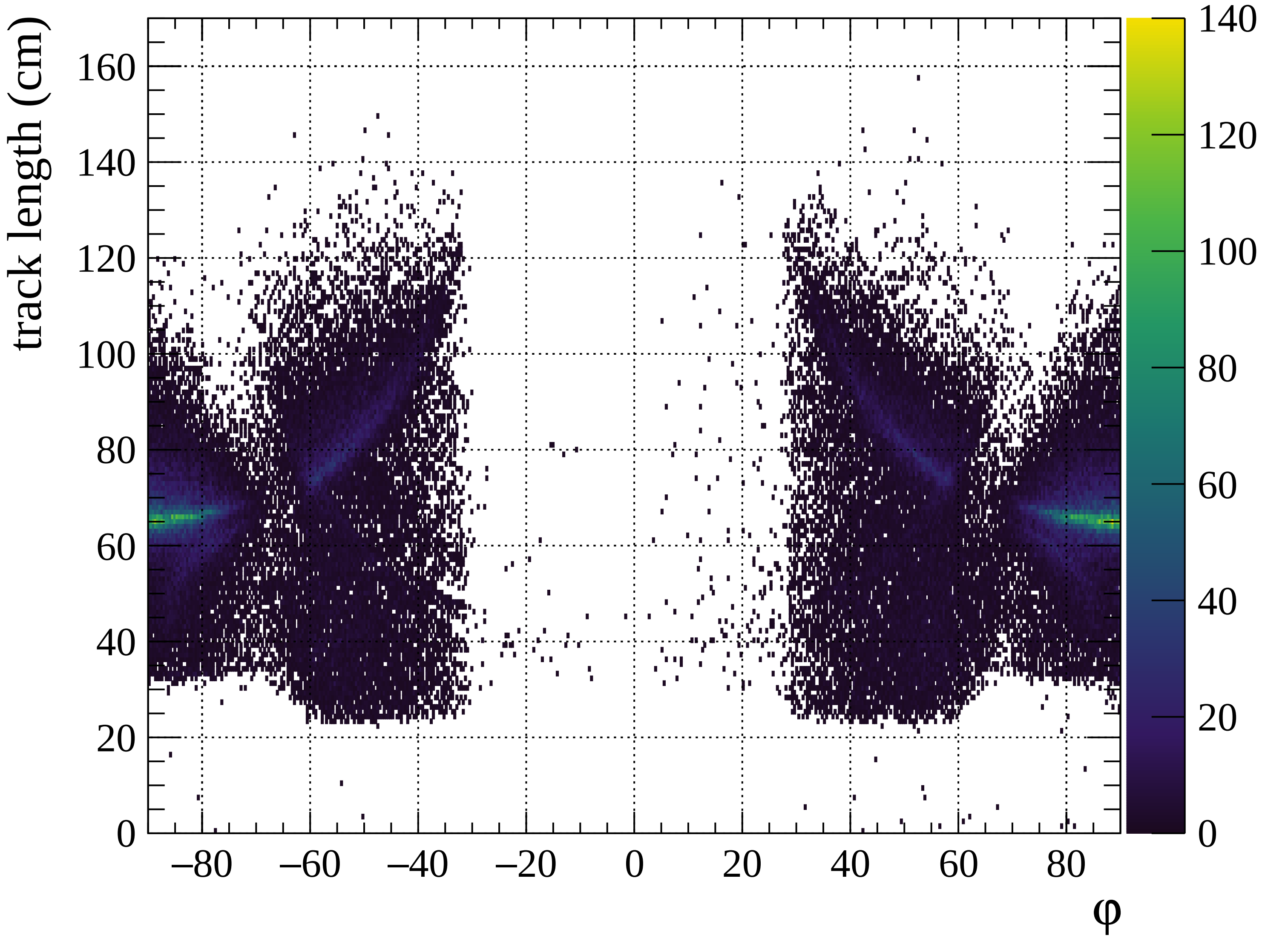
WF
XP
dE/dx resolution (%)
φ (°)
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80



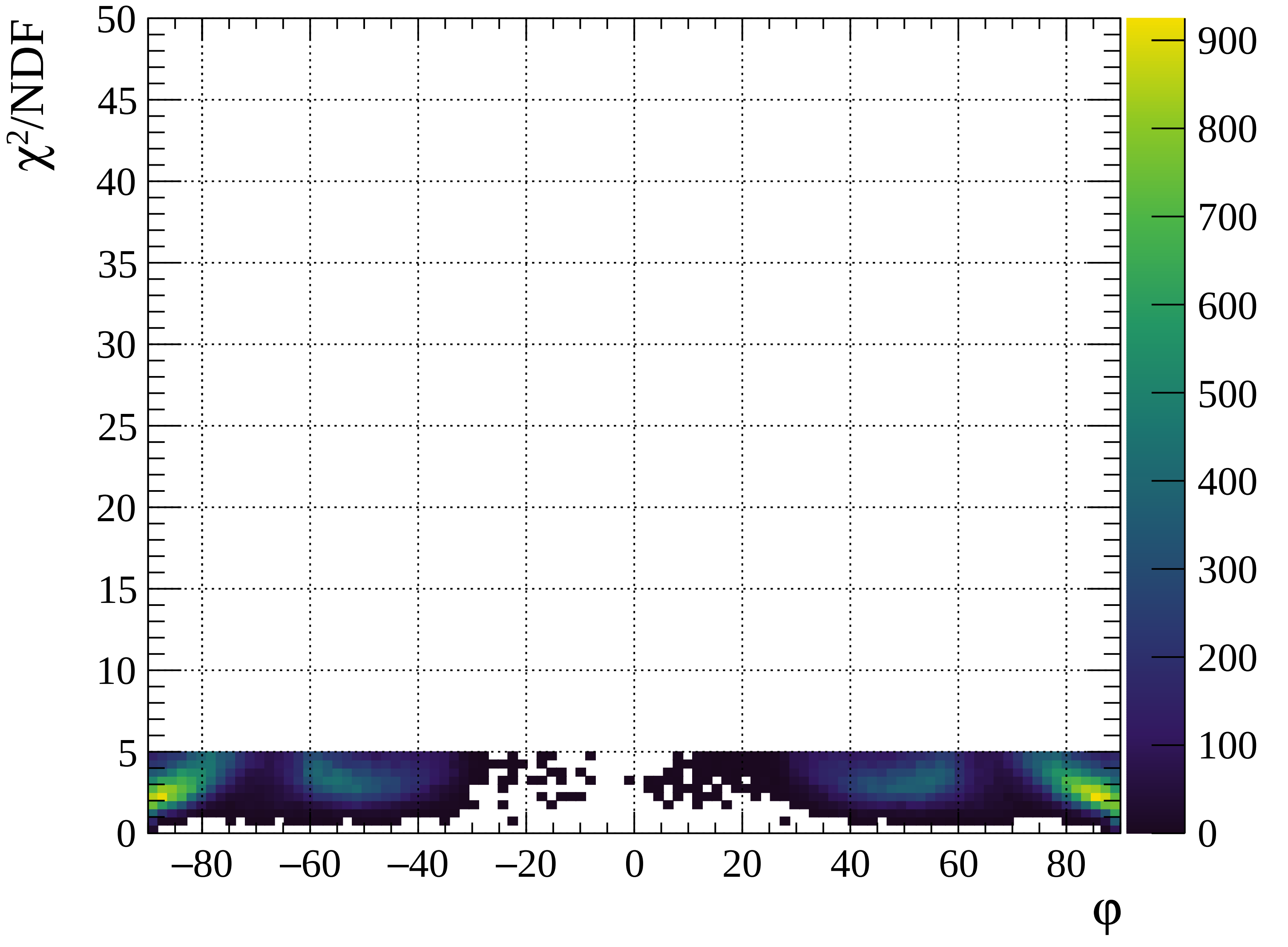
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)



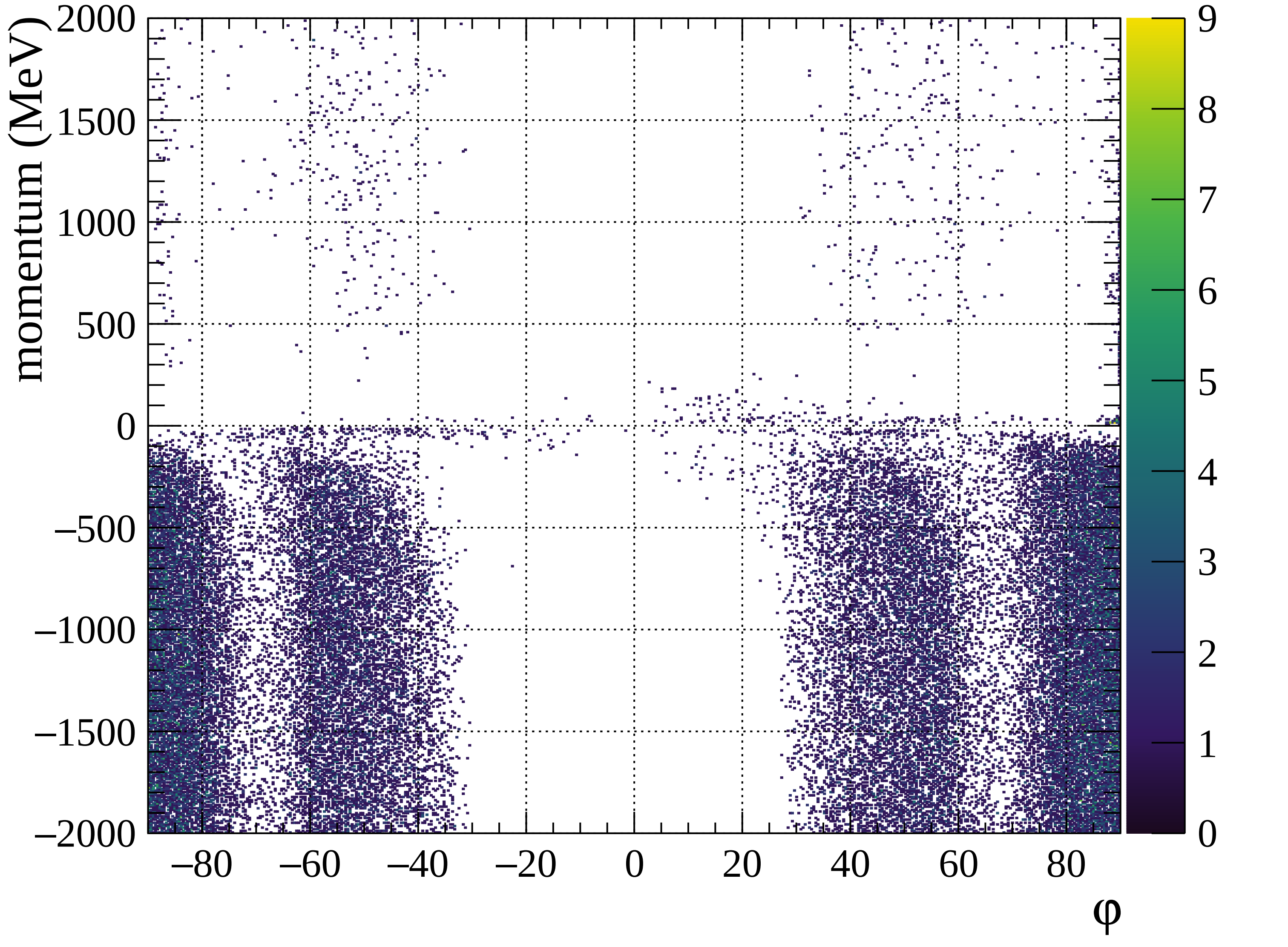
dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
24
22
20
18
16
14
12
10
8
6
4
2
0



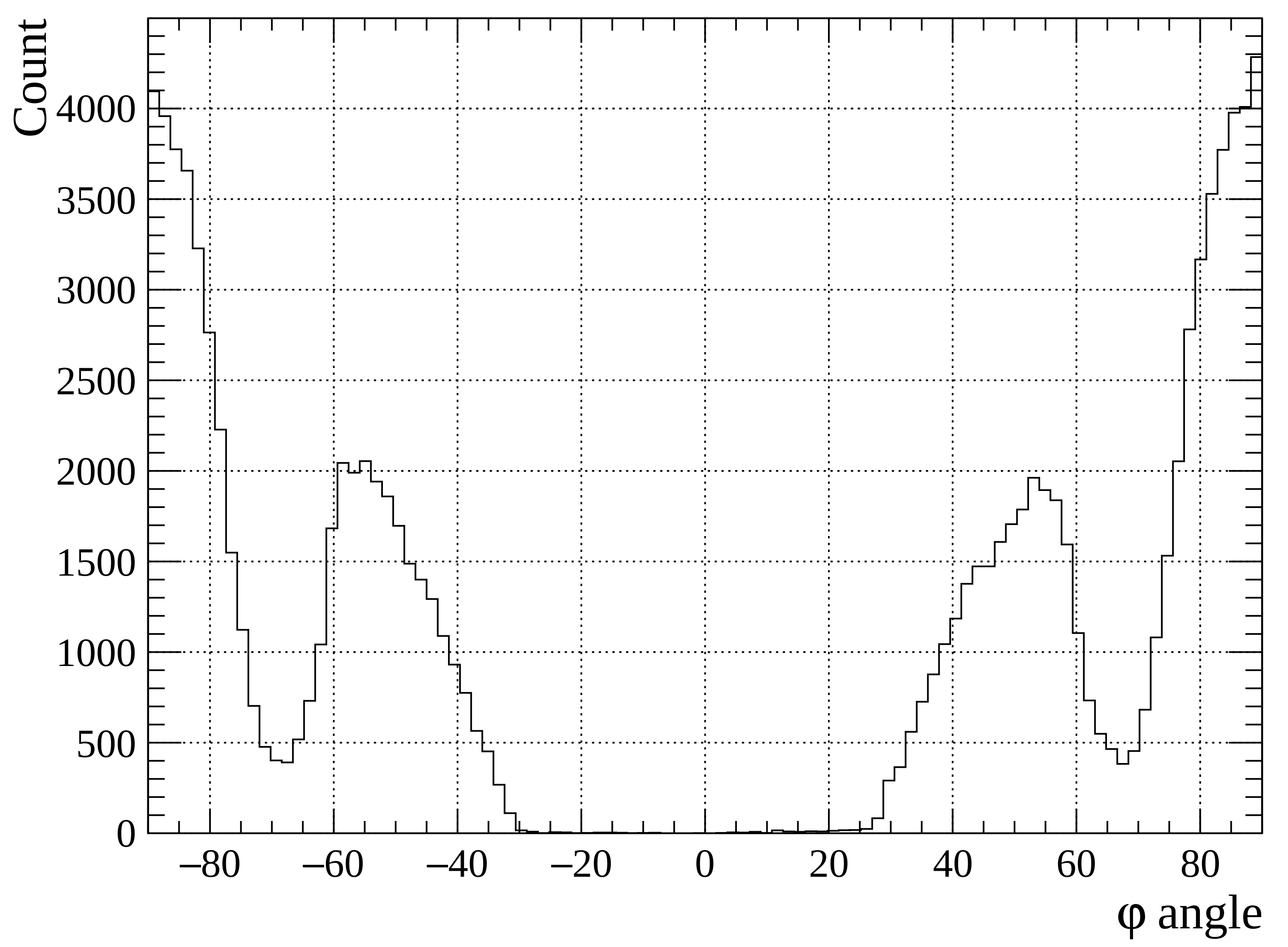
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
140
120
100
80
60
40
20
0

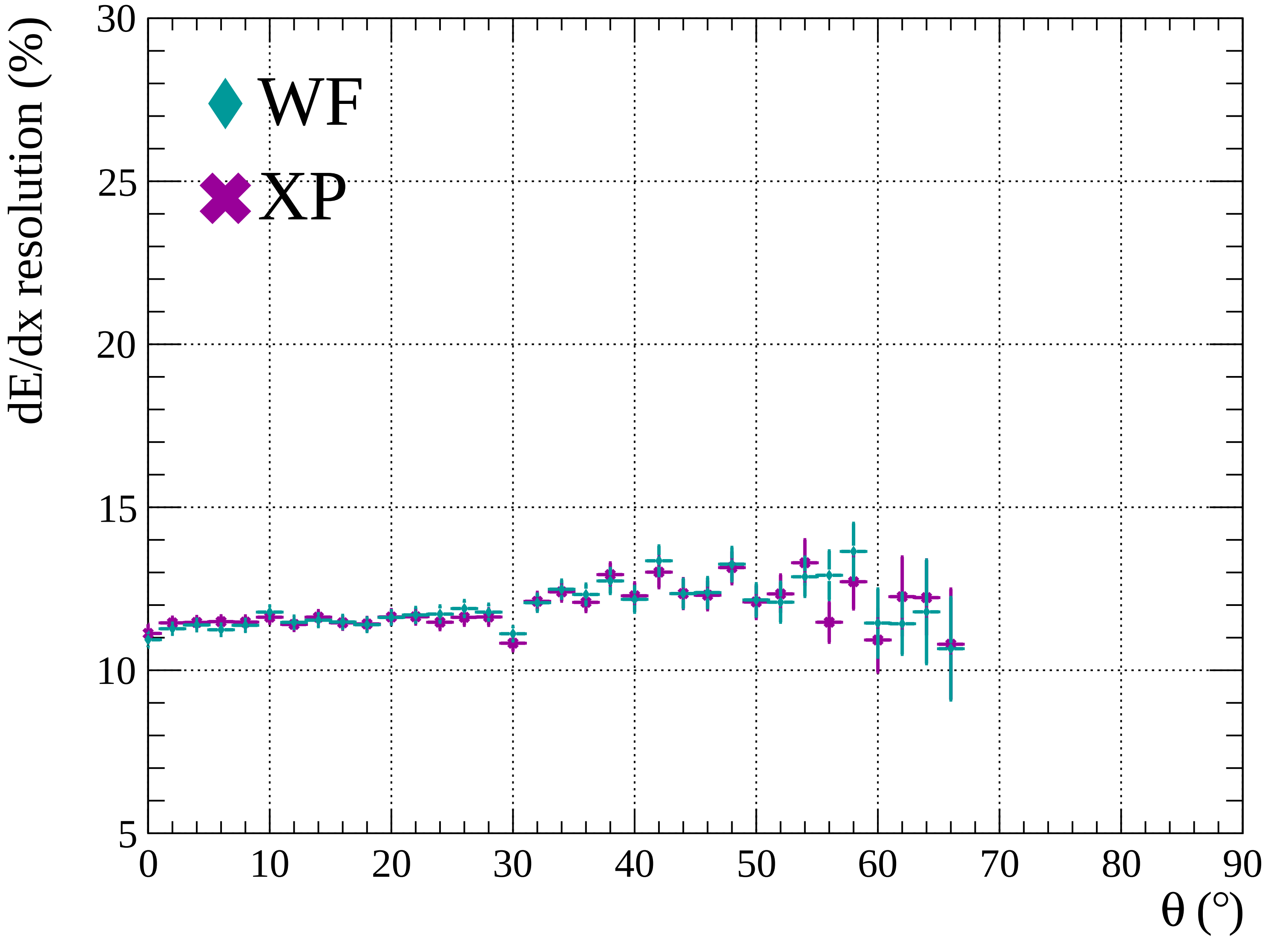


χ²/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
900
800
700
600
500
400
300
200
100
0

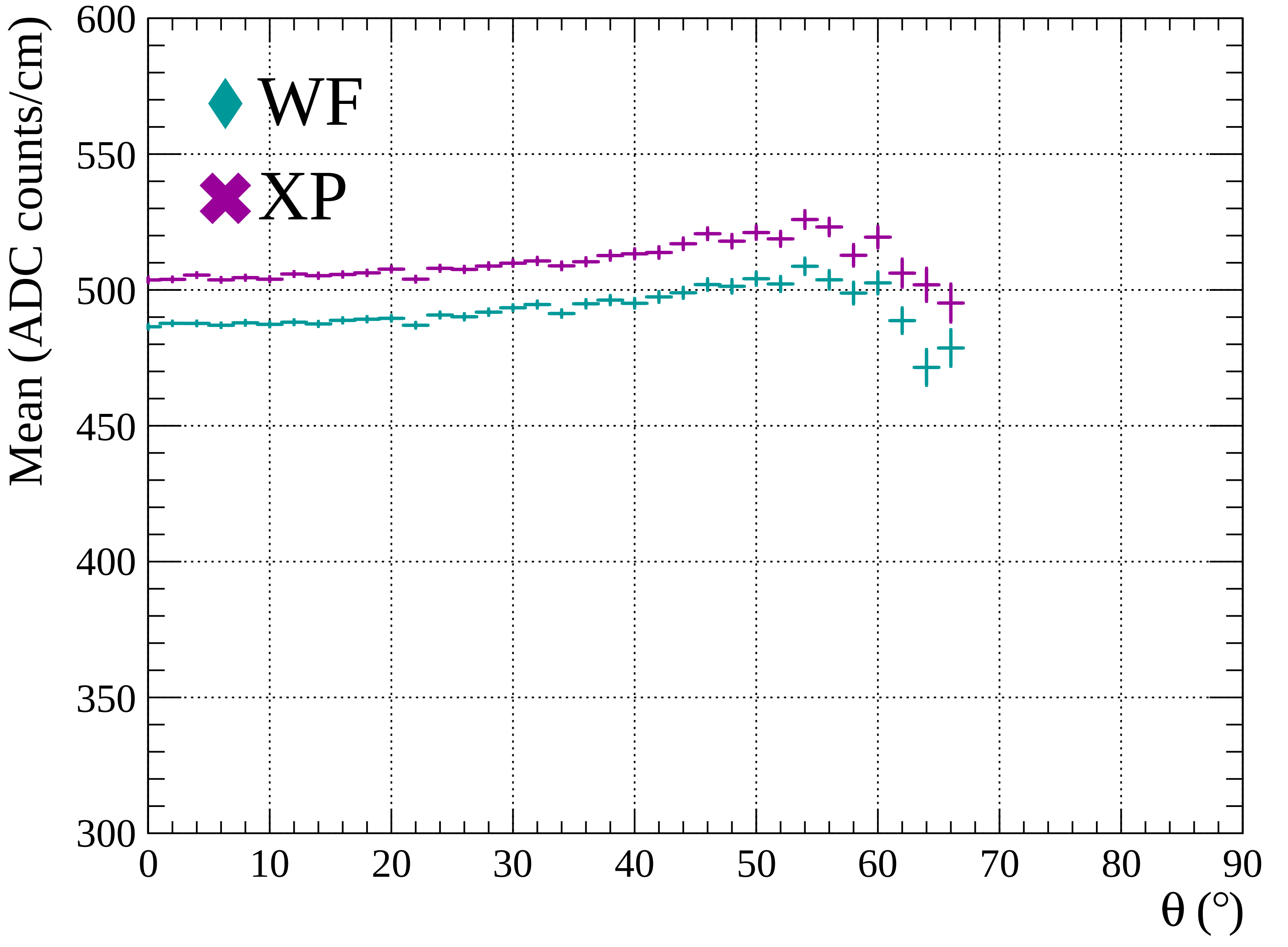


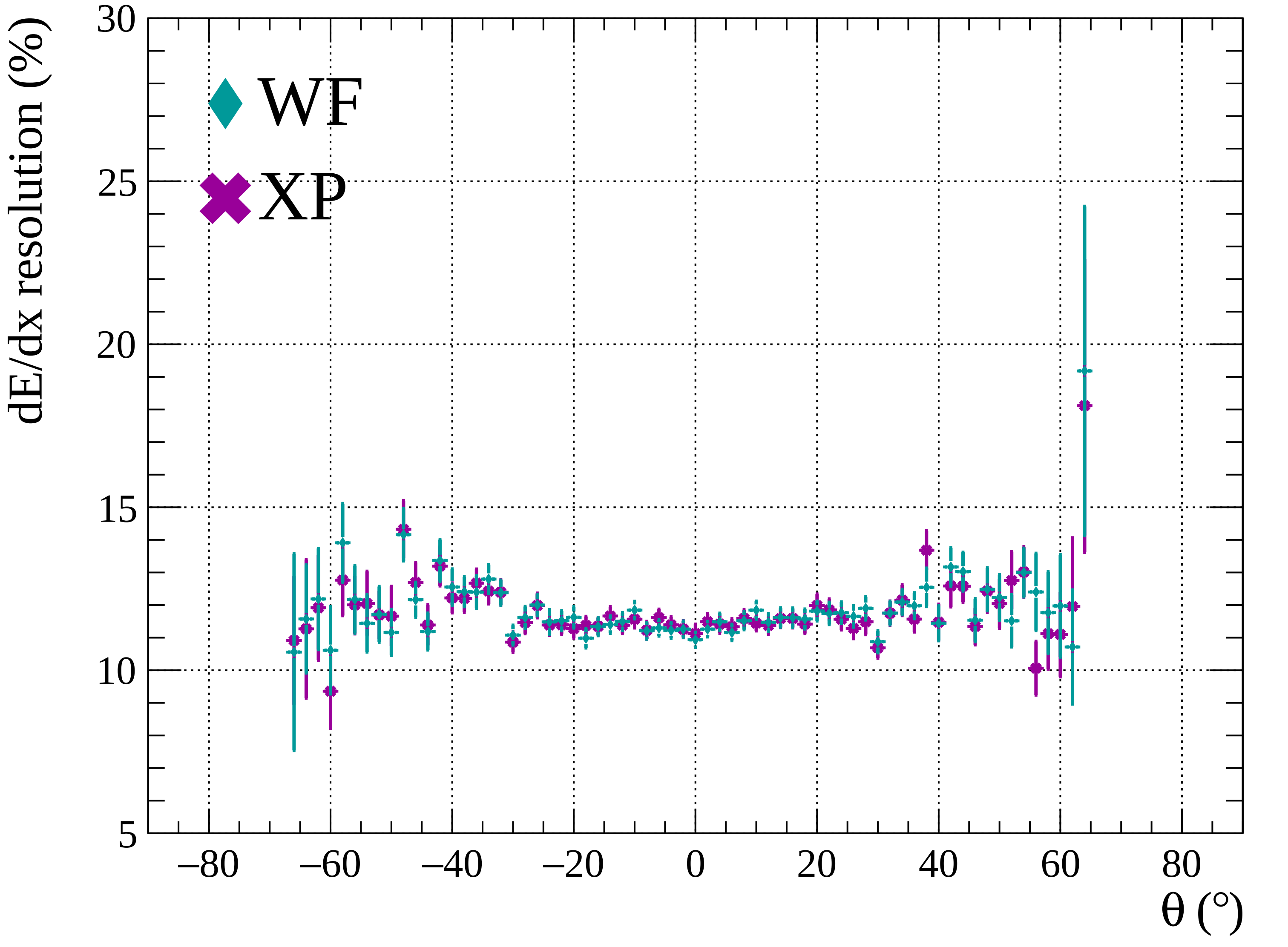
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
9
8
7
6
5
4
3
2
1
0


Count
4000
3500
3000
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

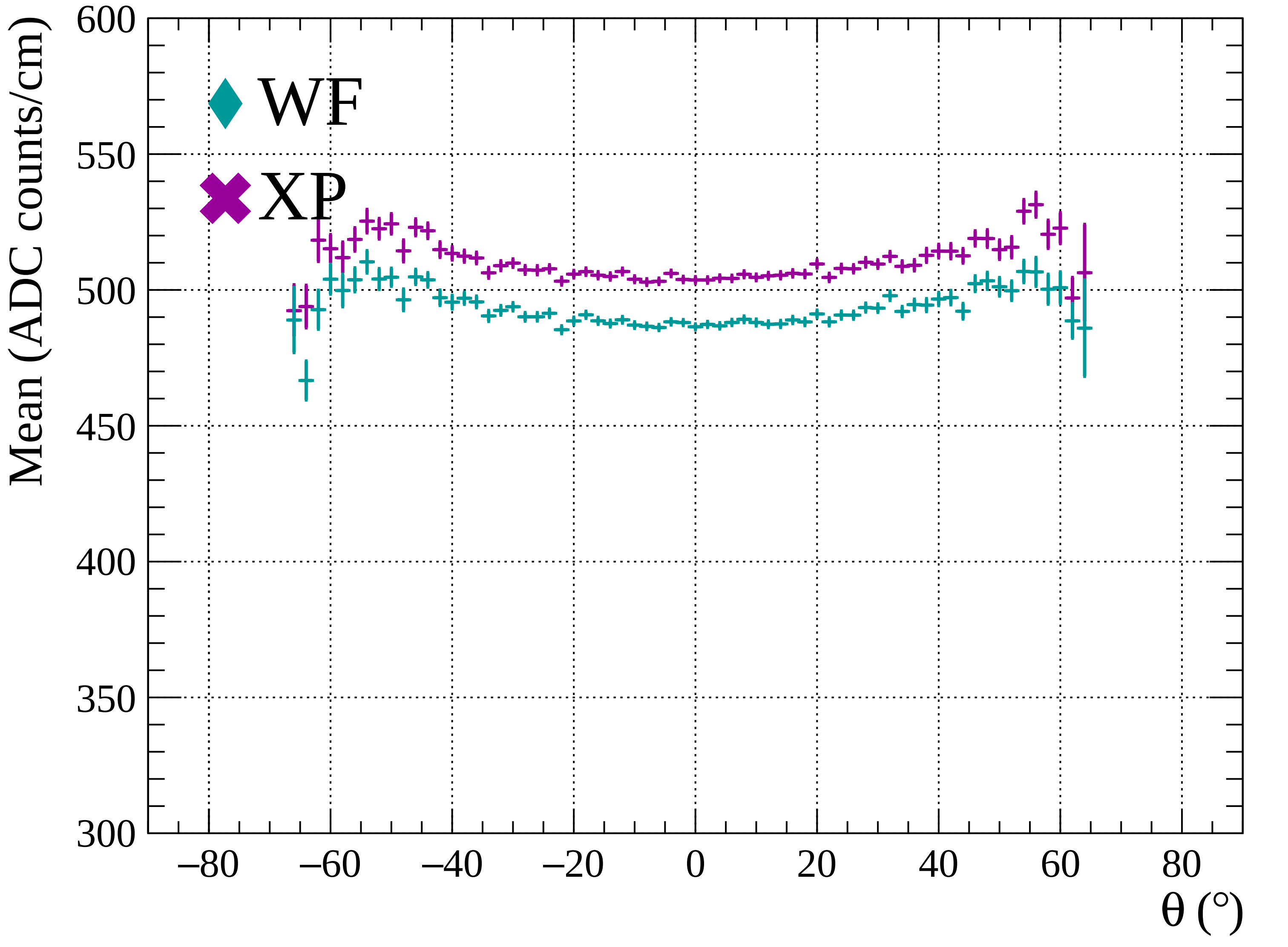


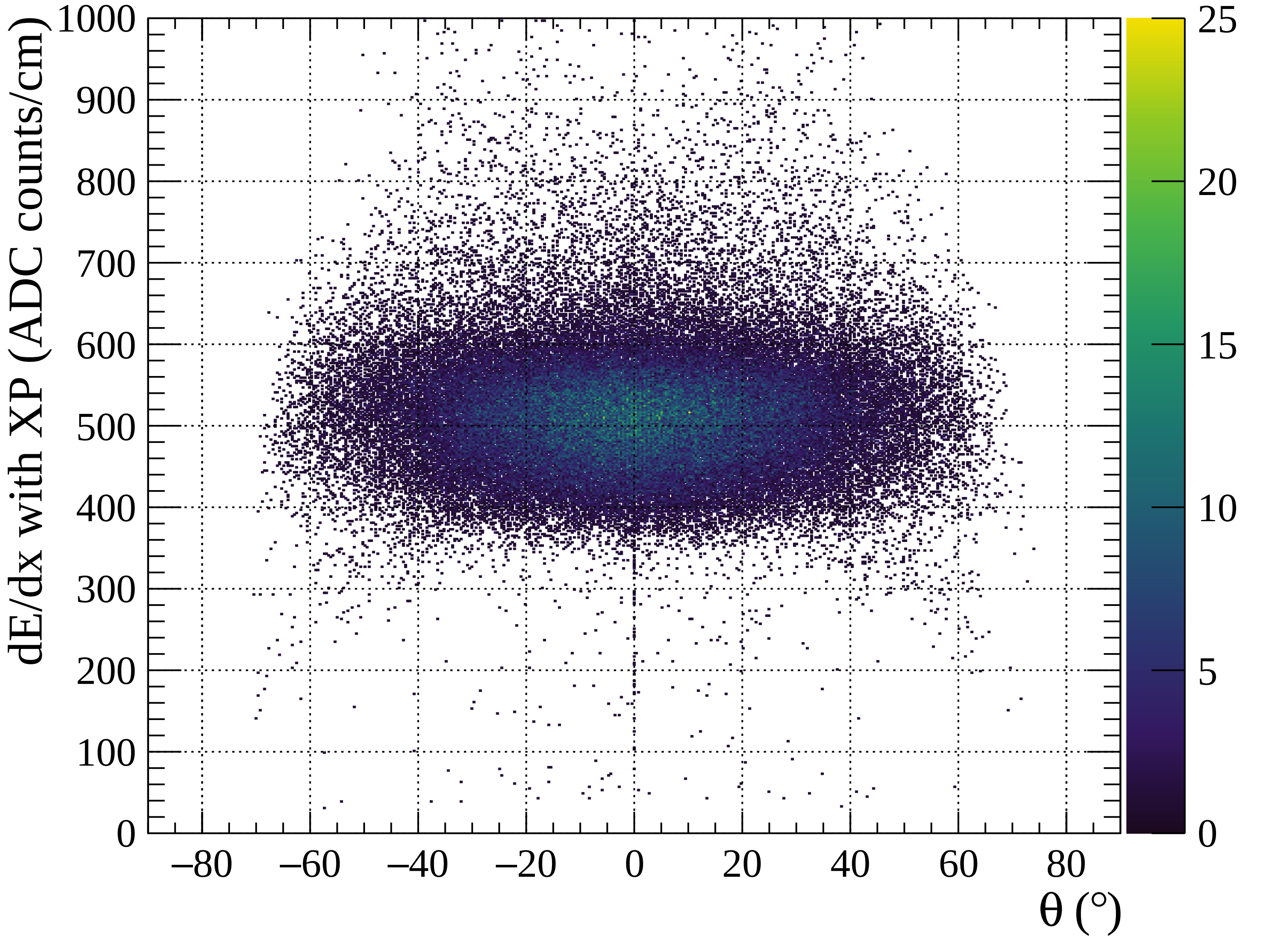
dE/dx resolution (%)
θ (°)
WF
XP
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90

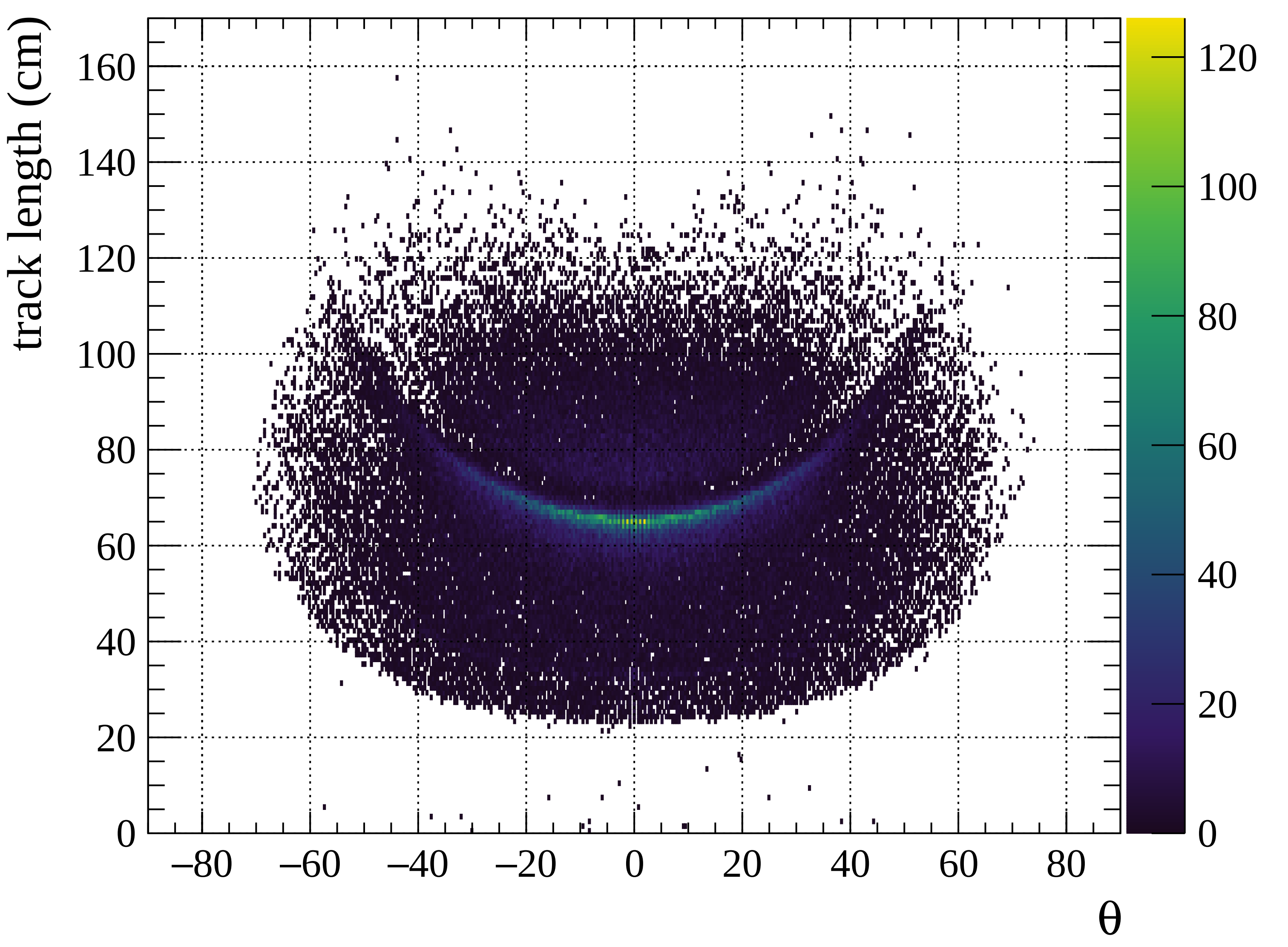

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)



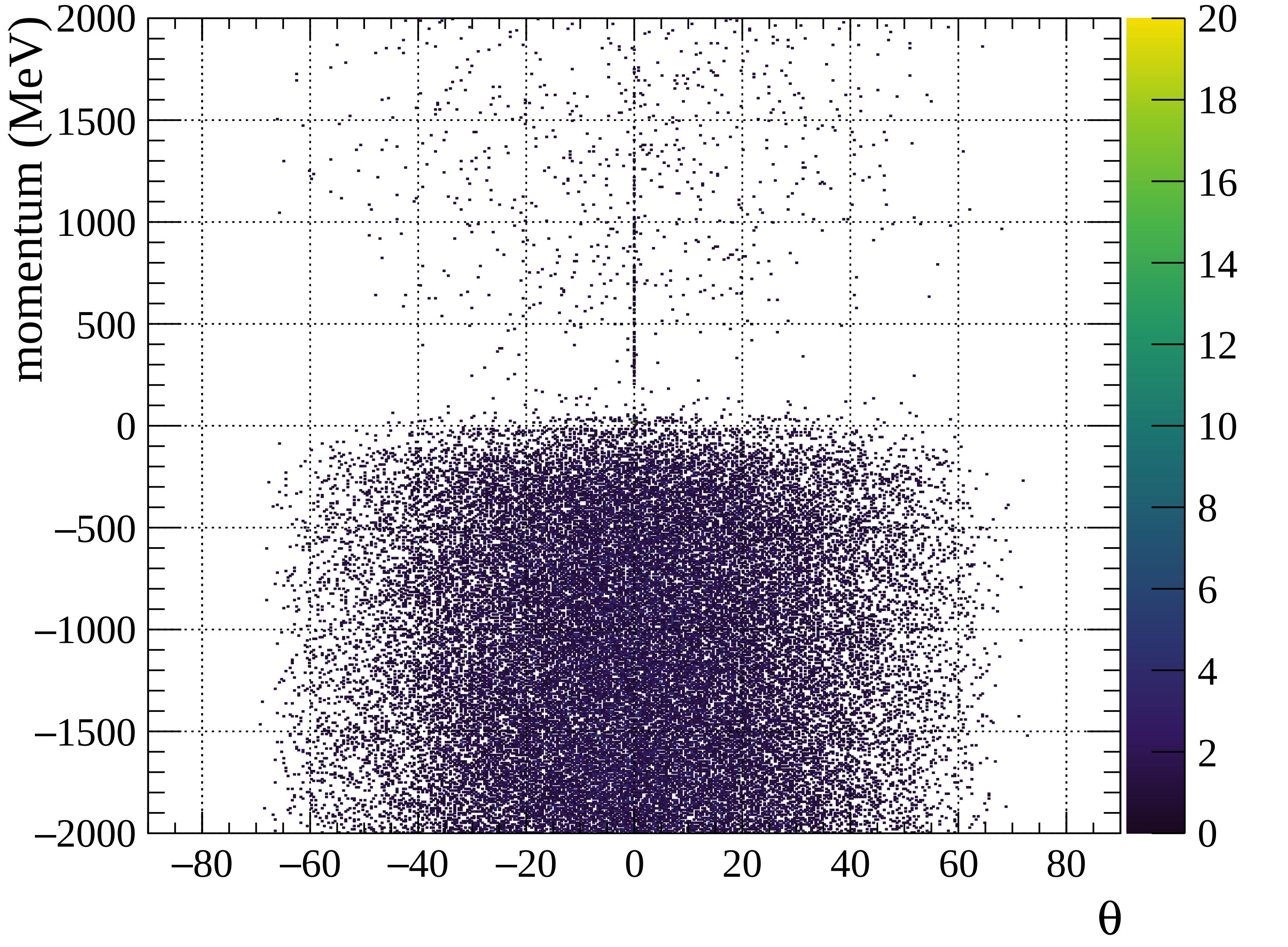
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)

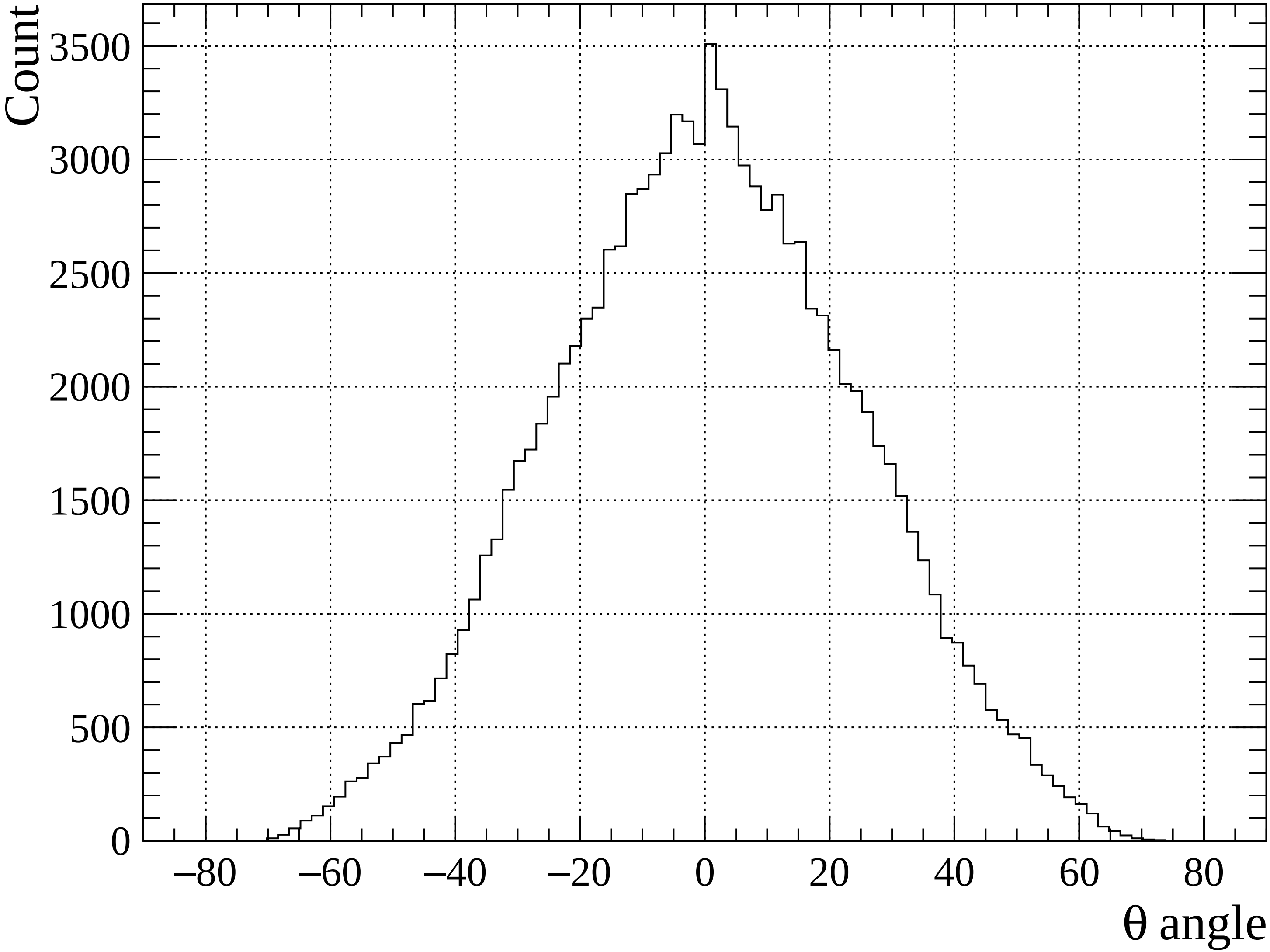

dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25



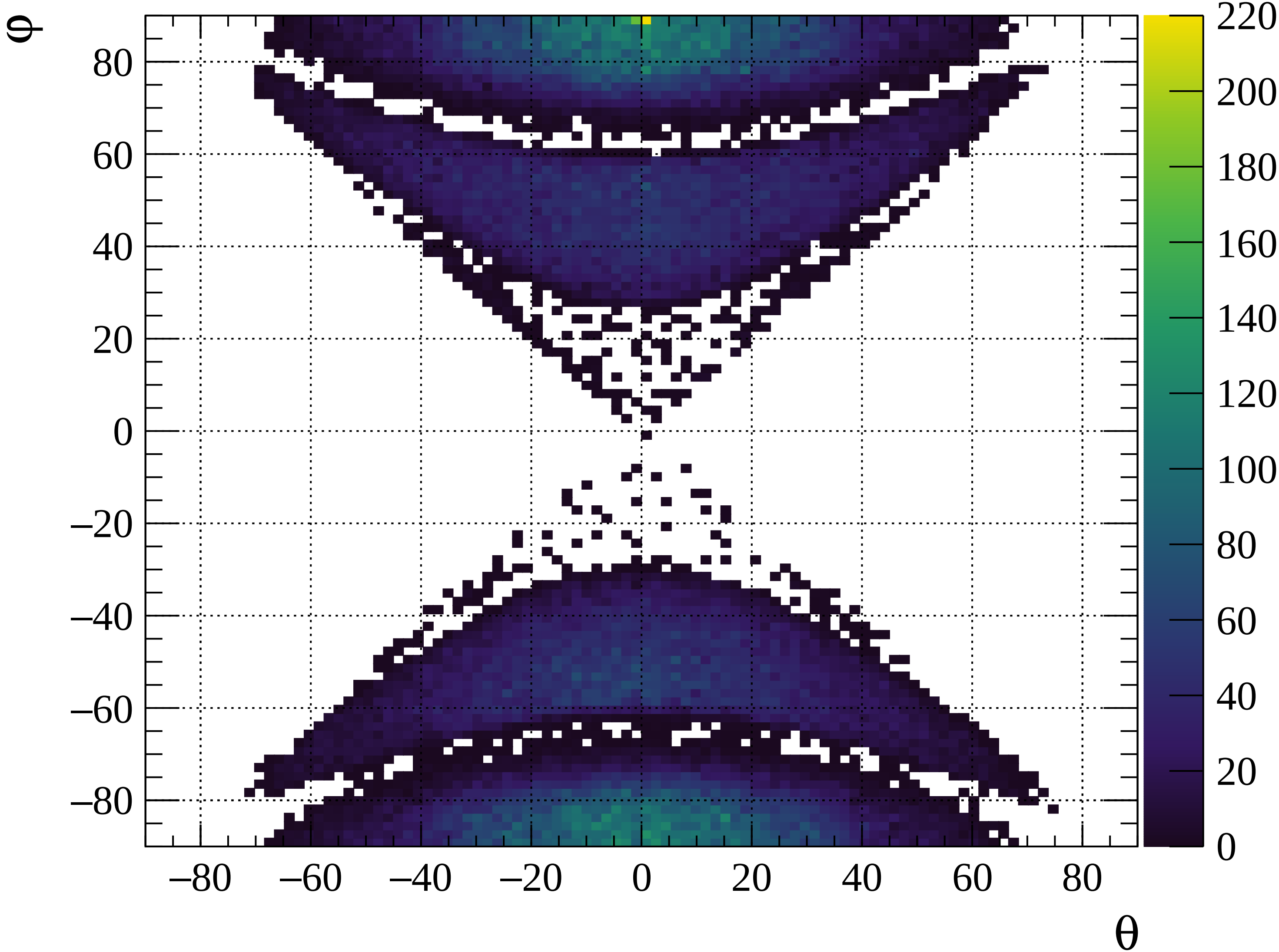
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
120
100
80
60
40
20
0



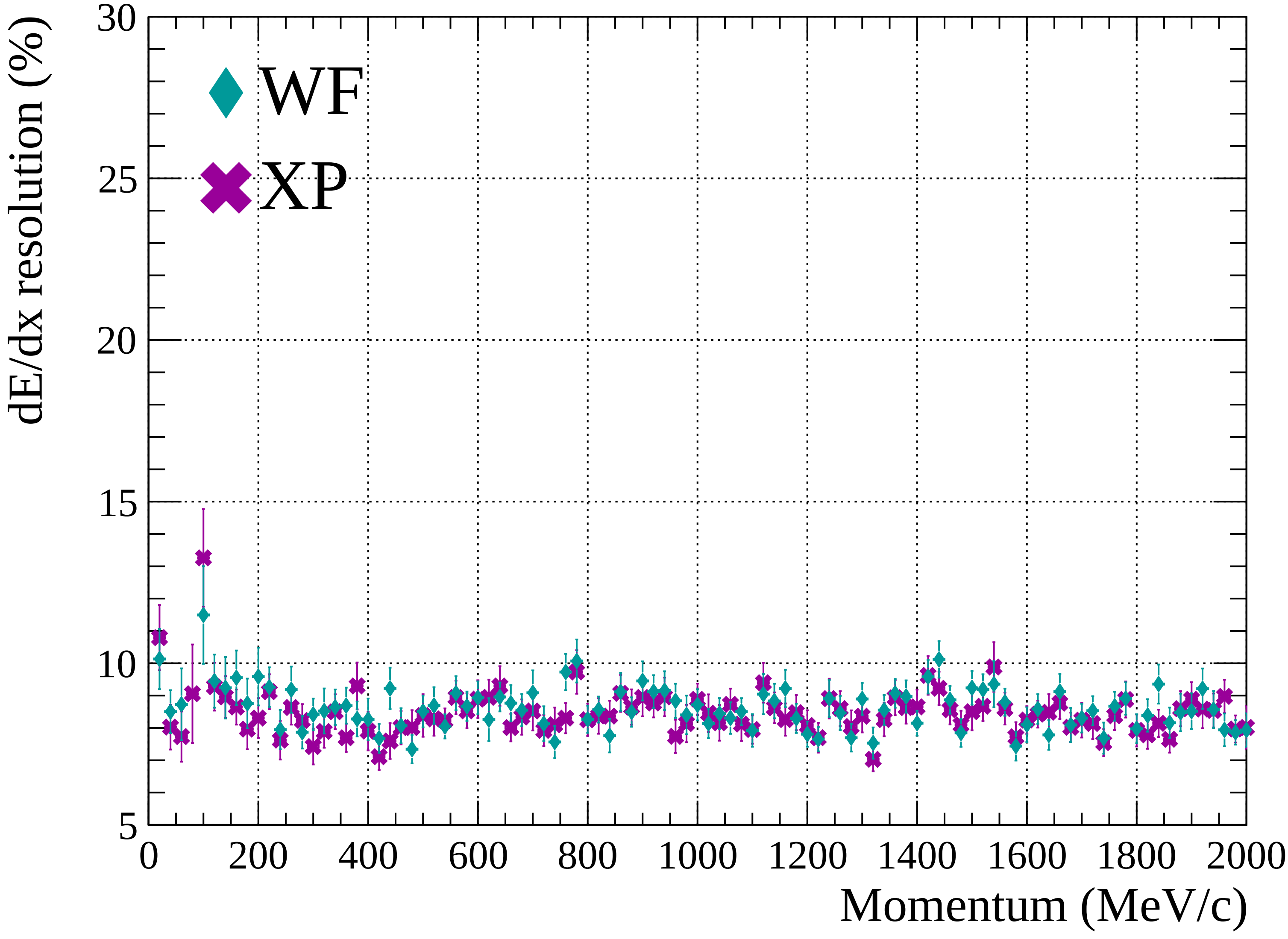
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
20
18
16
14
12
10
8
6
4
2
0



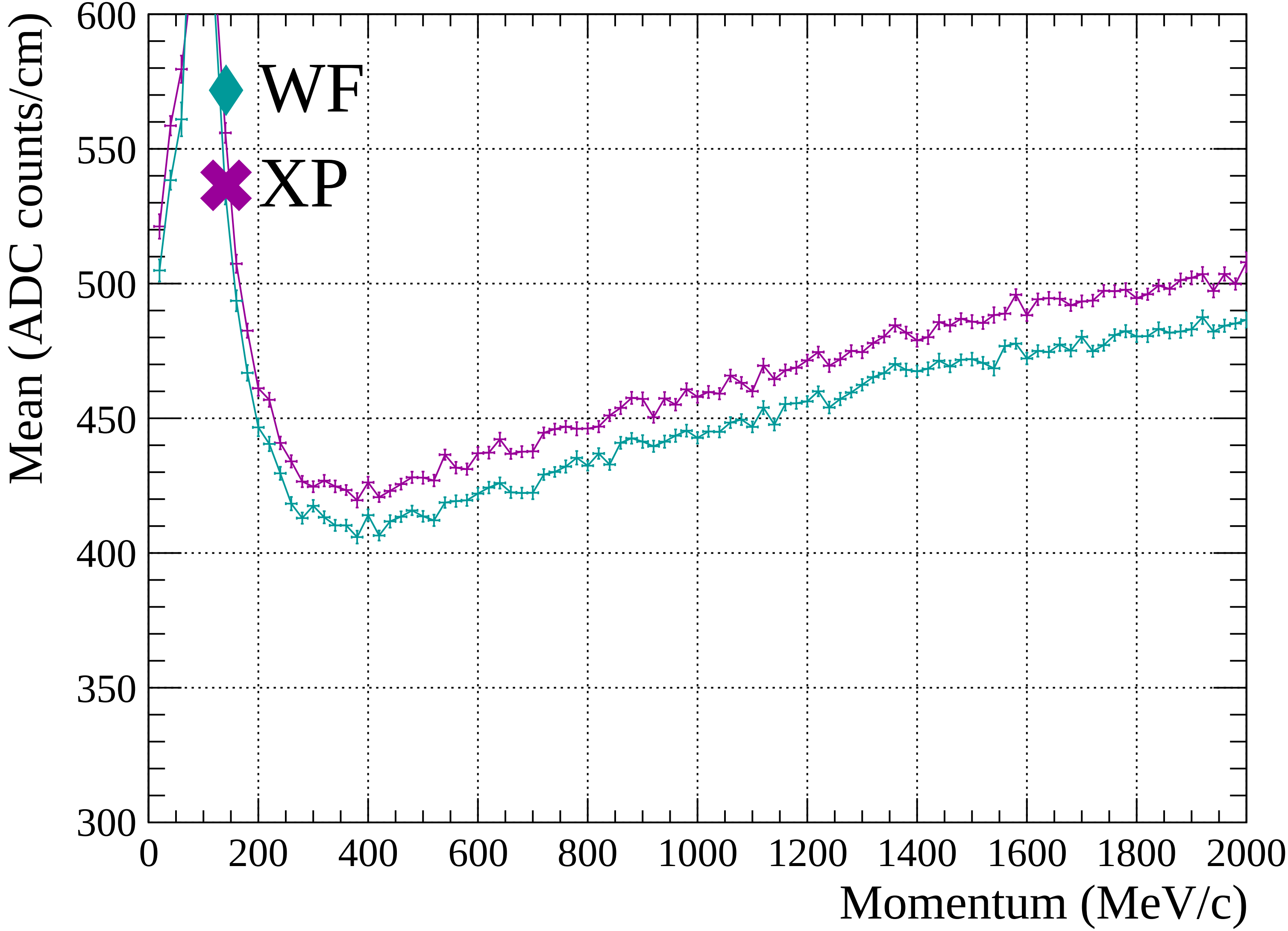
Count
3500
3000
2500
2000
1500
1000
500
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

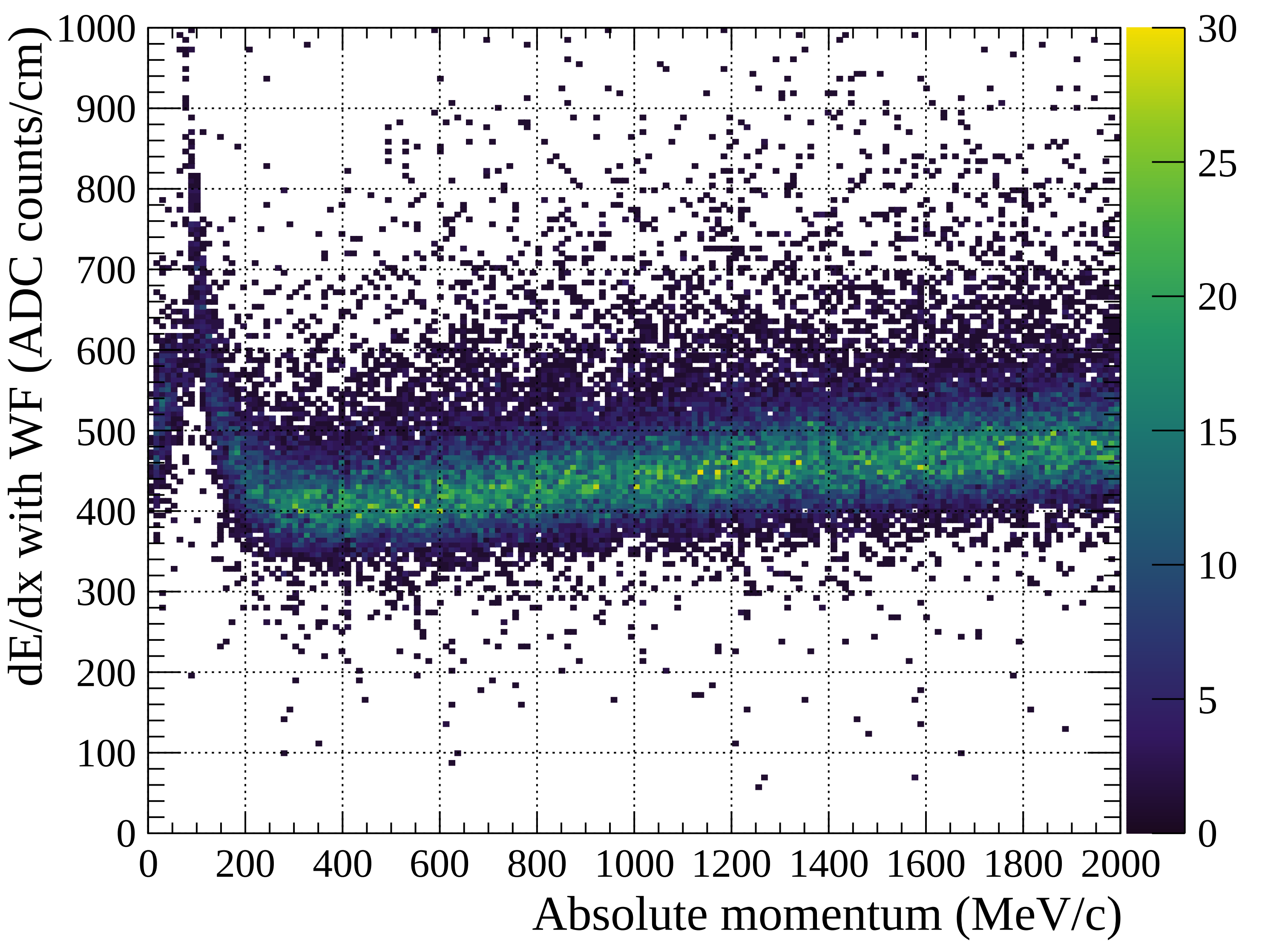


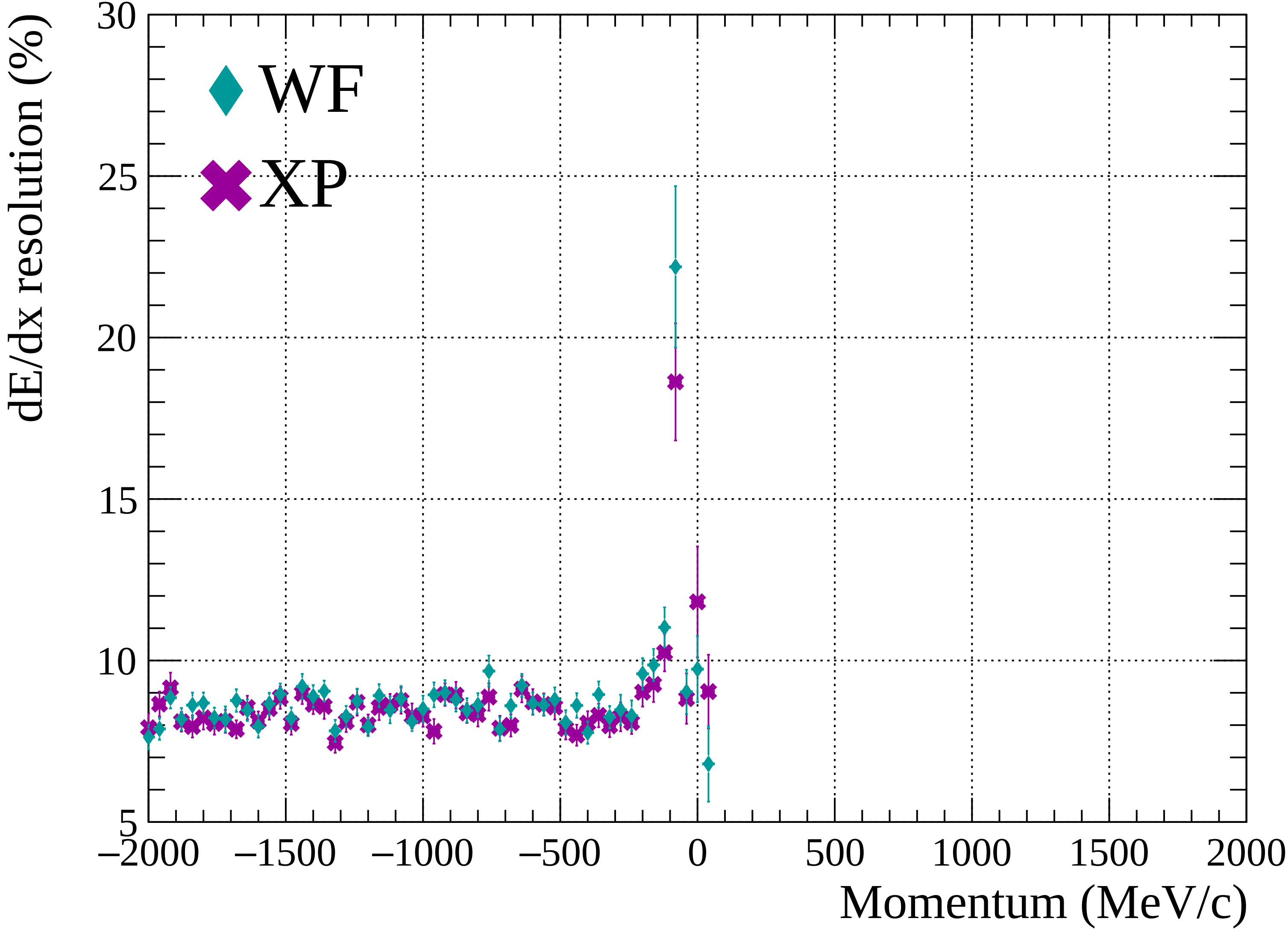
φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
220
200
180
160
140
120
100
80
60
40
20
0



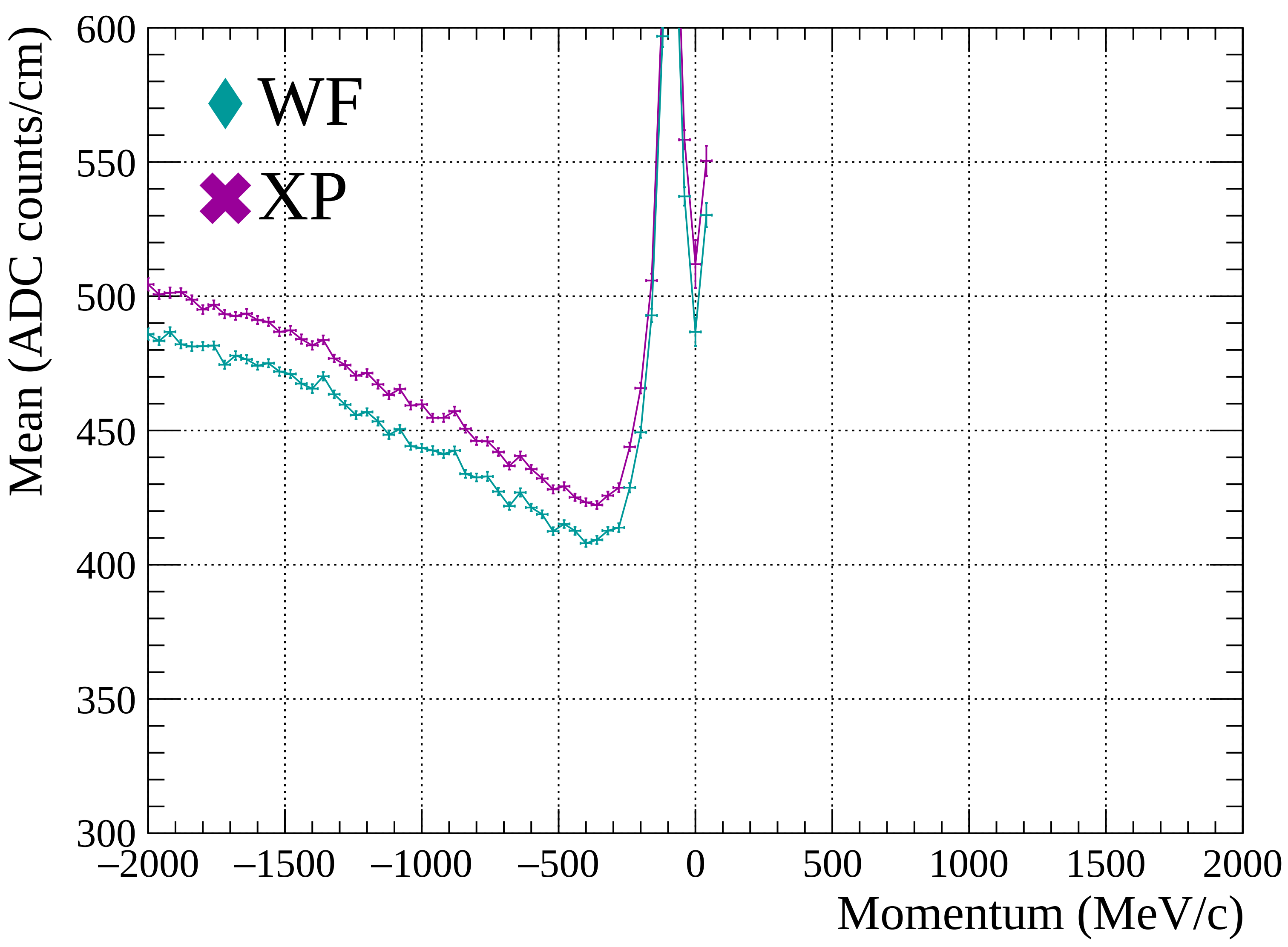
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000


Mean (ADC counts/cm)
Momentum (MeV/c)
WF
XP
300
350
400
450
500
550
600
0
200
400
600
800
1000
1200
1400
1600
1800
2000

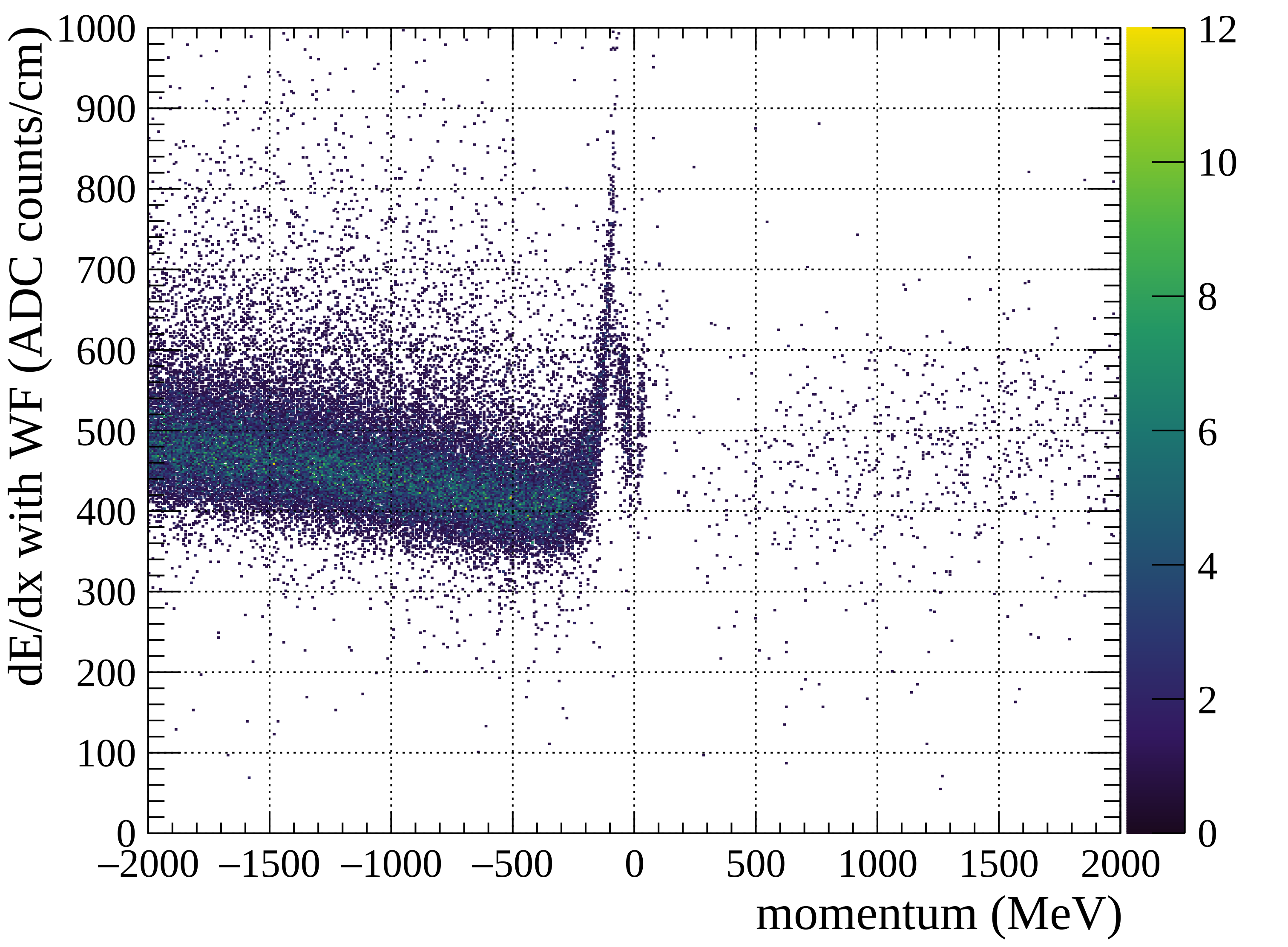

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30

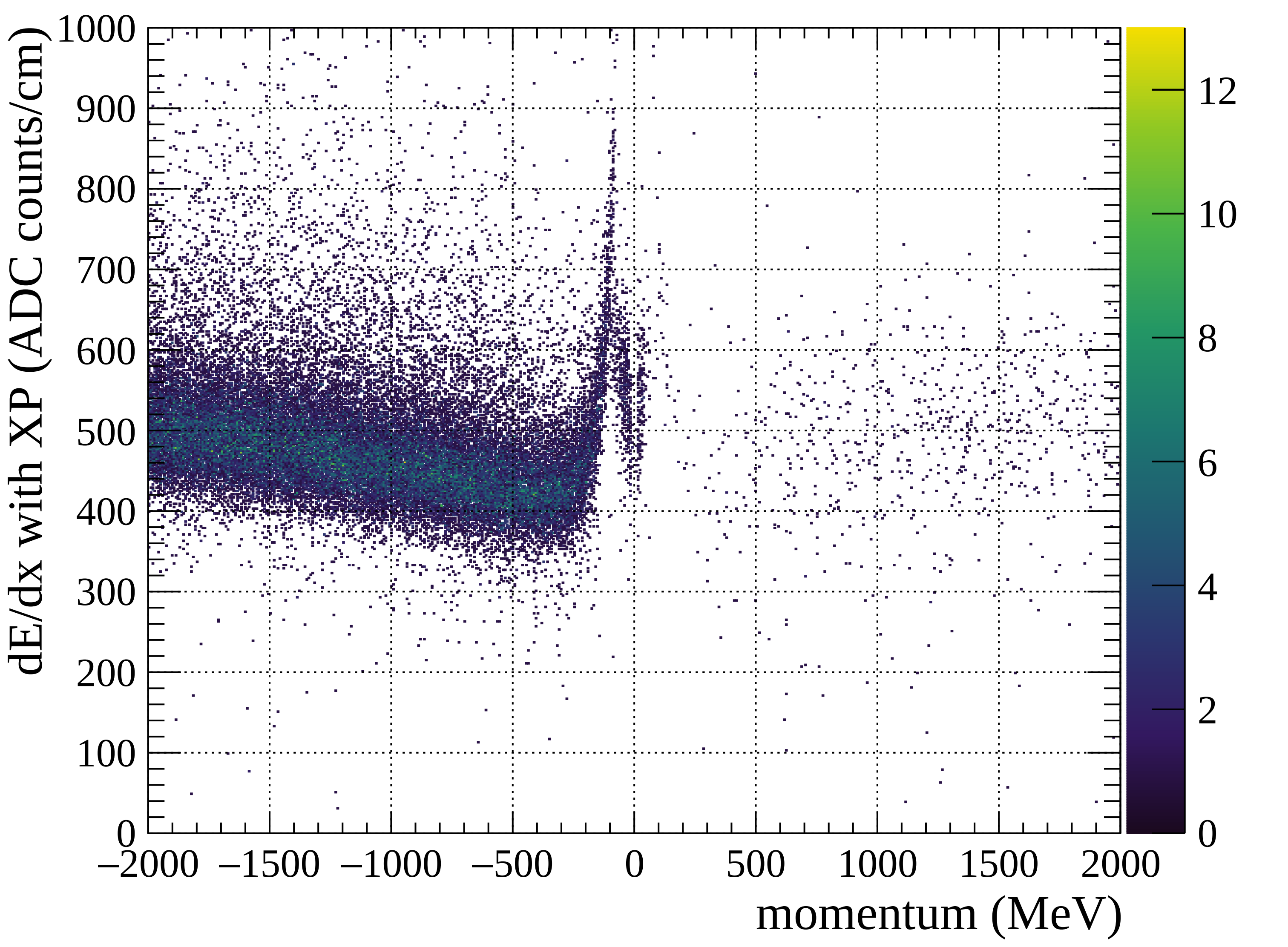


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

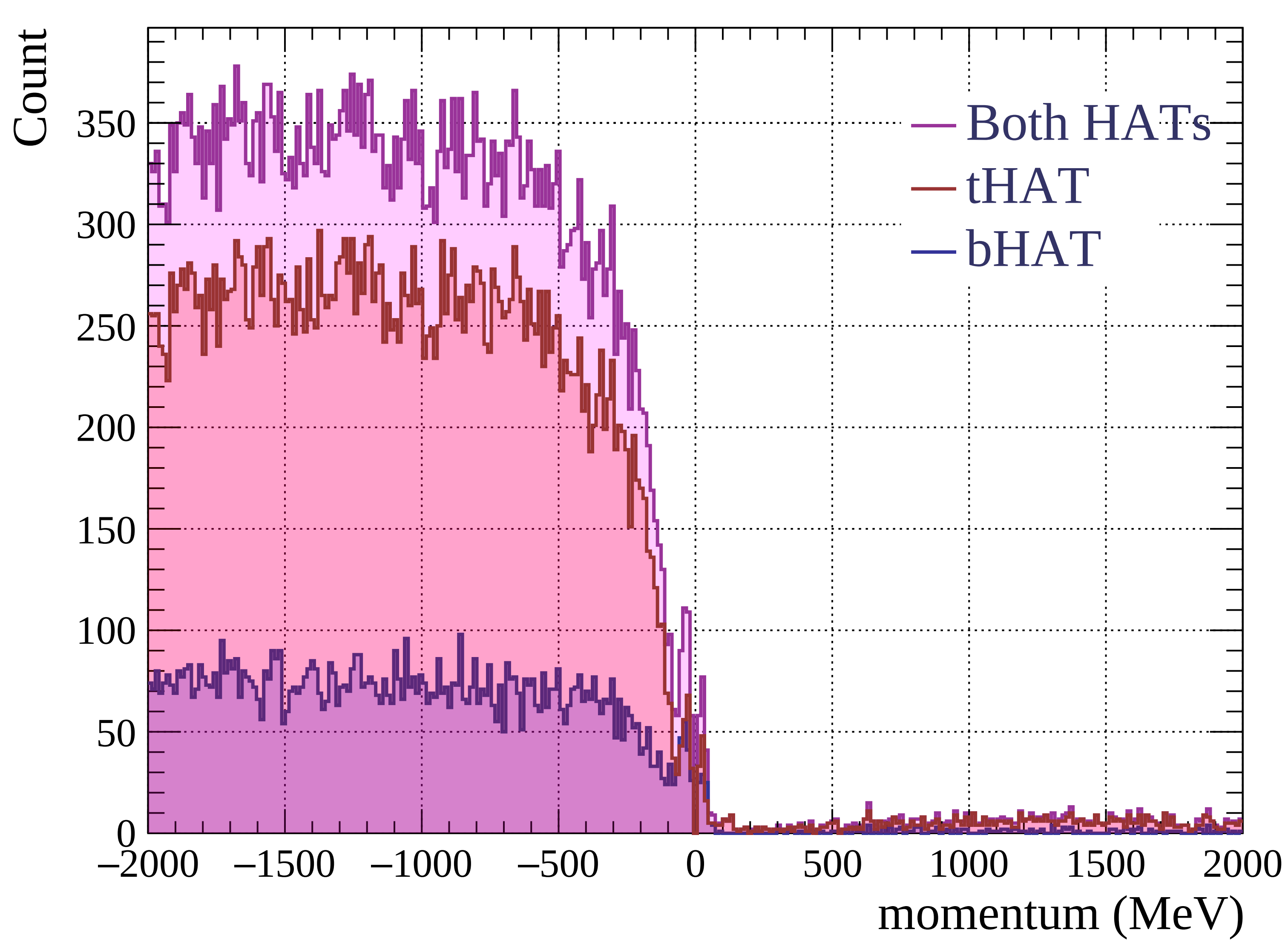


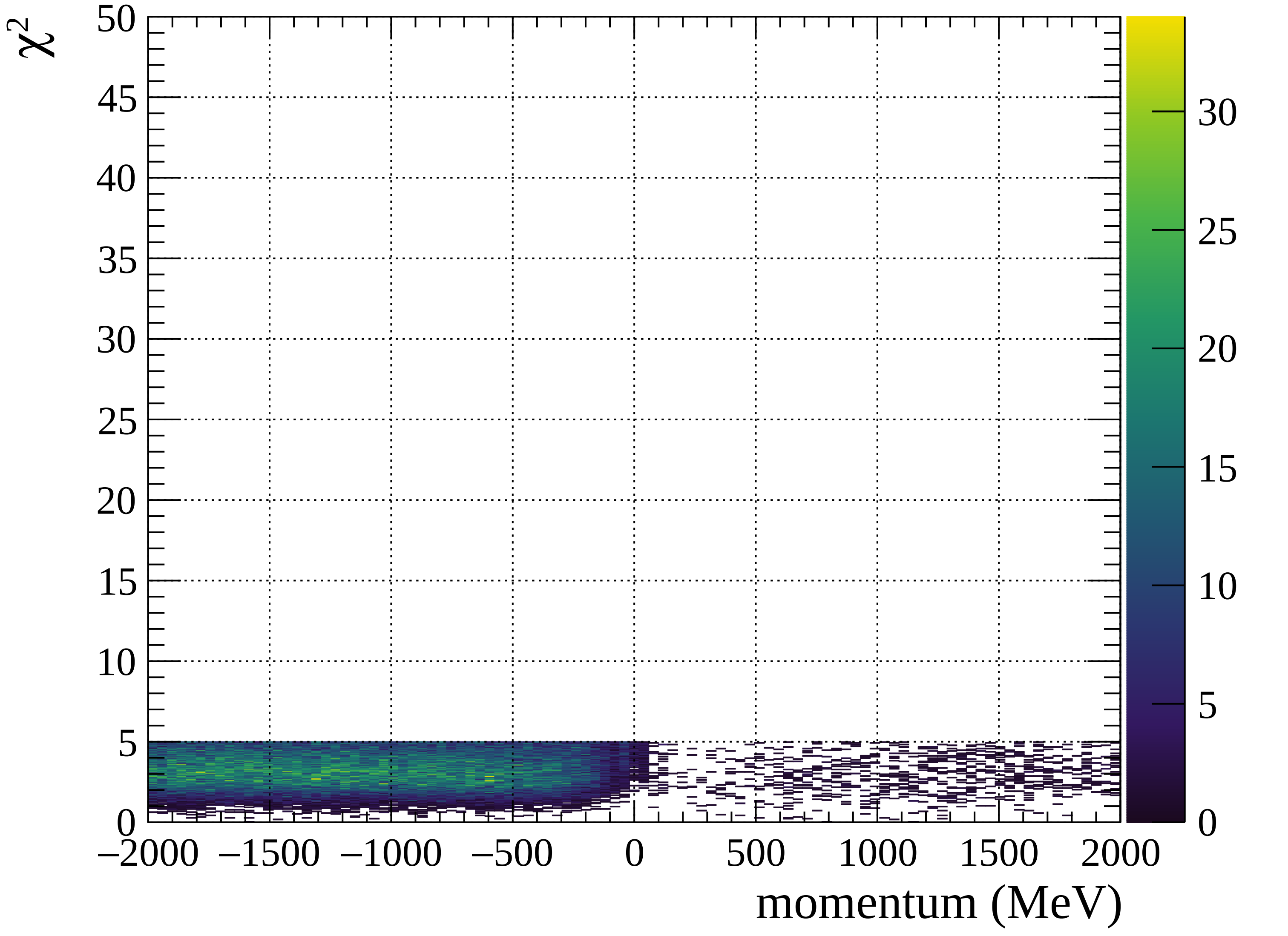
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000


dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
12
10
8
6
4
2
0

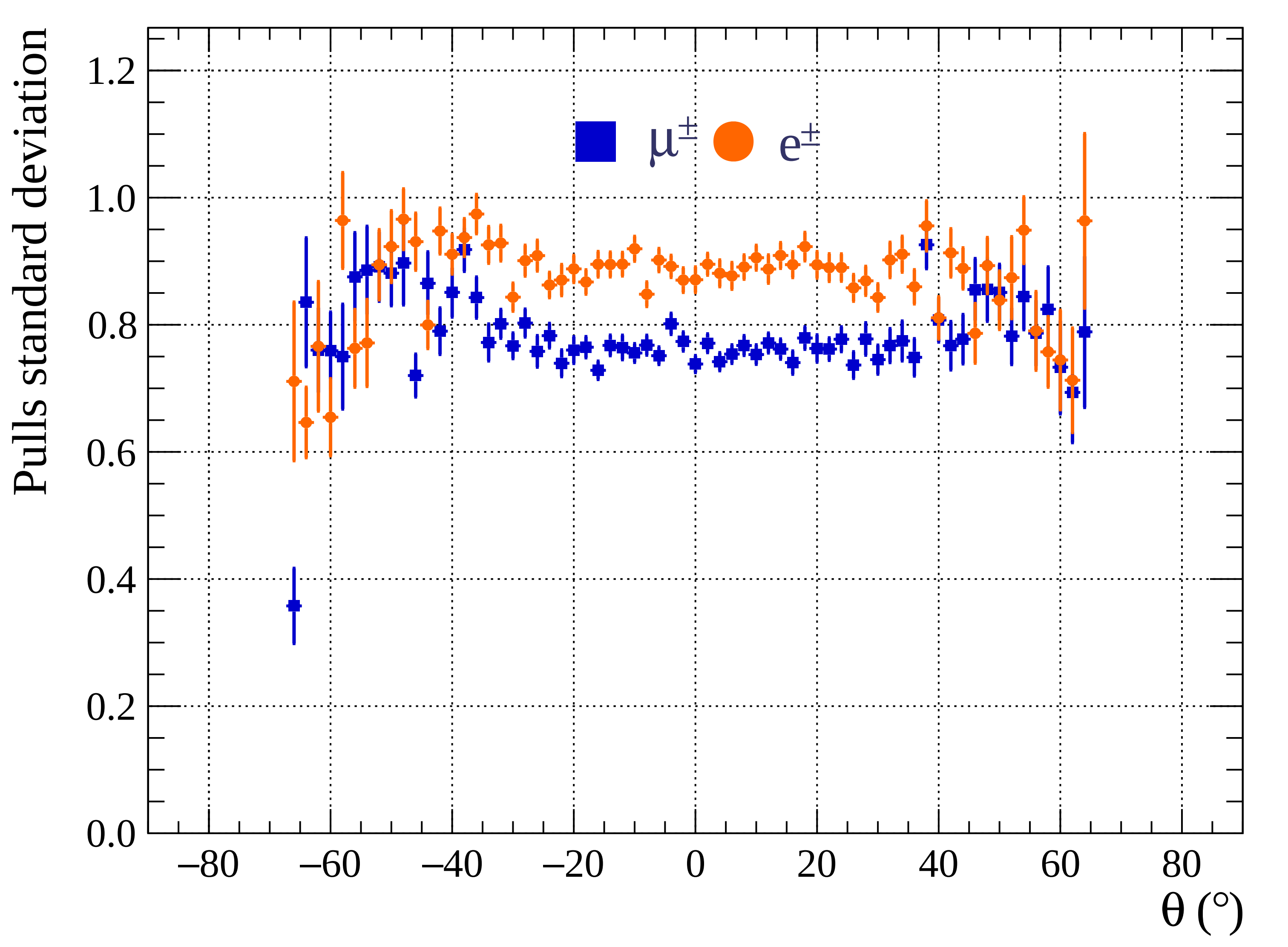


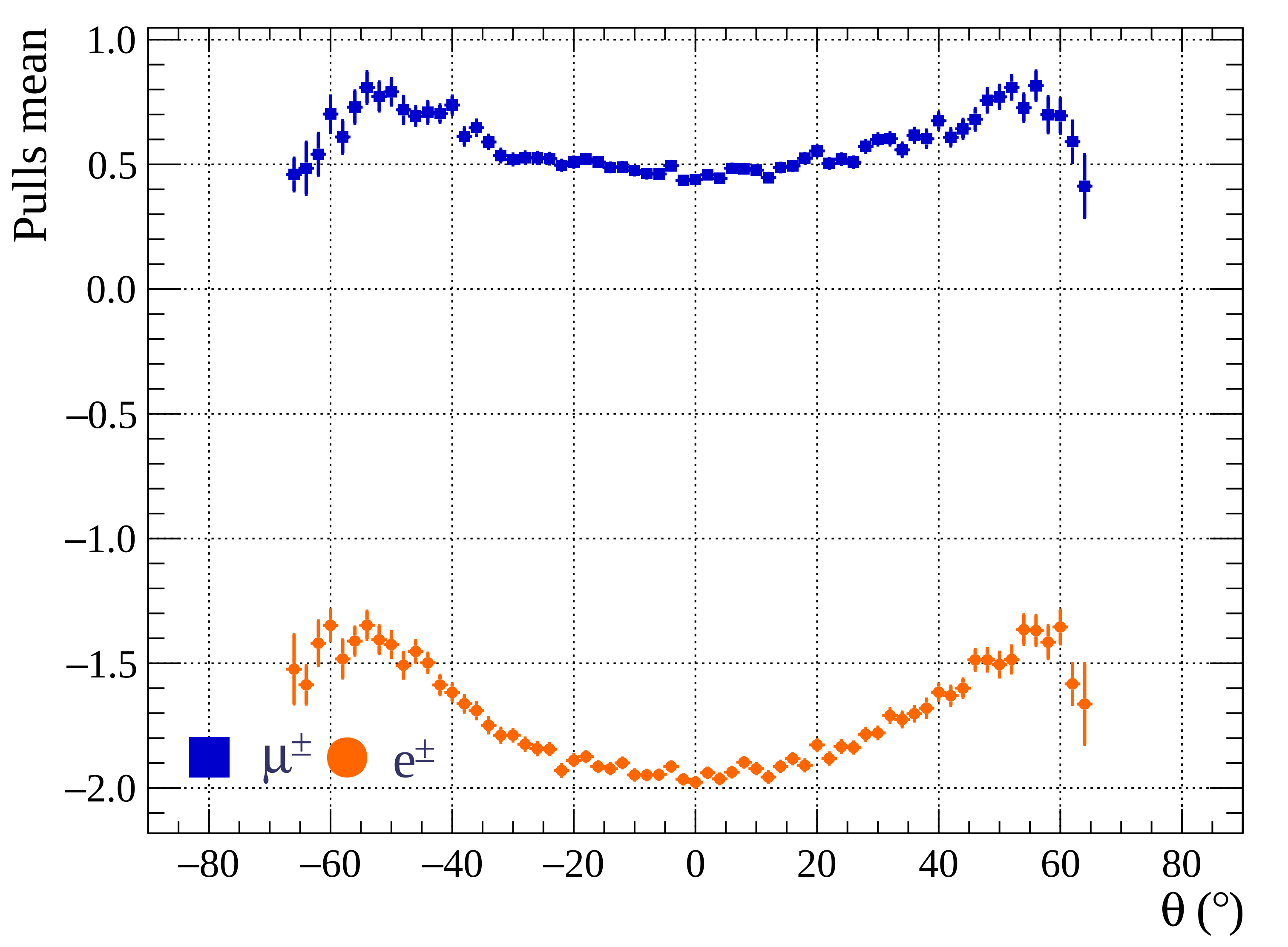
dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12


Count
momentum (MeV)
Both HATs
tHAT
bHAT
0
50
100
150
200
250
300
350
−2000
−1500
−1000
−500
0
500
1000
1500
2000

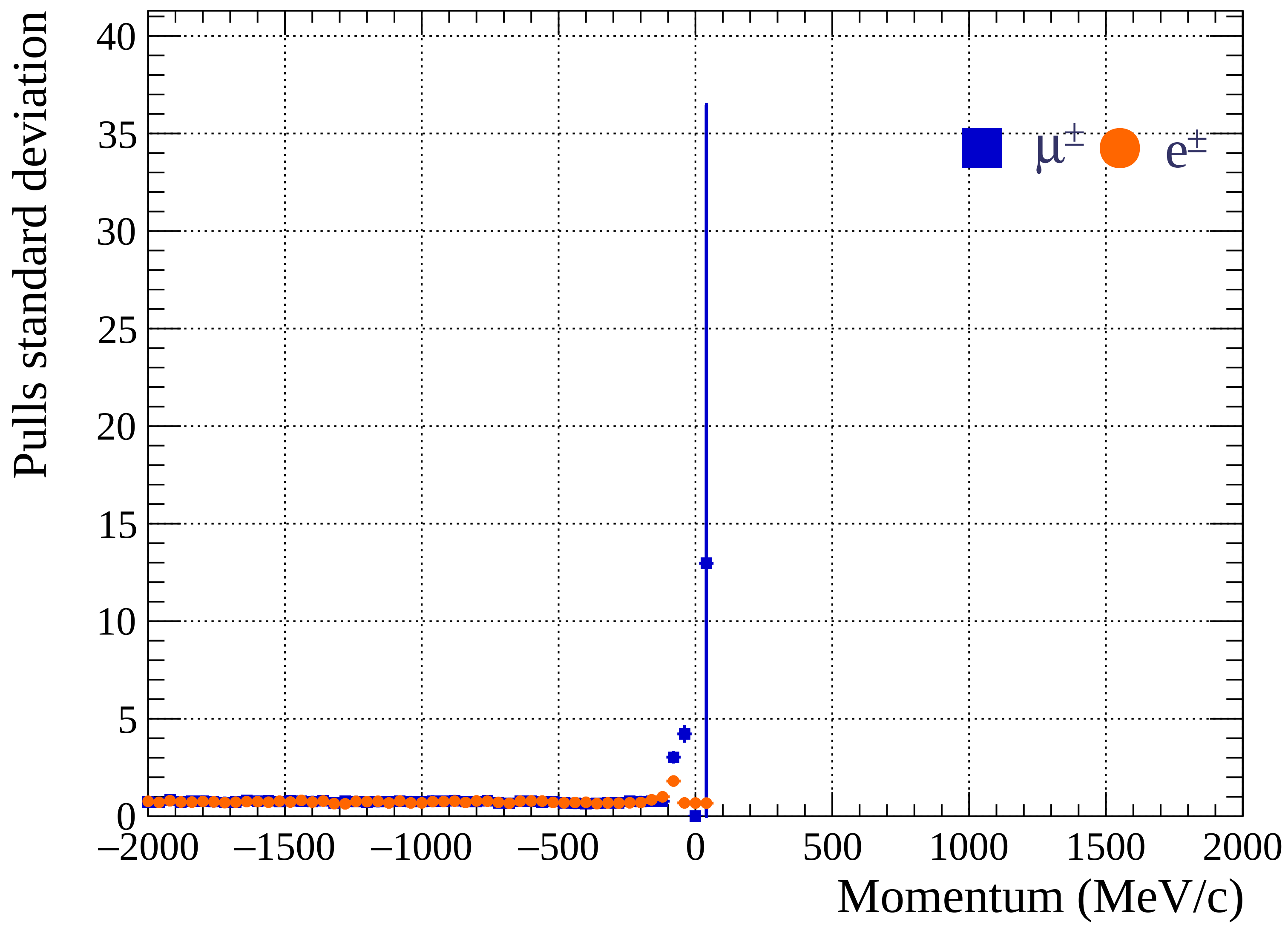


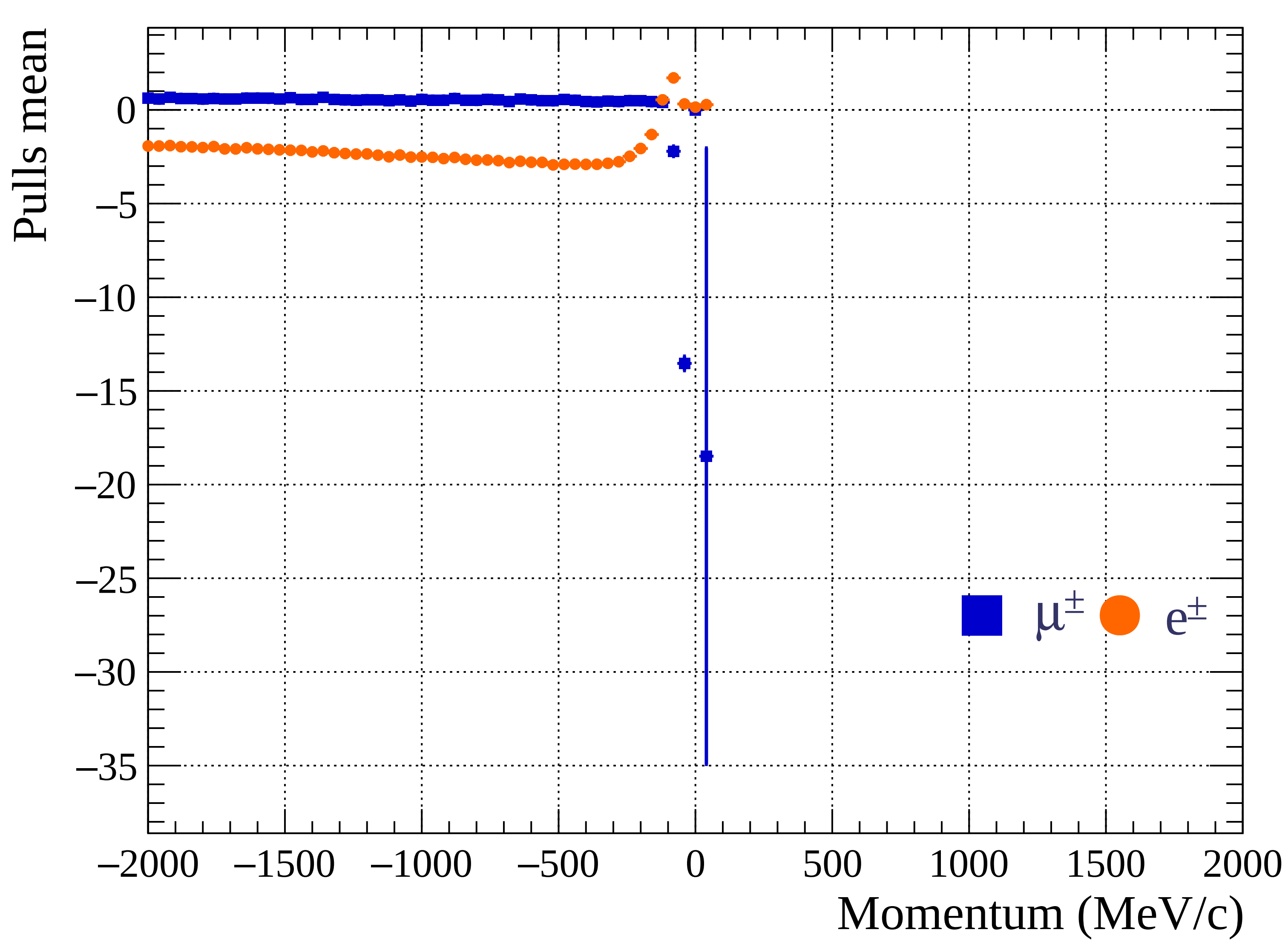
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
30
25
20
15
10
5
0

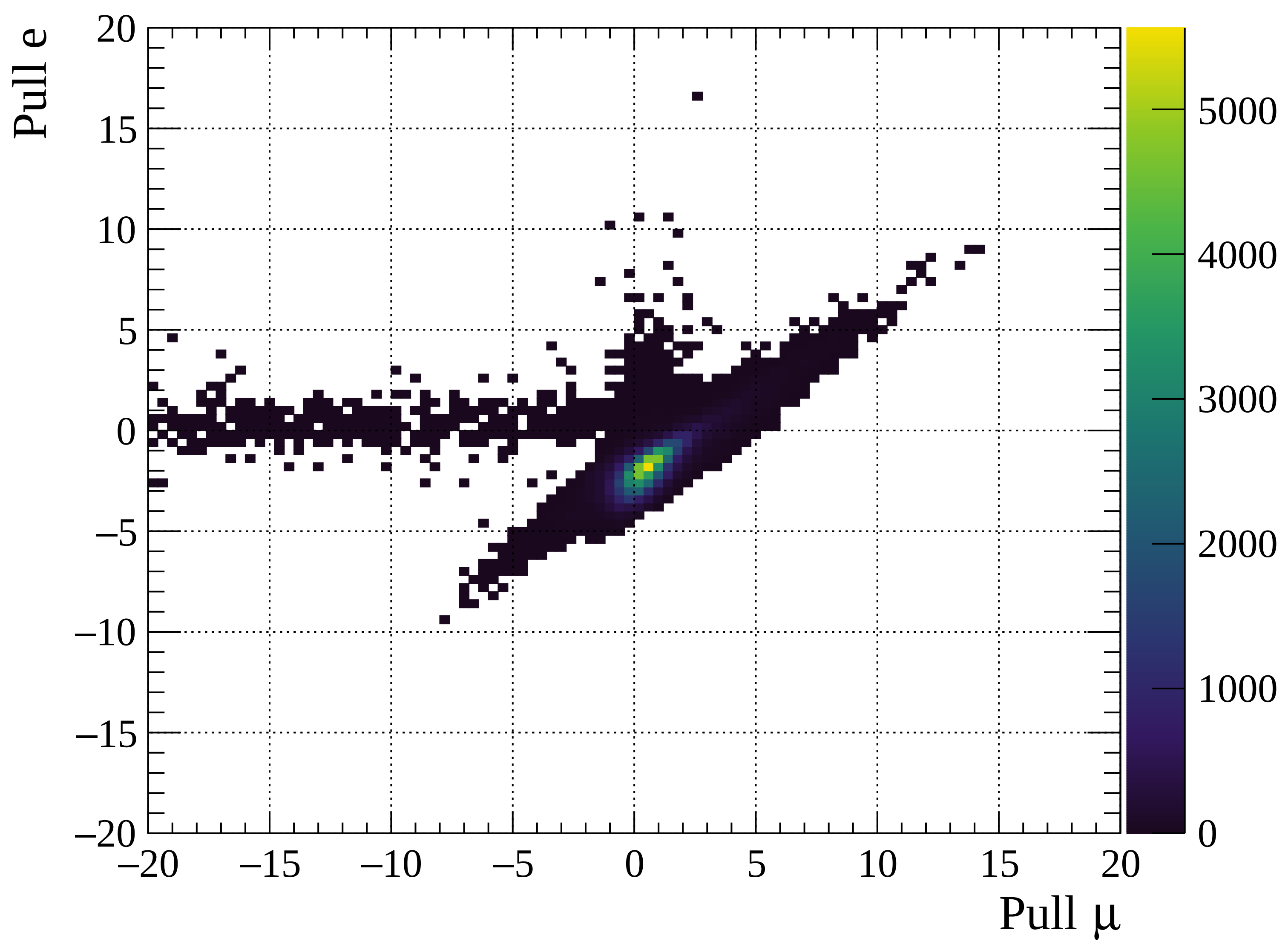

Pulls standard deviation
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
$\mu^{\pm}$
$e^{\pm}$



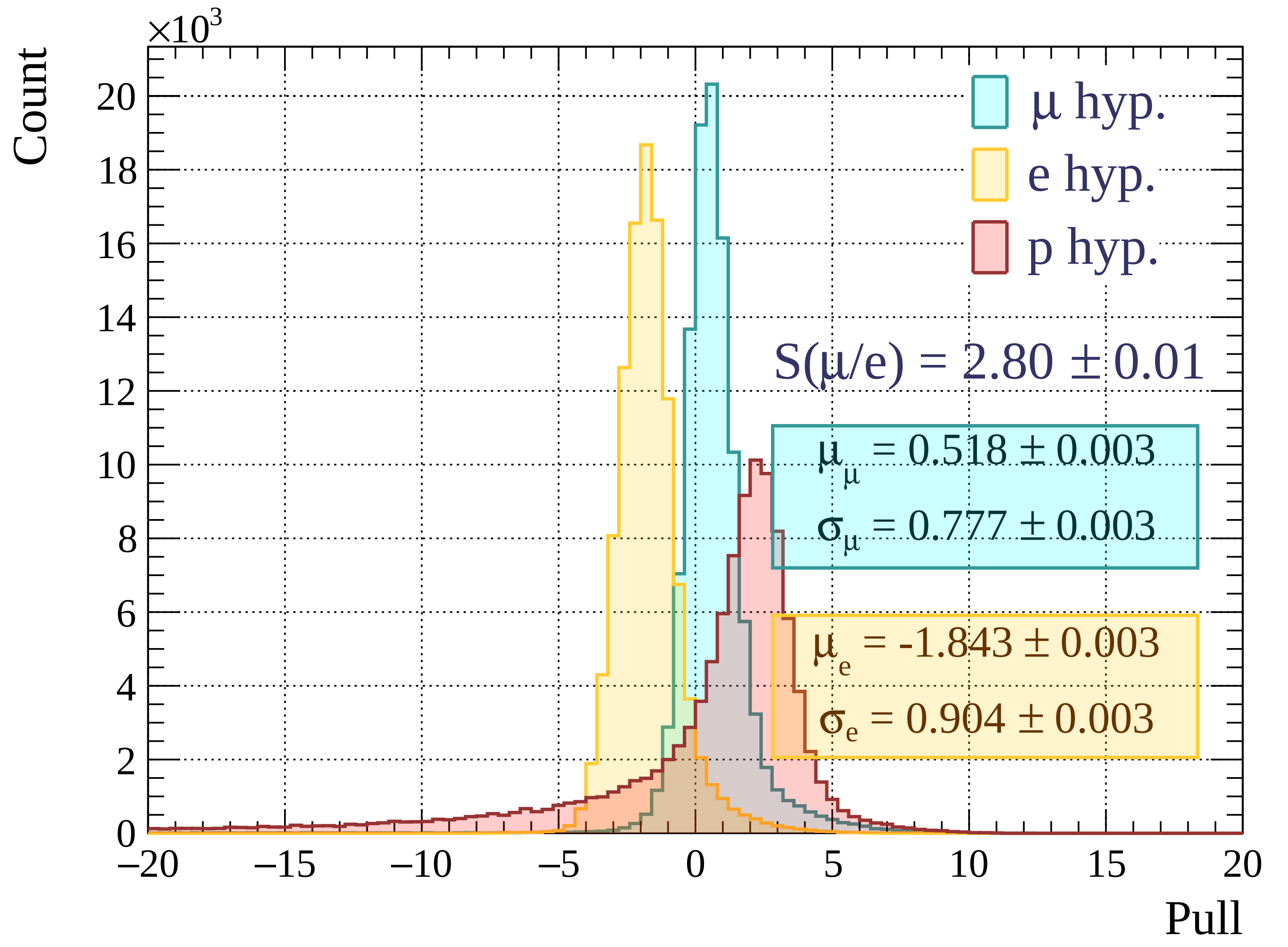
Pulls mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±


Pulls standard deviation
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

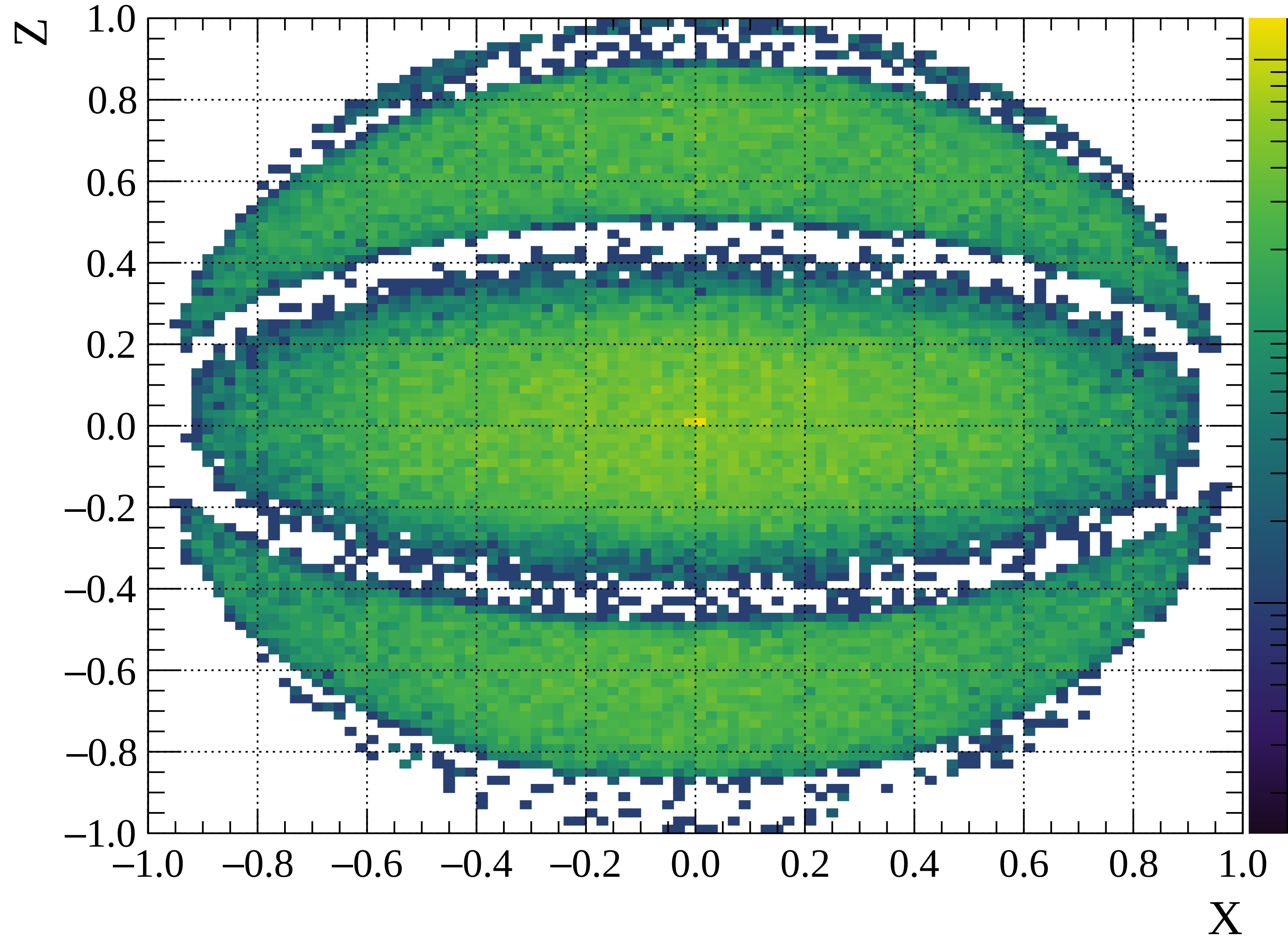

Pulls mean
0
−5
−10
−15
−20
−25
−30
−35
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

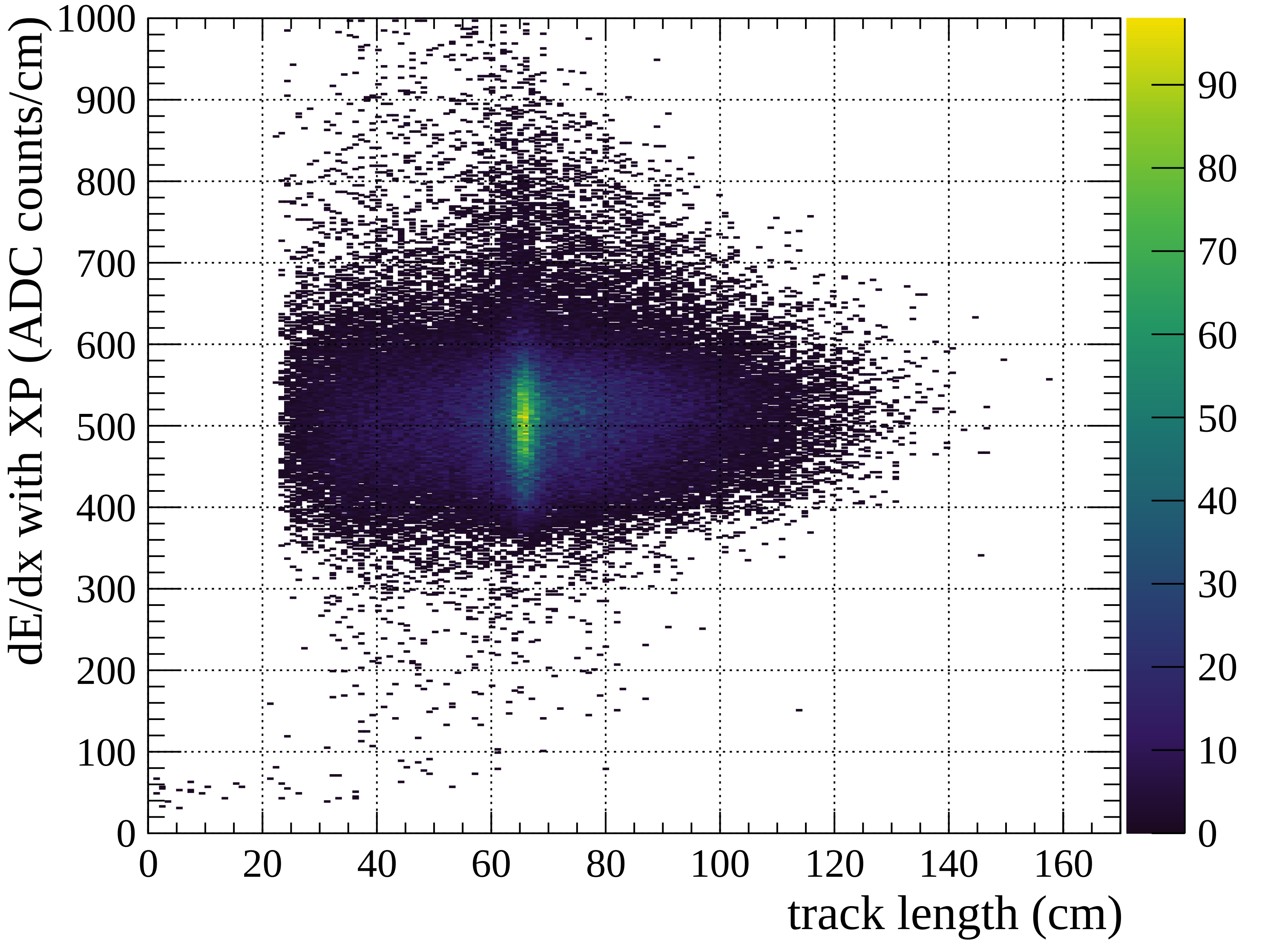


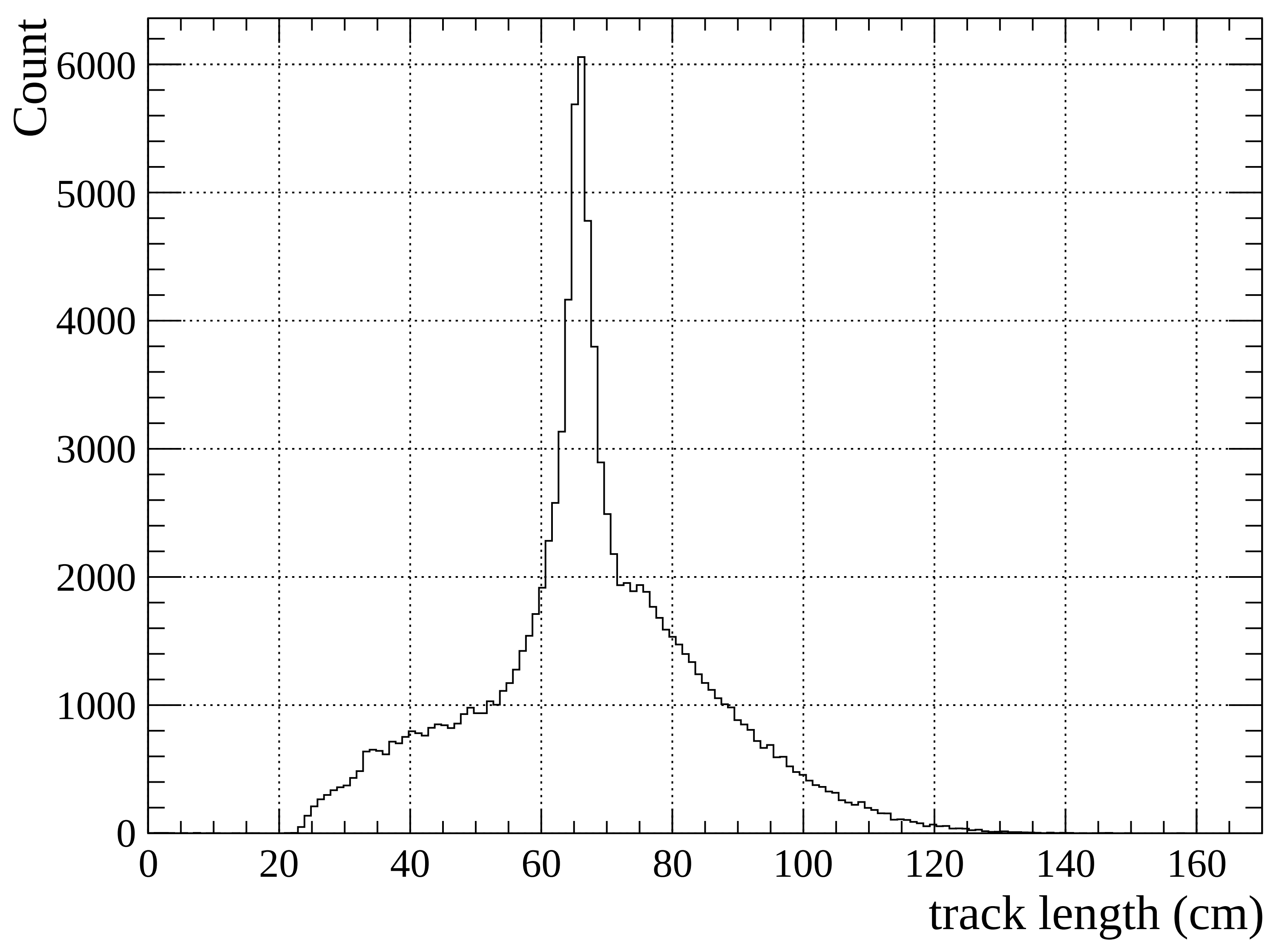
Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
5000
4000
3000
2000
1000
0



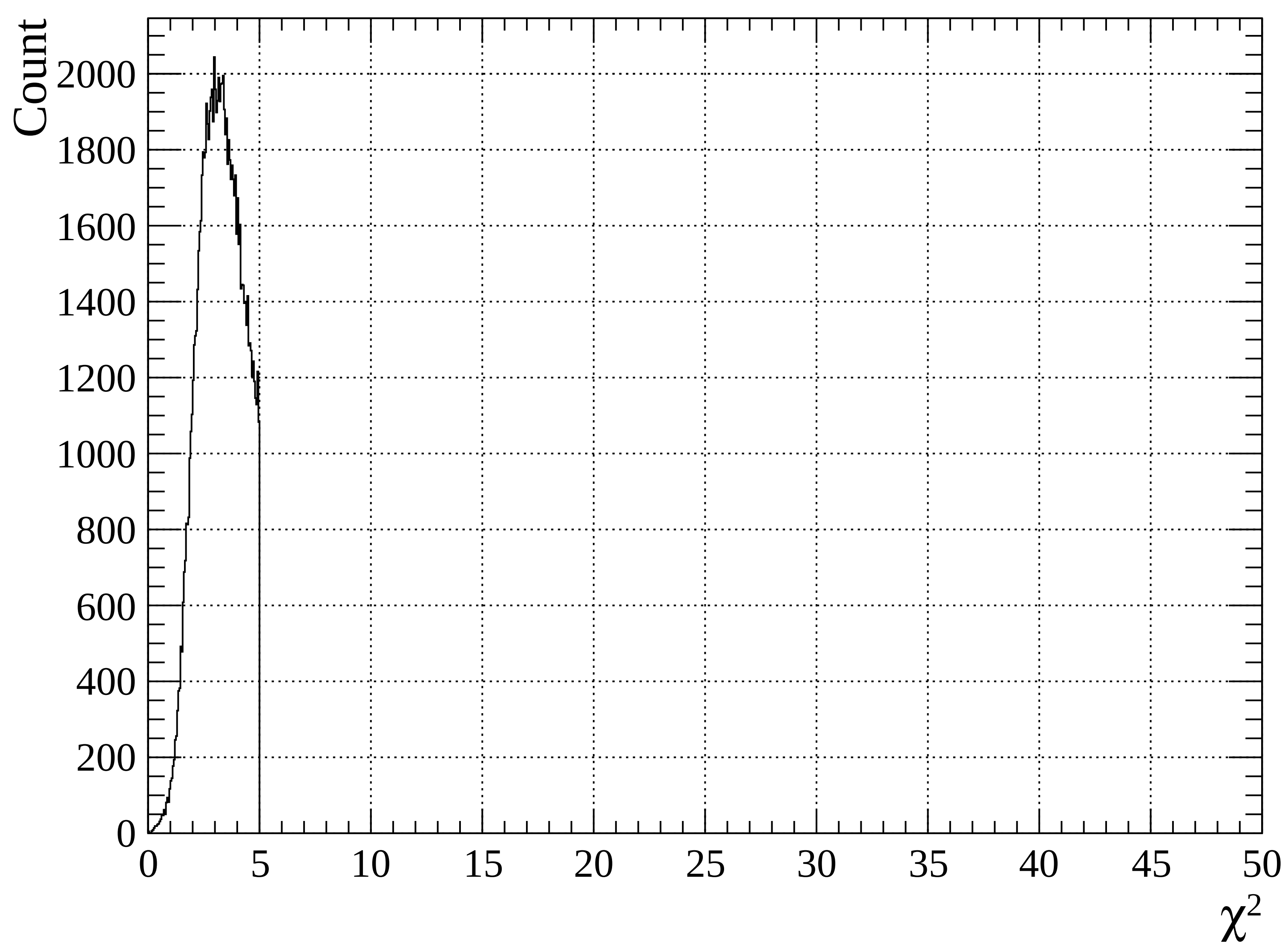
×10^3
Count
20
18
16
14
12
10
8
6
4
2
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.80 ± 0.01
μ_μ = 0.518 ± 0.003
σ_μ = 0.777 ± 0.003
μ_e = -1.843 ± 0.003
σ_e = 0.904 ± 0.003

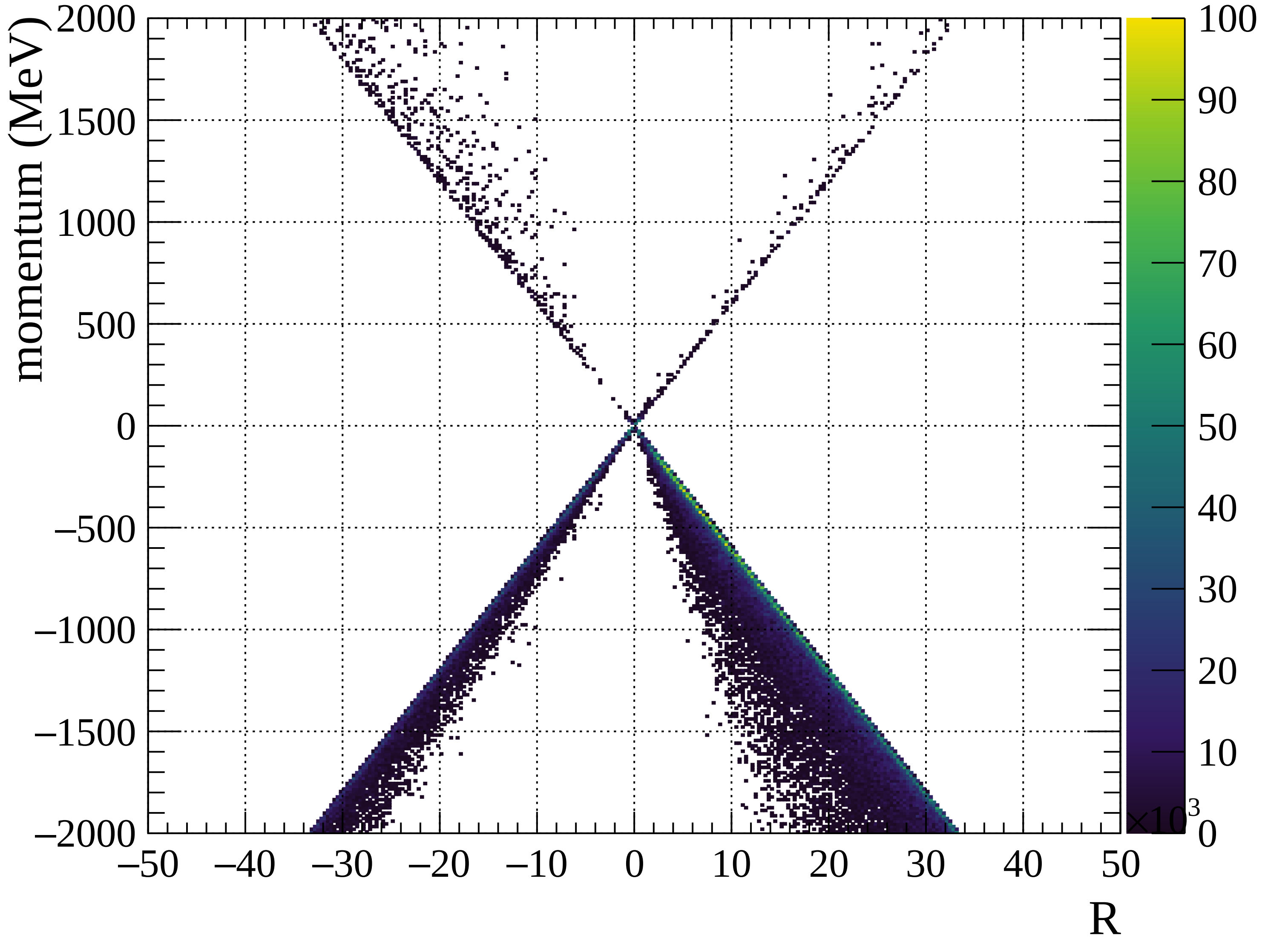

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X


dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
10
20
30
40
50
60
70
80
90

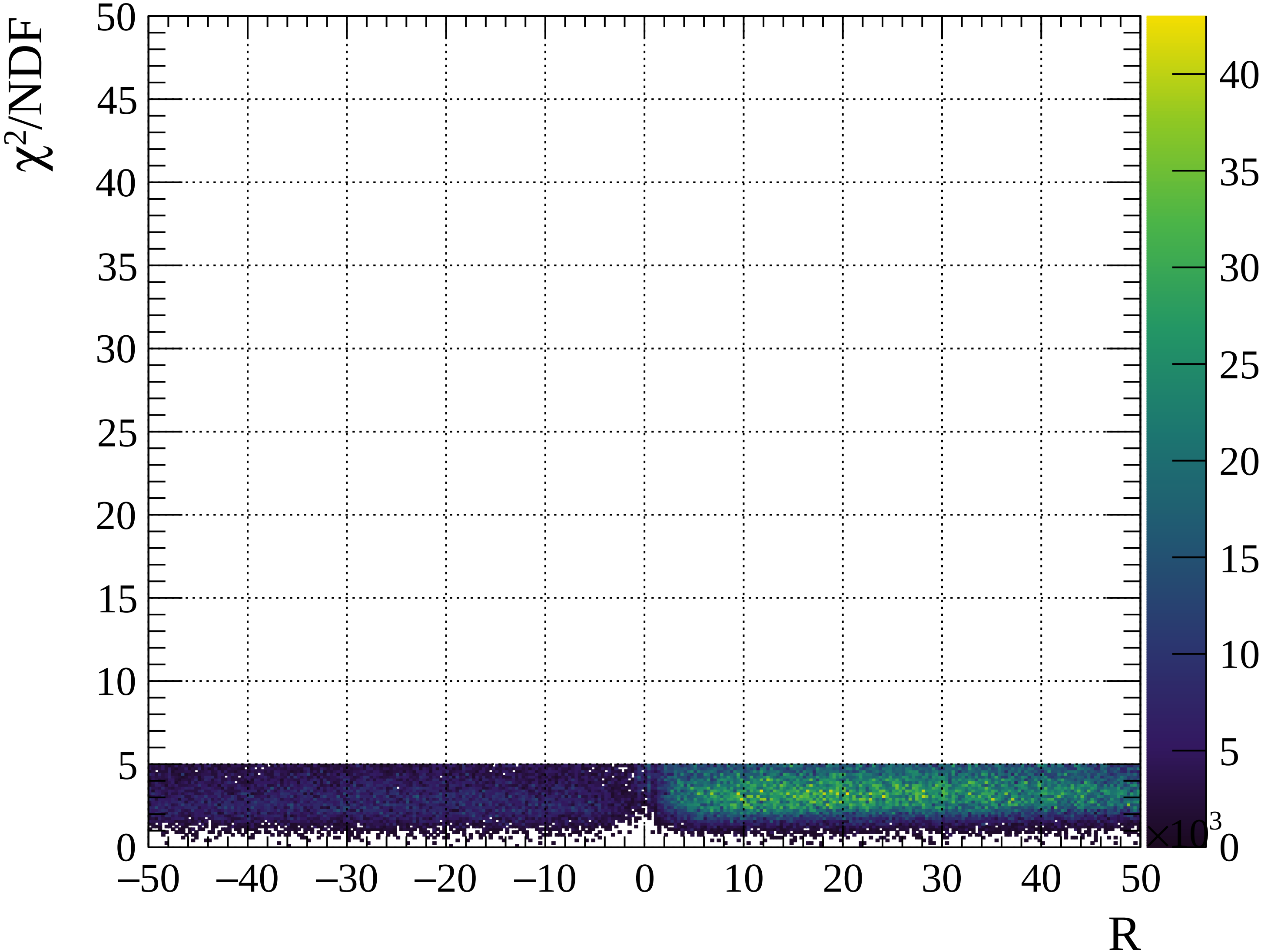


Count
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

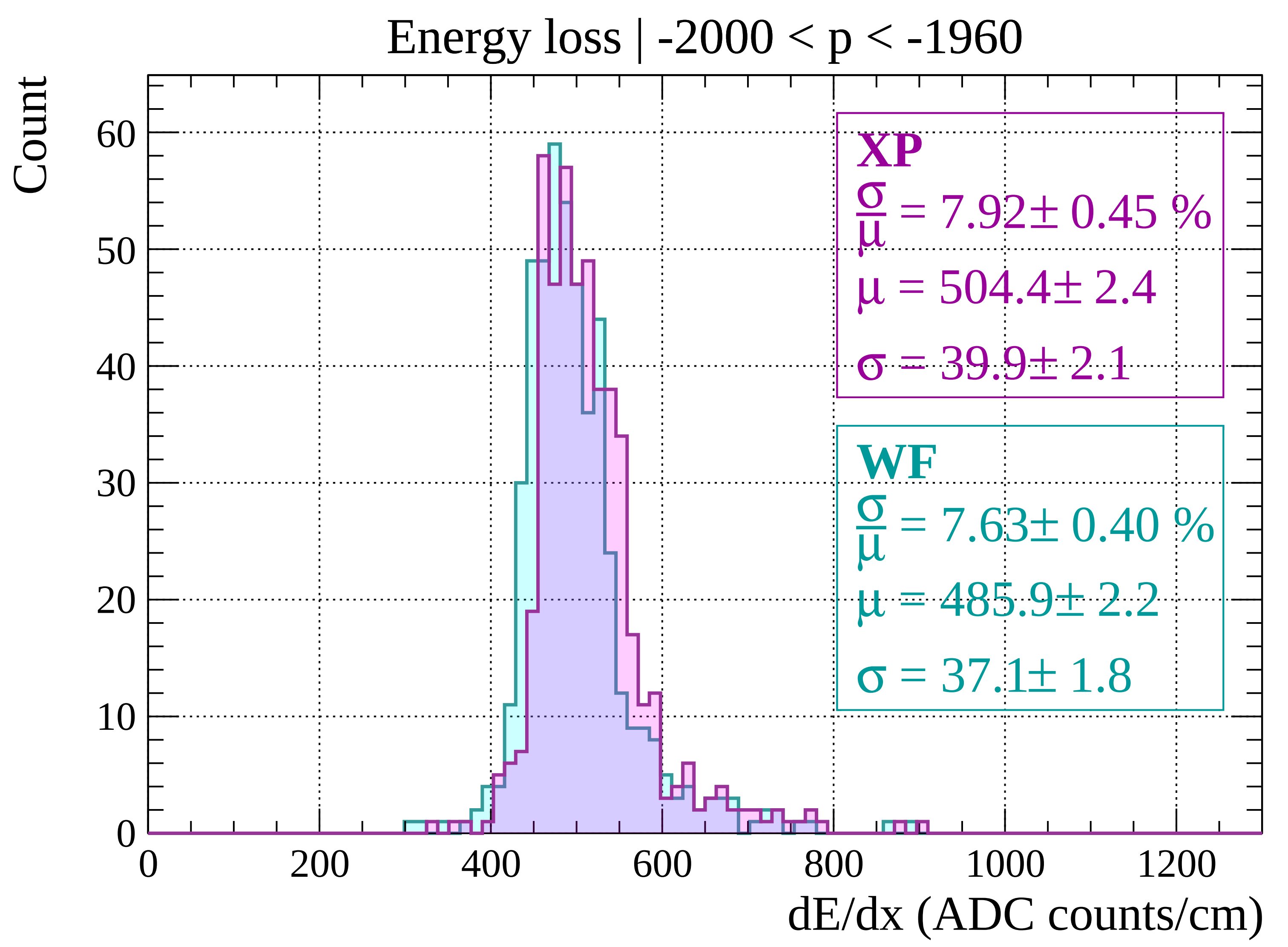

Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



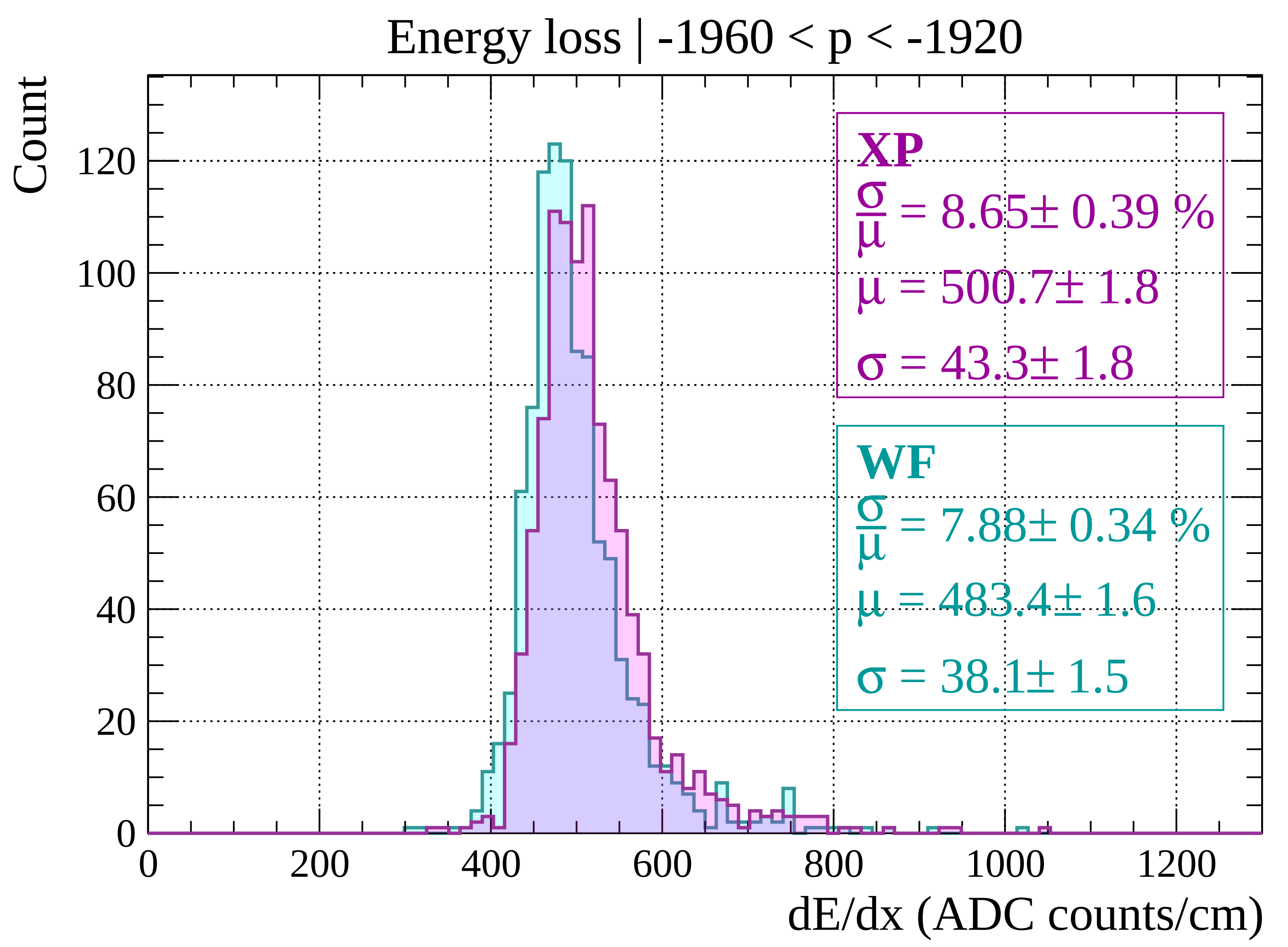
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
100
90
80
70
60
50
40
30
20
10
0
×10^3

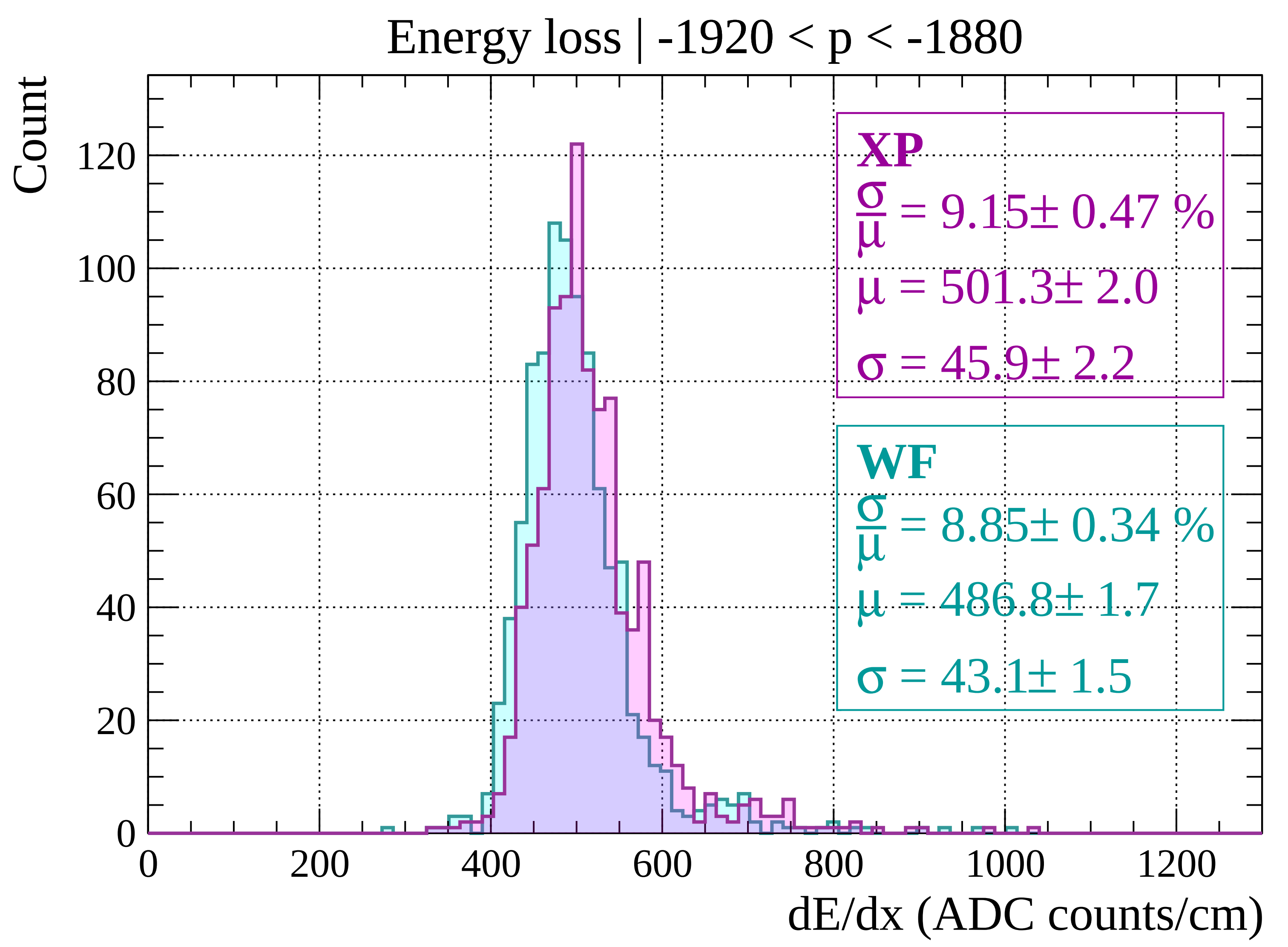


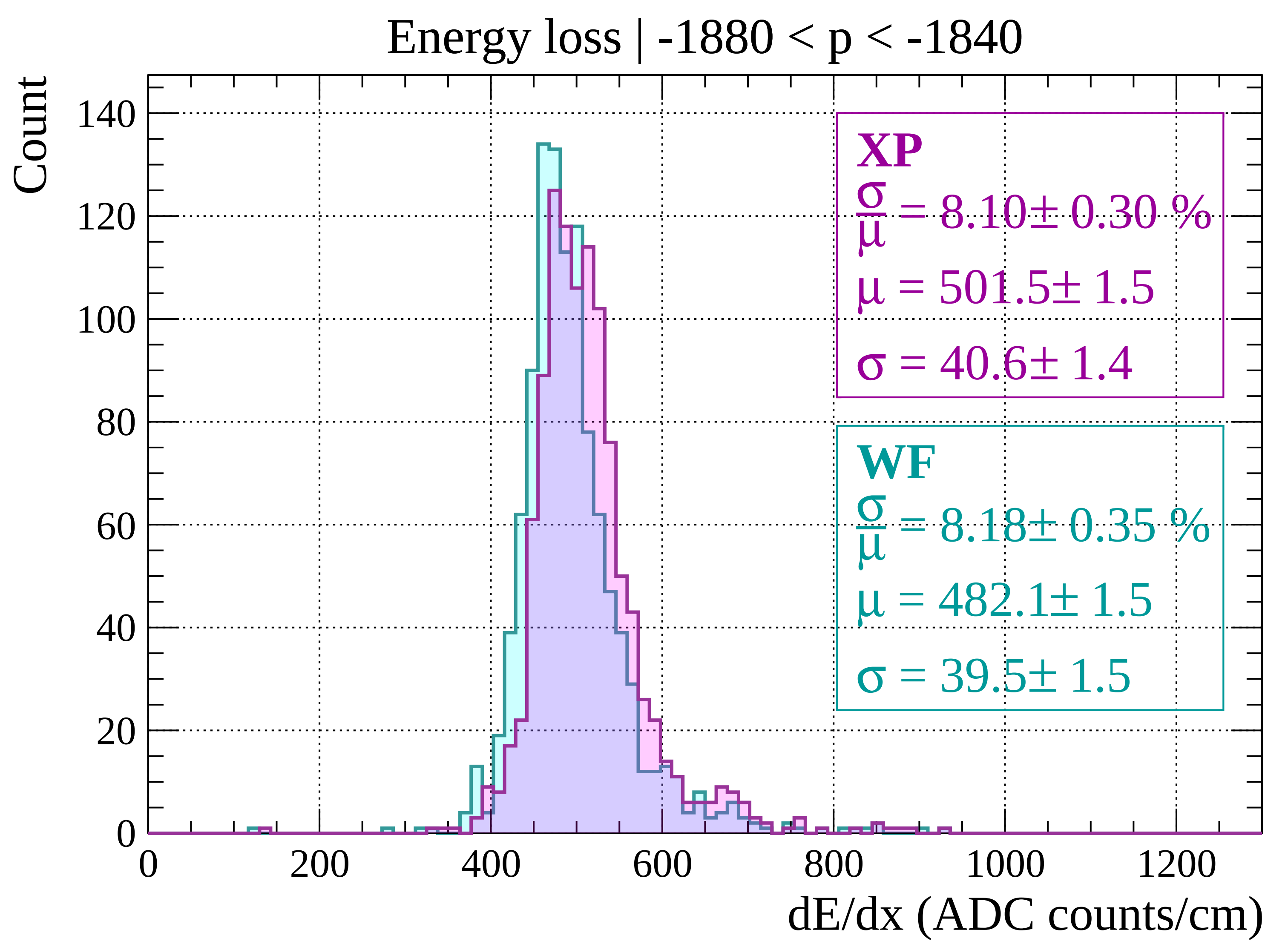
χ^2/NDF
R
$\times 10^3$



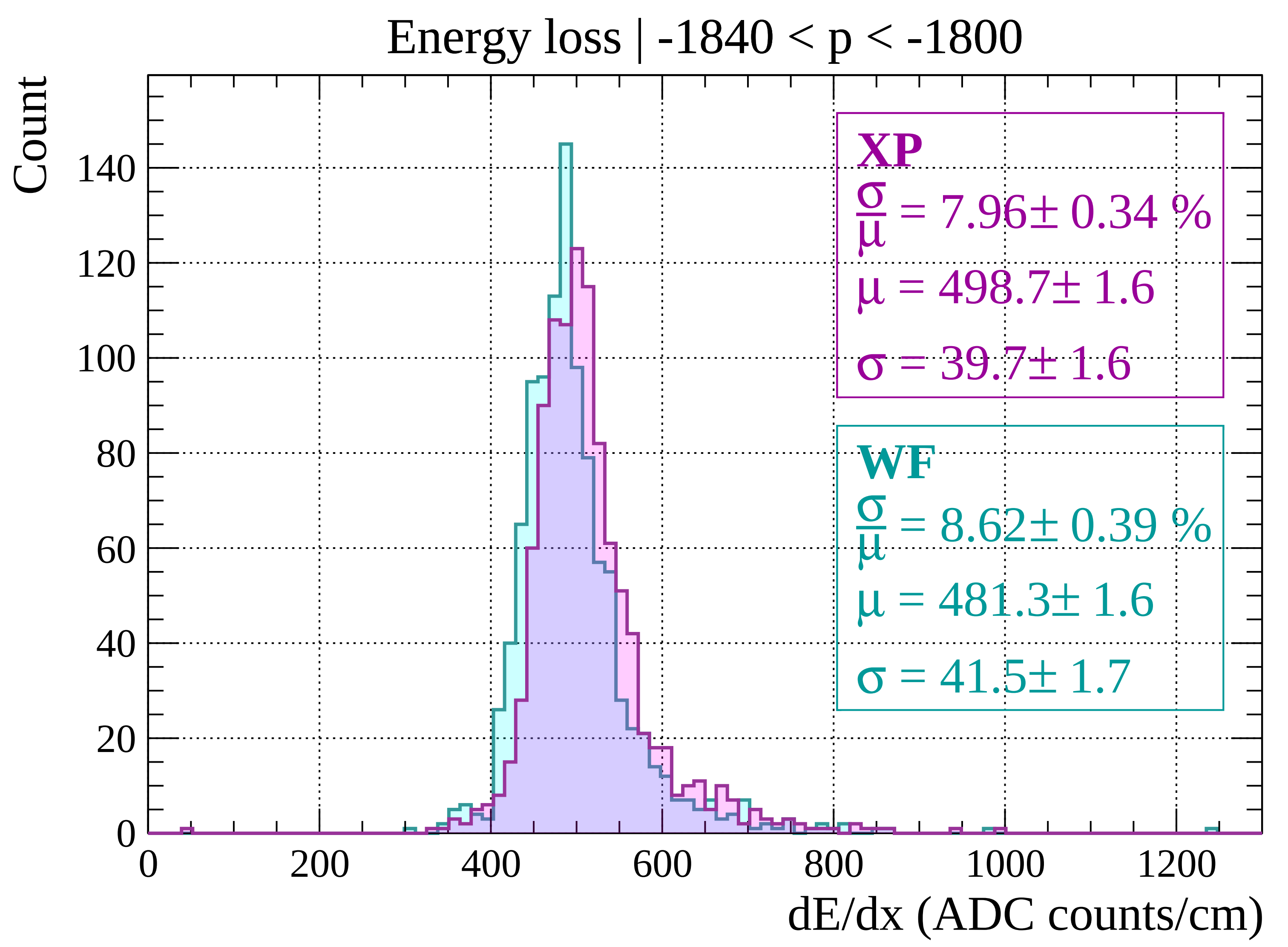
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 7.92± 0.45 %
μ = 504.4± 2.4
σ = 39.9± 2.1
WF
σ/μ = 7.63± 0.40 %
μ = 485.9± 2.2
σ = 37.1± 1.8

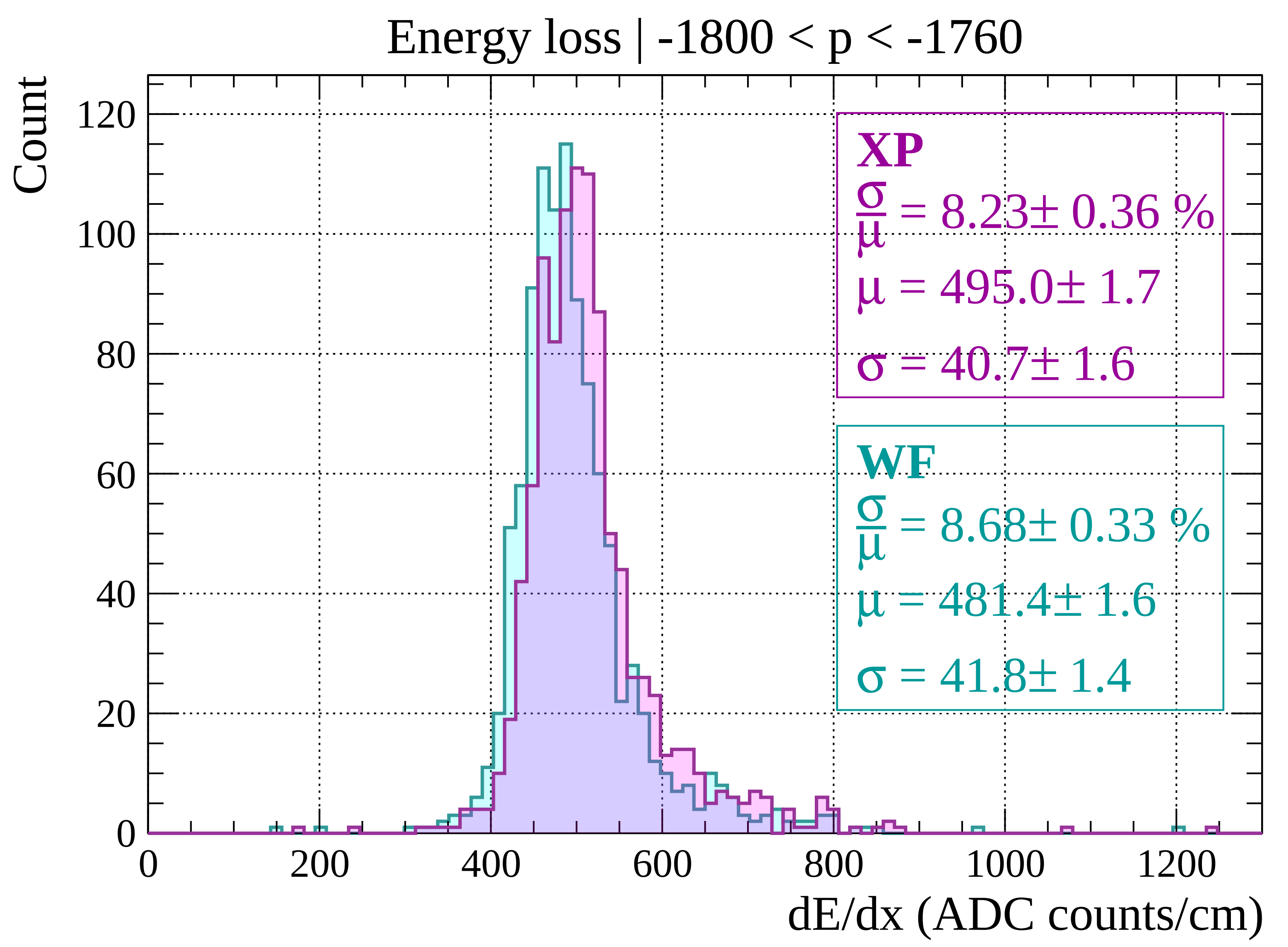

Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.65± 0.39 %
μ = 500.7± 1.8
σ = 43.3± 1.8
WF
σ/μ = 7.88± 0.34 %
μ = 483.4± 1.6
σ = 38.1± 1.5

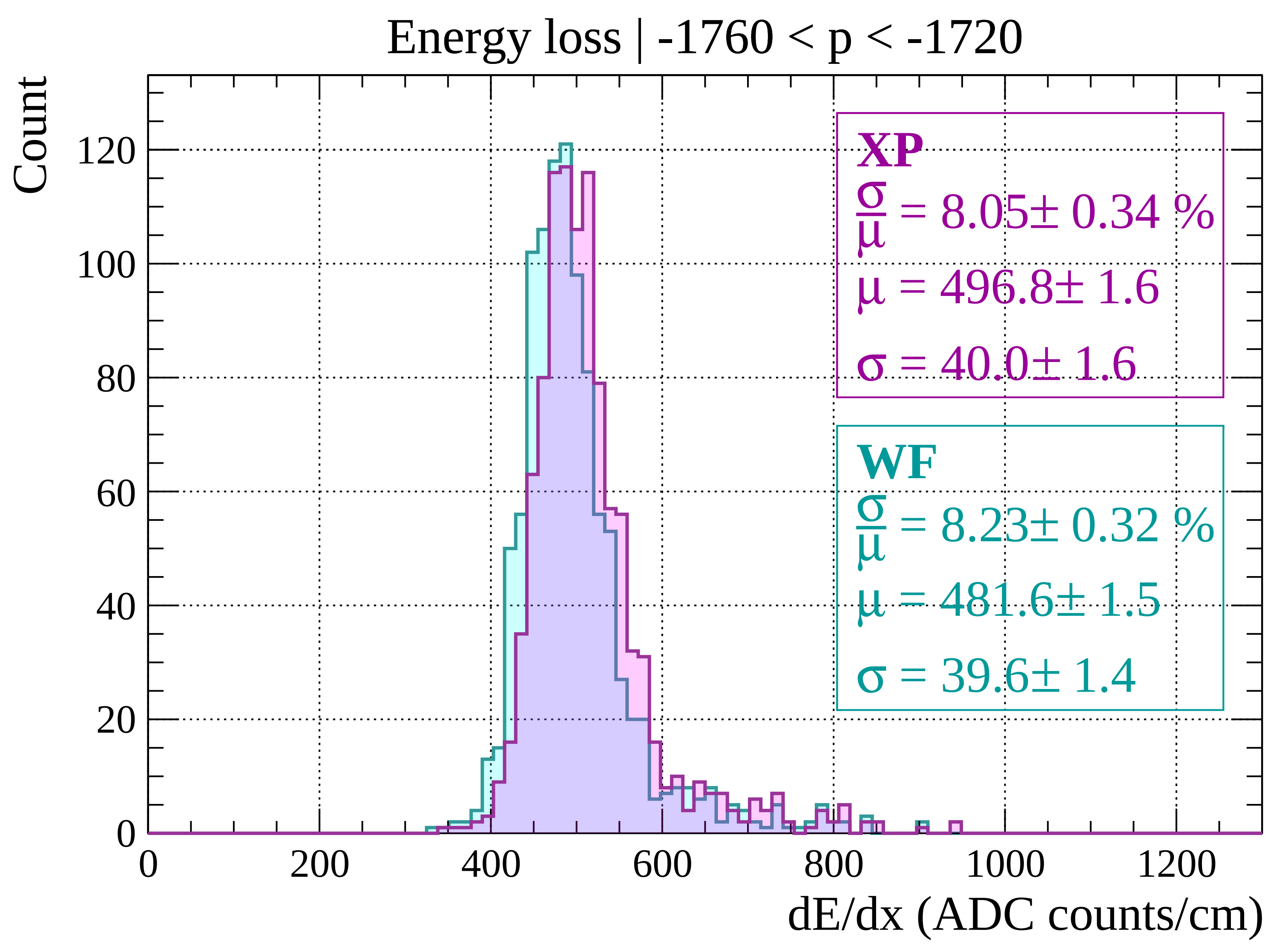

Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 9.15± 0.47 %
μ = 501.3± 2.0
σ = 45.9± 2.2
WF
$\frac{\sigma}{\mu}$ = 8.85± 0.34 %
μ = 486.8± 1.7
σ = 43.1± 1.5

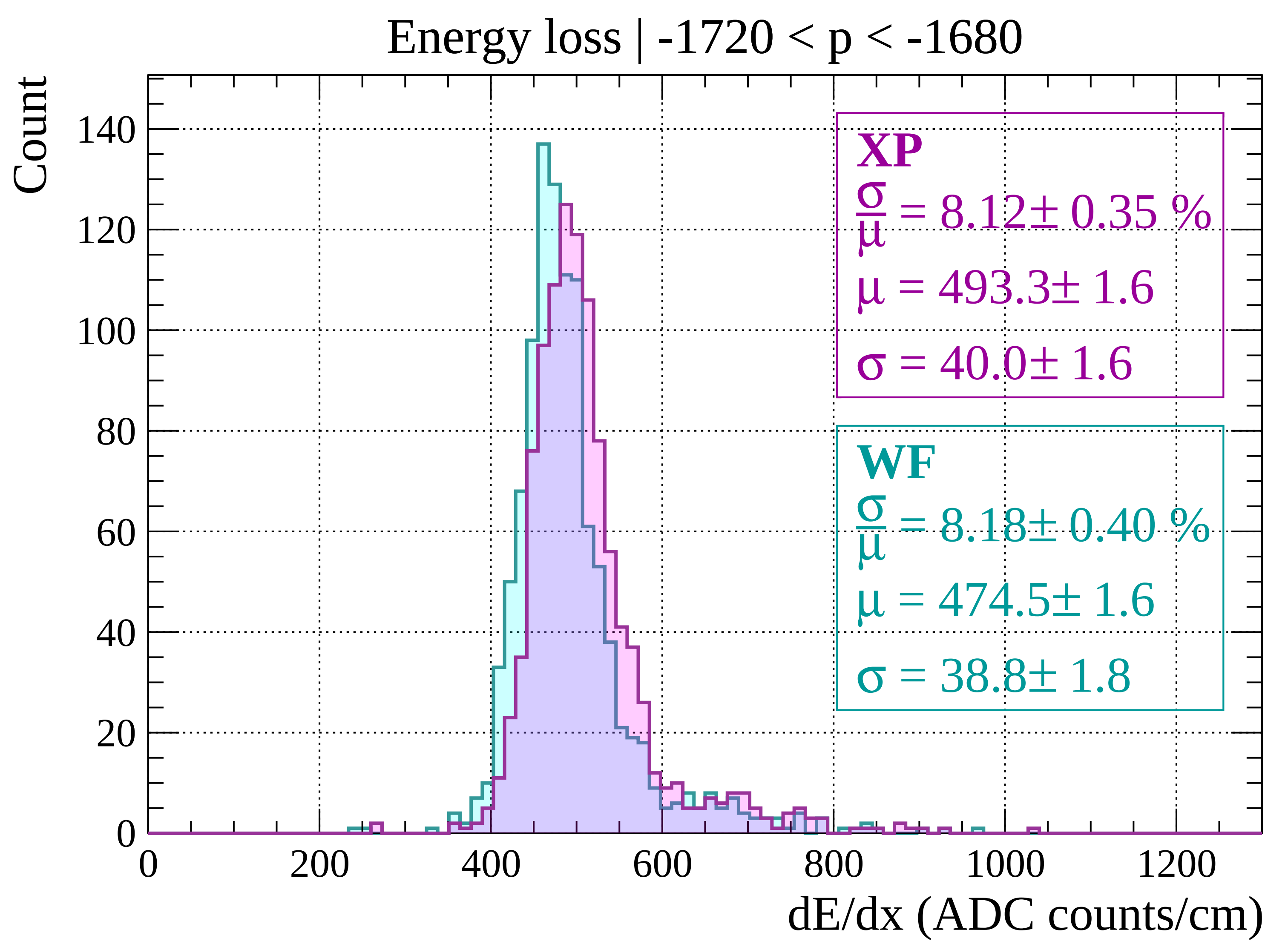


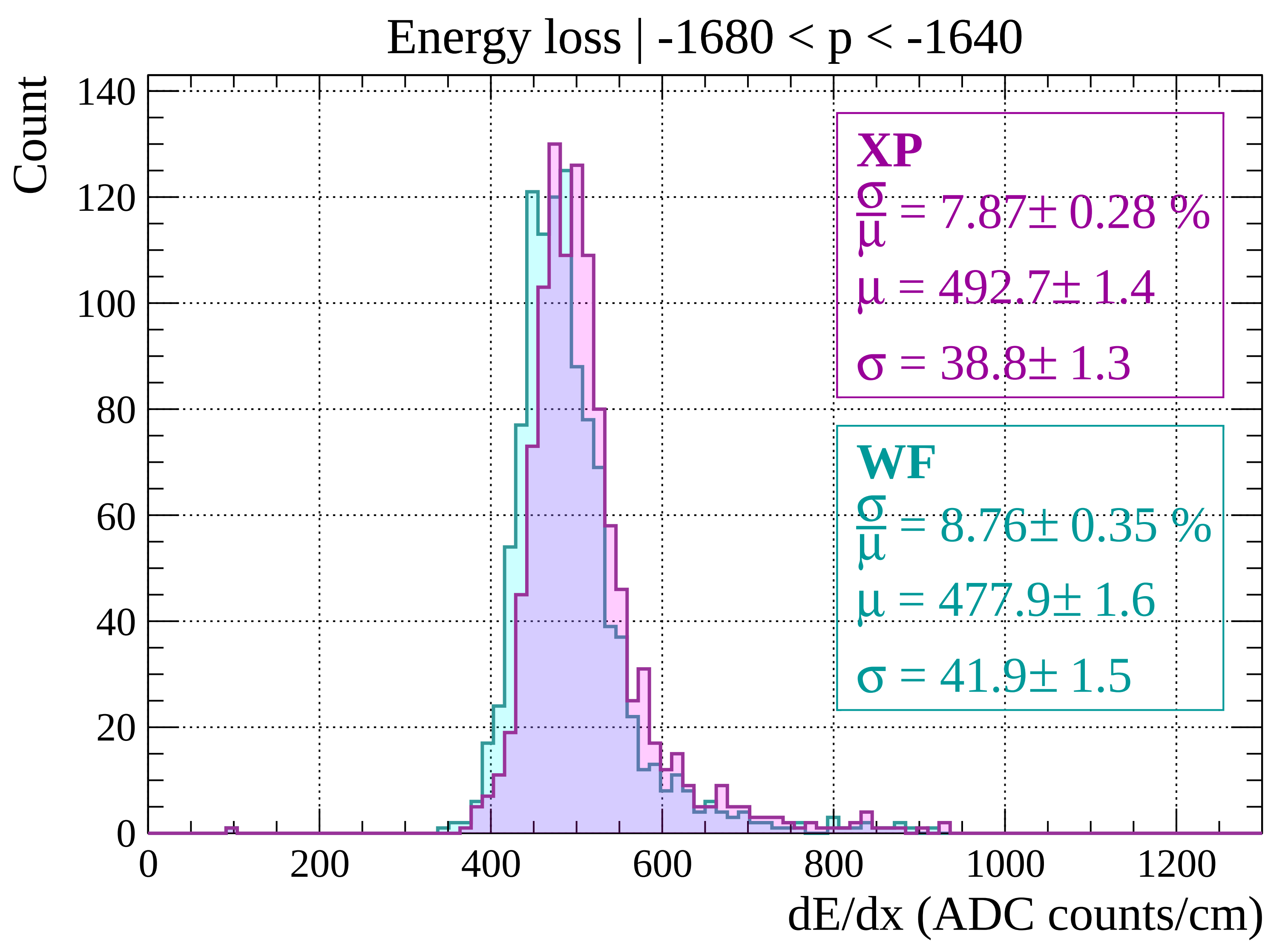
Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.10 \pm 0.30\ \%$
$\mu = 501.5 \pm 1.5$
$\sigma = 40.6 \pm 1.4$
WF
$\frac{\sigma}{\mu} = 8.18 \pm 0.35\ \%$
$\mu = 482.1 \pm 1.5$
$\sigma = 39.5 \pm 1.5$
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

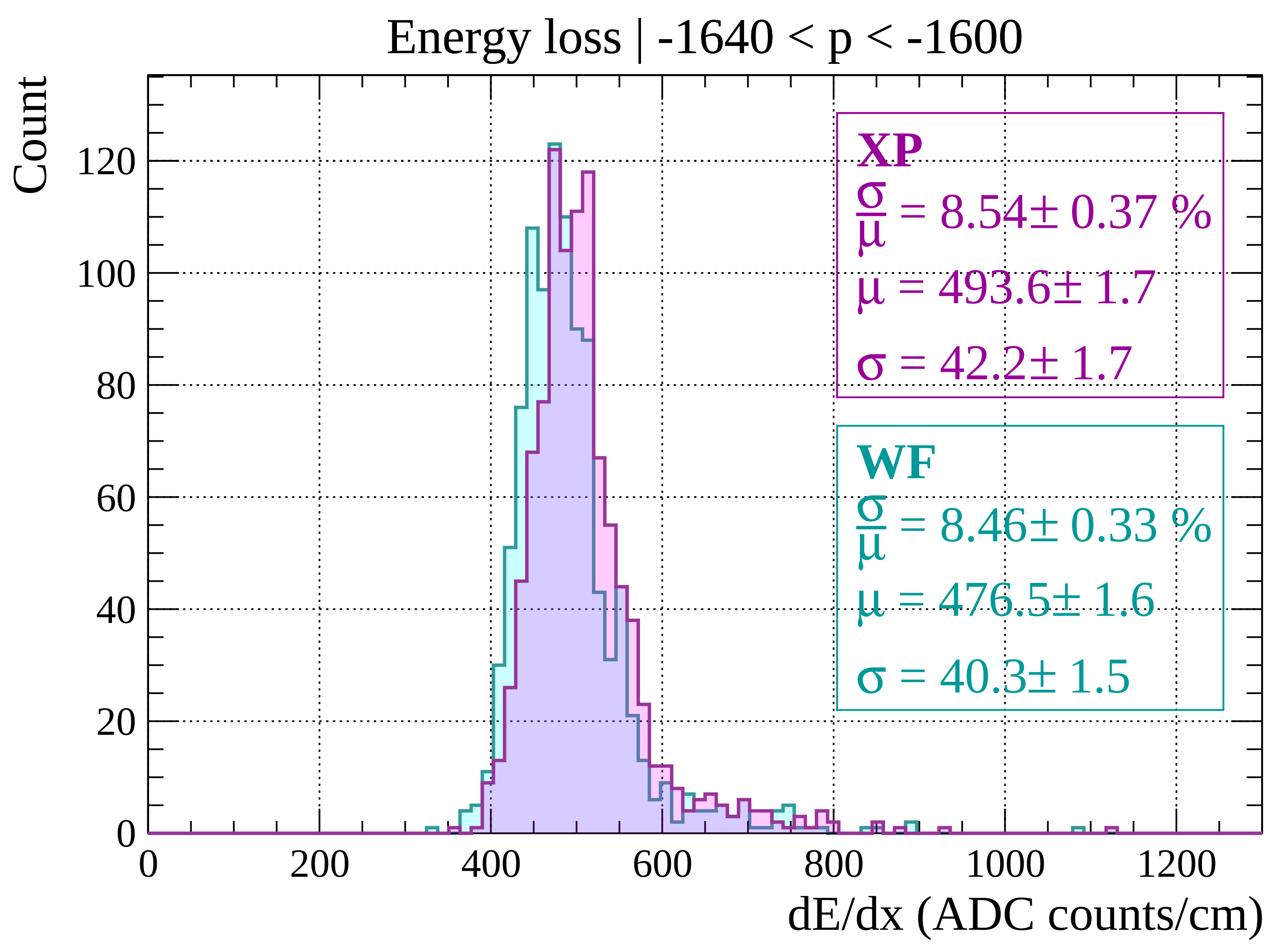

Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 7.96± 0.34 %
μ = 498.7± 1.6
σ = 39.7± 1.6
WF
σ/μ = 8.62± 0.39 %
μ = 481.3± 1.6
σ = 41.5± 1.7


Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.23± 0.36 %
μ = 495.0± 1.7
σ = 40.7± 1.6
WF
σ/μ = 8.68± 0.33 %
μ = 481.4± 1.6
σ = 41.8± 1.4

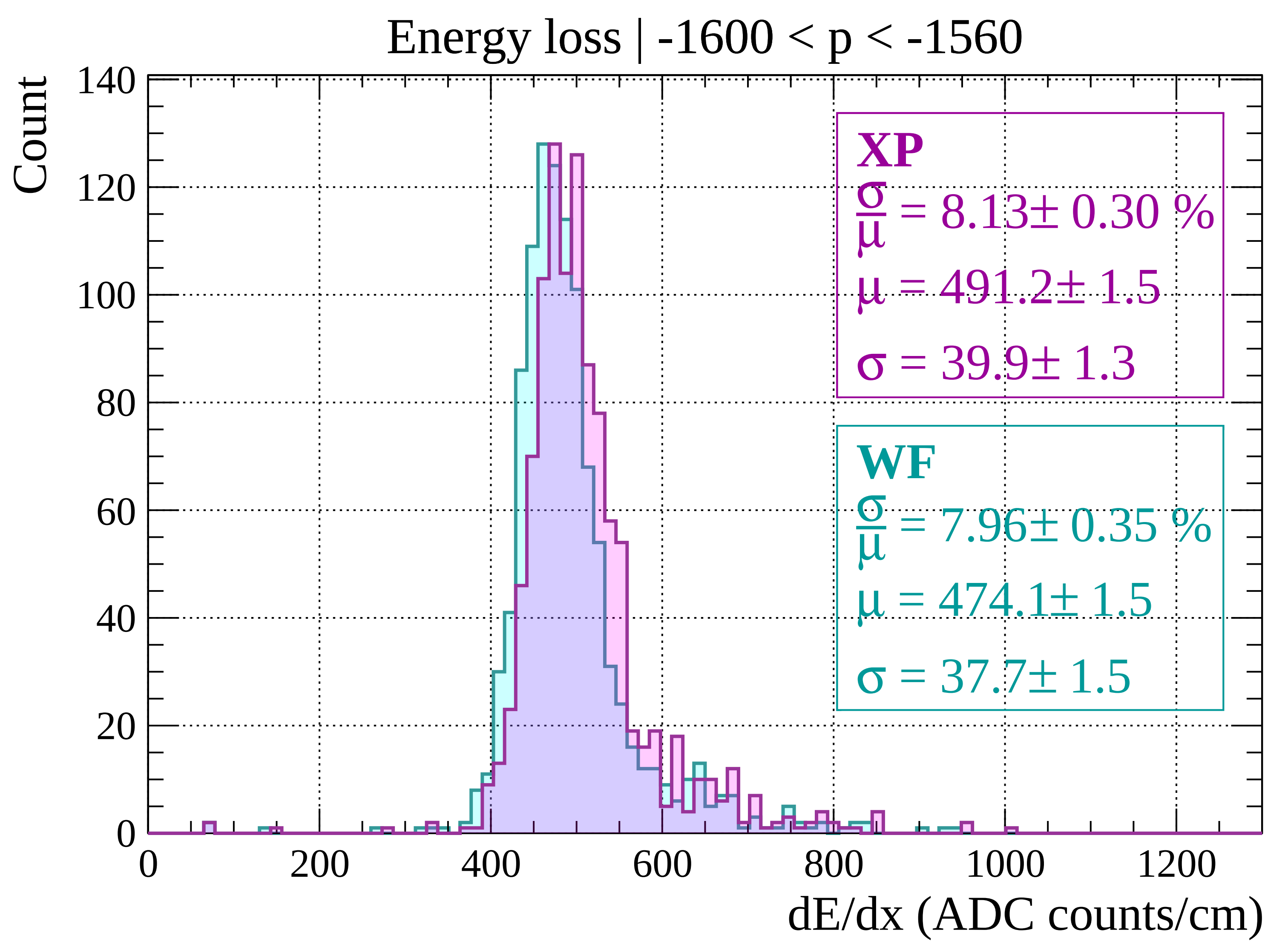

Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
XP
$\frac{\sigma}{\mu}$ = 8.05± 0.34 %
μ = 496.8± 1.6
σ = 40.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.23± 0.32 %
μ = 481.6± 1.5
σ = 39.6± 1.4

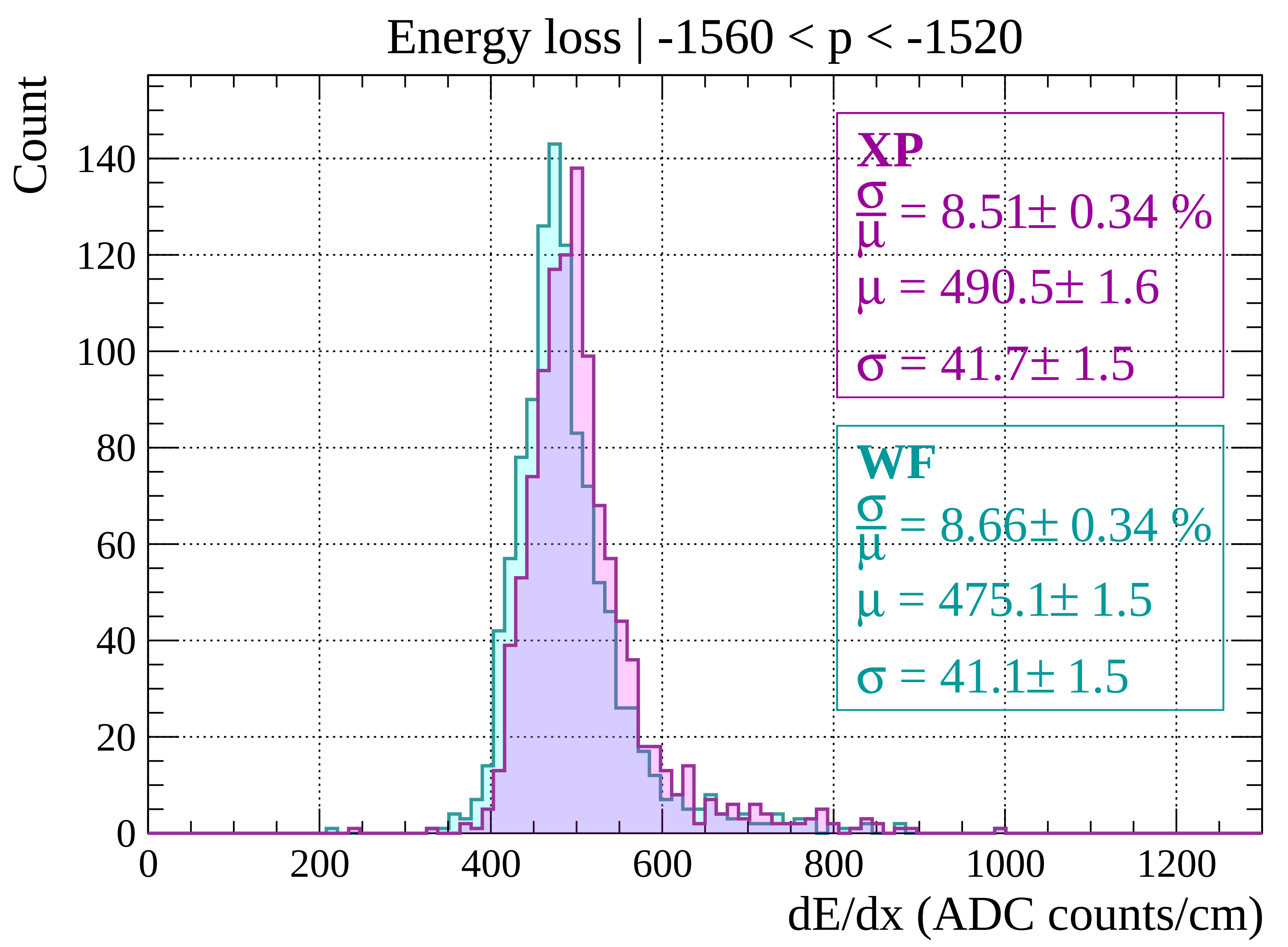

Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.12± 0.35 %
μ = 493.3± 1.6
σ = 40.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.18± 0.40 %
μ = 474.5± 1.6
σ = 38.8± 1.8

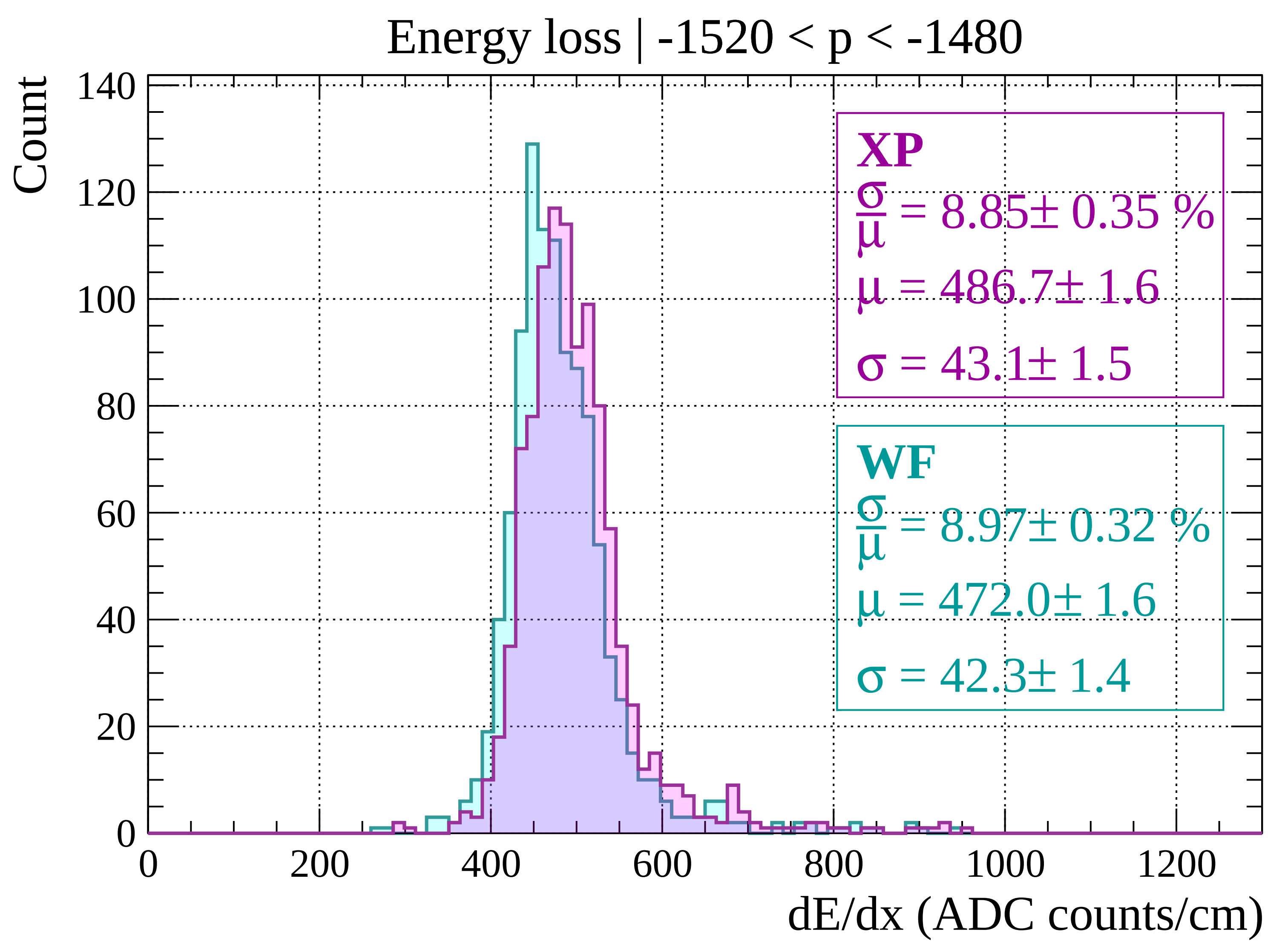

Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 7.87± 0.28 %
μ = 492.7± 1.4
σ = 38.8± 1.3
WF
σ/μ = 8.76± 0.35 %
μ = 477.9± 1.6
σ = 41.9± 1.5



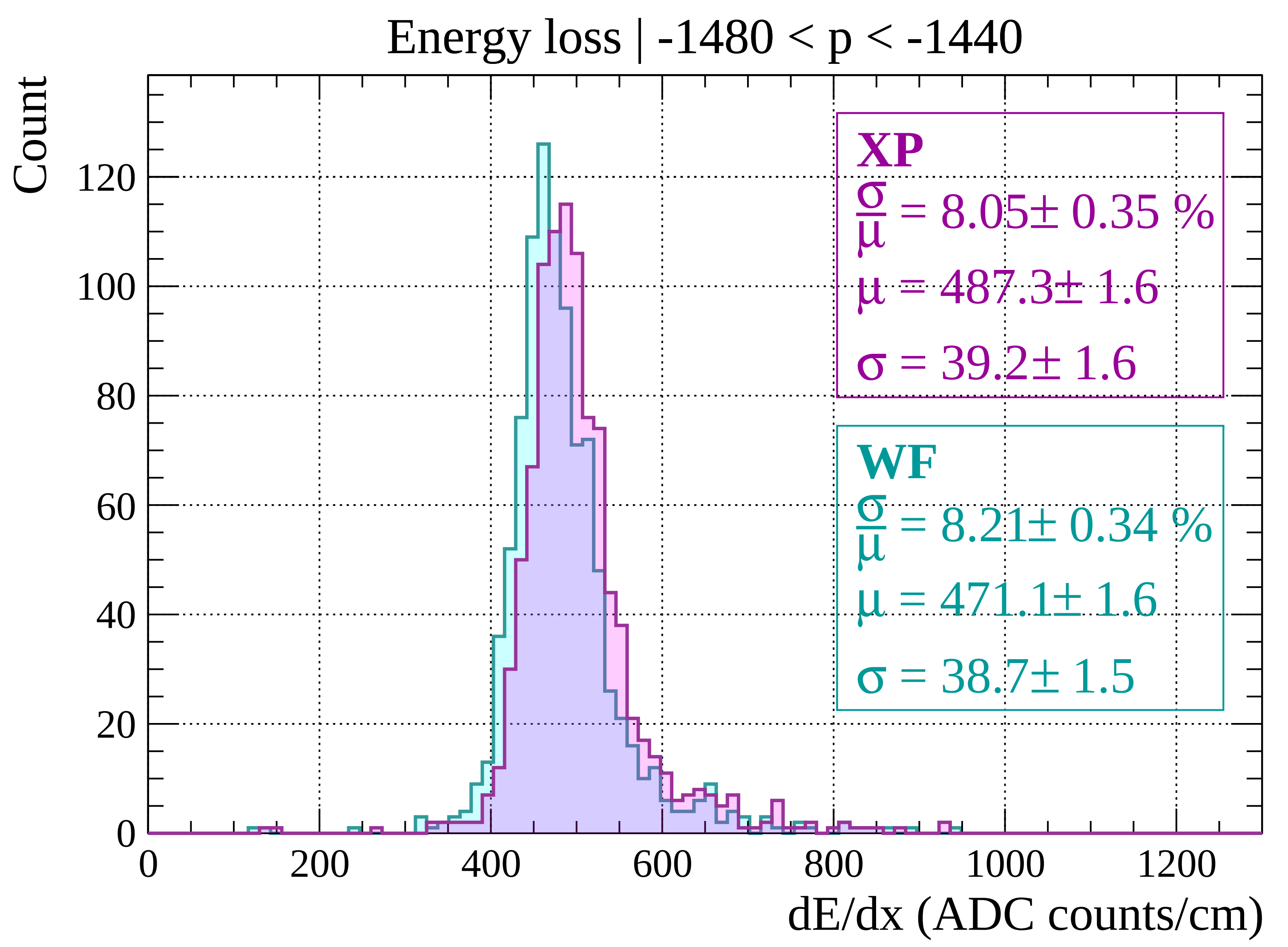
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.54± 0.37 %
μ = 493.6± 1.7
σ = 42.2± 1.7
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.33 %
μ = 476.5± 1.6
σ = 40.3± 1.5

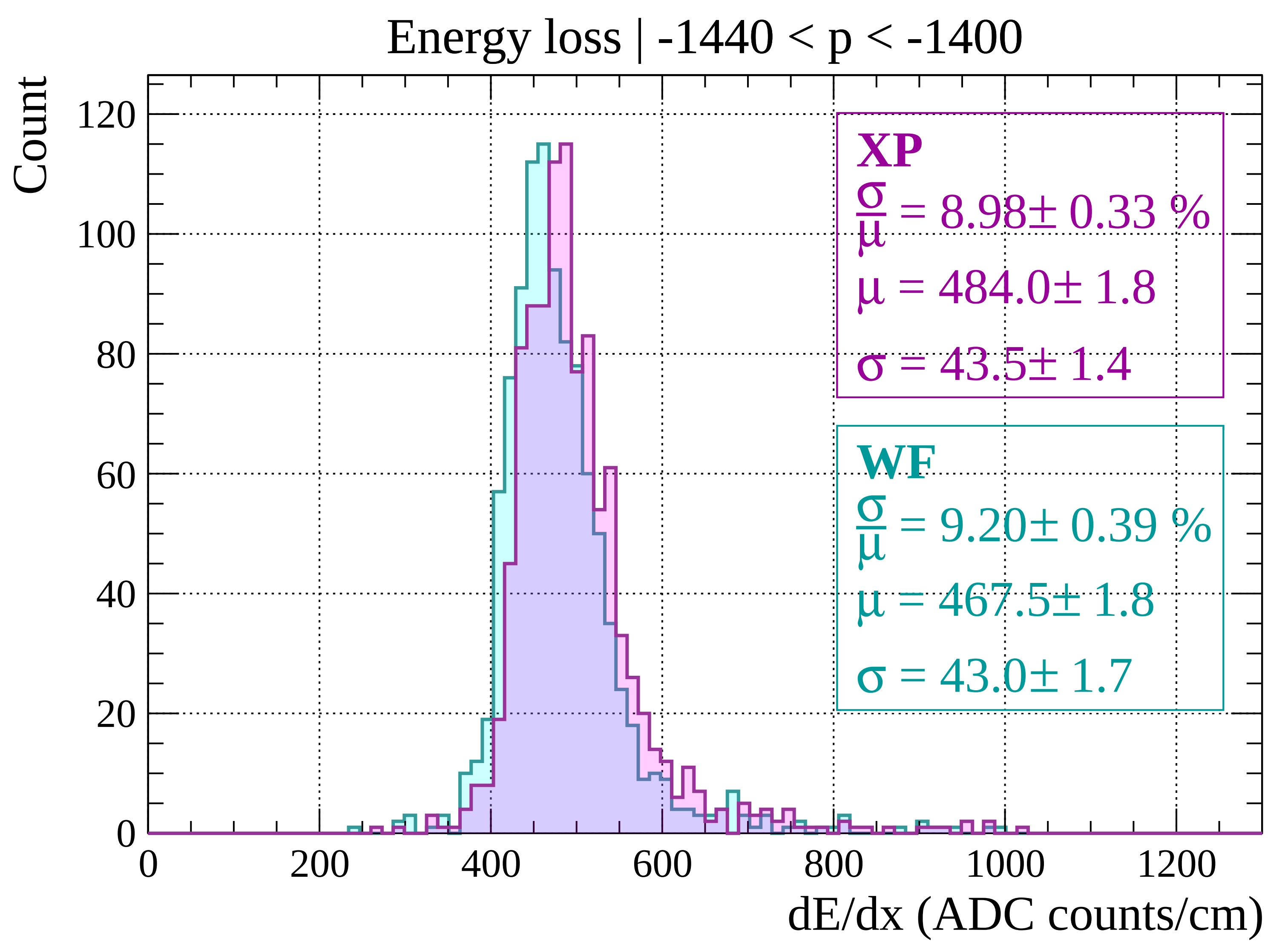

Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 8.13± 0.30 %
μ = 491.2± 1.5
σ = 39.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 7.96± 0.35 %
μ = 474.1± 1.5
σ = 37.7± 1.5


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.51± 0.34 %
μ = 490.5± 1.6
σ = 41.7± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.66± 0.34 %
μ = 475.1± 1.5
σ = 41.1± 1.5

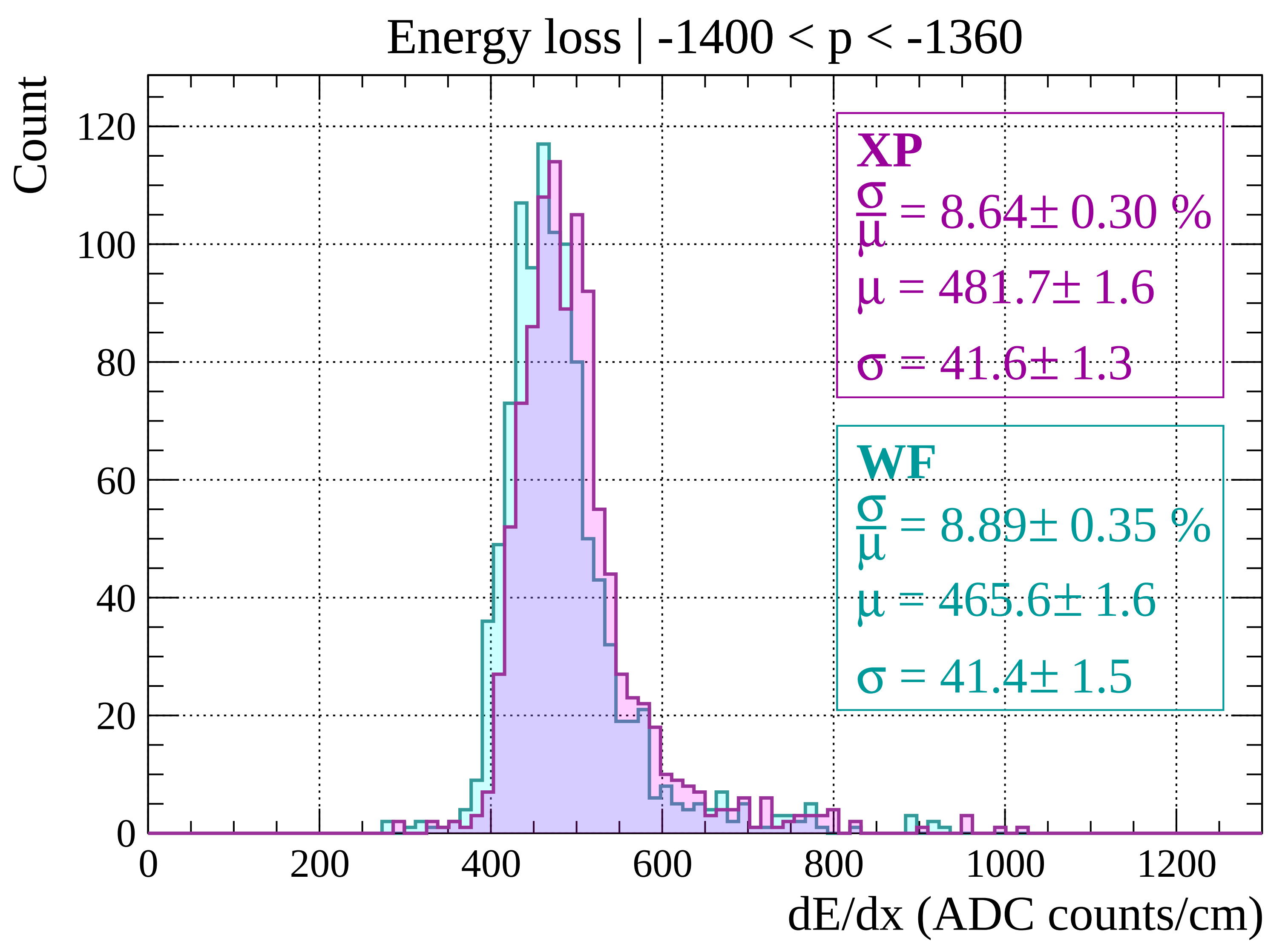


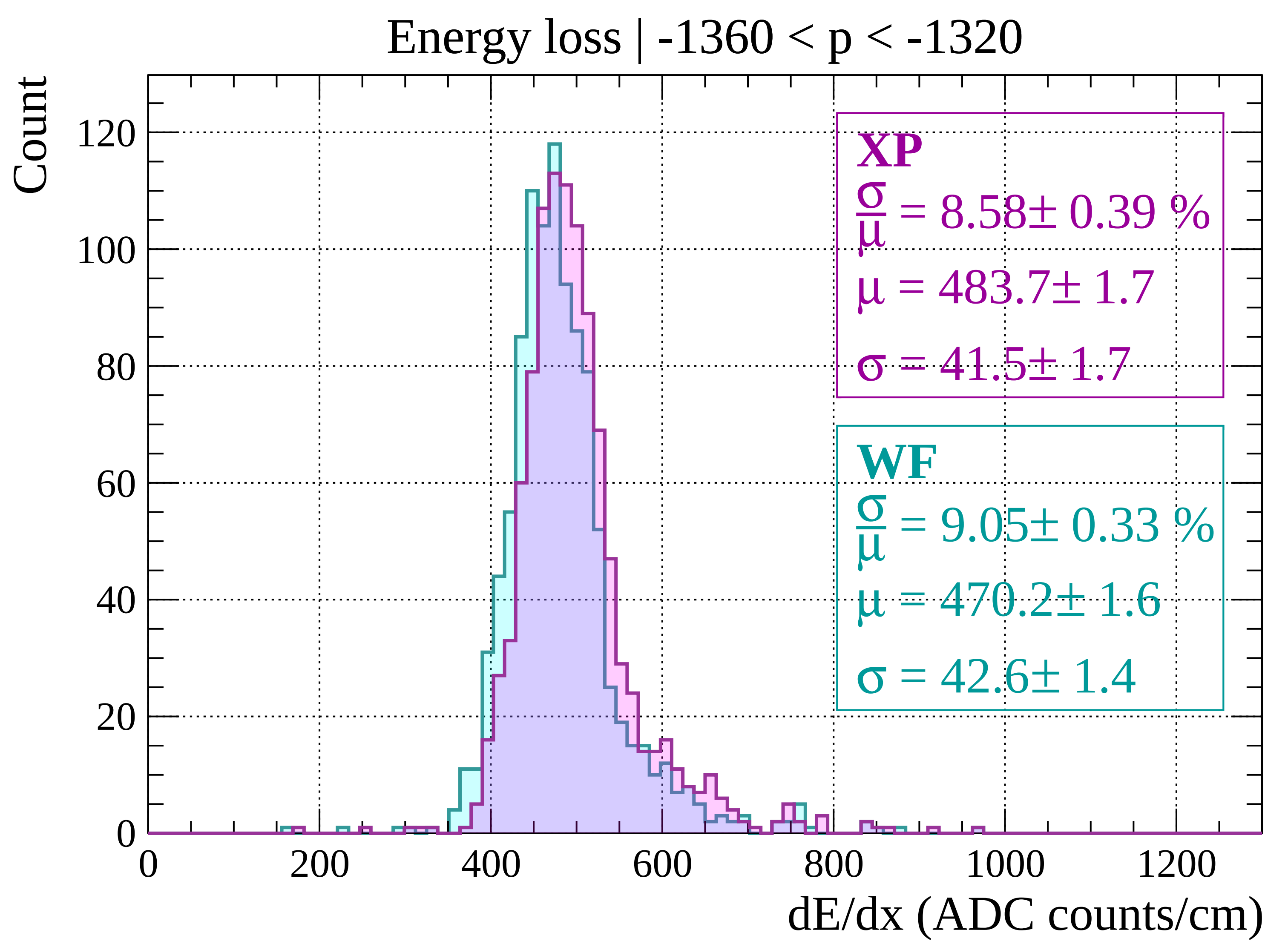
Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.85± 0.35 %
μ = 486.7± 1.6
σ = 43.1± 1.5
WF
σ/μ = 8.97± 0.32 %
μ = 472.0± 1.6
σ = 42.3± 1.4


Energy loss | -1480 < p < -1440
Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.05± 0.35 %
μ = 487.3± 1.6
σ = 39.2± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.21± 0.34 %
μ = 471.1± 1.6
σ = 38.7± 1.5

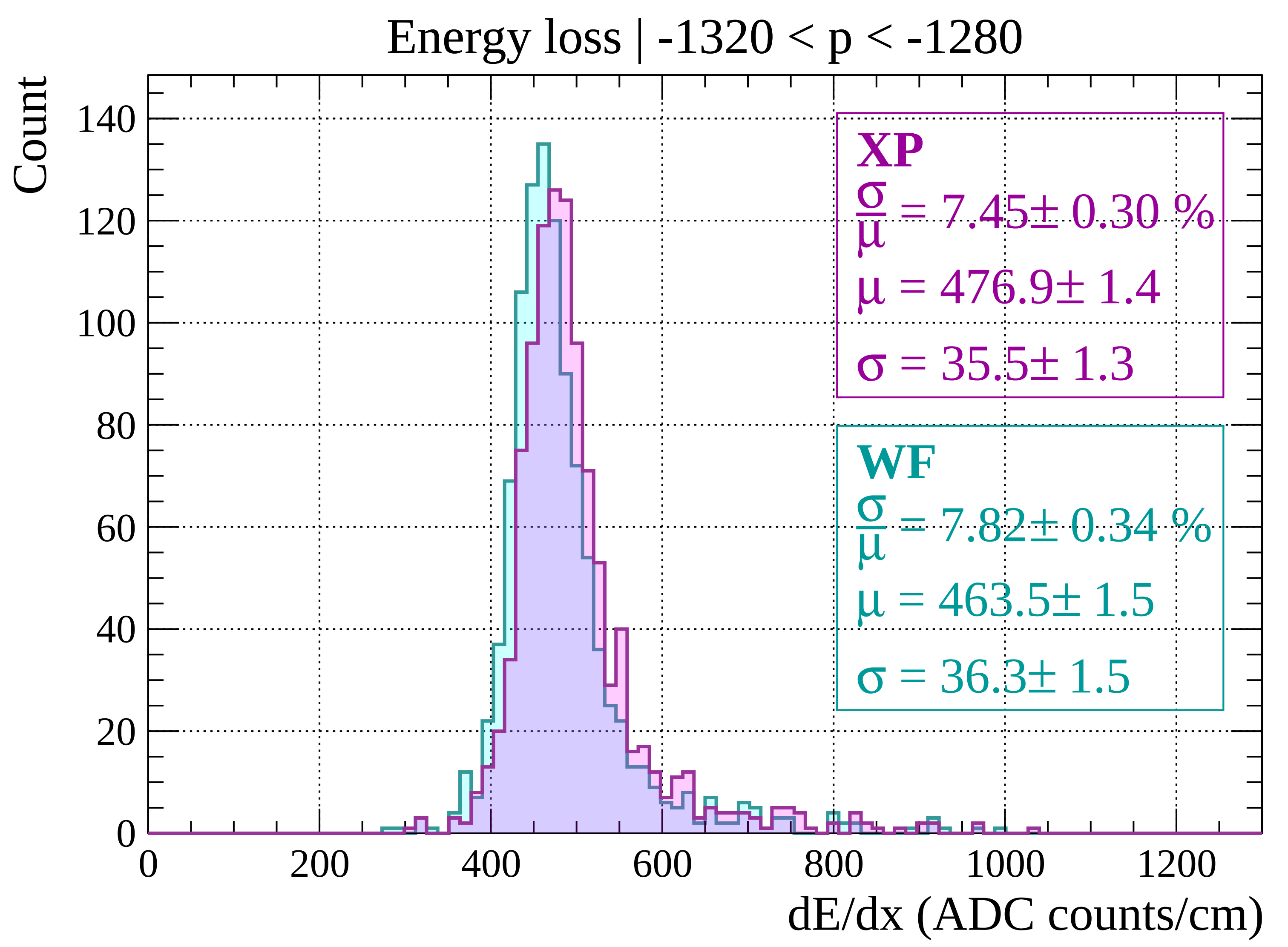


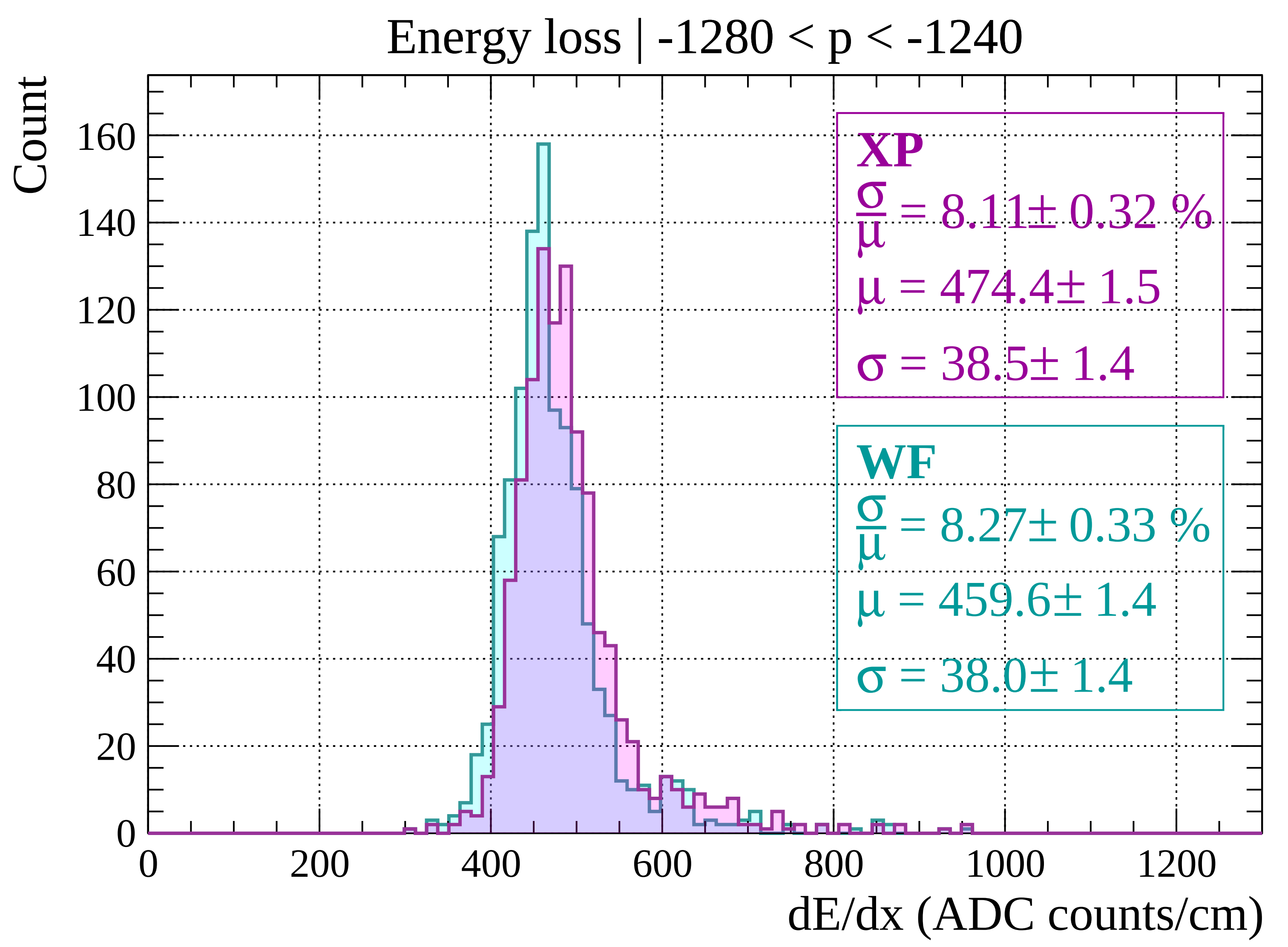
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.98± 0.33 %
μ = 484.0± 1.8
σ = 43.5± 1.4
WF
σ/μ = 9.20± 0.39 %
μ = 467.5± 1.8
σ = 43.0± 1.7

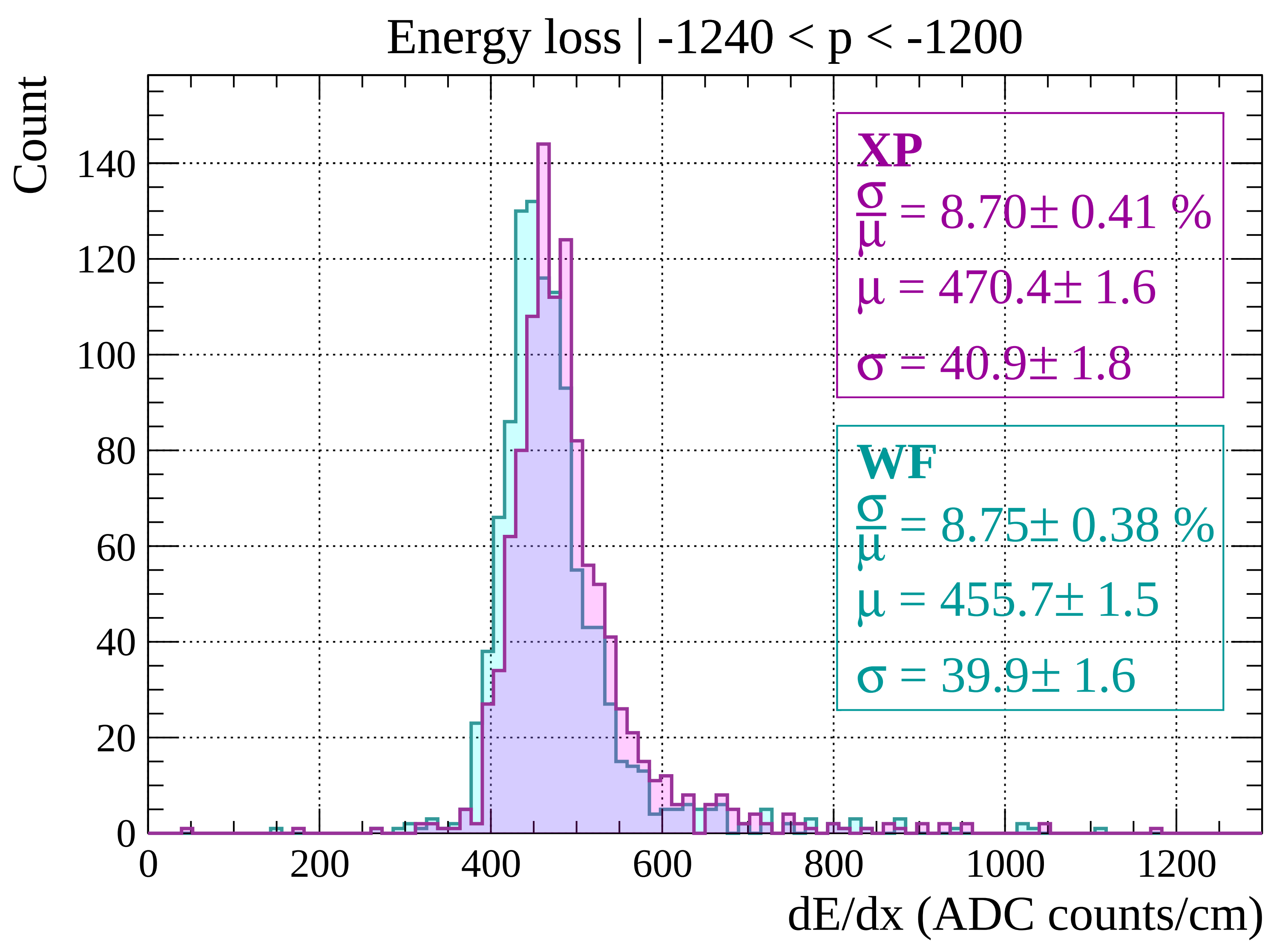

Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.64± 0.30 %
μ = 481.7± 1.6
σ = 41.6± 1.3
WF
σ/μ = 8.89± 0.35 %
μ = 465.6± 1.6
σ = 41.4± 1.5

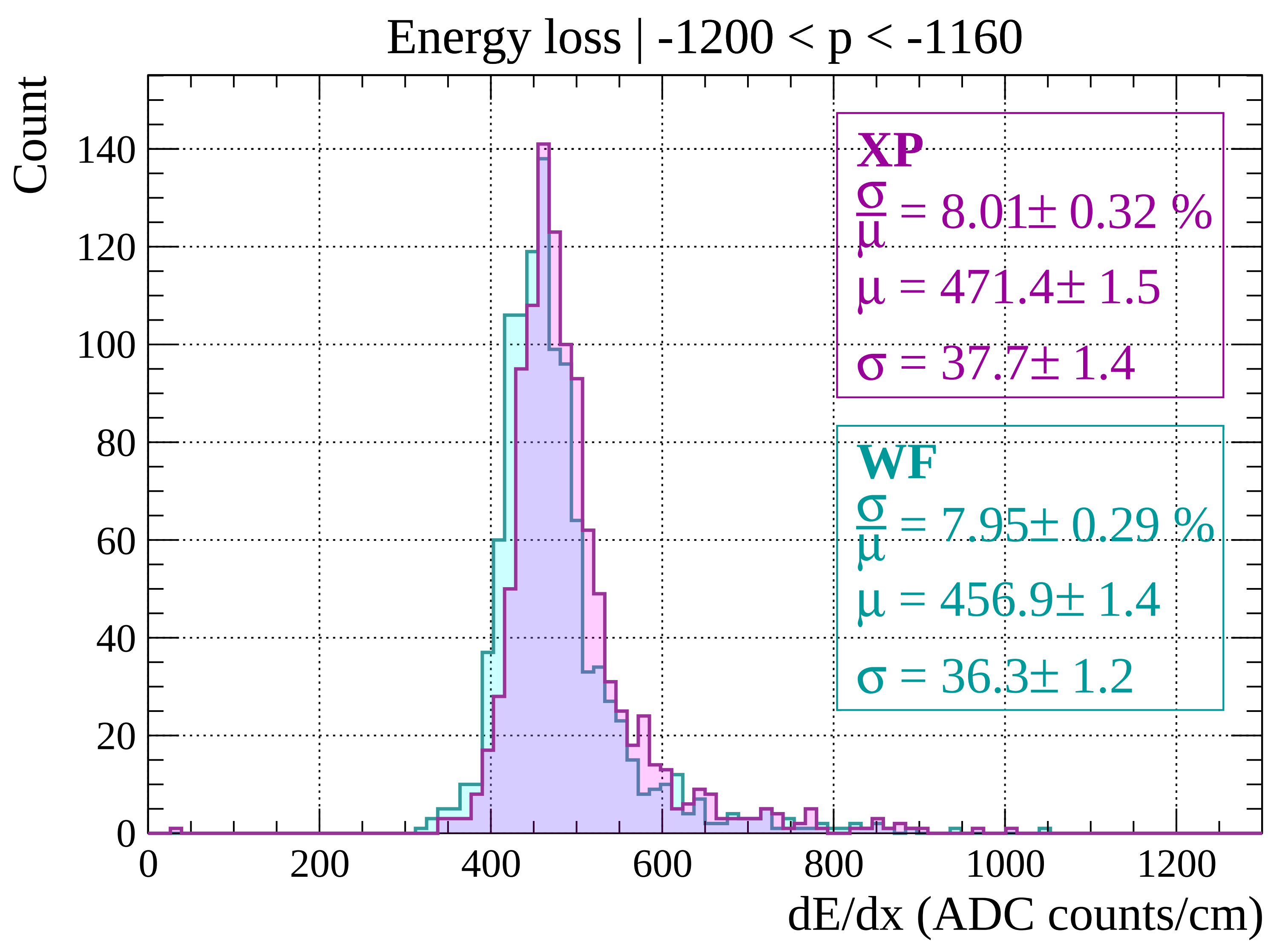


Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.58± 0.39 %
μ = 483.7± 1.7
σ = 41.5± 1.7
WF
σ/μ = 9.05± 0.33 %
μ = 470.2± 1.6
σ = 42.6± 1.4

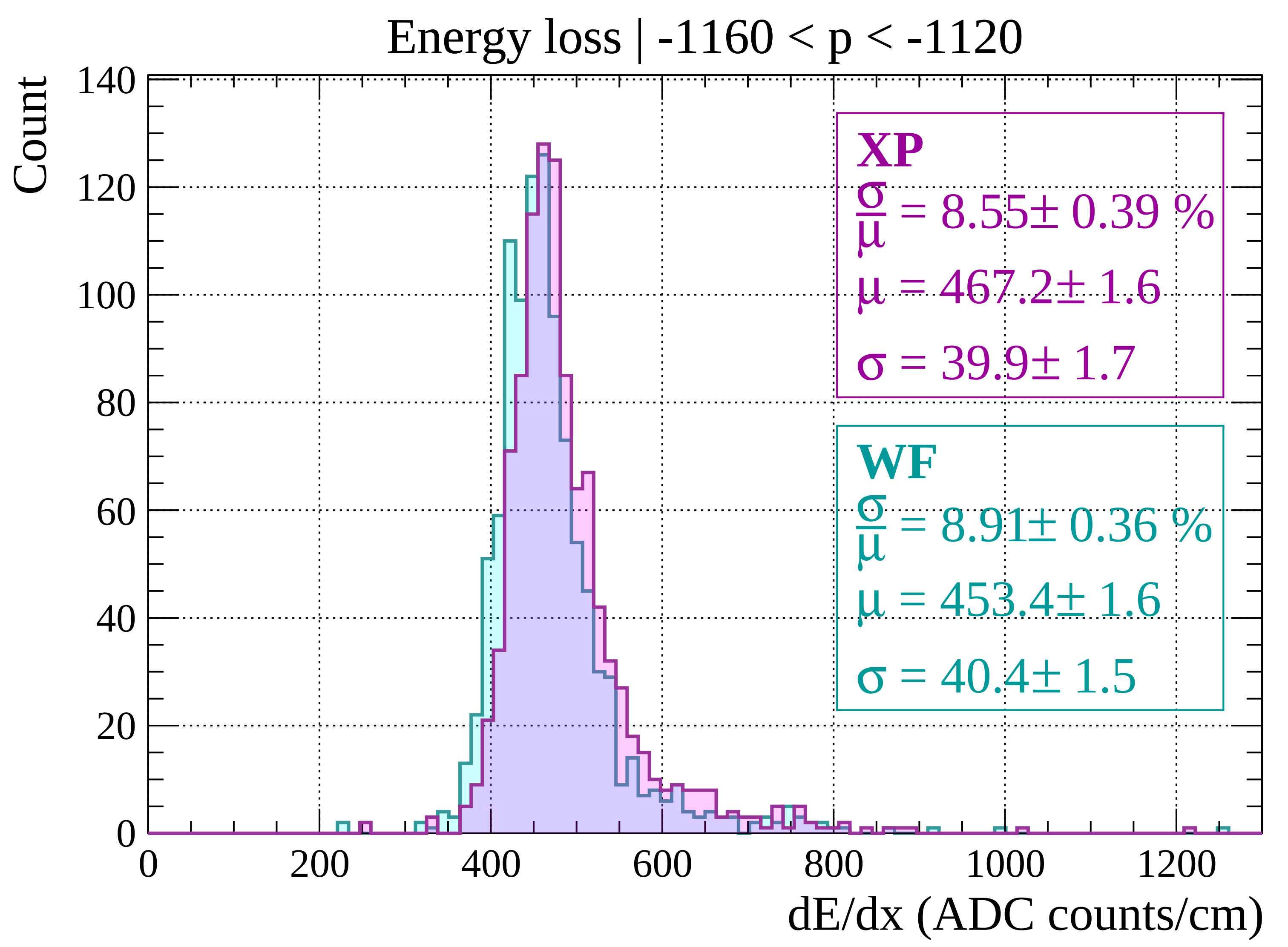

Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.45± 0.30 %
μ = 476.9± 1.4
σ = 35.5± 1.3
WF
σ/μ = 7.82± 0.34 %
μ = 463.5± 1.5
σ = 36.3± 1.5


Energy loss | -1280 < p < -1240
Count
160
140
120
100
80
60
40
20
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.11± 0.32 %
μ = 474.4± 1.5
σ = 38.5± 1.4
WF
σ/μ = 8.27± 0.33 %
μ = 459.6± 1.4
σ = 38.0± 1.4

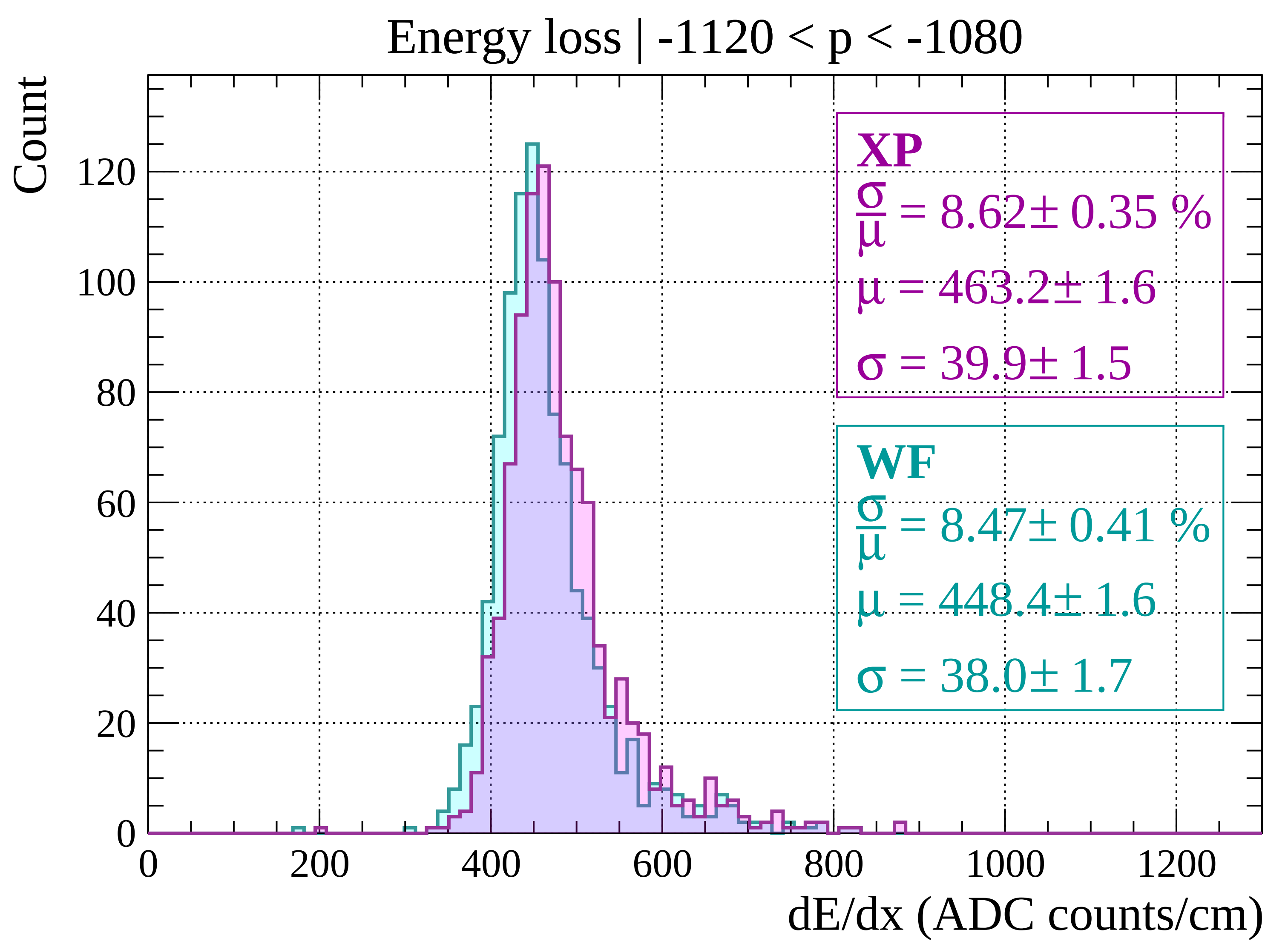

Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
XP
σ/μ = 8.70± 0.41 %
μ = 470.4± 1.6
σ = 40.9± 1.8
WF
σ/μ = 8.75± 0.38 %
μ = 455.7± 1.5
σ = 39.9± 1.6

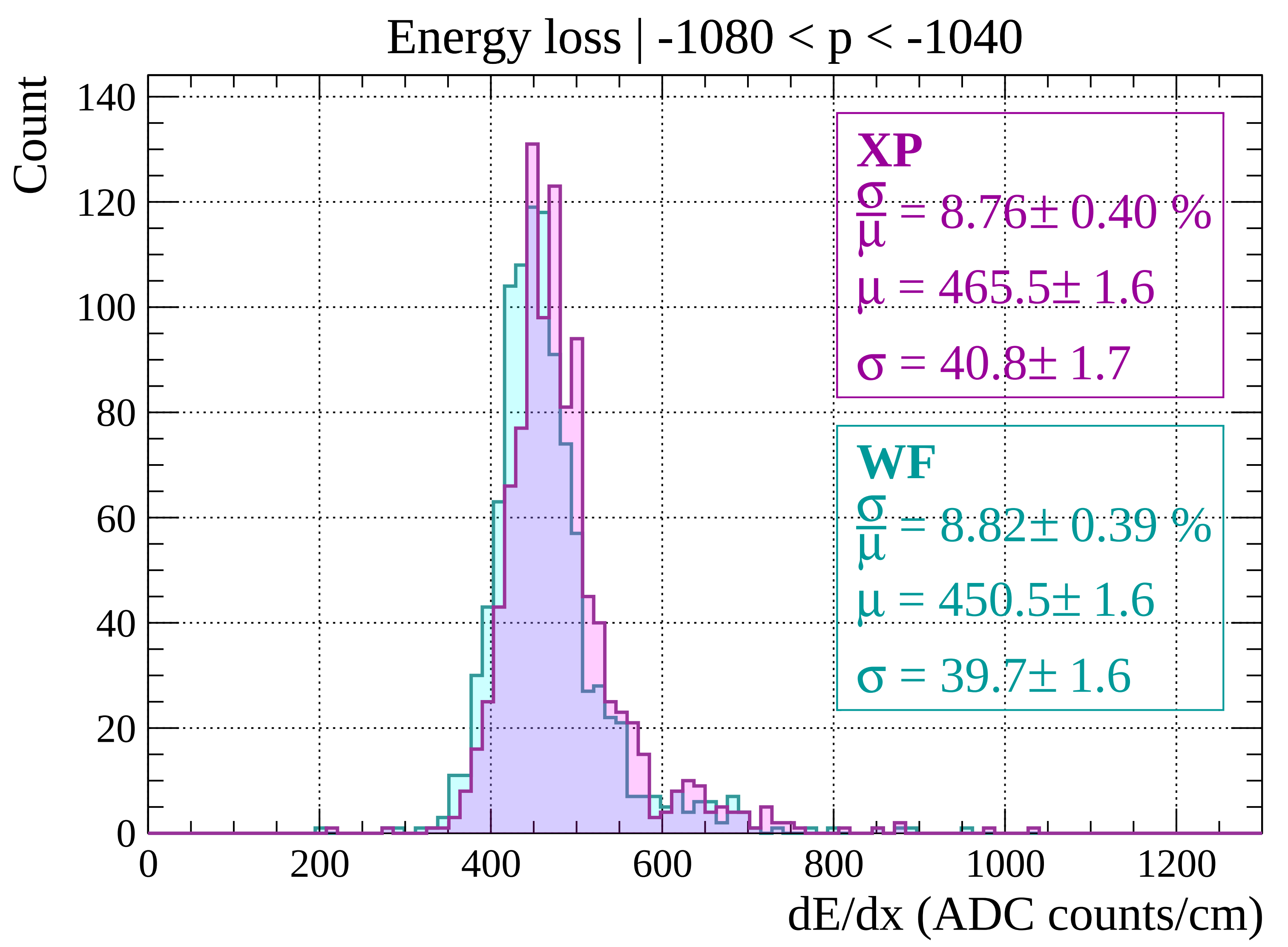


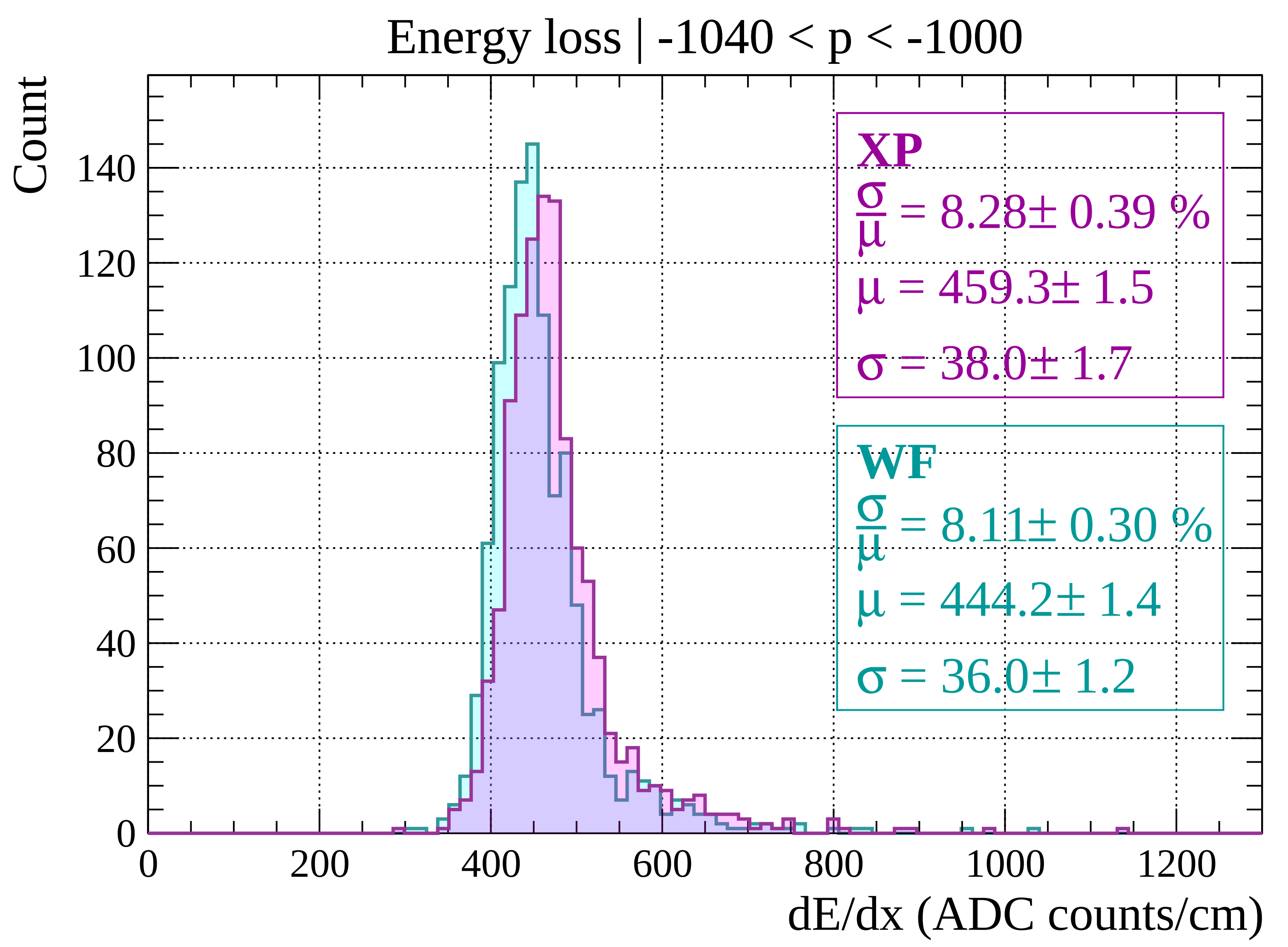
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.01 \pm 0.32$ %
$\mu = 471.4 \pm 1.5$
$\sigma = 37.7 \pm 1.4$
WF
$\frac{\sigma}{\mu} = 7.95 \pm 0.29$ %
$\mu = 456.9 \pm 1.4$
$\sigma = 36.3 \pm 1.2$

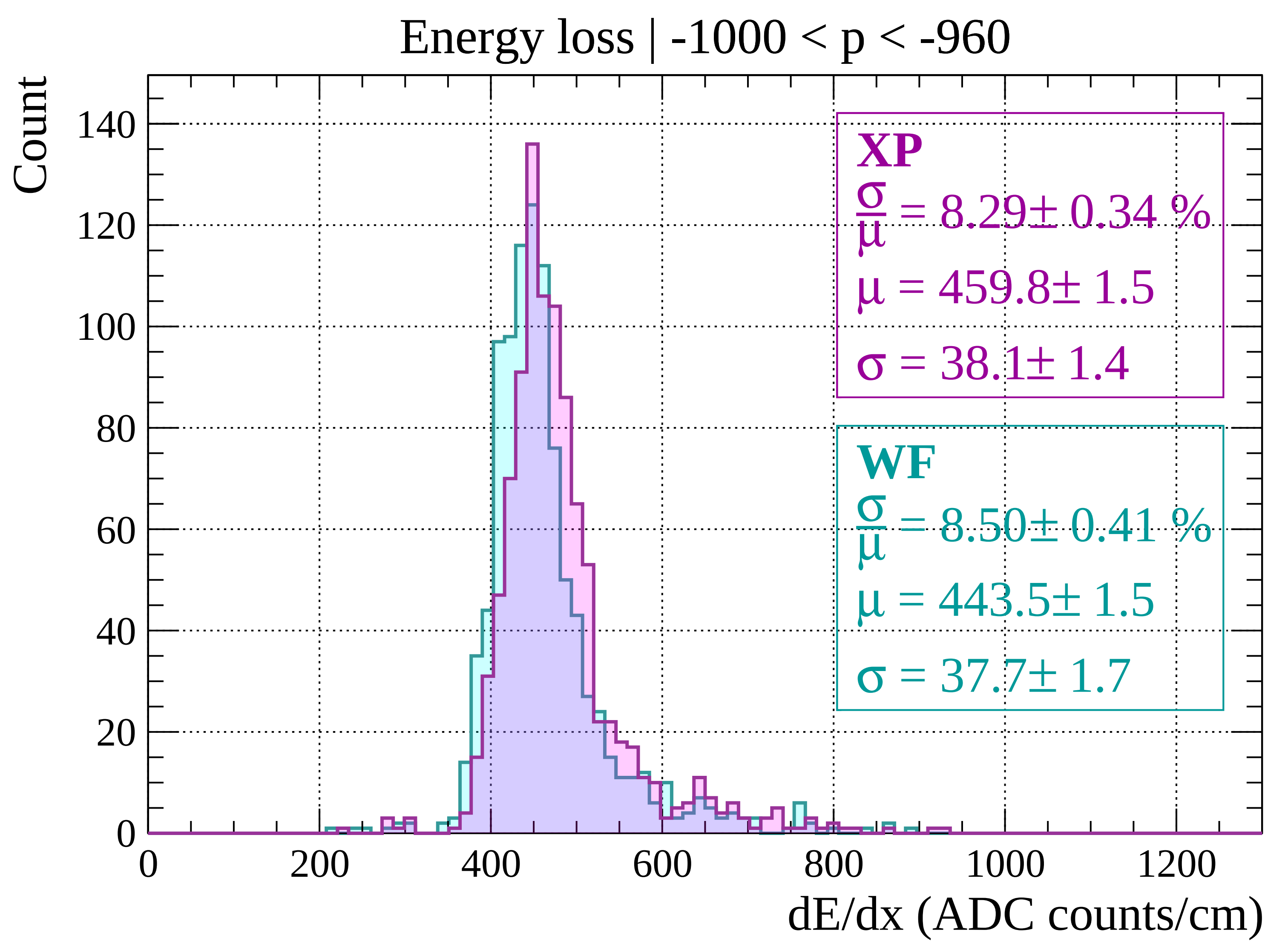


Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
XP
σ/μ = 8.55± 0.39 %
μ = 467.2± 1.6
σ = 39.9± 1.7
WF
σ/μ = 8.91± 0.36 %
μ = 453.4± 1.6
σ = 40.4± 1.5

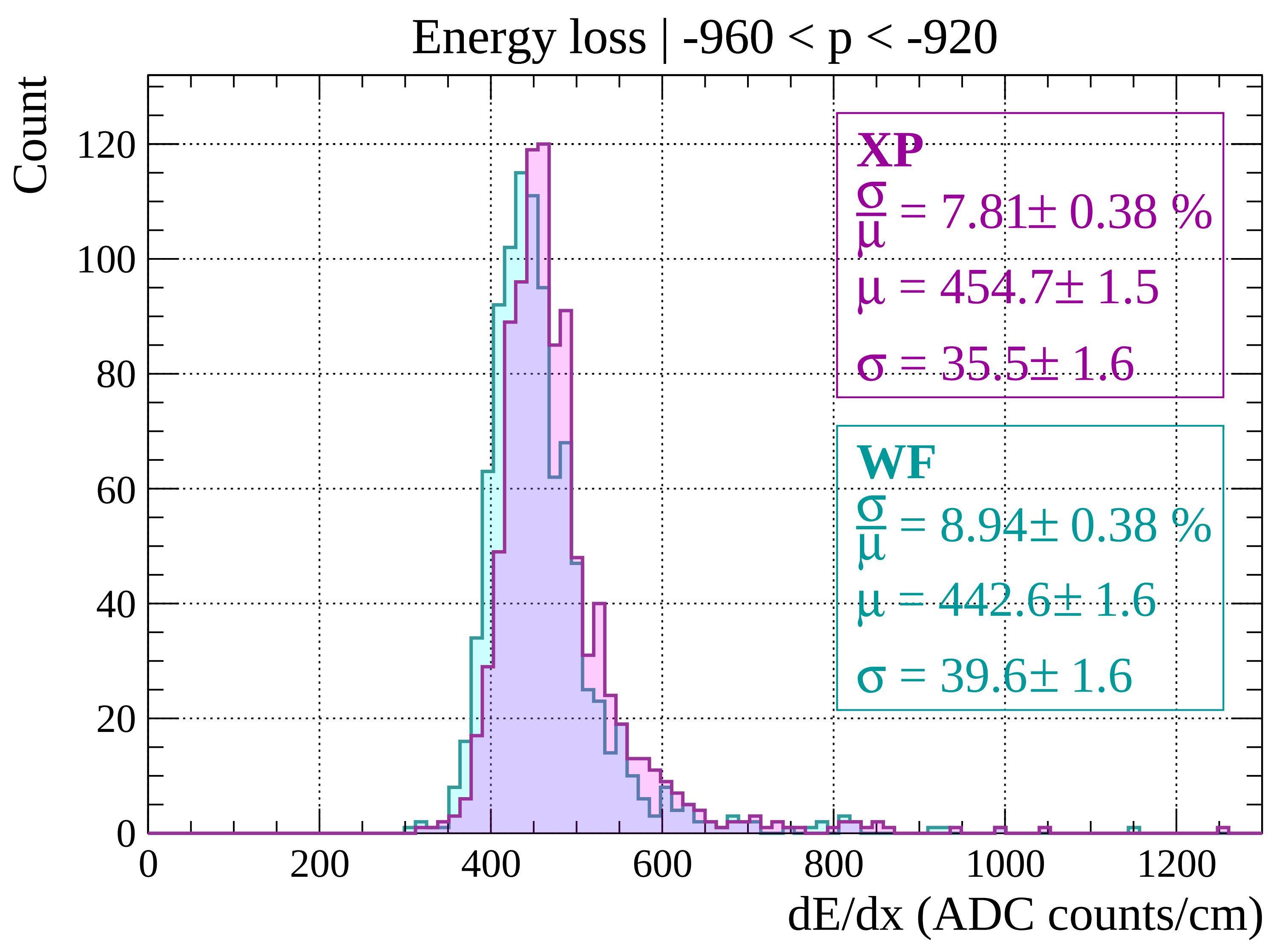

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 8.62± 0.35 %
μ = 463.2± 1.6
σ = 39.9± 1.5
WF
σ/μ = 8.47± 0.41 %
μ = 448.4± 1.6
σ = 38.0± 1.7


Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.76± 0.40 %
μ = 465.5± 1.6
σ = 40.8± 1.7
WF
σ/μ = 8.82± 0.39 %
μ = 450.5± 1.6
σ = 39.7± 1.6
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

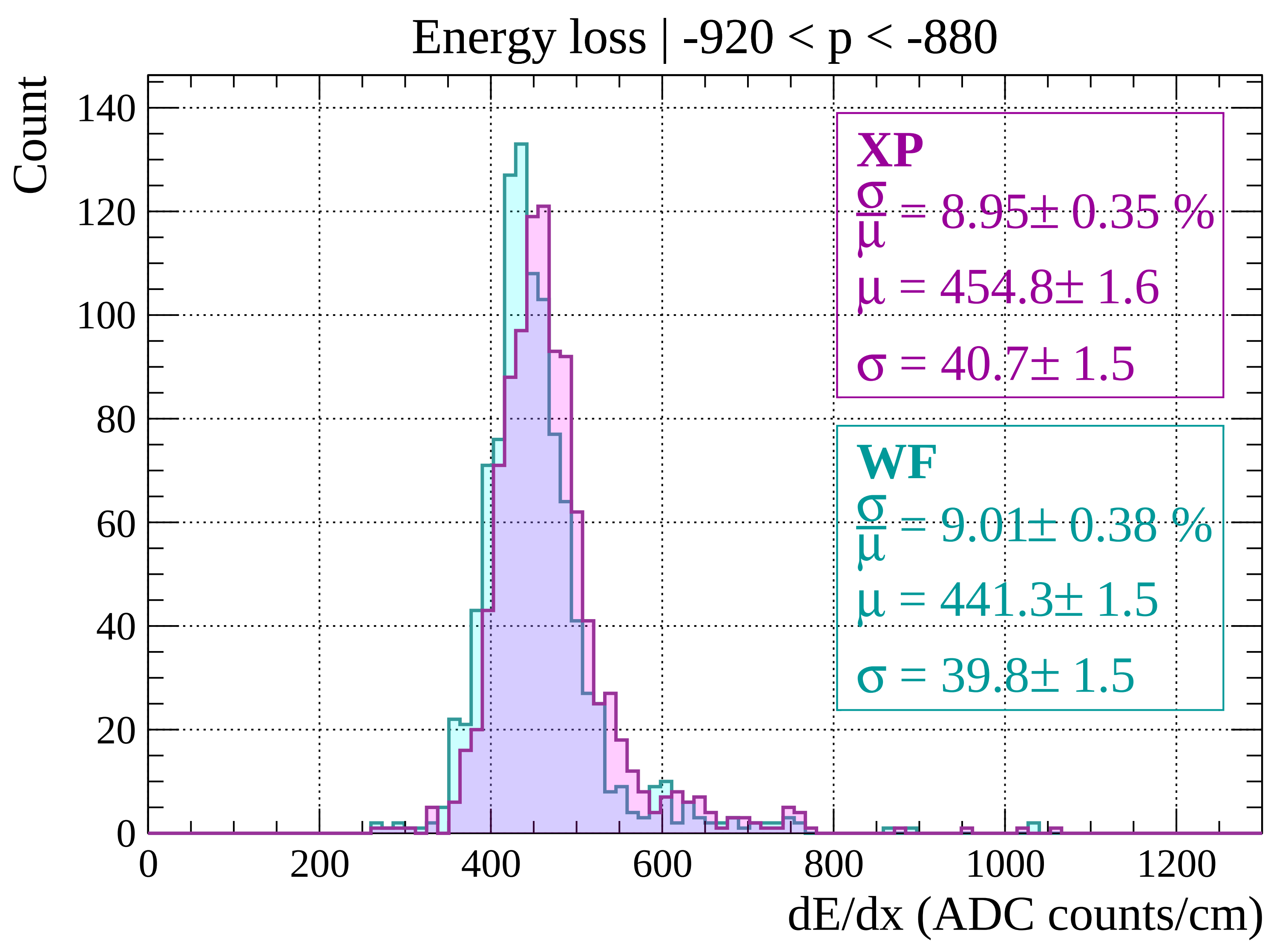

Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.28± 0.39 %
μ = 459.3± 1.5
σ = 38.0± 1.7
WF
σ/μ = 8.11± 0.30 %
μ = 444.2± 1.4
σ = 36.0± 1.2
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

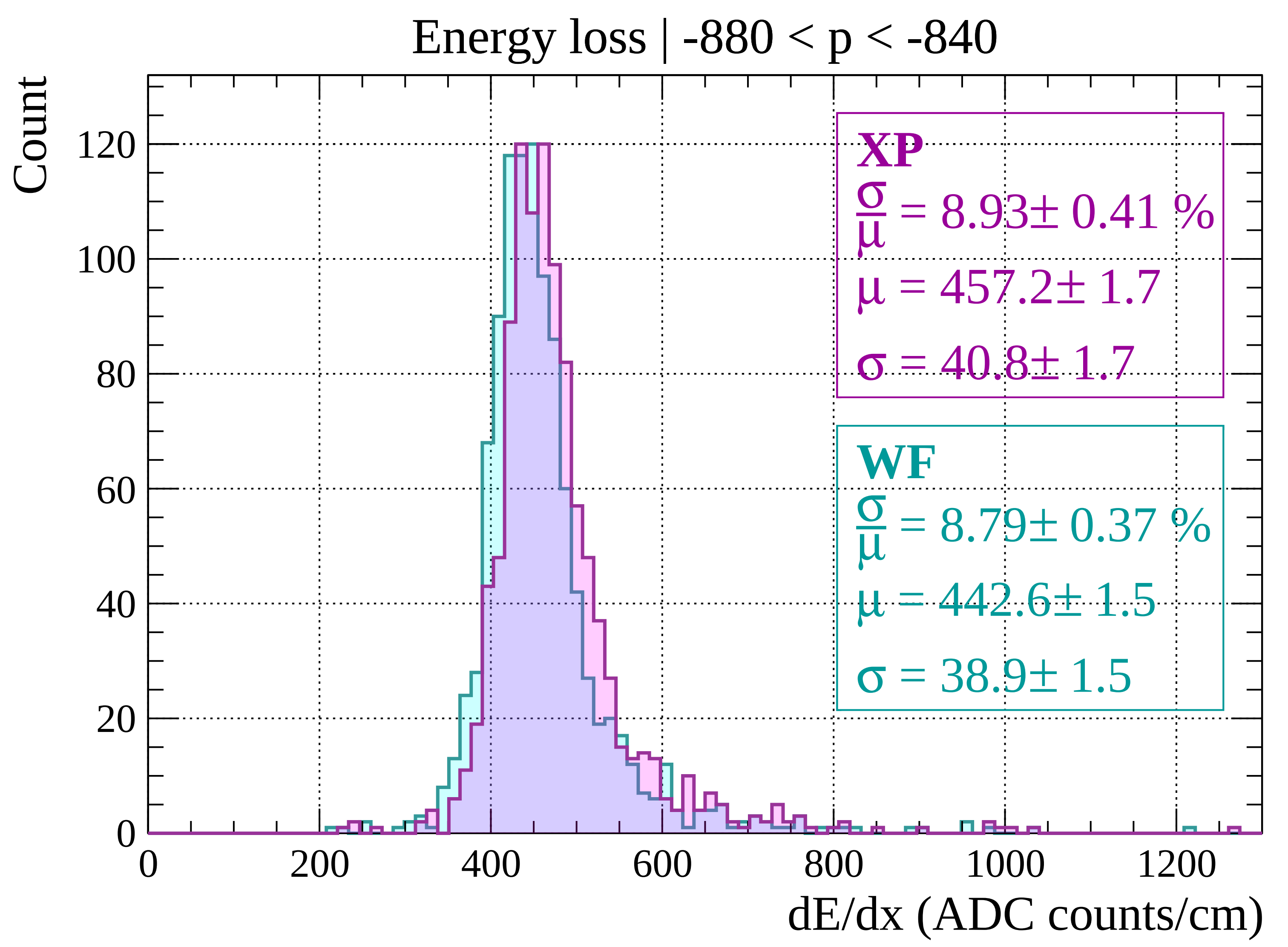


Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.29± 0.34 %
μ = 459.8± 1.5
σ = 38.1± 1.4
WF
σ/μ = 8.50± 0.41 %
μ = 443.5± 1.5
σ = 37.7± 1.7

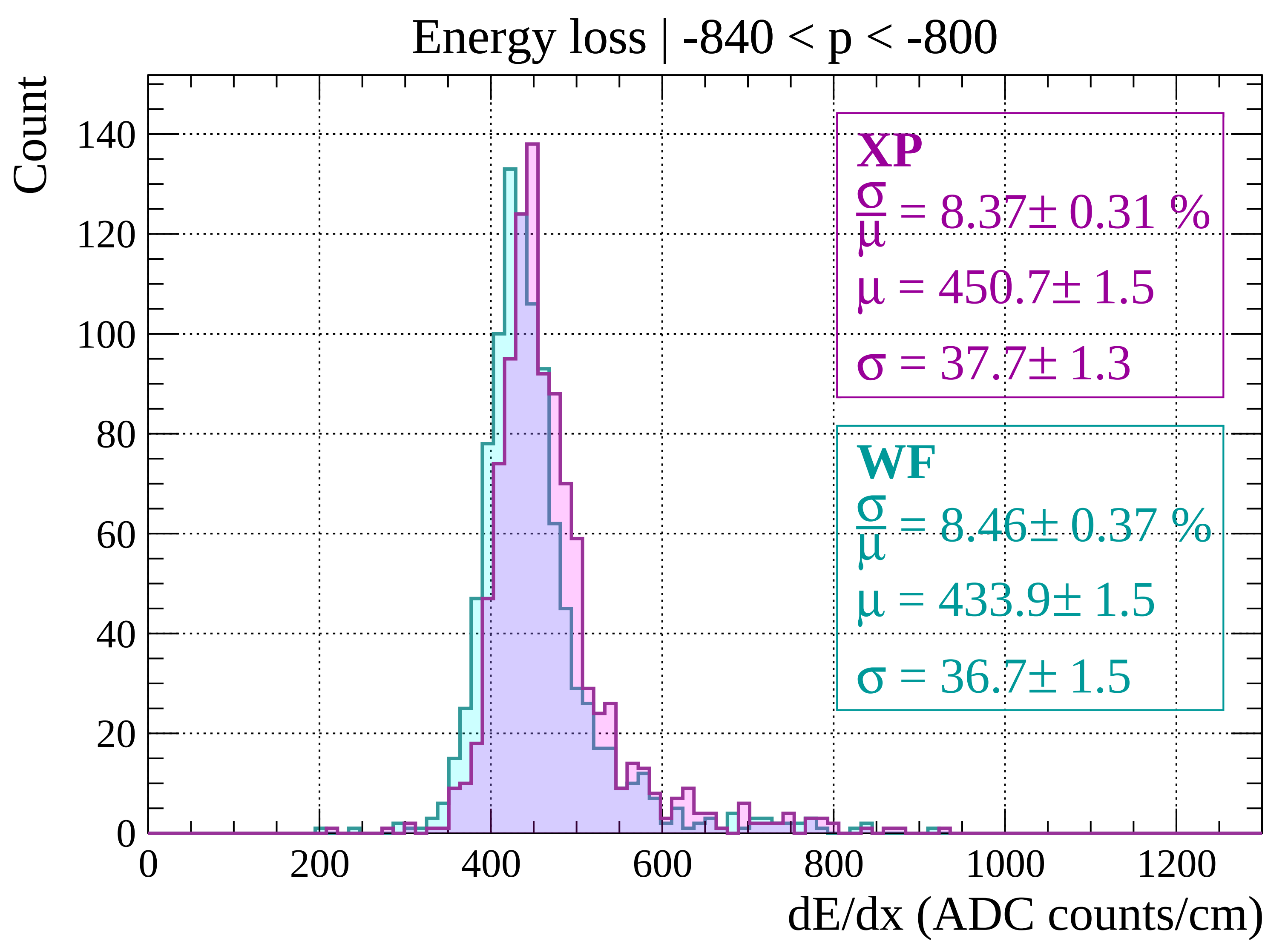


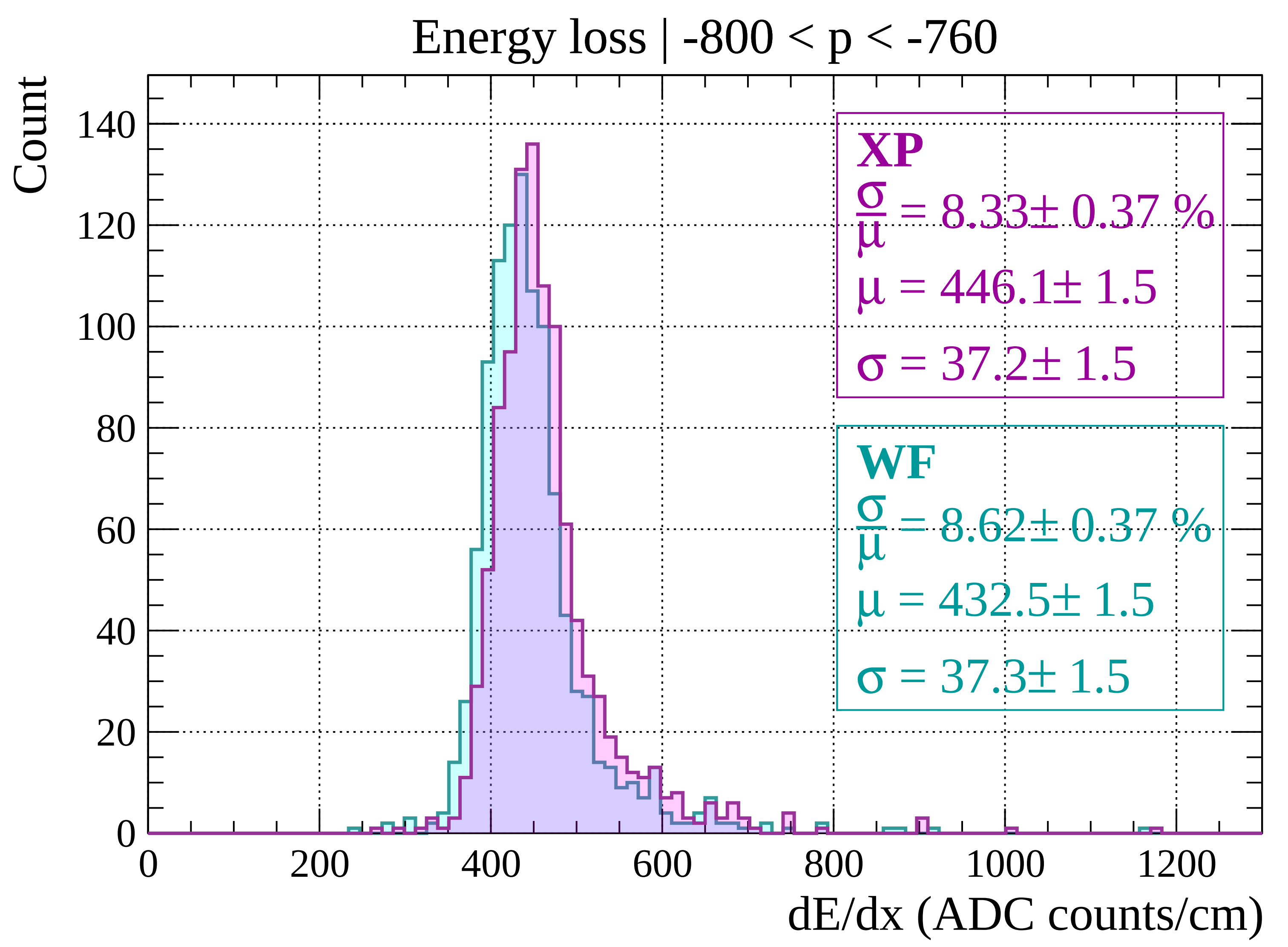
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.81± 0.38 %
μ = 454.7± 1.5
σ = 35.5± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.94± 0.38 %
μ = 442.6± 1.6
σ = 39.6± 1.6


Energy loss | -920 < p < -880
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.95± 0.35 %
μ = 454.8± 1.6
σ = 40.7± 1.5
WF
$\frac{\sigma}{\mu}$ = 9.01± 0.38 %
μ = 441.3± 1.5
σ = 39.8± 1.5

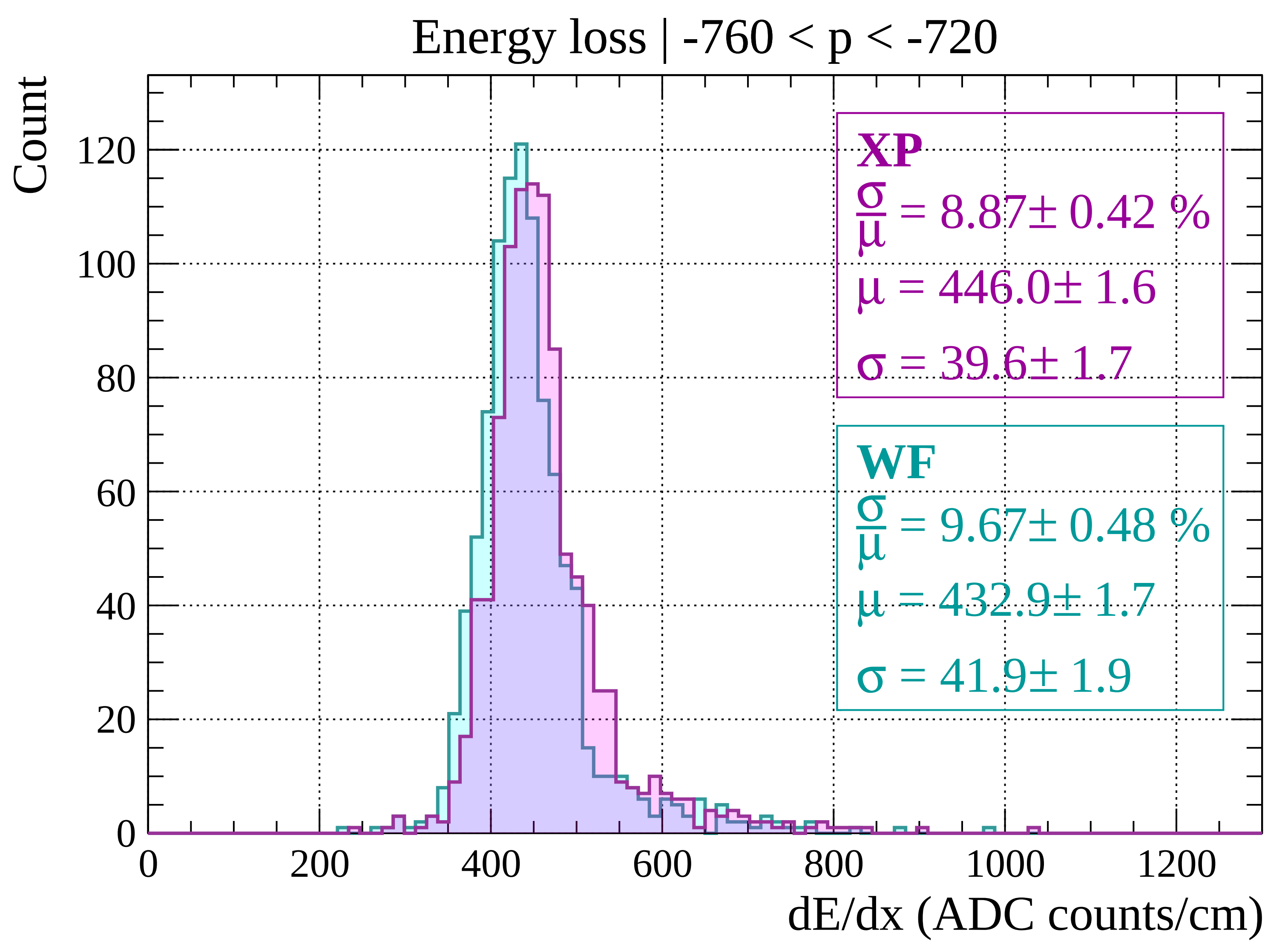


Energy loss | -880 < p < -840
Count
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.93± 0.41 %
μ = 457.2± 1.7
σ = 40.8± 1.7
WF
σ/μ = 8.79± 0.37 %
μ = 442.6± 1.5
σ = 38.9± 1.5

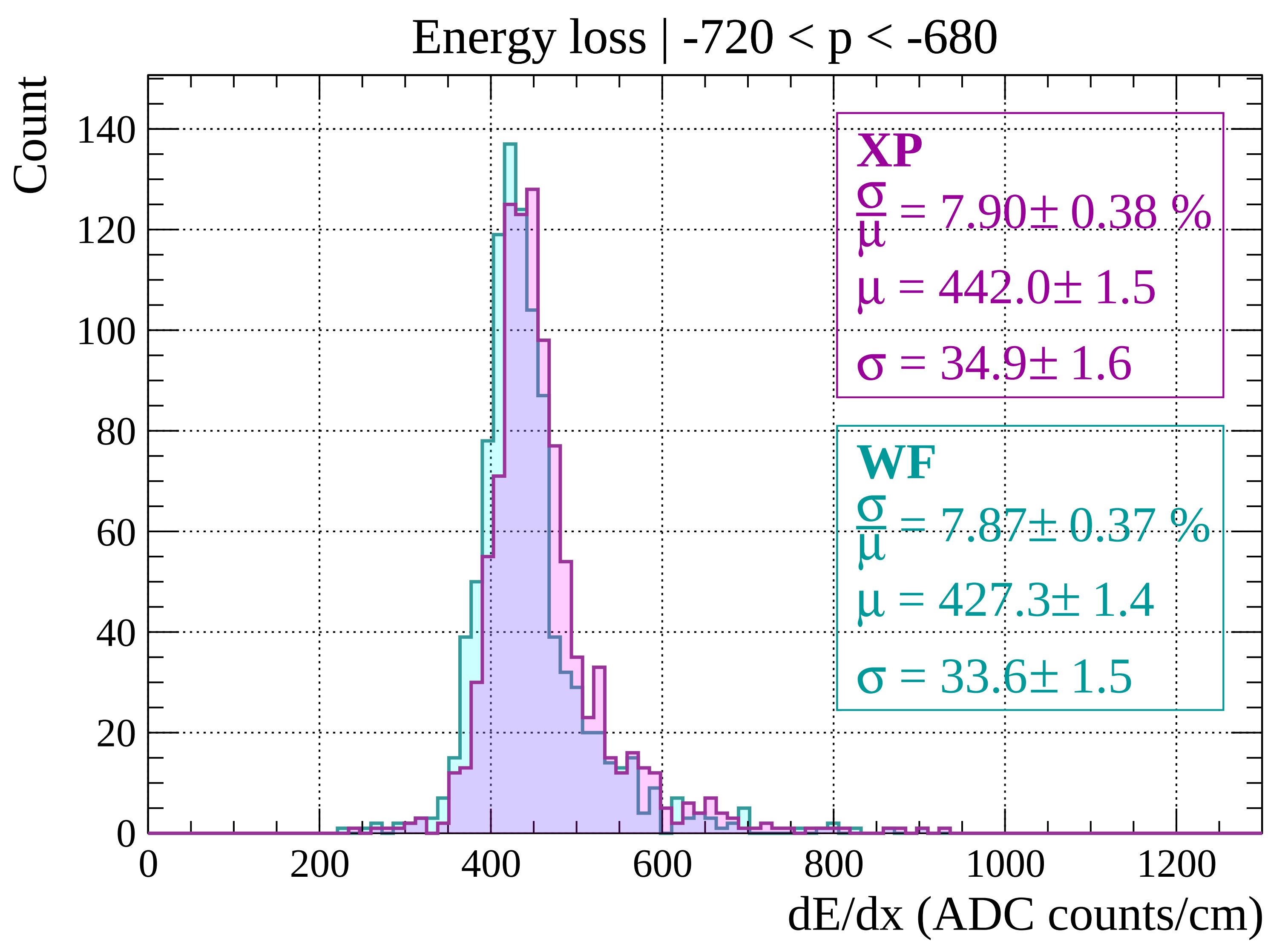

Energy loss | -840 < p < -800
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.37± 0.31 %
μ = 450.7± 1.5
σ = 37.7± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.37 %
μ = 433.9± 1.5
σ = 36.7± 1.5



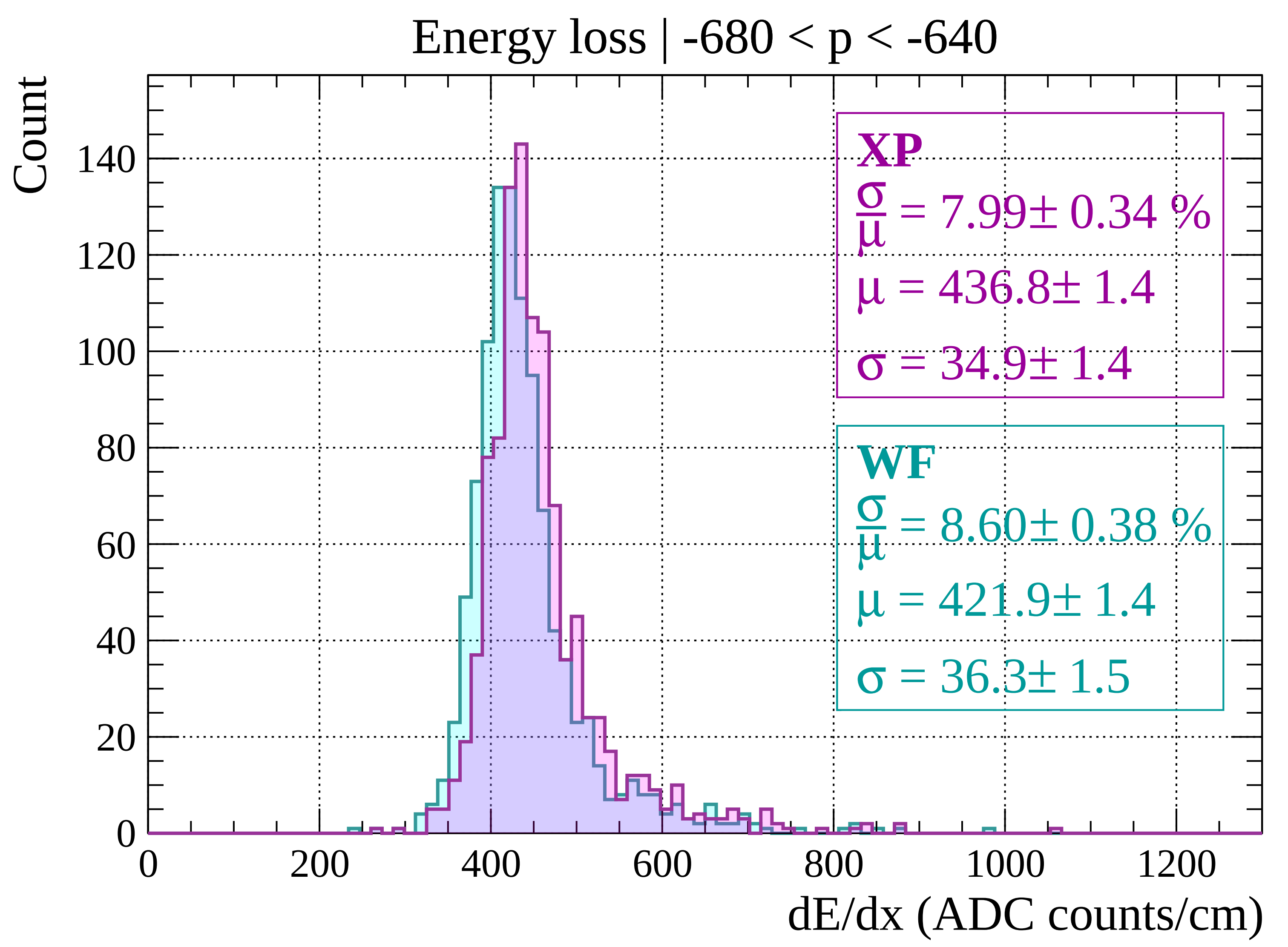
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.33± 0.37 %
μ = 446.1± 1.5
σ = 37.2± 1.5
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.37 %
μ = 432.5± 1.5
σ = 37.3± 1.5



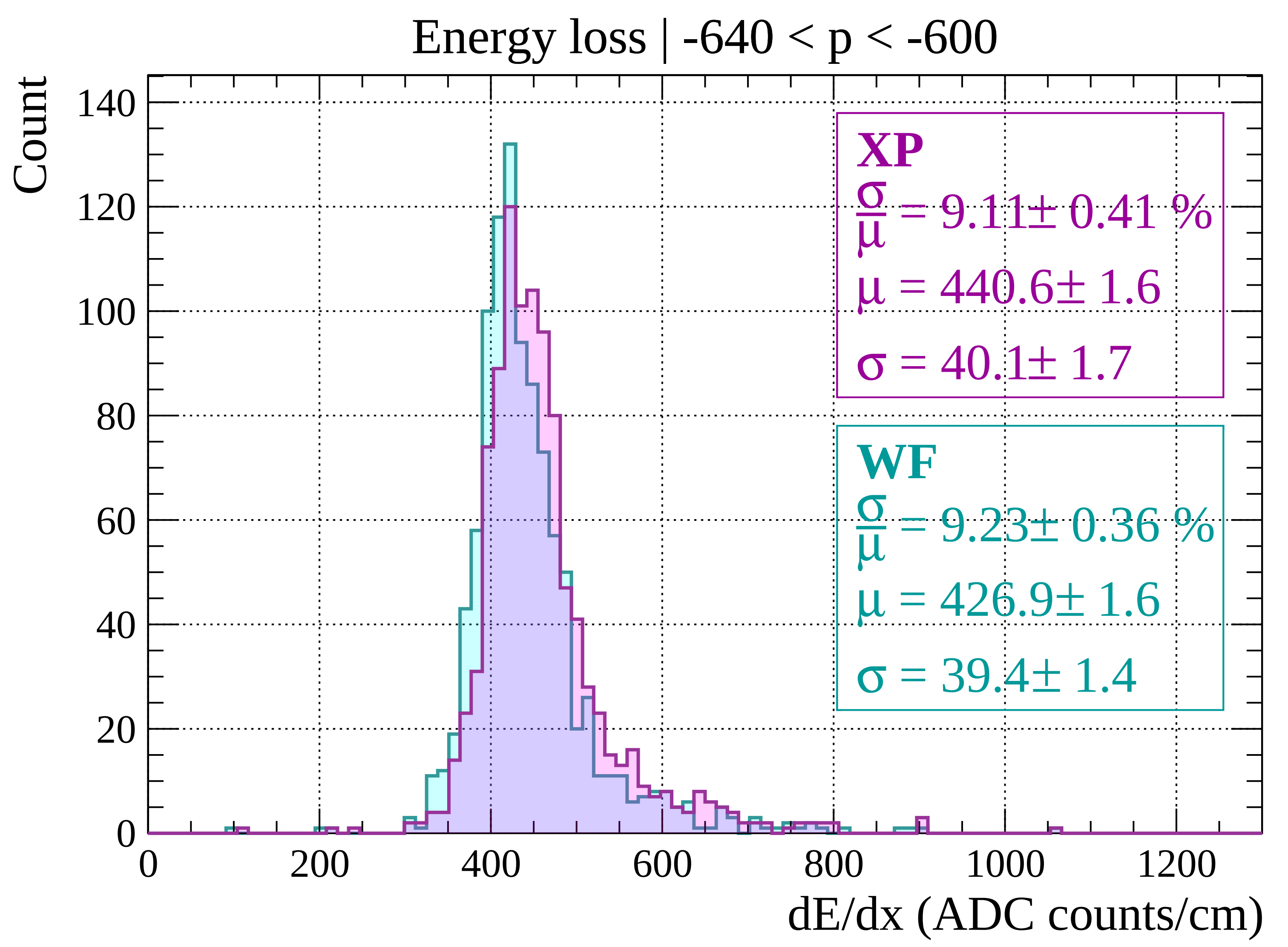
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 8.87± 0.42 %
μ = 446.0± 1.6
σ = 39.6± 1.7
WF
σ/μ = 9.67± 0.48 %
μ = 432.9± 1.7
σ = 41.9± 1.9



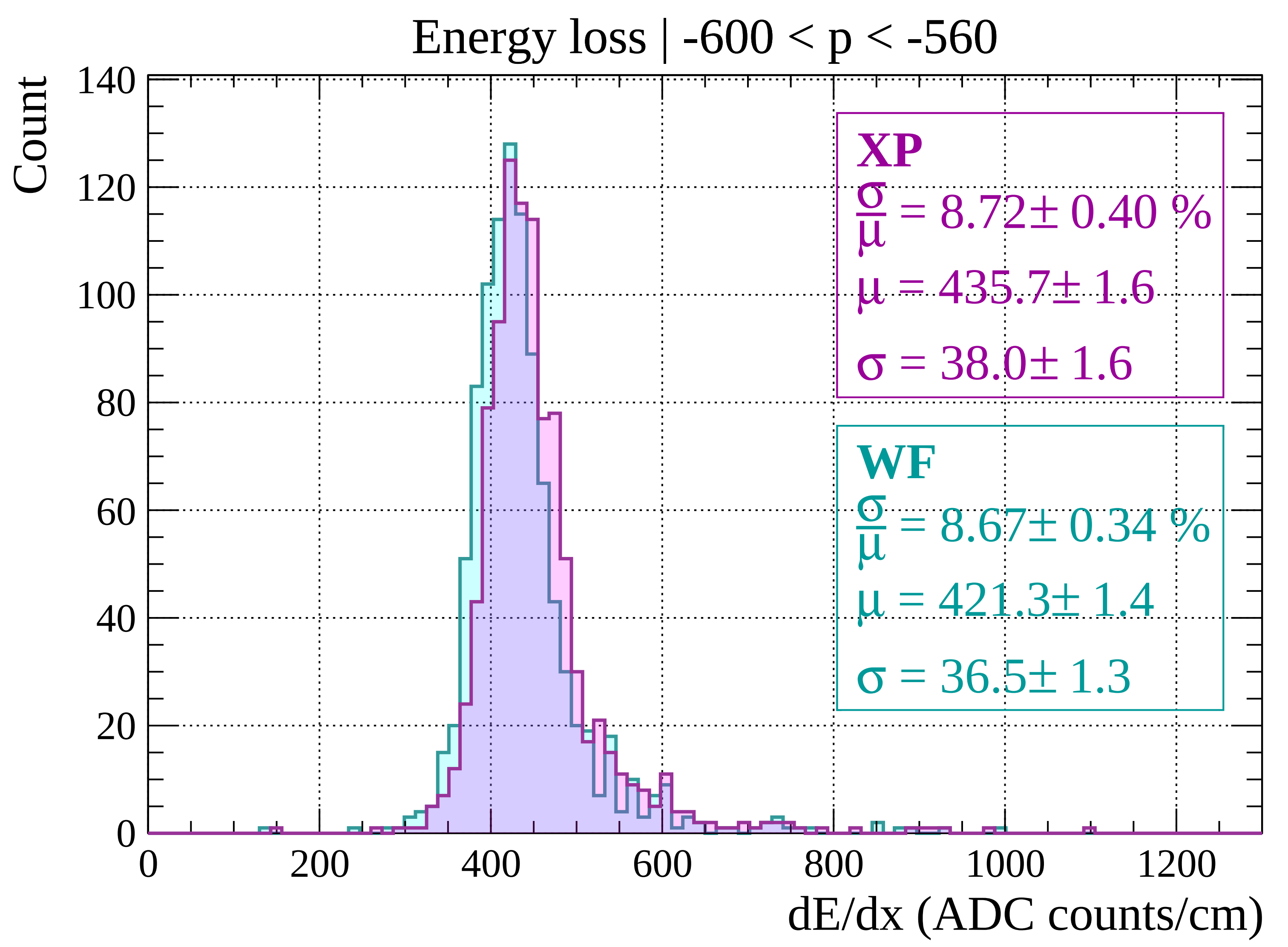
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 7.90± 0.38 %
μ = 442.0± 1.5
σ = 34.9± 1.6
WF
σ/μ = 7.87± 0.37 %
μ = 427.3± 1.4
σ = 33.6± 1.5

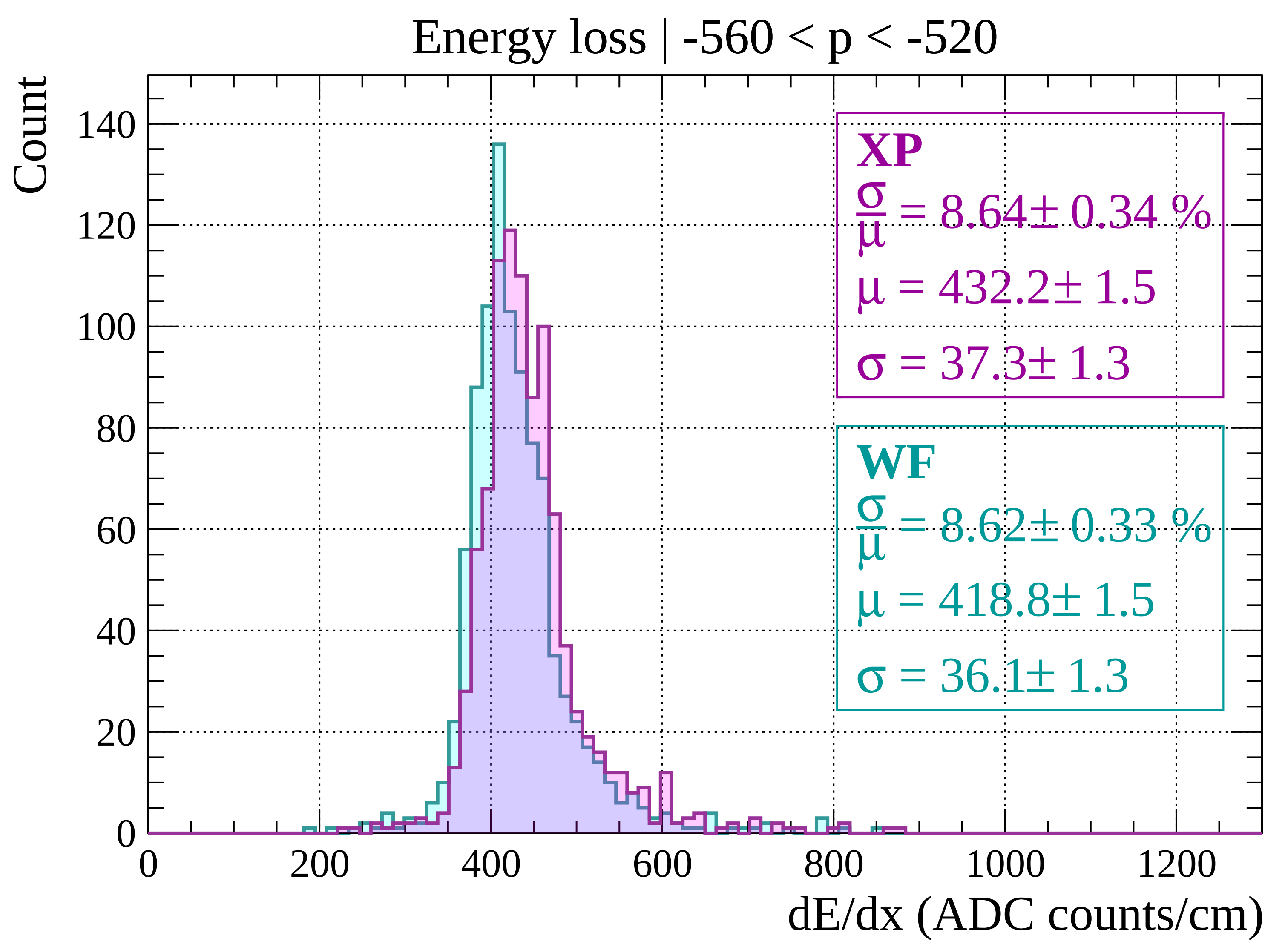


Energy loss | -680 < p < -640
Count
XP
σ/μ = 7.99± 0.34 %
μ = 436.8± 1.4
σ = 34.9± 1.4
WF
σ/μ = 8.60± 0.38 %
μ = 421.9± 1.4
σ = 36.3± 1.5
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

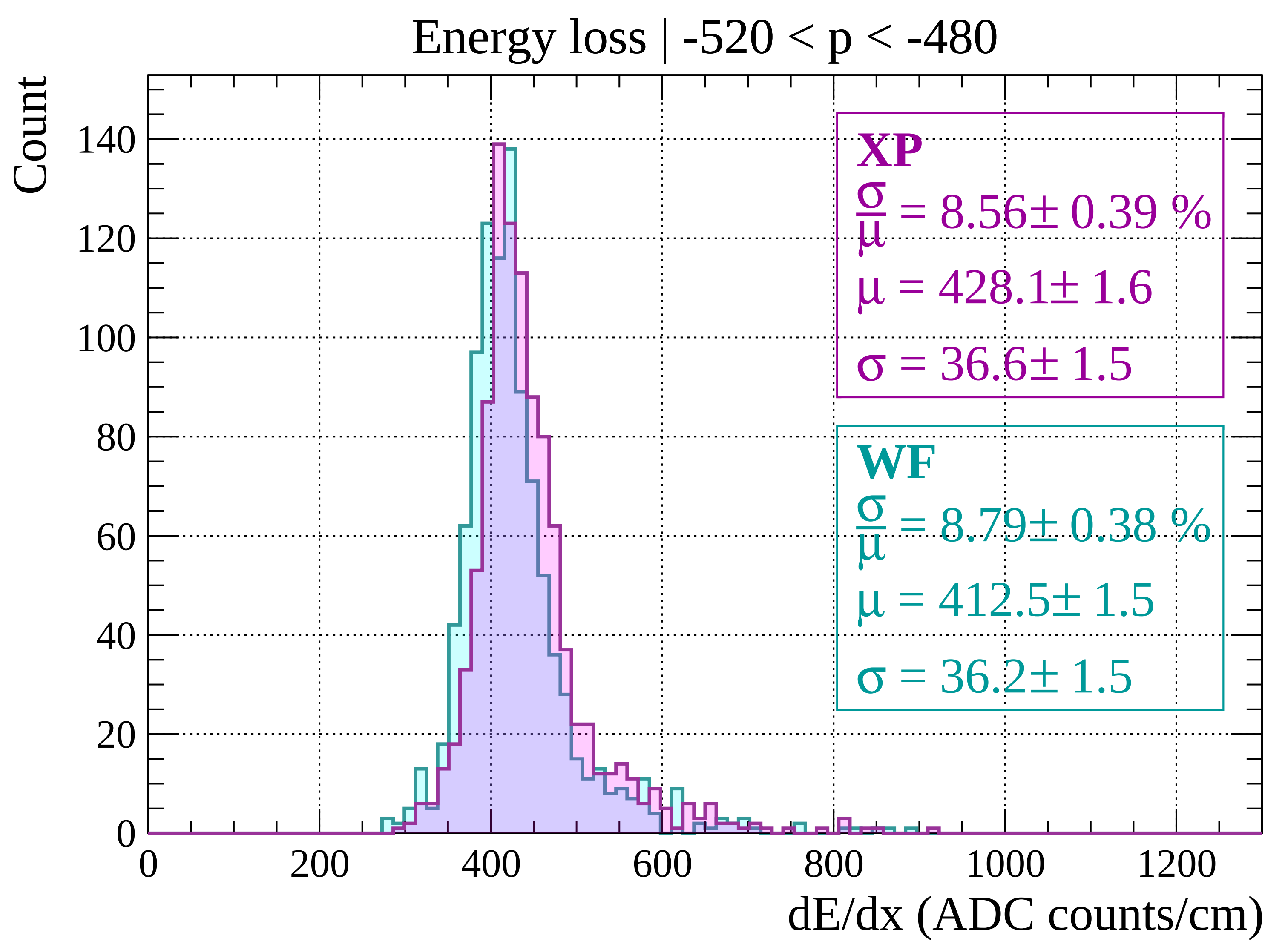


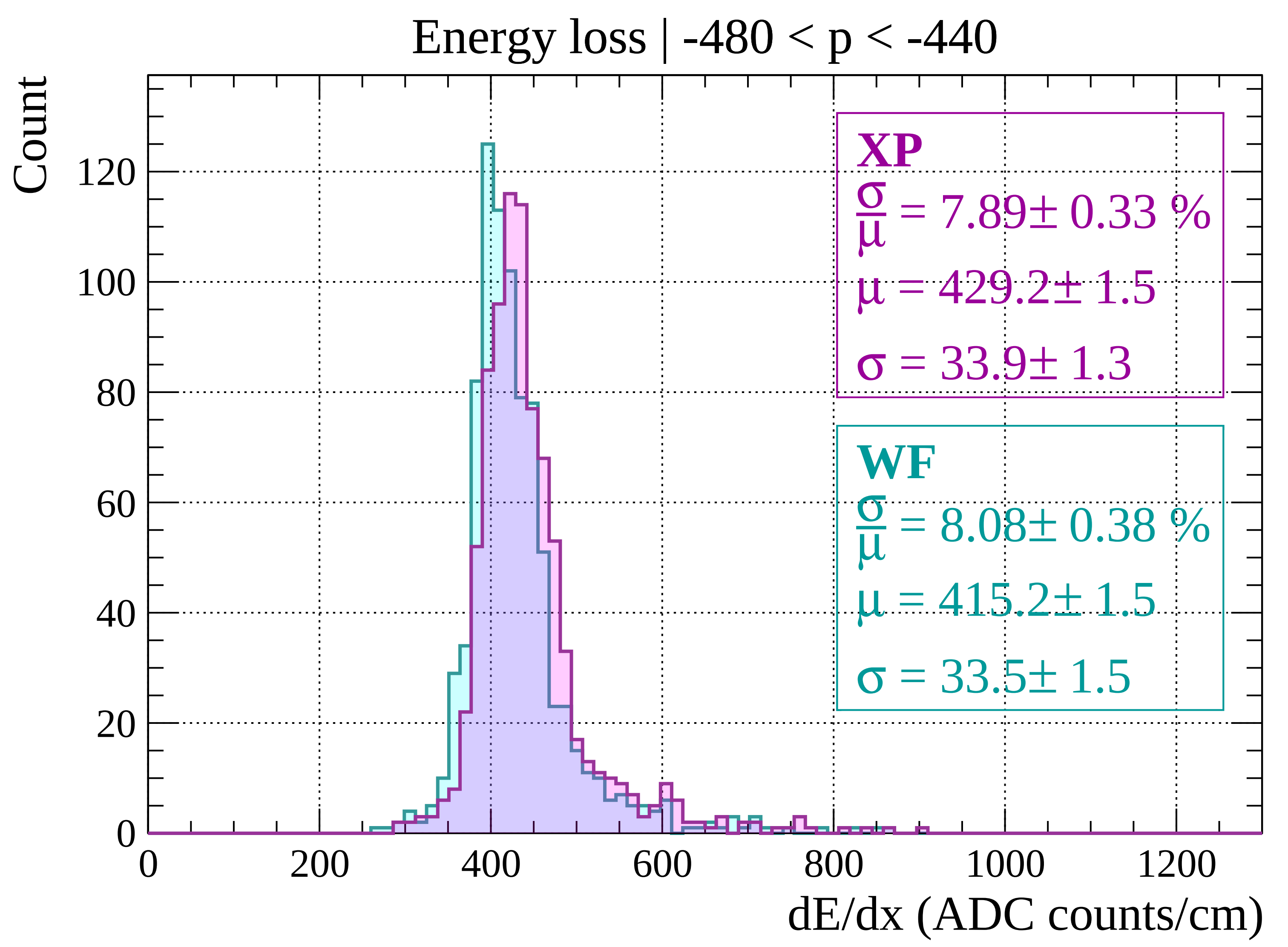
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 9.11± 0.41 %
μ = 440.6± 1.6
σ = 40.1± 1.7
WF
$\frac{\sigma}{\mu}$ = 9.23± 0.36 %
μ = 426.9± 1.6
σ = 39.4± 1.4

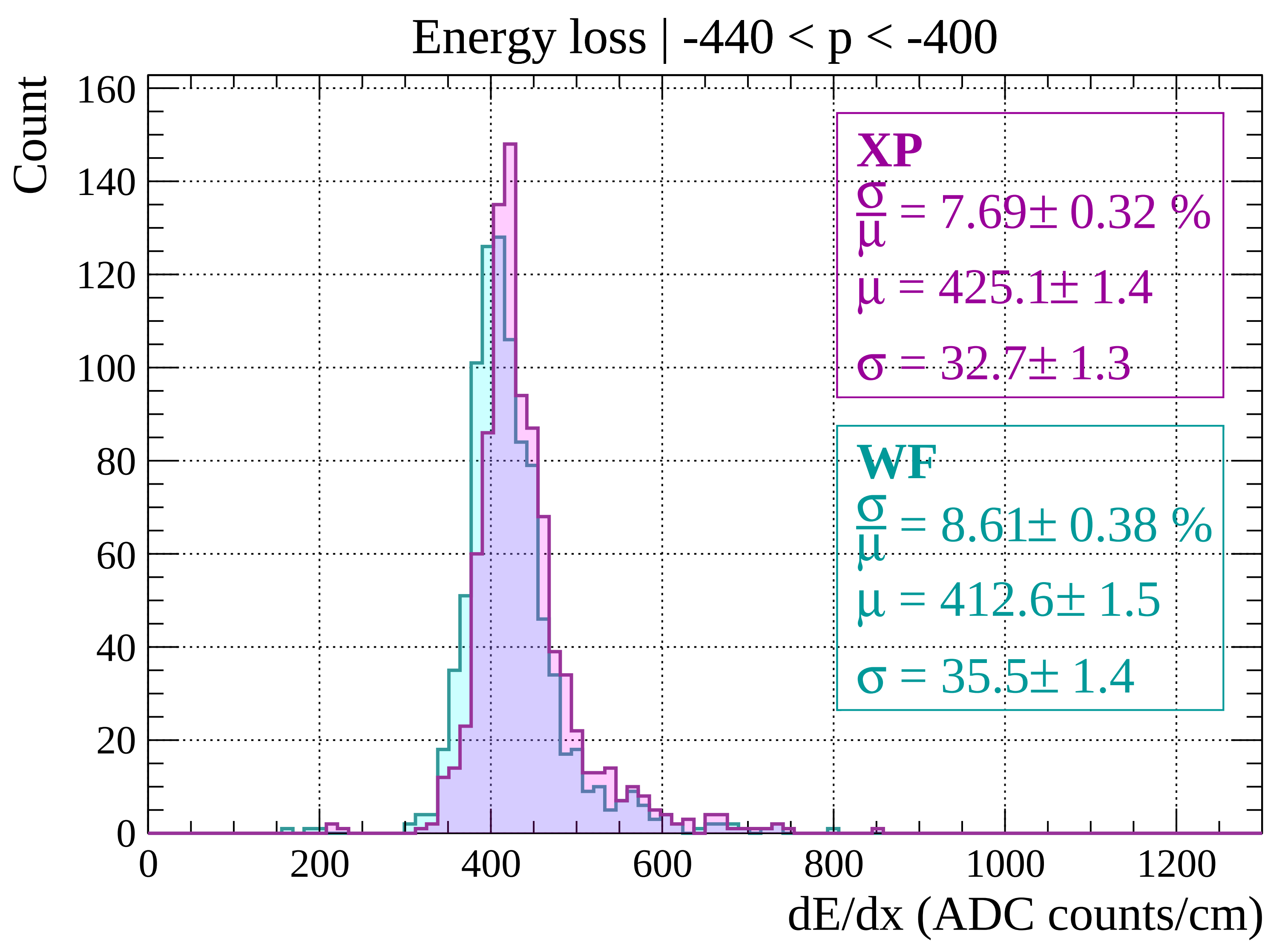

Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.72± 0.40 %
μ = 435.7± 1.6
σ = 38.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 8.67± 0.34 %
μ = 421.3± 1.4
σ = 36.5± 1.3

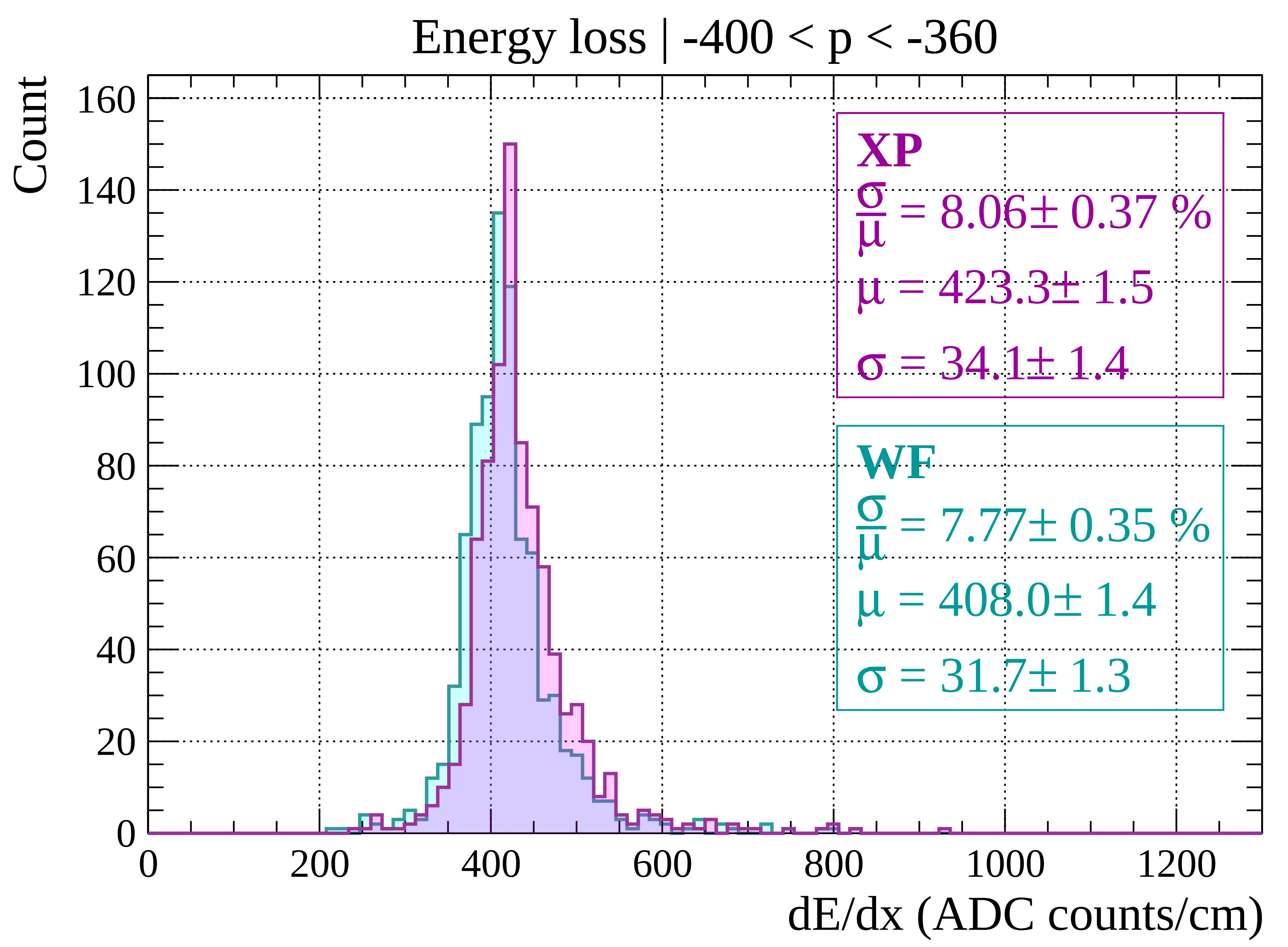


Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.64± 0.34 %
μ = 432.2± 1.5
σ = 37.3± 1.3
WF
σ/μ = 8.62± 0.33 %
μ = 418.8± 1.5
σ = 36.1± 1.3

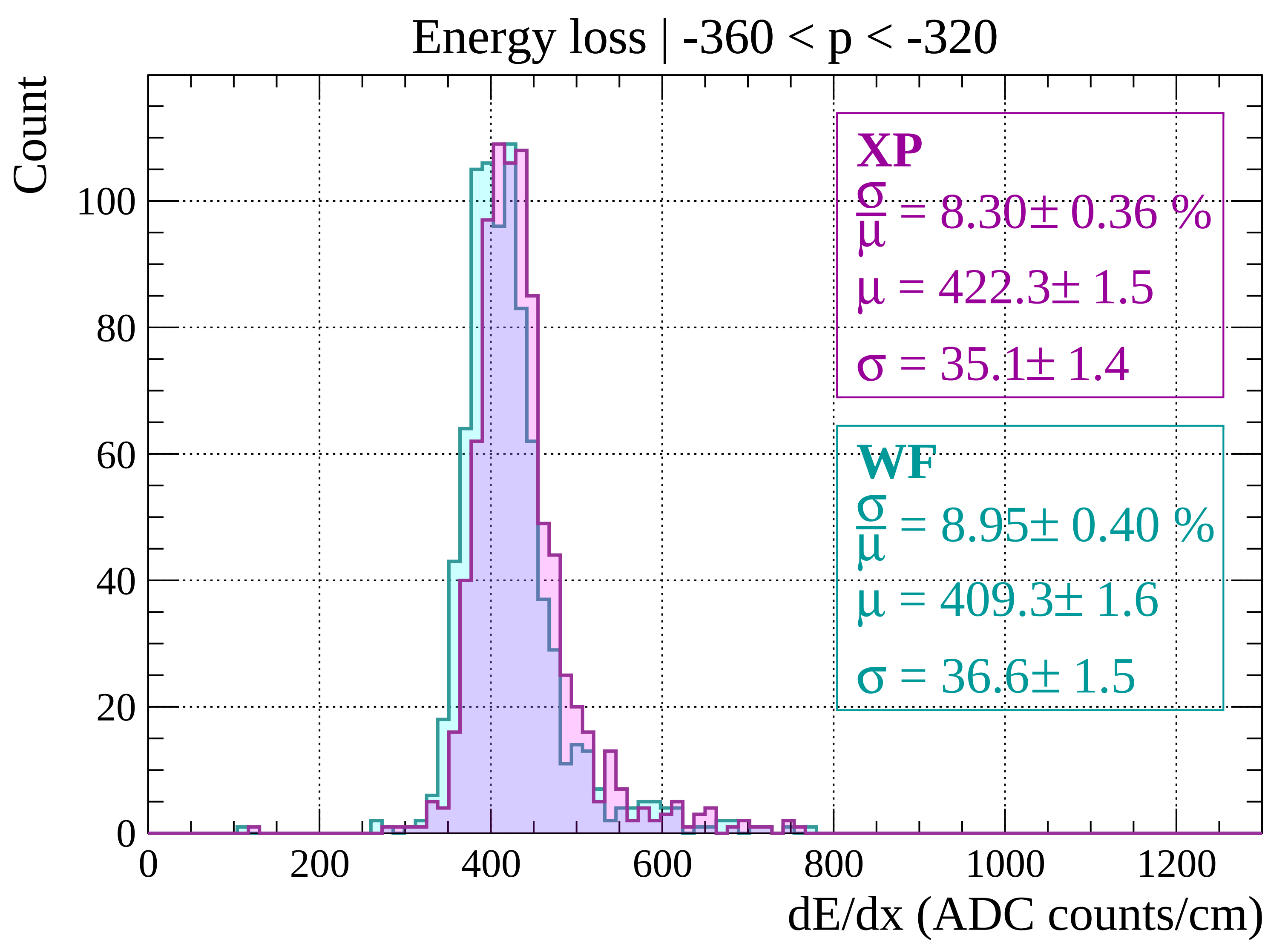

Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.56± 0.39 %
μ = 428.1± 1.6
σ = 36.6± 1.5
WF
σ/μ = 8.79± 0.38 %
μ = 412.5± 1.5
σ = 36.2± 1.5
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200


Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 7.89± 0.33 %
μ = 429.2± 1.5
σ = 33.9± 1.3
WF
σ/μ = 8.08± 0.38 %
μ = 415.2± 1.5
σ = 33.5± 1.5

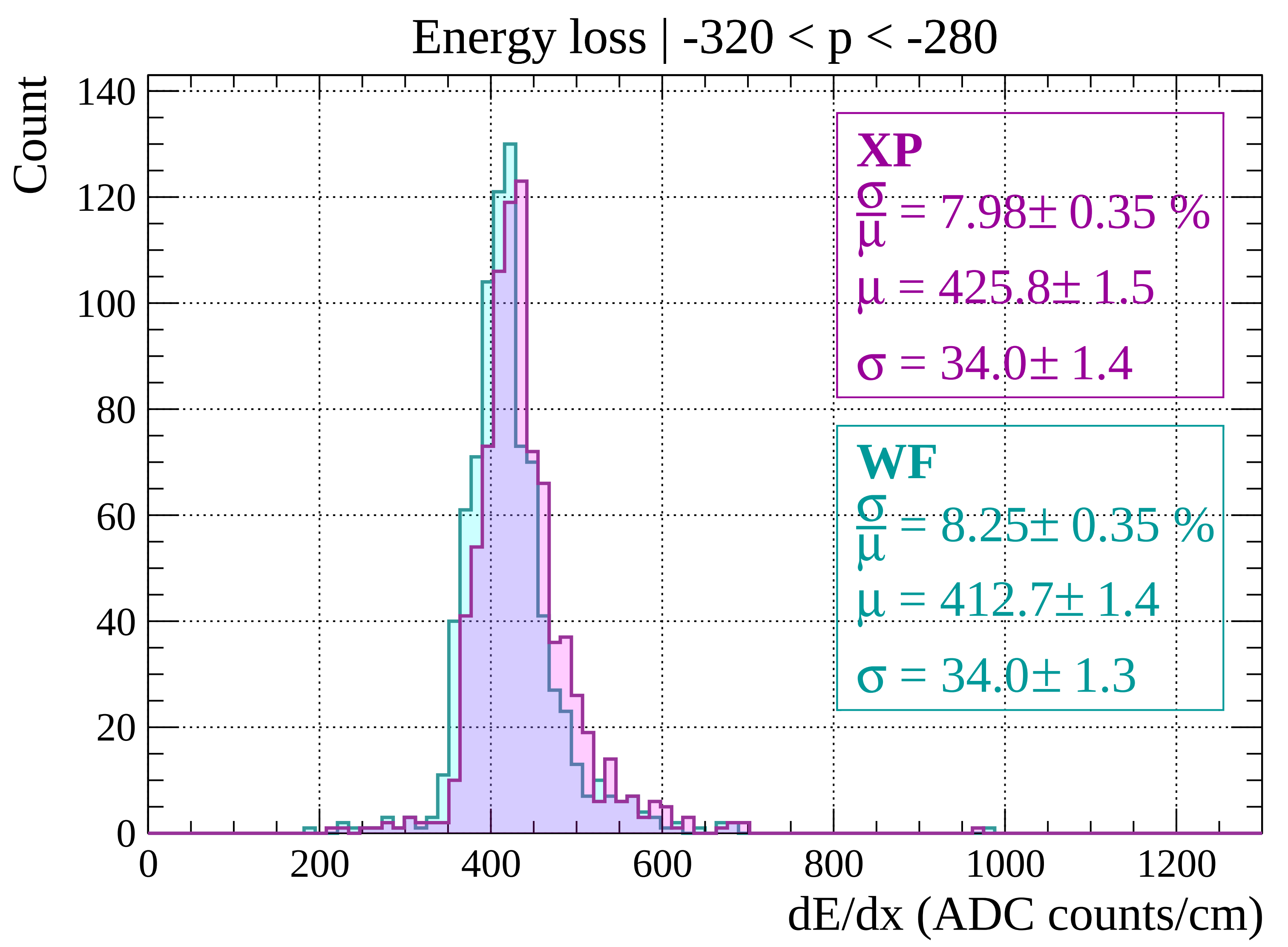

Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.69± 0.32 %
μ = 425.1± 1.4
σ = 32.7± 1.3
WF
σ/μ = 8.61± 0.38 %
μ = 412.6± 1.5
σ = 35.5± 1.4
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



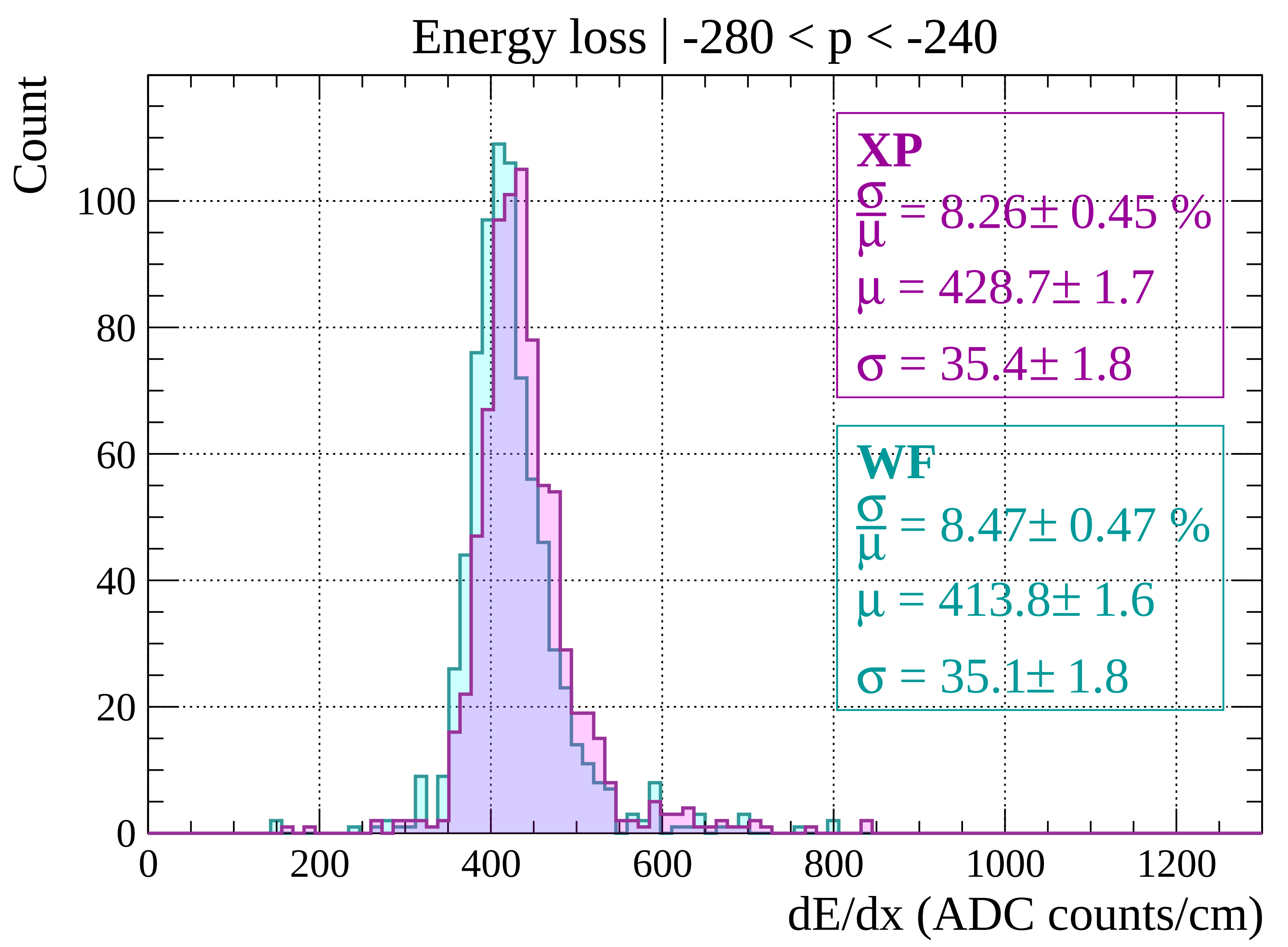
Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 8.06± 0.37 %
μ = 423.3± 1.5
σ = 34.1± 1.4
WF
$\frac{\sigma}{\mu}$ = 7.77± 0.35 %
μ = 408.0± 1.4
σ = 31.7± 1.3



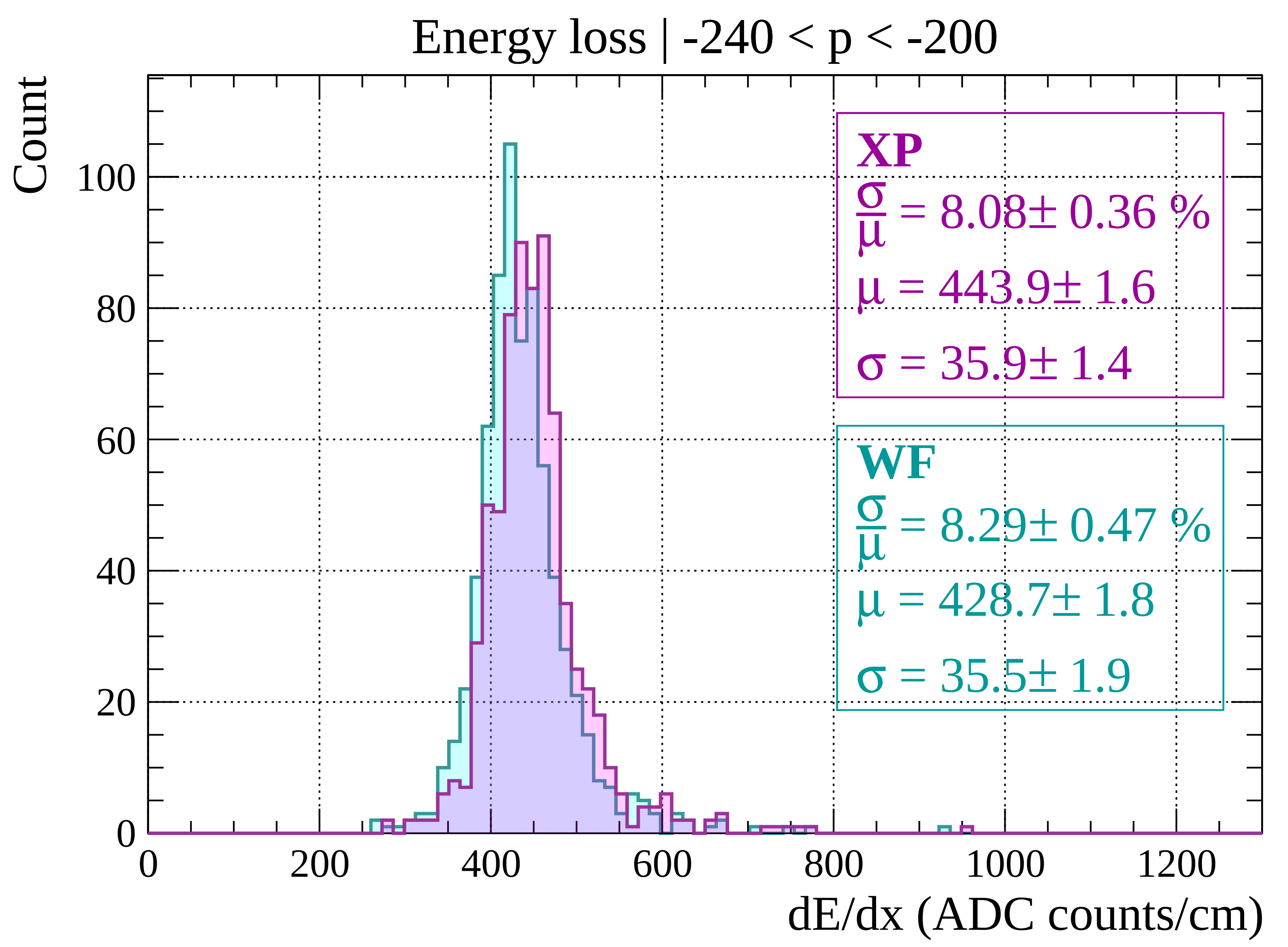
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.30± 0.36 %
μ = 422.3± 1.5
σ = 35.1± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.95± 0.40 %
μ = 409.3± 1.6
σ = 36.6± 1.5



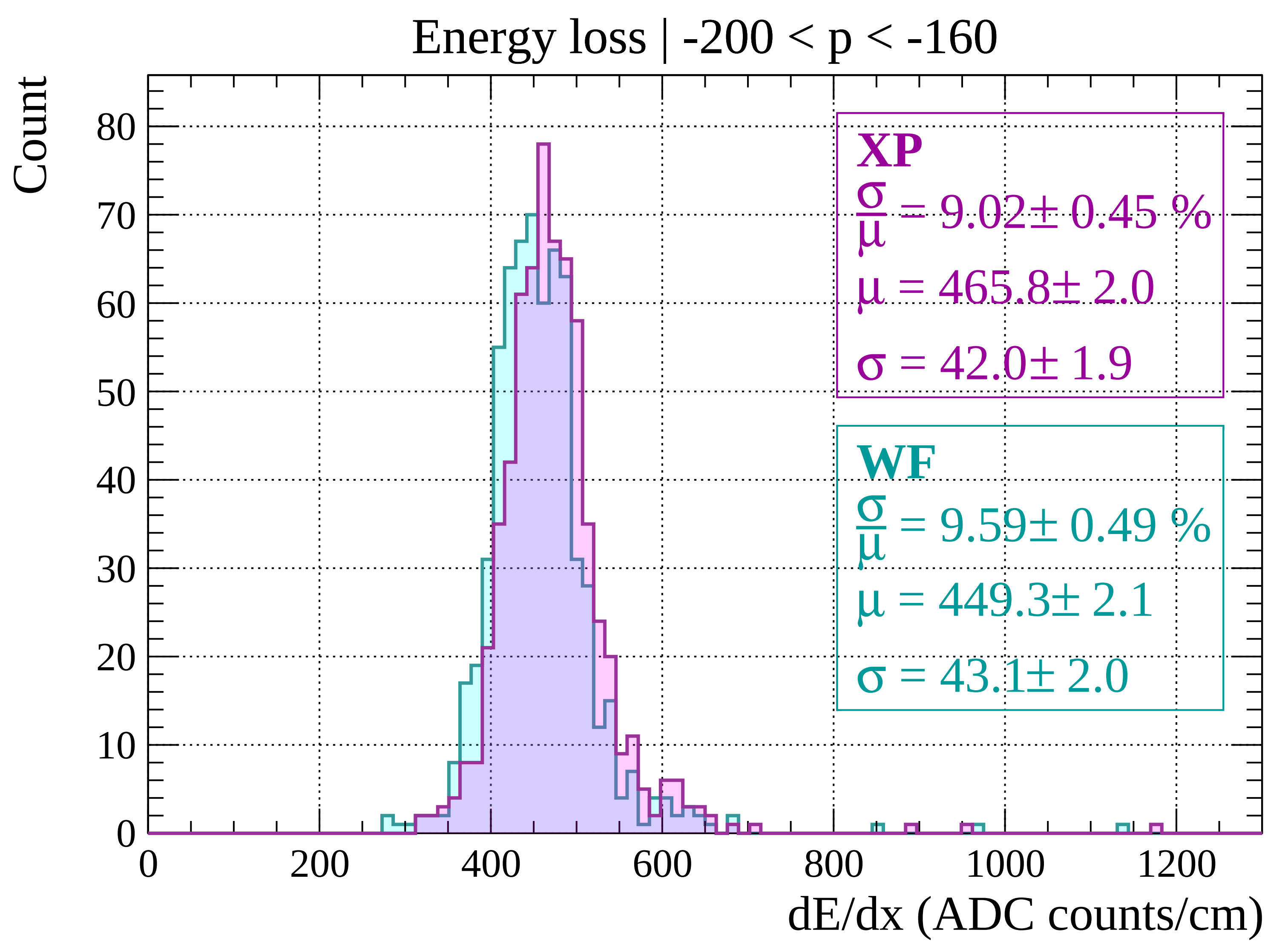
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.98± 0.35 %
μ = 425.8± 1.5
σ = 34.0± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.25± 0.35 %
μ = 412.7± 1.4
σ = 34.0± 1.3
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



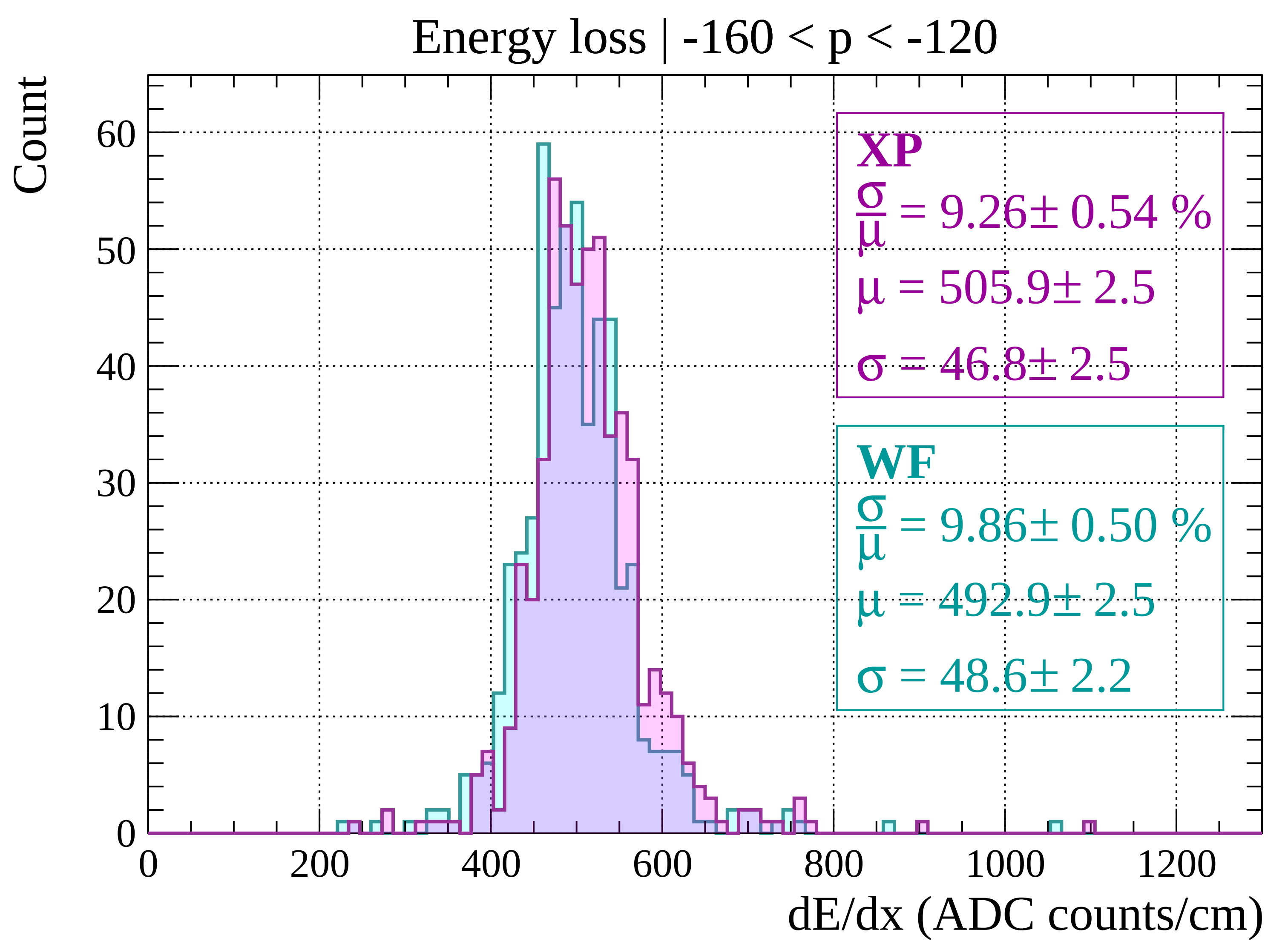
Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 8.26± 0.45 %
μ = 428.7± 1.7
σ = 35.4± 1.8
WF
σ/μ = 8.47± 0.47 %
μ = 413.8± 1.6
σ = 35.1± 1.8

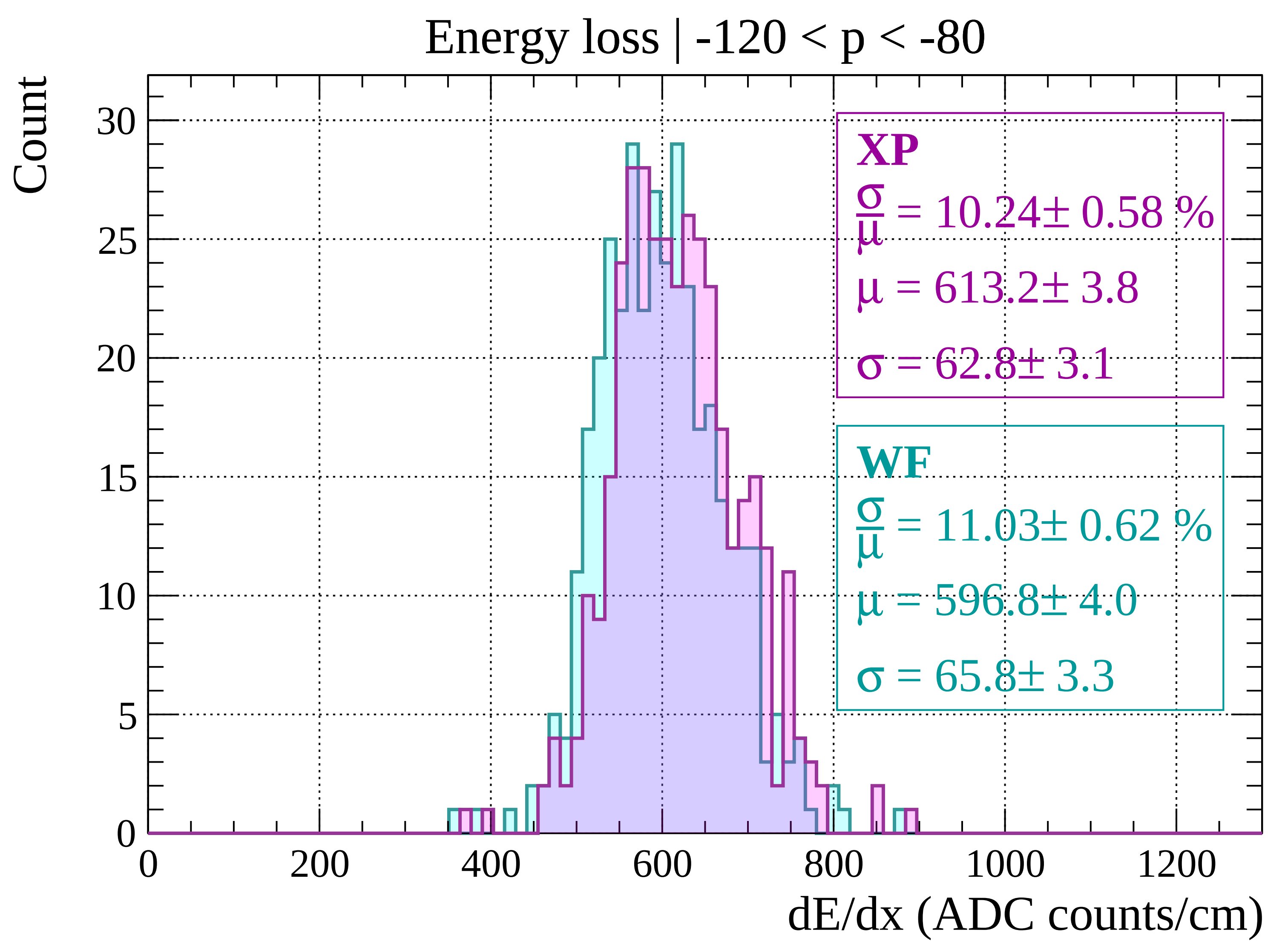


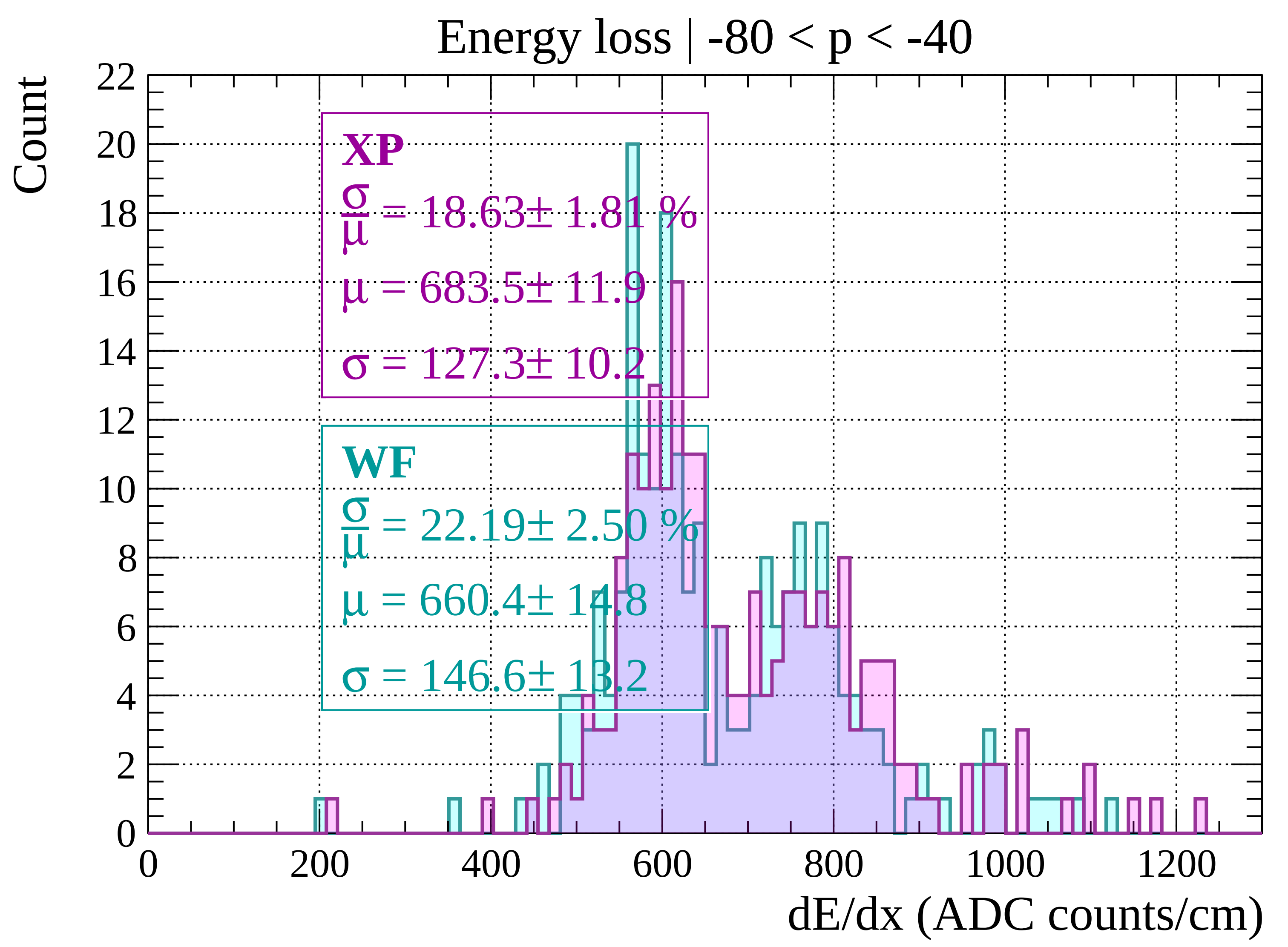
Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.08± 0.36 %
μ = 443.9± 1.6
σ = 35.9± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.29± 0.47 %
μ = 428.7± 1.8
σ = 35.5± 1.9



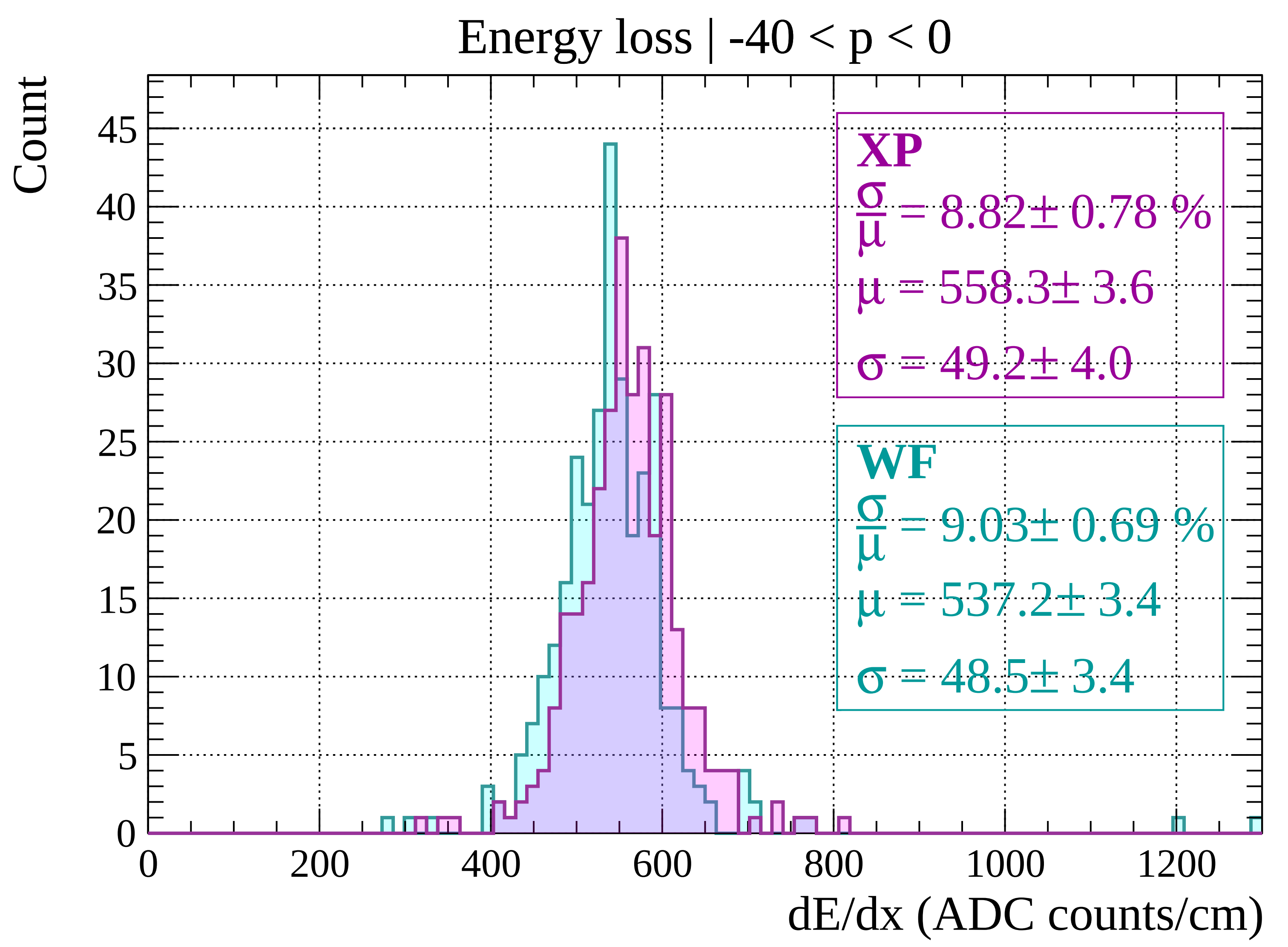
Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.02± 0.45 %
μ = 465.8± 2.0
σ = 42.0± 1.9
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.49 %
μ = 449.3± 2.1
σ = 43.1± 2.0


Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
$\frac{\sigma}{\mu}$ = 9.26± 0.54 %
μ = 505.9± 2.5
σ = 46.8± 2.5
WF
$\frac{\sigma}{\mu}$ = 9.86± 0.50 %
μ = 492.9± 2.5
σ = 48.6± 2.2

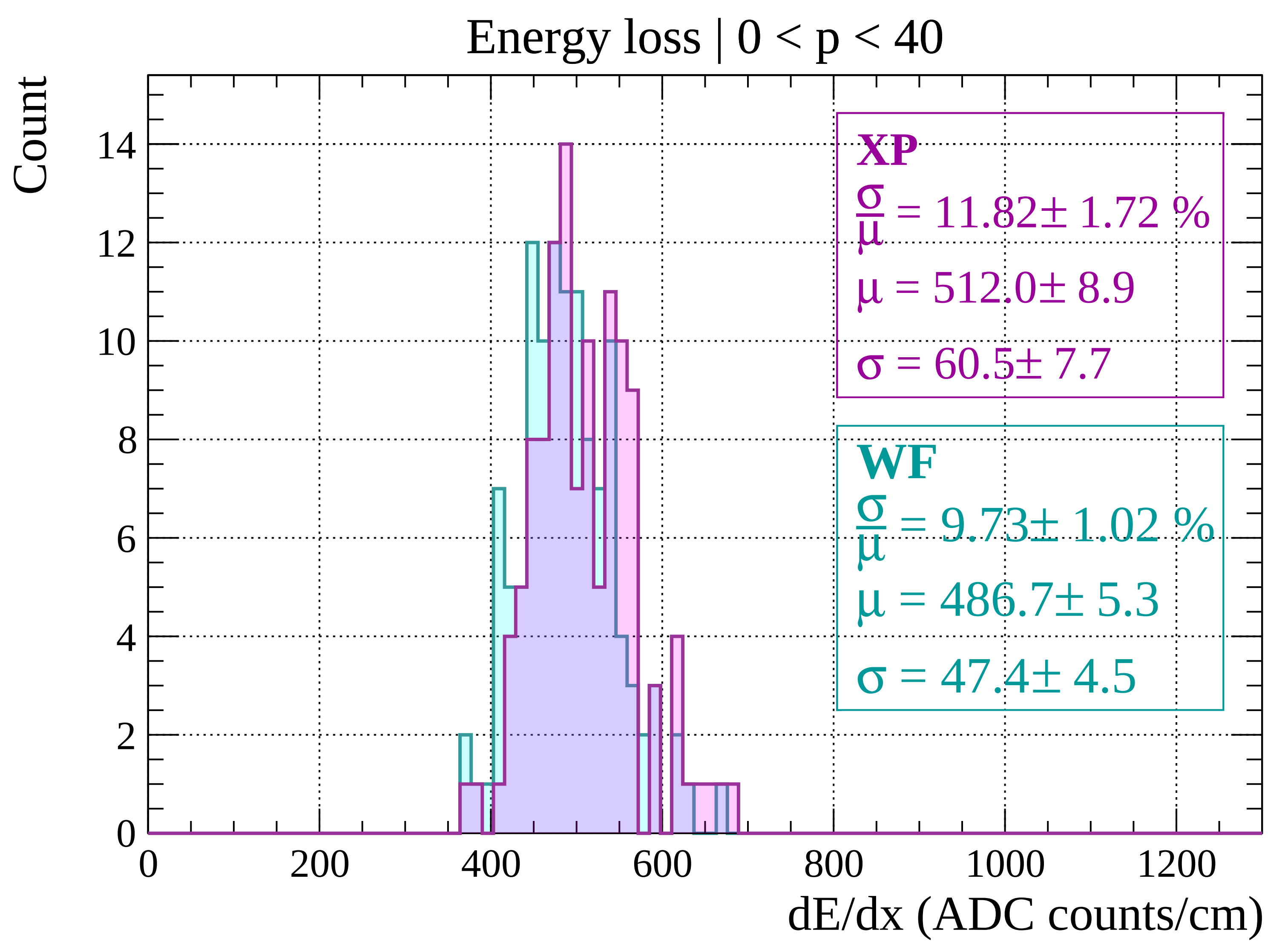

Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 10.24± 0.58 %
μ = 613.2± 3.8
σ = 62.8± 3.1
WF
σ/μ = 11.03± 0.62 %
μ = 596.8± 4.0
σ = 65.8± 3.3

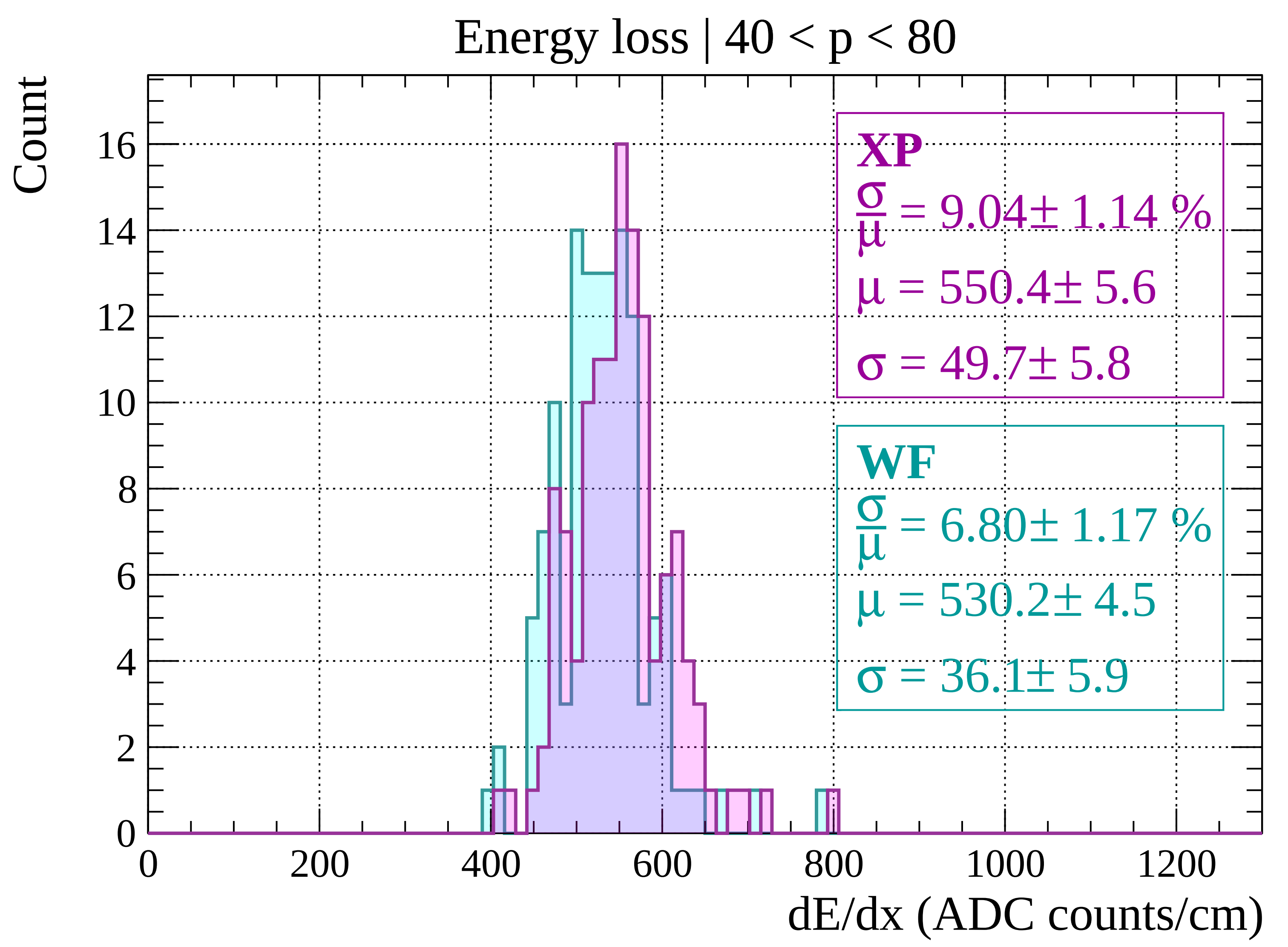


Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
XP
$\frac{\sigma}{\mu}$ = 18.63± 1.81 %
μ = 683.5± 11.9
σ = 127.3± 10.2
WF
$\frac{\sigma}{\mu}$ = 22.19± 2.50 %
μ = 660.4± 14.8
σ = 146.6± 13.2

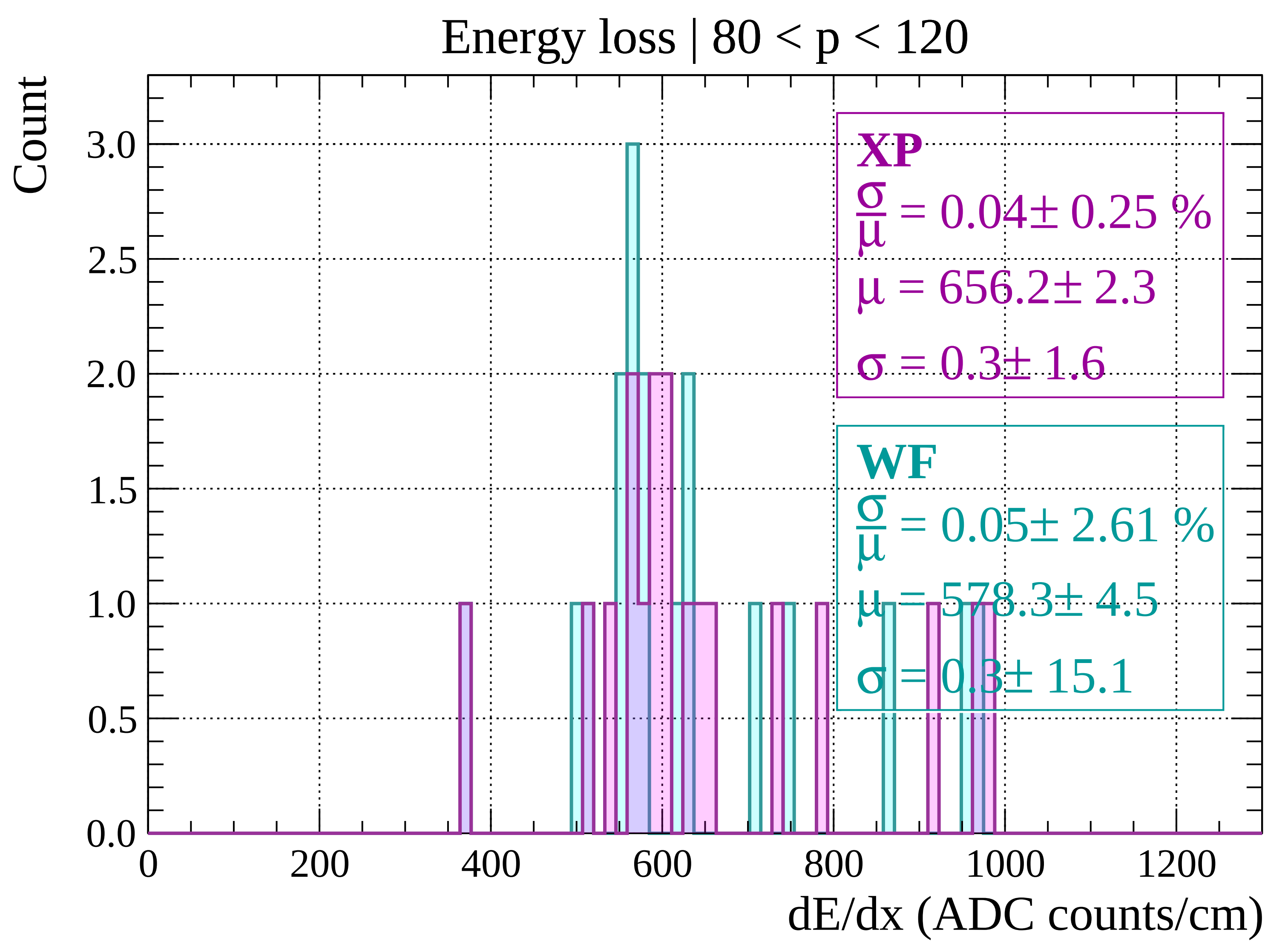


Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.82± 0.78 %
μ = 558.3± 3.6
σ = 49.2± 4.0
WF
$\frac{\sigma}{\mu}$ = 9.03± 0.69 %
μ = 537.2± 3.4
σ = 48.5± 3.4
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200

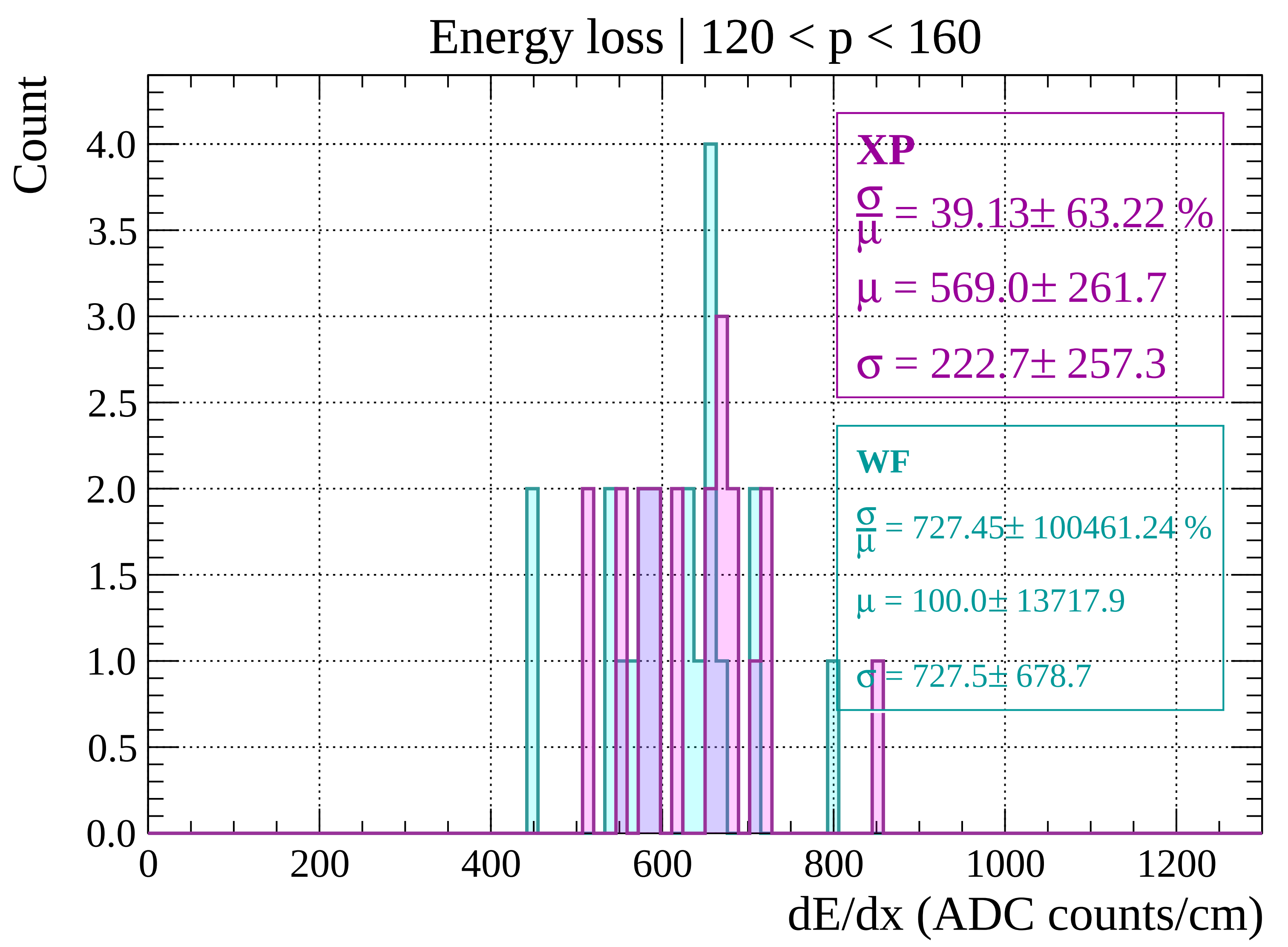

Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.82± 1.72 %
μ = 512.0± 8.9
σ = 60.5± 7.7
WF
$\frac{\sigma}{\mu}$ = 9.73± 1.02 %
μ = 486.7± 5.3
σ = 47.4± 4.5



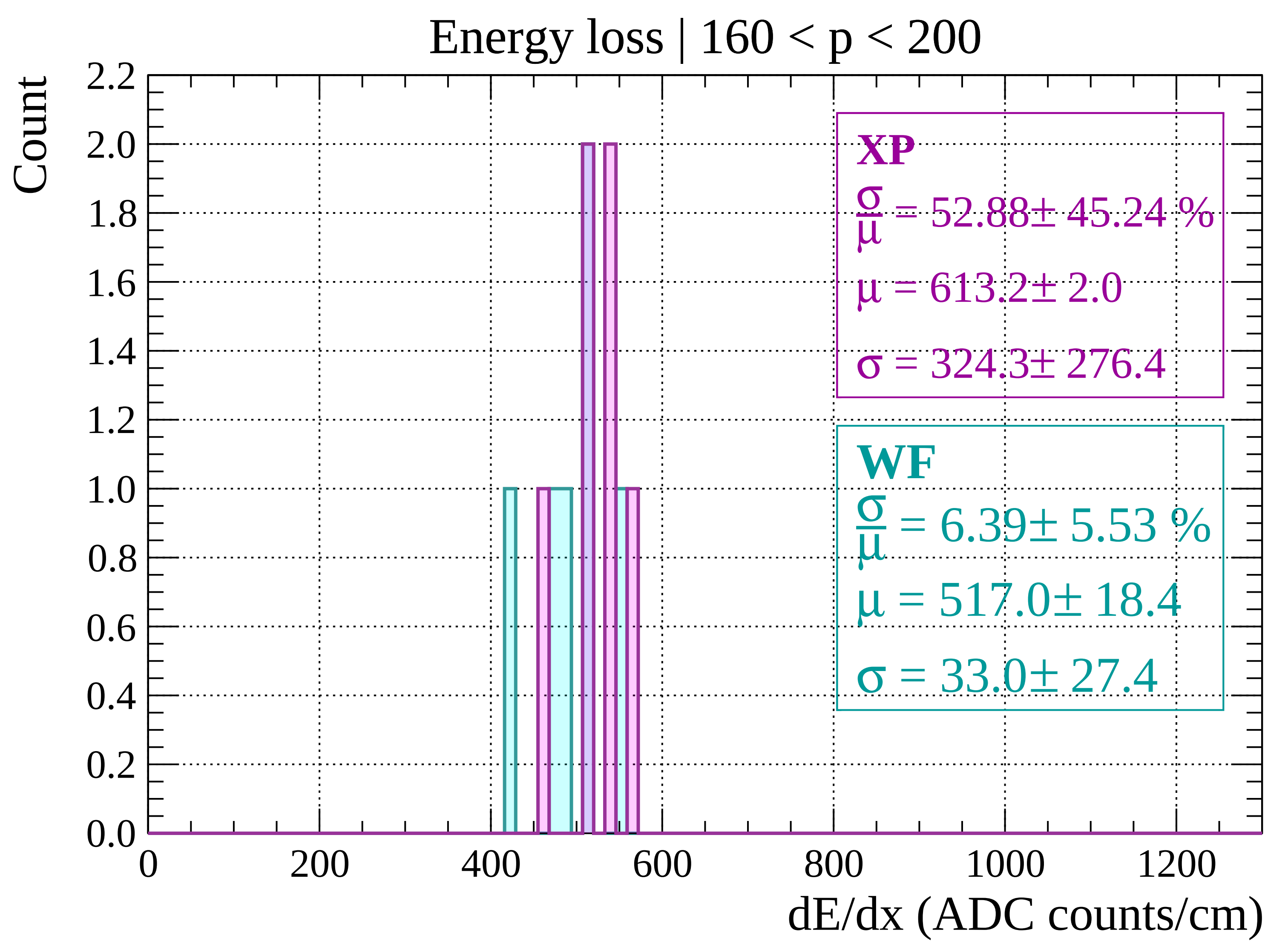
Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.04± 1.14 %
μ = 550.4± 5.6
σ = 49.7± 5.8
WF
$\frac{\sigma}{\mu}$ = 6.80± 1.17 %
μ = 530.2± 4.5
σ = 36.1± 5.9



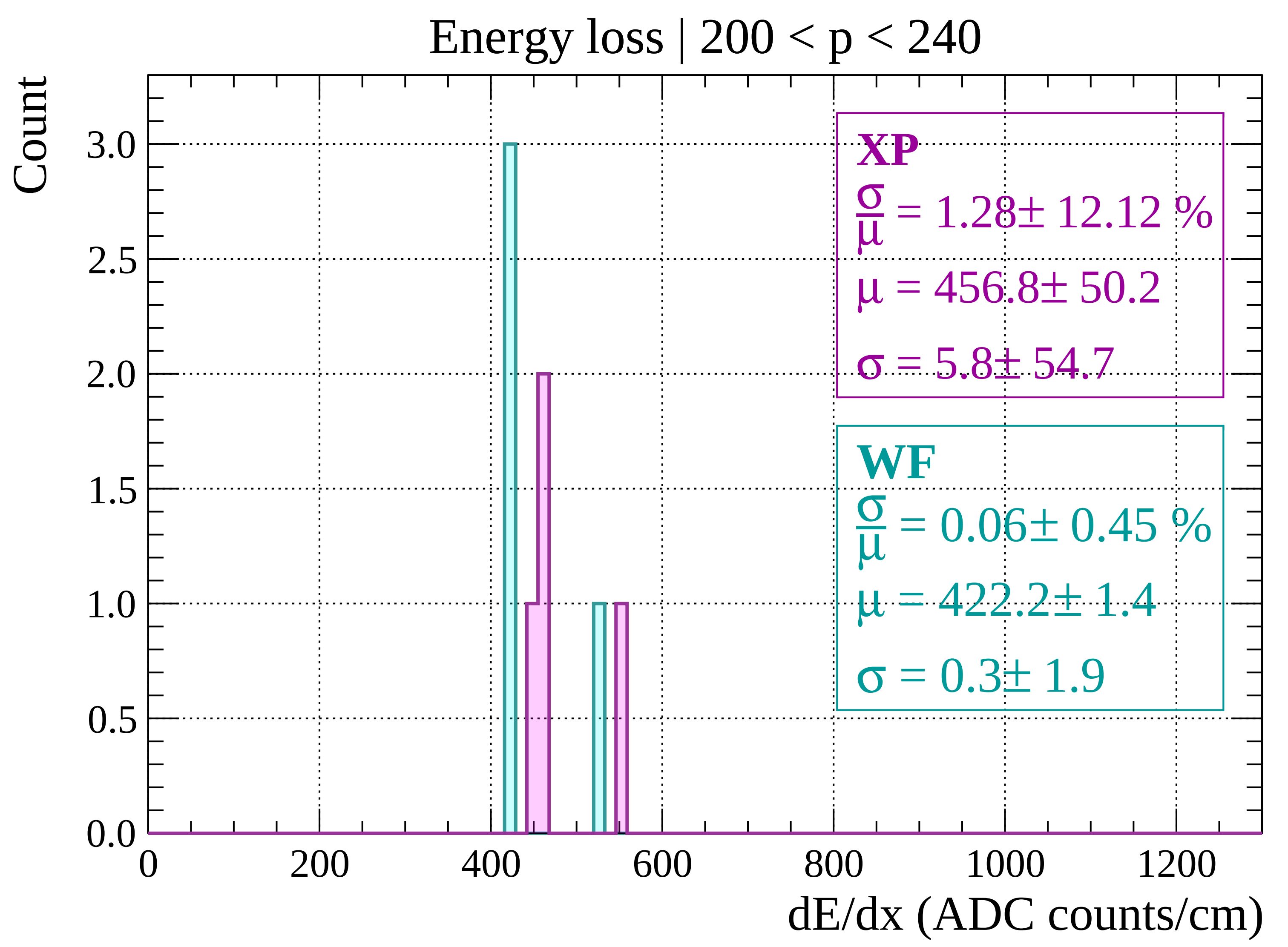
Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.04± 0.25 %
μ = 656.2± 2.3
σ = 0.3± 1.6
WF
$\frac{\sigma}{\mu}$ = 0.05± 2.61 %
μ = 578.3± 4.5
σ = 0.3± 15.1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200



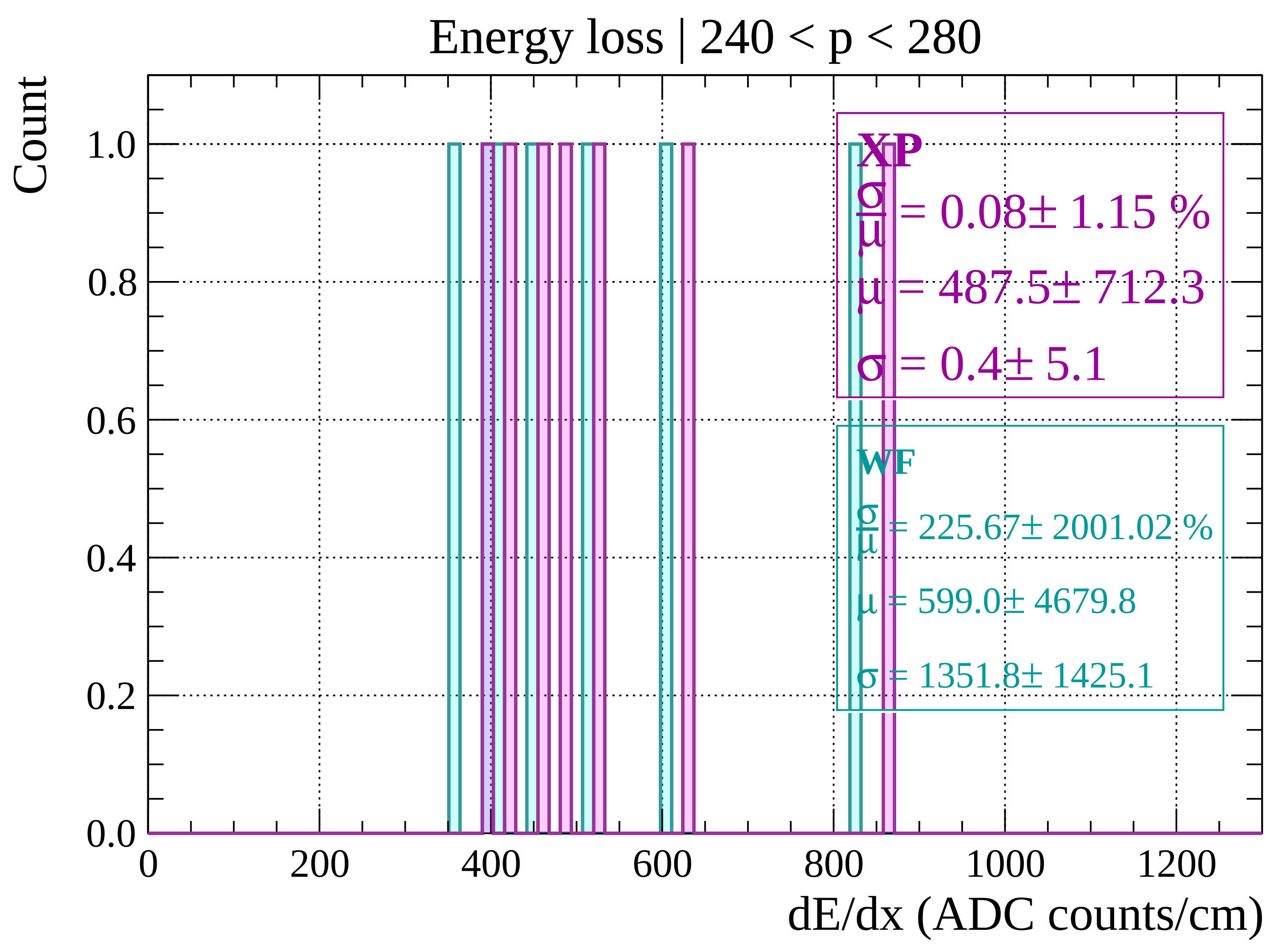
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 39.13± 63.22 %
μ = 569.0± 261.7
σ = 222.7± 257.3
WF
σ/μ = 727.45± 100461.24 %
μ = 100.0± 13717.9
σ = 727.5± 678.7



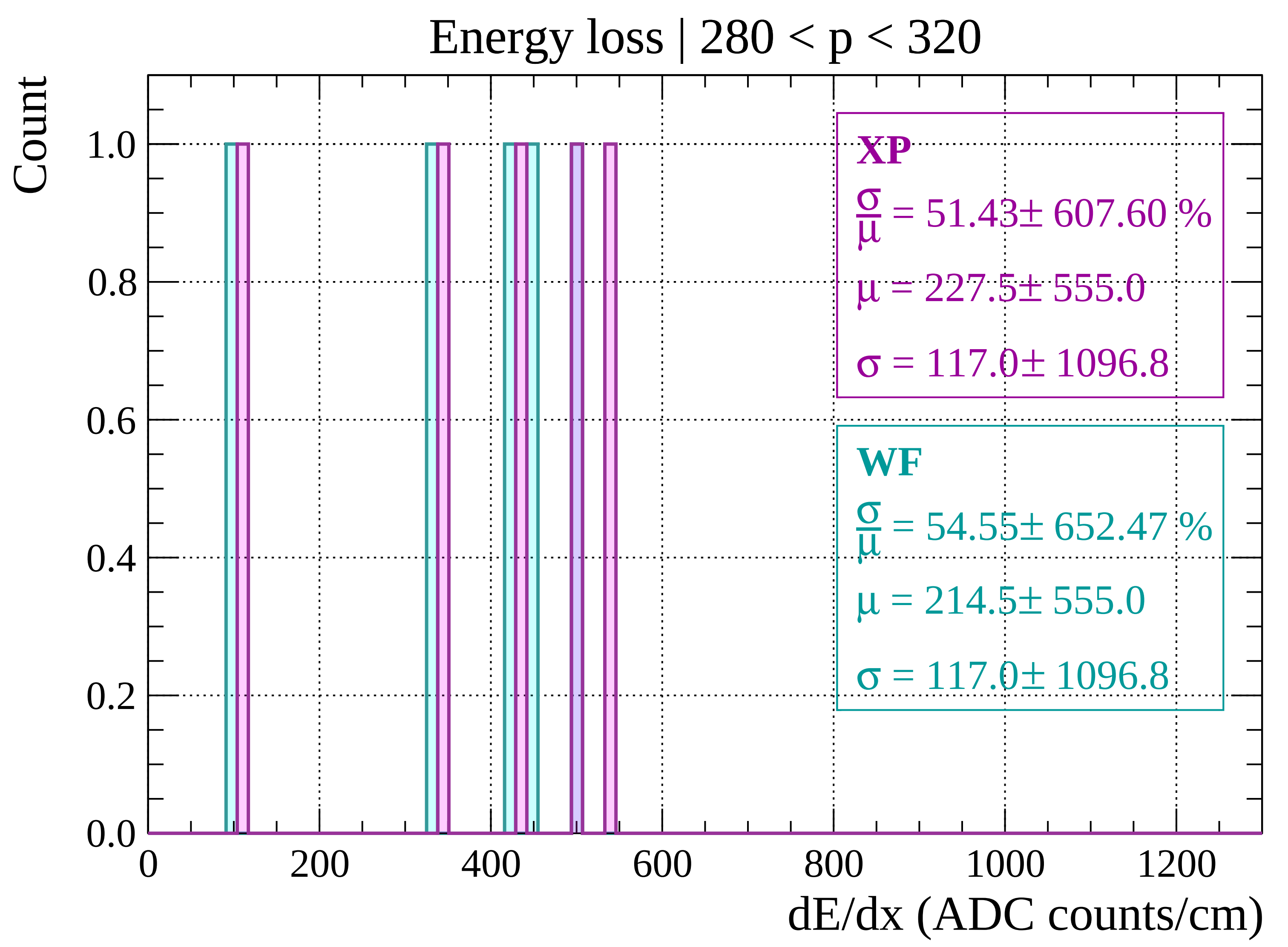
Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 52.88± 45.24 %
μ = 613.2± 2.0
σ = 324.3± 276.4
WF
$\frac{\sigma}{\mu}$ = 6.39± 5.53 %
μ = 517.0± 18.4
σ = 33.0± 27.4

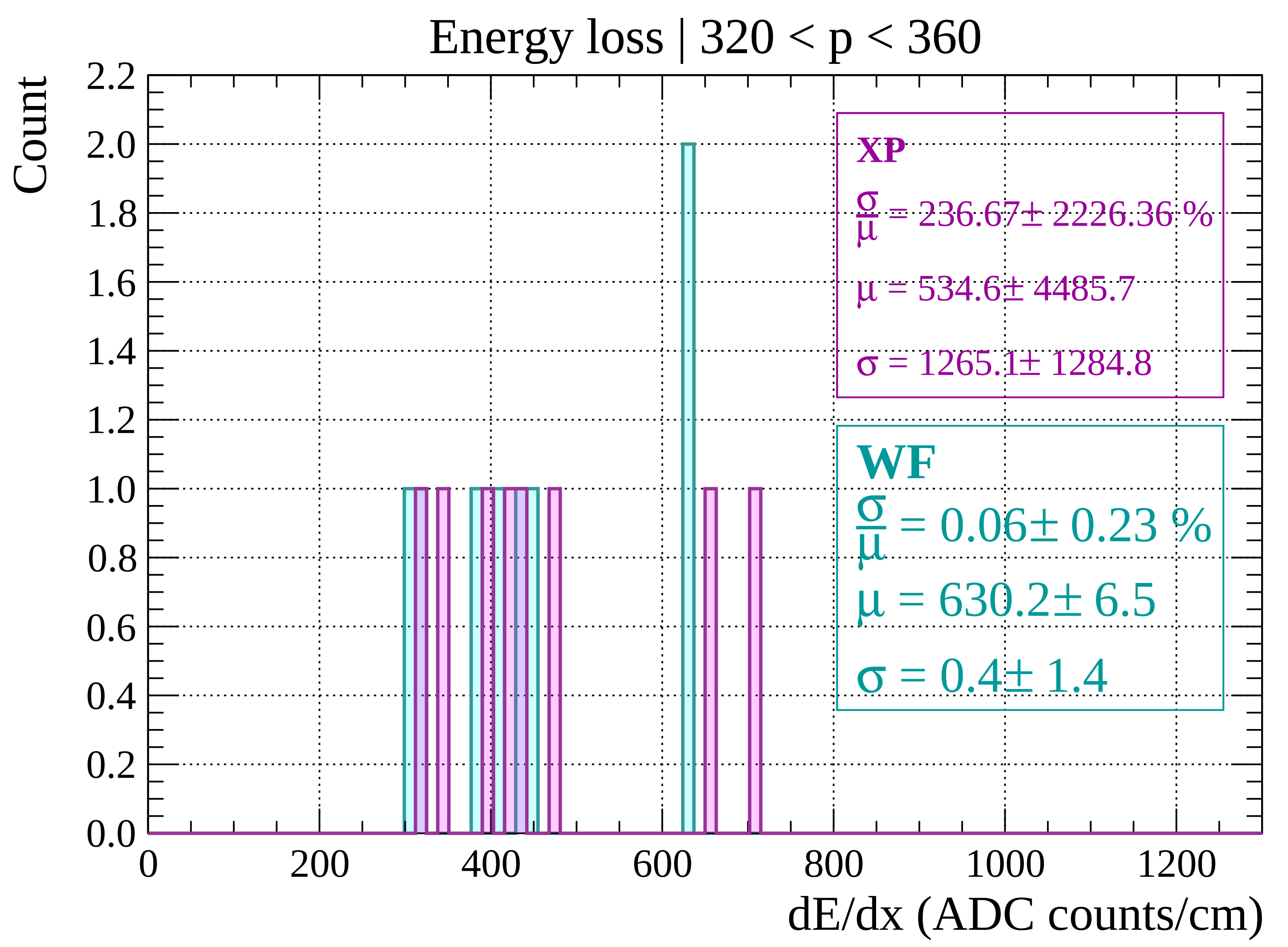


Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 1.28± 12.12 %
μ = 456.8± 50.2
σ = 5.8± 54.7
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.45 %
μ = 422.2± 1.4
σ = 0.3± 1.9

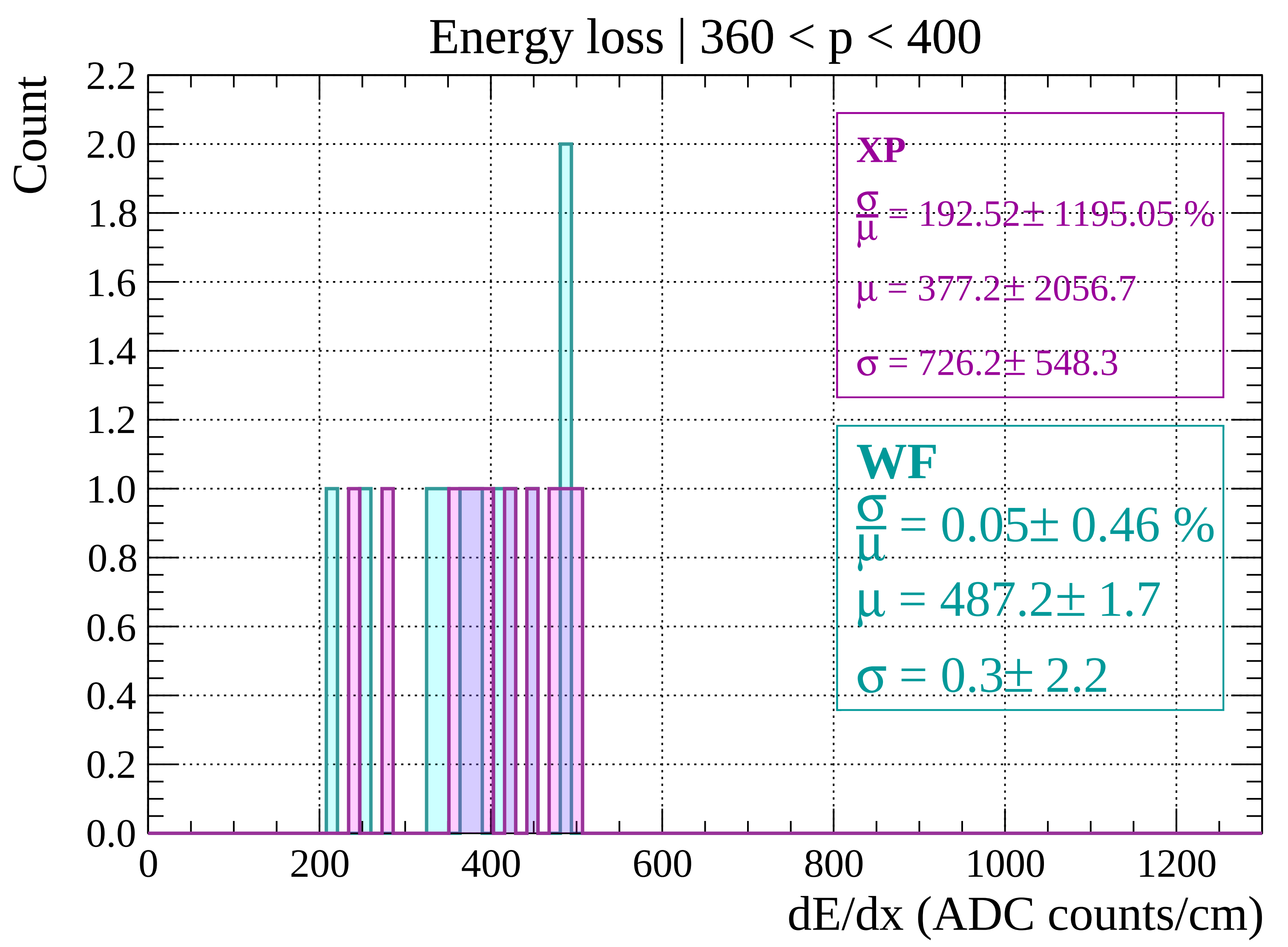


Energy loss | 240 < p < 280
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.08± 1.15 %
μ = 487.5± 712.3
σ = 0.4± 5.1
WF
$\frac{\sigma}{\mu}$ = 225.67± 2001.02 %
μ = 599.0± 4679.8
σ = 1351.8± 1425.1

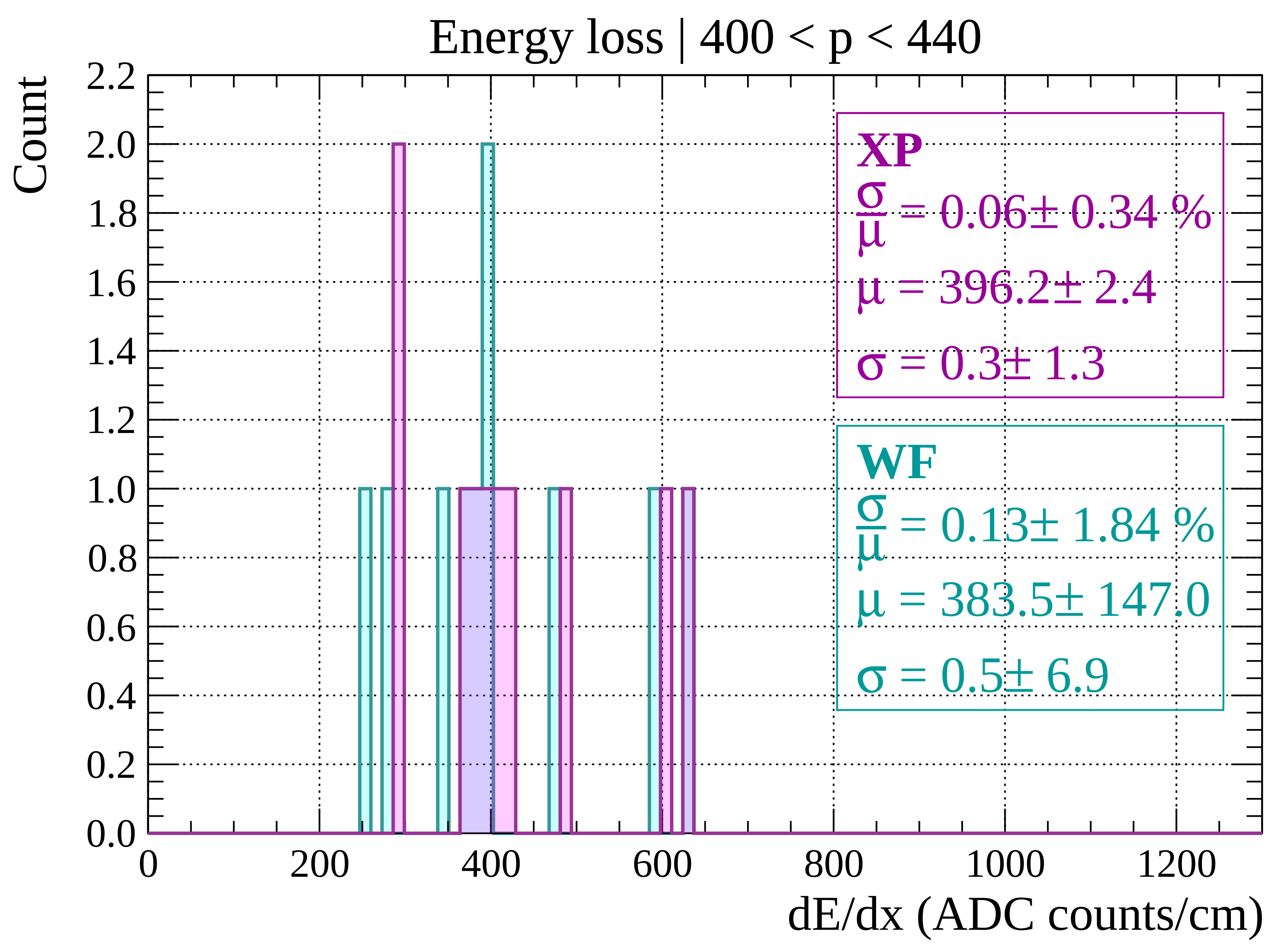

Energy loss | 280 < p < 320
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 51.43± 607.60 %
μ = 227.5± 555.0
σ = 117.0± 1096.8
WF
σ/μ = 54.55± 652.47 %
μ = 214.5± 555.0
σ = 117.0± 1096.8



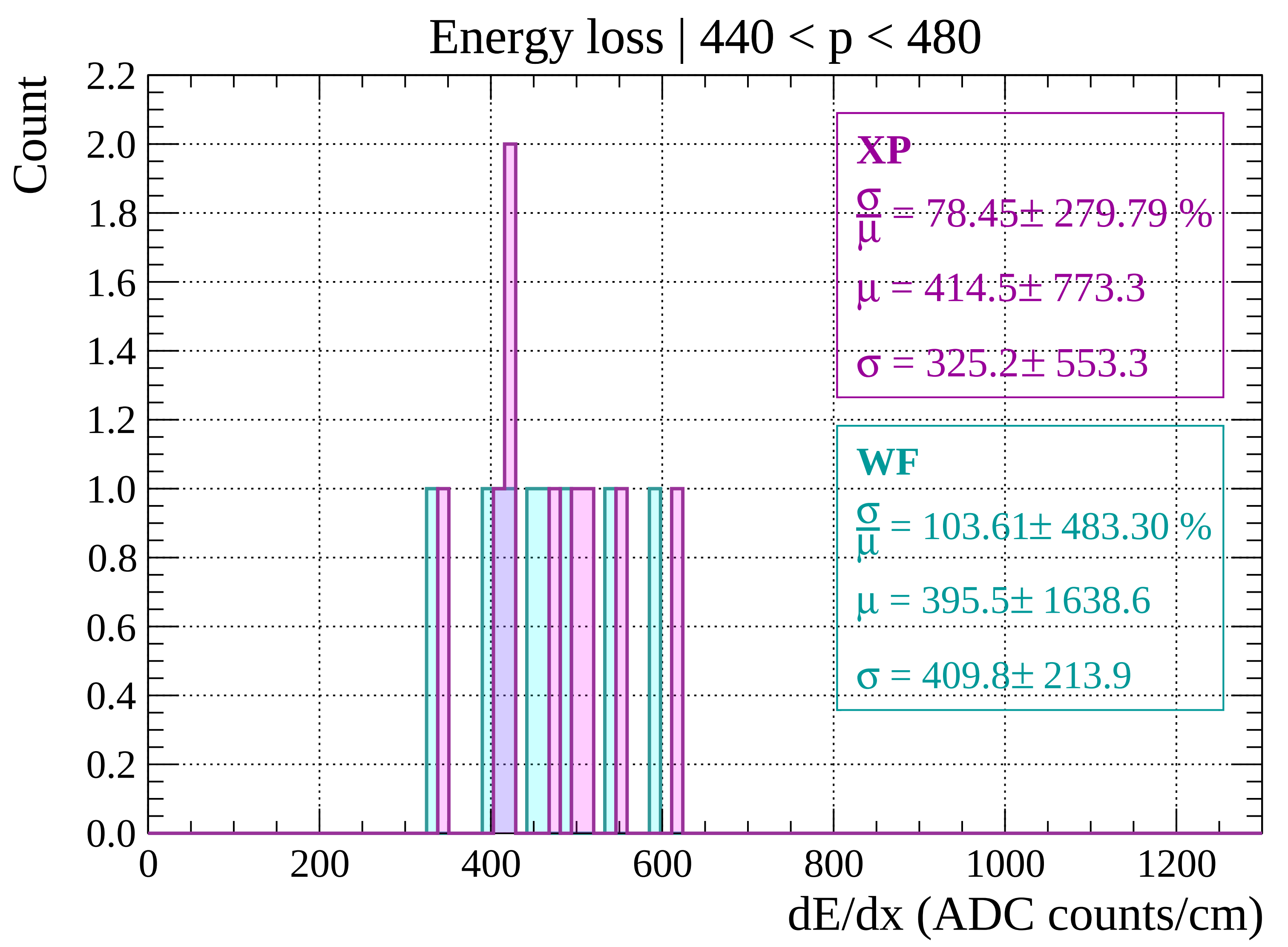
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 236.67± 2226.36 %
μ = 534.6± 4485.7
σ = 1265.1± 1284.8
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.23 %
μ = 630.2± 6.5
σ = 0.4± 1.4



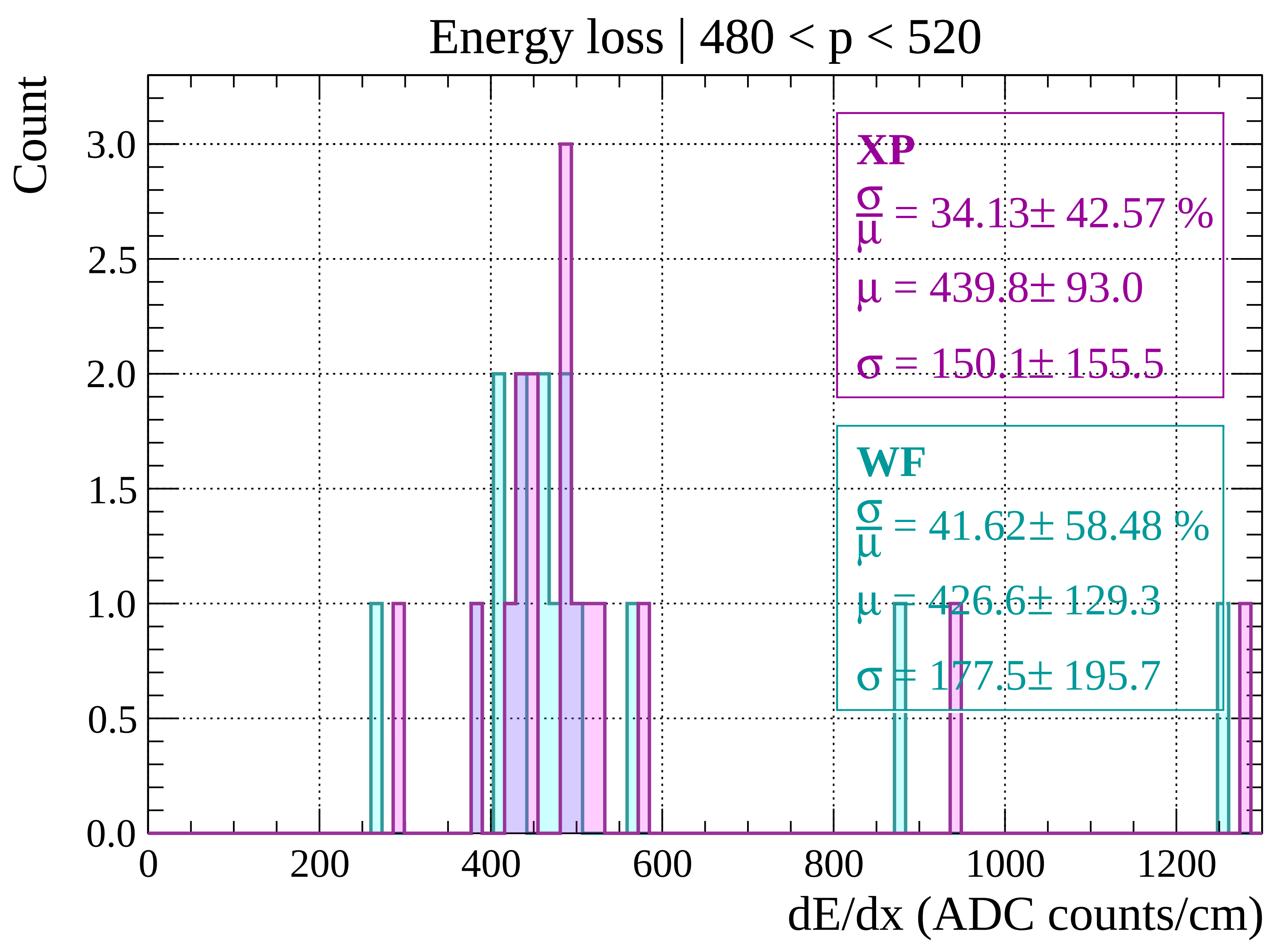
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 192.52± 1195.05 %
μ = 377.2± 2056.7
σ = 726.2± 548.3
WF
$\frac{\sigma}{\mu}$ = 0.05± 0.46 %
μ = 487.2± 1.7
σ = 0.3± 2.2

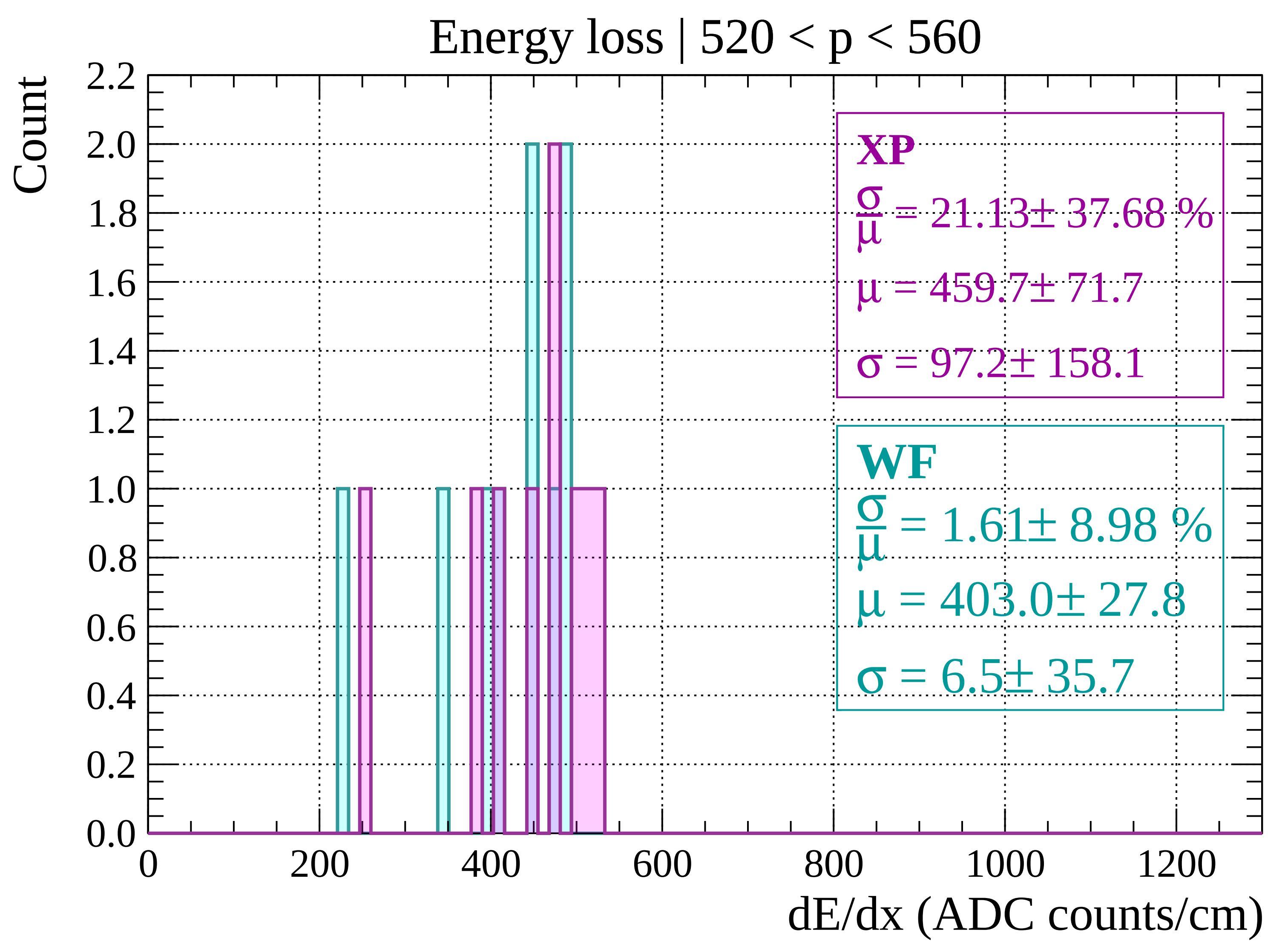


Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.06± 0.34 %
μ = 396.2± 2.4
σ = 0.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 0.13± 1.84 %
μ = 383.5± 147.0
σ = 0.5± 6.9
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200

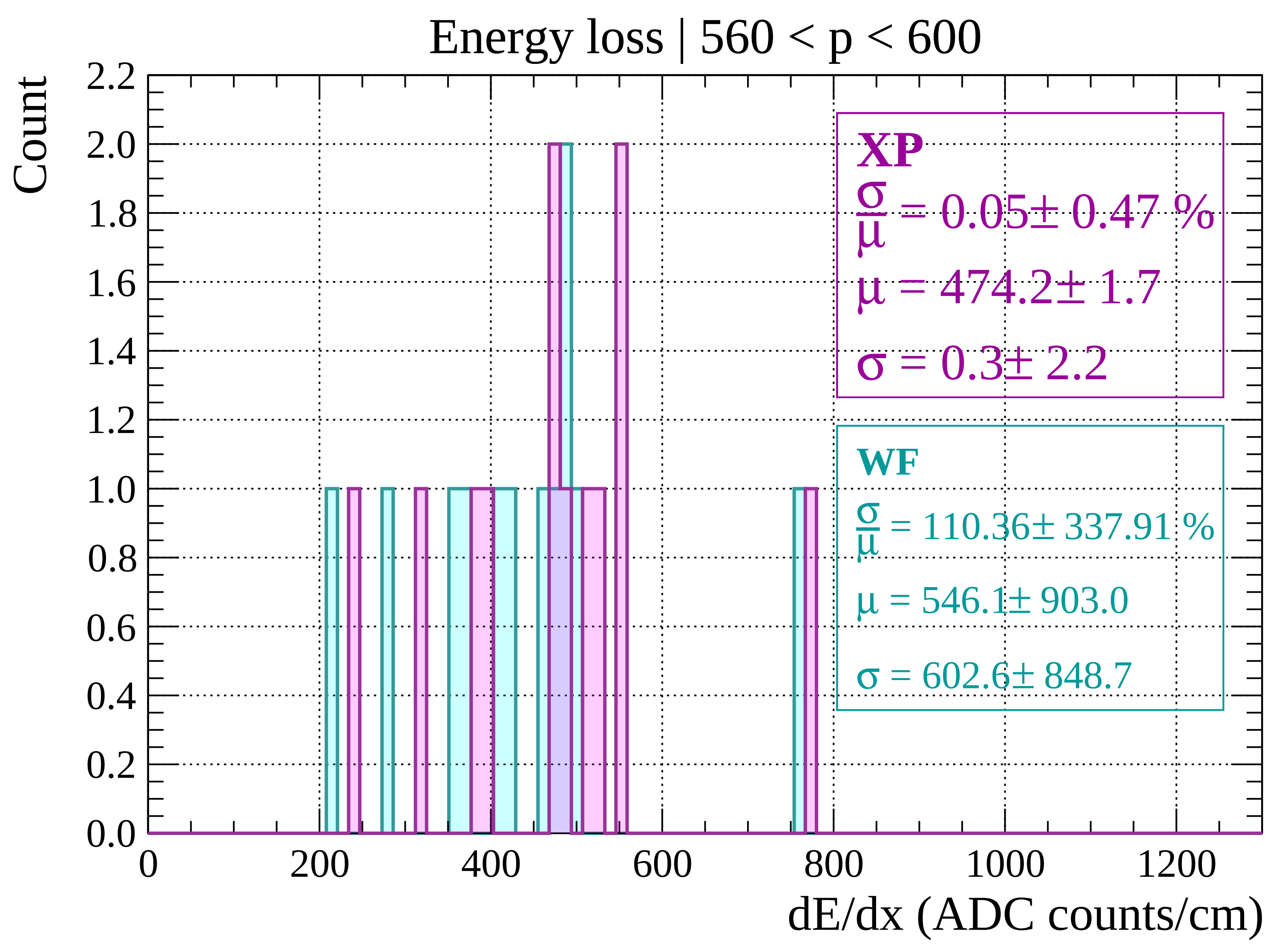


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 78.45± 279.79 %
μ = 414.5± 773.3
σ = 325.2± 553.3
WF
σ/μ = 103.61± 483.30 %
μ = 395.5± 1638.6
σ = 409.8± 213.9

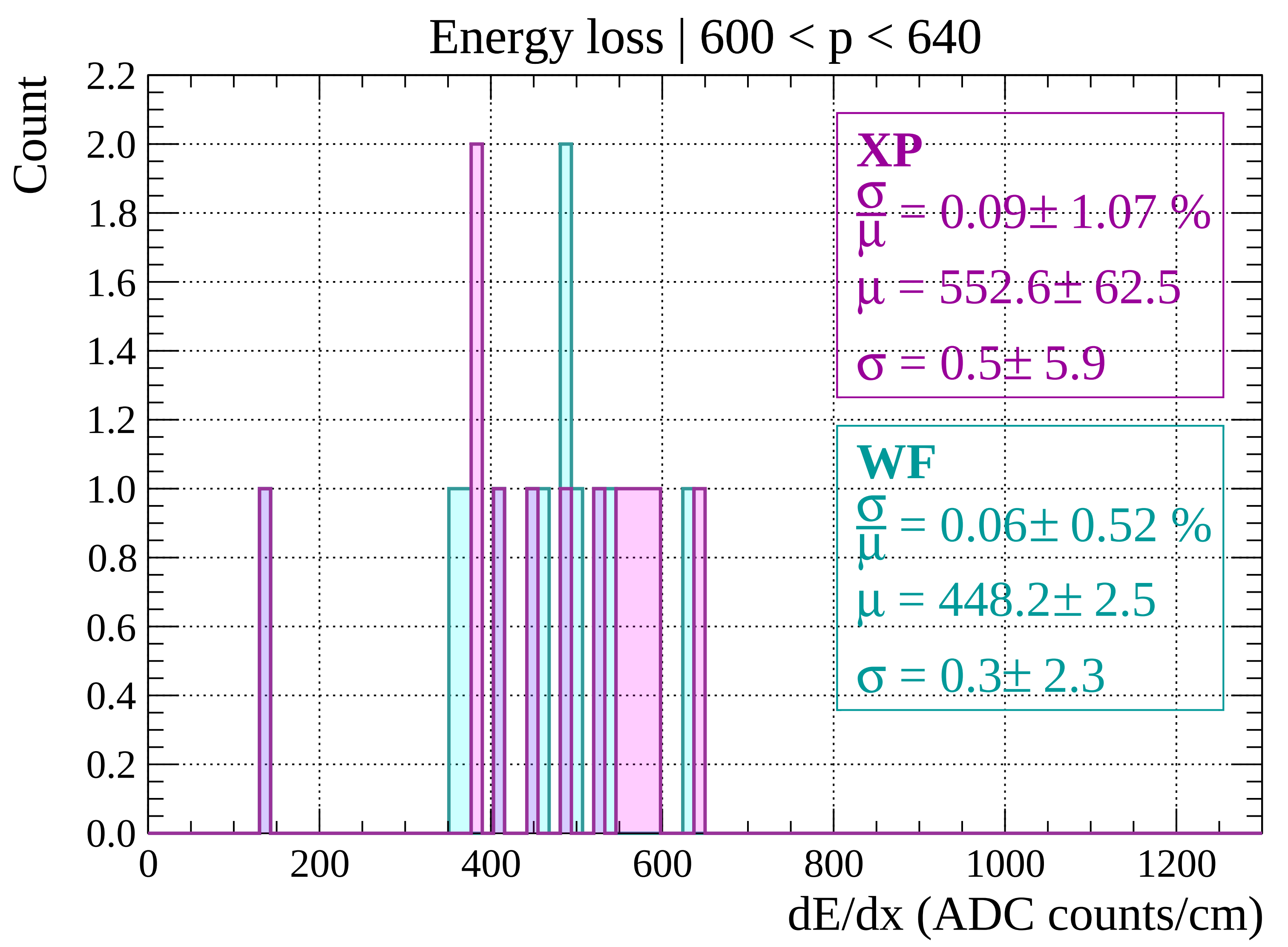

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = 34.13± 42.57 %
μ = 439.8± 93.0
σ = 150.1± 155.5
WF
σ/μ = 41.62± 58.48 %
μ = 426.6± 129.3
σ = 177.5± 195.7



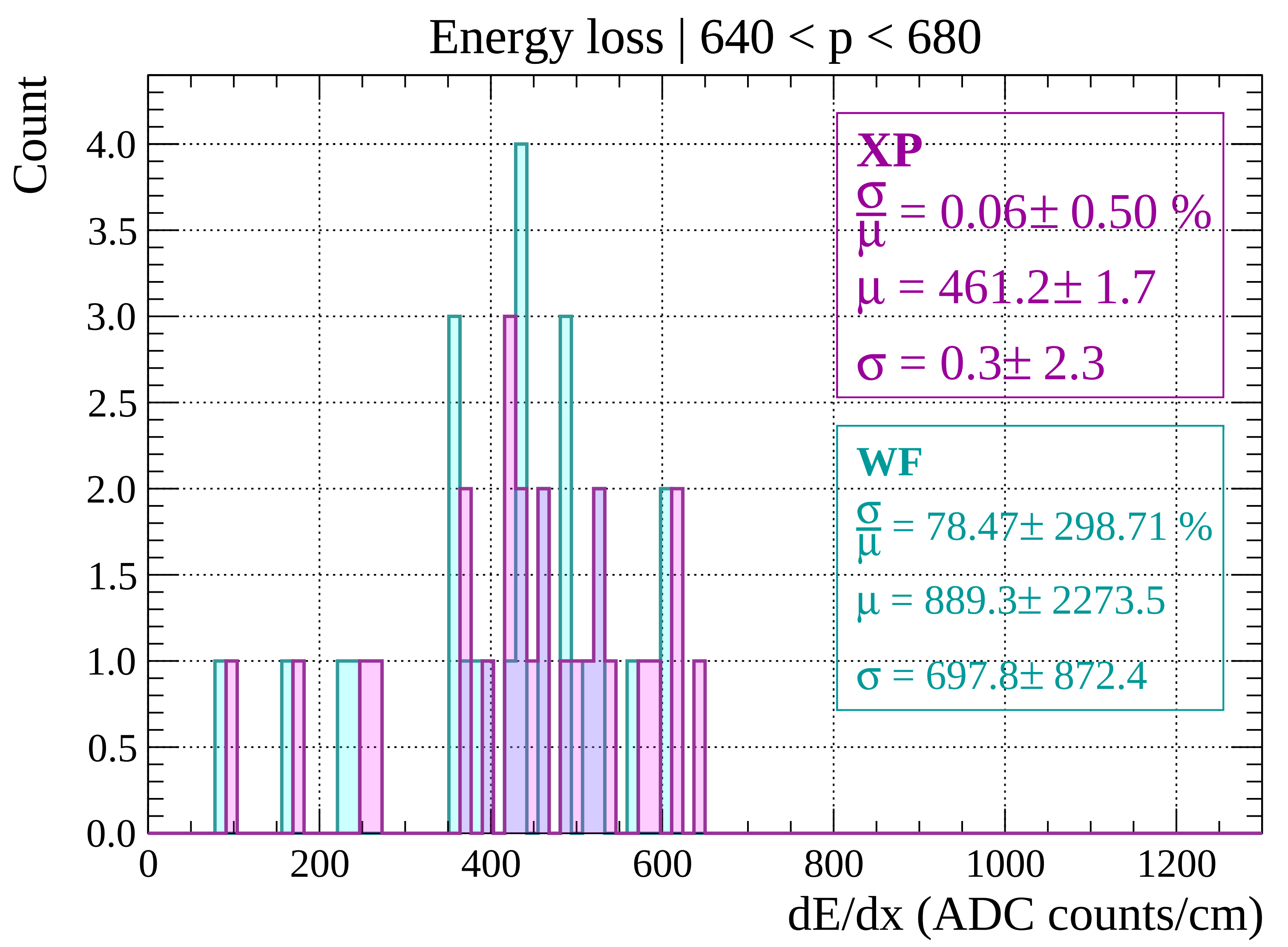
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 21.13± 37.68 %
μ = 459.7± 71.7
σ = 97.2± 158.1
WF
σ/μ = 1.61± 8.98 %
μ = 403.0± 27.8
σ = 6.5± 35.7



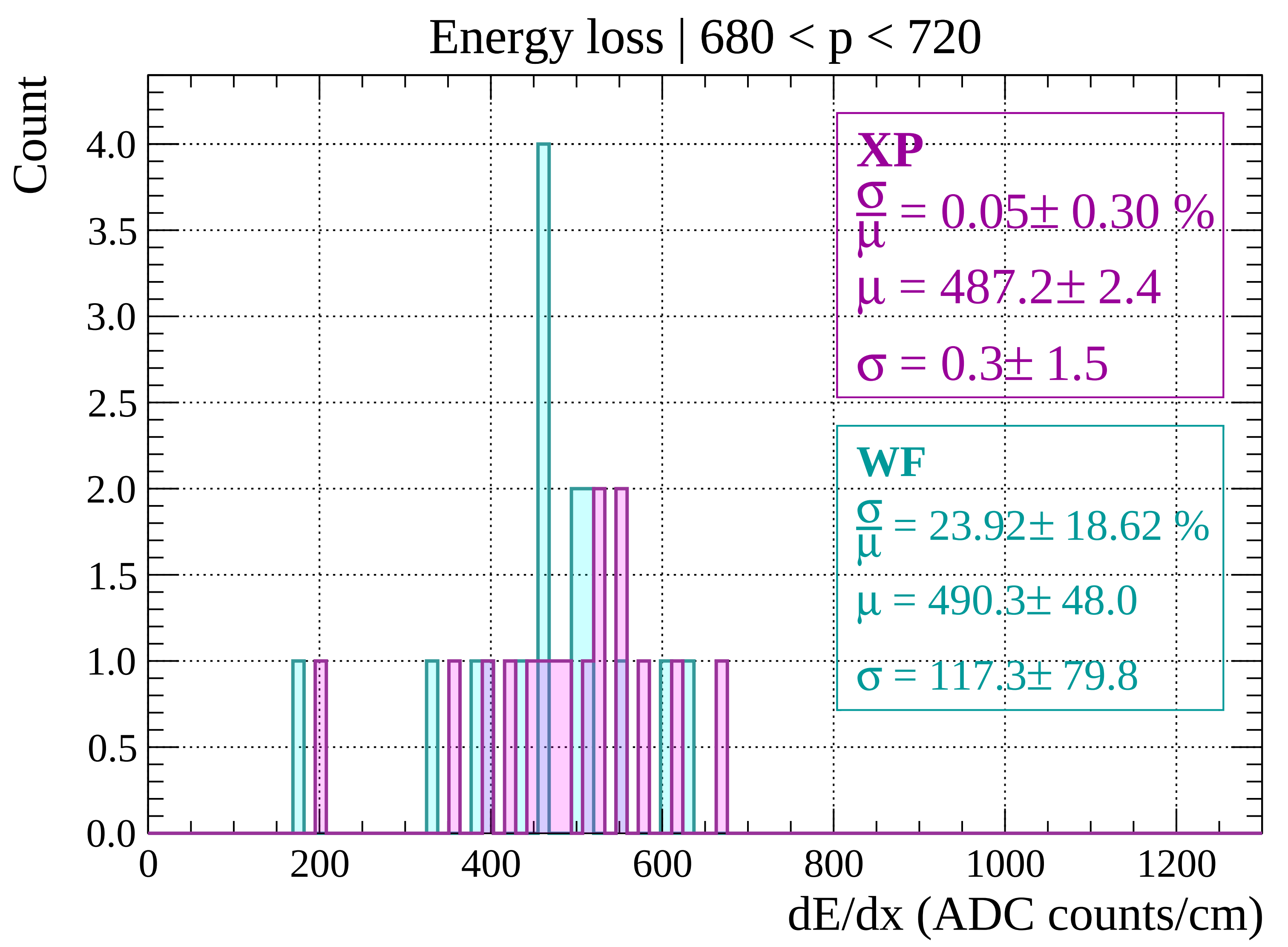
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 0.05± 0.47 %
μ = 474.2± 1.7
σ = 0.3± 2.2
WF
σ/μ = 110.36± 337.91 %
μ = 546.1± 903.0
σ = 602.6± 848.7

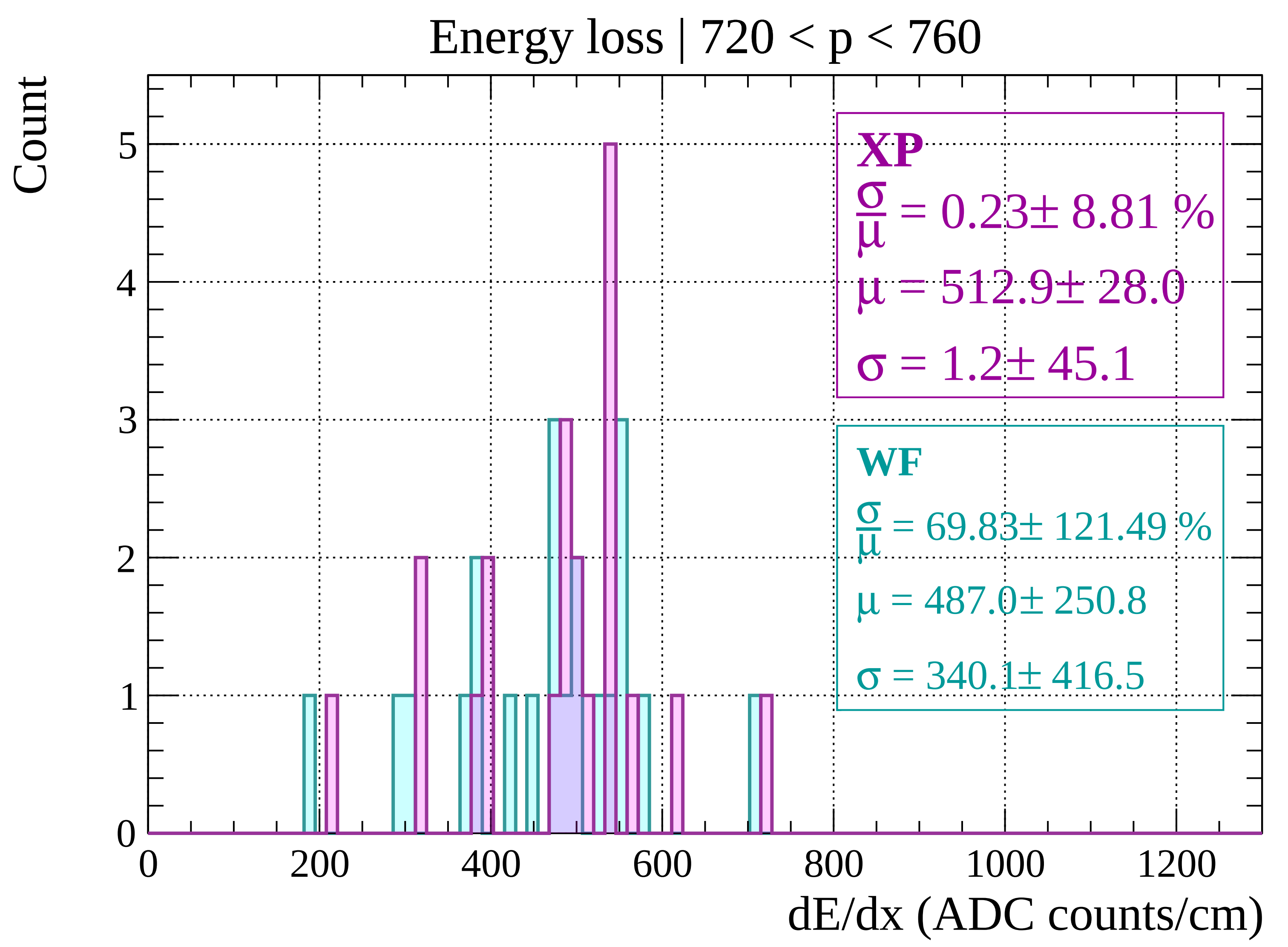


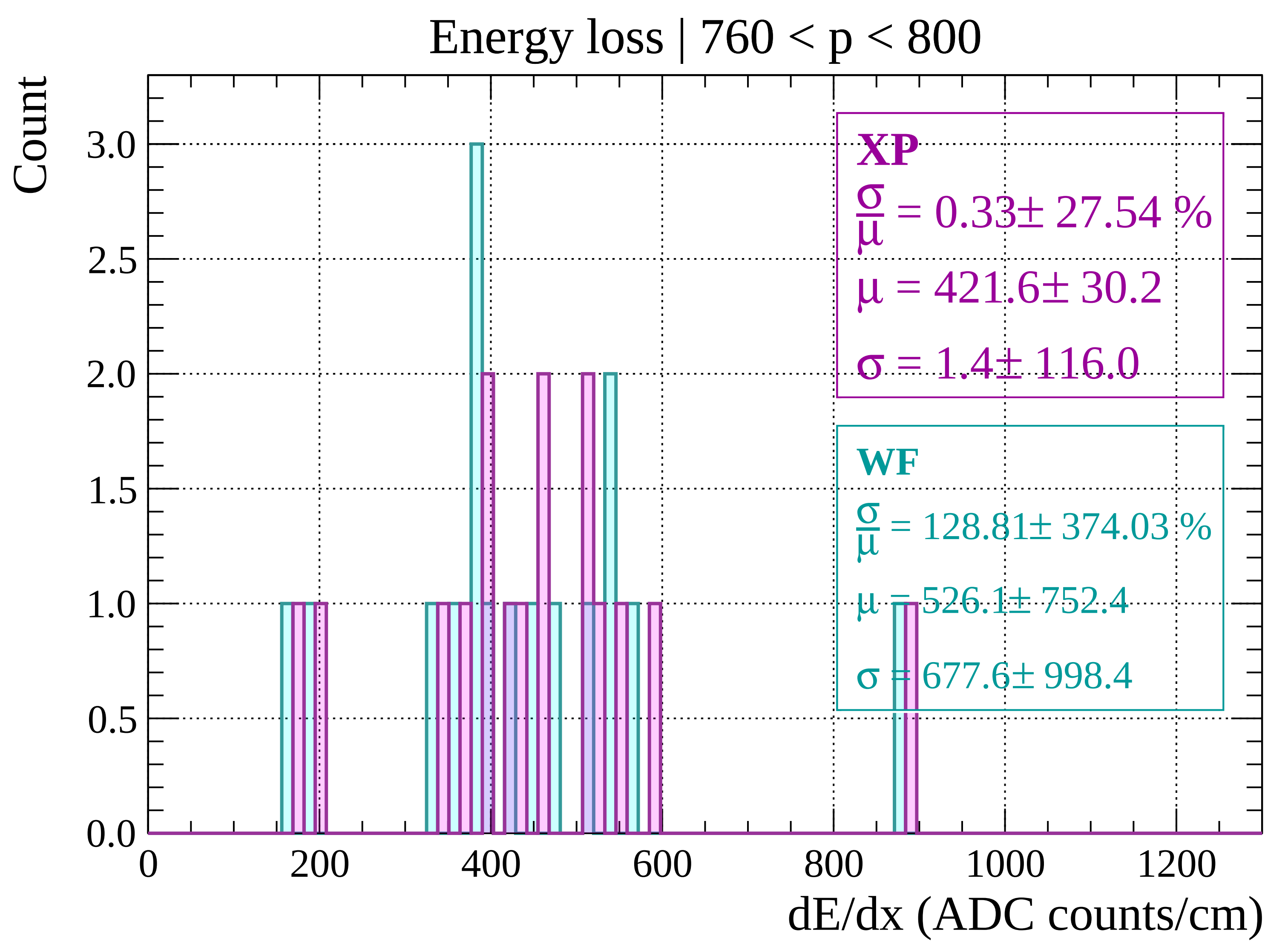
Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.09± 1.07 %
μ = 552.6± 62.5
σ = 0.5± 5.9
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.52 %
μ = 448.2± 2.5
σ = 0.3± 2.3



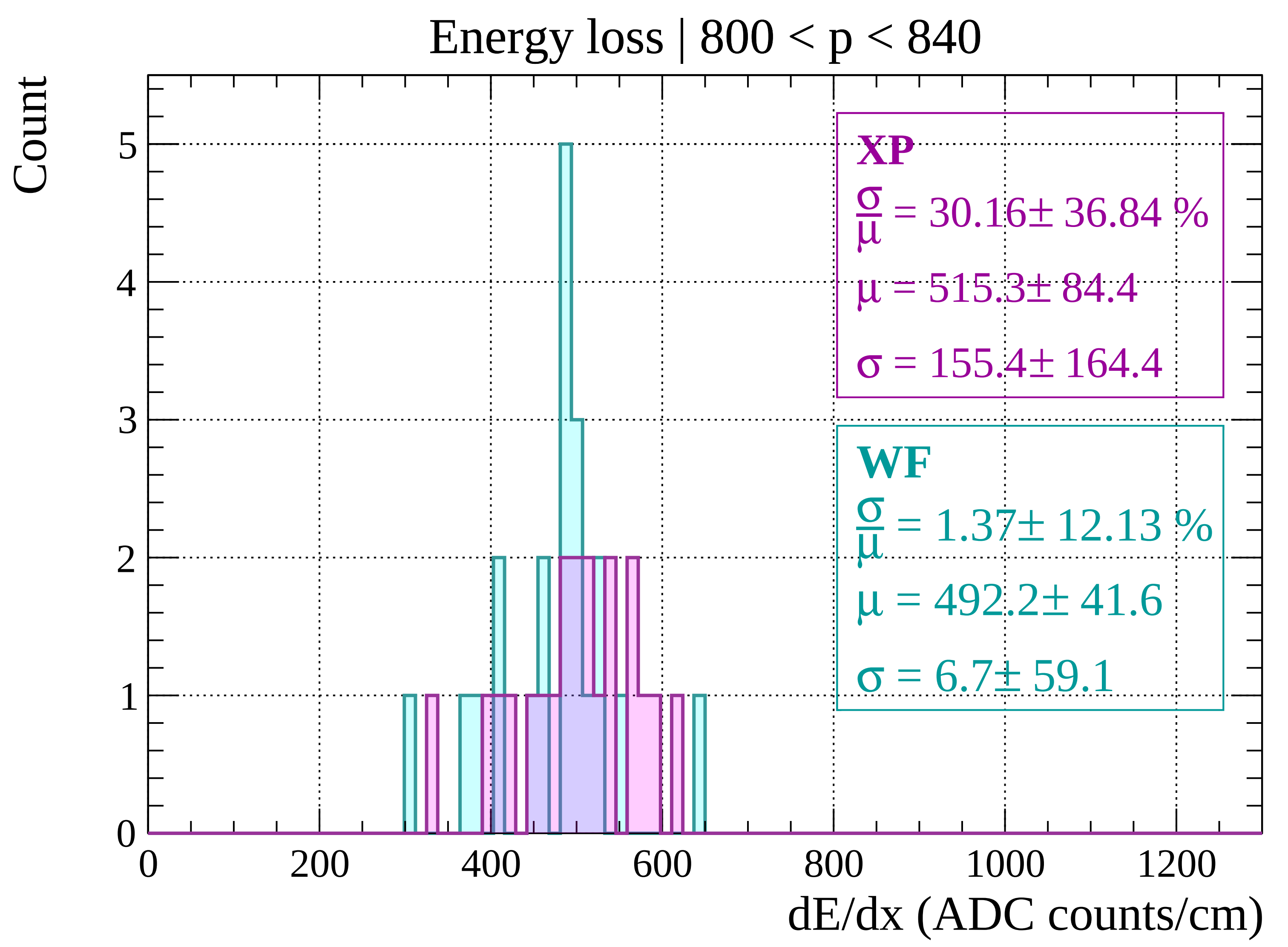
Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.50 %
μ = 461.2± 1.7
σ = 0.3± 2.3
WF
σ/μ = 78.47± 298.71 %
μ = 889.3± 2273.5
σ = 697.8± 872.4


Energy loss | 680 < p < 720
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.05± 0.30 %
μ = 487.2± 2.4
σ = 0.3± 1.5
WF
σ/μ = 23.92± 18.62 %
μ = 490.3± 48.0
σ = 117.3± 79.8

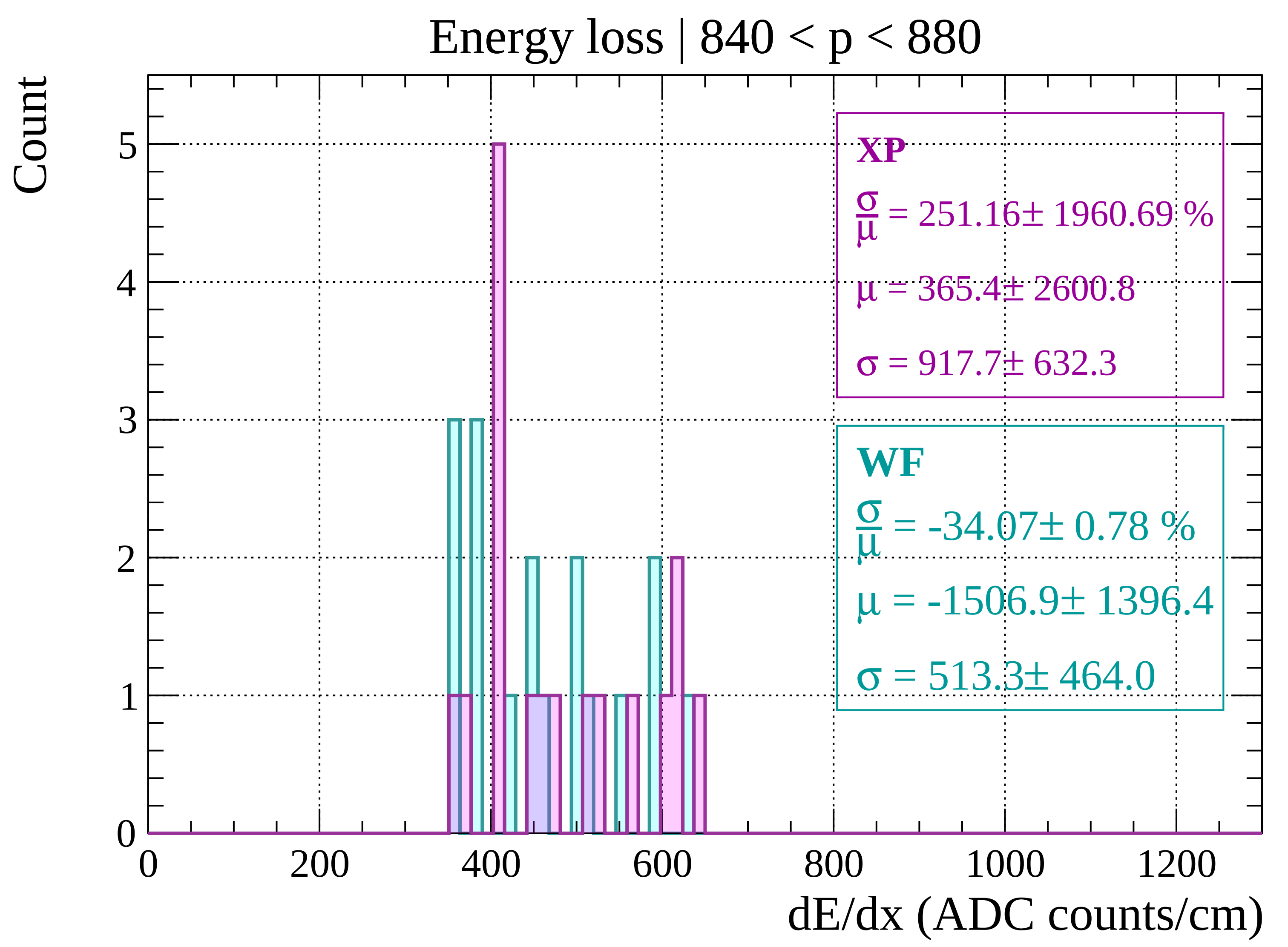

Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 0.23± 8.81 %
μ = 512.9± 28.0
σ = 1.2± 45.1
WF
σ/μ = 69.83± 121.49 %
μ = 487.0± 250.8
σ = 340.1± 416.5



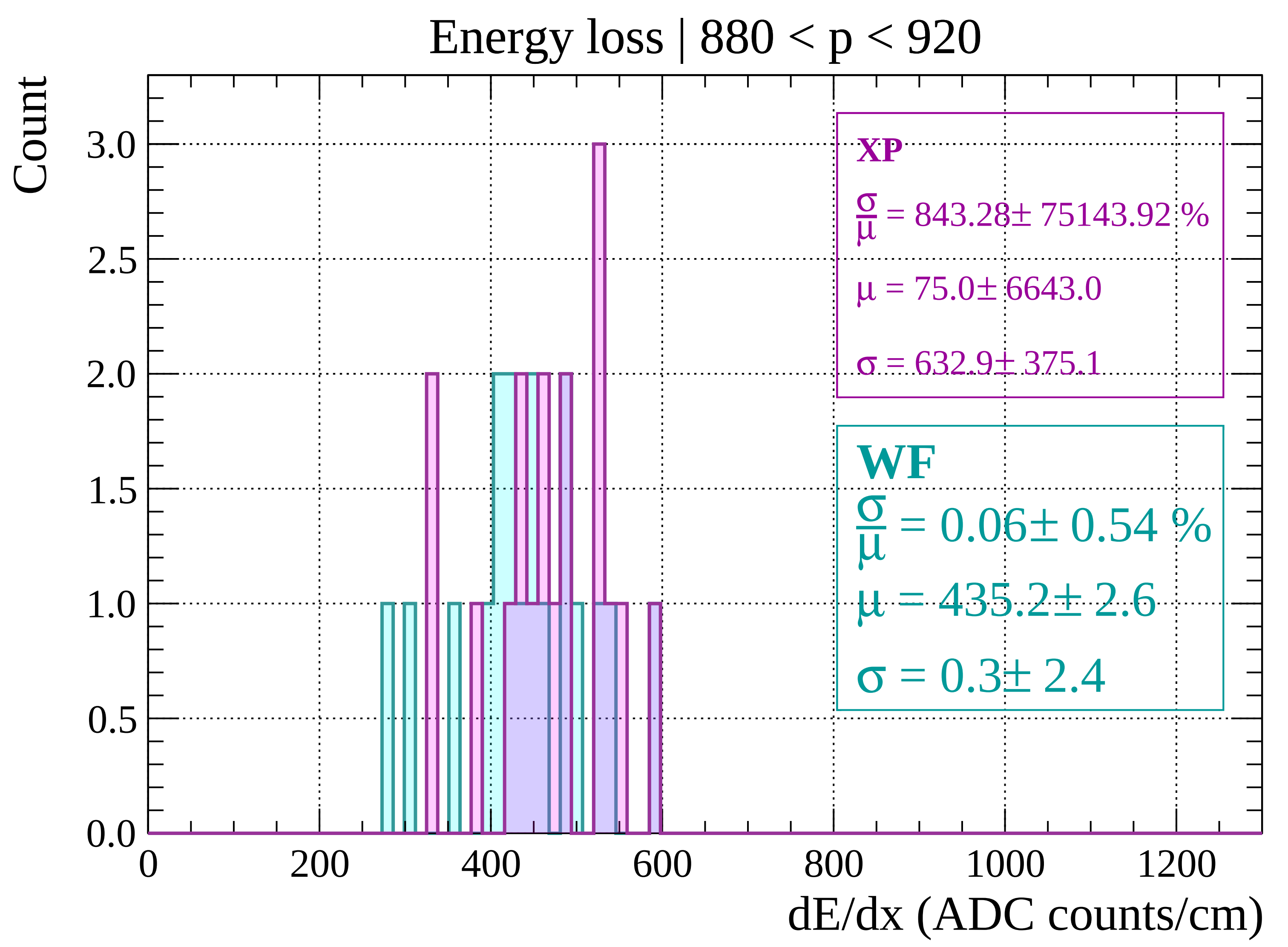
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.33± 27.54 %
μ = 421.6± 30.2
σ = 1.4± 116.0
WF
$\frac{\sigma}{\mu}$ = 128.81± 374.03 %
μ = 526.1± 752.4
σ = 677.6± 998.4



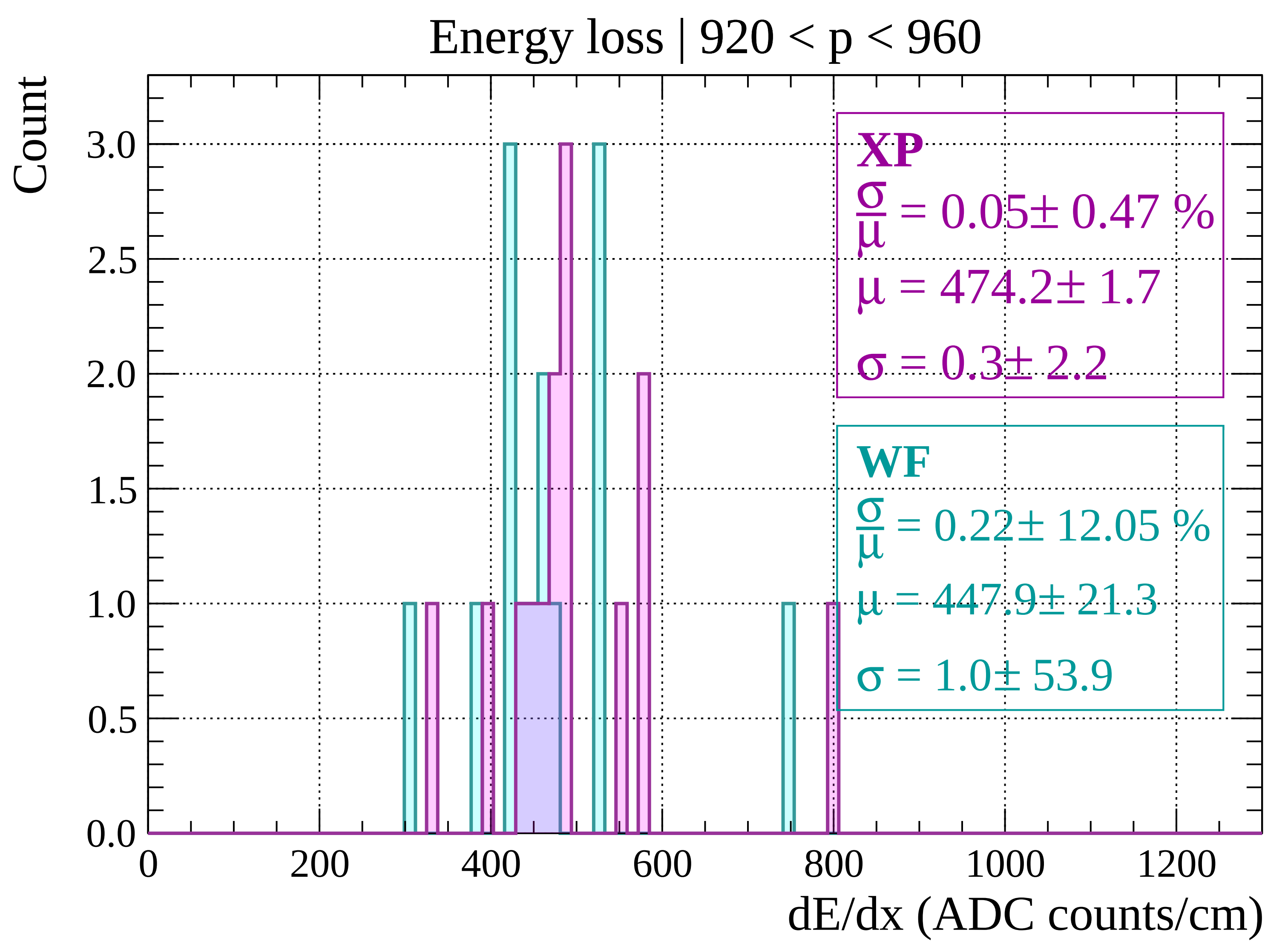
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 30.16± 36.84 %
μ = 515.3± 84.4
σ = 155.4± 164.4
WF
$\frac{\sigma}{\mu}$ = 1.37± 12.13 %
μ = 492.2± 41.6
σ = 6.7± 59.1



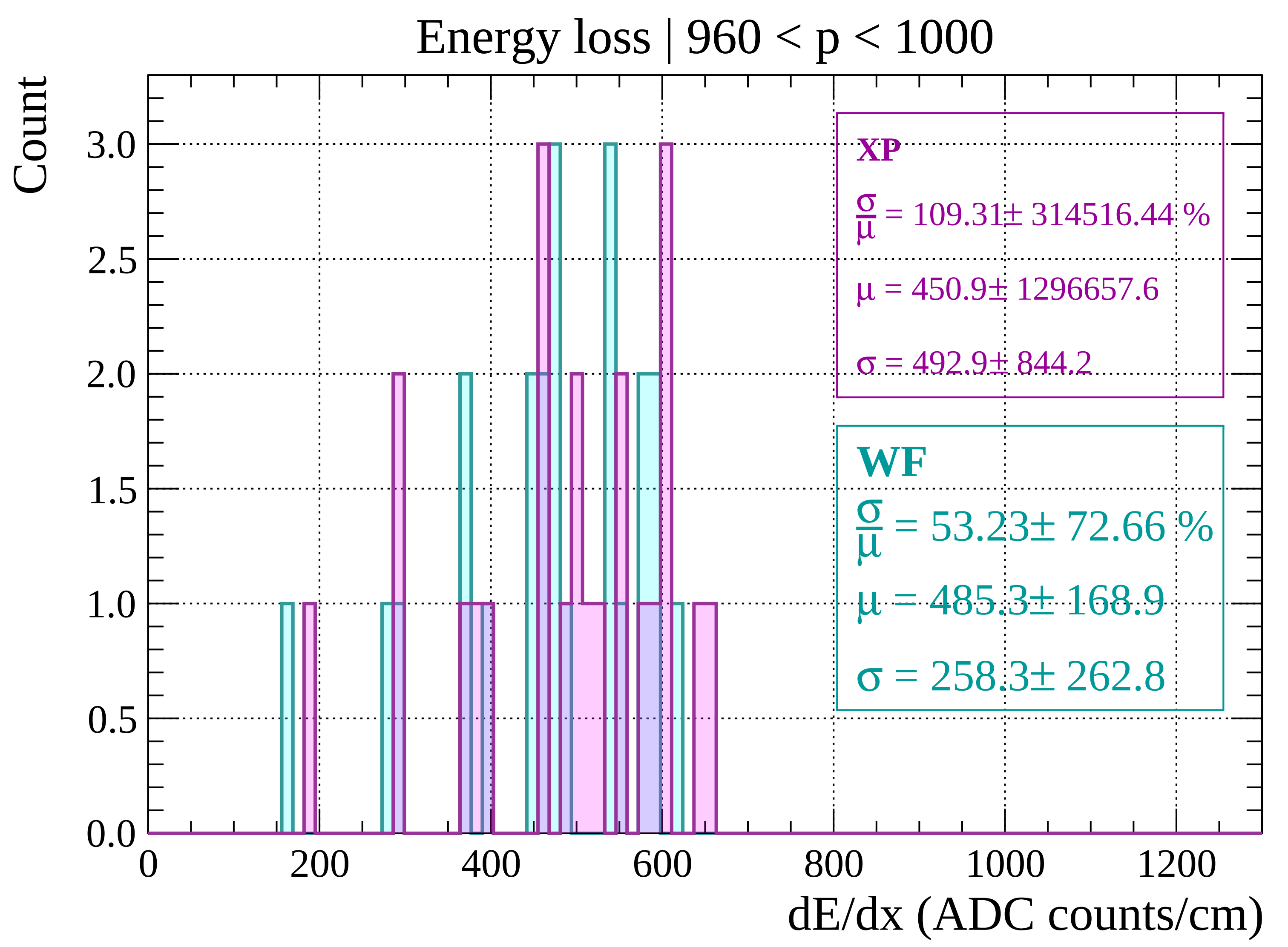
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 251.16± 1960.69 %
μ = 365.4± 2600.8
σ = 917.7± 632.3
WF
$\frac{\sigma}{\mu}$ = -34.07± 0.78 %
μ = -1506.9± 1396.4
σ = 513.3± 464.0



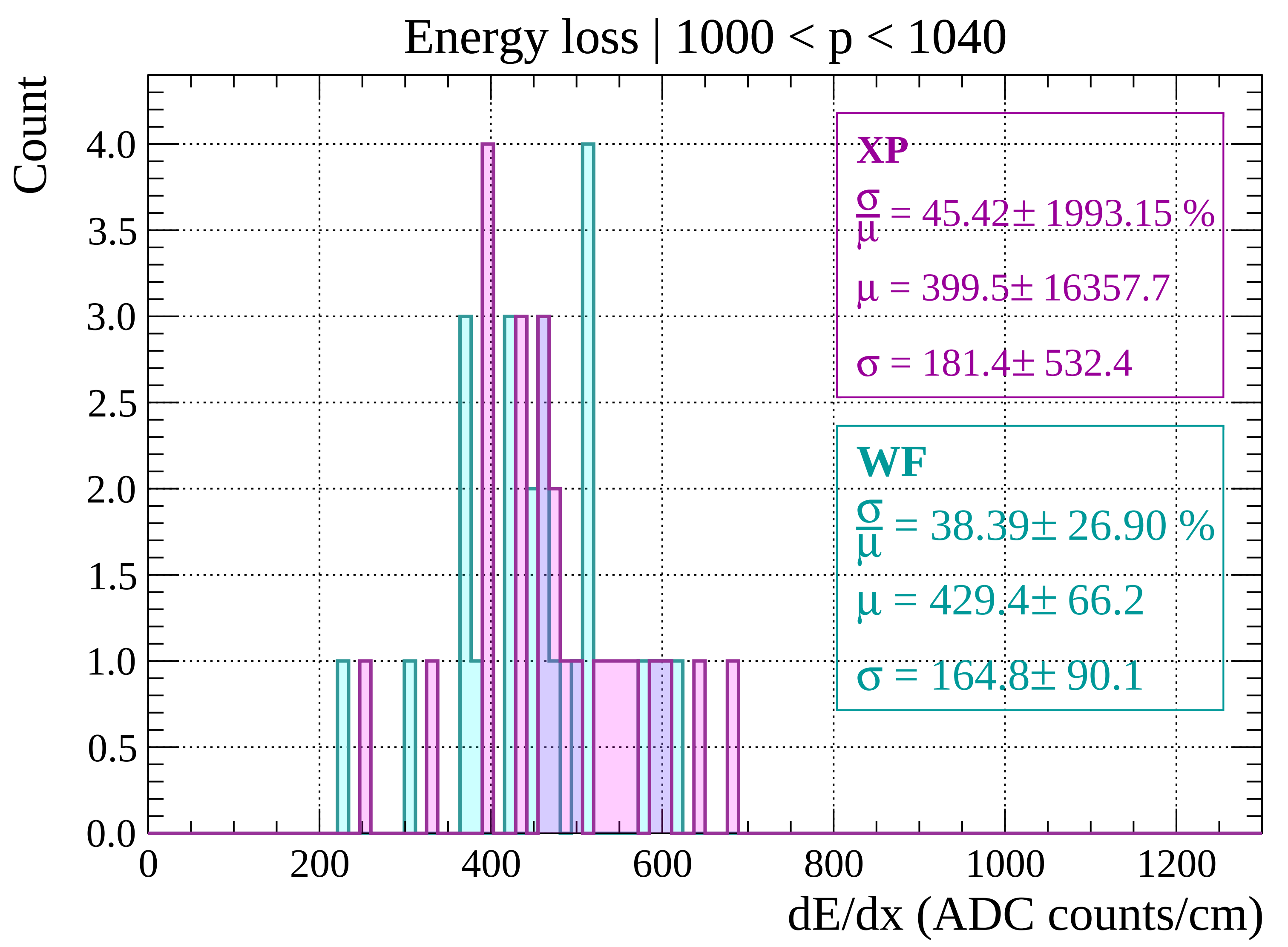
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 843.28± 75143.92 %
μ = 75.0± 6643.0
σ = 632.9± 375.1
WF
$\frac{\sigma}{\mu}$ = 0.06± 0.54 %
μ = 435.2± 2.6
σ = 0.3± 2.4



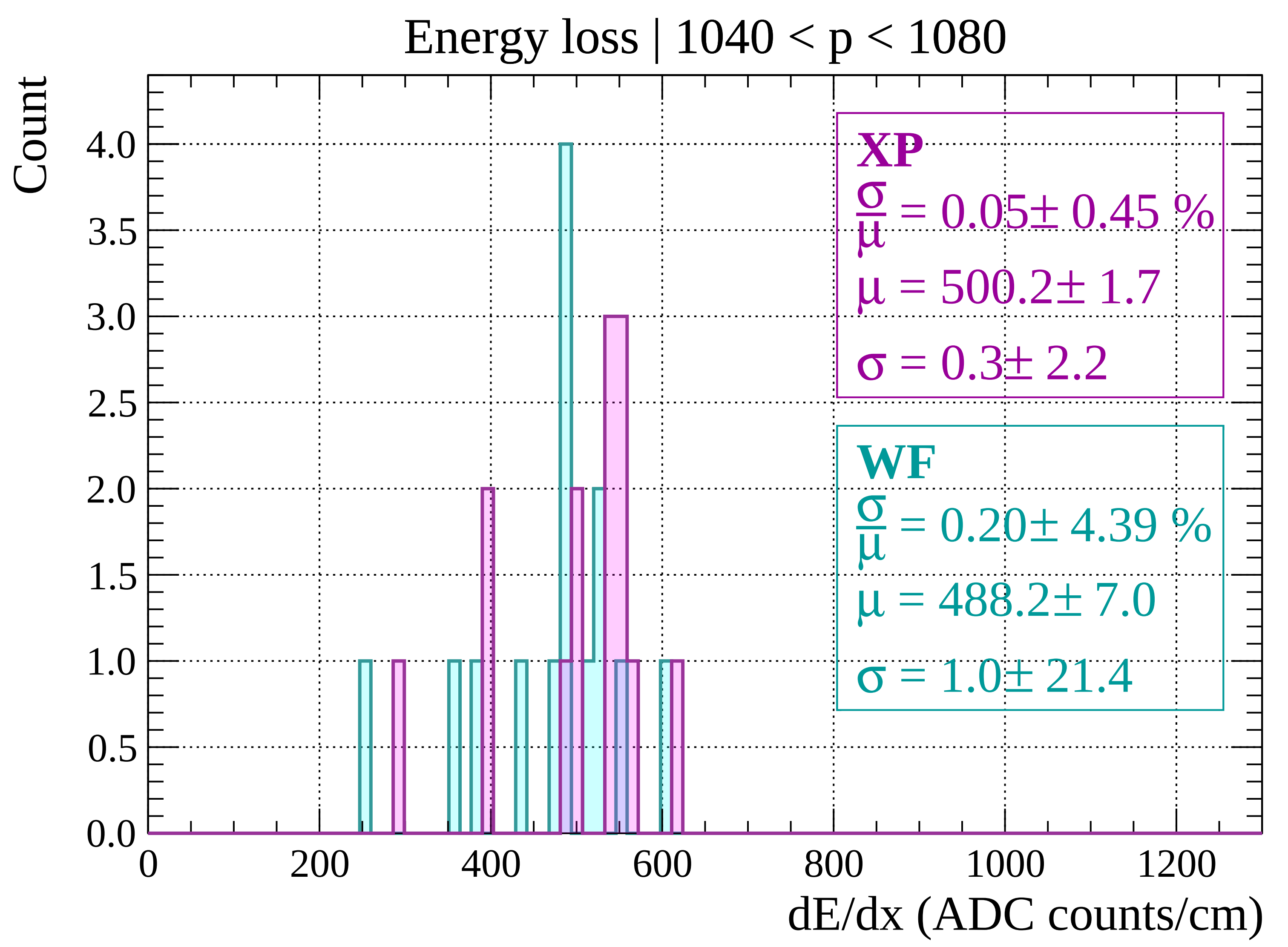
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.47 %
μ = 474.2± 1.7
σ = 0.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 0.22± 12.05 %
μ = 447.9± 21.3
σ = 1.0± 53.9



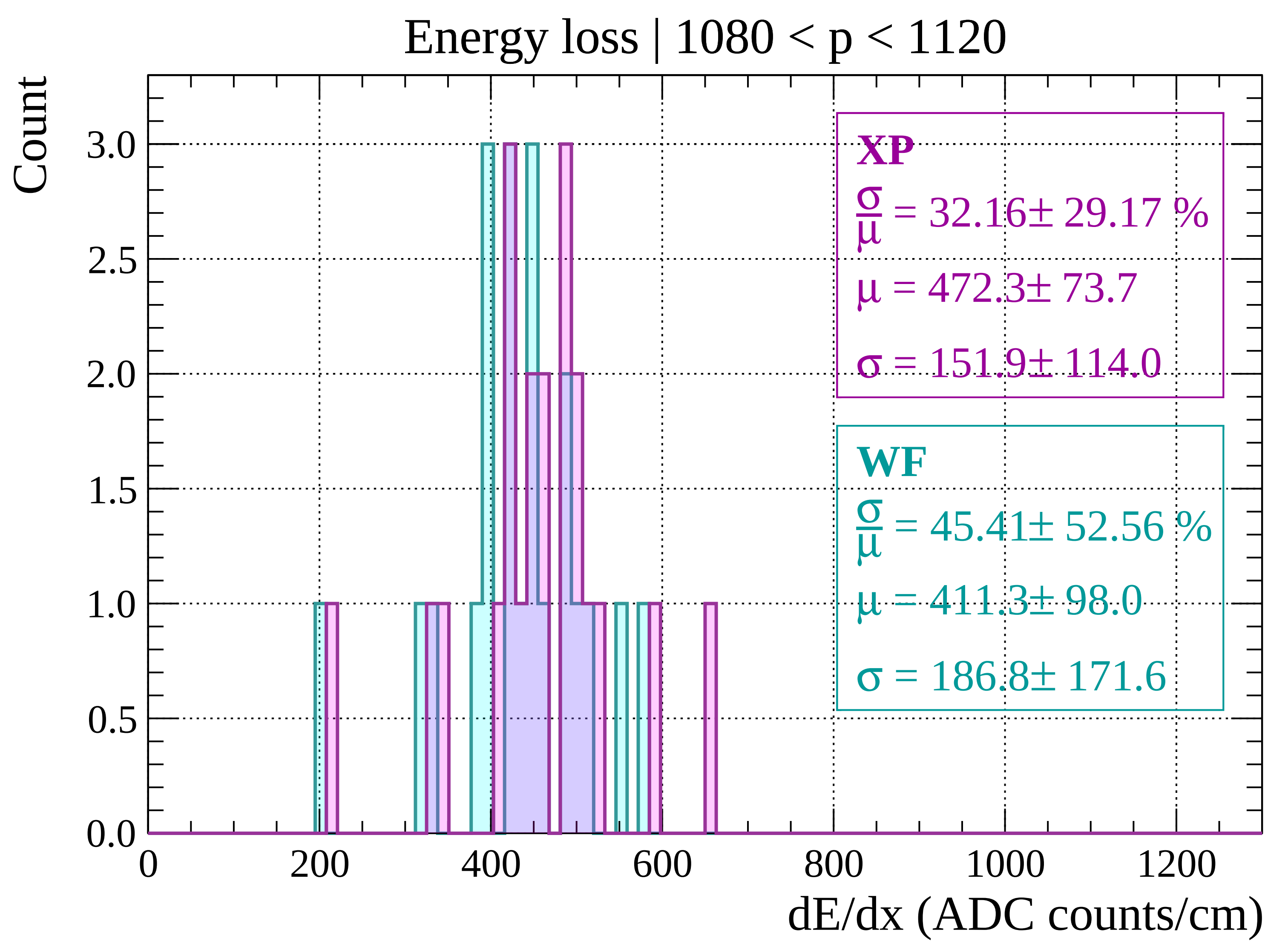
Energy loss | 960 < p < 1000
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 109.31± 314516.44 %
μ = 450.9± 1296657.6
σ = 492.9± 844.2
WF
$\frac{\sigma}{\mu}$ = 53.23± 72.66 %
μ = 485.3± 168.9
σ = 258.3± 262.8



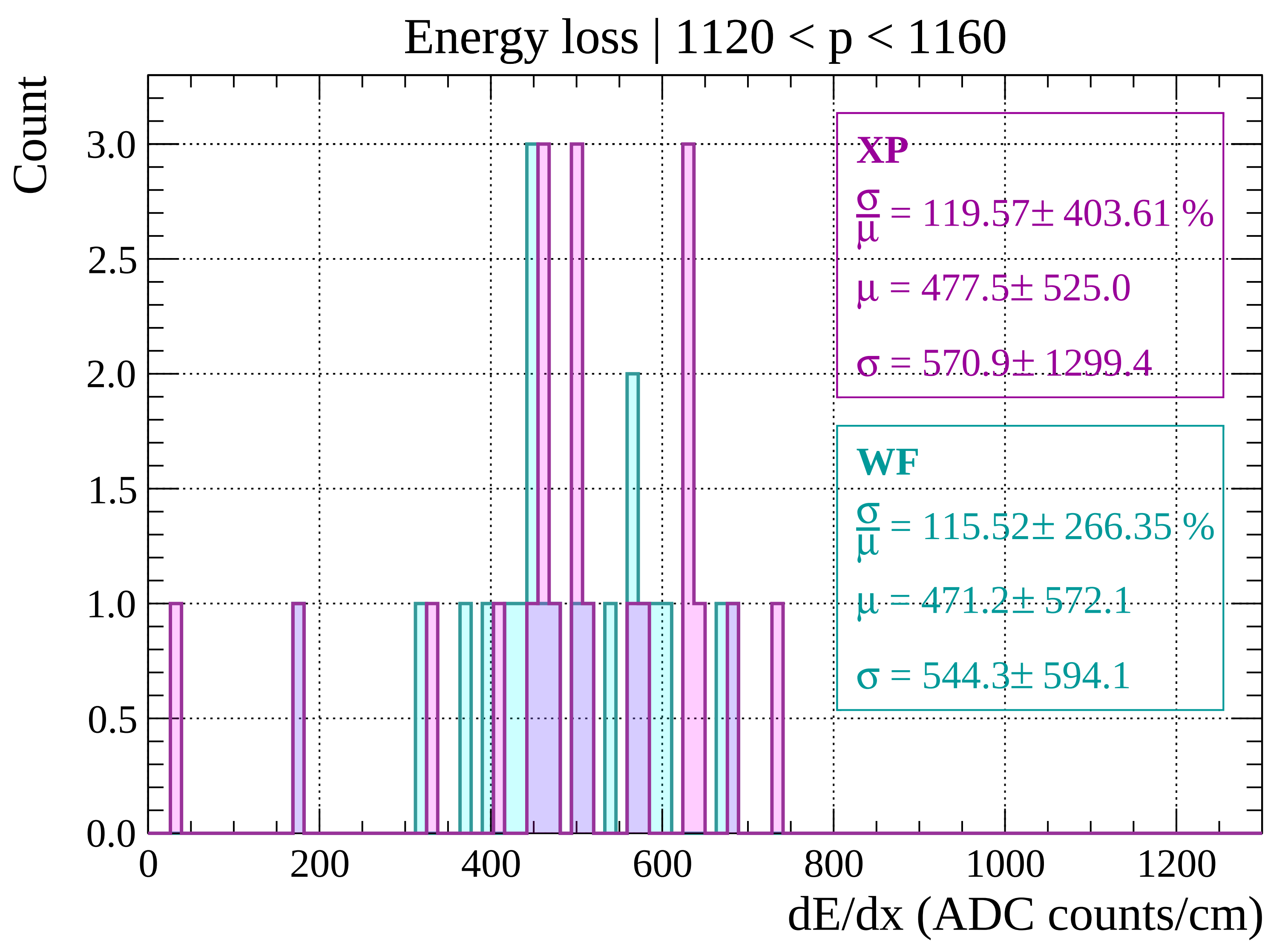
Energy loss | 1000 < p < 1040
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 45.42± 1993.15 %
μ = 399.5± 16357.7
σ = 181.4± 532.4
WF
σ/μ = 38.39± 26.90 %
μ = 429.4± 66.2
σ = 164.8± 90.1



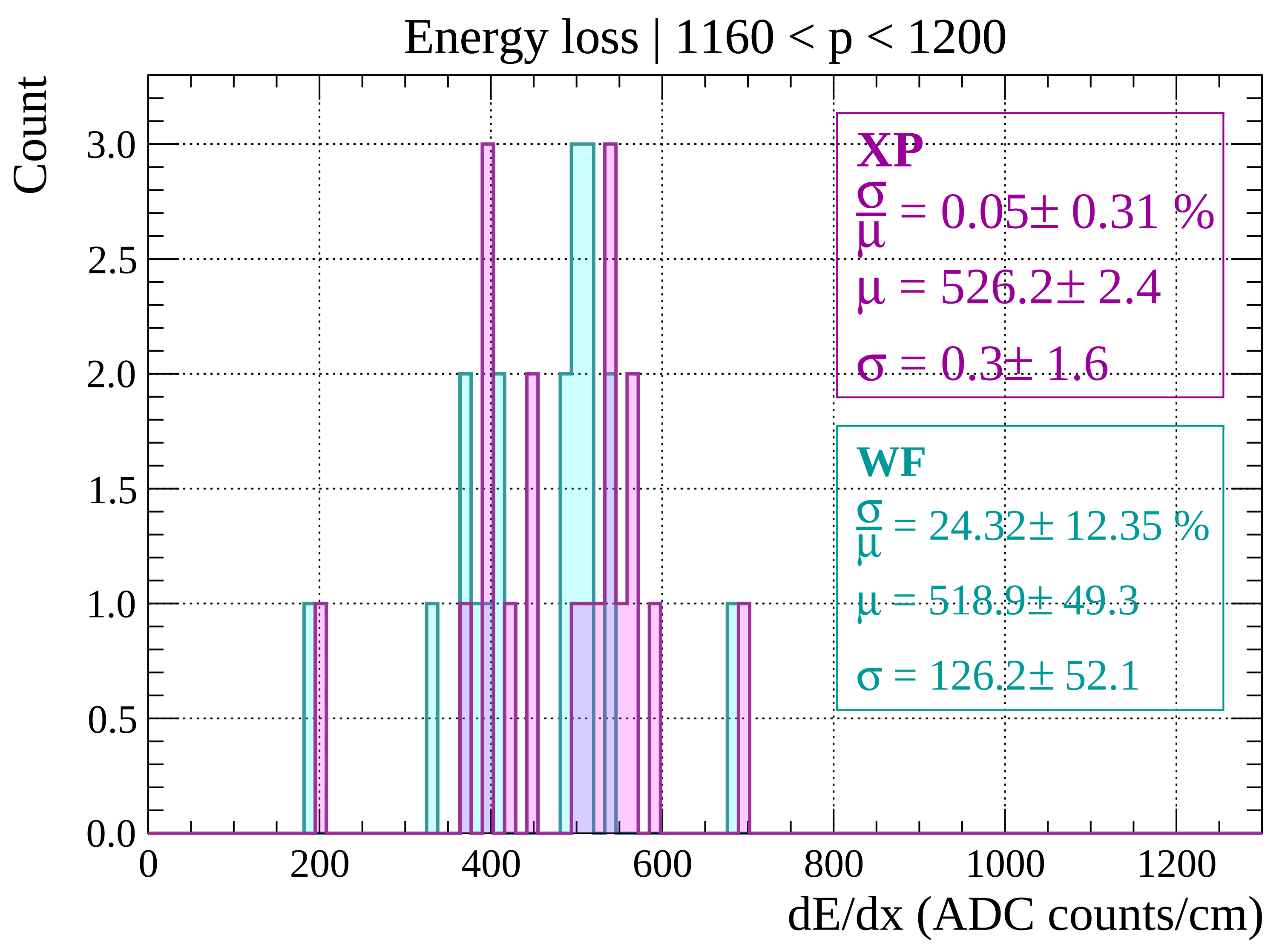
Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.45 %
μ = 500.2± 1.7
σ = 0.3± 2.2
WF
$\frac{\sigma}{\mu}$ = 0.20± 4.39 %
μ = 488.2± 7.0
σ = 1.0± 21.4



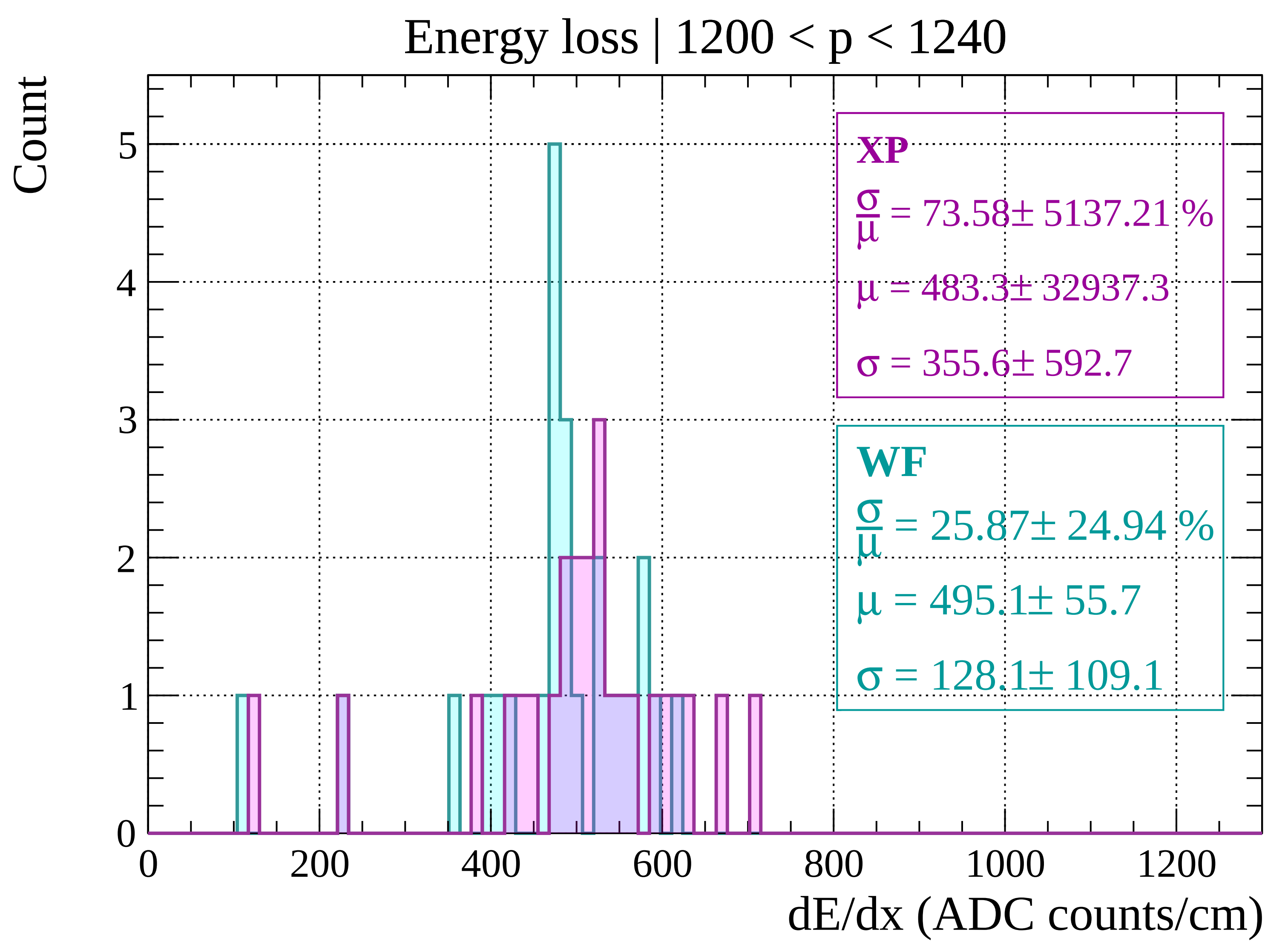
Energy loss | 1080 < p < 1120
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 32.16± 29.17 %
μ = 472.3± 73.7
σ = 151.9± 114.0
WF
$\frac{\sigma}{\mu}$ = 45.41± 52.56 %
μ = 411.3± 98.0
σ = 186.8± 171.6



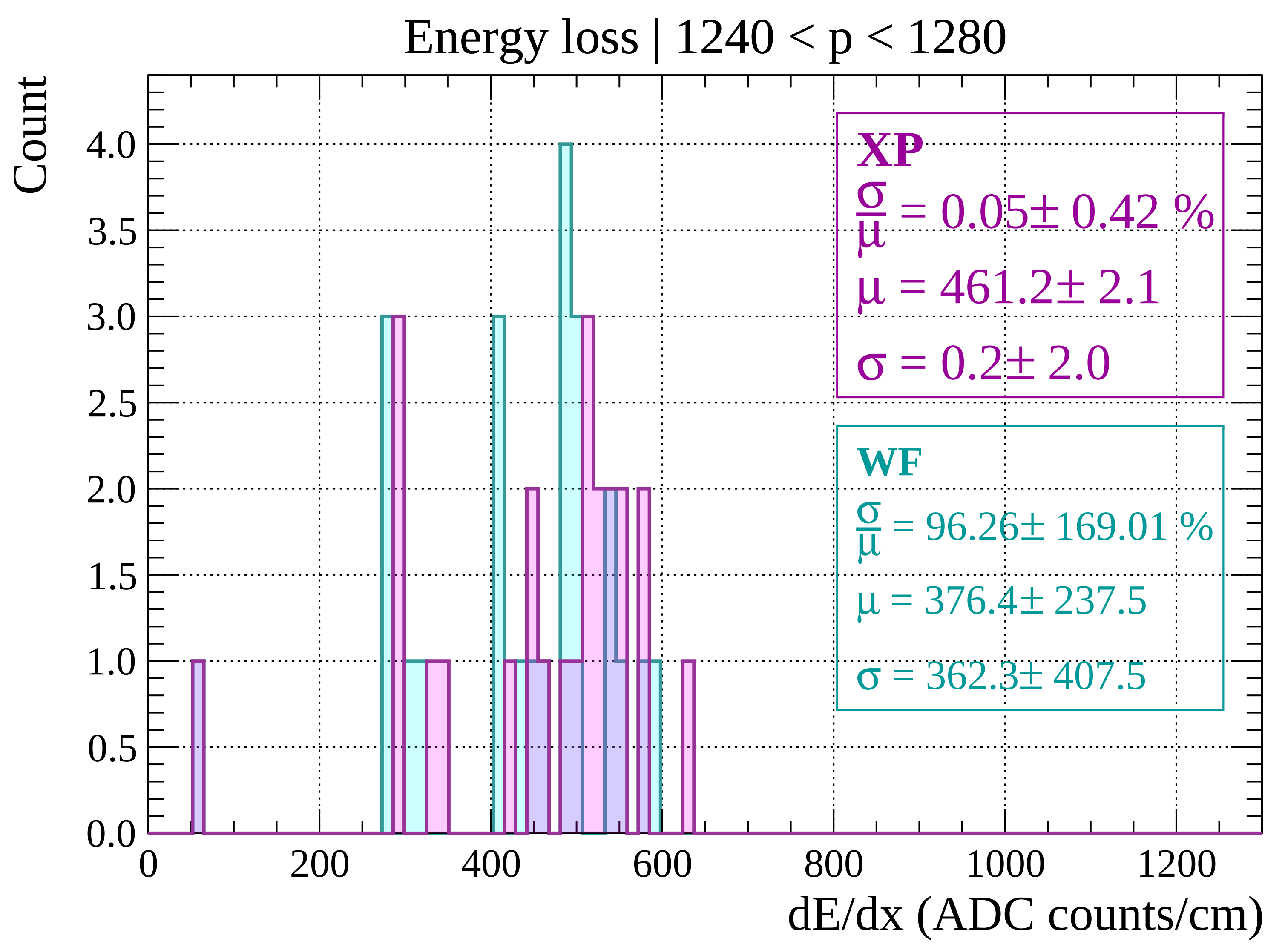
Energy loss | 1120 < p < 1160
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 119.57± 403.61 %
μ = 477.5± 525.0
σ = 570.9± 1299.4
WF
σ/μ = 115.52± 266.35 %
μ = 471.2± 572.1
σ = 544.3± 594.1



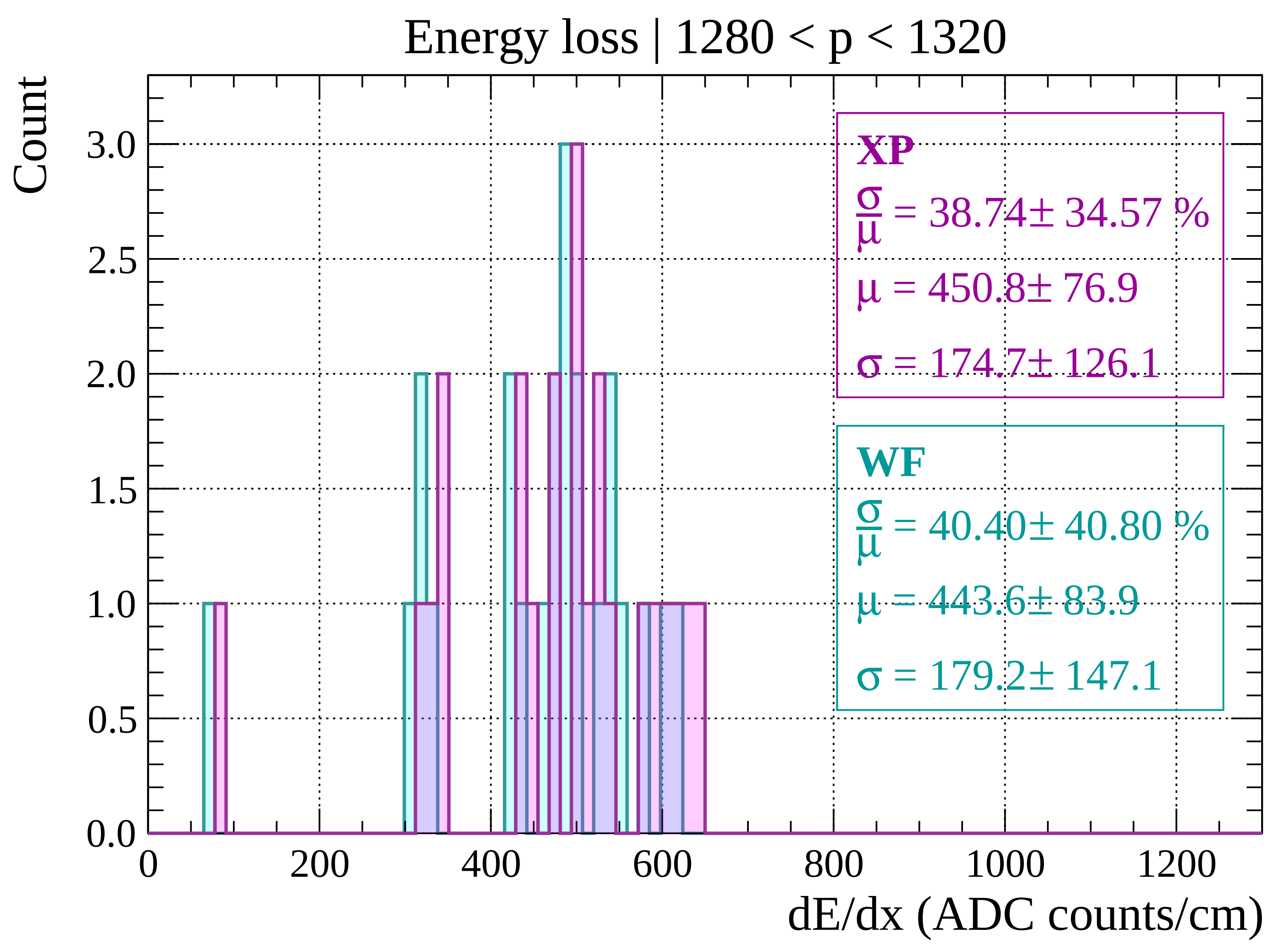
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.05± 0.31 %
μ = 526.2± 2.4
σ = 0.3± 1.6
WF
σ/μ = 24.32± 12.35 %
μ = 518.9± 49.3
σ = 126.2± 52.1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200



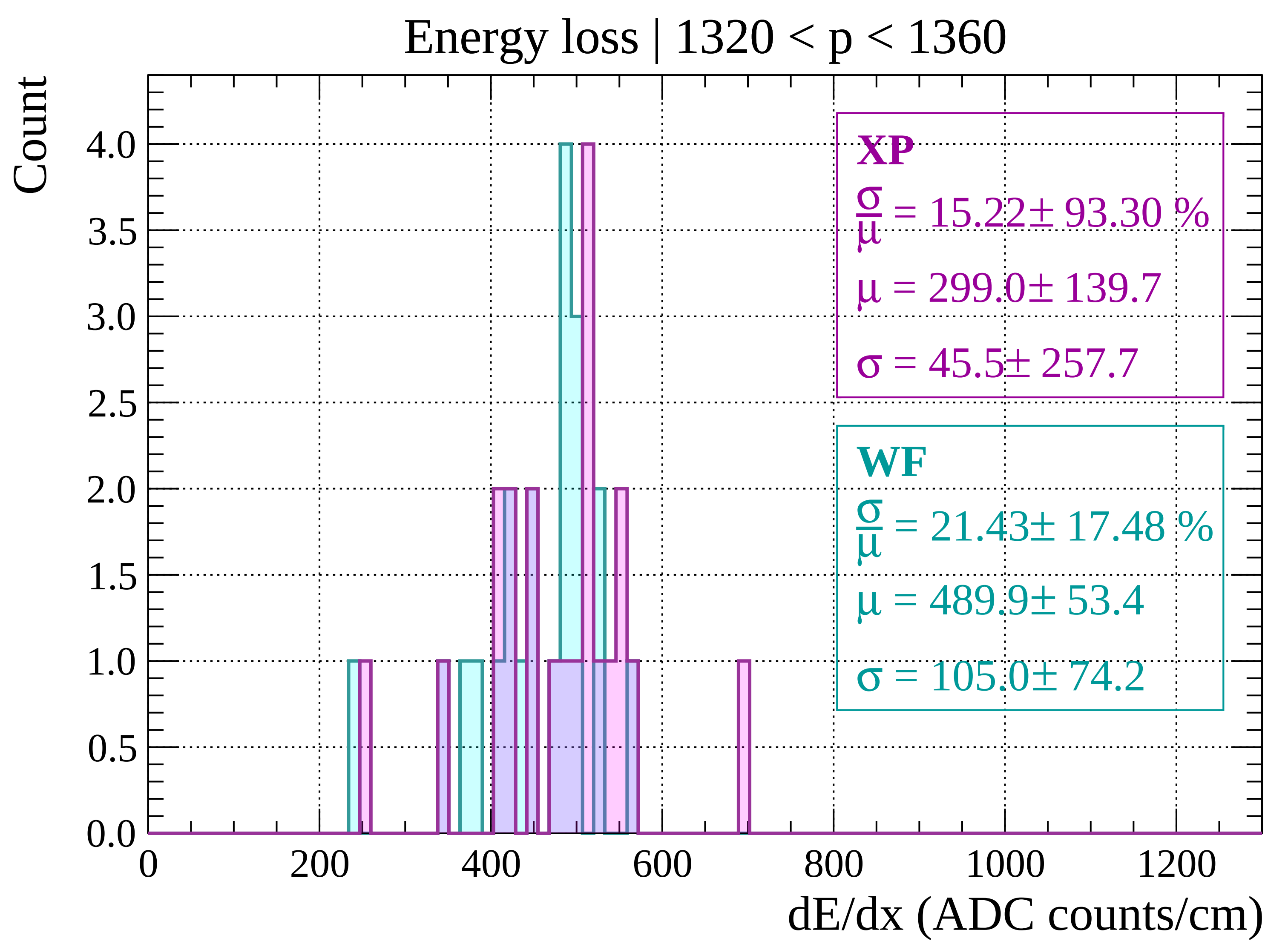
Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 73.58± 5137.21 %
μ = 483.3± 32937.3
σ = 355.6± 592.7
WF
σ/μ = 25.87± 24.94 %
μ = 495.1± 55.7
σ = 128.1± 109.1



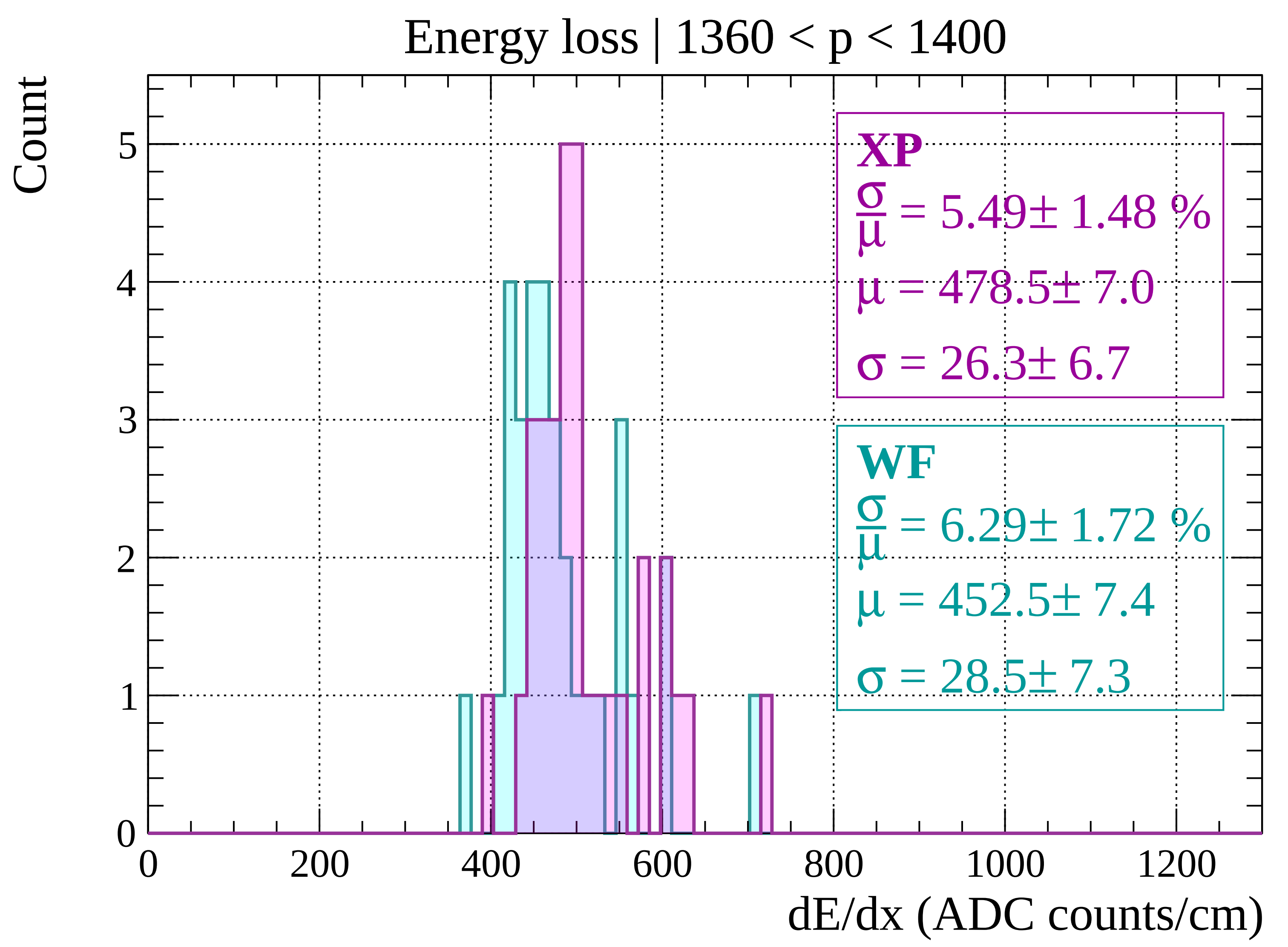
Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.42 %
μ = 461.2± 2.1
σ = 0.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 96.26± 169.01 %
μ = 376.4± 237.5
σ = 362.3± 407.5

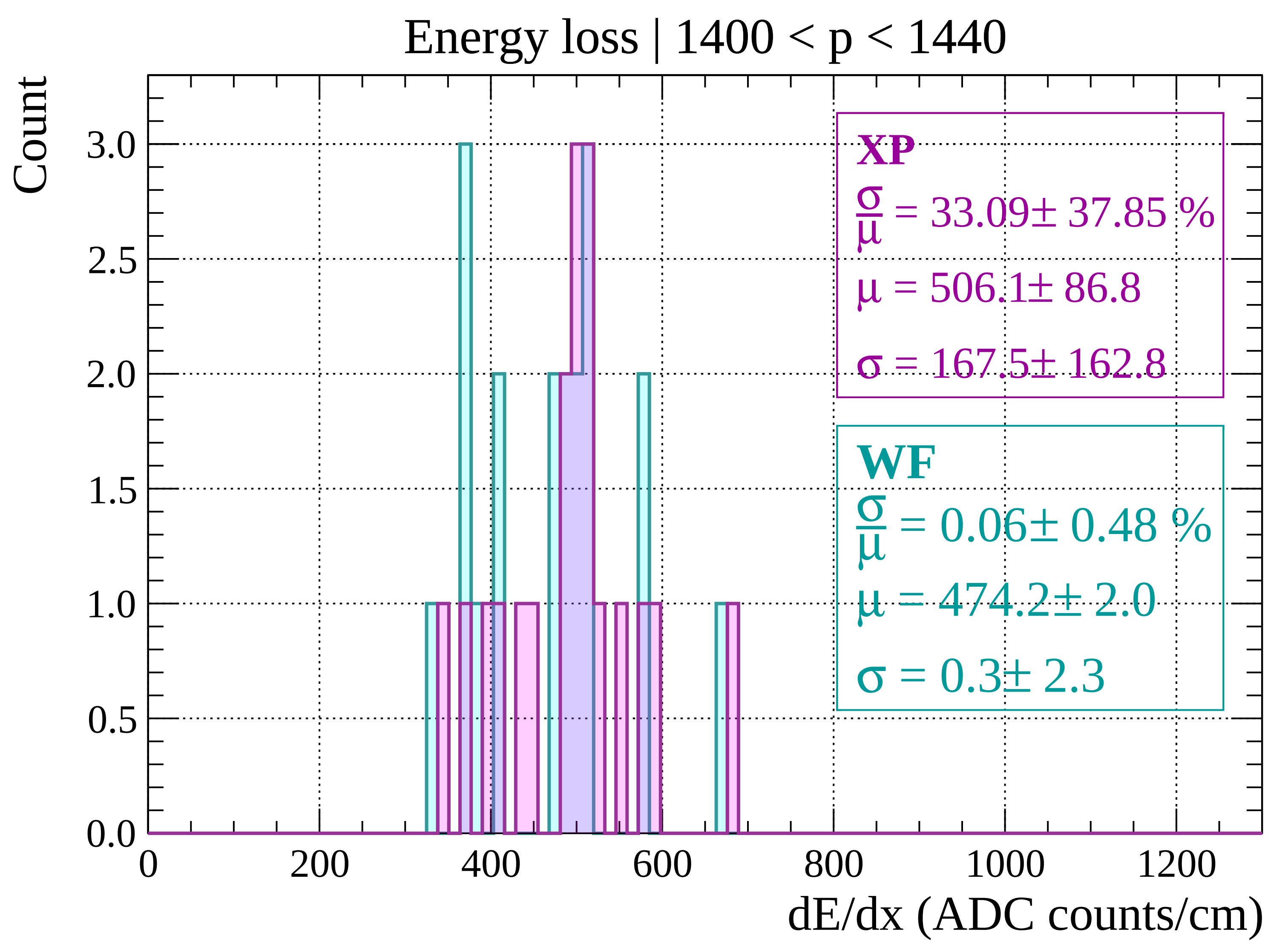


Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 38.74± 34.57 %
μ = 450.8± 76.9
σ = 174.7± 126.1
WF
$\frac{\sigma}{\mu}$ = 40.40± 40.80 %
μ = 443.6± 83.9
σ = 179.2± 147.1

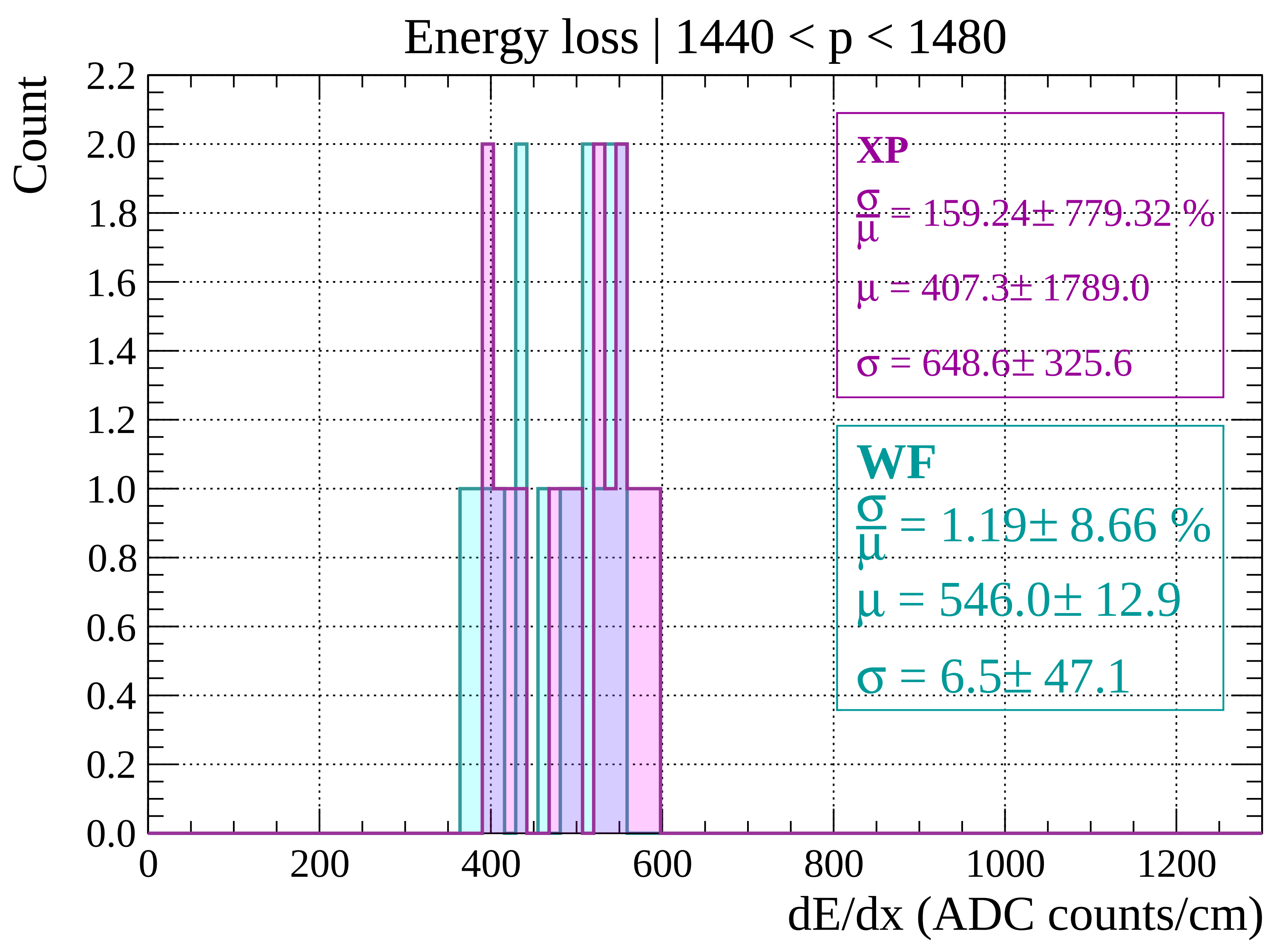


Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.22± 93.30 %
μ = 299.0± 139.7
σ = 45.5± 257.7
WF
$\frac{\sigma}{\mu}$ = 21.43± 17.48 %
μ = 489.9± 53.4
σ = 105.0± 74.2

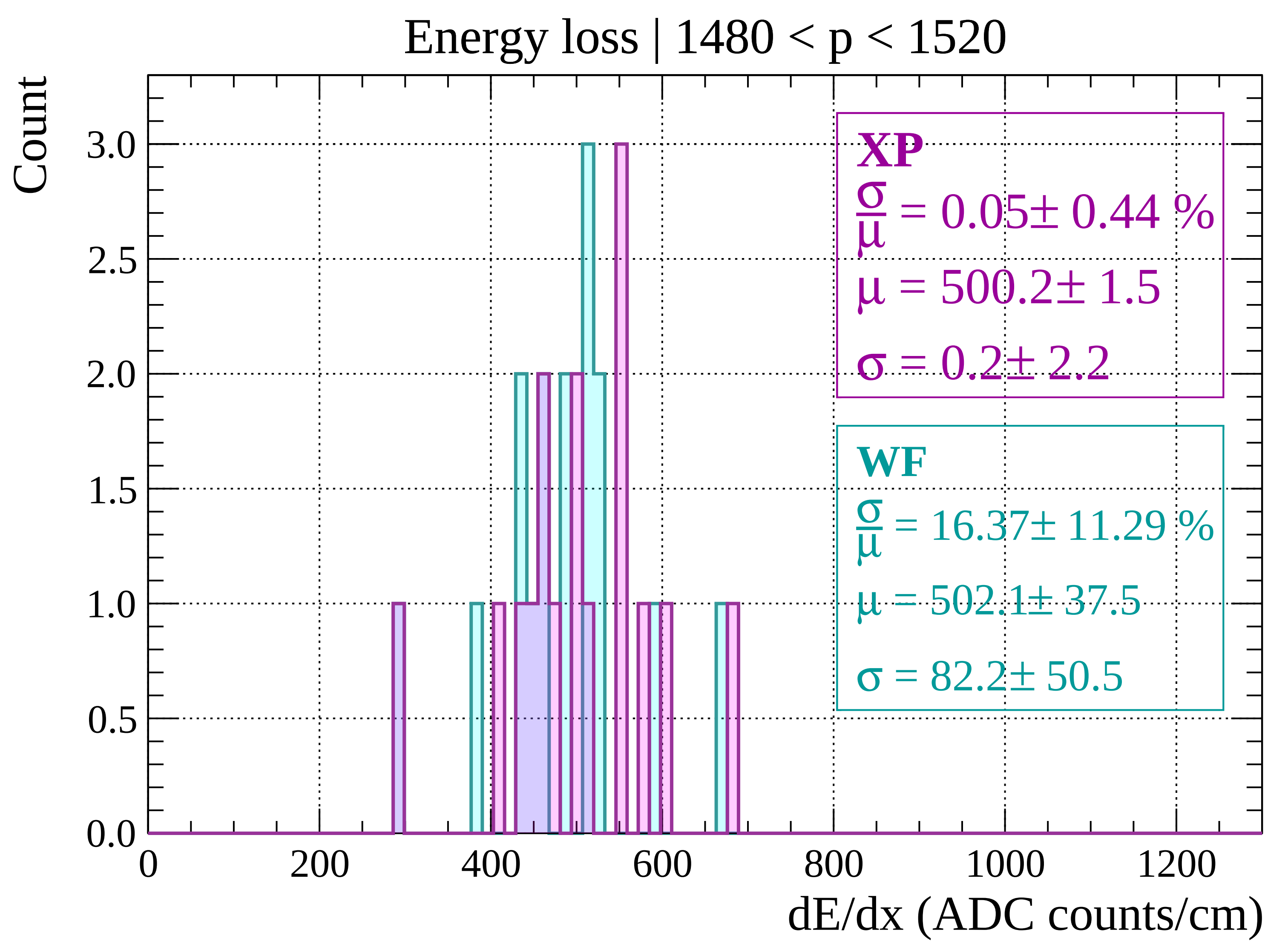

Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 5.49± 1.48 %
μ = 478.5± 7.0
σ = 26.3± 6.7
WF
$\frac{\sigma}{\mu}$ = 6.29± 1.72 %
μ = 452.5± 7.4
σ = 28.5± 7.3

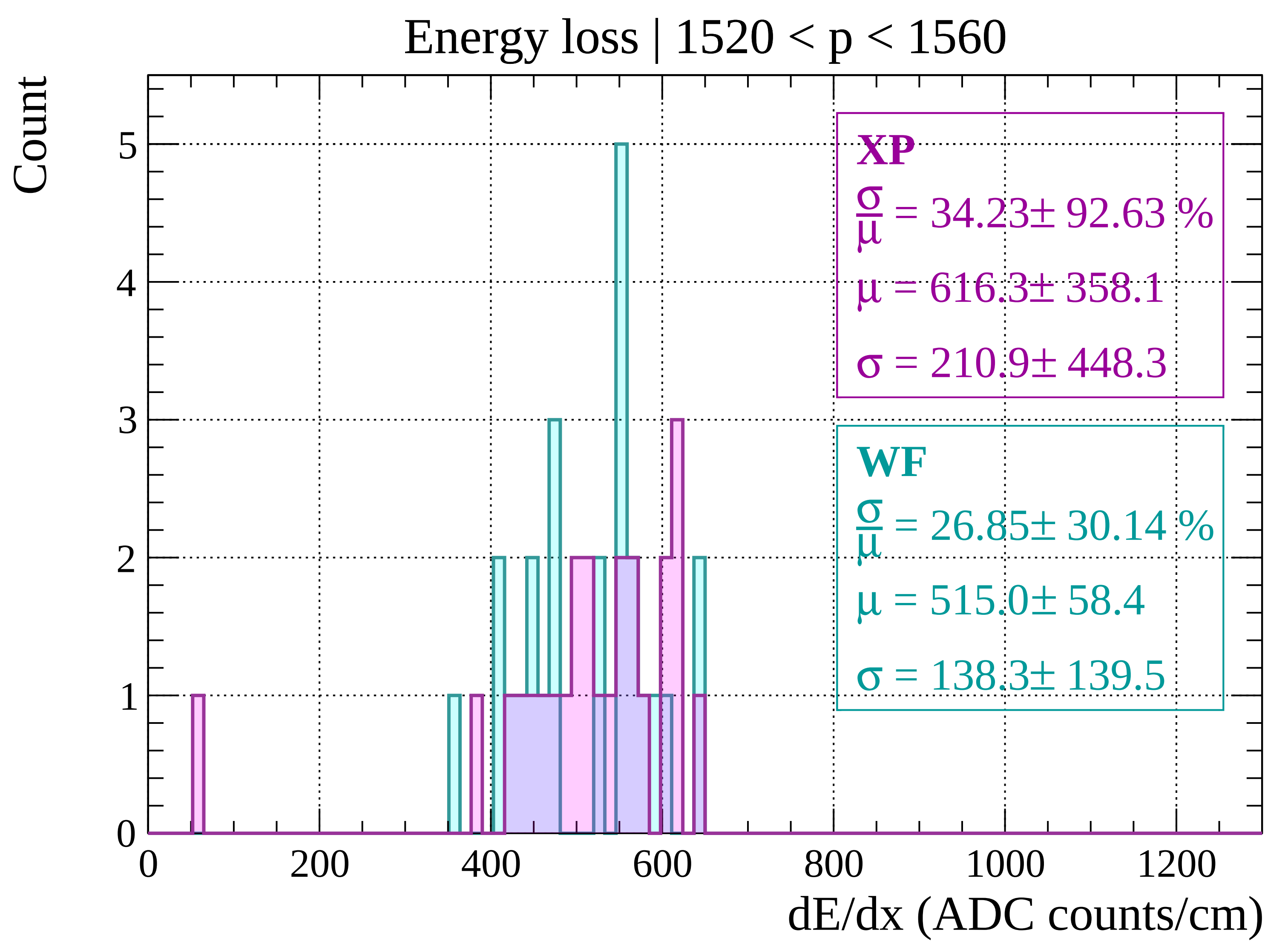


Energy loss | 1400 < p < 1440
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 33.09± 37.85 %
μ = 506.1± 86.8
σ = 167.5± 162.8
WF
σ/μ = 0.06± 0.48 %
μ = 474.2± 2.0
σ = 0.3± 2.3

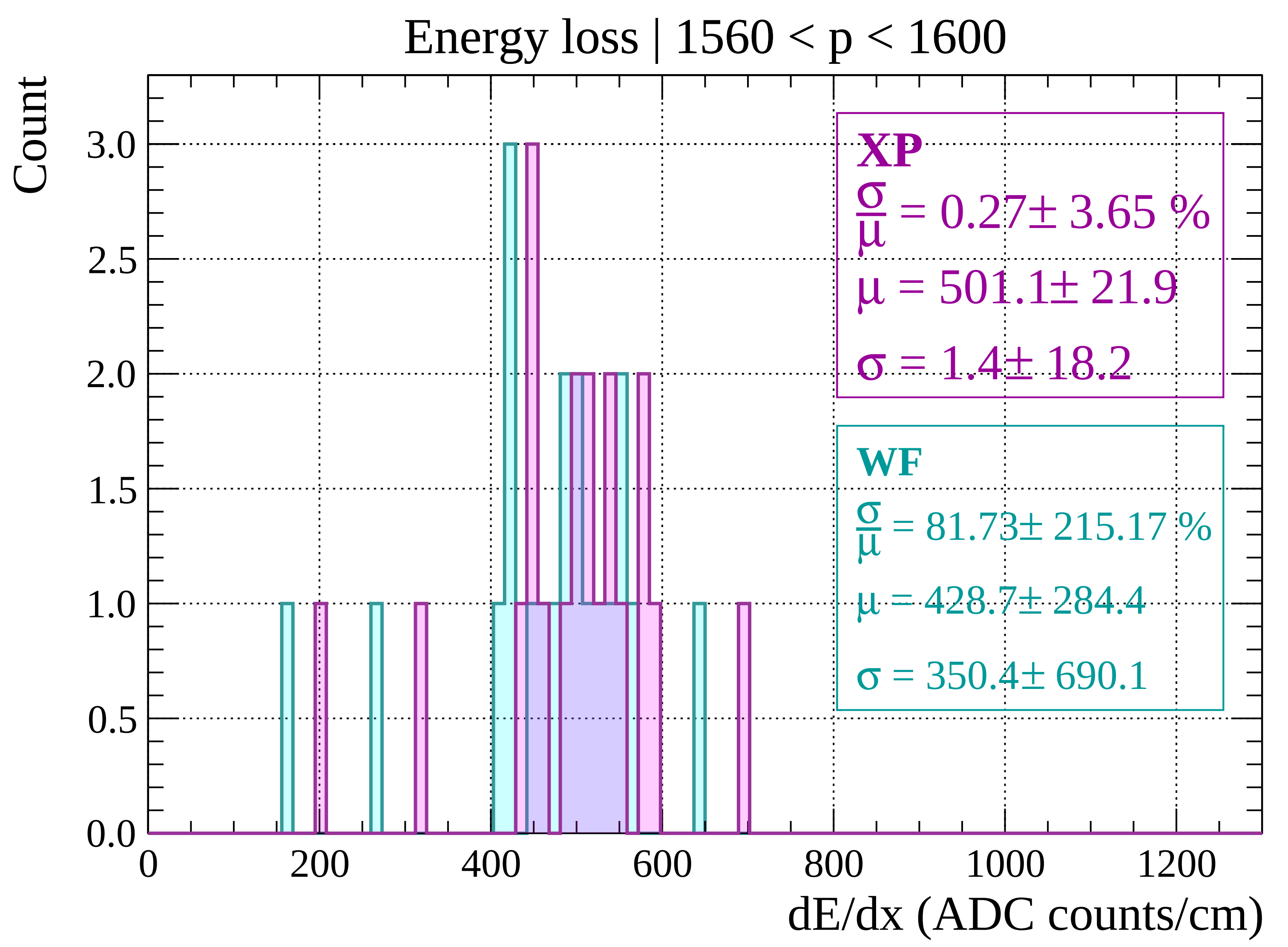


Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 159.24± 779.32 %
μ = 407.3± 1789.0
σ = 648.6± 325.6
WF
$\frac{\sigma}{\mu}$ = 1.19± 8.66 %
μ = 546.0± 12.9
σ = 6.5± 47.1

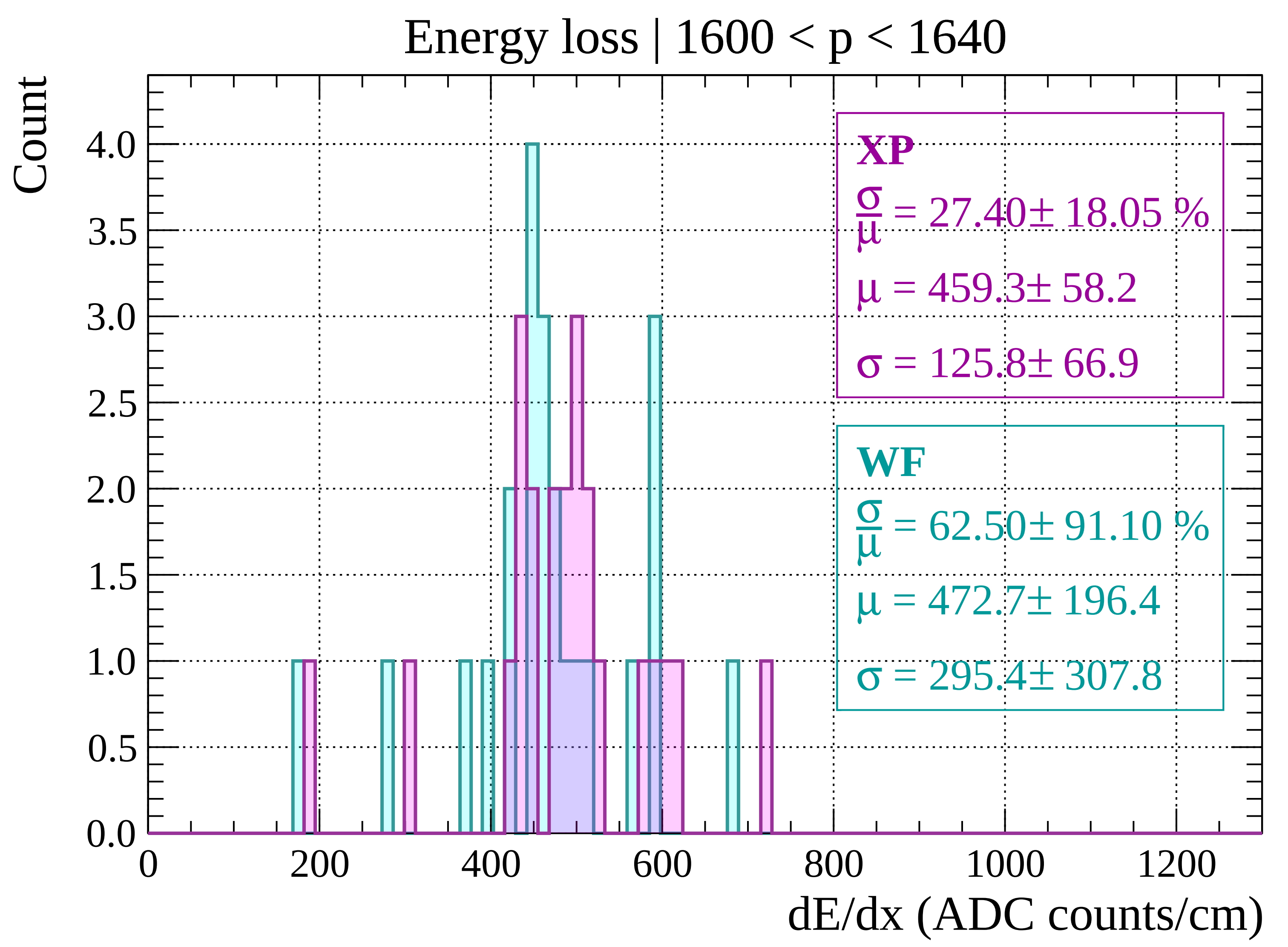

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.44 %
μ = 500.2± 1.5
σ = 0.2± 2.2
WF
$\frac{\sigma}{\mu}$ = 16.37± 11.29 %
μ = 502.1± 37.5
σ = 82.2± 50.5



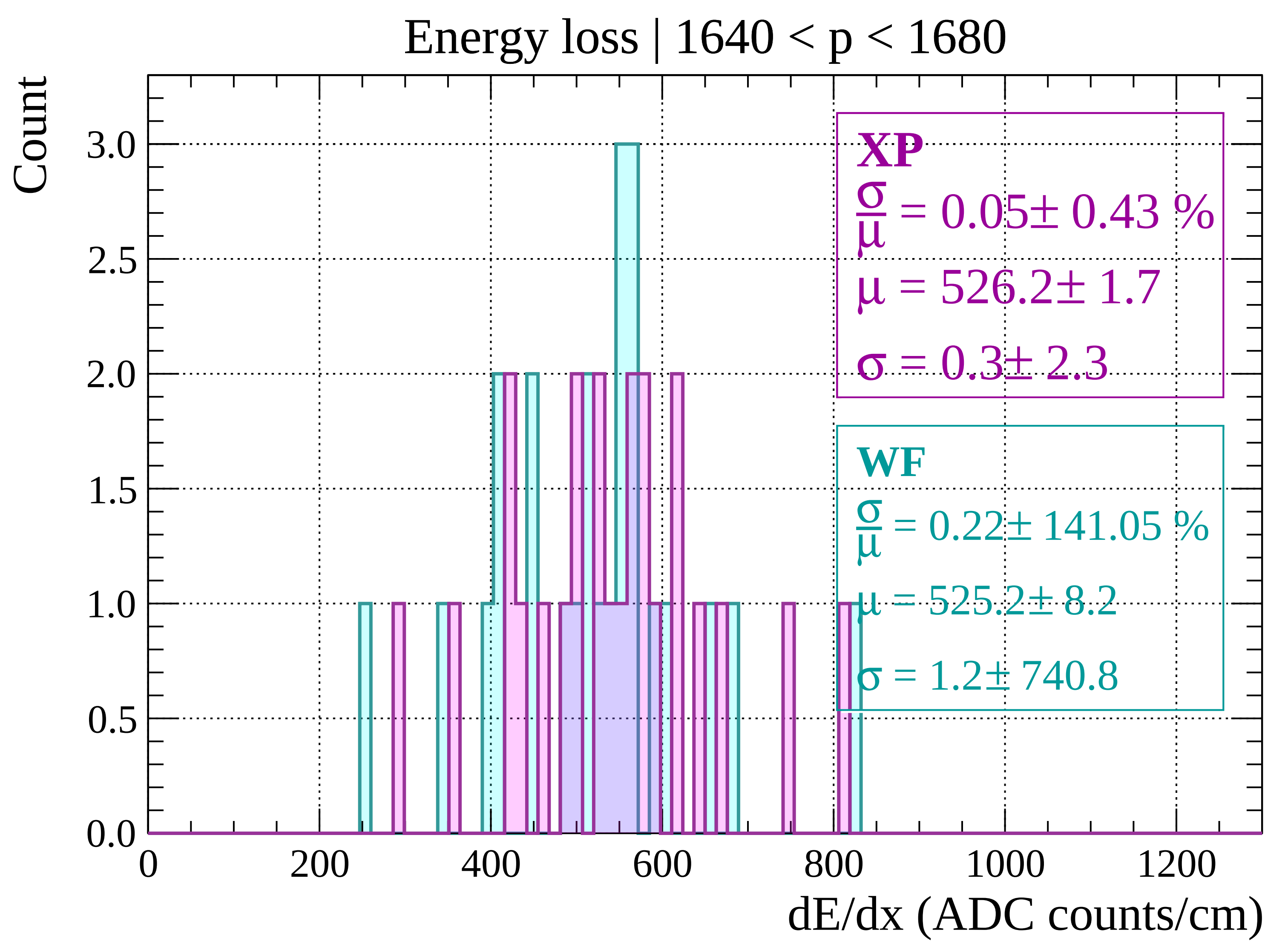
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 34.23± 92.63 %
μ = 616.3± 358.1
σ = 210.9± 448.3
WF
$\frac{\sigma}{\mu}$ = 26.85± 30.14 %
μ = 515.0± 58.4
σ = 138.3± 139.5



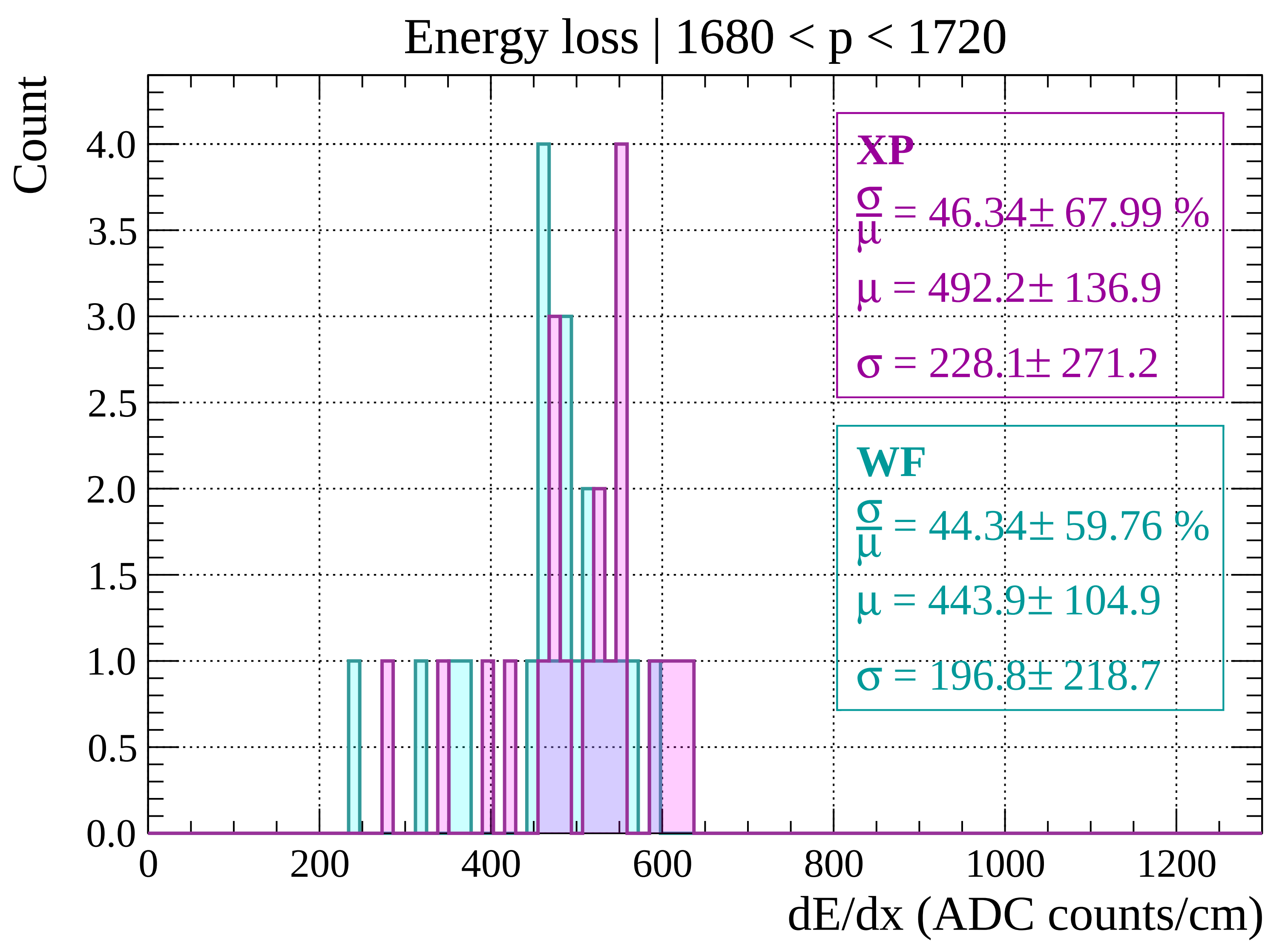
Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.5
1.0
1.5
2.0
2.5
3.0
XP
σ/μ = 0.27± 3.65 %
μ = 501.1± 21.9
σ = 1.4± 18.2
WF
σ/μ = 81.73± 215.17 %
μ = 428.7± 284.4
σ = 350.4± 690.1



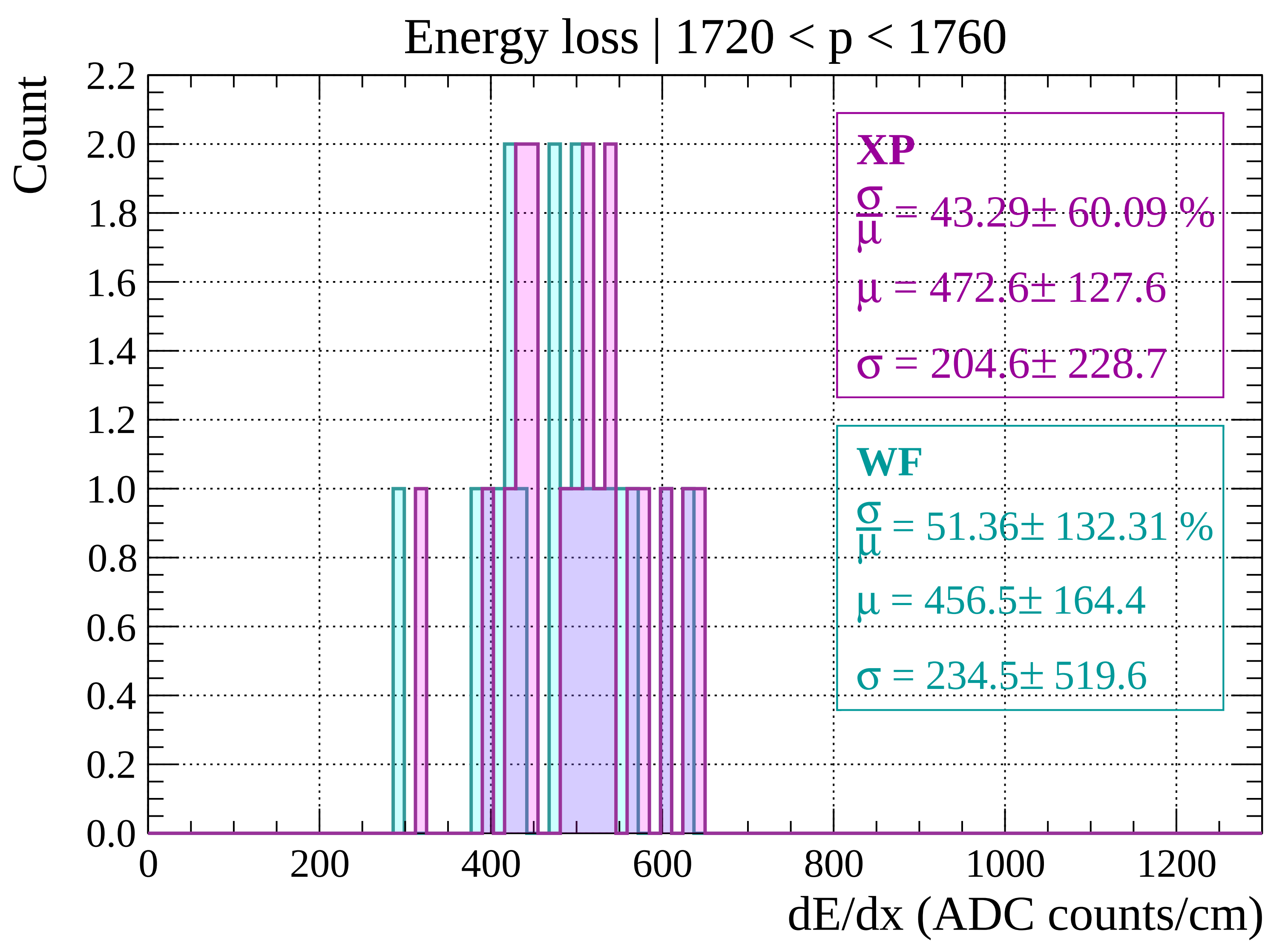
Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 27.40± 18.05 %
μ = 459.3± 58.2
σ = 125.8± 66.9
WF
$\frac{\sigma}{\mu}$ = 62.50± 91.10 %
μ = 472.7± 196.4
σ = 295.4± 307.8

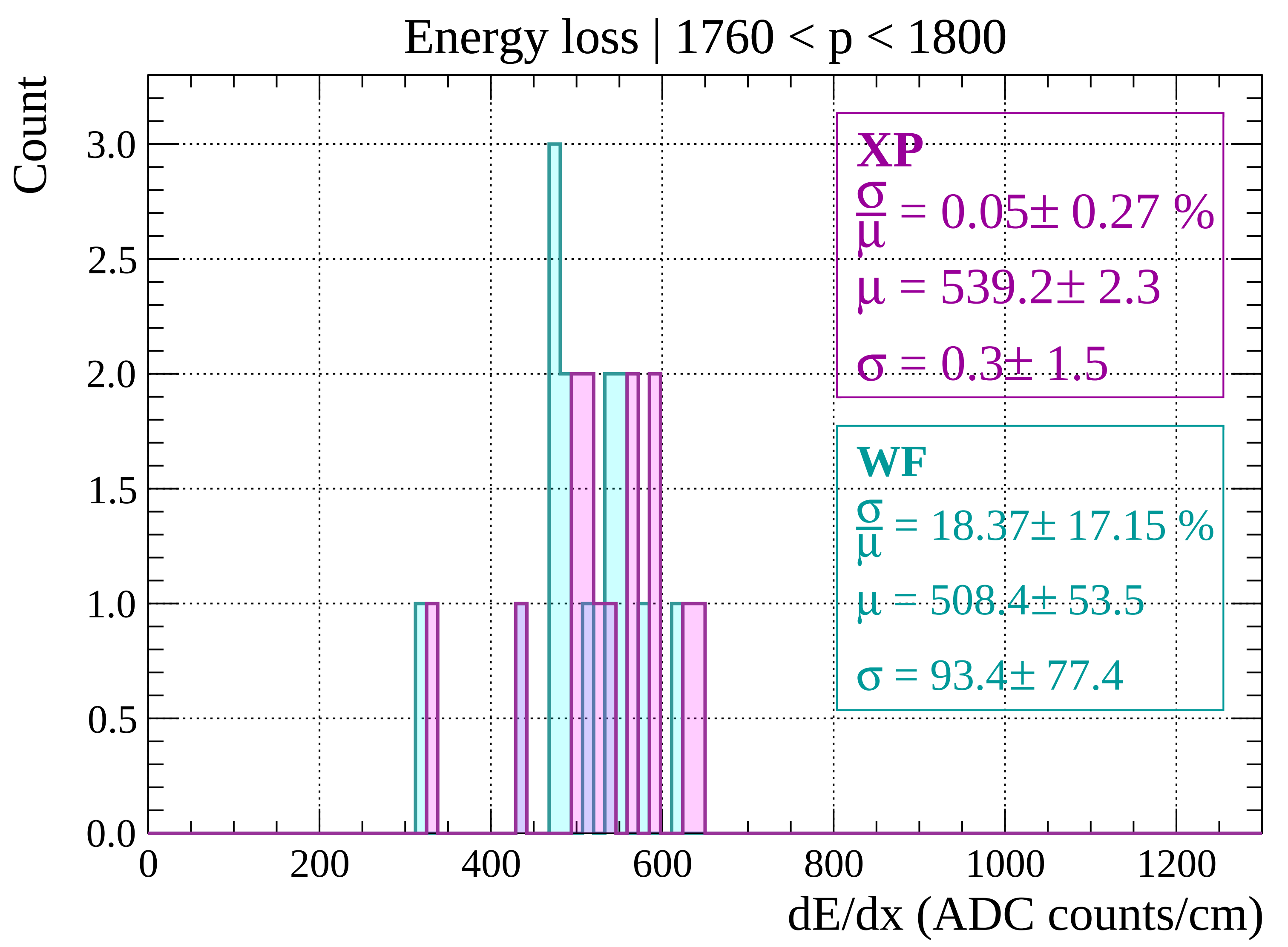


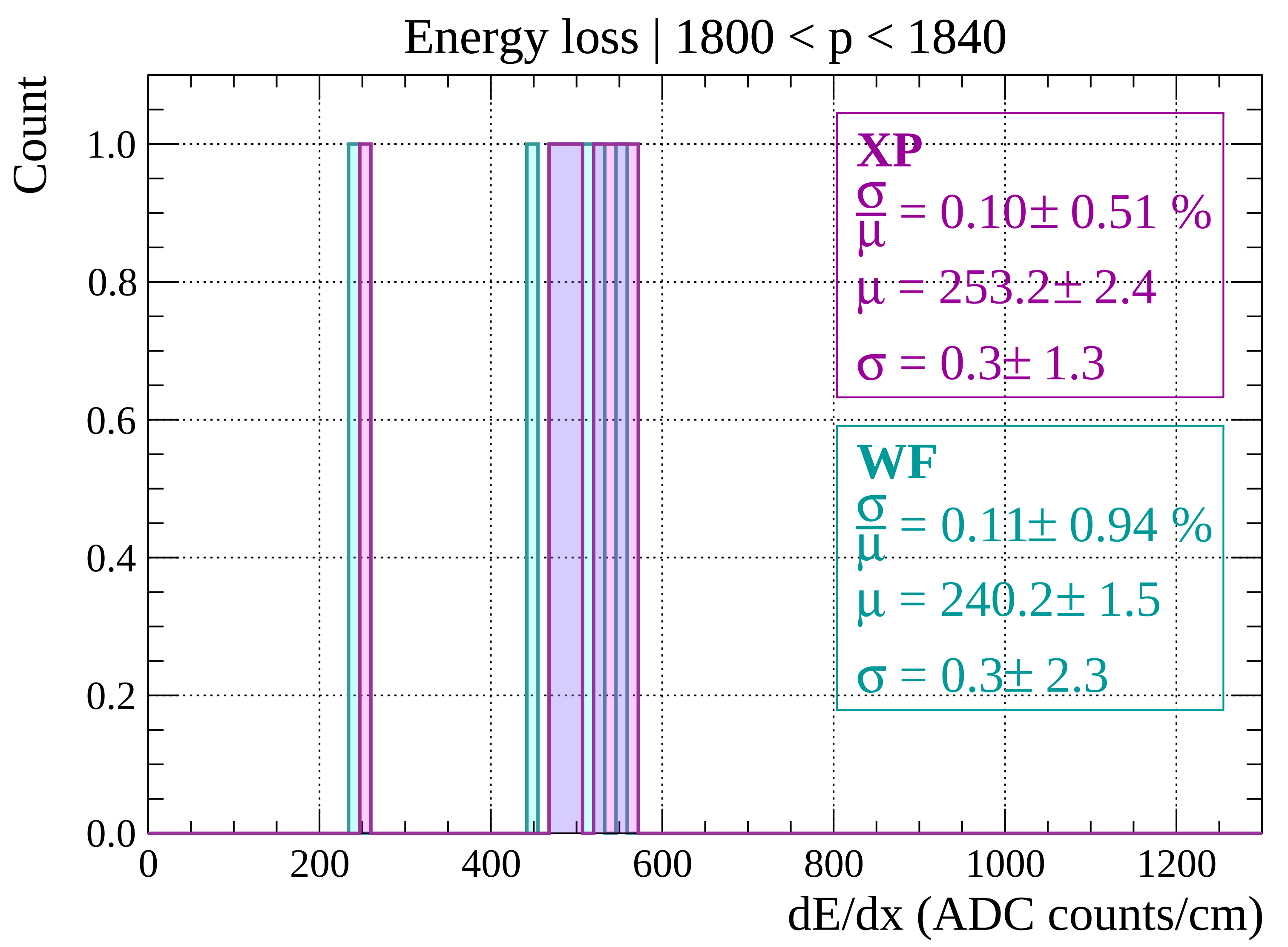
Energy loss | 1640 < p < 1680
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.05± 0.43 %
μ = 526.2± 1.7
σ = 0.3± 2.3
WF
σ/μ = 0.22± 141.05 %
μ = 525.2± 8.2
σ = 1.2± 740.8



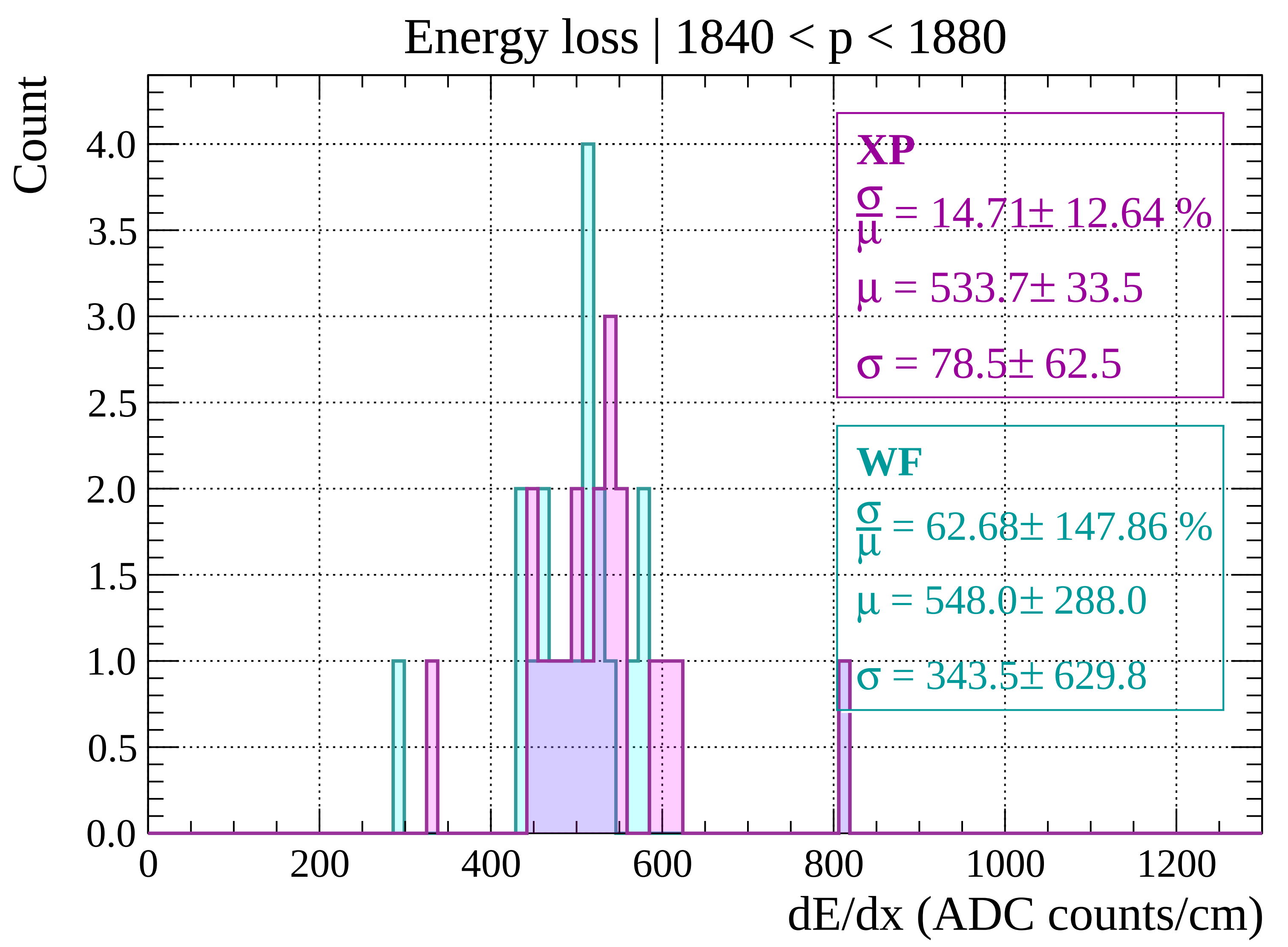
Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 46.34± 67.99 %
μ = 492.2± 136.9
σ = 228.1± 271.2
WF
σ/μ = 44.34± 59.76 %
μ = 443.9± 104.9
σ = 196.8± 218.7
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200

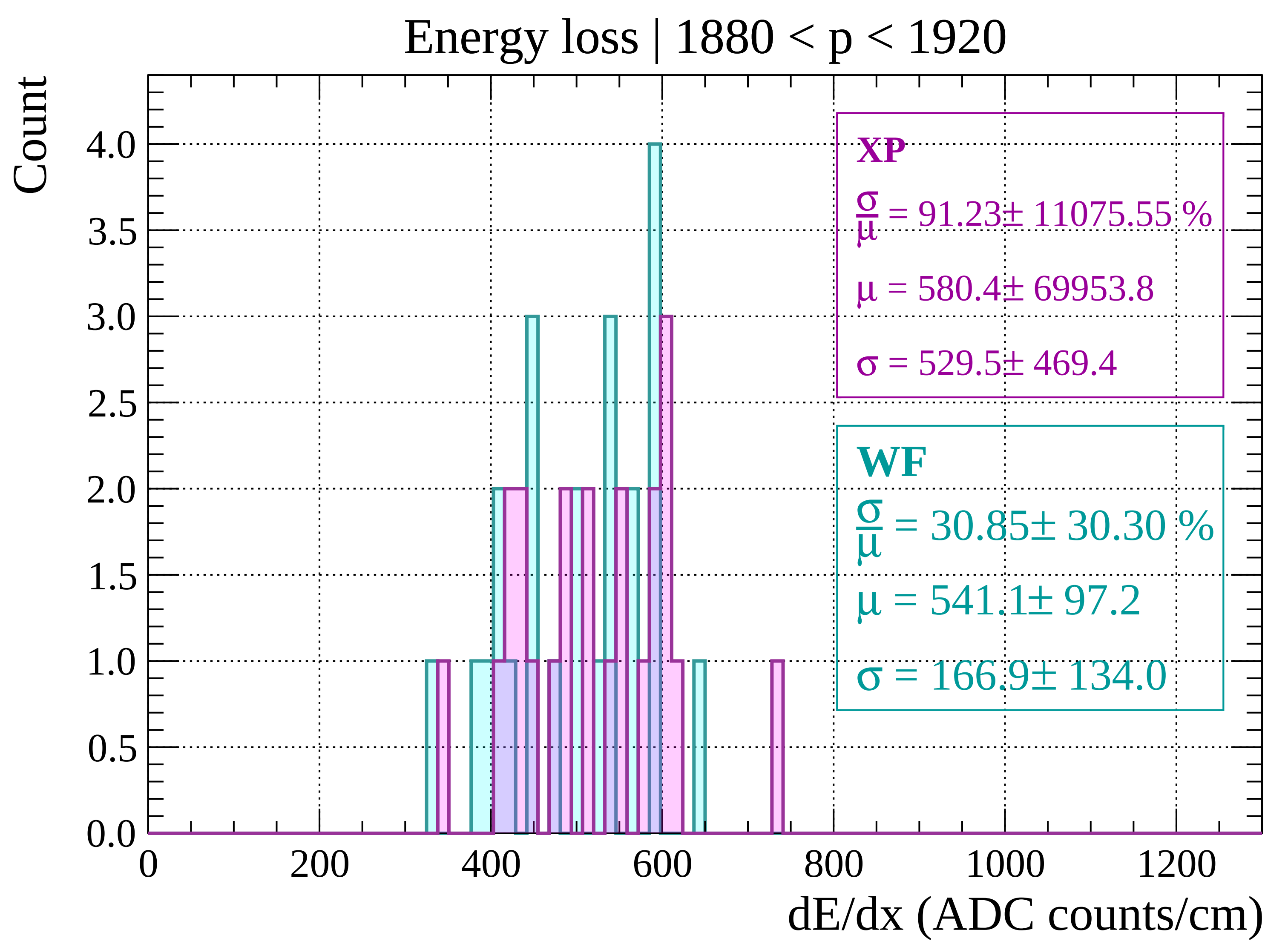

Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 43.29± 60.09 %
μ = 472.6± 127.6
σ = 204.6± 228.7
WF
σ/μ = 51.36± 132.31 %
μ = 456.5± 164.4
σ = 234.5± 519.6


Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 0.05± 0.27 %
μ = 539.2± 2.3
σ = 0.3± 1.5
WF
σ/μ = 18.37± 17.15 %
μ = 508.4± 53.5
σ = 93.4± 77.4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200

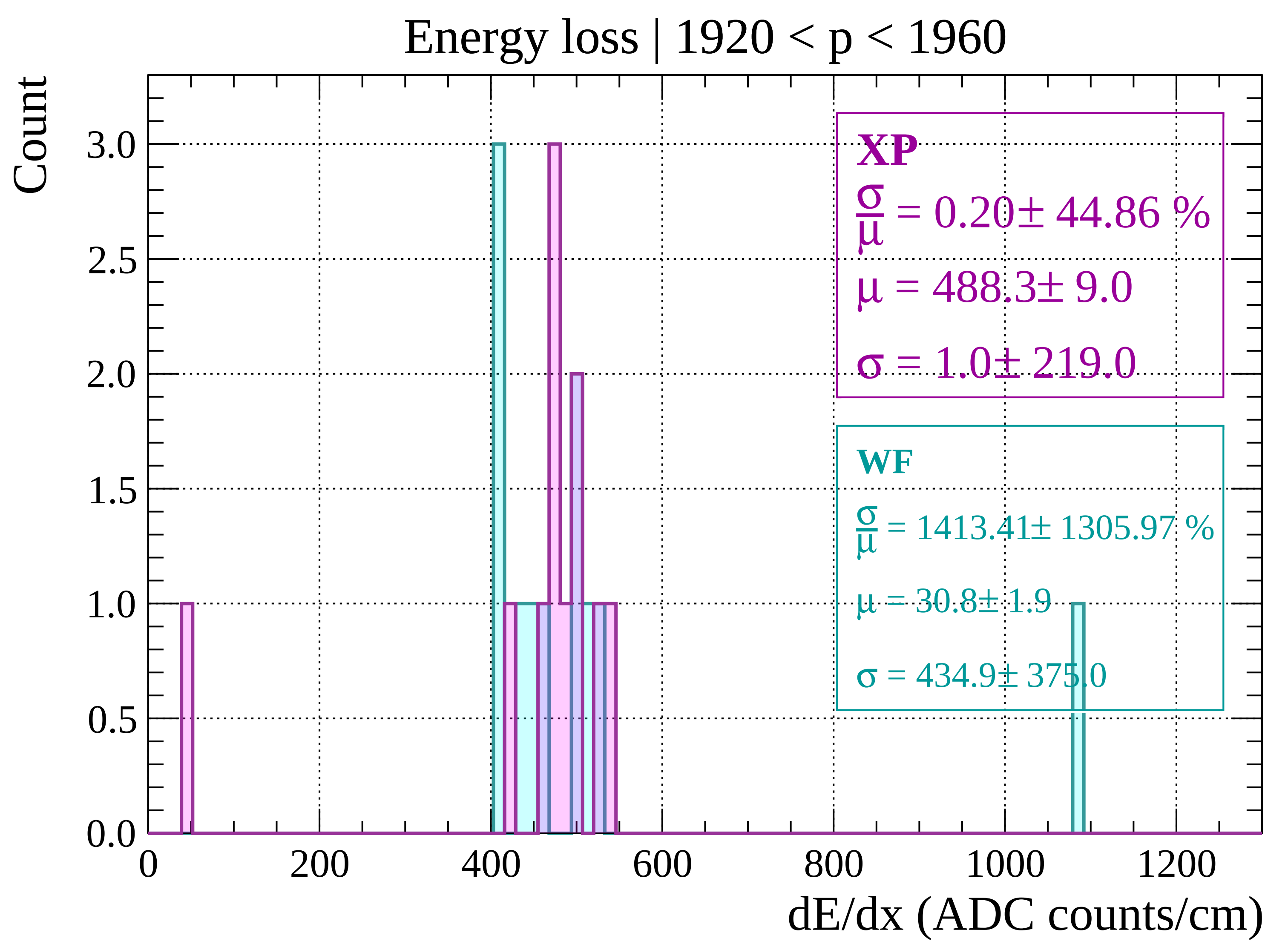


Energy loss | 1800 < p < 1840
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 0.10 \pm 0.51$ %
$\mu = 253.2 \pm 2.4$
$\sigma = 0.3 \pm 1.3$
WF
$\frac{\sigma}{\mu} = 0.11 \pm 0.94$ %
$\mu = 240.2 \pm 1.5$
$\sigma = 0.3 \pm 2.3$

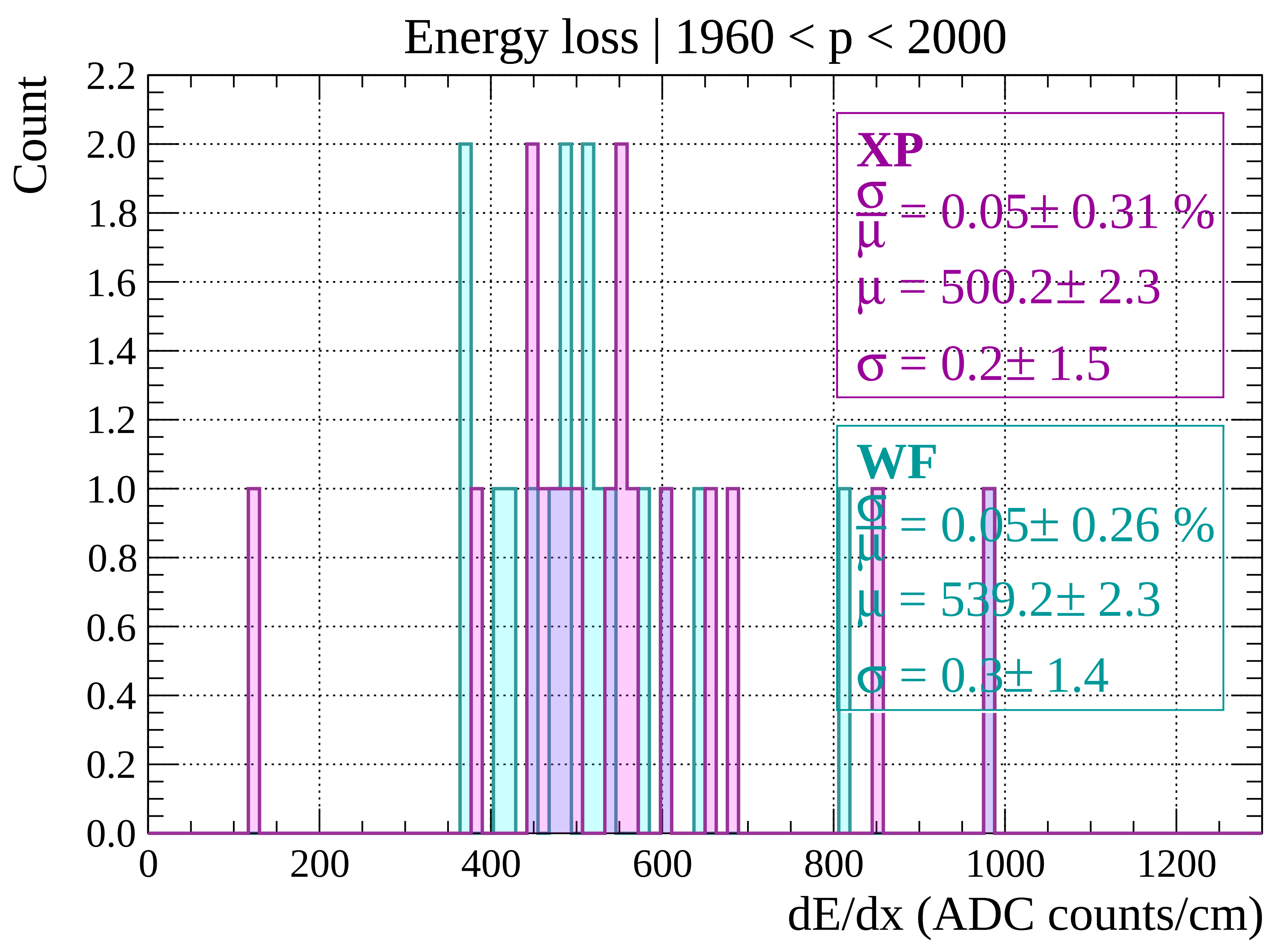

Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.71± 12.64 %
μ = 533.7± 33.5
σ = 78.5± 62.5
WF
$\frac{\sigma}{\mu}$ = 62.68± 147.86 %
μ = 548.0± 288.0
σ = 343.5± 629.8



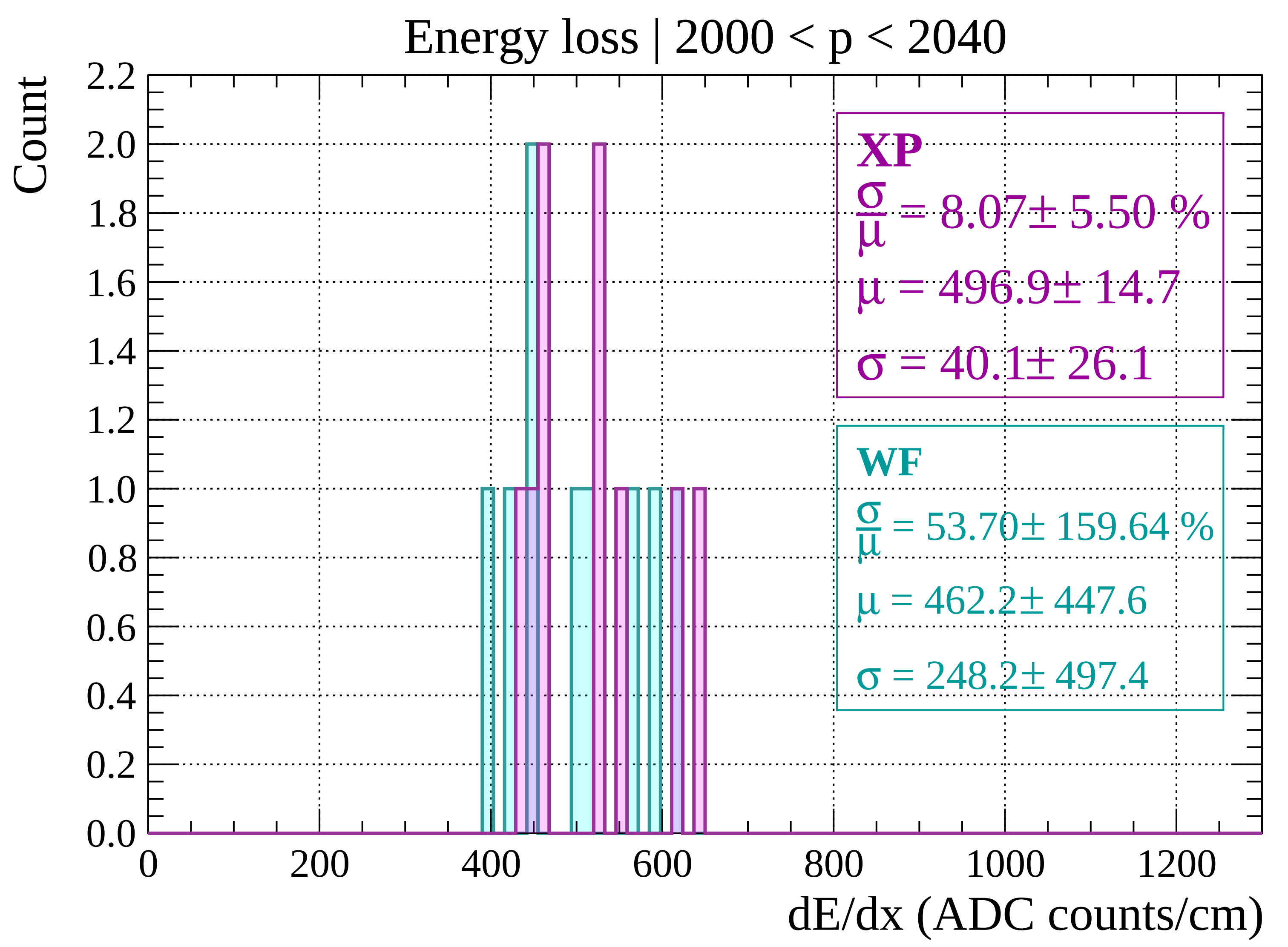
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
XP
σ/μ = 91.23± 11075.55 %
μ = 580.4± 69953.8
σ = 529.5± 469.4
WF
σ/μ = 30.85± 30.30 %
μ = 541.1± 97.2
σ = 166.9± 134.0



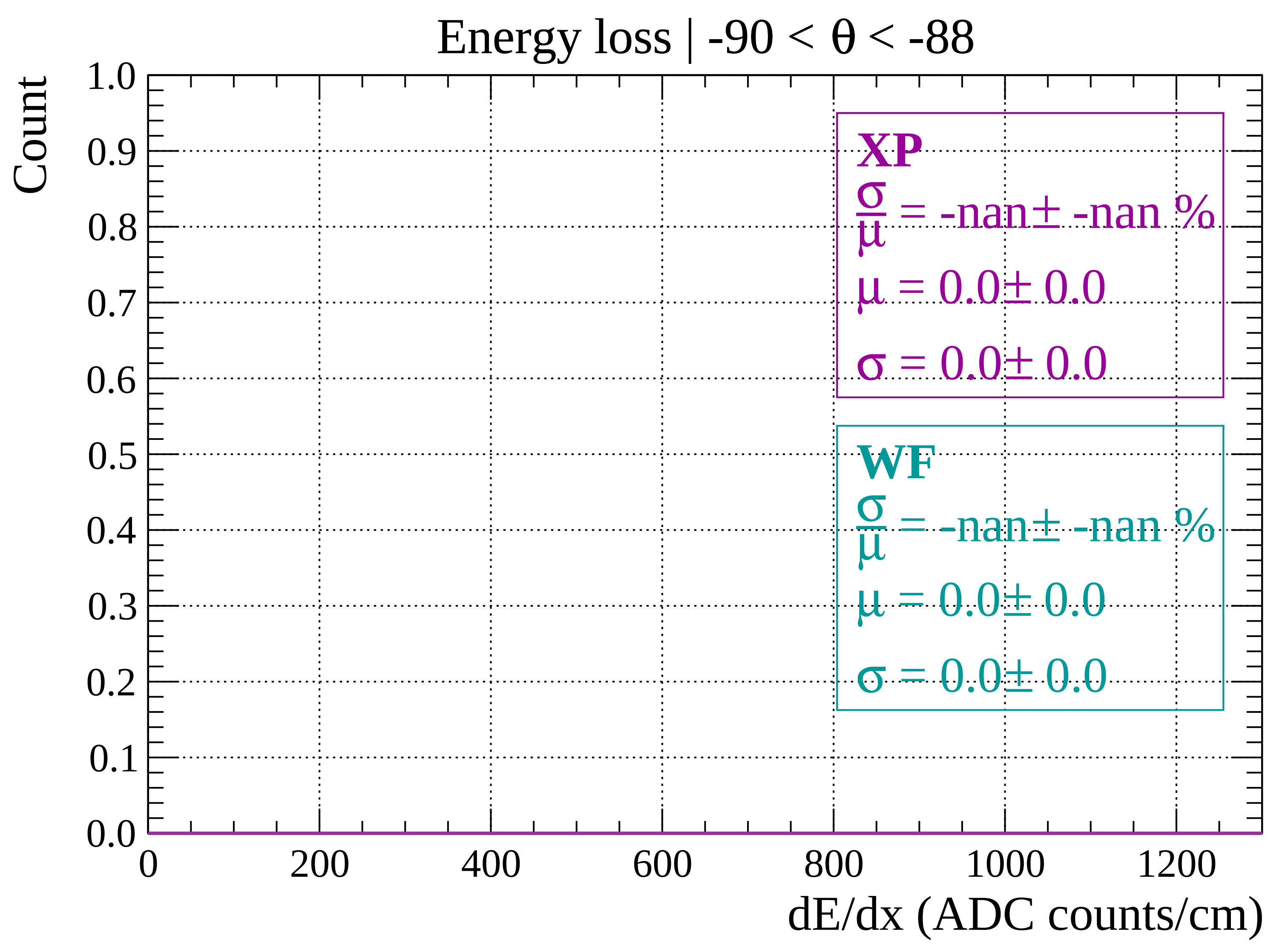
Energy loss | 1920 < p < 1960
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.20± 44.86 %
μ = 488.3± 9.0
σ = 1.0± 219.0
WF
σ/μ = 1413.41± 1305.97 %
μ = 30.8± 1.9
σ = 434.9± 375.0

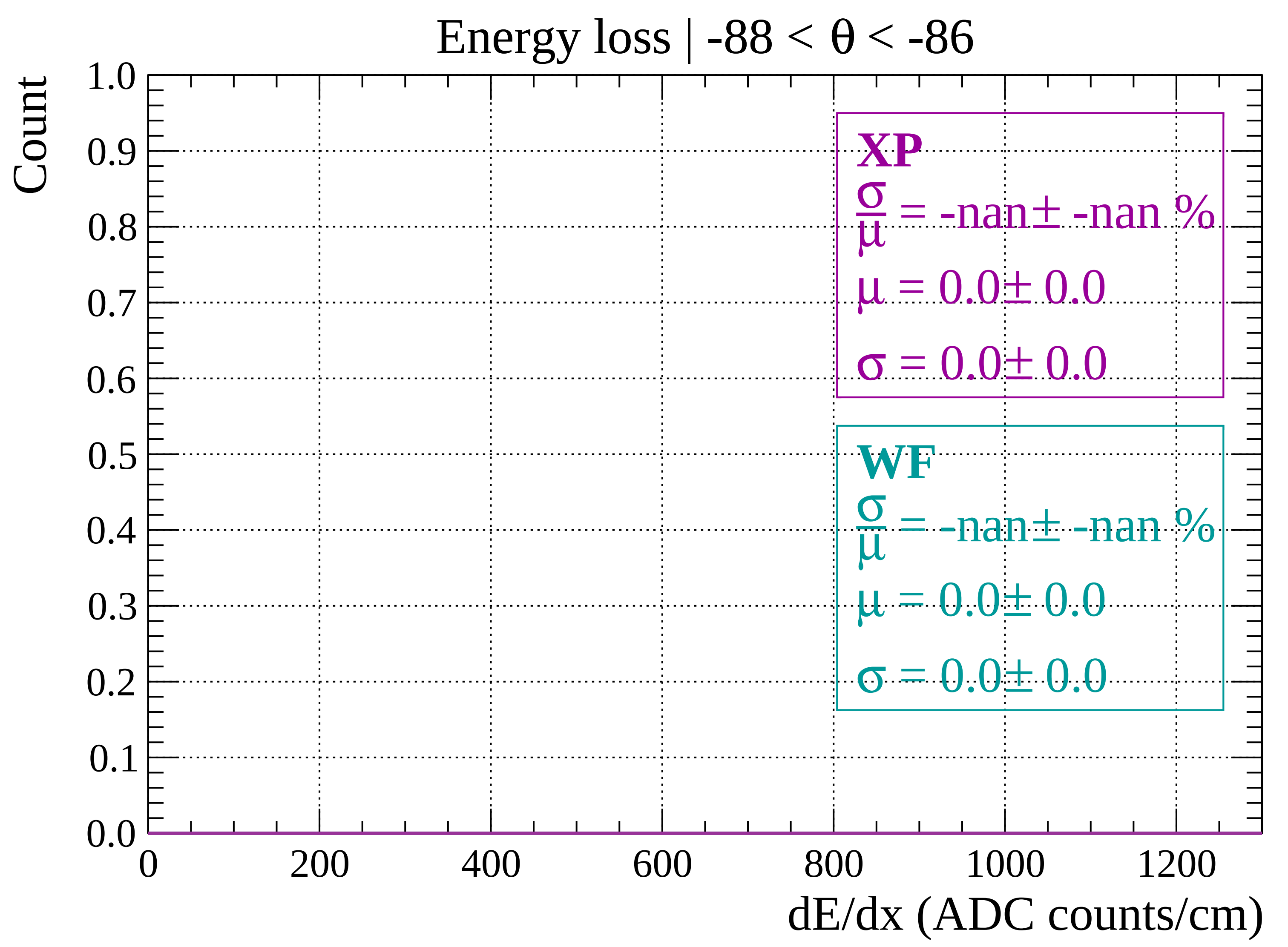


Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 0.05± 0.31 %
μ = 500.2± 2.3
σ = 0.2± 1.5
WF
σ/μ = 0.05± 0.26 %
μ = 539.2± 2.3
σ = 0.3± 1.4

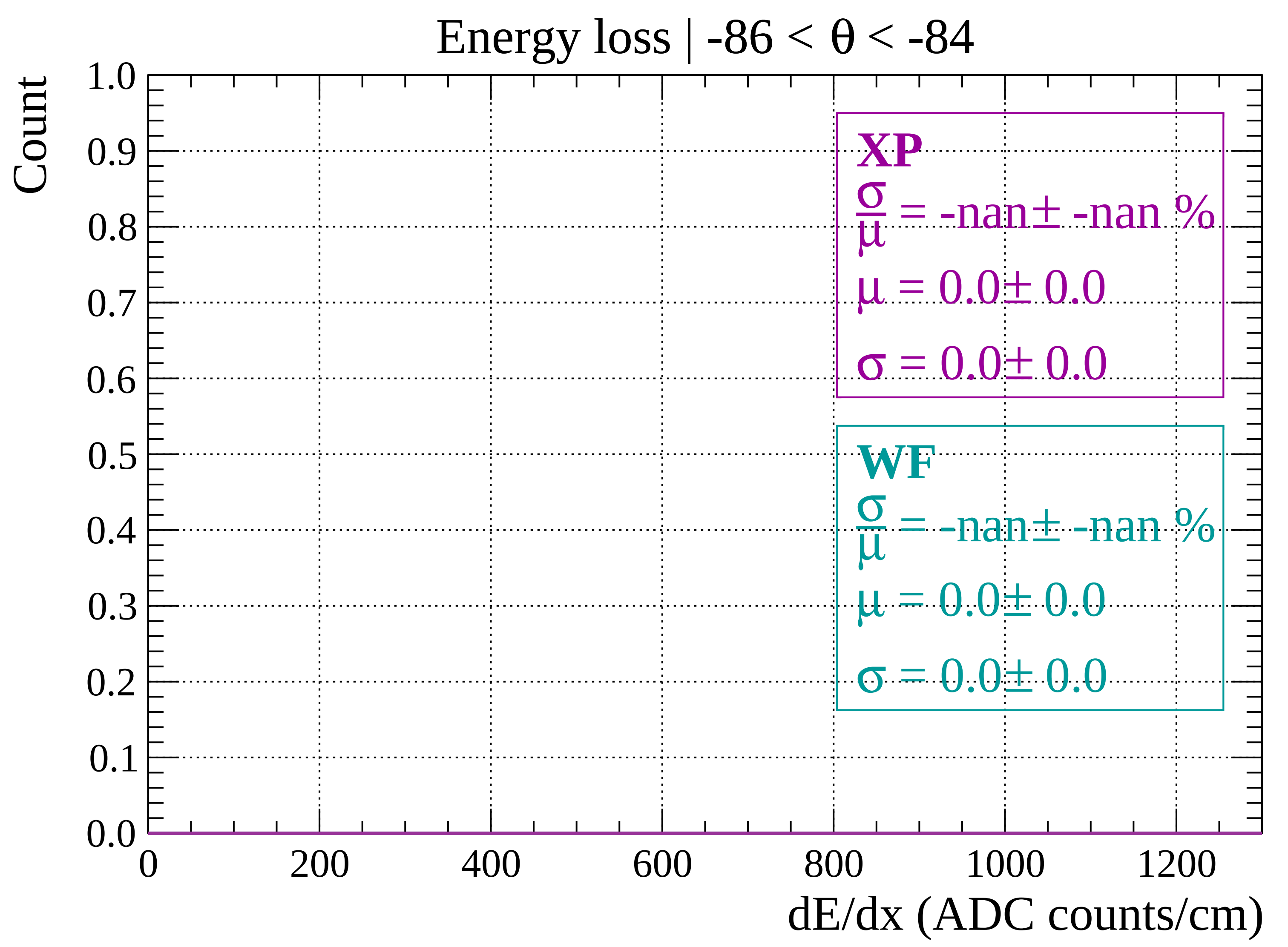


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.07± 5.50 %
μ = 496.9± 14.7
σ = 40.1± 26.1
WF
$\frac{\sigma}{\mu}$ = 53.70± 159.64 %
μ = 462.2± 447.6
σ = 248.2± 497.4

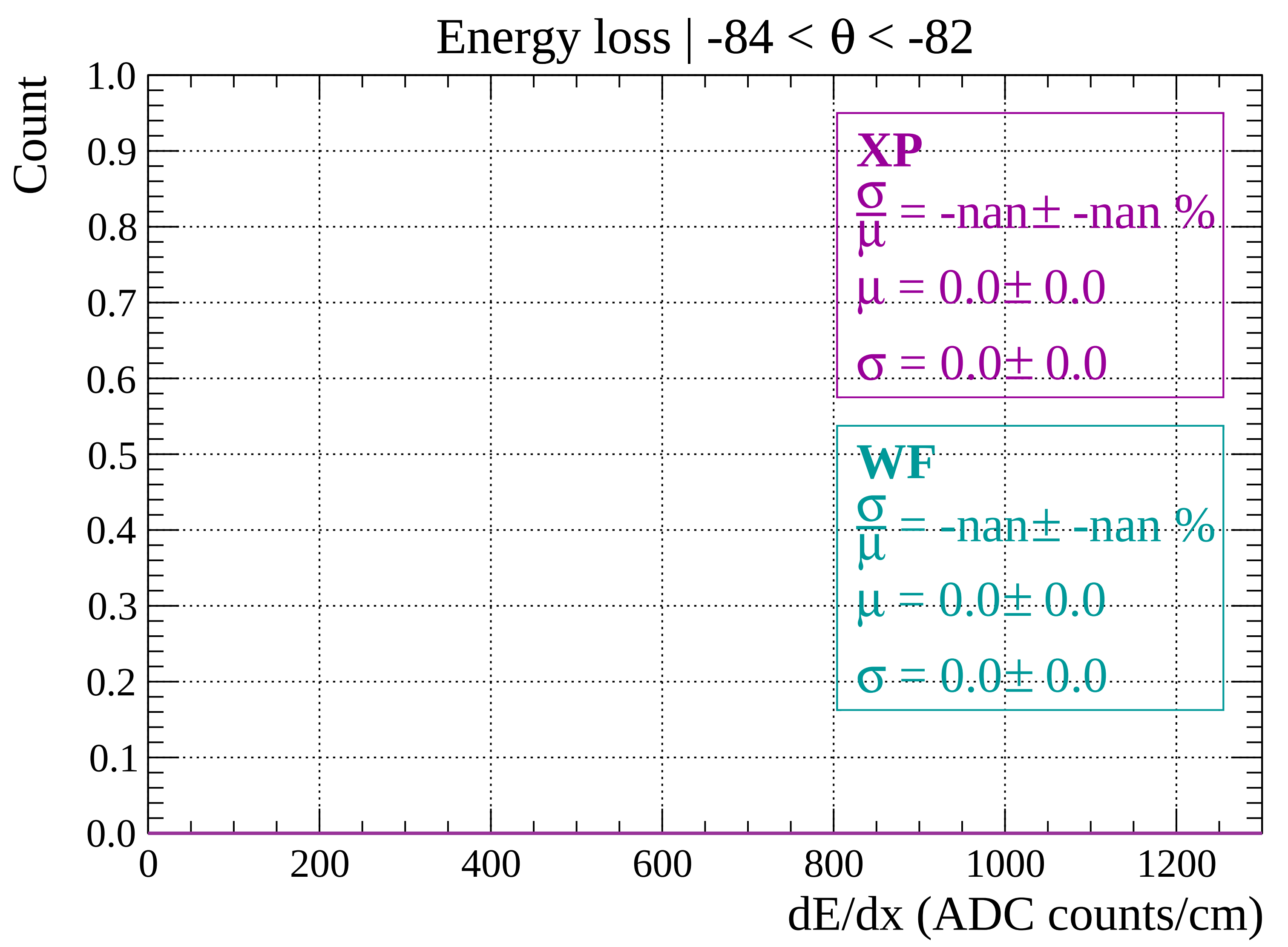

Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



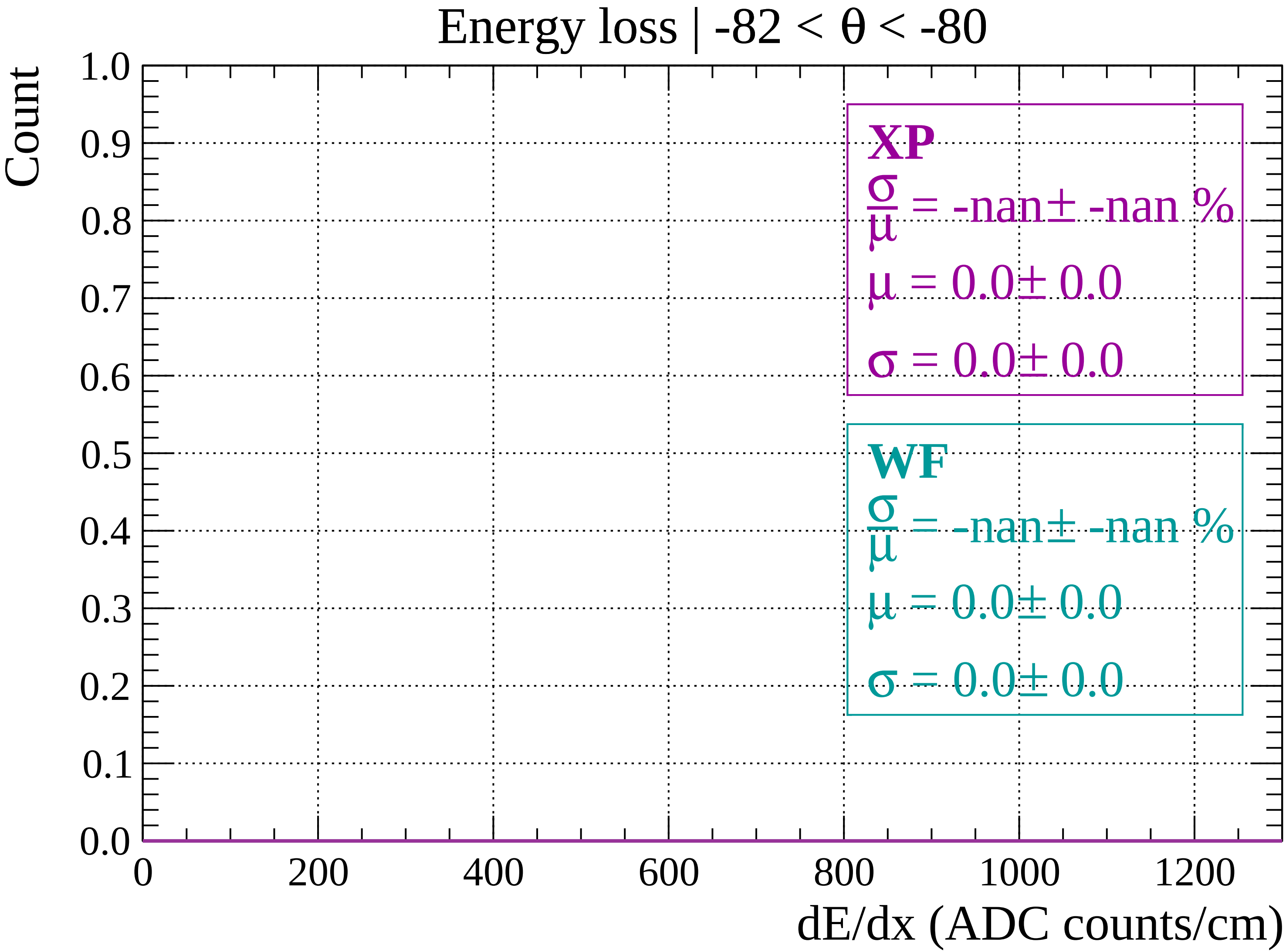
Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



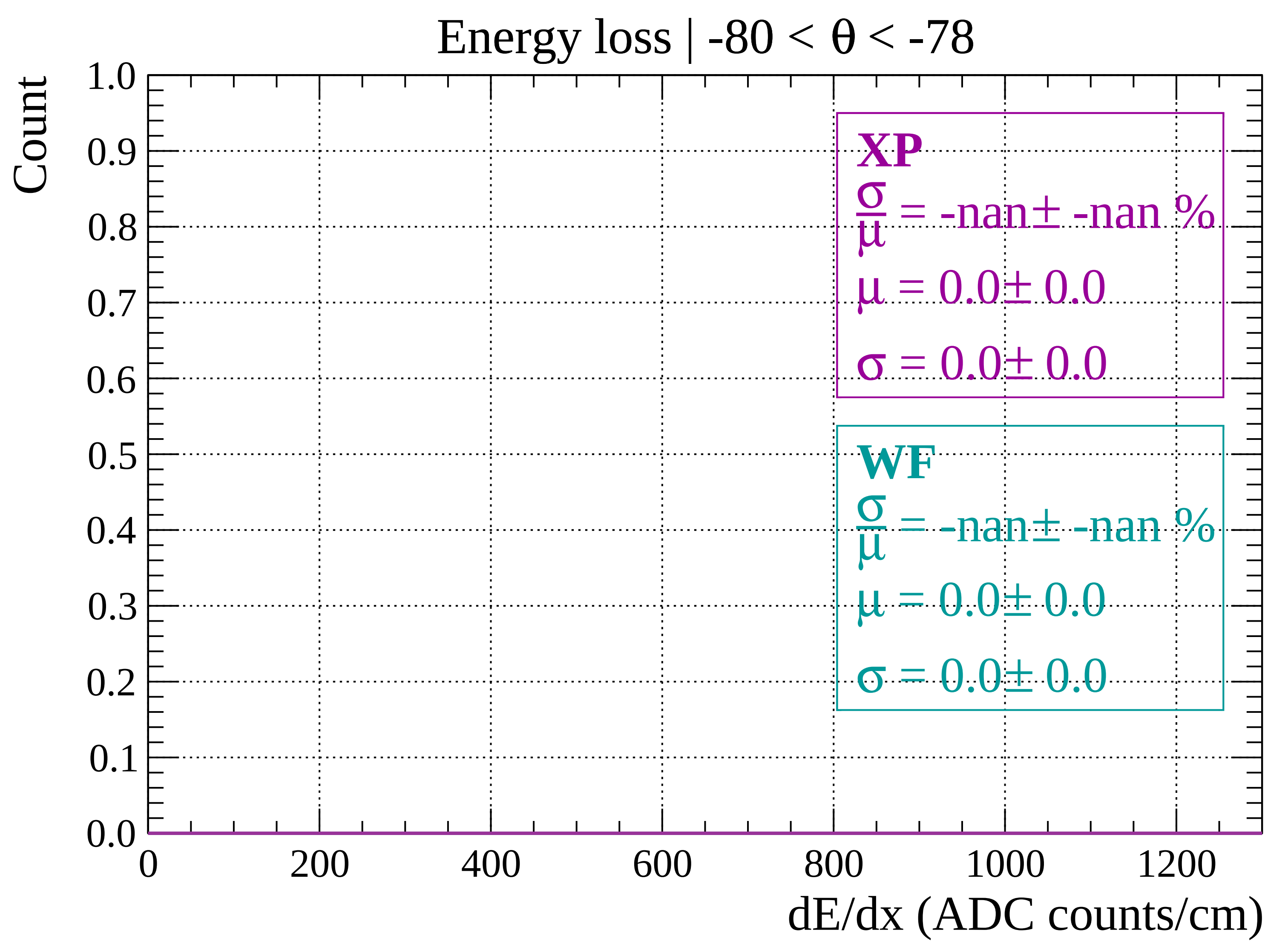
Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



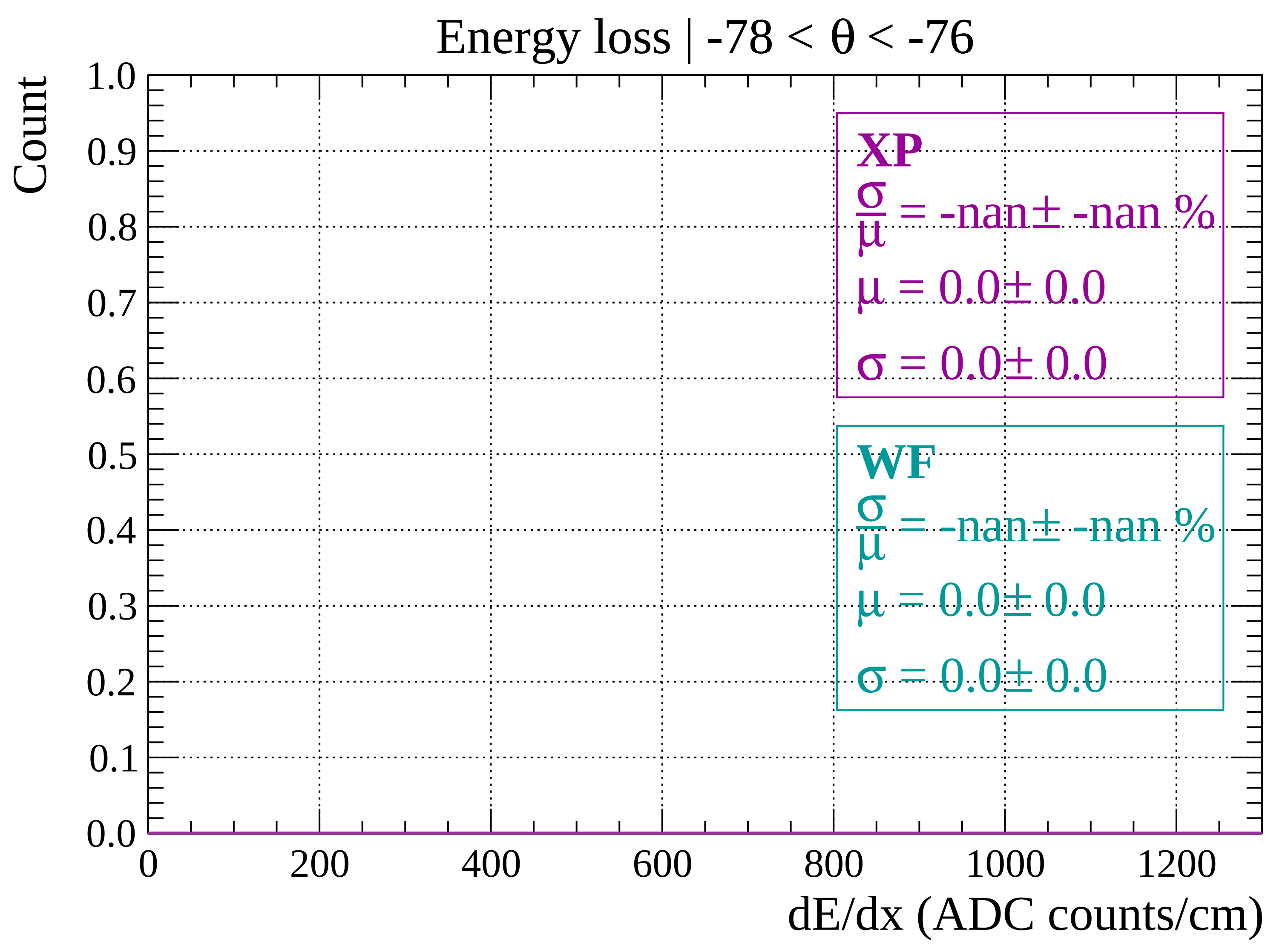
Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



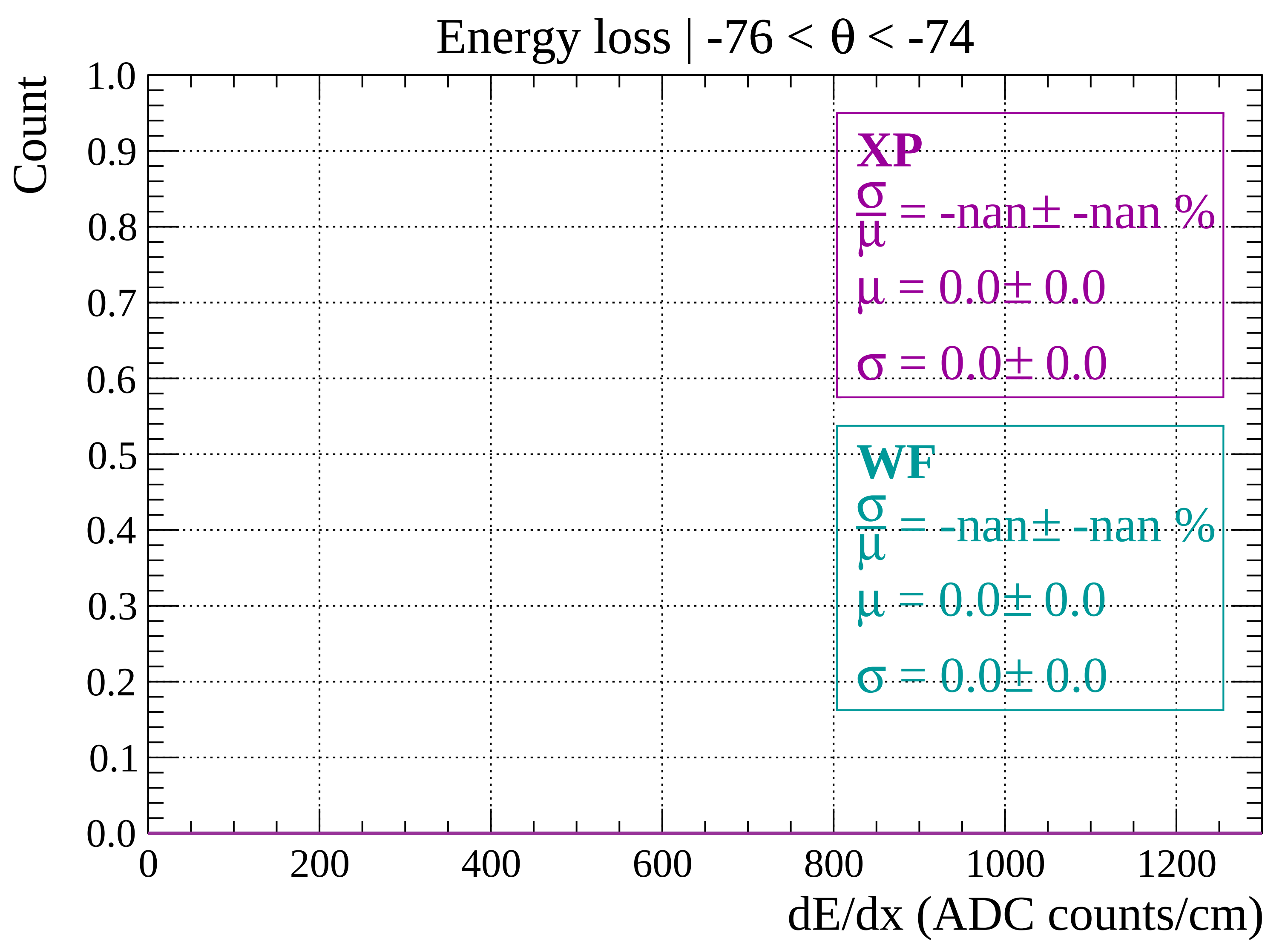
Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



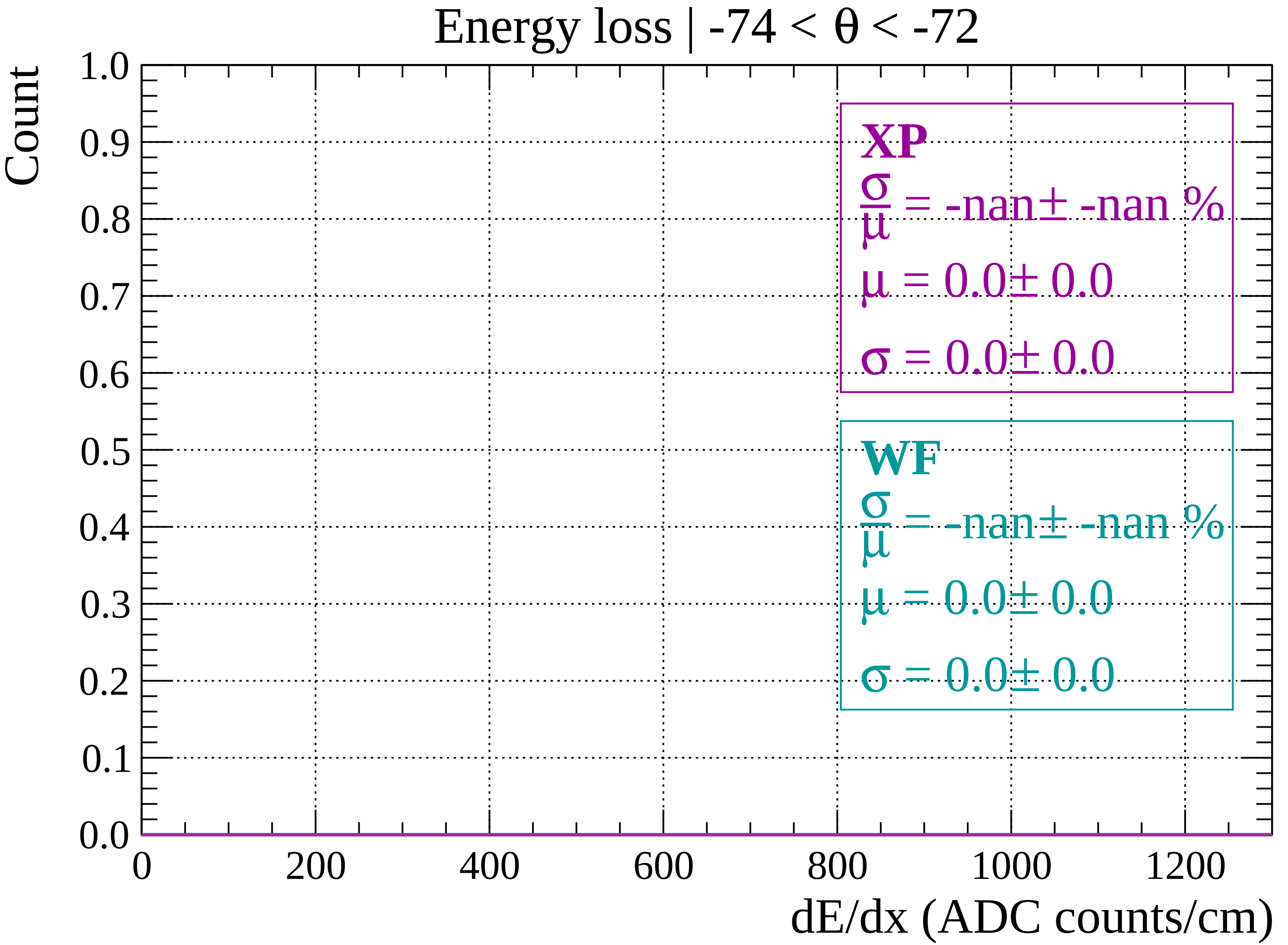
Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

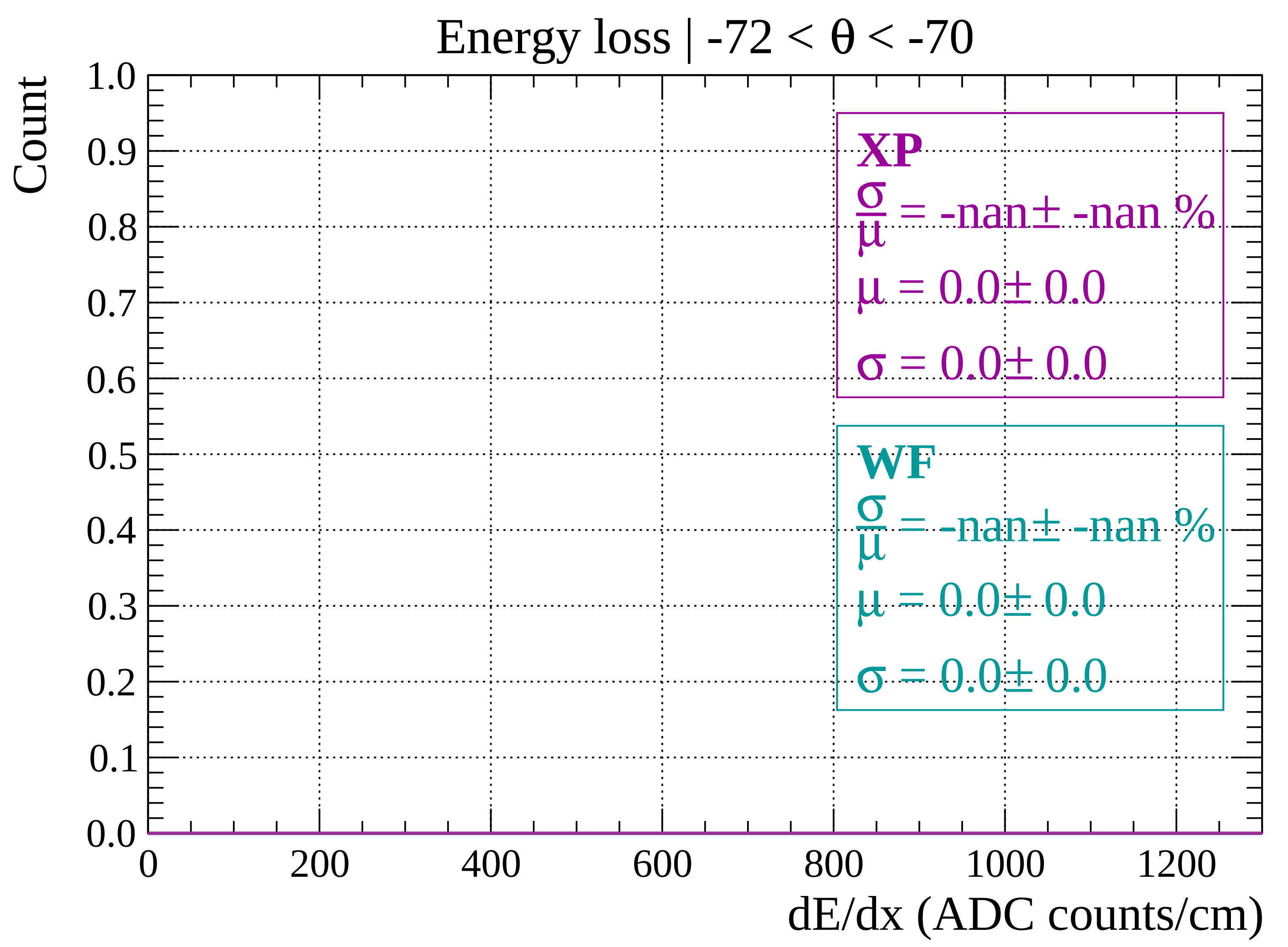


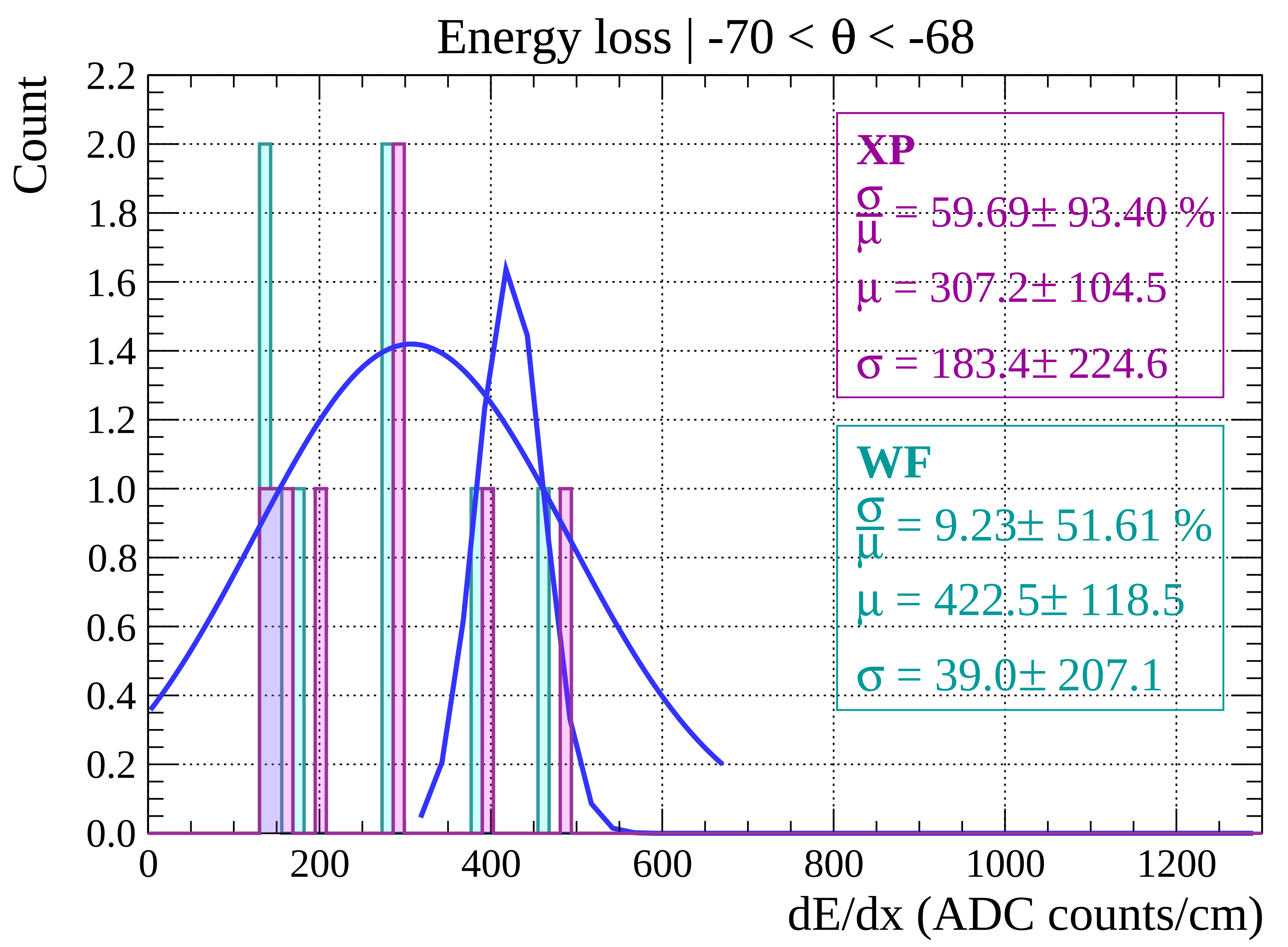
Energy loss | -78 < θ < -76
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



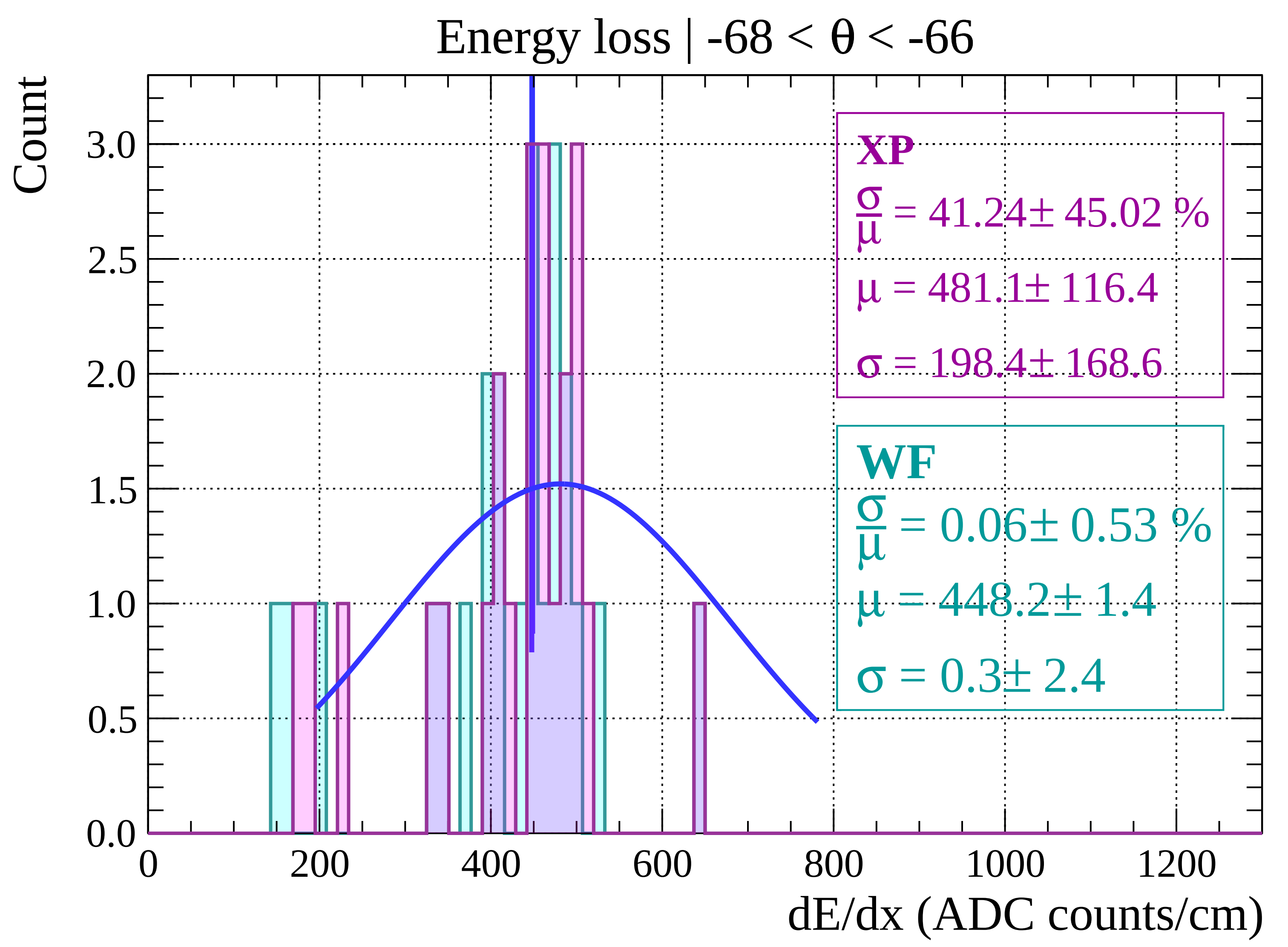
Energy loss | -76 < θ < -74
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -74 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

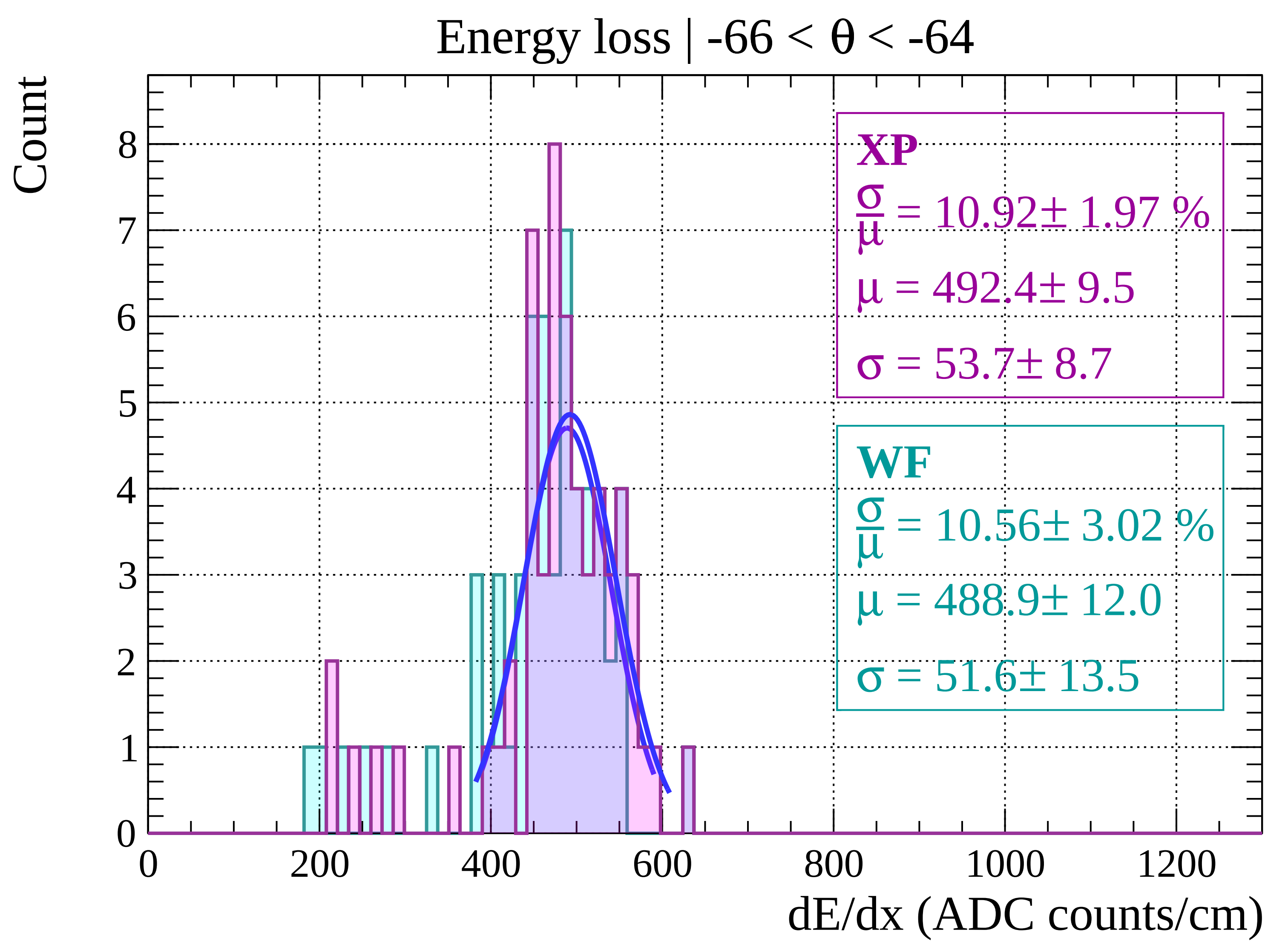

Energy loss | -72 < θ < -70
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



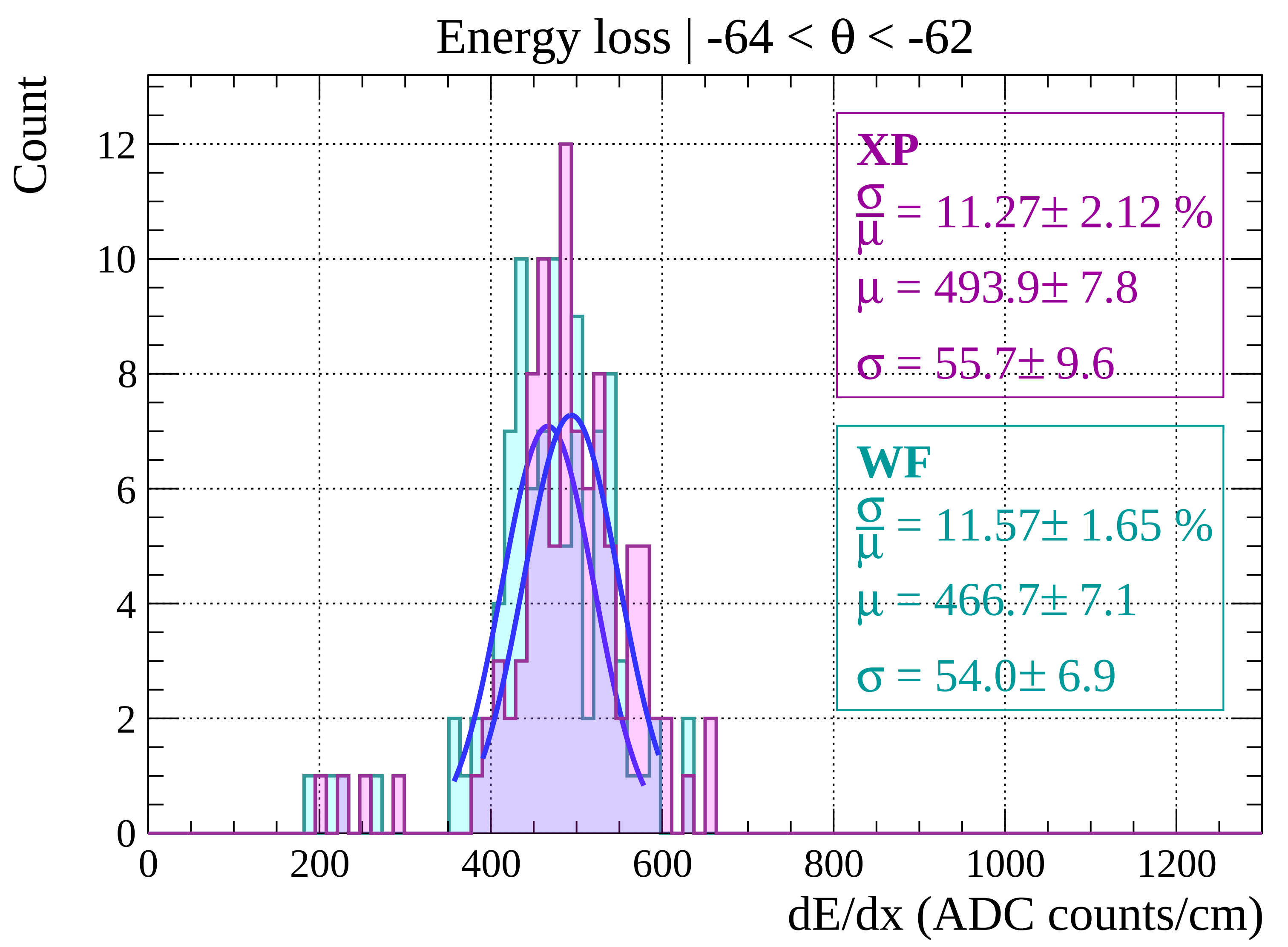
Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 59.69± 93.40 %
μ = 307.2± 104.5
σ = 183.4± 224.6
WF
σ/μ = 9.23± 51.61 %
μ = 422.5± 118.5
σ = 39.0± 207.1
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



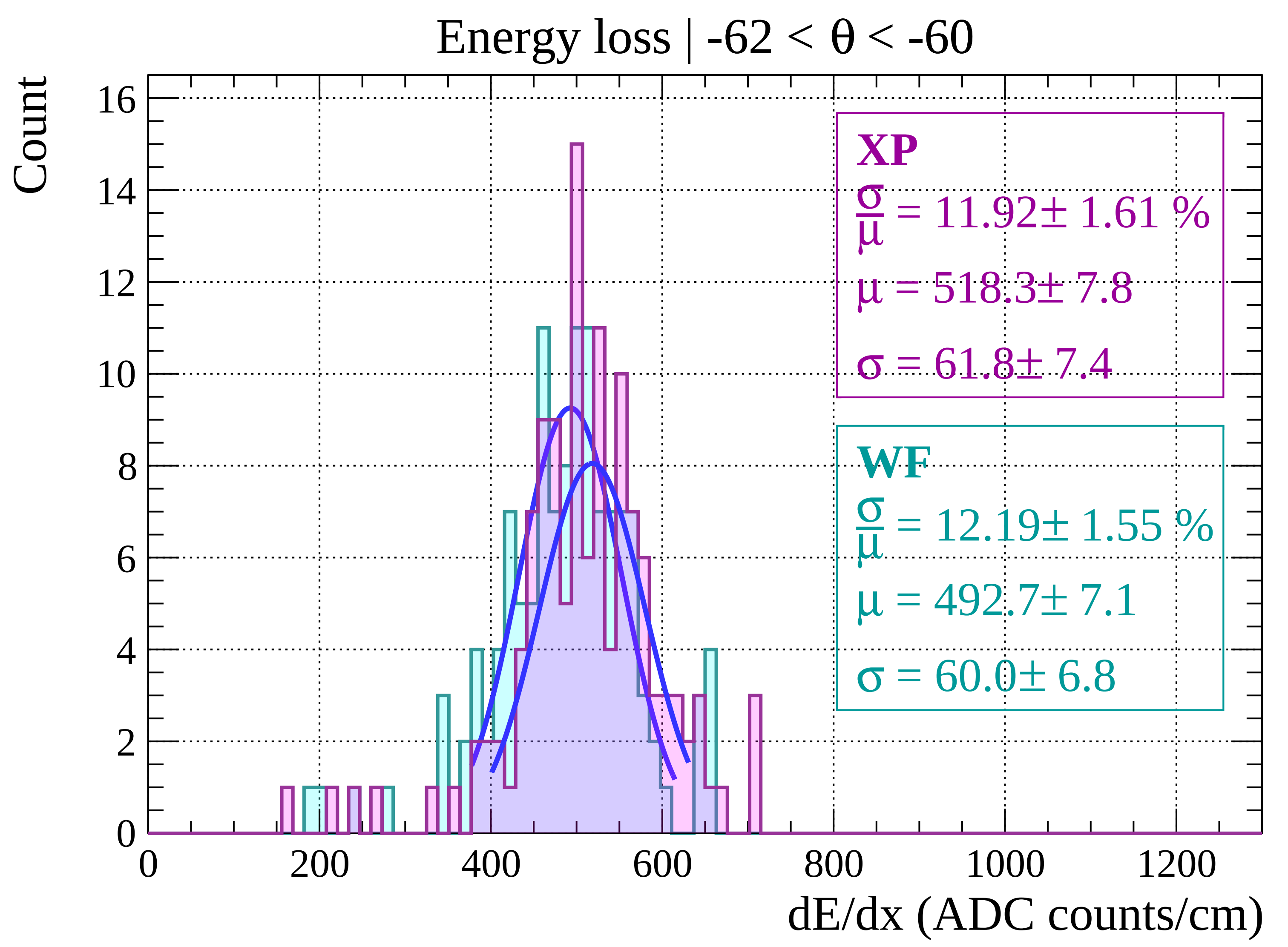
Energy loss | -68 < θ < -66
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = 41.24± 45.02 %
μ = 481.1± 116.4
σ = 198.4± 168.6
WF
σ/μ = 0.06± 0.53 %
μ = 448.2± 1.4
σ = 0.3± 2.4

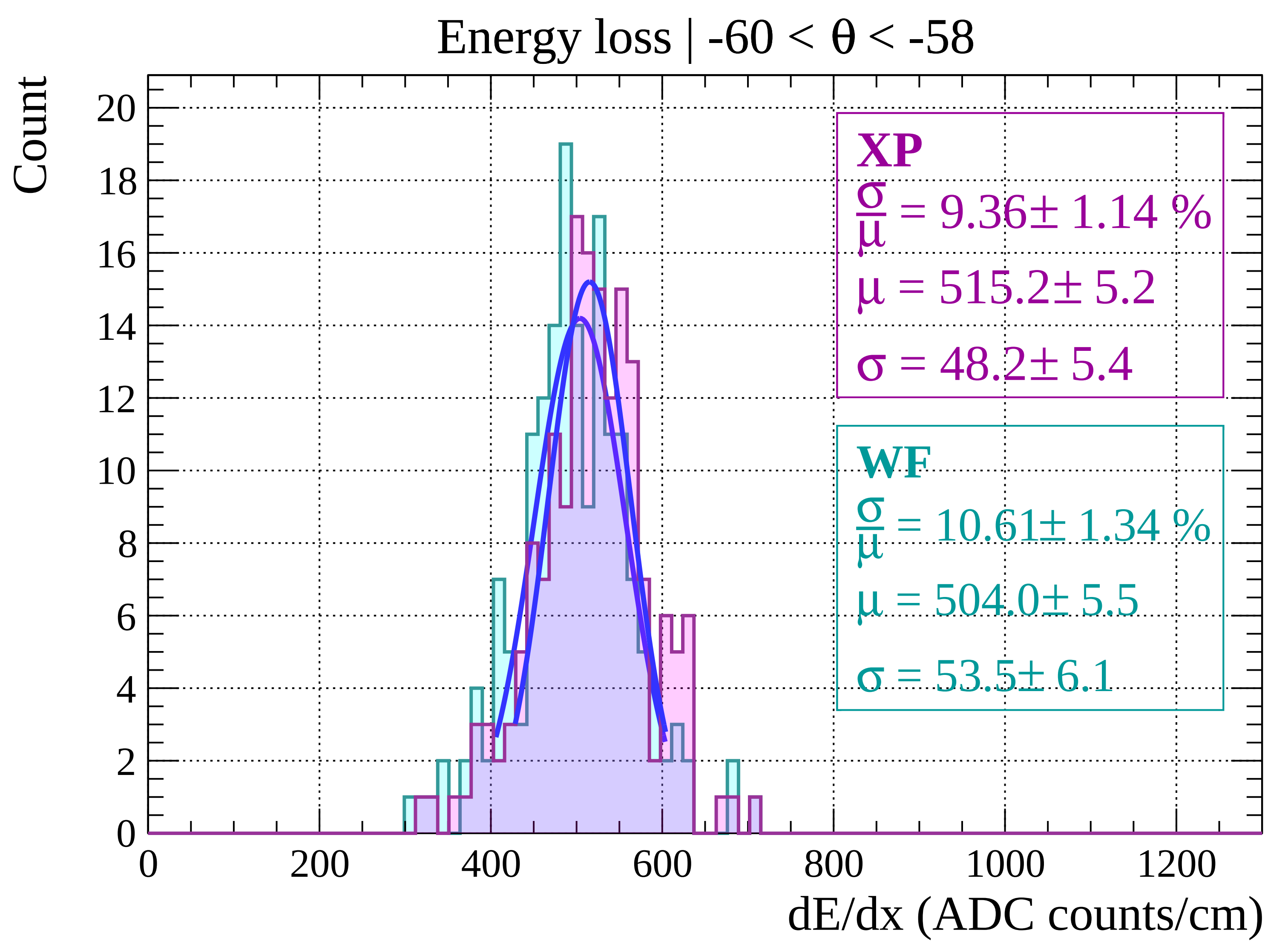


Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.92± 1.97 %
μ = 492.4± 9.5
σ = 53.7± 8.7
WF
σ/μ = 10.56± 3.02 %
μ = 488.9± 12.0
σ = 51.6± 13.5

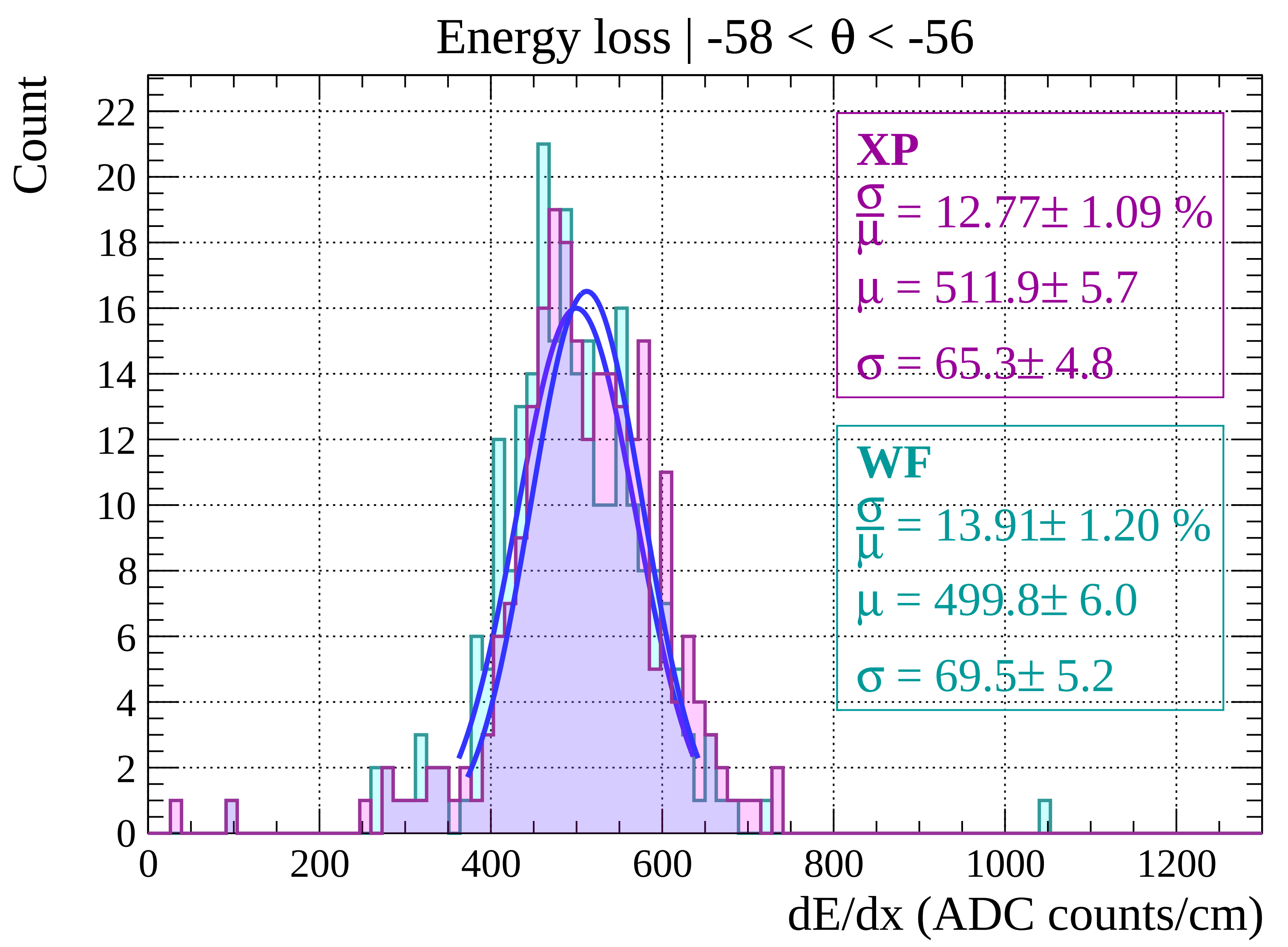


Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 11.27± 2.12 %
μ = 493.9± 7.8
σ = 55.7± 9.6
WF
σ/μ = 11.57± 1.65 %
μ = 466.7± 7.1
σ = 54.0± 6.9

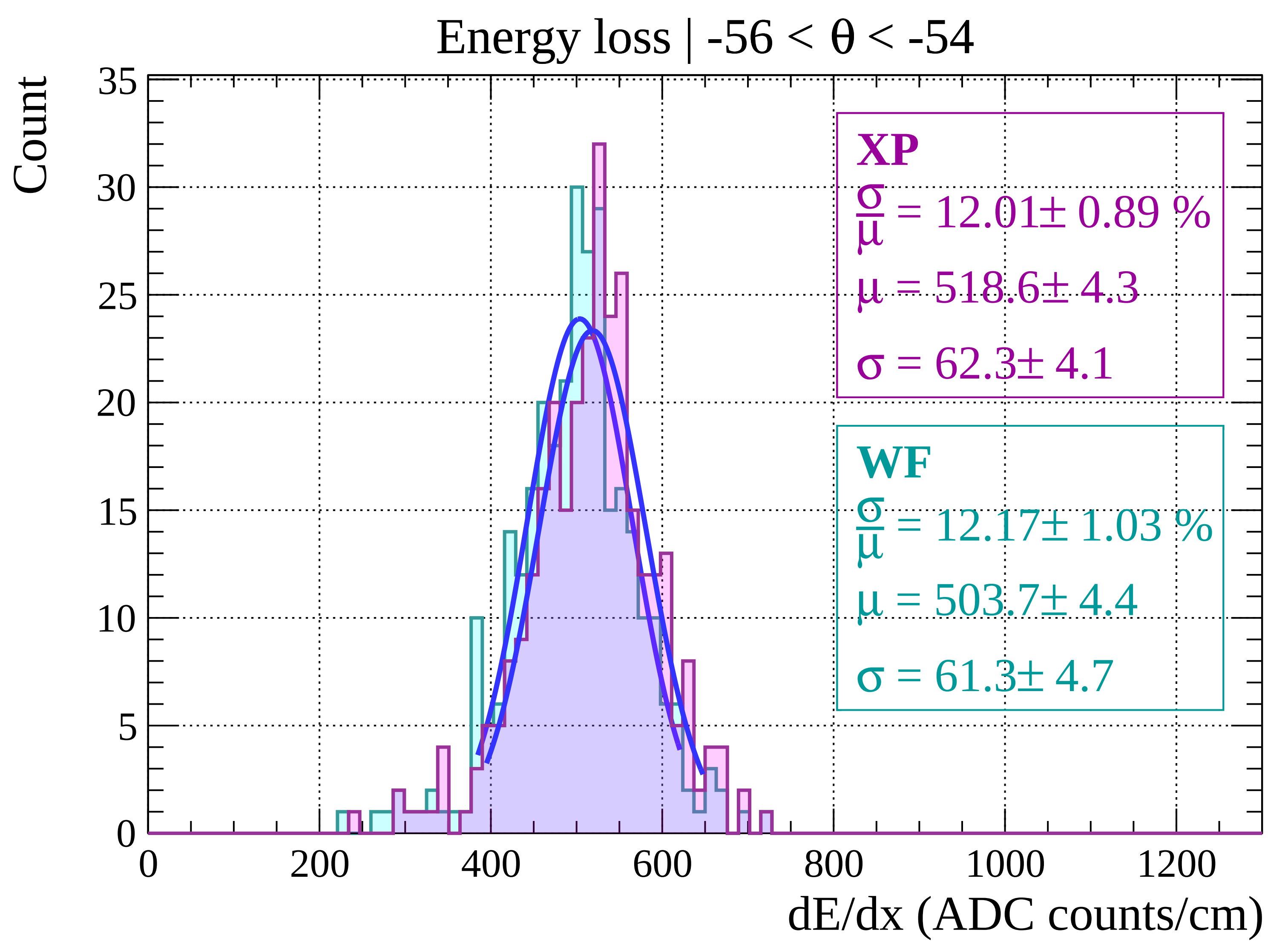

Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
σ/μ = 11.92± 1.61 %
μ = 518.3± 7.8
σ = 61.8± 7.4
WF
σ/μ = 12.19± 1.55 %
μ = 492.7± 7.1
σ = 60.0± 6.8

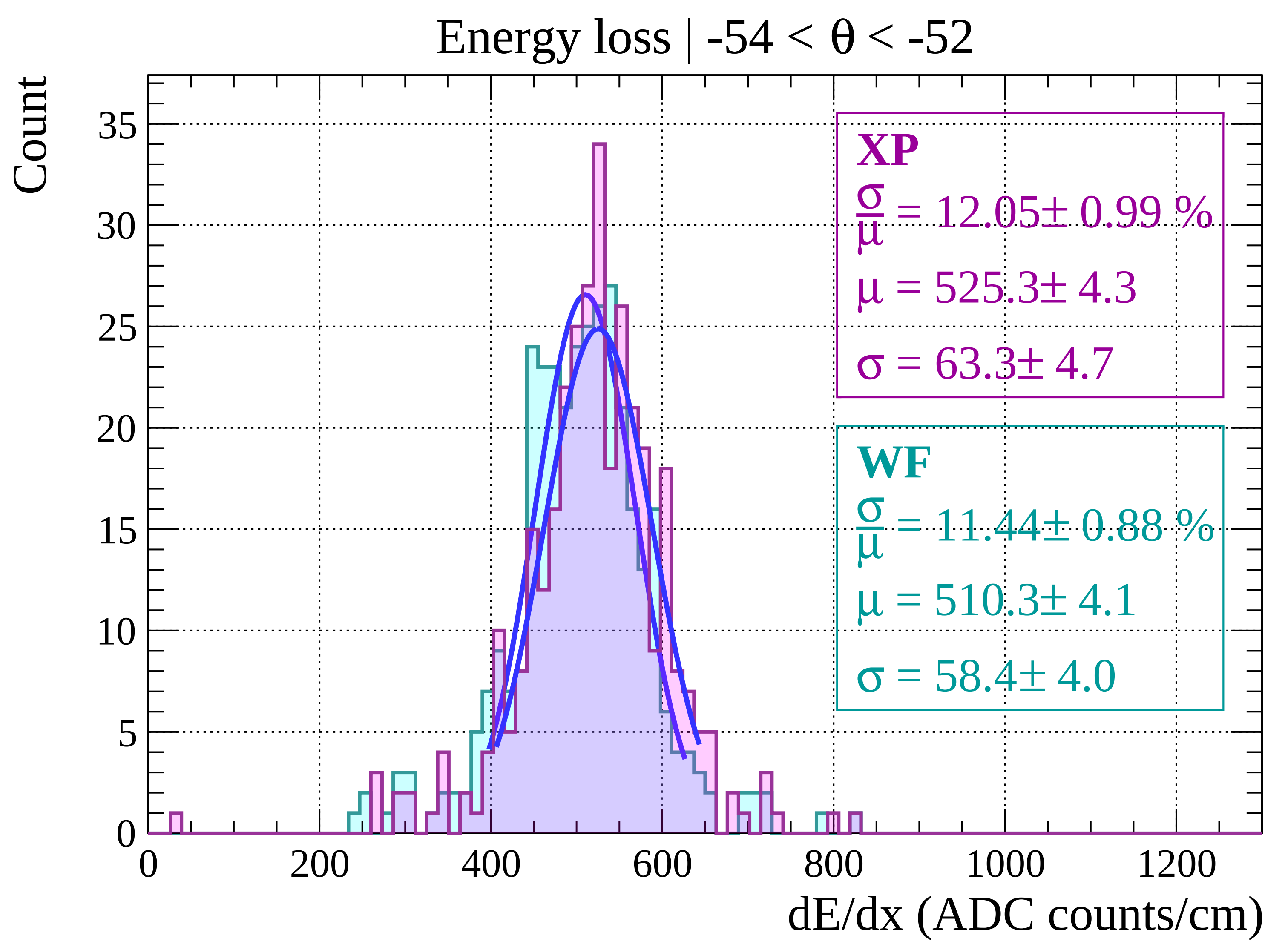


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
XP
σ/μ = 9.36± 1.14 %
μ = 515.2± 5.2
σ = 48.2± 5.4
WF
σ/μ = 10.61± 1.34 %
μ = 504.0± 5.5
σ = 53.5± 6.1

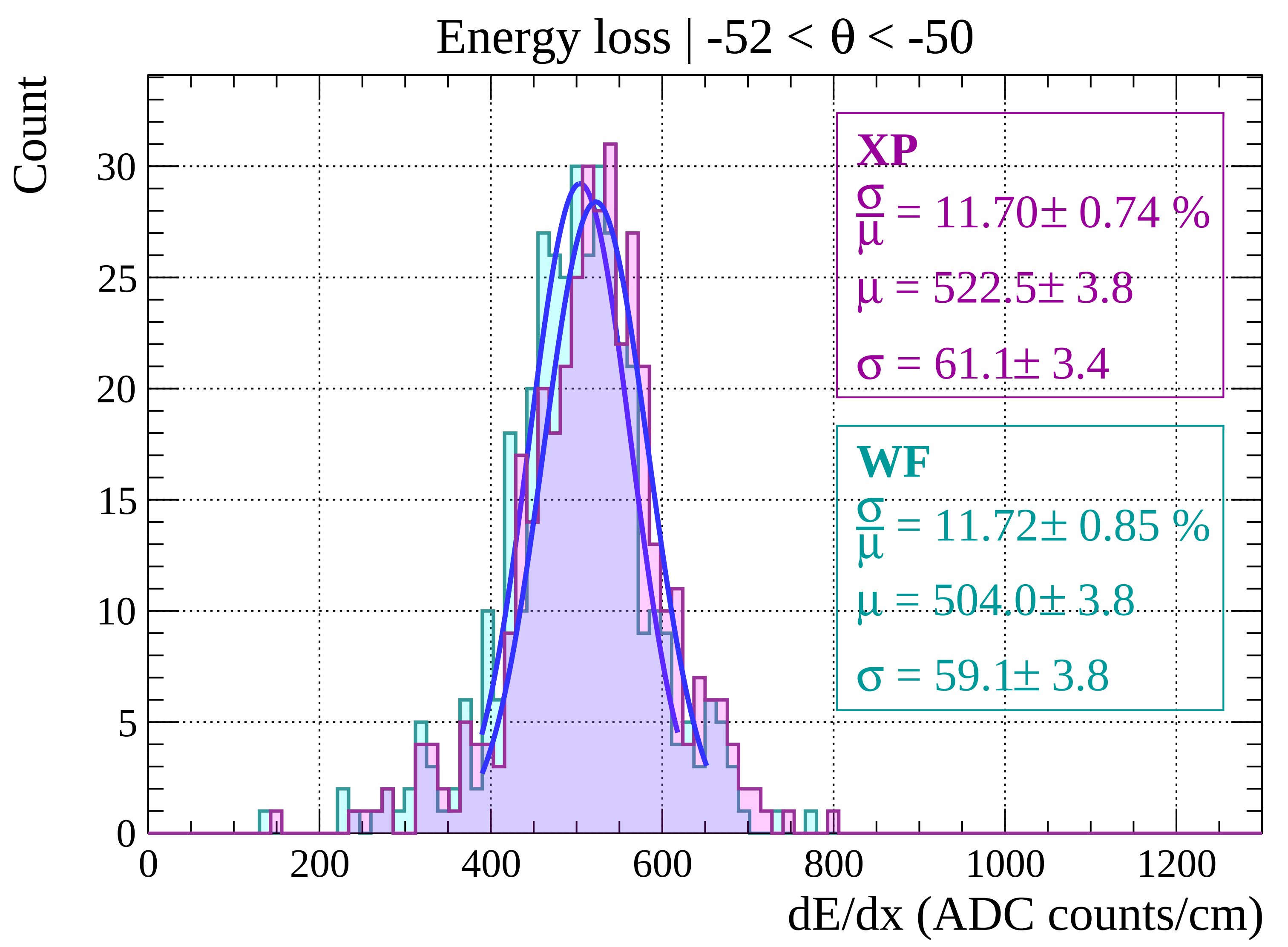


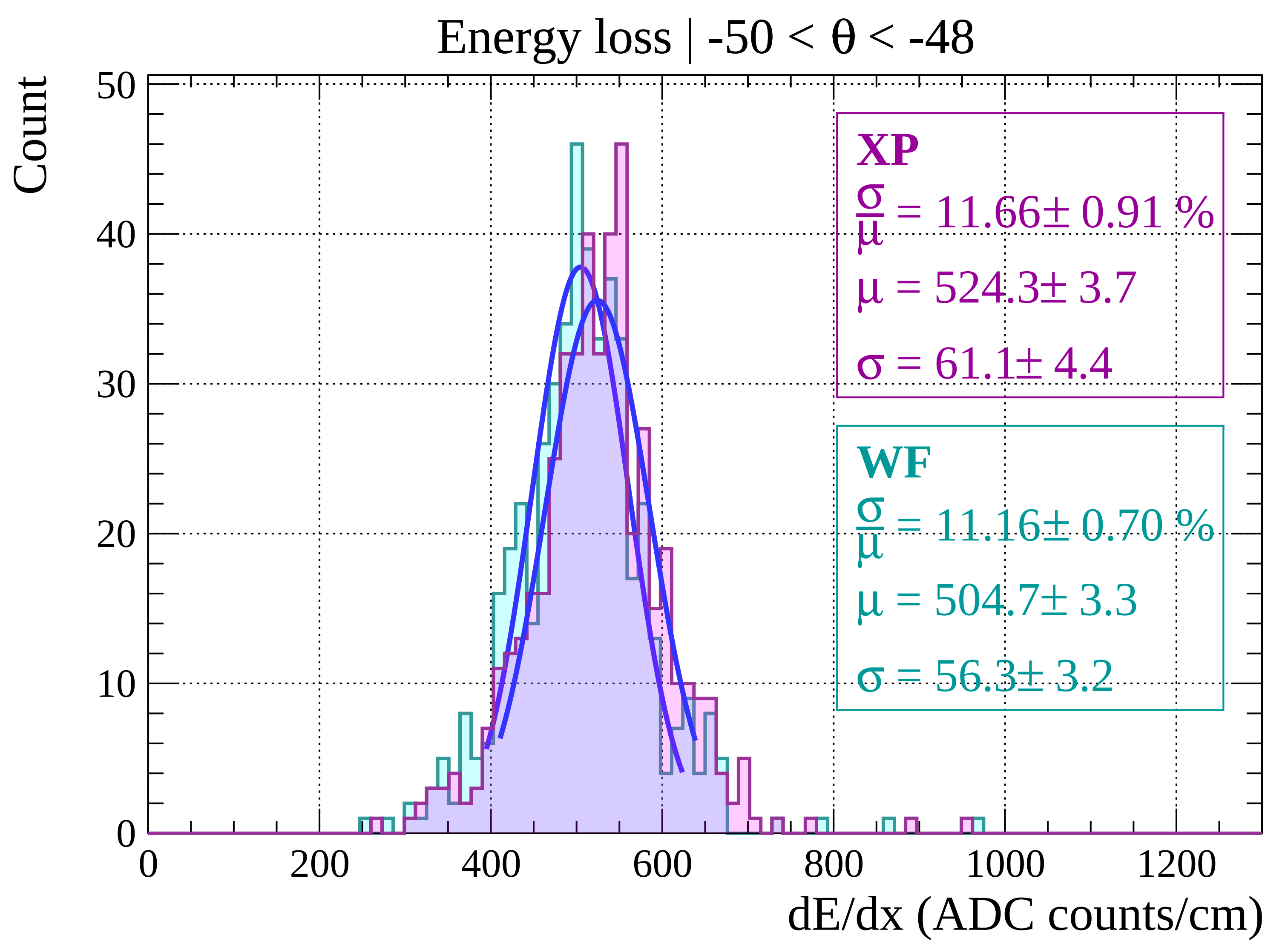
Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
XP
σ/μ = 12.77± 1.09 %
μ = 511.9± 5.7
σ = 65.3± 4.8
WF
σ/μ = 13.91± 1.20 %
μ = 499.8± 6.0
σ = 69.5± 5.2

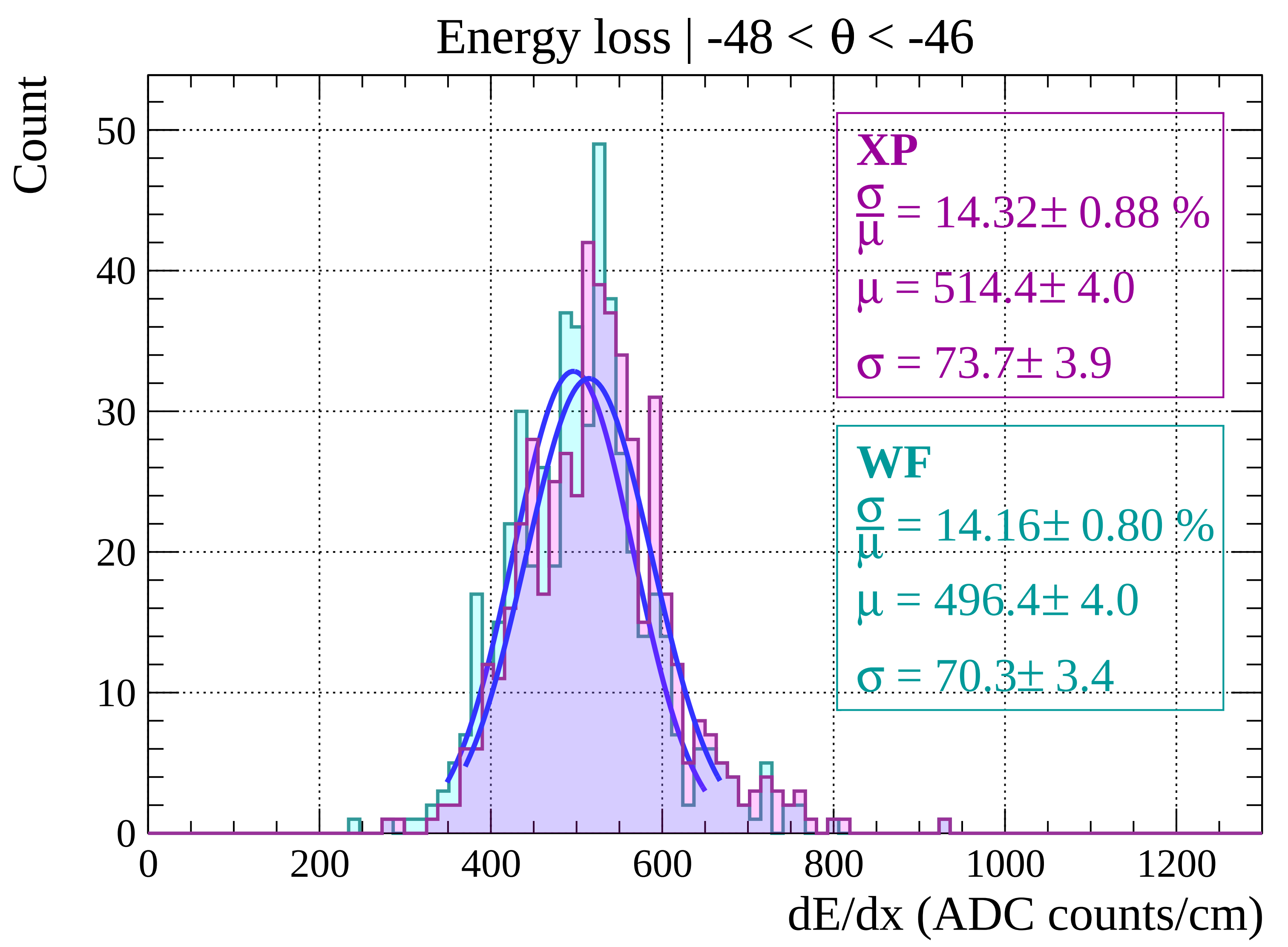

Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 12.01± 0.89 %
μ = 518.6± 4.3
σ = 62.3± 4.1
WF
σ/μ = 12.17± 1.03 %
μ = 503.7± 4.4
σ = 61.3± 4.7



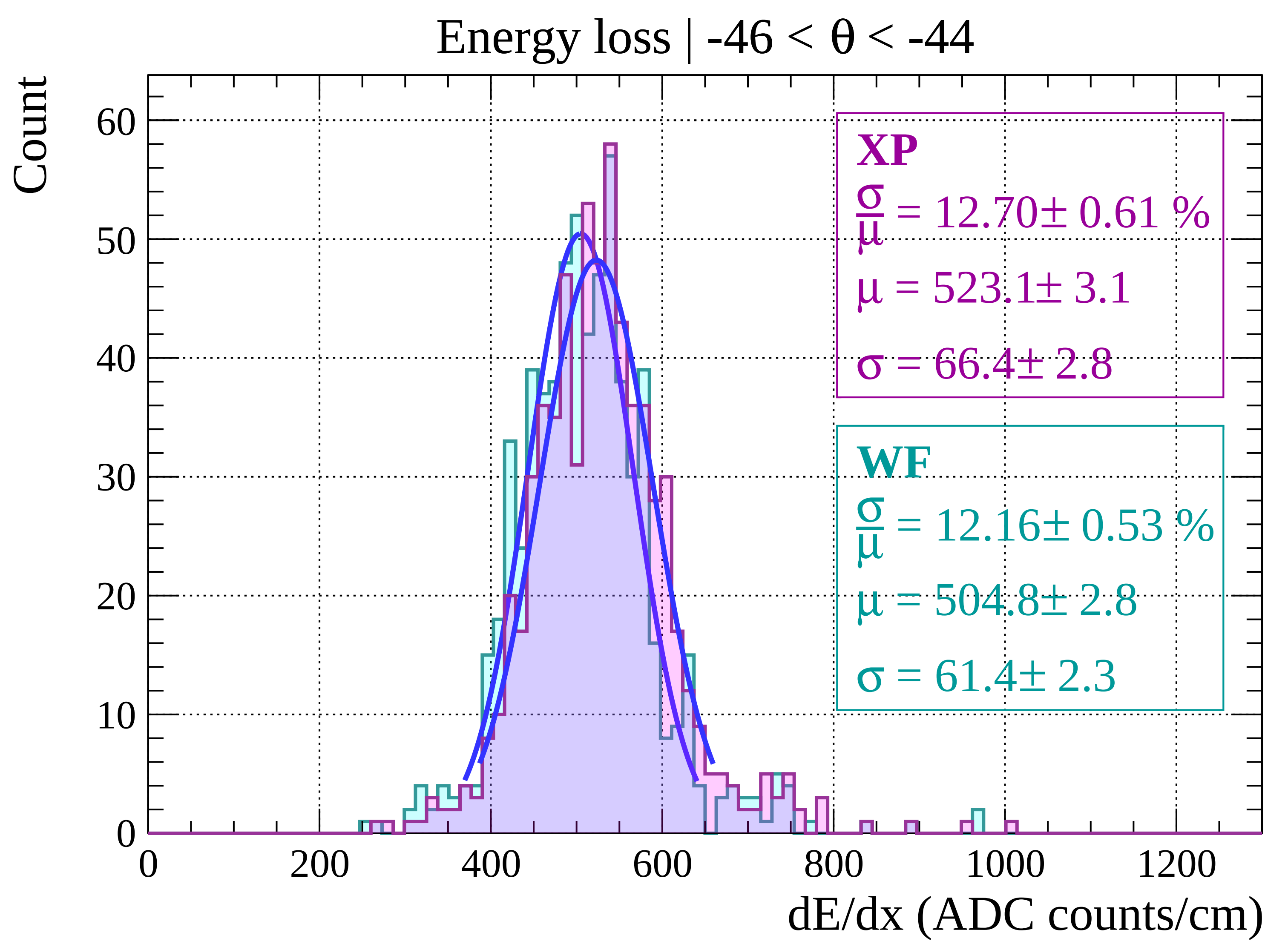
Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.05± 0.99 %
μ = 525.3± 4.3
σ = 63.3± 4.7
WF
σ/μ = 11.44± 0.88 %
μ = 510.3± 4.1
σ = 58.4± 4.0
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35

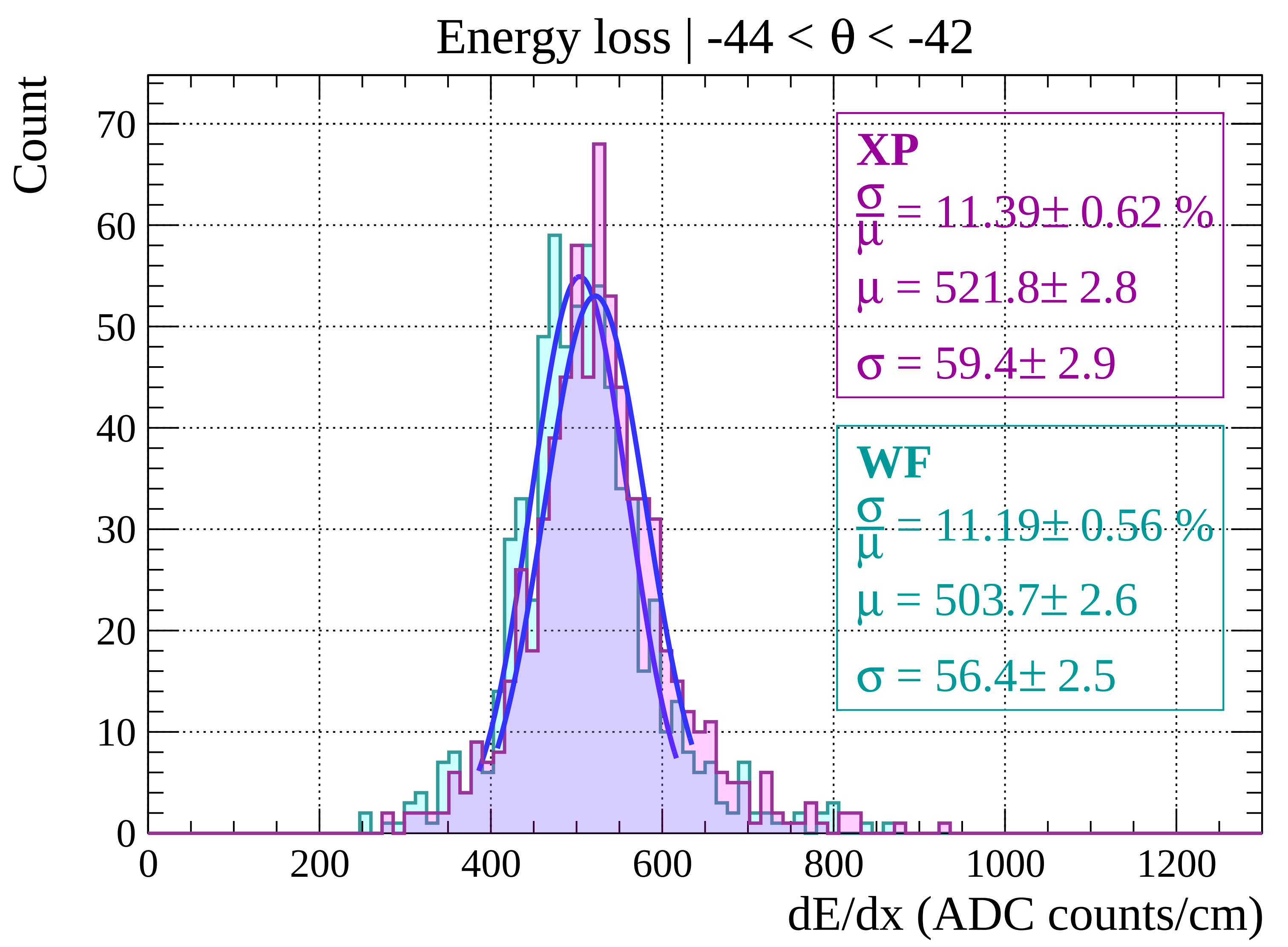

Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.70± 0.74 %
μ = 522.5± 3.8
σ = 61.1± 3.4
WF
σ/μ = 11.72± 0.85 %
μ = 504.0± 3.8
σ = 59.1± 3.8
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200


Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 11.66± 0.91 %
μ = 524.3± 3.7
σ = 61.1± 4.4
WF
σ/μ = 11.16± 0.70 %
μ = 504.7± 3.3
σ = 56.3± 3.2

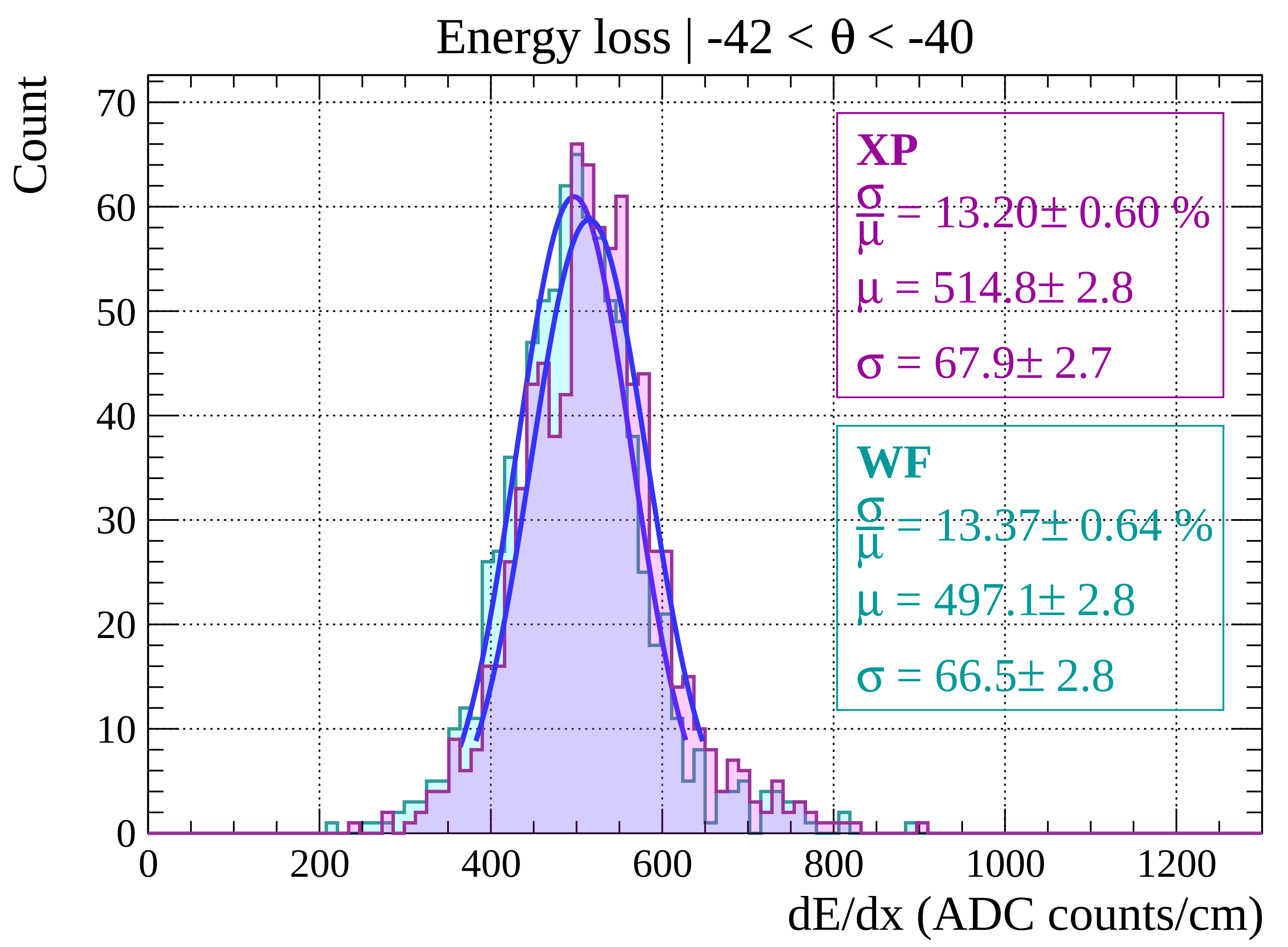


Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.32± 0.88 %
μ = 514.4± 4.0
σ = 73.7± 3.9
WF
$\frac{\sigma}{\mu}$ = 14.16± 0.80 %
μ = 496.4± 4.0
σ = 70.3± 3.4

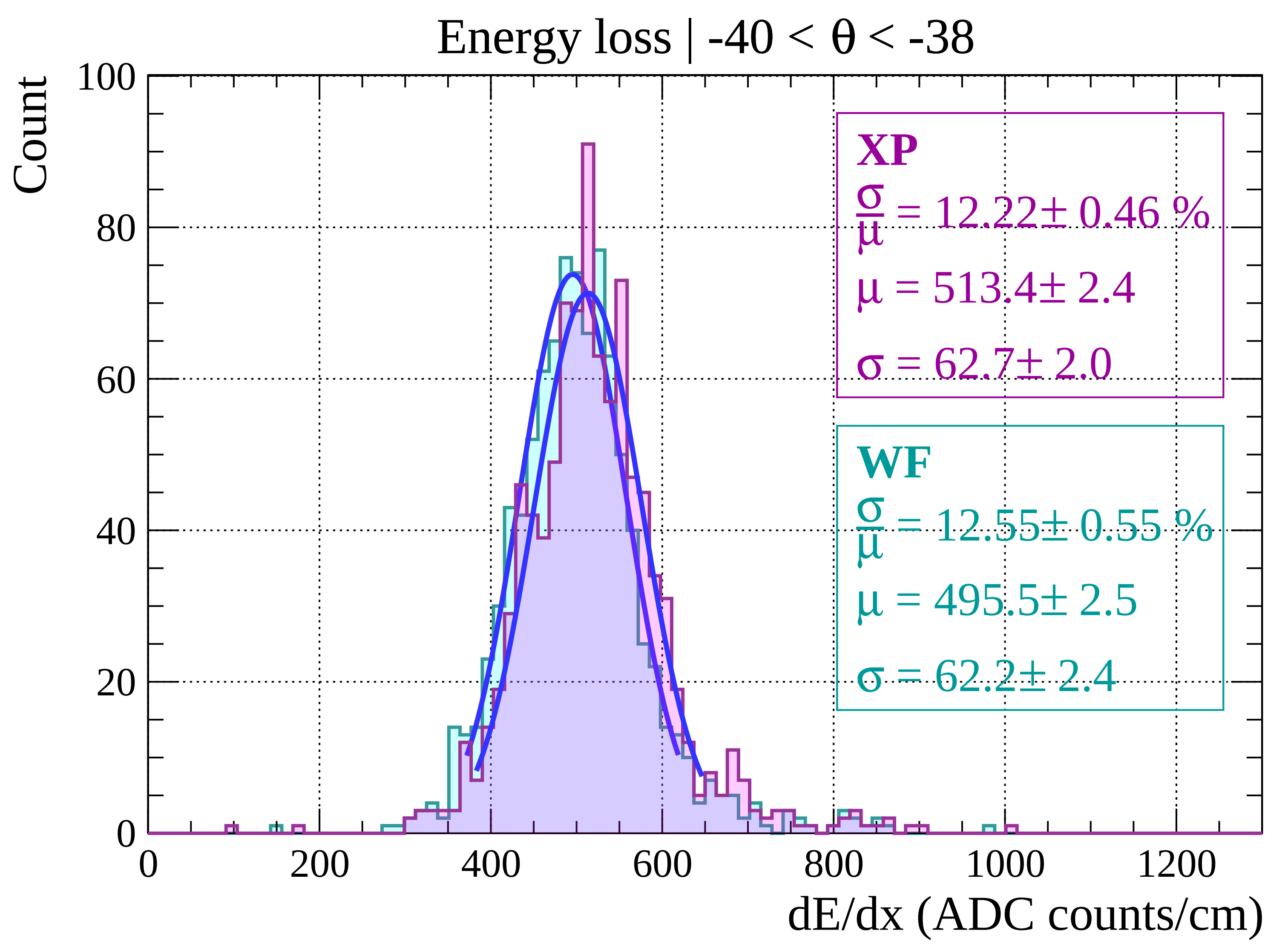

Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 12.70± 0.61 %
μ = 523.1± 3.1
σ = 66.4± 2.8
WF
σ/μ = 12.16± 0.53 %
μ = 504.8± 2.8
σ = 61.4± 2.3



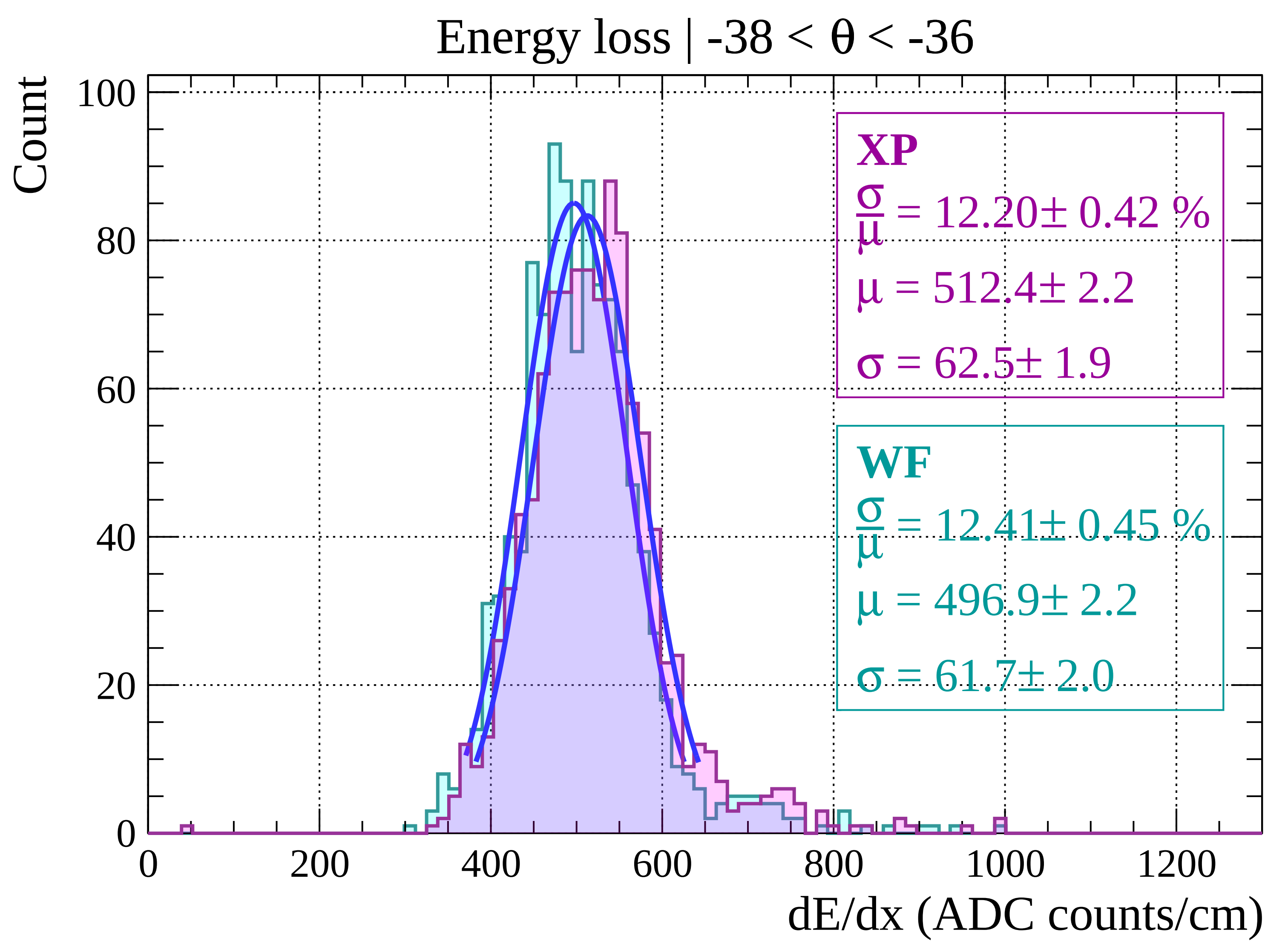
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.39± 0.62 %
μ = 521.8± 2.8
σ = 59.4± 2.9
WF
σ/μ = 11.19± 0.56 %
μ = 503.7± 2.6
σ = 56.4± 2.5
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

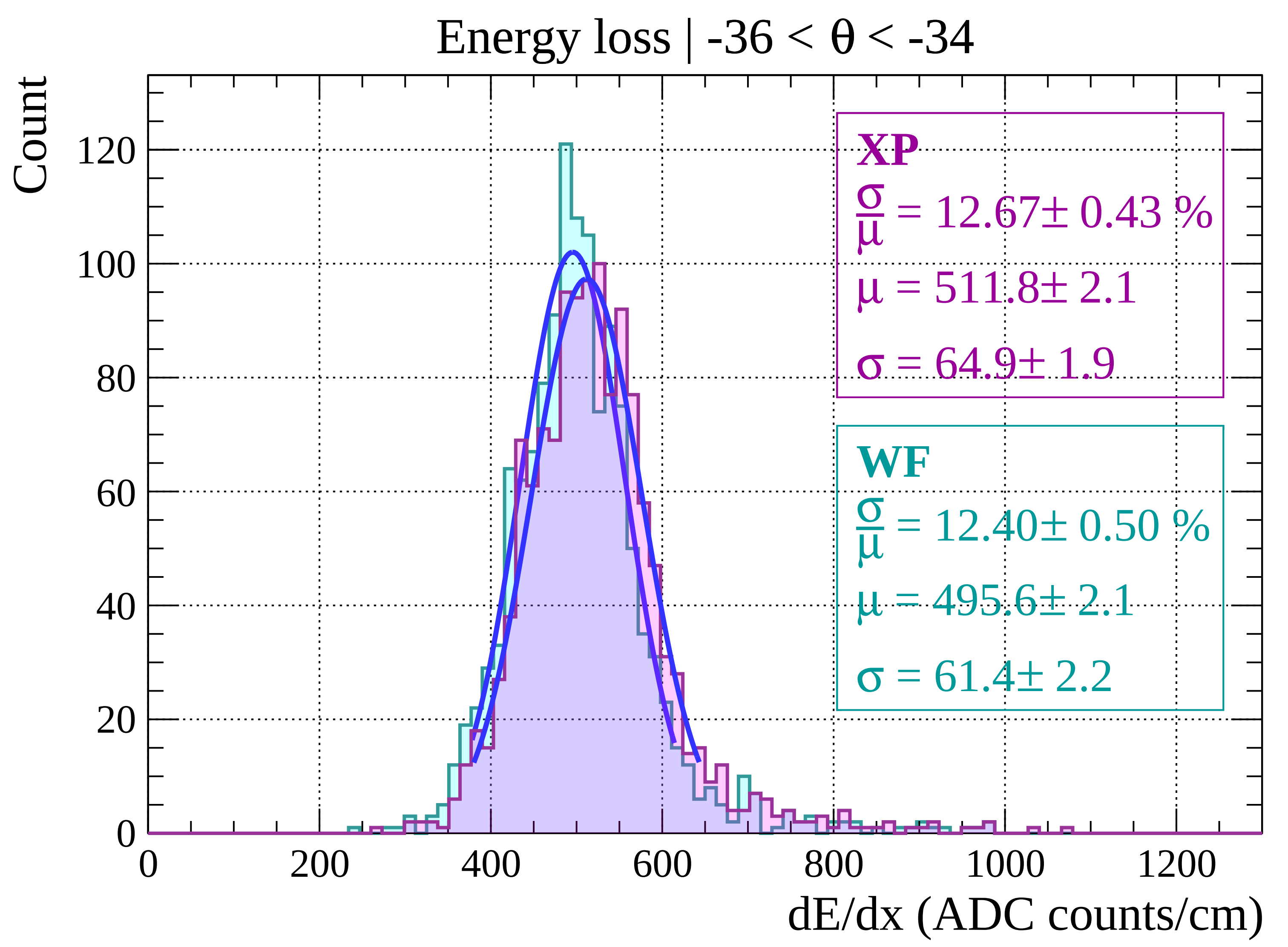


Energy loss | -42 < θ < -40
Count
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.20± 0.60 %
μ = 514.8± 2.8
σ = 67.9± 2.7
WF
σ/μ = 13.37± 0.64 %
μ = 497.1± 2.8
σ = 66.5± 2.8

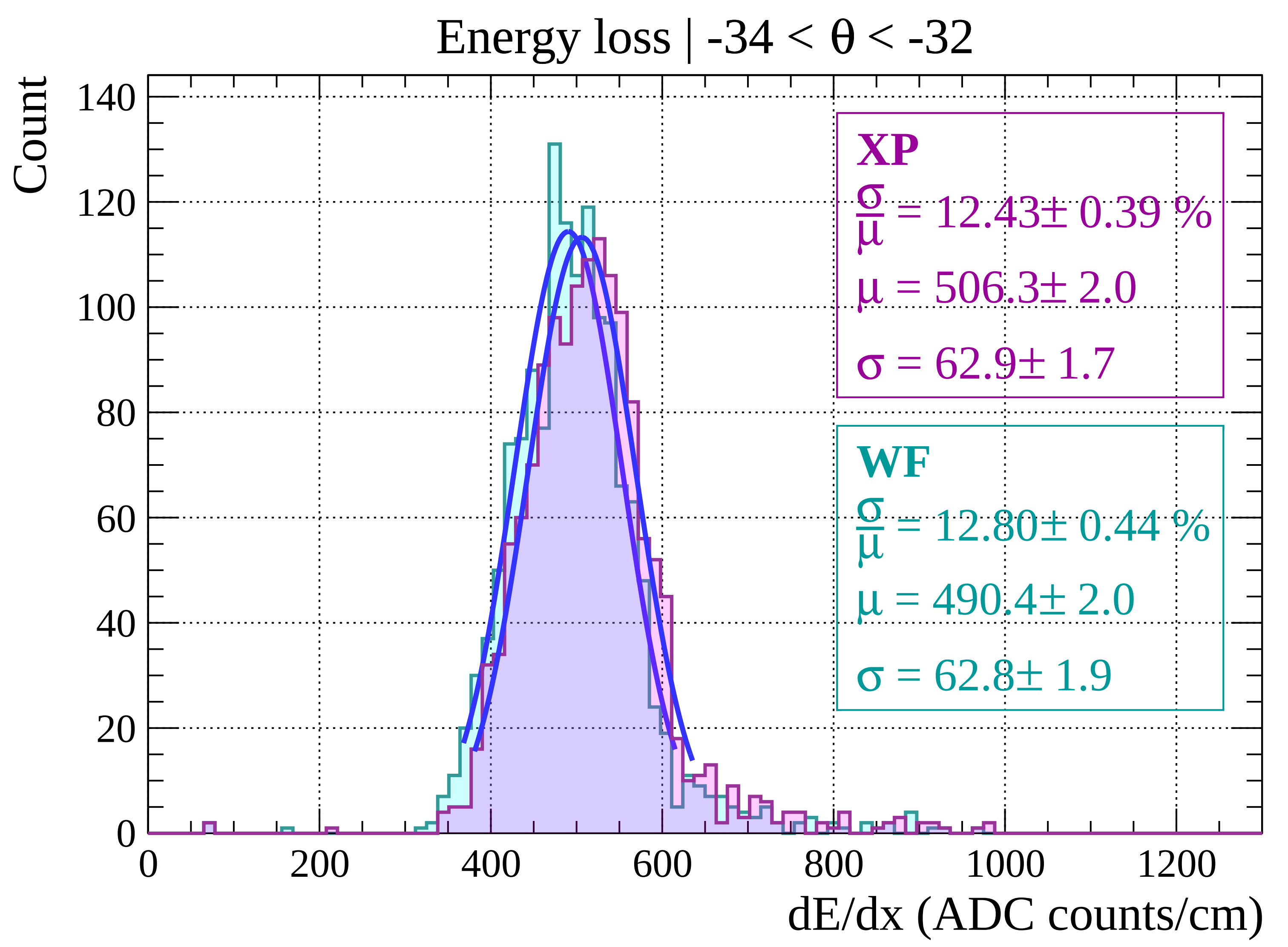


Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 12.22 ± 0.46 %
μ = 513.4 ± 2.4
σ = 62.7 ± 2.0
WF
σ/μ = 12.55 ± 0.55 %
μ = 495.5 ± 2.5
σ = 62.2 ± 2.4

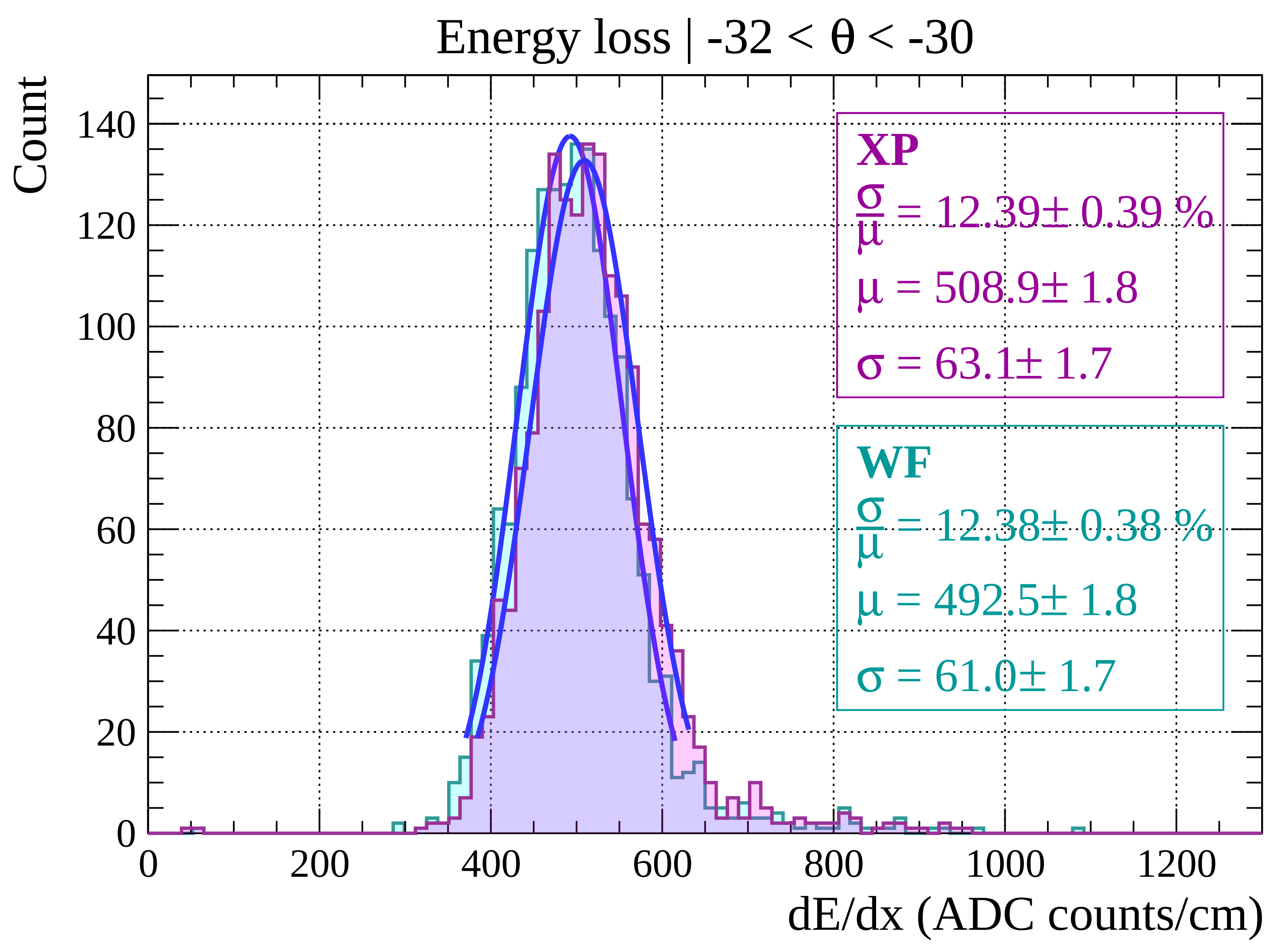

Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 12.20± 0.42 %
μ = 512.4± 2.2
σ = 62.5± 1.9
WF
σ/μ = 12.41± 0.45 %
μ = 496.9± 2.2
σ = 61.7± 2.0



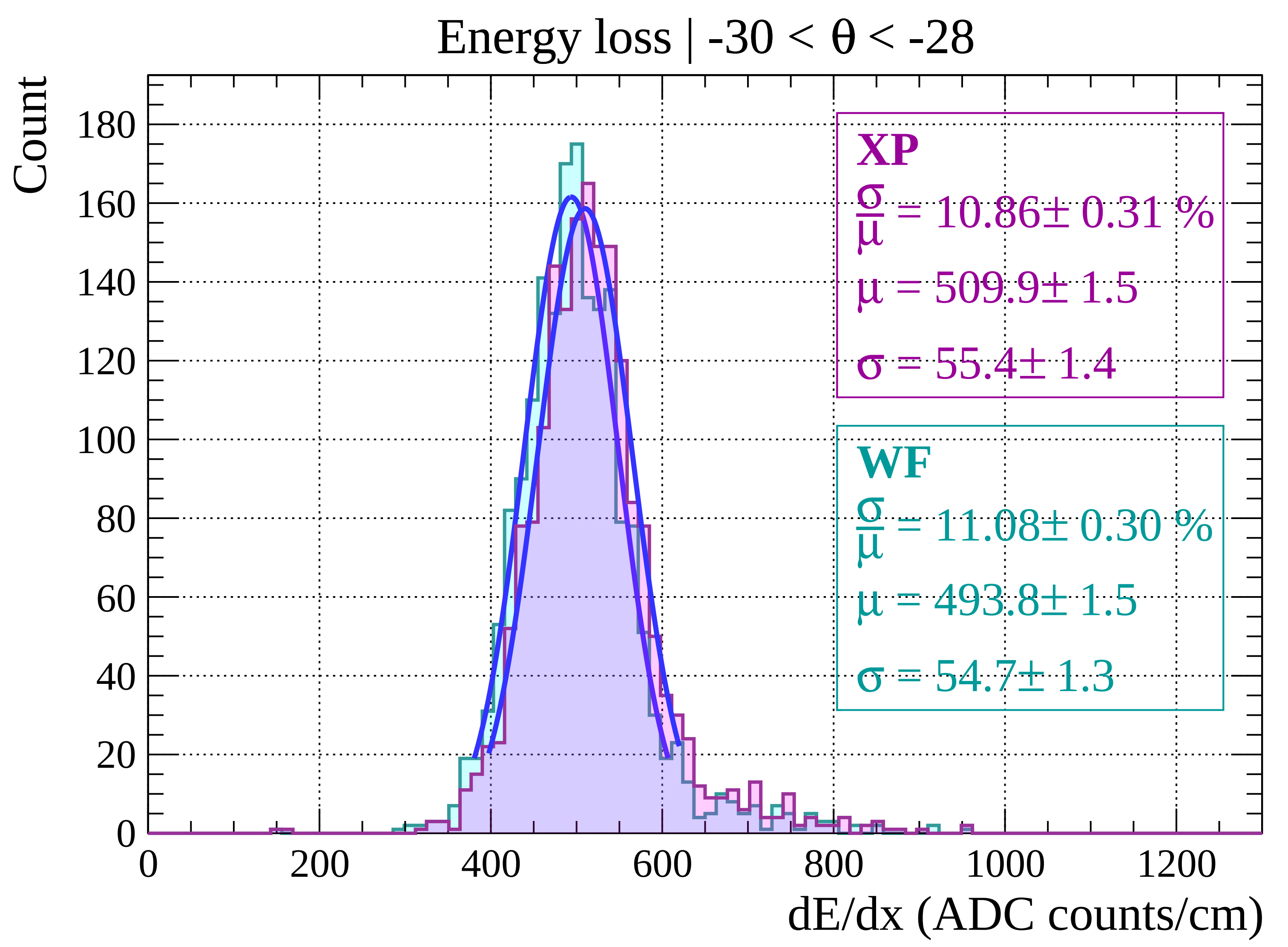
Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 12.67± 0.43 %
μ = 511.8± 2.1
σ = 64.9± 1.9
WF
σ/μ = 12.40± 0.50 %
μ = 495.6± 2.1
σ = 61.4± 2.2



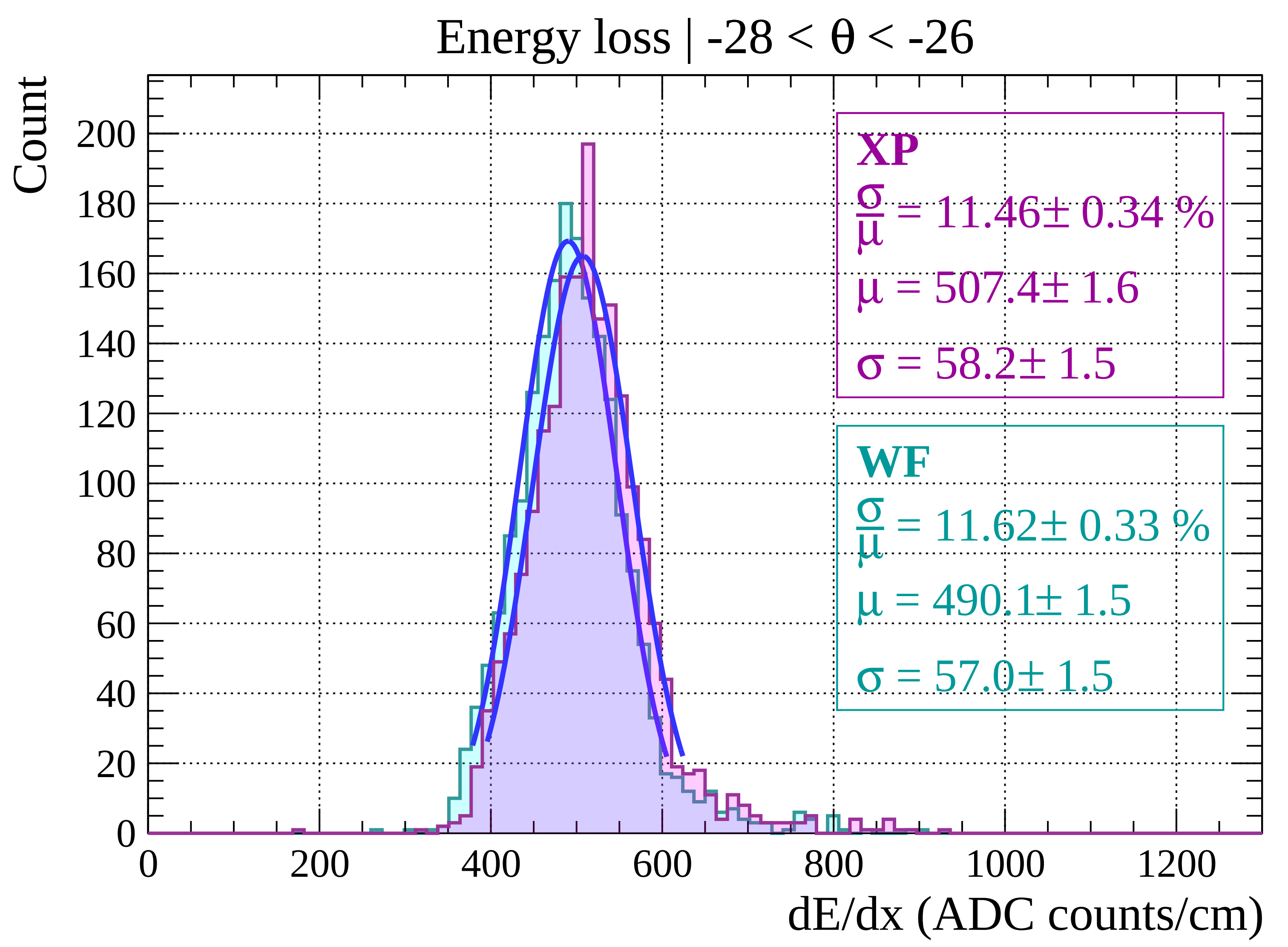
Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 12.43± 0.39 %
μ = 506.3± 2.0
σ = 62.9± 1.7
WF
σ/μ = 12.80± 0.44 %
μ = 490.4± 2.0
σ = 62.8± 1.9



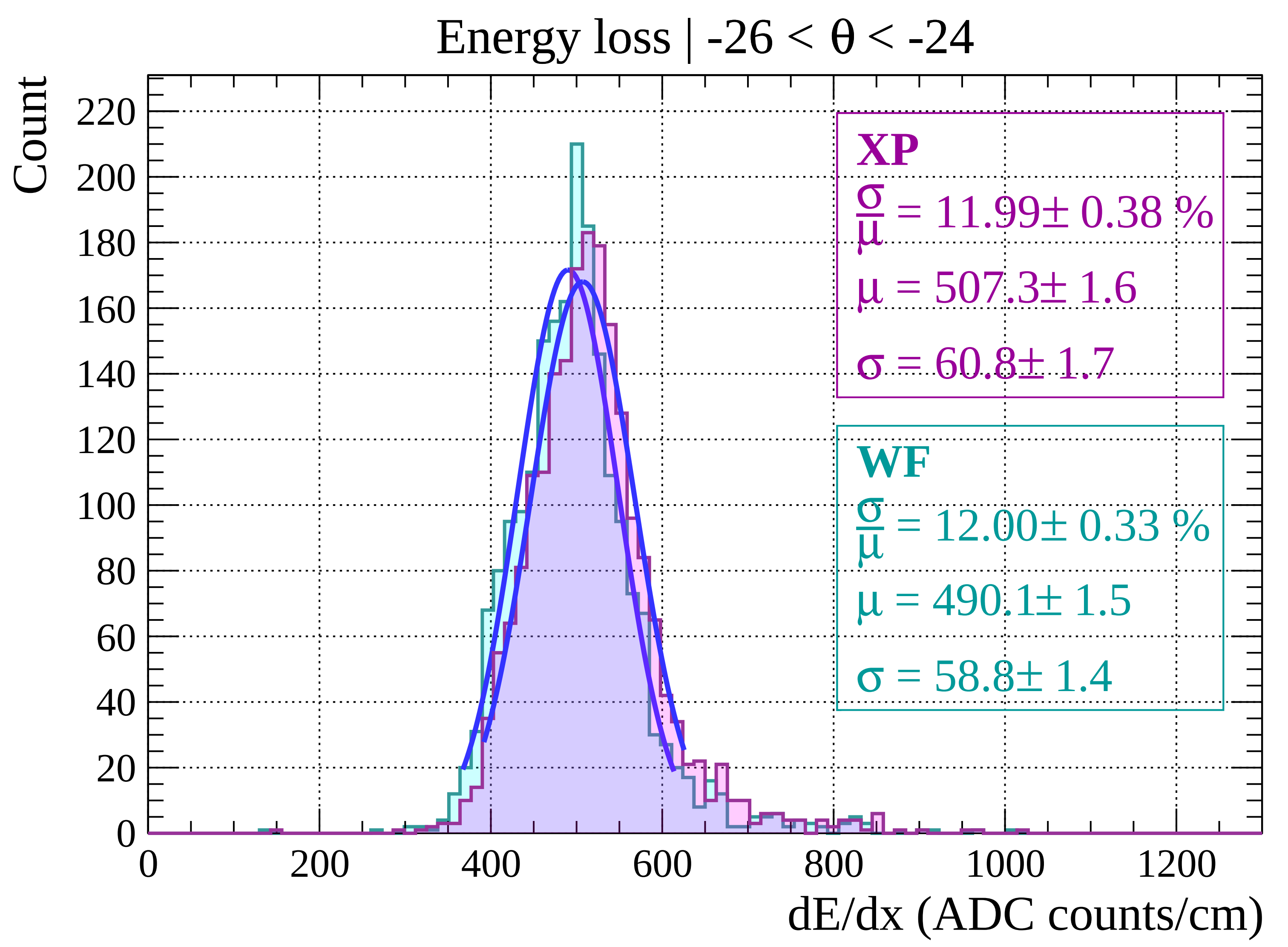
Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 12.39± 0.39 %
μ = 508.9± 1.8
σ = 63.1± 1.7
WF
σ/μ = 12.38± 0.38 %
μ = 492.5± 1.8
σ = 61.0± 1.7



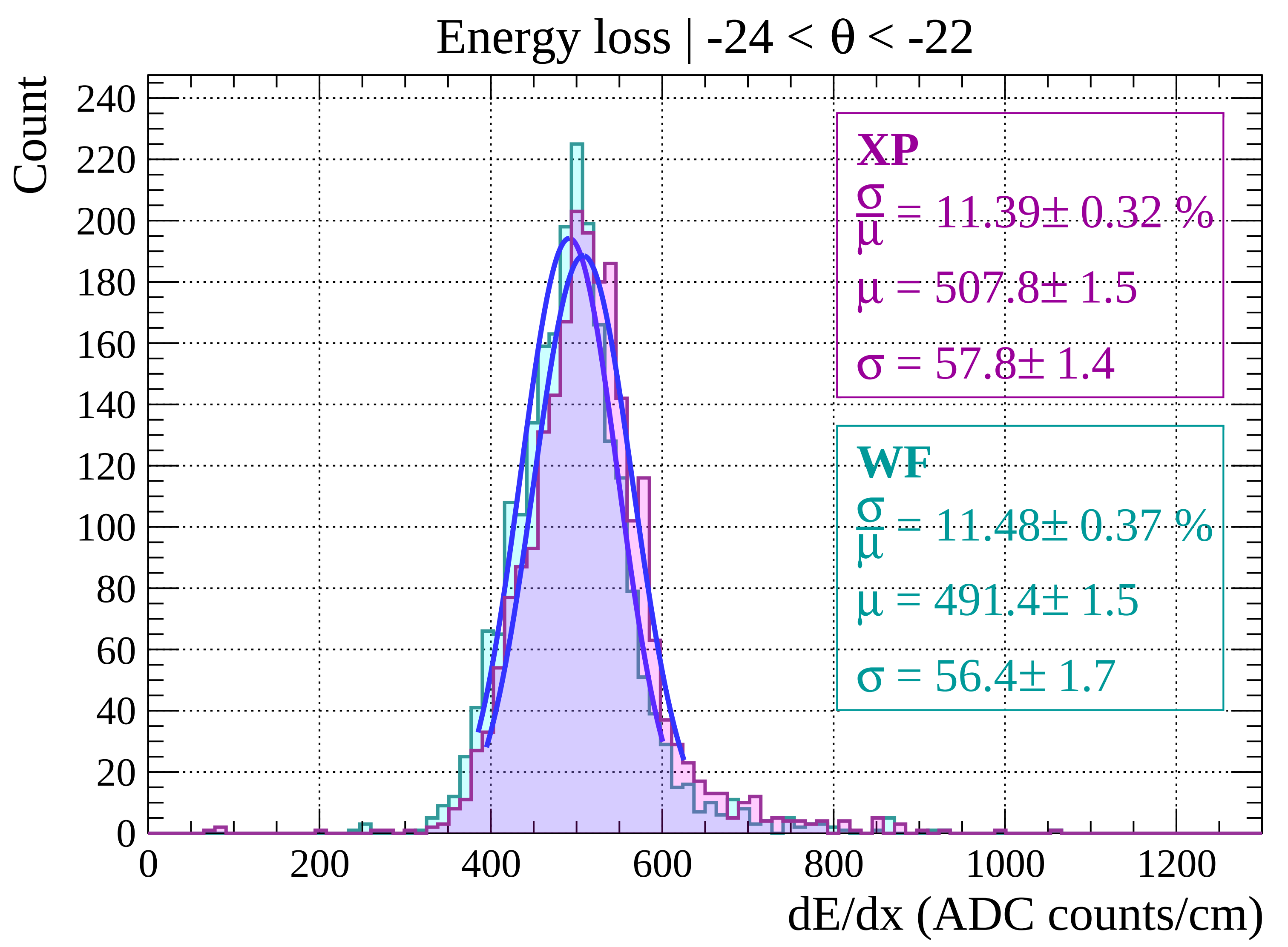
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 10.86± 0.31 %
μ = 509.9± 1.5
σ = 55.4± 1.4
WF
σ/μ = 11.08± 0.30 %
μ = 493.8± 1.5
σ = 54.7± 1.3

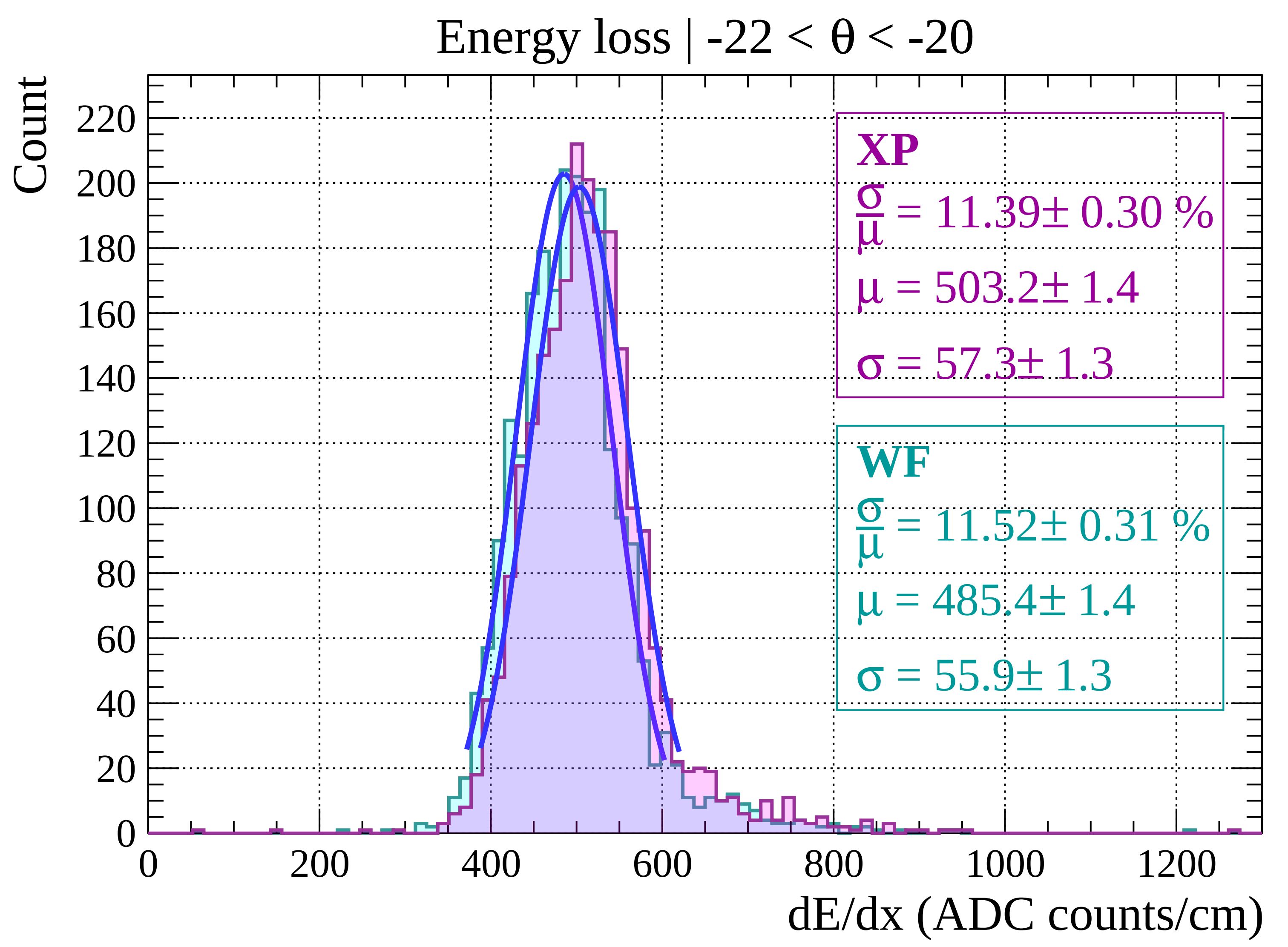


Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.46± 0.34 %
μ = 507.4± 1.6
σ = 58.2± 1.5
WF
σ/μ = 11.62± 0.33 %
μ = 490.1± 1.5
σ = 57.0± 1.5
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200

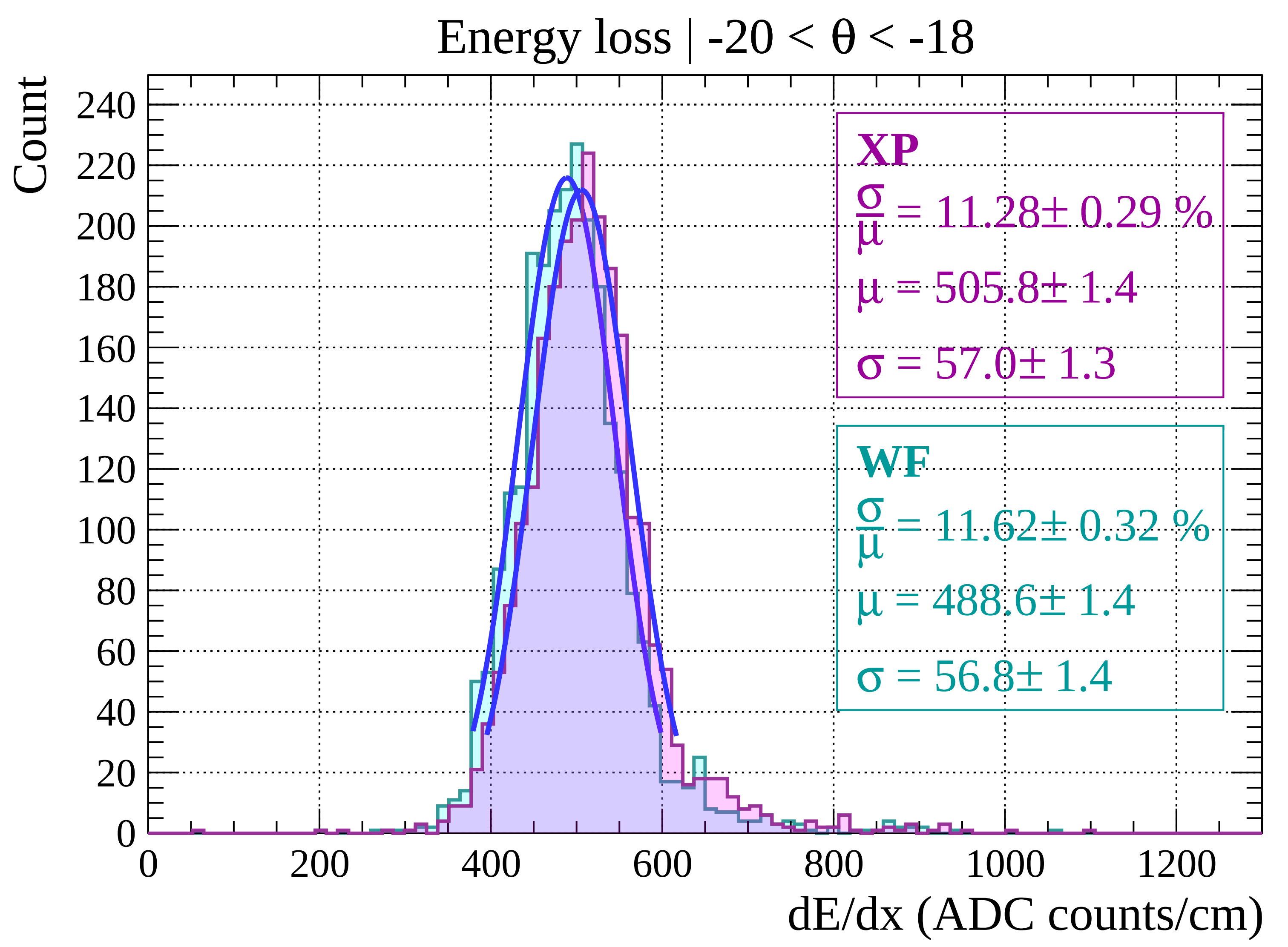


Energy loss | -26 < θ < -24
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.99± 0.38 %
μ = 507.3± 1.6
σ = 60.8± 1.7
WF
σ/μ = 12.00± 0.33 %
μ = 490.1± 1.5
σ = 58.8± 1.4

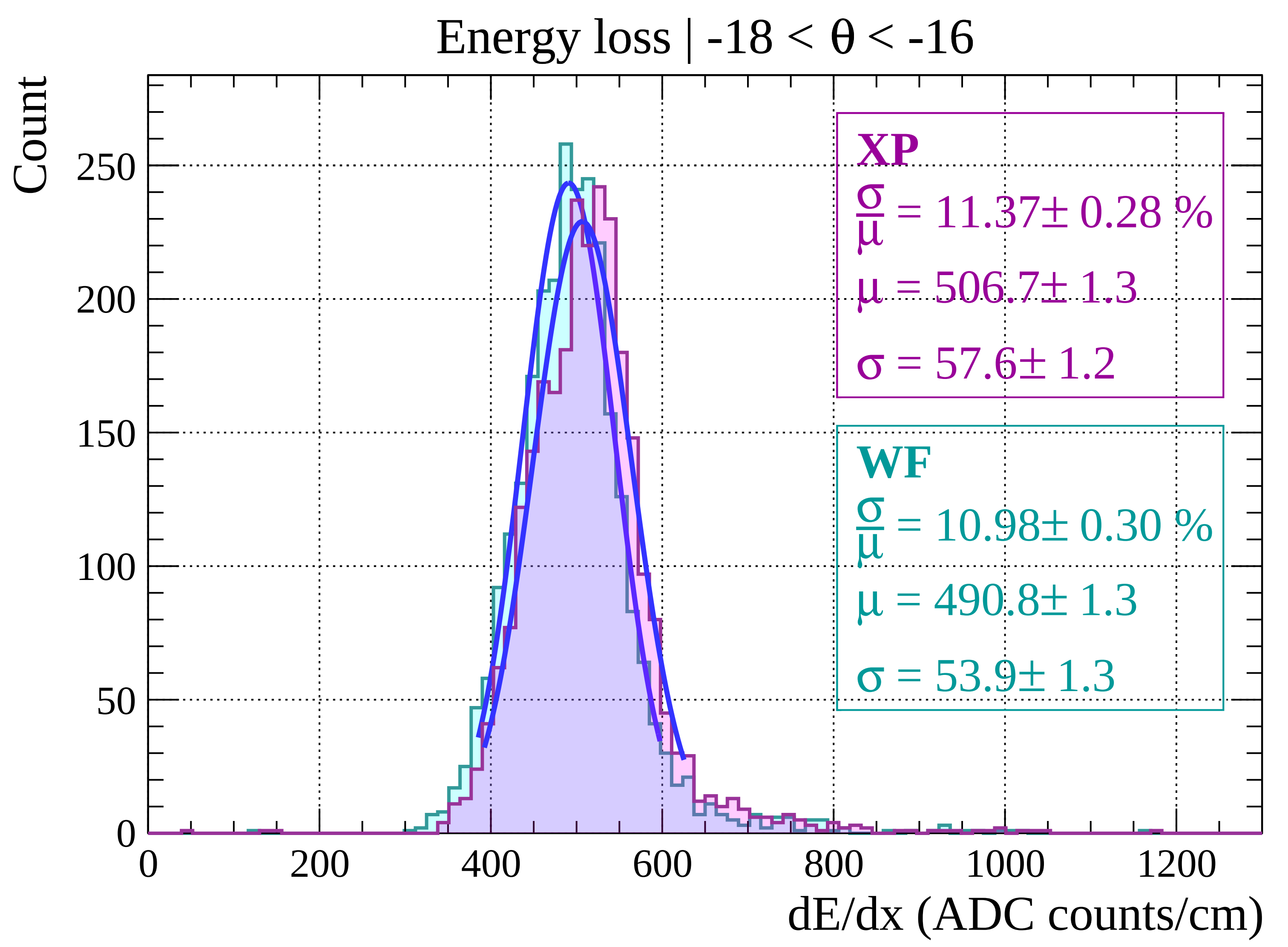

Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.39± 0.32 %
μ = 507.8± 1.5
σ = 57.8± 1.4
WF
σ/μ = 11.48± 0.37 %
μ = 491.4± 1.5
σ = 56.4± 1.7
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200



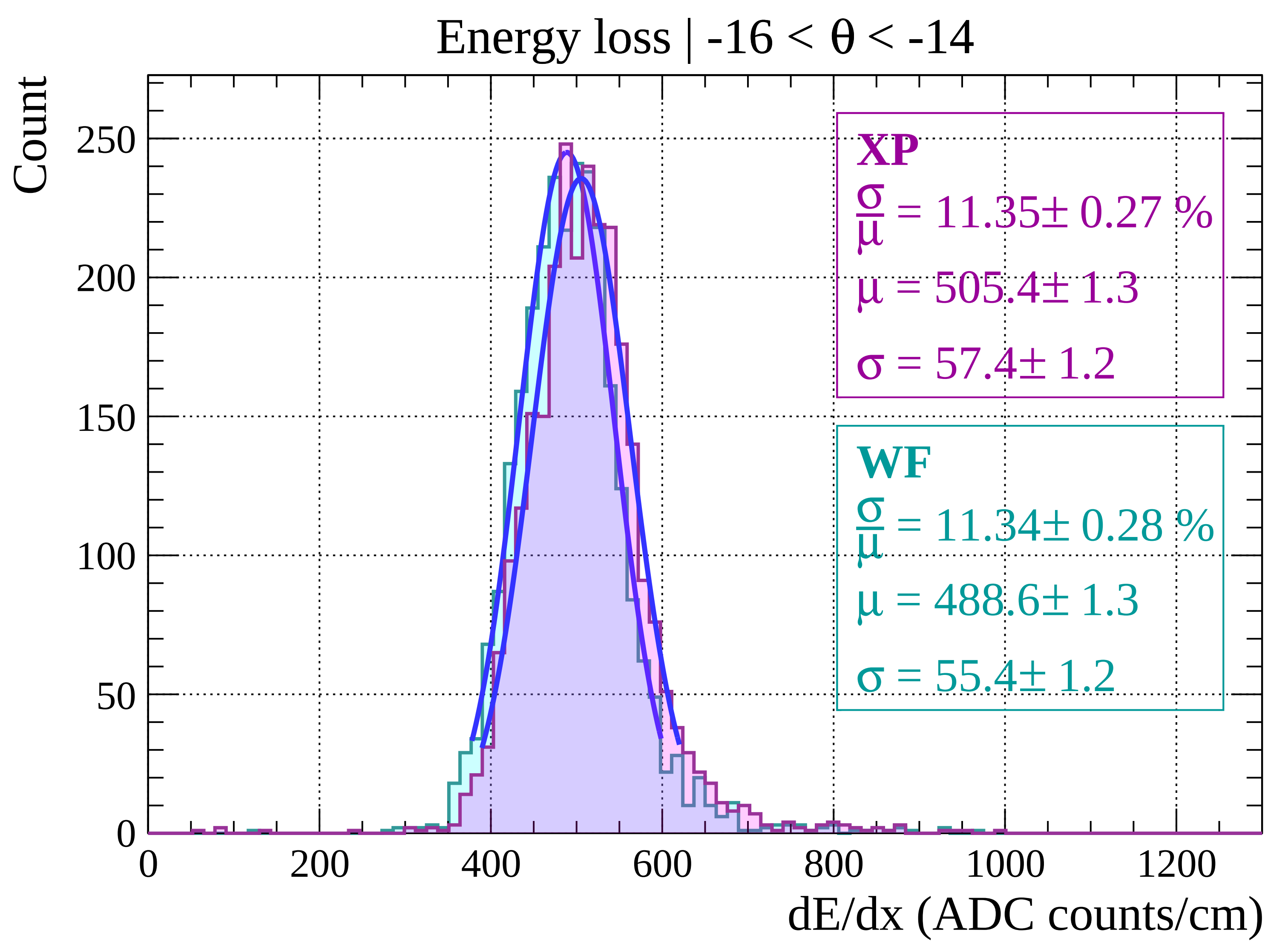
Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.39± 0.30 %
μ = 503.2± 1.4
σ = 57.3± 1.3
WF
σ/μ = 11.52± 0.31 %
μ = 485.4± 1.4
σ = 55.9± 1.3



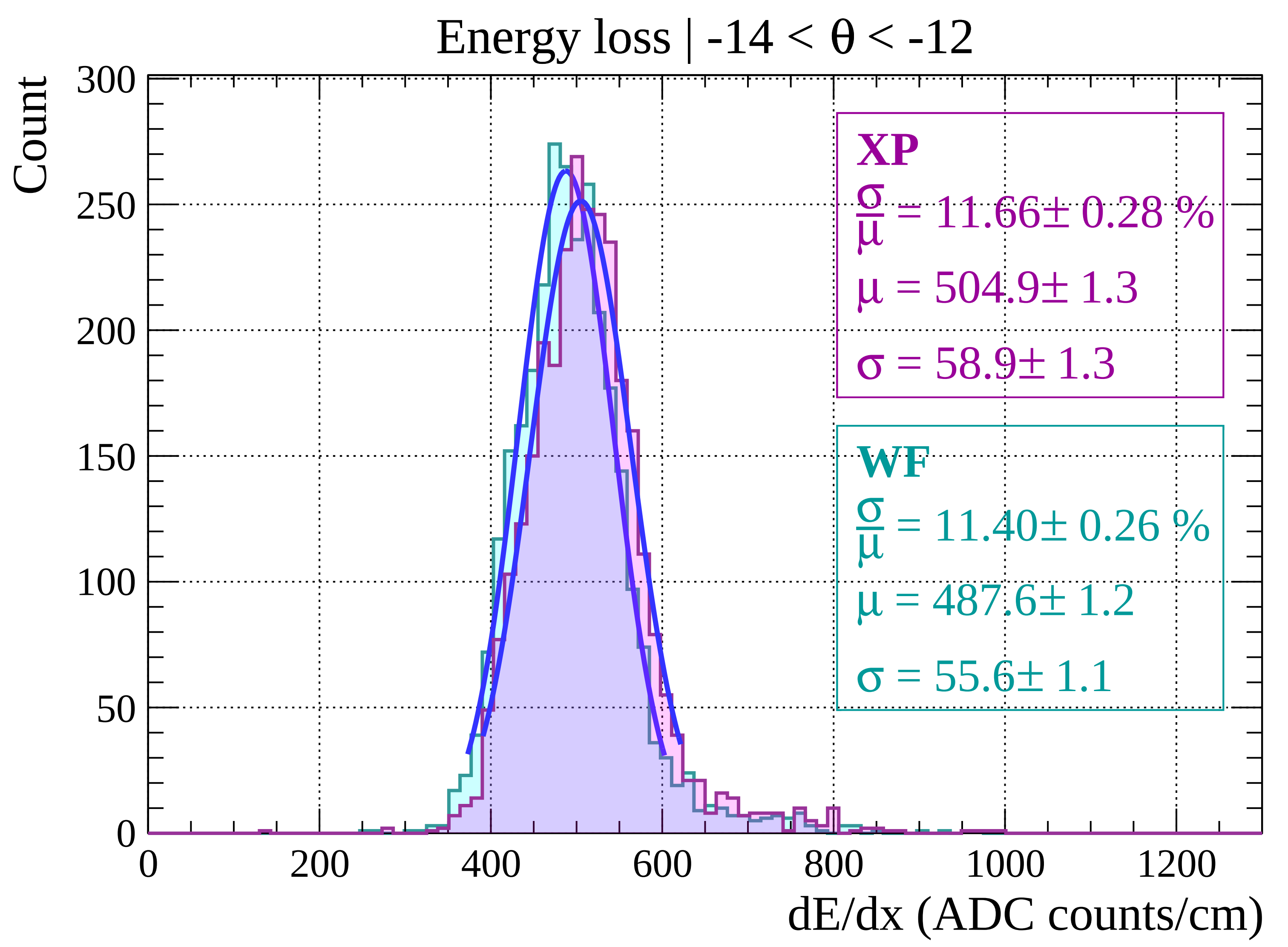
Energy loss | -20 < θ < -18
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.28± 0.29 %
μ = 505.8± 1.4
σ = 57.0± 1.3
WF
σ/μ = 11.62± 0.32 %
μ = 488.6± 1.4
σ = 56.8± 1.4



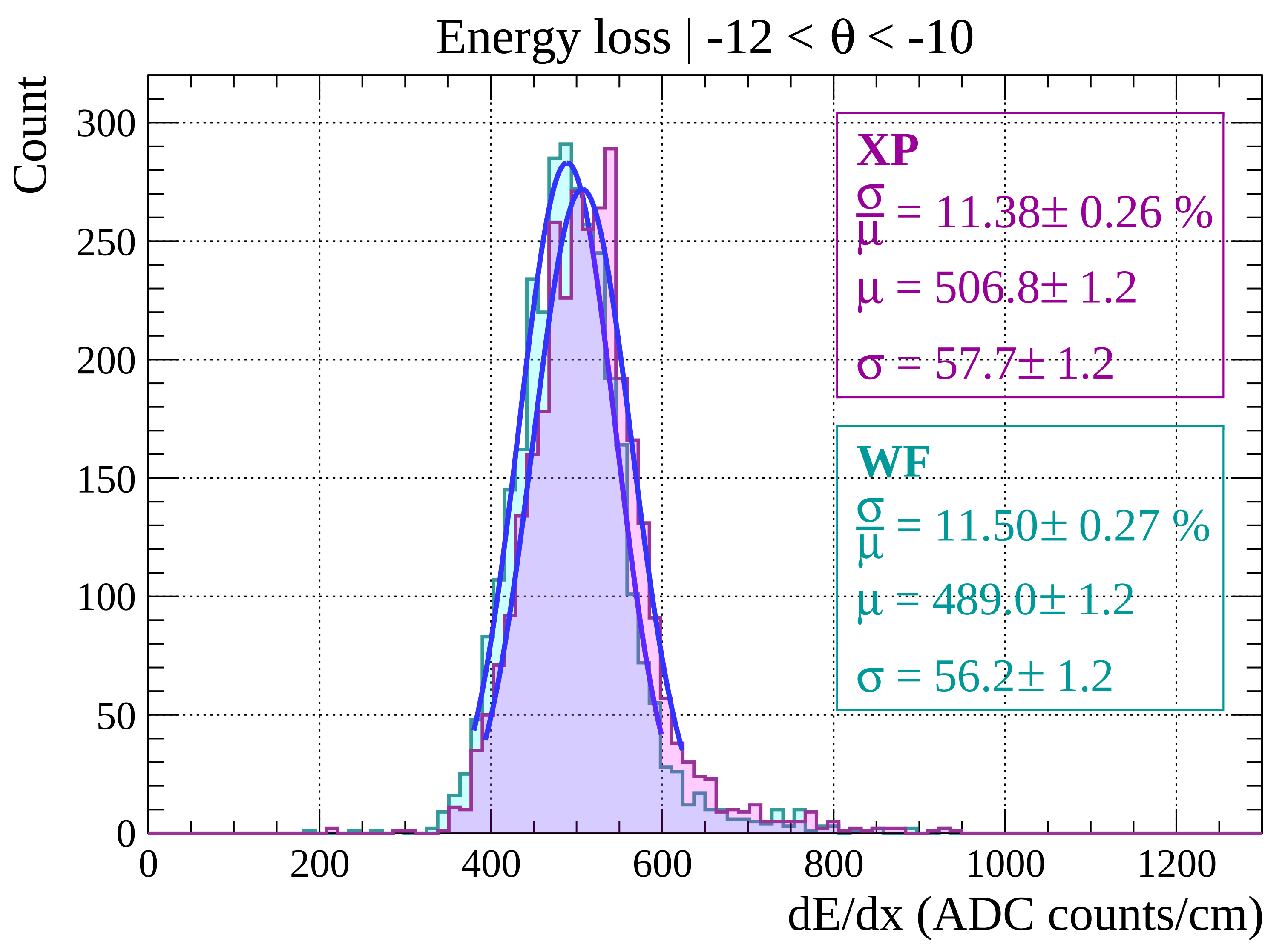
Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.37± 0.28 %
μ = 506.7± 1.3
σ = 57.6± 1.2
WF
σ/μ = 10.98± 0.30 %
μ = 490.8± 1.3
σ = 53.9± 1.3



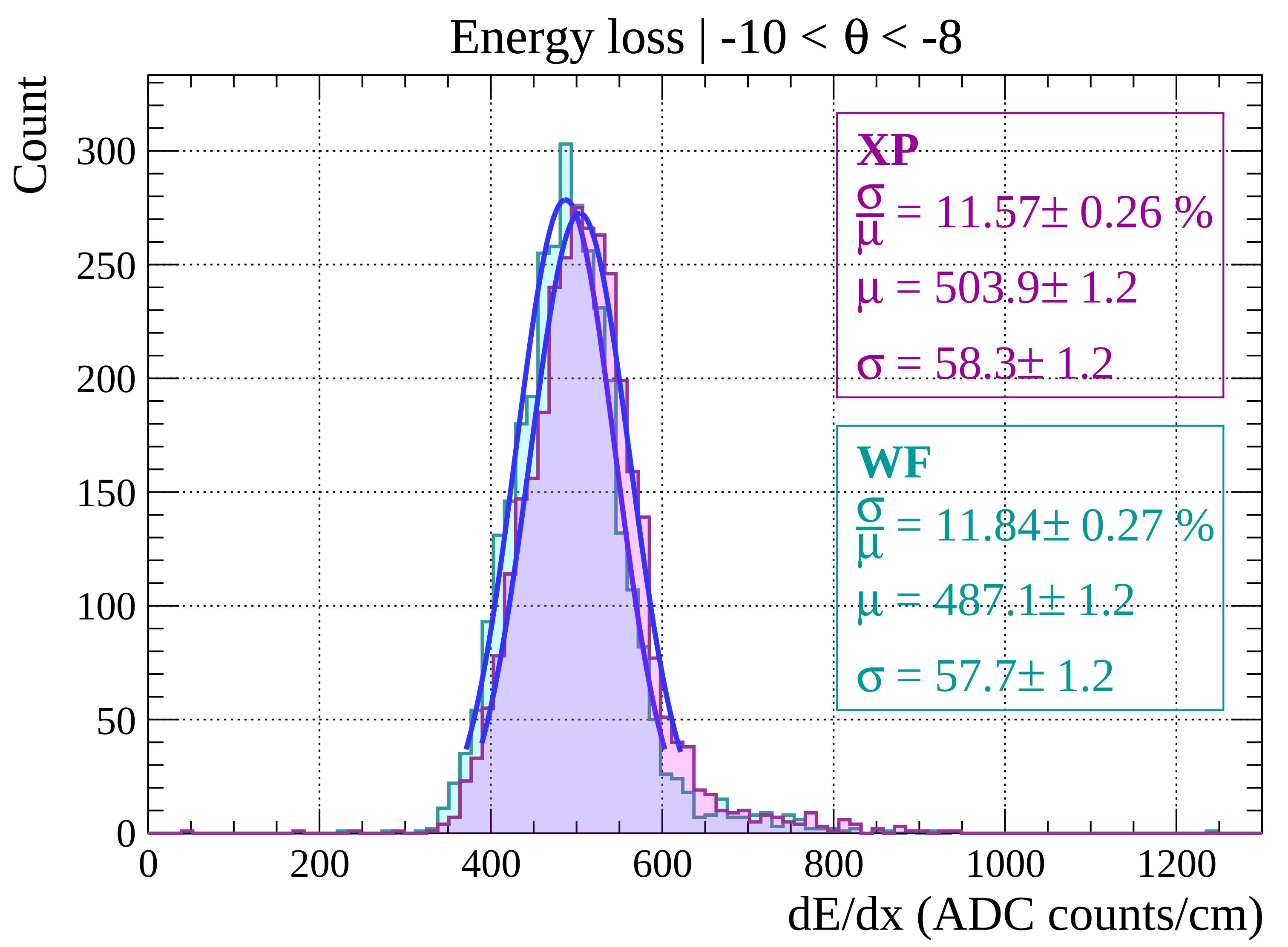
Energy loss | -16 < θ < -14
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.35± 0.27 %
μ = 505.4± 1.3
σ = 57.4± 1.2
WF
σ/μ = 11.34± 0.28 %
μ = 488.6± 1.3
σ = 55.4± 1.2



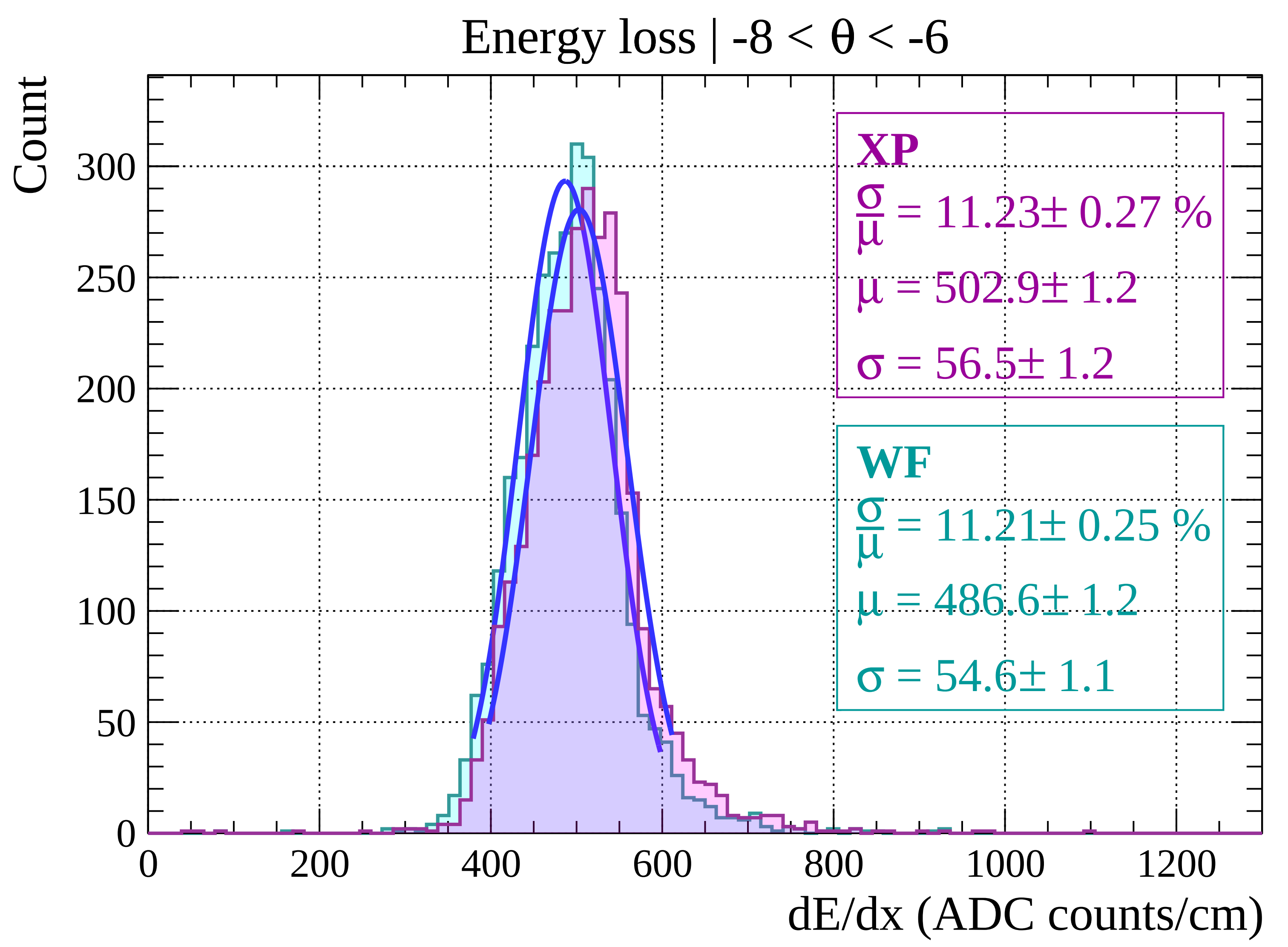
Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 11.66± 0.28 %
μ = 504.9± 1.3
σ = 58.9± 1.3
WF
σ/μ = 11.40± 0.26 %
μ = 487.6± 1.2
σ = 55.6± 1.1



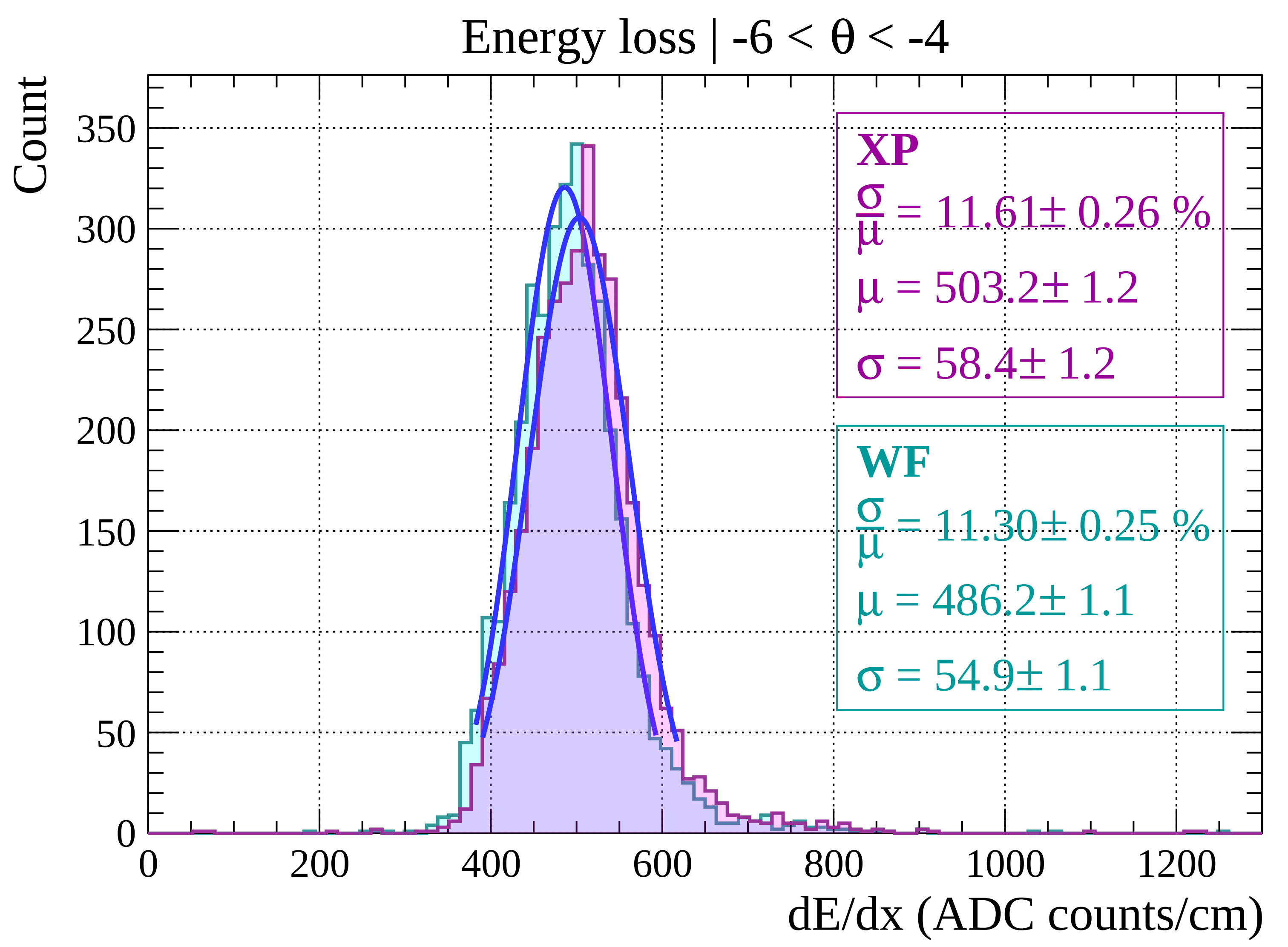
Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 11.38± 0.26 %
μ = 506.8± 1.2
σ = 57.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.50± 0.27 %
μ = 489.0± 1.2
σ = 56.2± 1.2



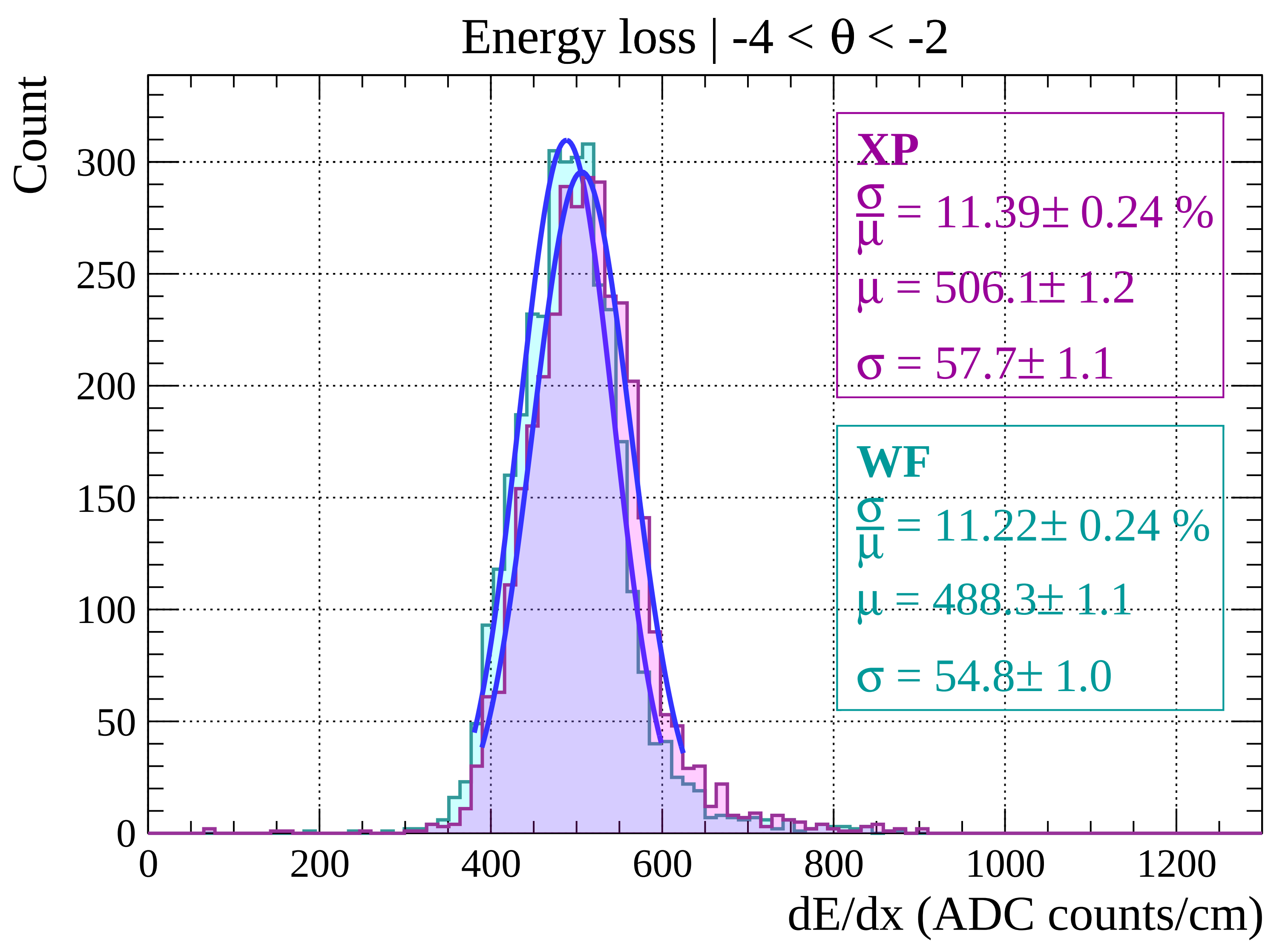
Energy loss | -10 < θ < -8
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.57± 0.26 %
μ = 503.9± 1.2
σ = 58.3± 1.2
WF
σ/μ = 11.84± 0.27 %
μ = 487.1± 1.2
σ = 57.7± 1.2



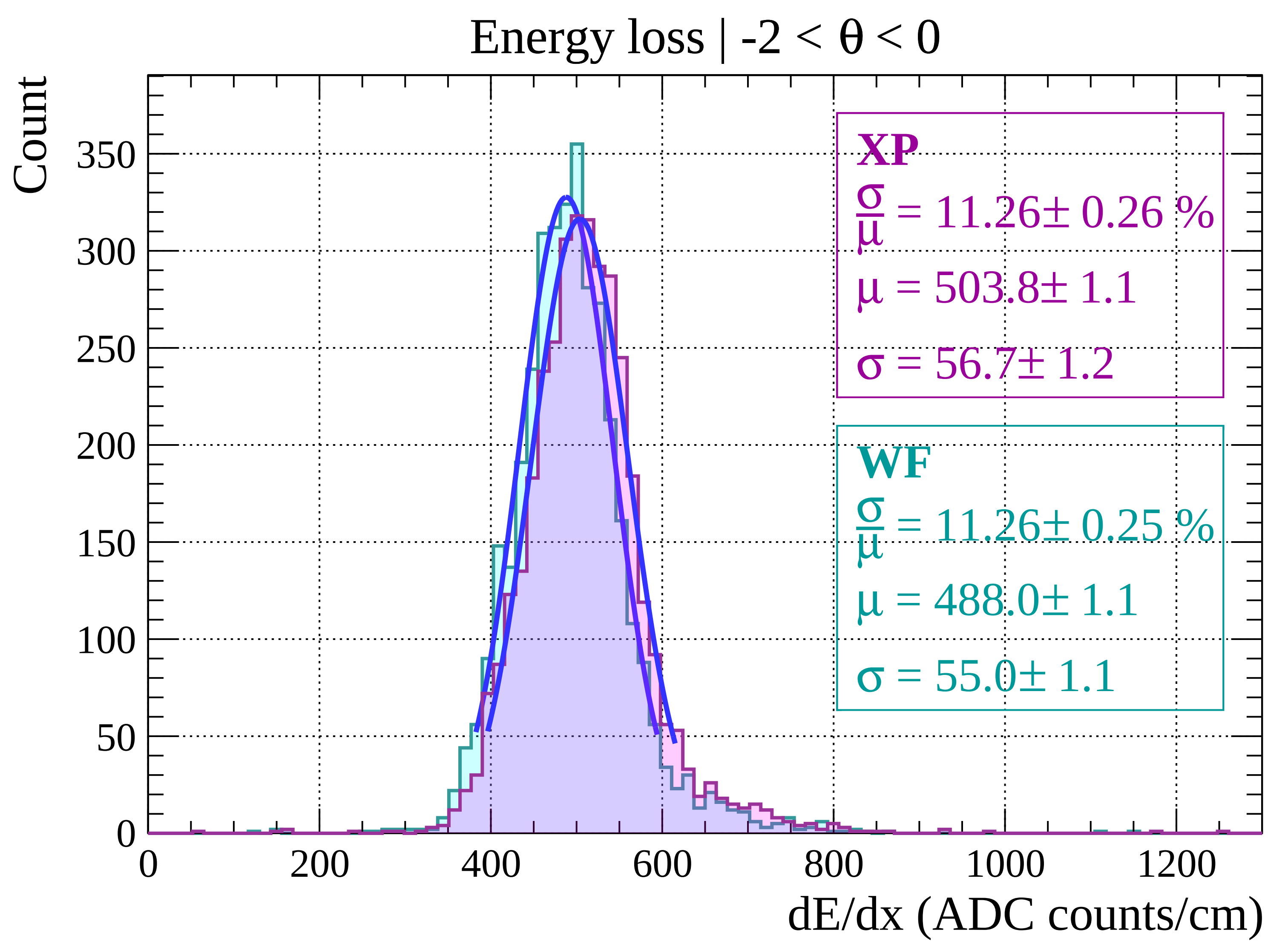
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 11.23± 0.27 %
μ = 502.9± 1.2
σ = 56.5± 1.2
WF
σ/μ = 11.21± 0.25 %
μ = 486.6± 1.2
σ = 54.6± 1.1



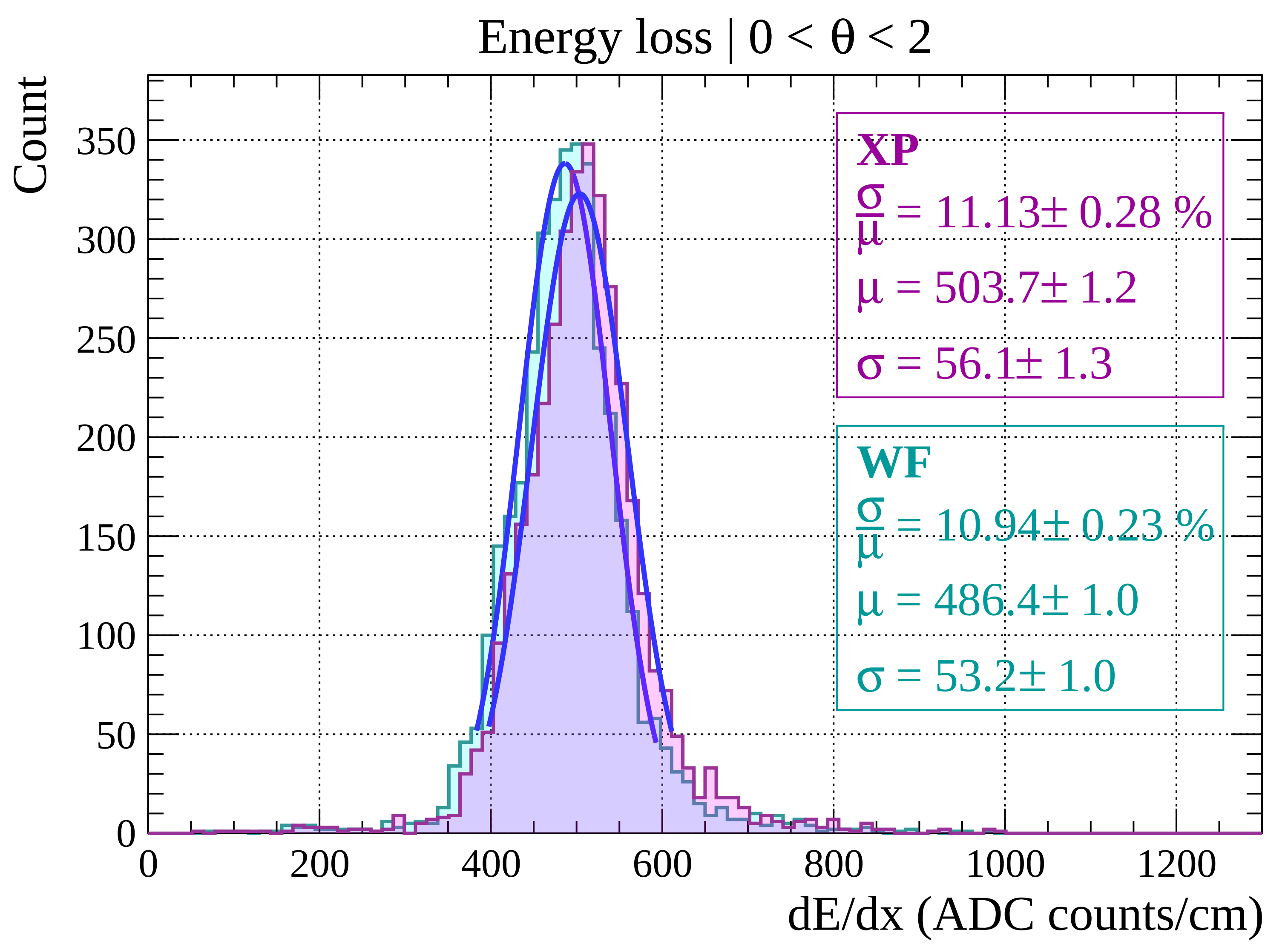
Energy loss | -6 < θ < -4
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.61± 0.26 %
μ = 503.2± 1.2
σ = 58.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.30± 0.25 %
μ = 486.2± 1.1
σ = 54.9± 1.1

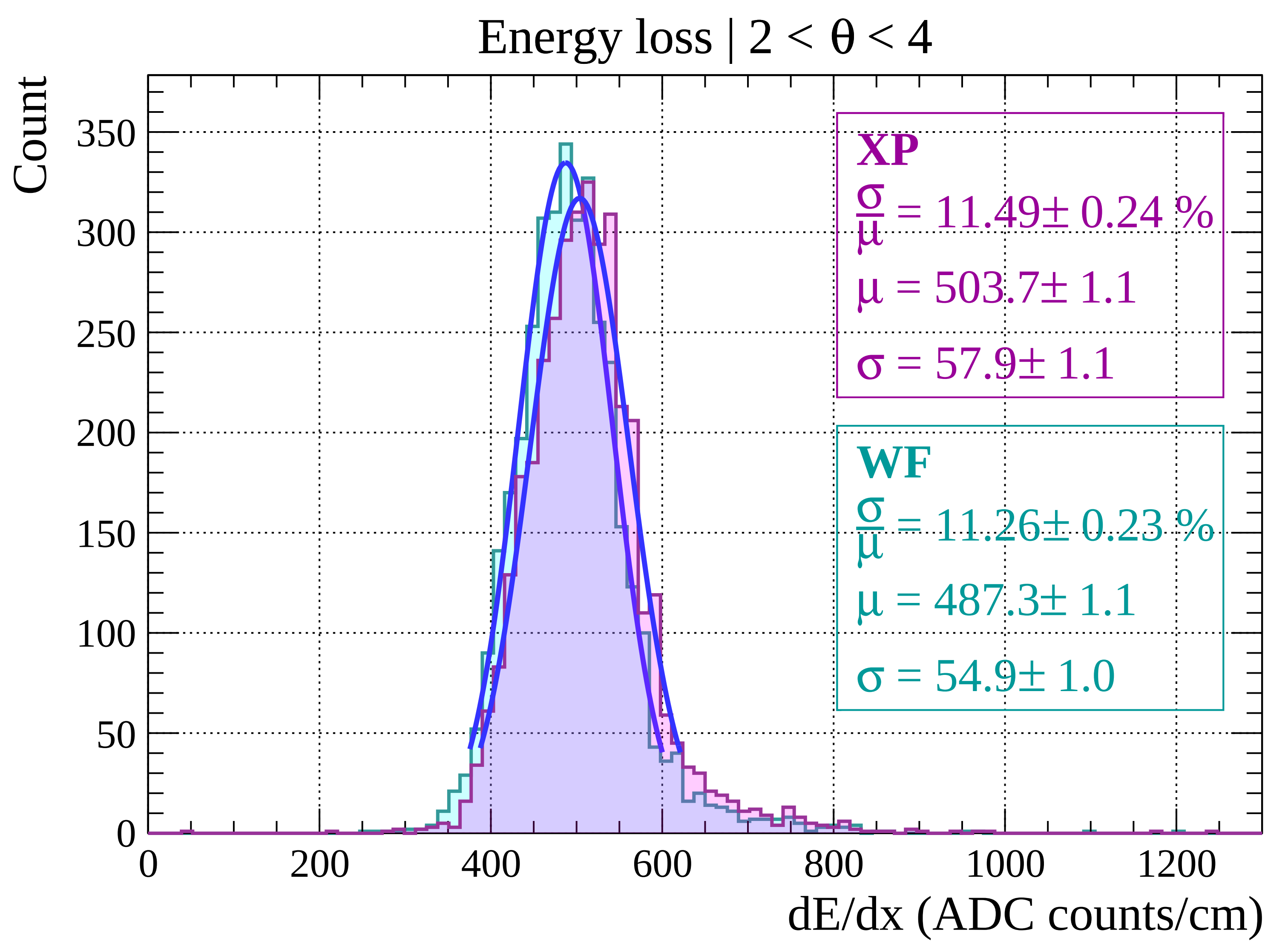


Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 11.39± 0.24 %
μ = 506.1± 1.2
σ = 57.7± 1.1
WF
σ/μ = 11.22± 0.24 %
μ = 488.3± 1.1
σ = 54.8± 1.0

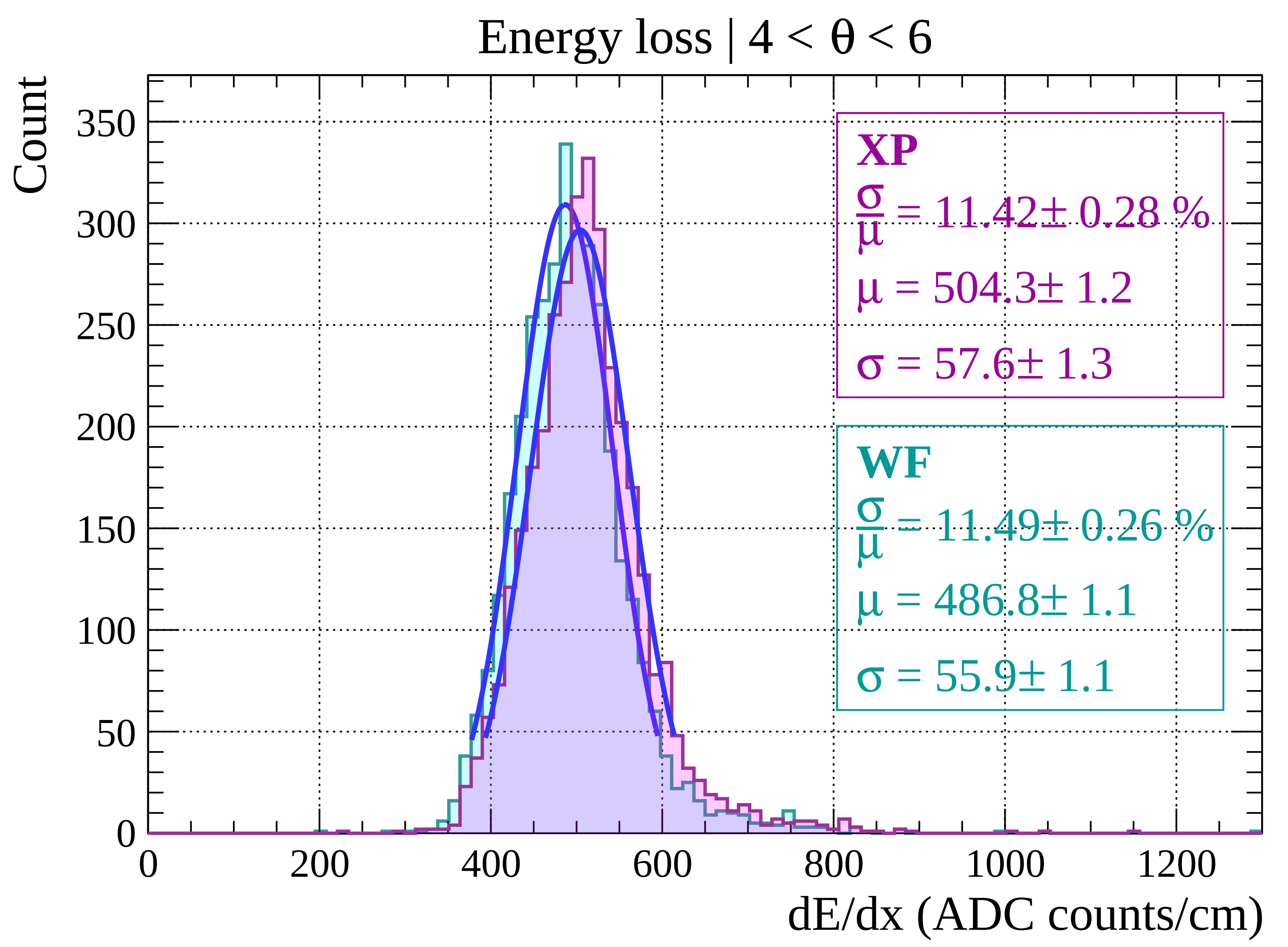


Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.26± 0.26 %
μ = 503.8± 1.1
σ = 56.7± 1.2
WF
σ/μ = 11.26± 0.25 %
μ = 488.0± 1.1
σ = 55.0± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

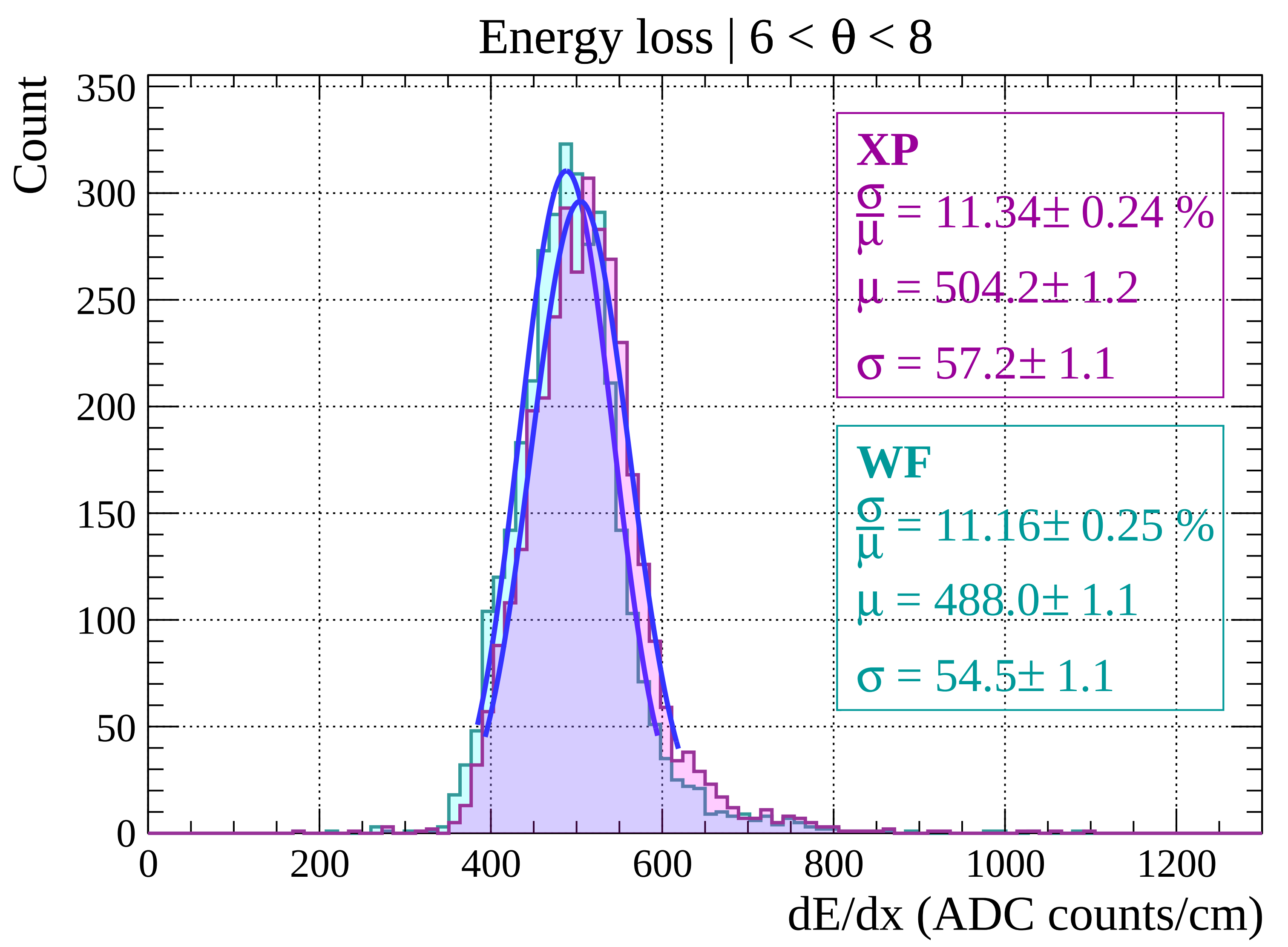

Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.13 ± 0.28 %
μ = 503.7 ± 1.2
σ = 56.1 ± 1.3
WF
σ/μ = 10.94 ± 0.23 %
μ = 486.4 ± 1.0
σ = 53.2 ± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



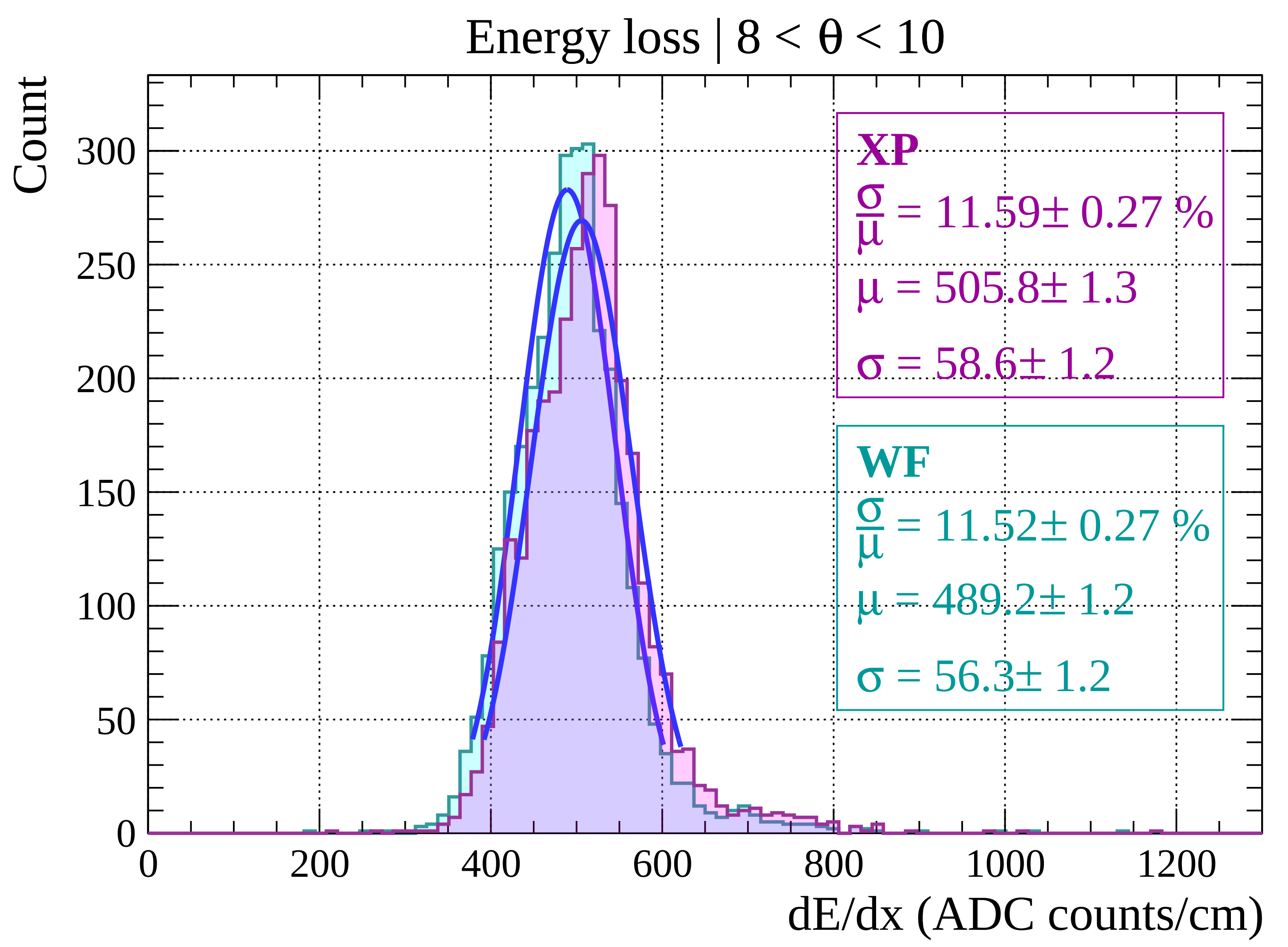
Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 11.49± 0.24 %
μ = 503.7± 1.1
σ = 57.9± 1.1
WF
σ/μ = 11.26± 0.23 %
μ = 487.3± 1.1
σ = 54.9± 1.0



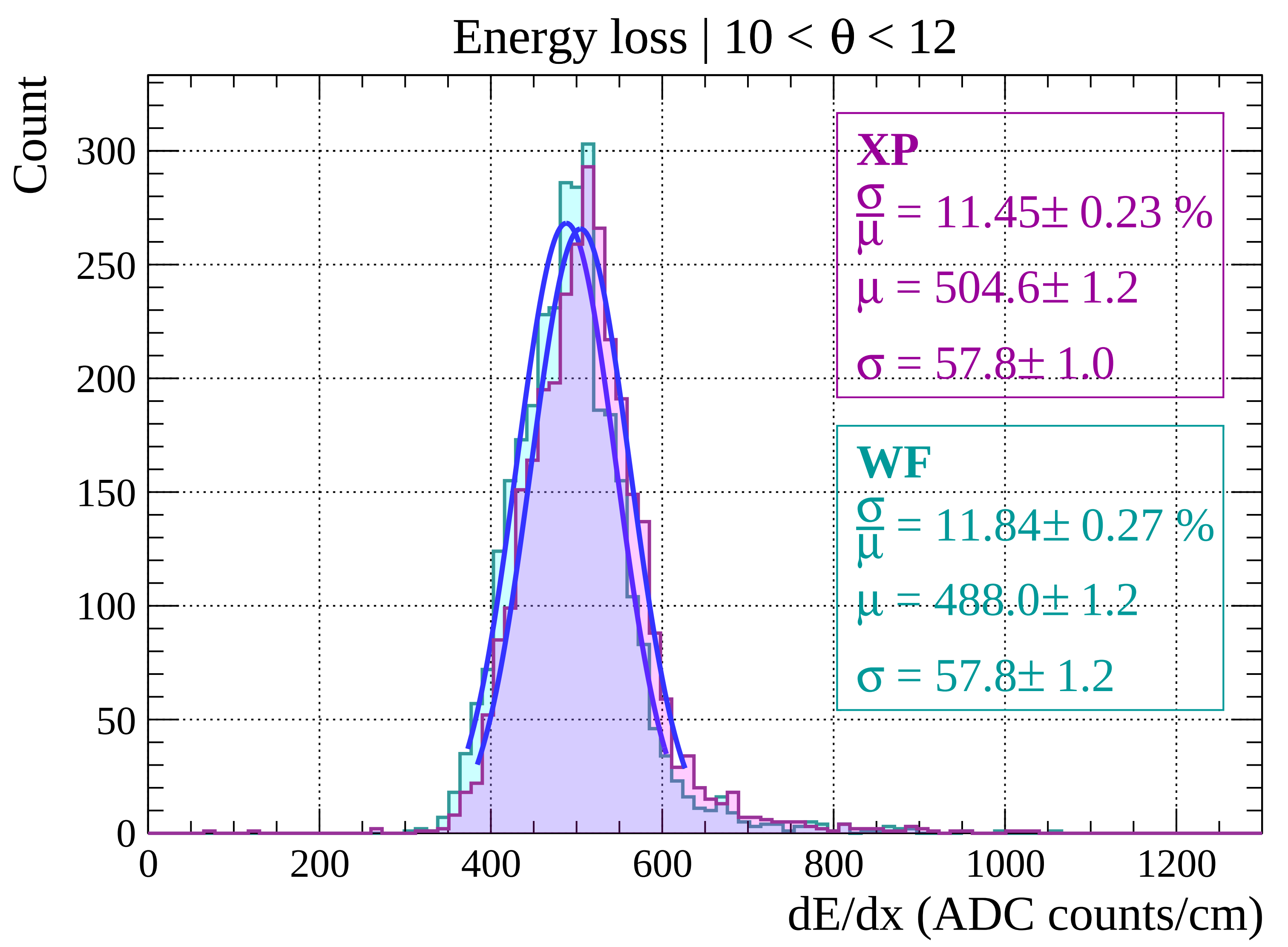
Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 11.42± 0.28 %
μ = 504.3± 1.2
σ = 57.6± 1.3
WF
σ/μ = 11.49± 0.26 %
μ = 486.8± 1.1
σ = 55.9± 1.1

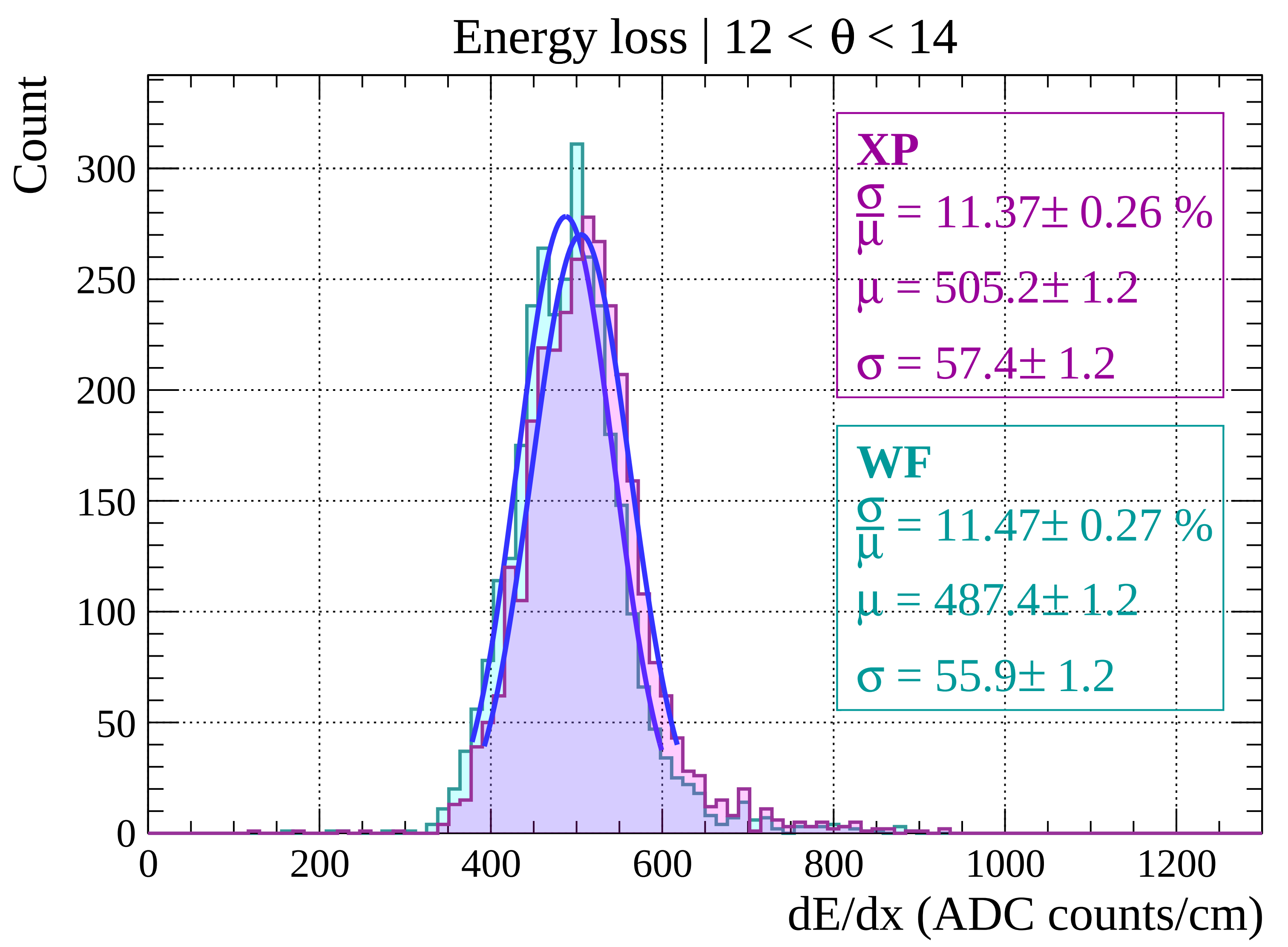


Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 11.34± 0.24 %
μ = 504.2± 1.2
σ = 57.2± 1.1
WF
σ/μ = 11.16± 0.25 %
μ = 488.0± 1.1
σ = 54.5± 1.1

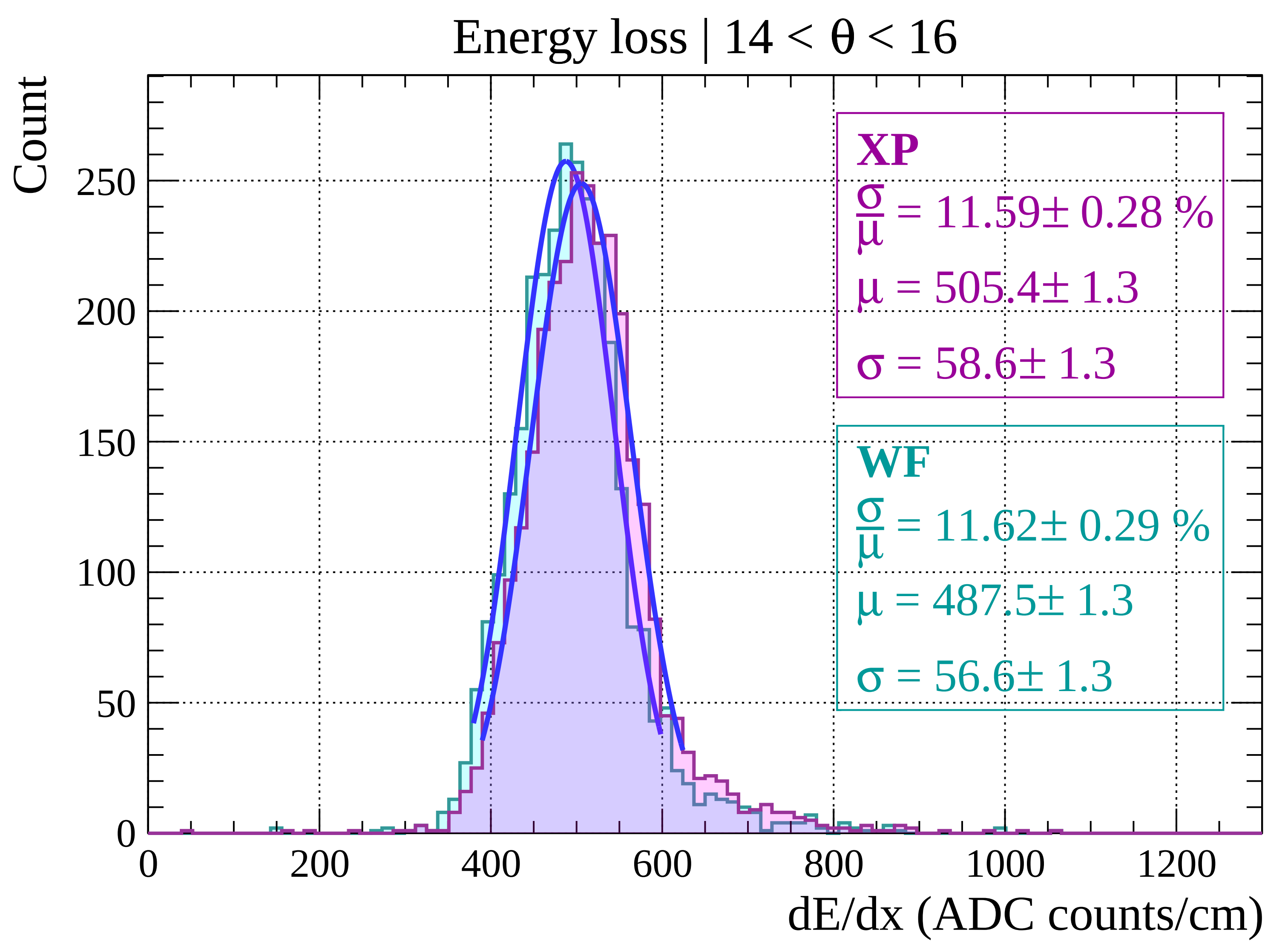


Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 11.59± 0.27 %
μ = 505.8± 1.3
σ = 58.6± 1.2
WF
σ/μ = 11.52± 0.27 %
μ = 489.2± 1.2
σ = 56.3± 1.2

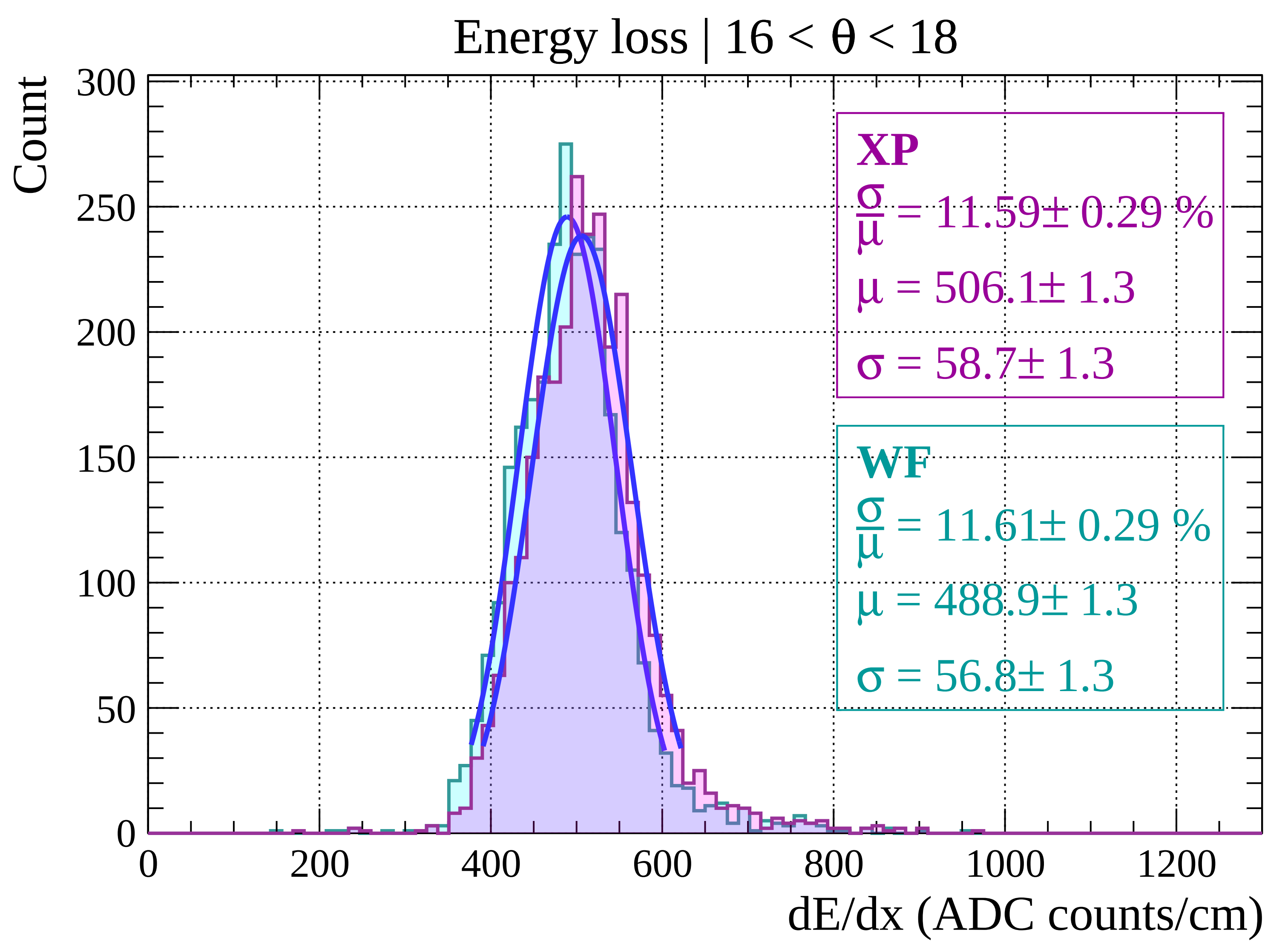

Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.45± 0.23 %
μ = 504.6± 1.2
σ = 57.8± 1.0
WF
σ/μ = 11.84± 0.27 %
μ = 488.0± 1.2
σ = 57.8± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



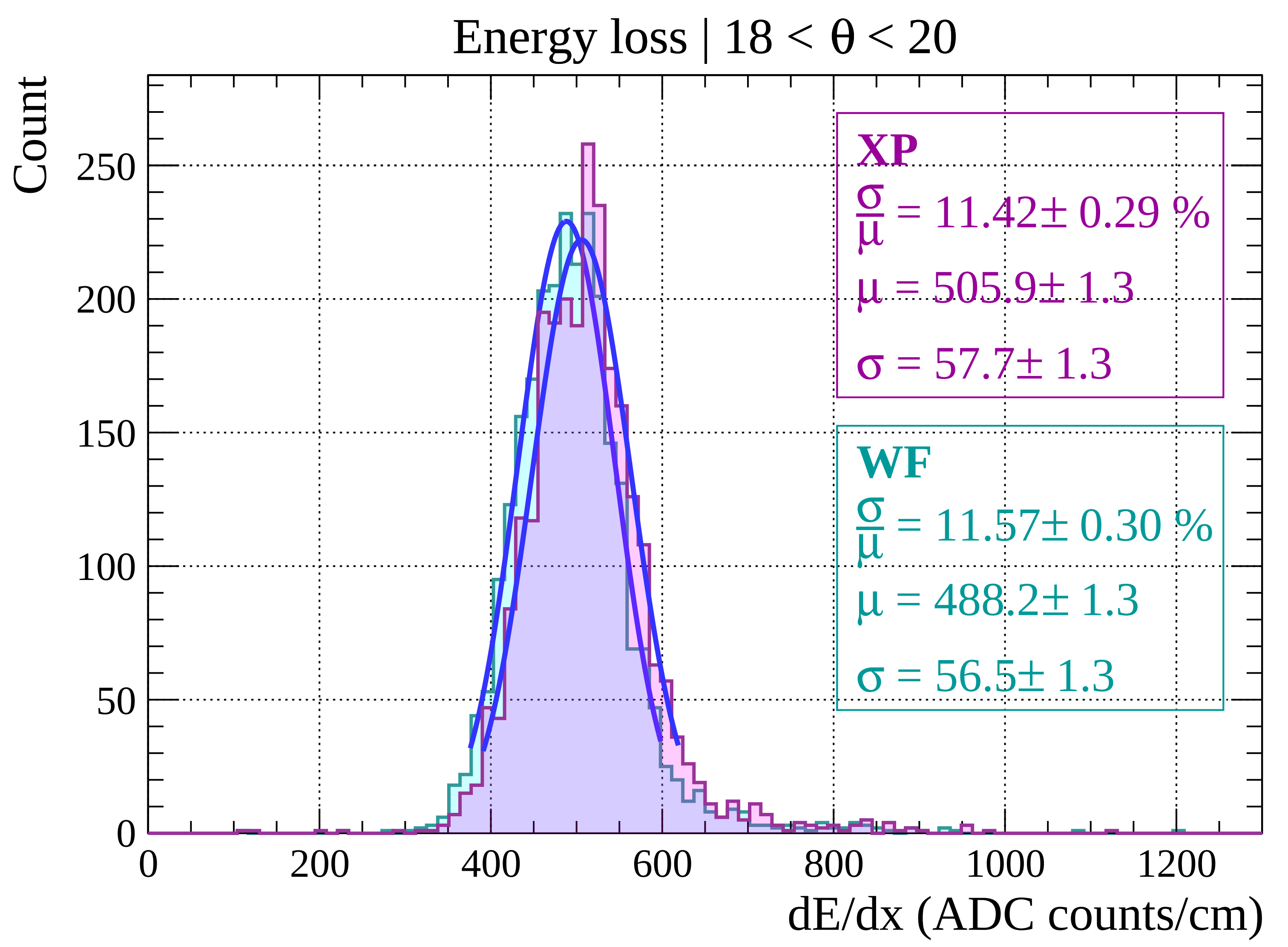
Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 11.37± 0.26 %
μ = 505.2± 1.2
σ = 57.4± 1.2
WF
σ/μ = 11.47± 0.27 %
μ = 487.4± 1.2
σ = 55.9± 1.2



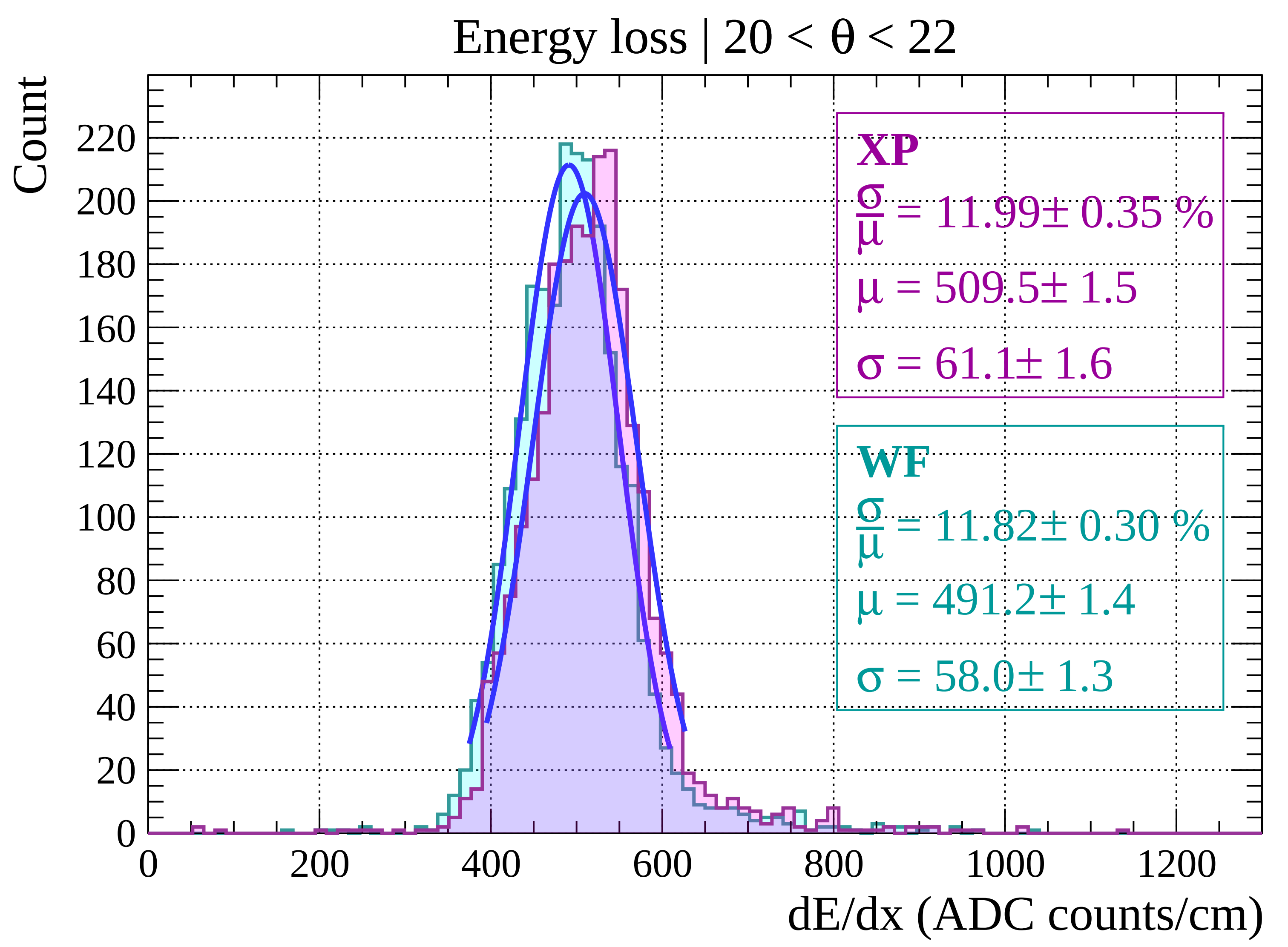
Energy loss | 14 < θ < 16
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.59± 0.28 %
μ = 505.4± 1.3
σ = 58.6± 1.3
WF
σ/μ = 11.62± 0.29 %
μ = 487.5± 1.3
σ = 56.6± 1.3



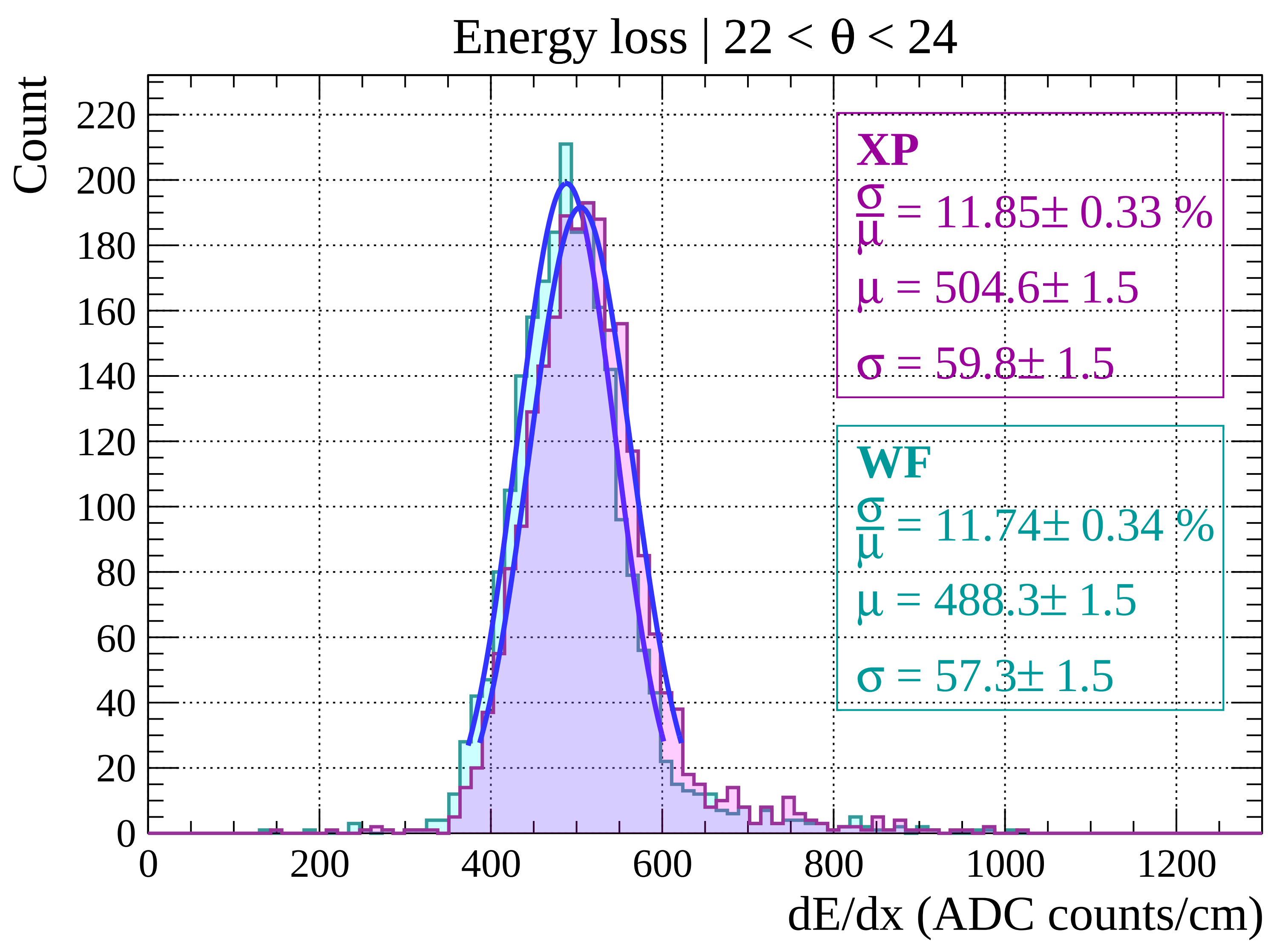
Energy loss | 16 < θ < 18
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.59± 0.29 %
μ = 506.1± 1.3
σ = 58.7± 1.3
WF
σ/μ = 11.61± 0.29 %
μ = 488.9± 1.3
σ = 56.8± 1.3



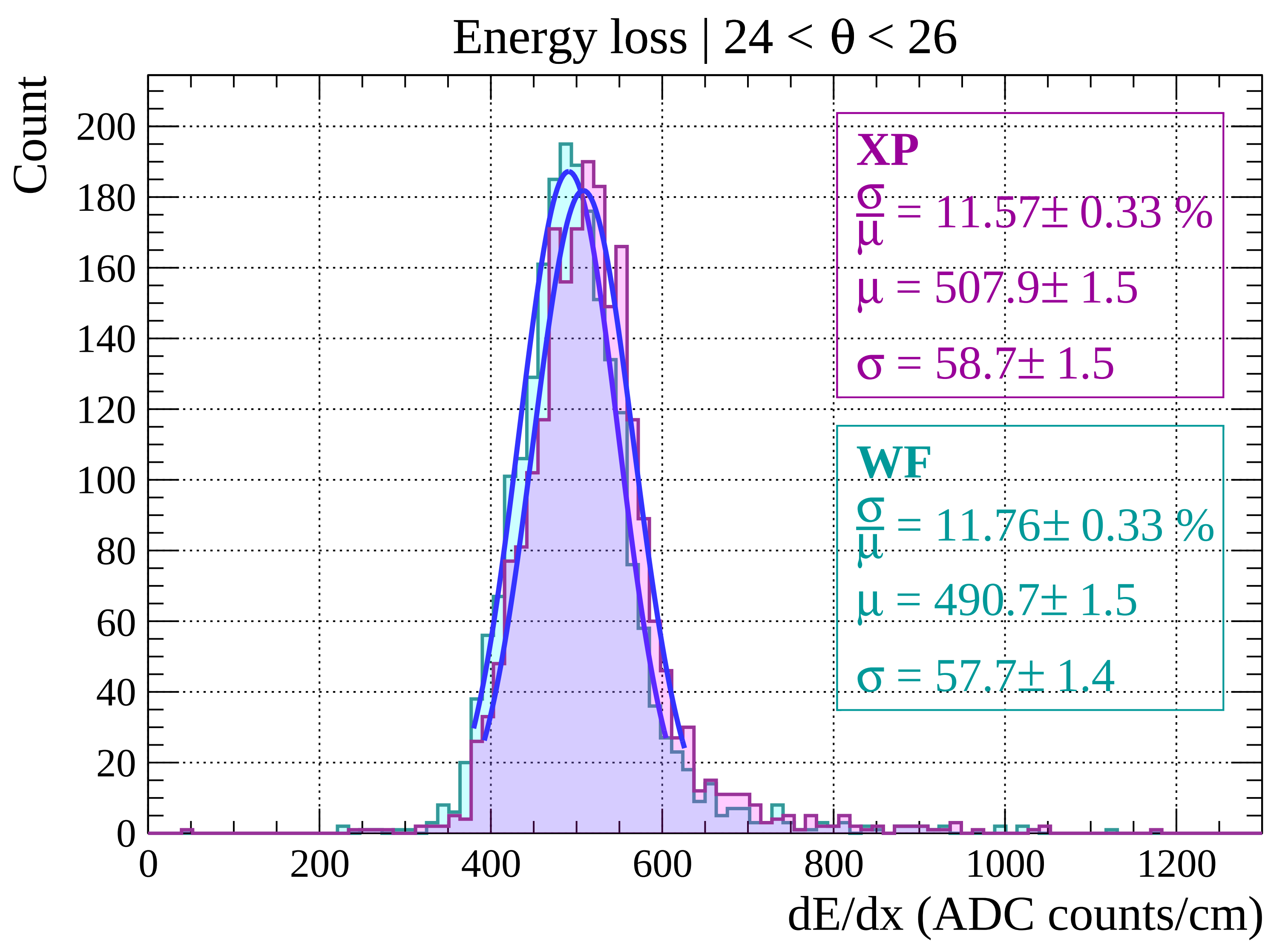
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.42± 0.29 %
μ = 505.9± 1.3
σ = 57.7± 1.3
WF
σ/μ = 11.57± 0.30 %
μ = 488.2± 1.3
σ = 56.5± 1.3



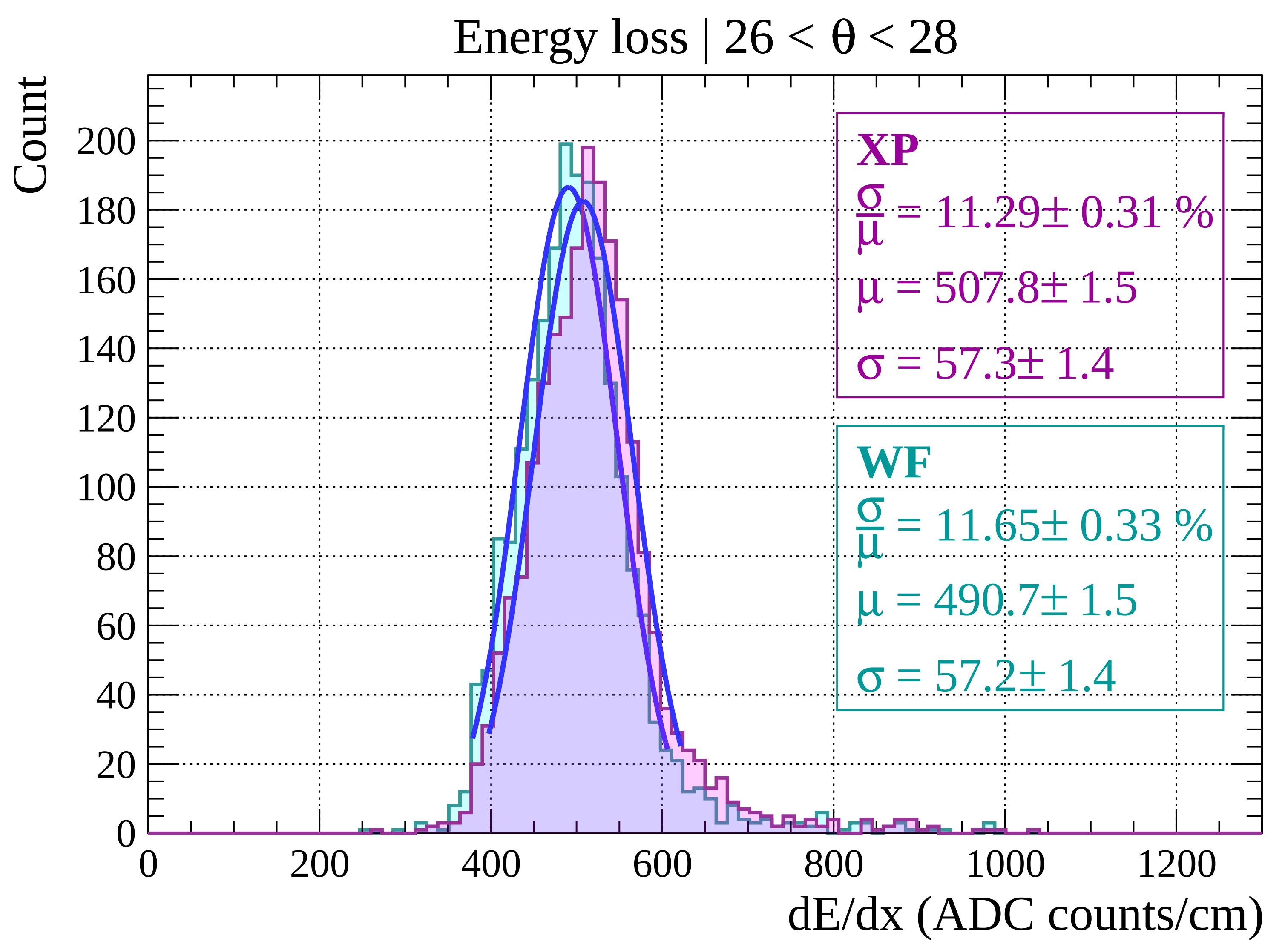
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 11.99± 0.35 %
μ = 509.5± 1.5
σ = 61.1± 1.6
WF
σ/μ = 11.82± 0.30 %
μ = 491.2± 1.4
σ = 58.0± 1.3

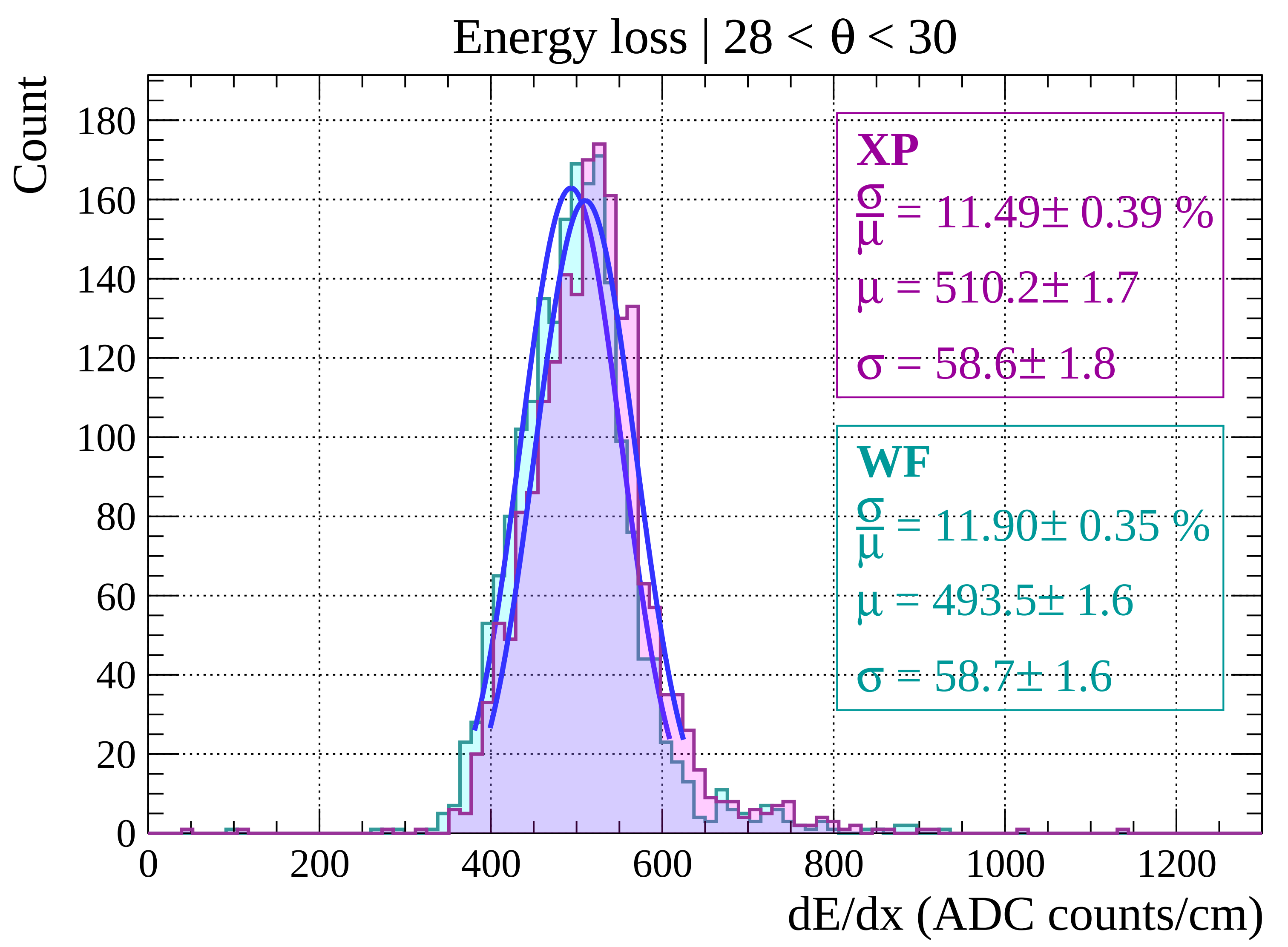


Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.85± 0.33 %
μ = 504.6± 1.5
σ = 59.8± 1.5
WF
$\frac{\sigma}{\mu}$ = 11.74± 0.34 %
μ = 488.3± 1.5
σ = 57.3± 1.5

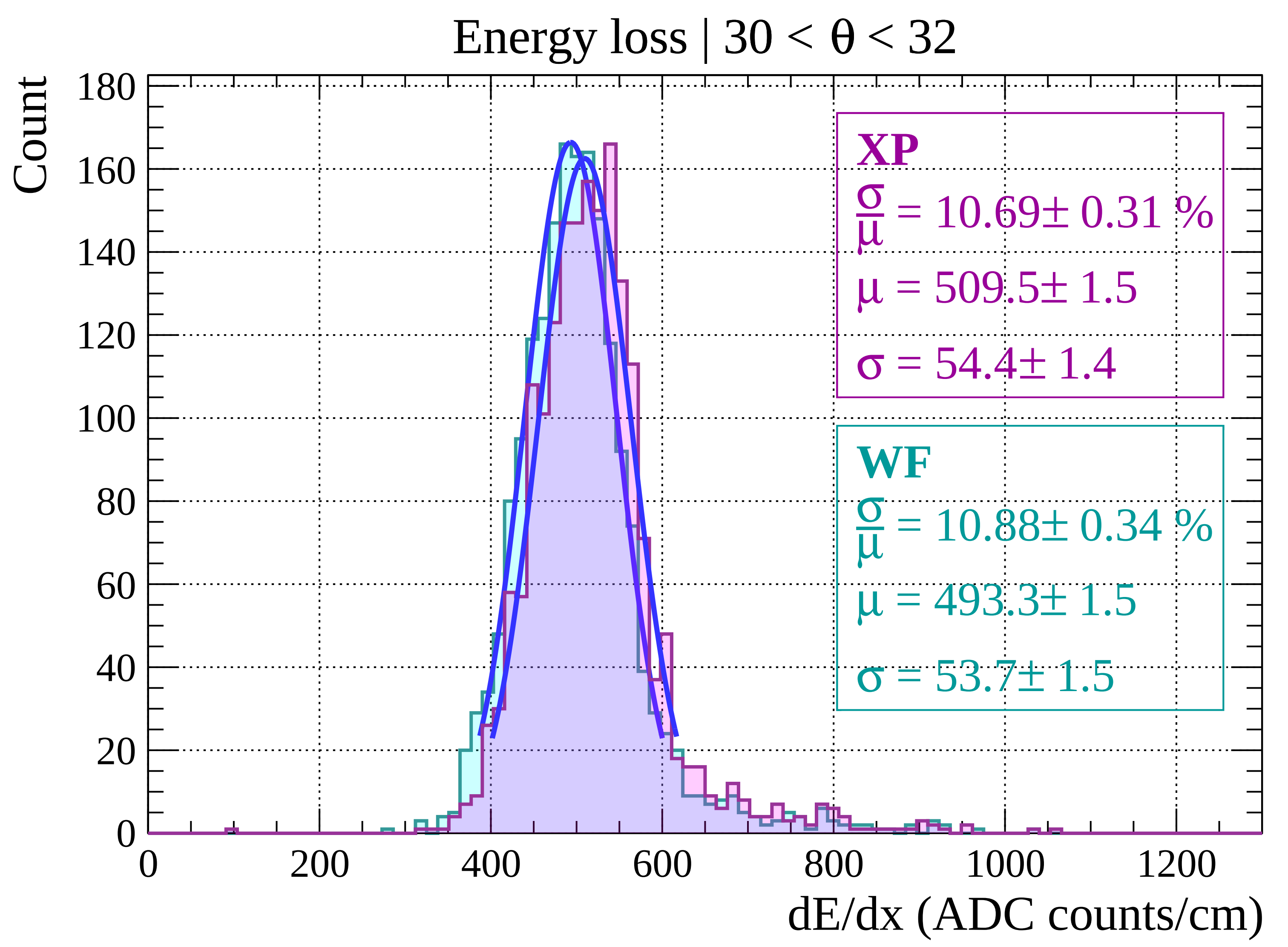


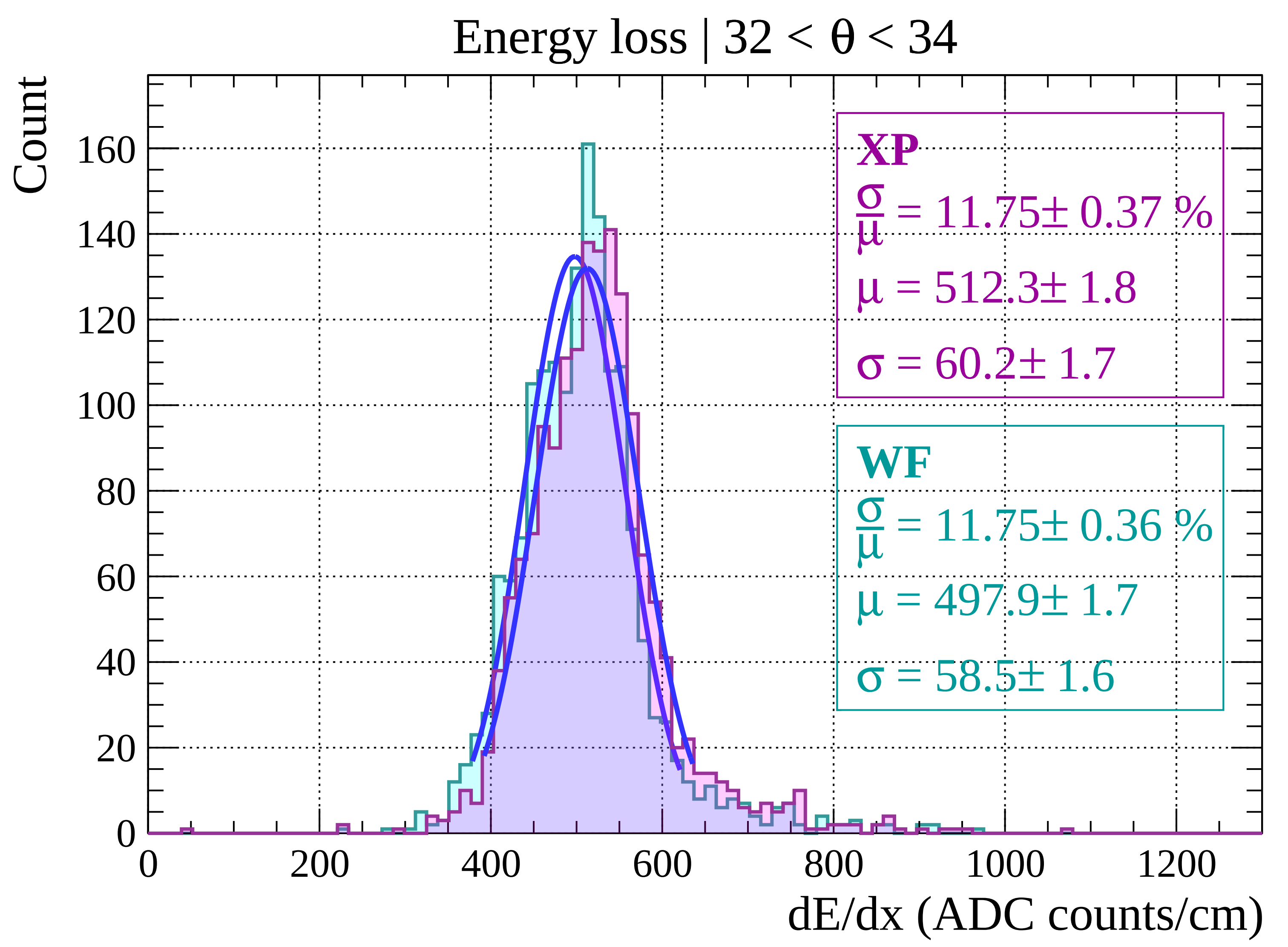
Energy loss | 24 < θ < 26
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.57± 0.33 %
μ = 507.9± 1.5
σ = 58.7± 1.5
WF
σ/μ = 11.76± 0.33 %
μ = 490.7± 1.5
σ = 57.7± 1.4

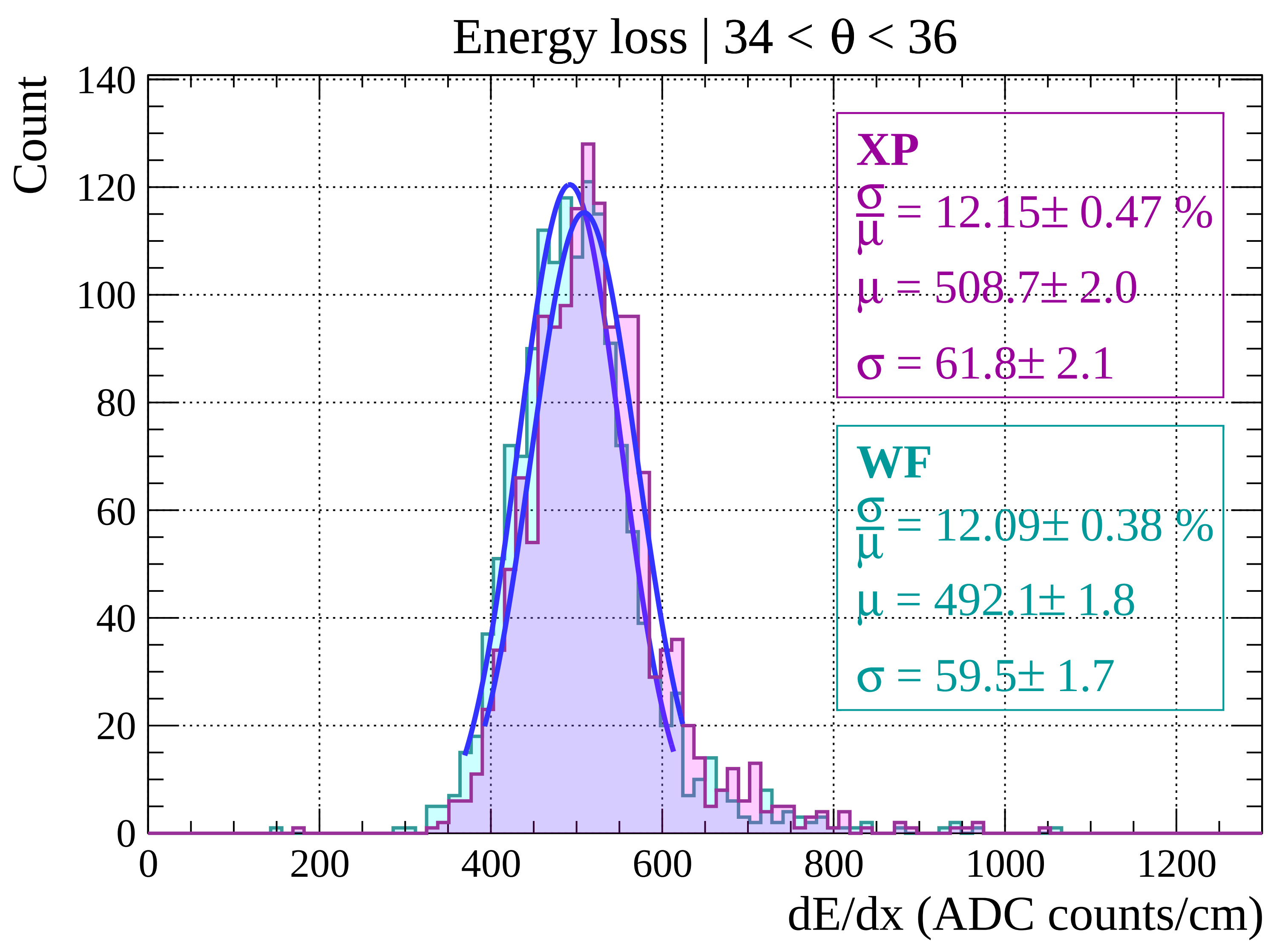

Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.29± 0.31 %
μ = 507.8± 1.5
σ = 57.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 11.65± 0.33 %
μ = 490.7± 1.5
σ = 57.2± 1.4
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



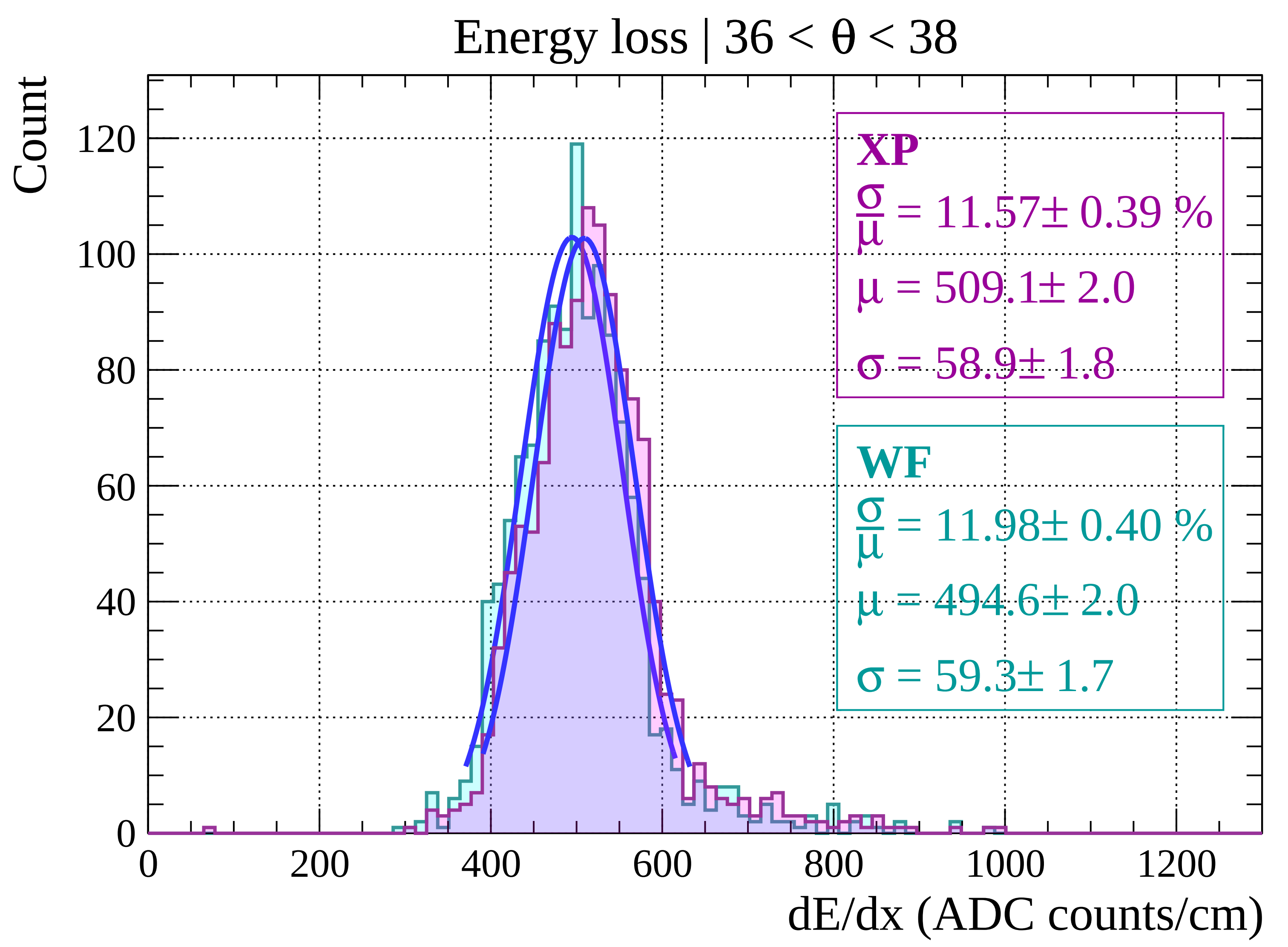
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 11.49± 0.39 %
μ = 510.2± 1.7
σ = 58.6± 1.8
WF
σ/μ = 11.90± 0.35 %
μ = 493.5± 1.6
σ = 58.7± 1.6

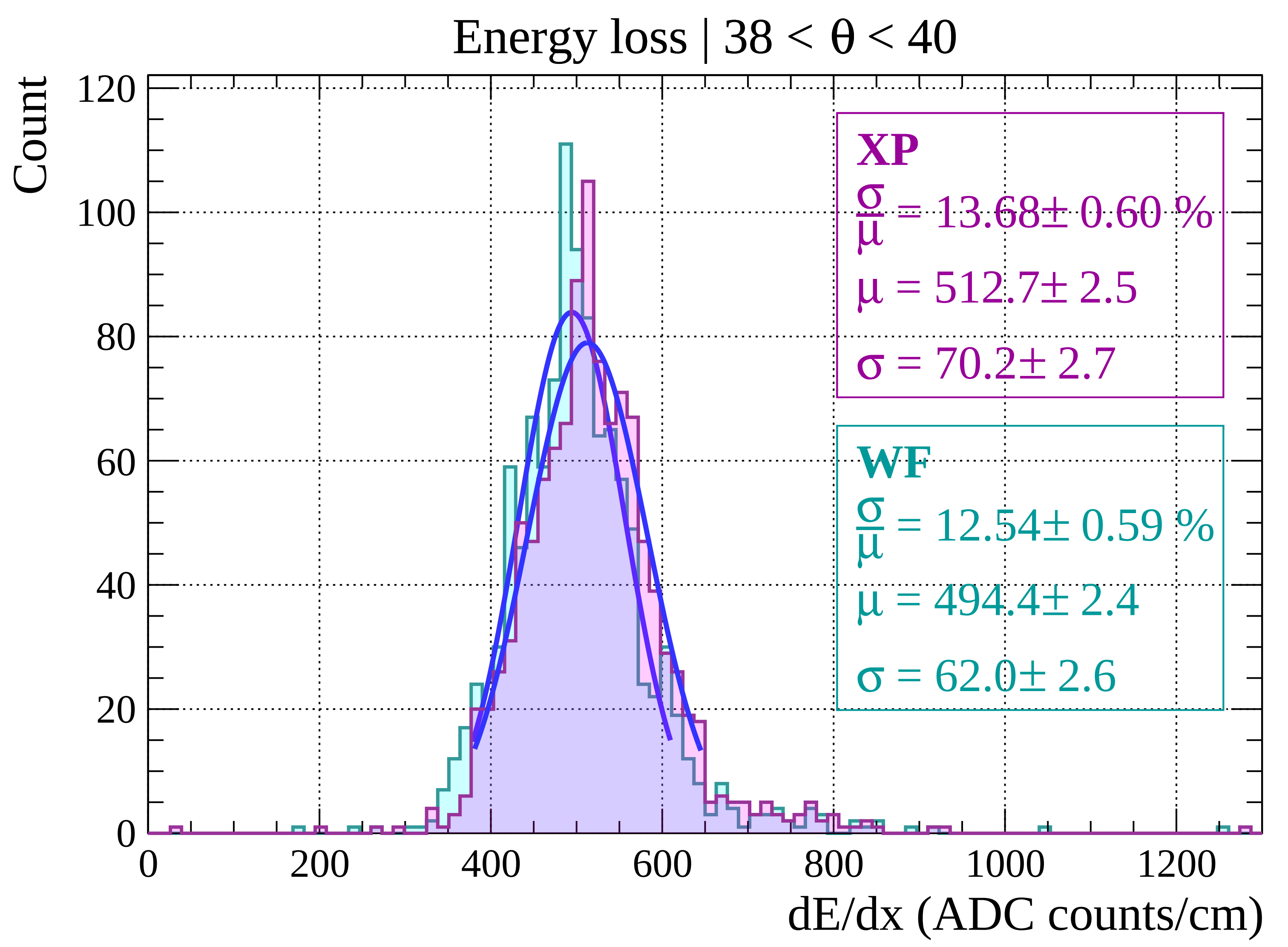

Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 10.69± 0.31 %
μ = 509.5± 1.5
σ = 54.4± 1.4
WF
σ/μ = 10.88± 0.34 %
μ = 493.3± 1.5
σ = 53.7± 1.5


Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
σ/μ = 11.75± 0.37 %
μ = 512.3± 1.8
σ = 60.2± 1.7
WF
σ/μ = 11.75± 0.36 %
μ = 497.9± 1.7
σ = 58.5± 1.6

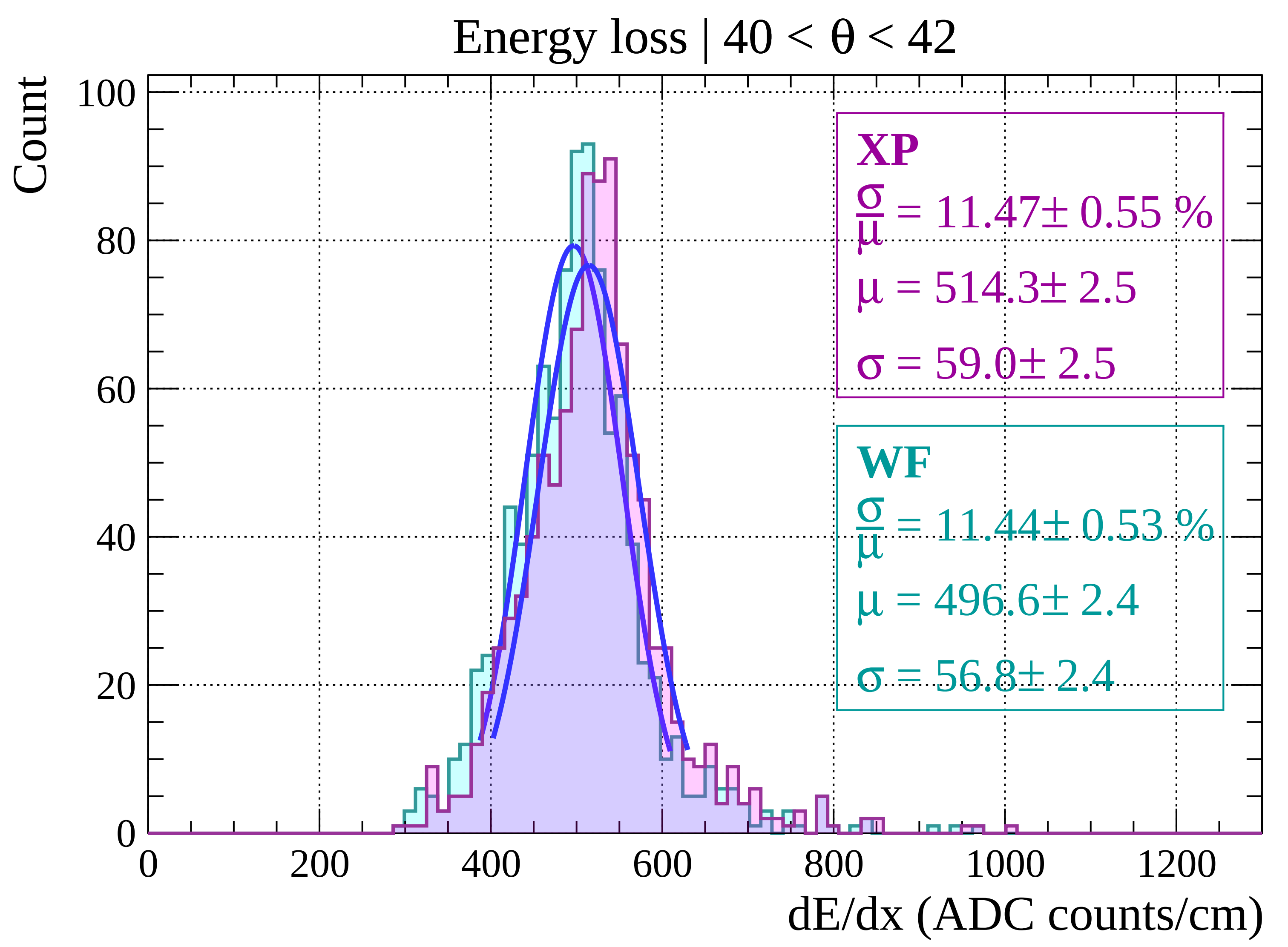


Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
XP
σ/μ = 12.15± 0.47 %
μ = 508.7± 2.0
σ = 61.8± 2.1
WF
σ/μ = 12.09± 0.38 %
μ = 492.1± 1.8
σ = 59.5± 1.7

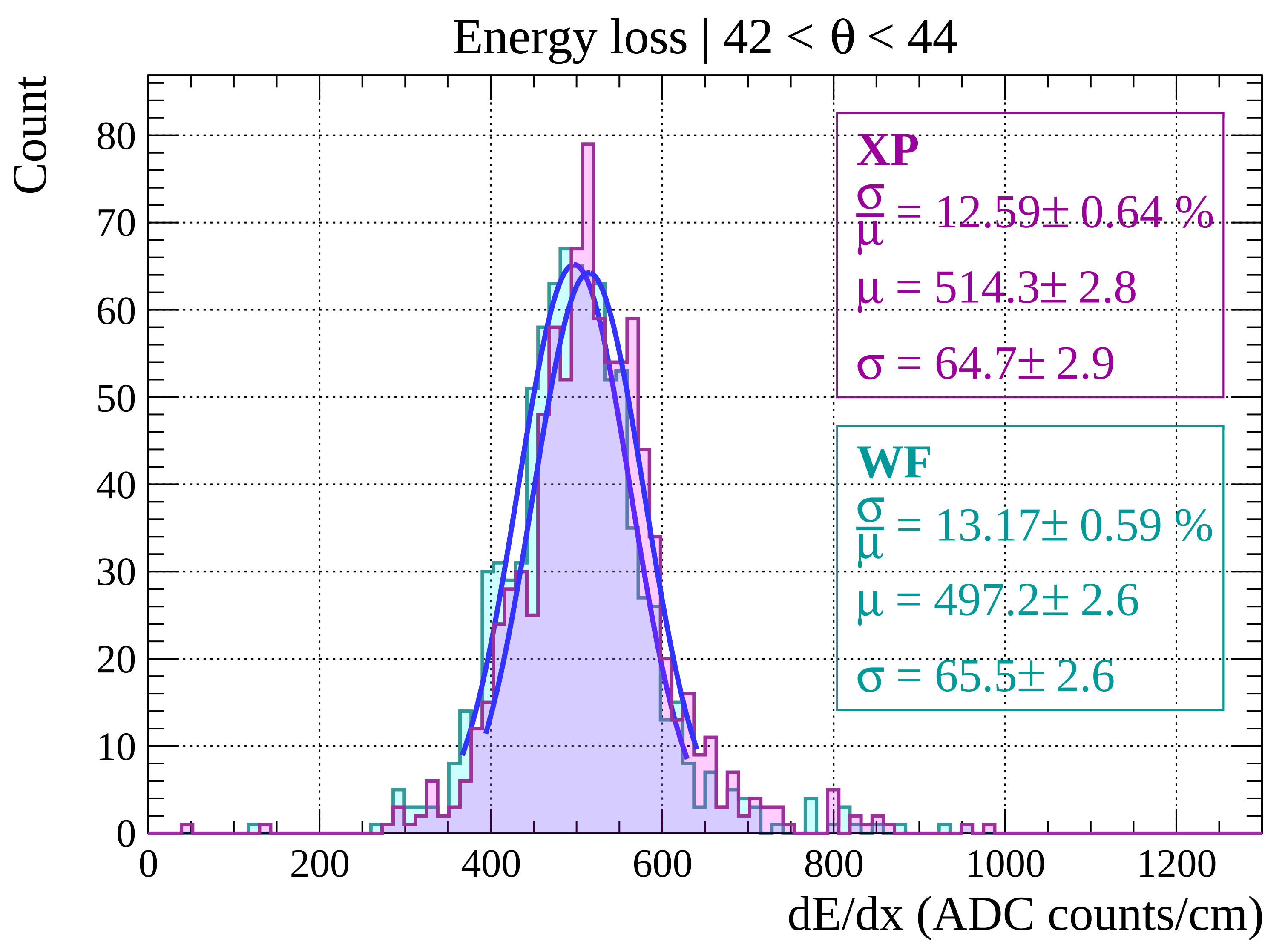

Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 11.57± 0.39 %
μ = 509.1± 2.0
σ = 58.9± 1.8
WF
σ/μ = 11.98± 0.40 %
μ = 494.6± 2.0
σ = 59.3± 1.7



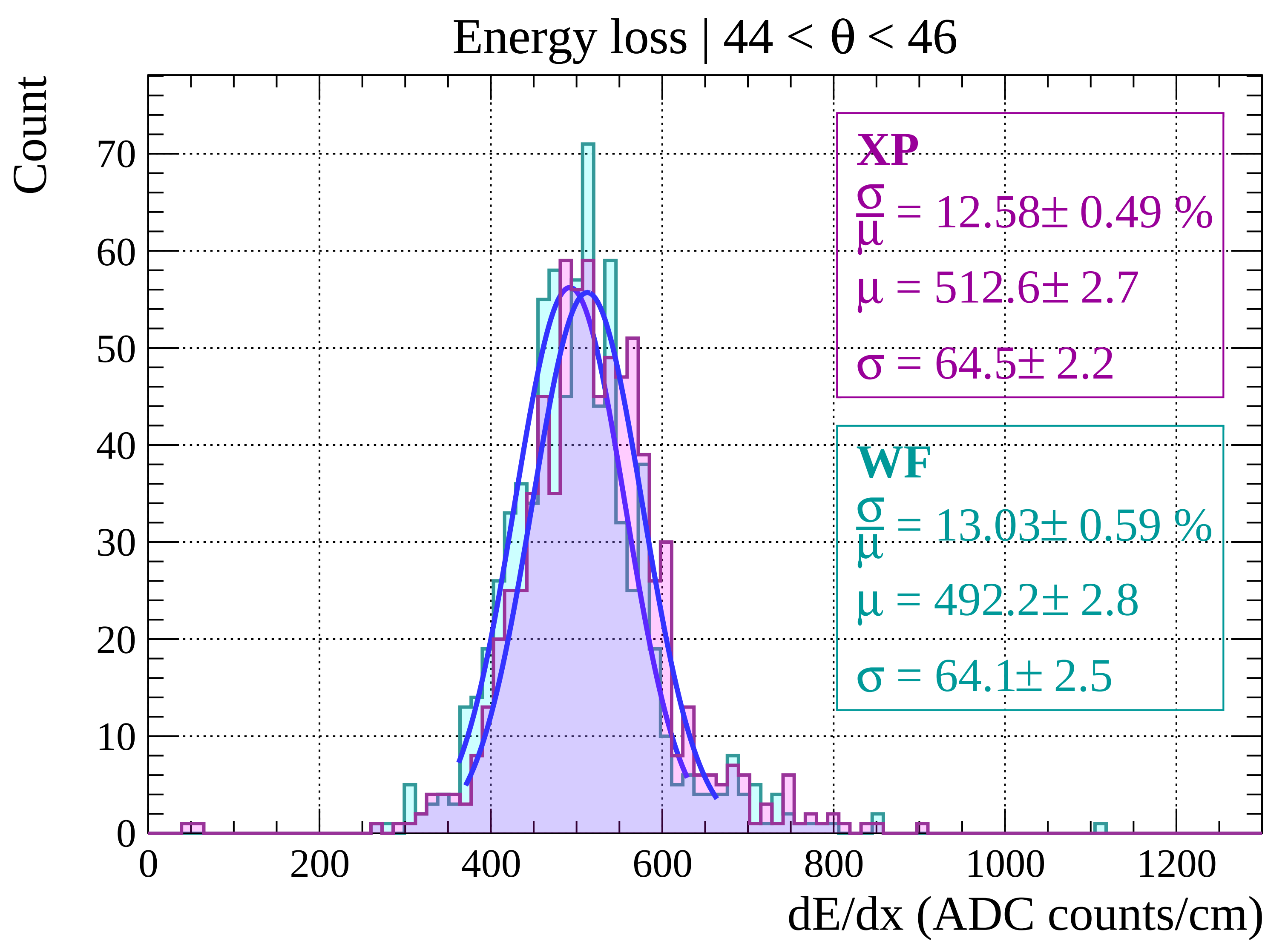
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.68± 0.60 %
μ = 512.7± 2.5
σ = 70.2± 2.7
WF
σ/μ = 12.54± 0.59 %
μ = 494.4± 2.4
σ = 62.0± 2.6
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

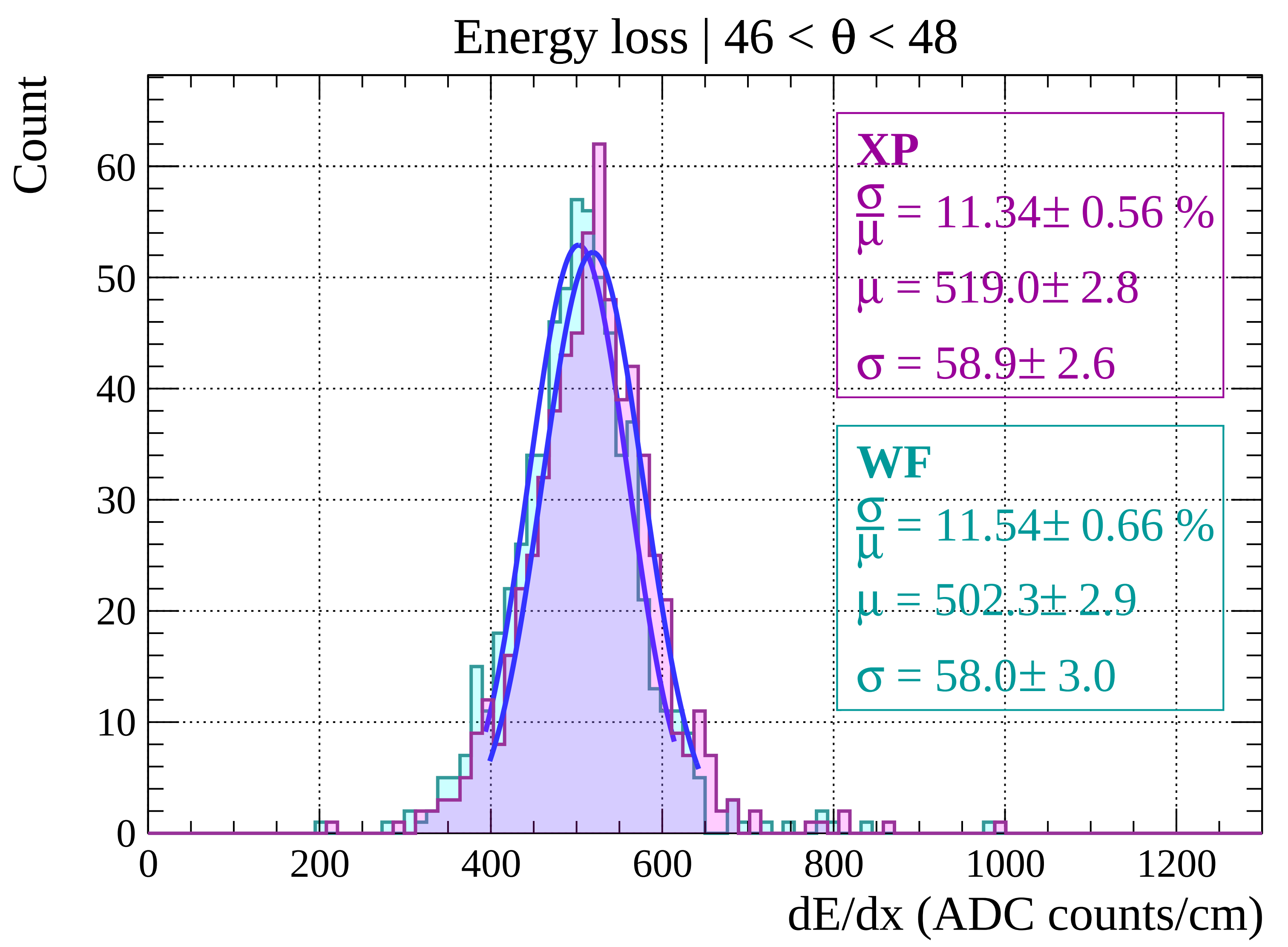


Energy loss | 40 < θ < 42
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.47± 0.55 %
μ = 514.3± 2.5
σ = 59.0± 2.5
WF
σ/μ = 11.44± 0.53 %
μ = 496.6± 2.4
σ = 56.8± 2.4

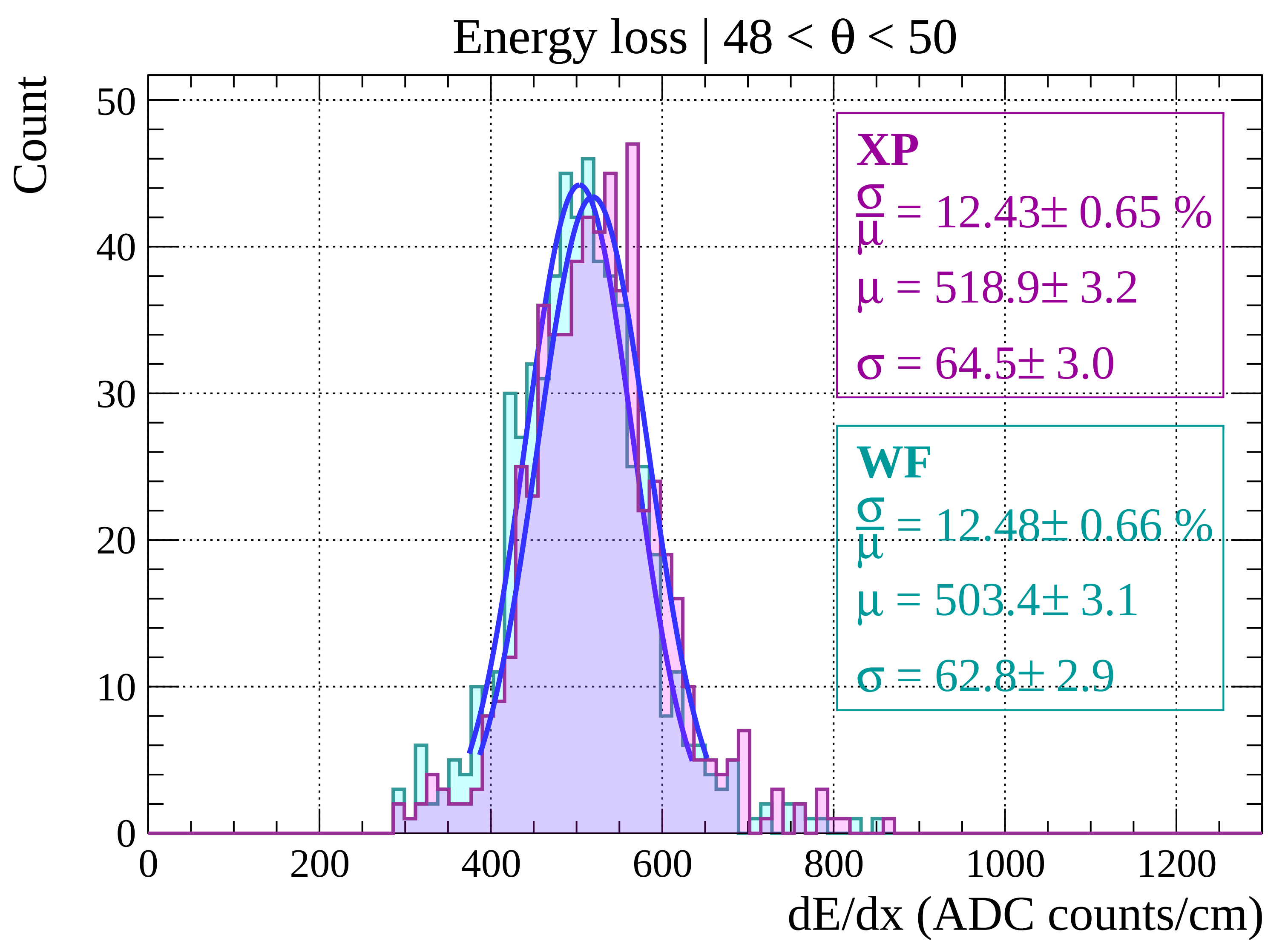


Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 12.59± 0.64 %
μ = 514.3± 2.8
σ = 64.7± 2.9
WF
σ/μ = 13.17± 0.59 %
μ = 497.2± 2.6
σ = 65.5± 2.6

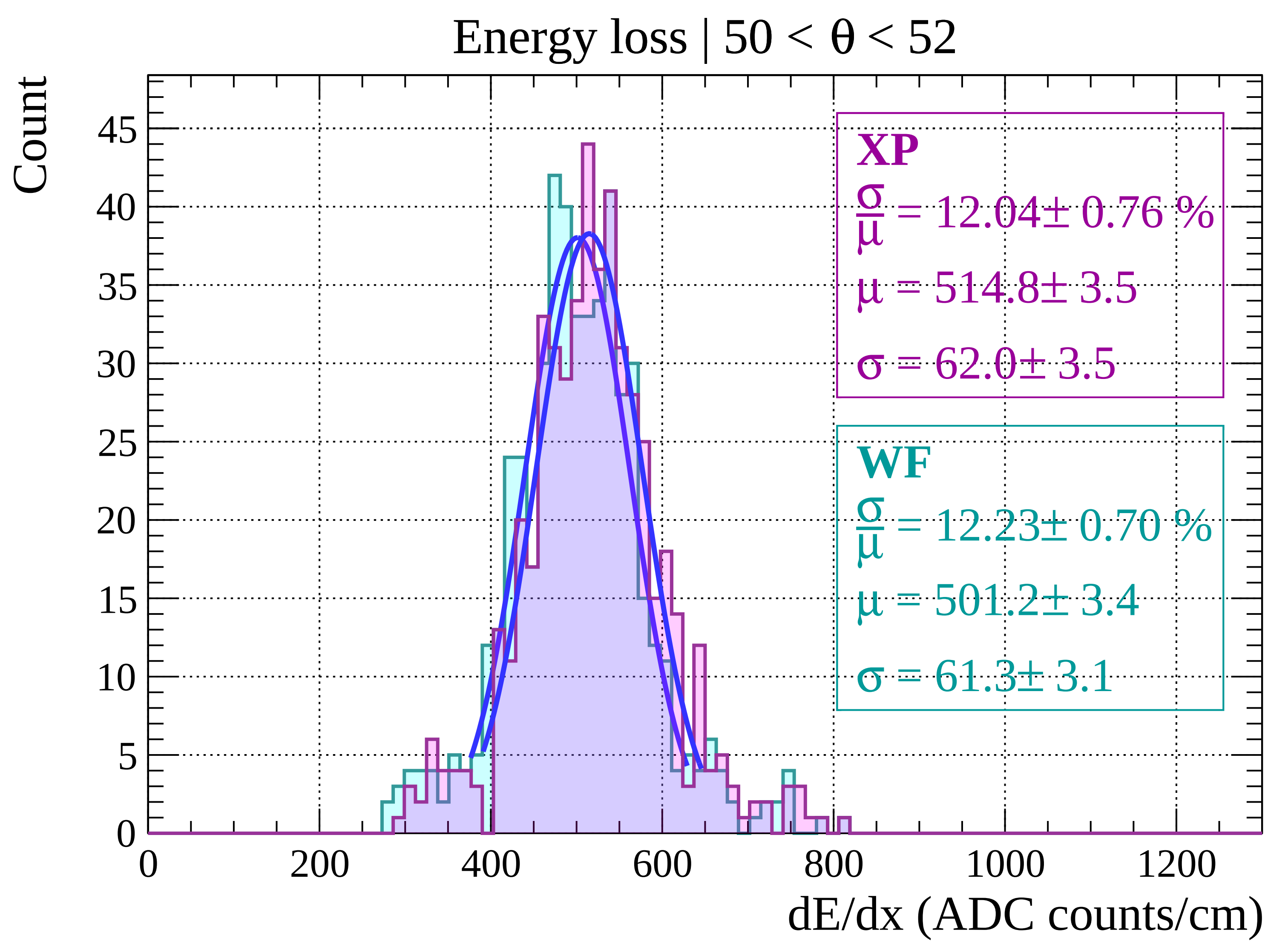

Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 12.58± 0.49 %
μ = 512.6± 2.7
σ = 64.5± 2.2
WF
$\frac{\sigma}{\mu}$ = 13.03± 0.59 %
μ = 492.2± 2.8
σ = 64.1± 2.5



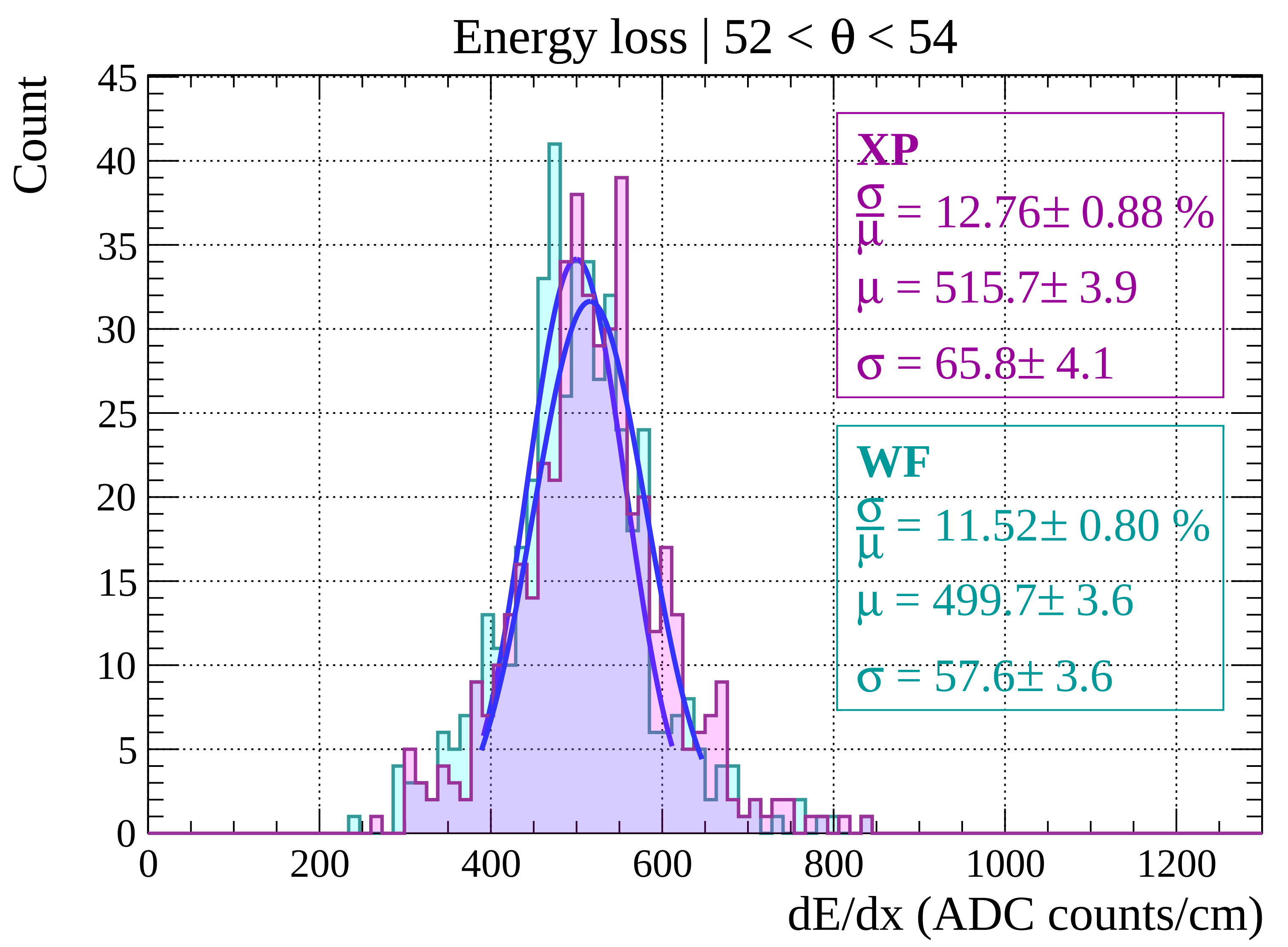
Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 11.34± 0.56 %
μ = 519.0± 2.8
σ = 58.9± 2.6
WF
σ/μ = 11.54± 0.66 %
μ = 502.3± 2.9
σ = 58.0± 3.0



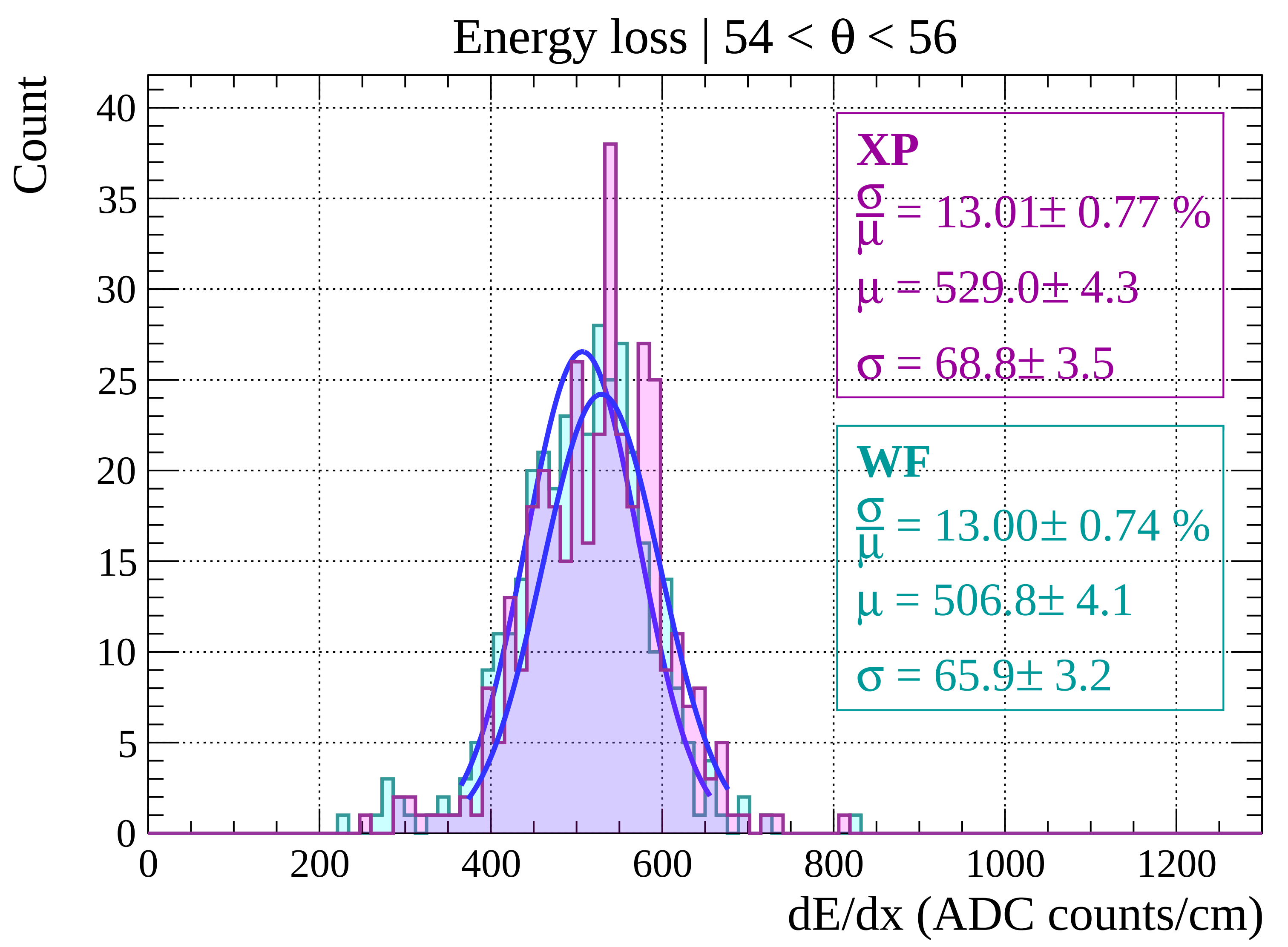
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 12.43± 0.65 %
μ = 518.9± 3.2
σ = 64.5± 3.0
WF
σ/μ = 12.48± 0.66 %
μ = 503.4± 3.1
σ = 62.8± 2.9



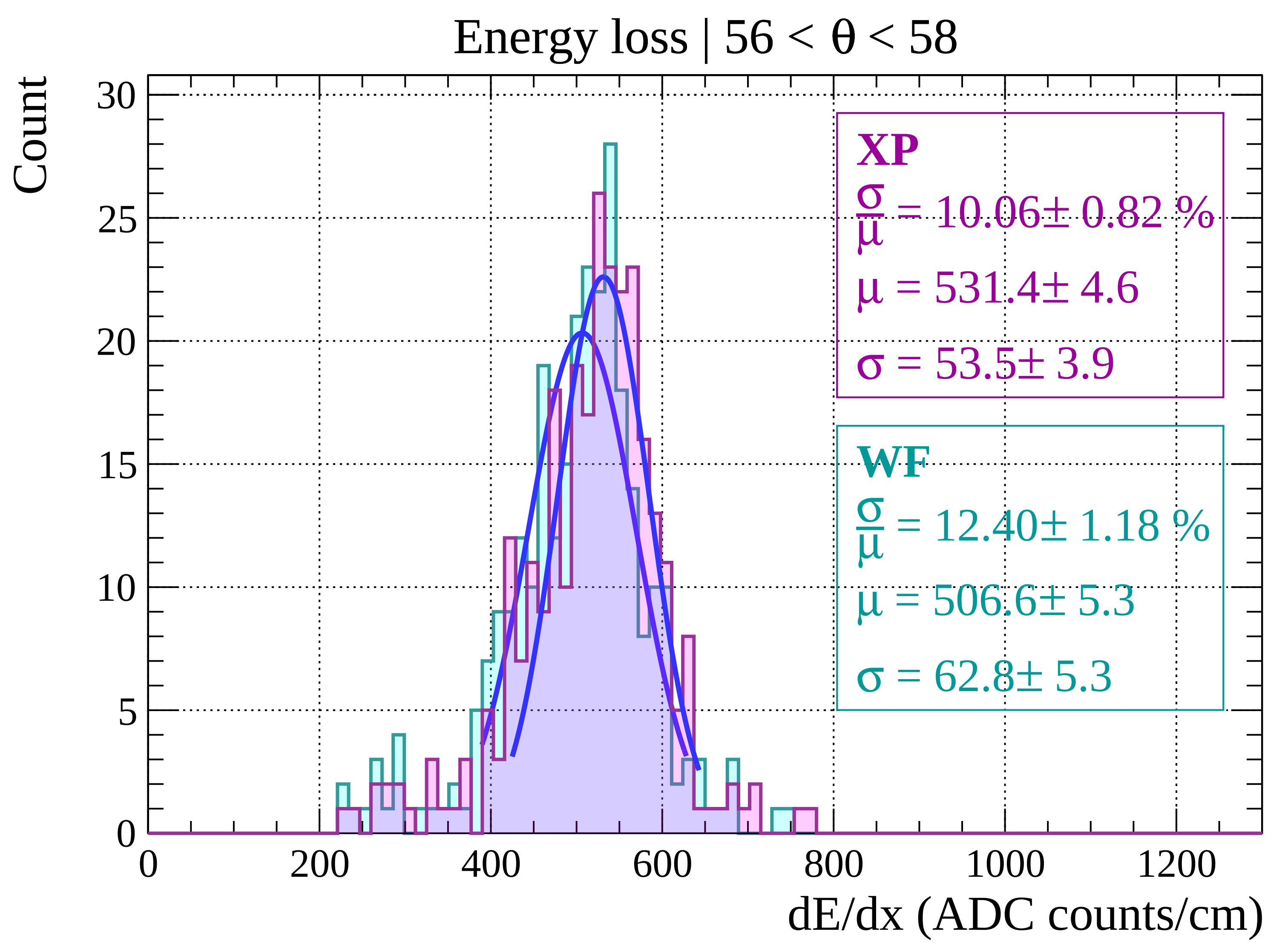
Energy loss | 50 < θ < 52
Count
XP
$\frac{\sigma}{\mu}$ = 12.04± 0.76 %
μ = 514.8± 3.5
σ = 62.0± 3.5
WF
$\frac{\sigma}{\mu}$ = 12.23± 0.70 %
μ = 501.2± 3.4
σ = 61.3± 3.1
dE/dx (ADC counts/cm)

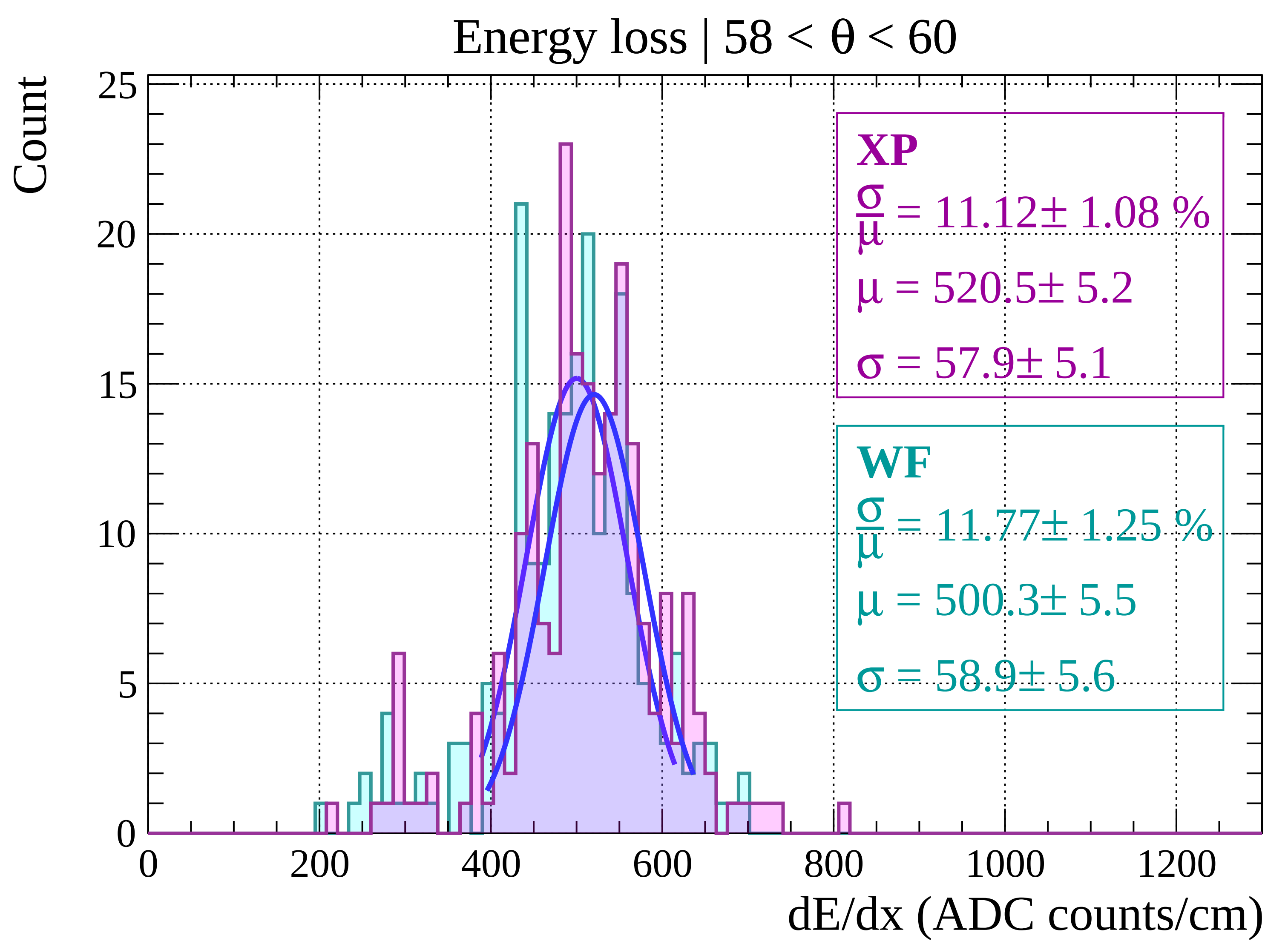


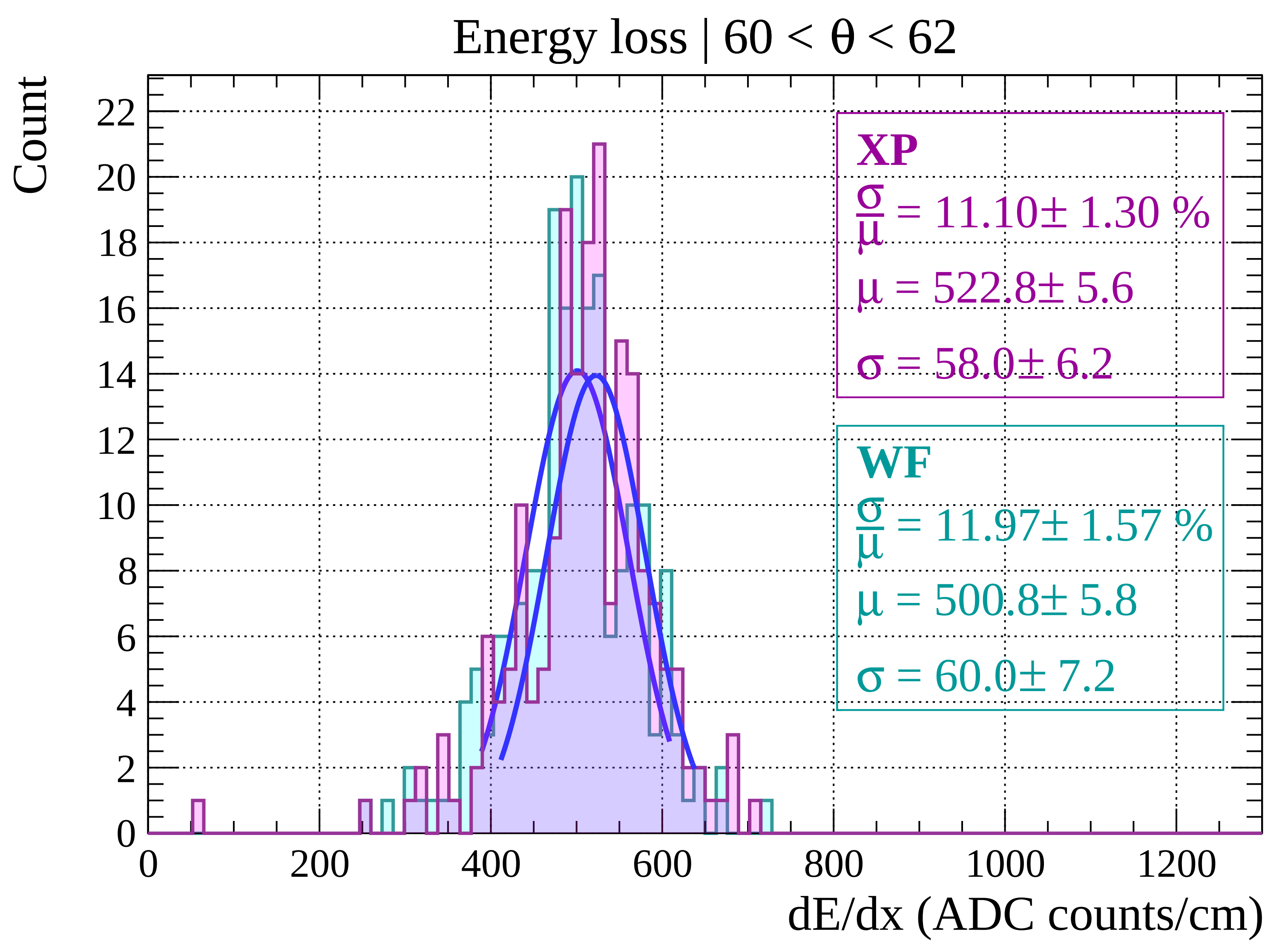
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.76± 0.88 %
μ = 515.7± 3.9
σ = 65.8± 4.1
WF
σ/μ = 11.52± 0.80 %
μ = 499.7± 3.6
σ = 57.6± 3.6
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200



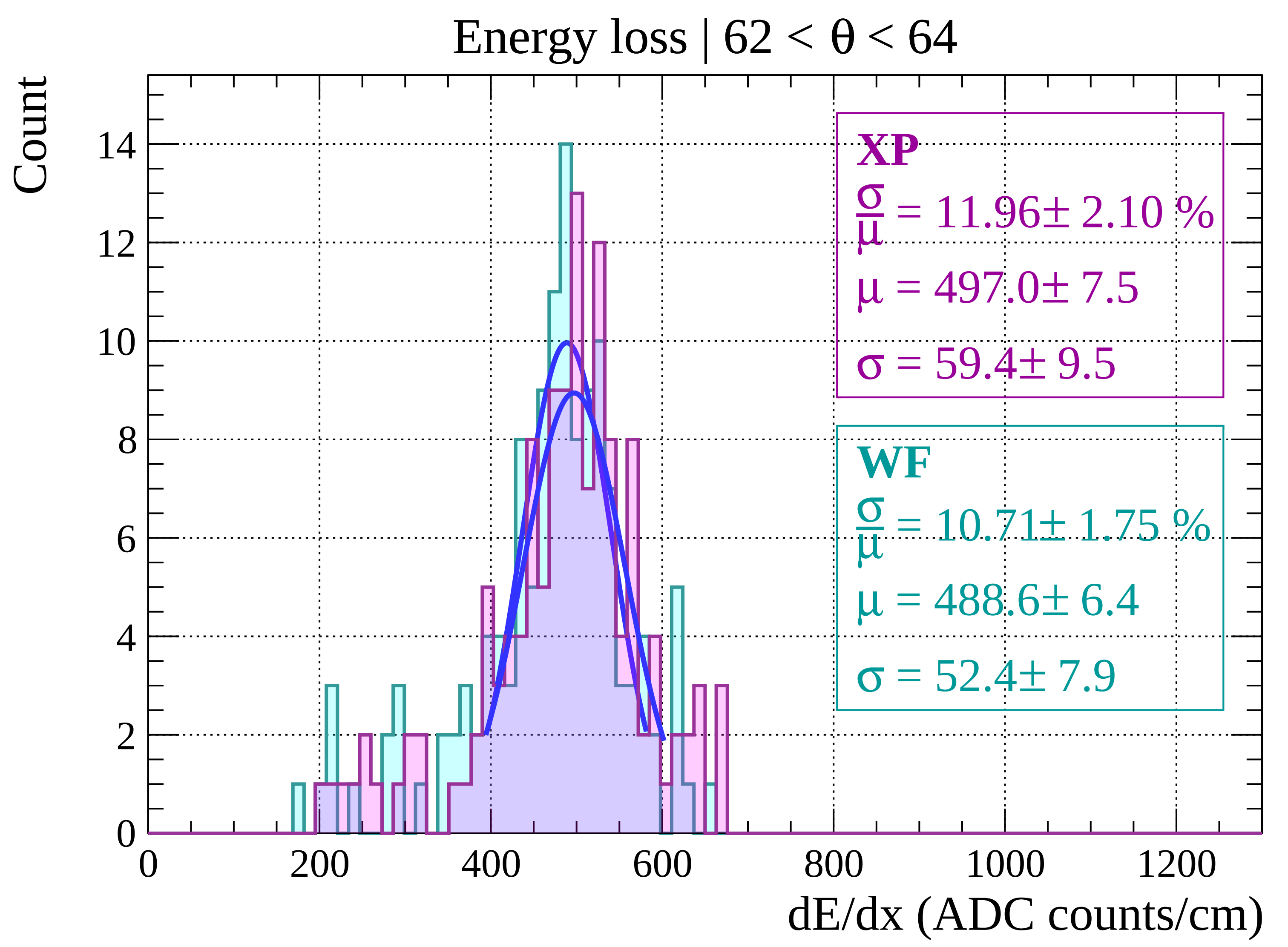
Energy loss | 54 < θ < 56
Count
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 13.01± 0.77 %
μ = 529.0± 4.3
σ = 68.8± 3.5
WF
$\frac{\sigma}{\mu}$ = 13.00± 0.74 %
μ = 506.8± 4.1
σ = 65.9± 3.2


Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 10.06± 0.82 %
μ = 531.4± 4.6
σ = 53.5± 3.9
WF
σ/μ = 12.40± 1.18 %
μ = 506.6± 5.3
σ = 62.8± 5.3

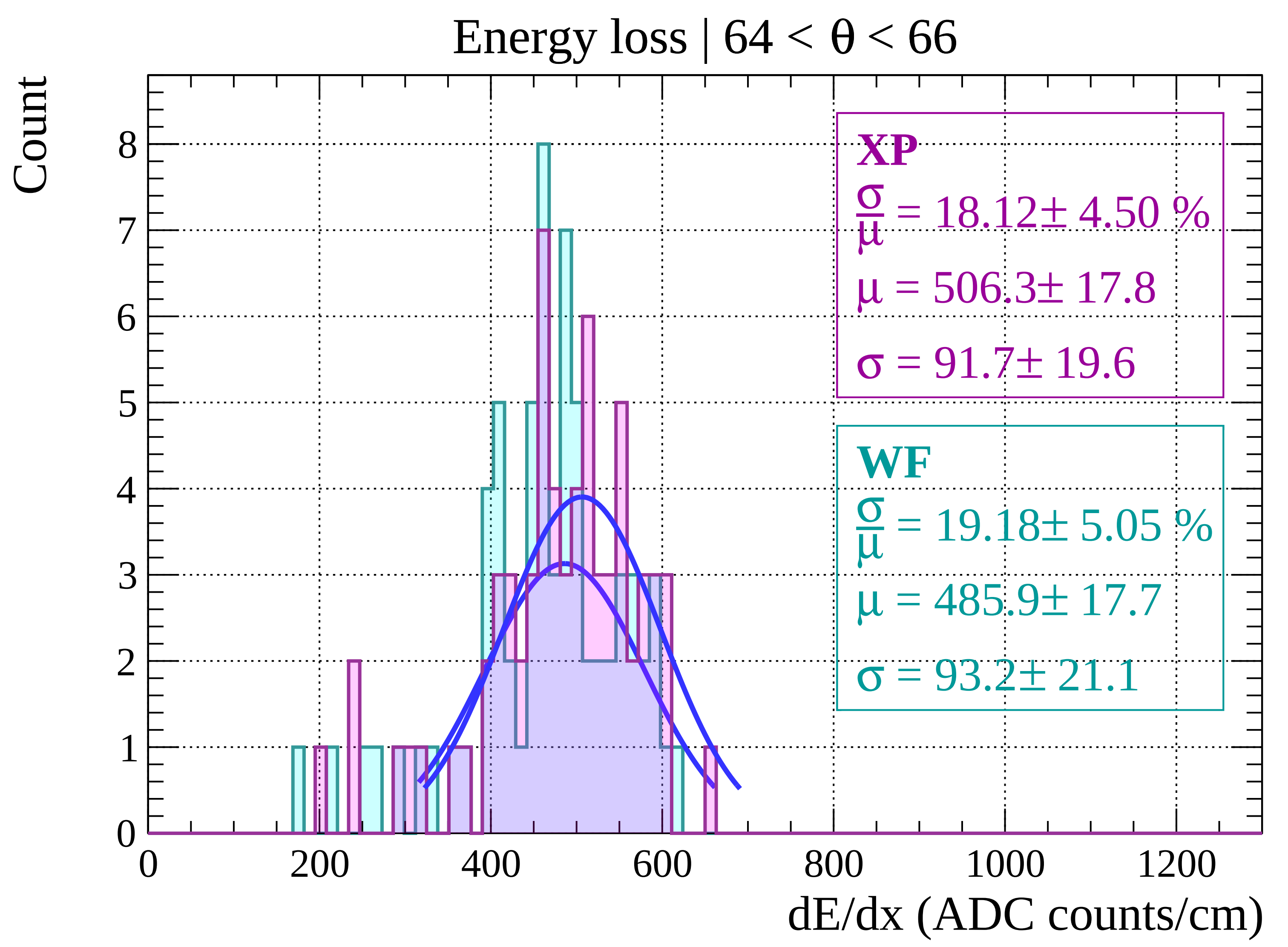

Energy loss | 58 < θ < 60
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.12± 1.08 %
μ = 520.5± 5.2
σ = 57.9± 5.1
WF
σ/μ = 11.77± 1.25 %
μ = 500.3± 5.5
σ = 58.9± 5.6



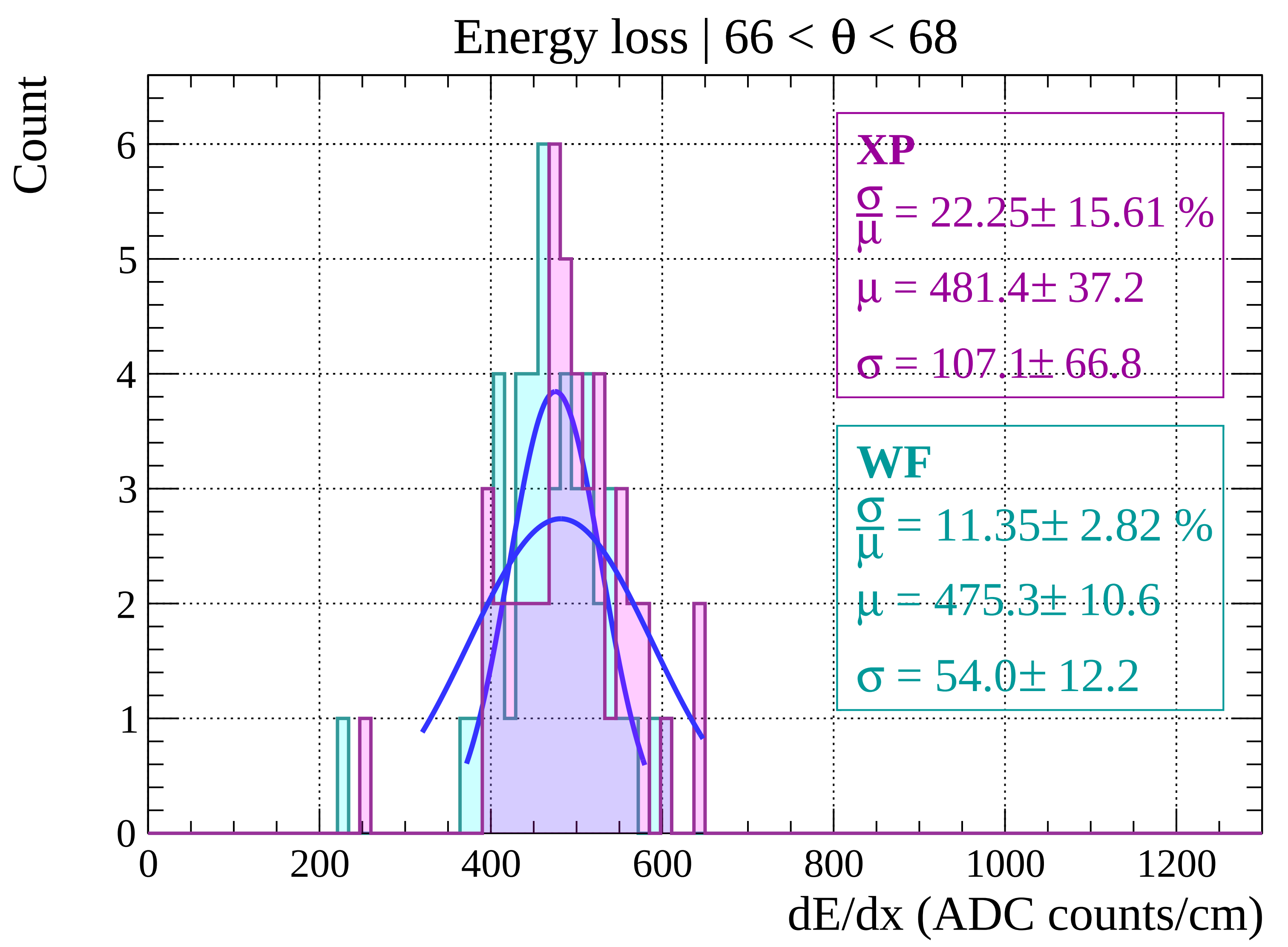
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22
XP
σ/μ = 11.10± 1.30 %
μ = 522.8± 5.6
σ = 58.0± 6.2
WF
σ/μ = 11.97± 1.57 %
μ = 500.8± 5.8
σ = 60.0± 7.2



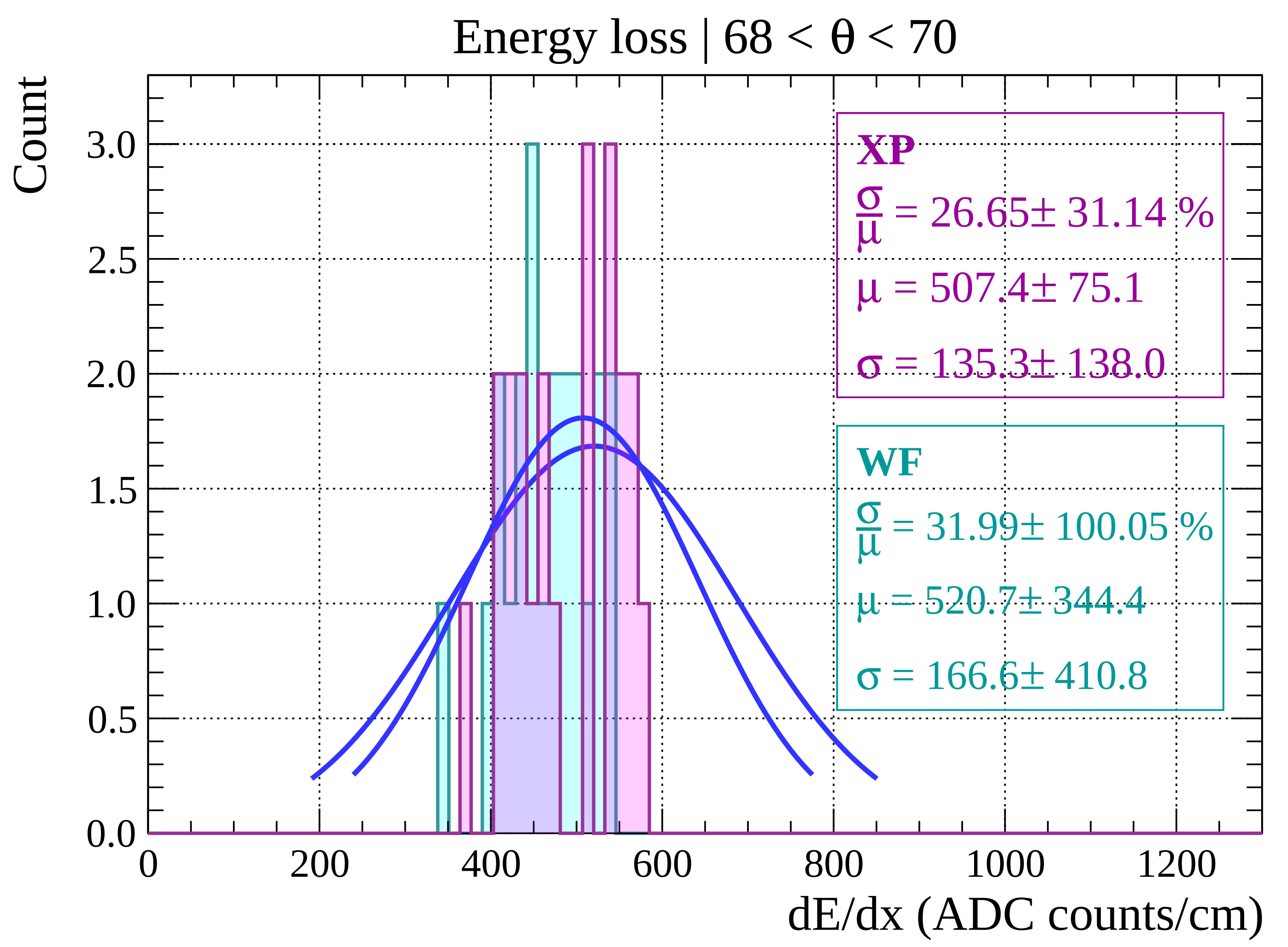
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
XP
σ/μ = 11.96± 2.10 %
μ = 497.0± 7.5
σ = 59.4± 9.5
WF
σ/μ = 10.71± 1.75 %
μ = 488.6± 6.4
σ = 52.4± 7.9



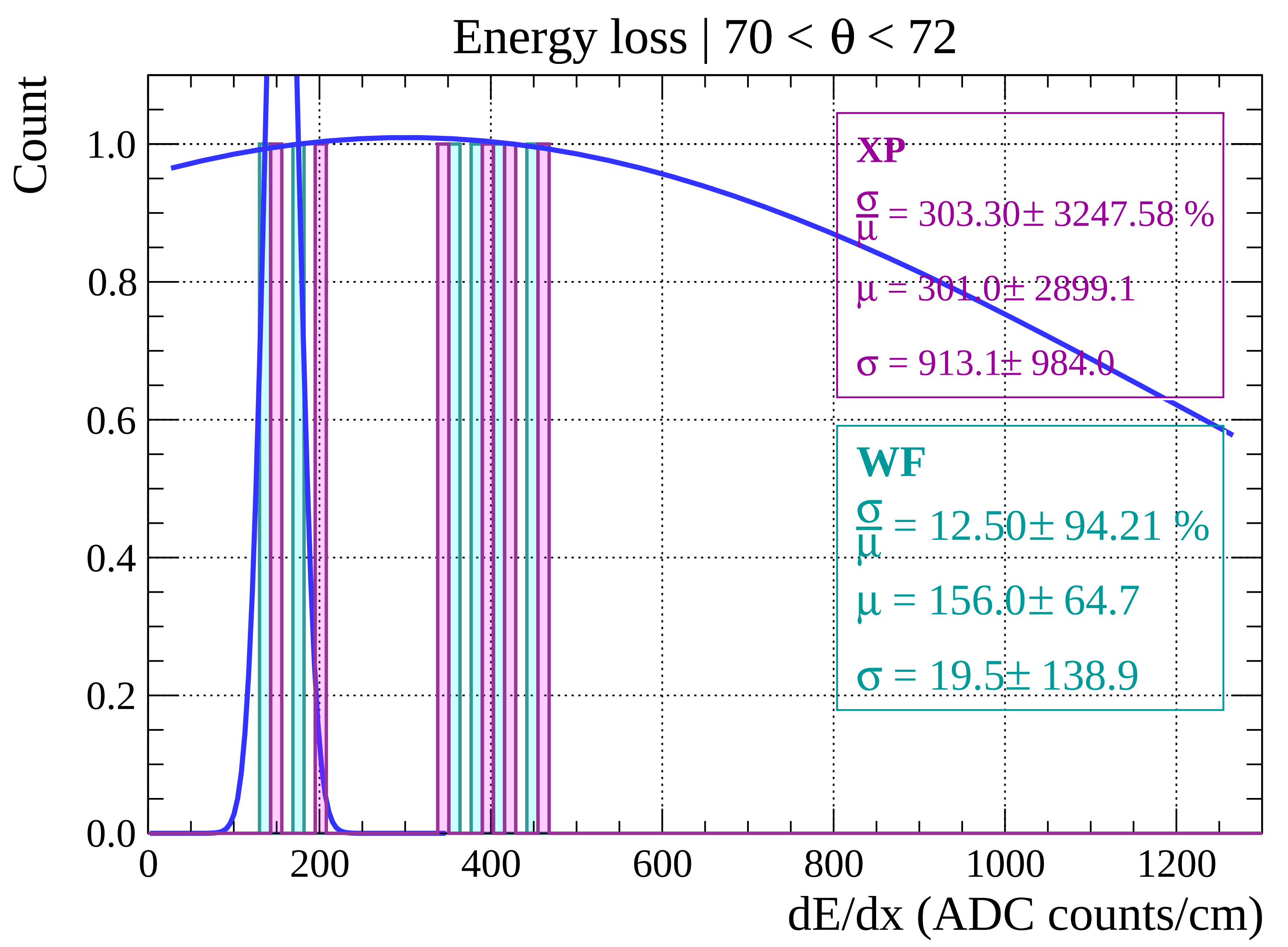
Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 18.12± 4.50 %
μ = 506.3± 17.8
σ = 91.7± 19.6
WF
σ/μ = 19.18± 5.05 %
μ = 485.9± 17.7
σ = 93.2± 21.1



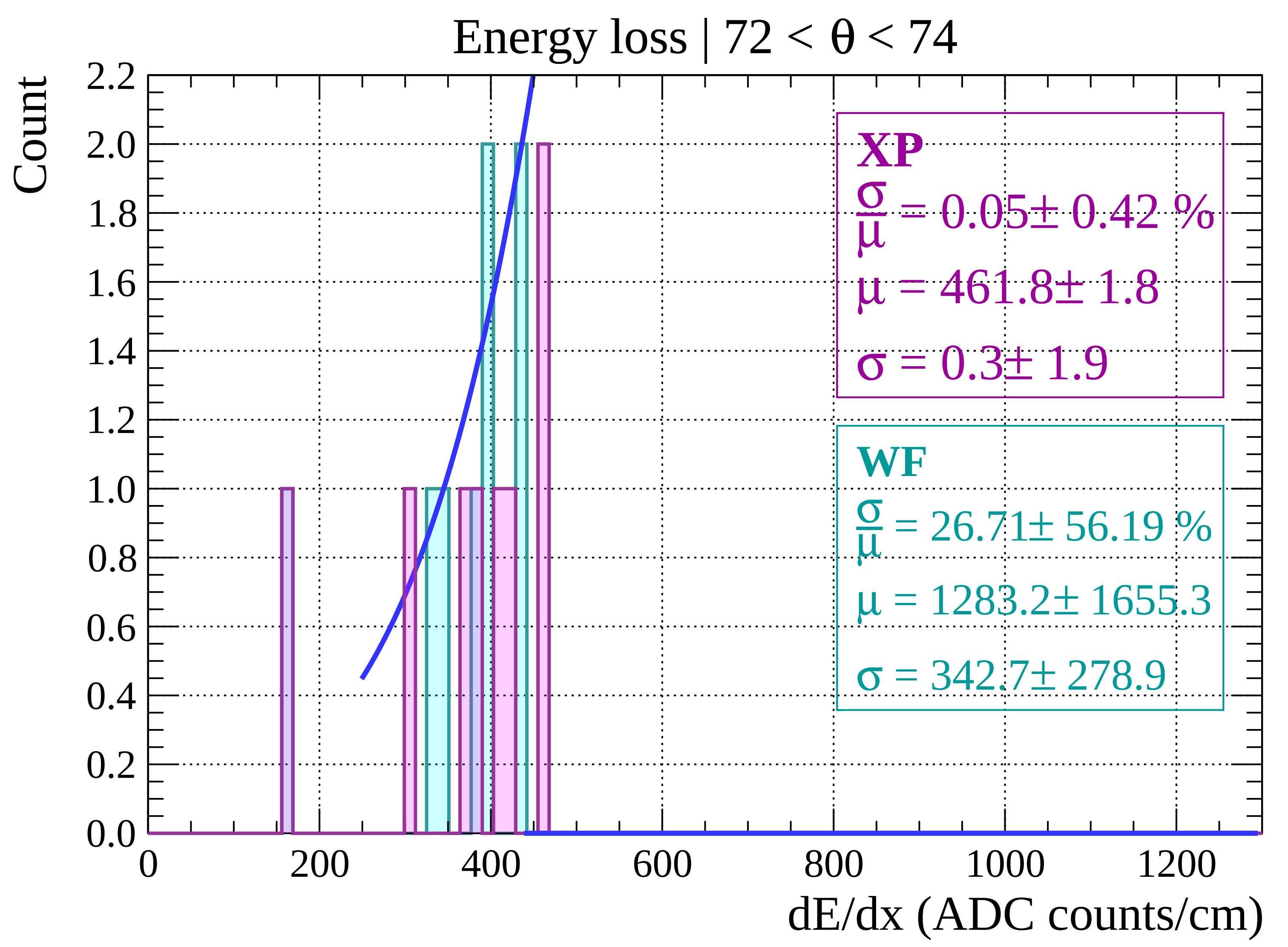
Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
XP
σ/μ = 22.25± 15.61 %
μ = 481.4± 37.2
σ = 107.1± 66.8
WF
σ/μ = 11.35± 2.82 %
μ = 475.3± 10.6
σ = 54.0± 12.2

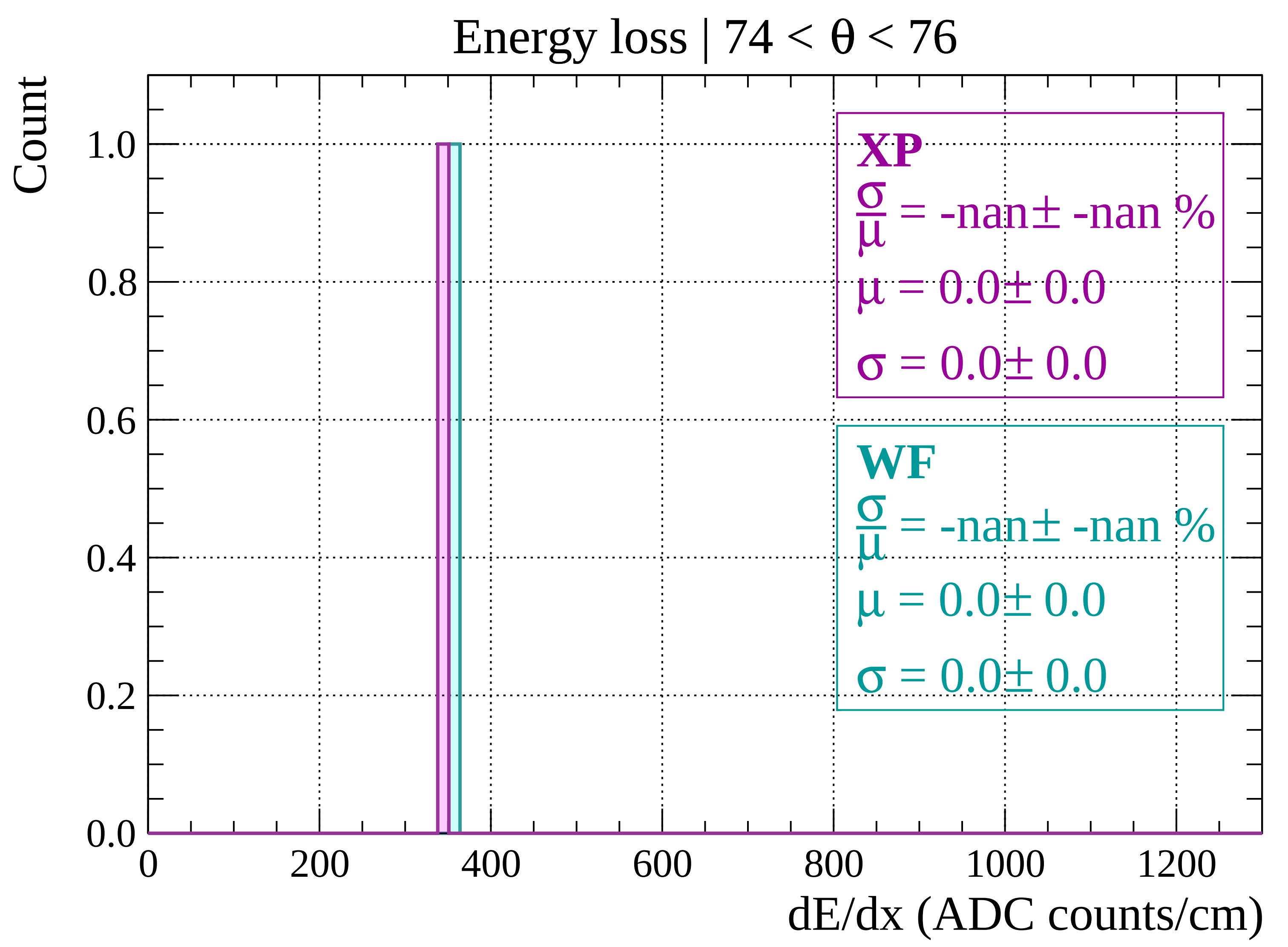


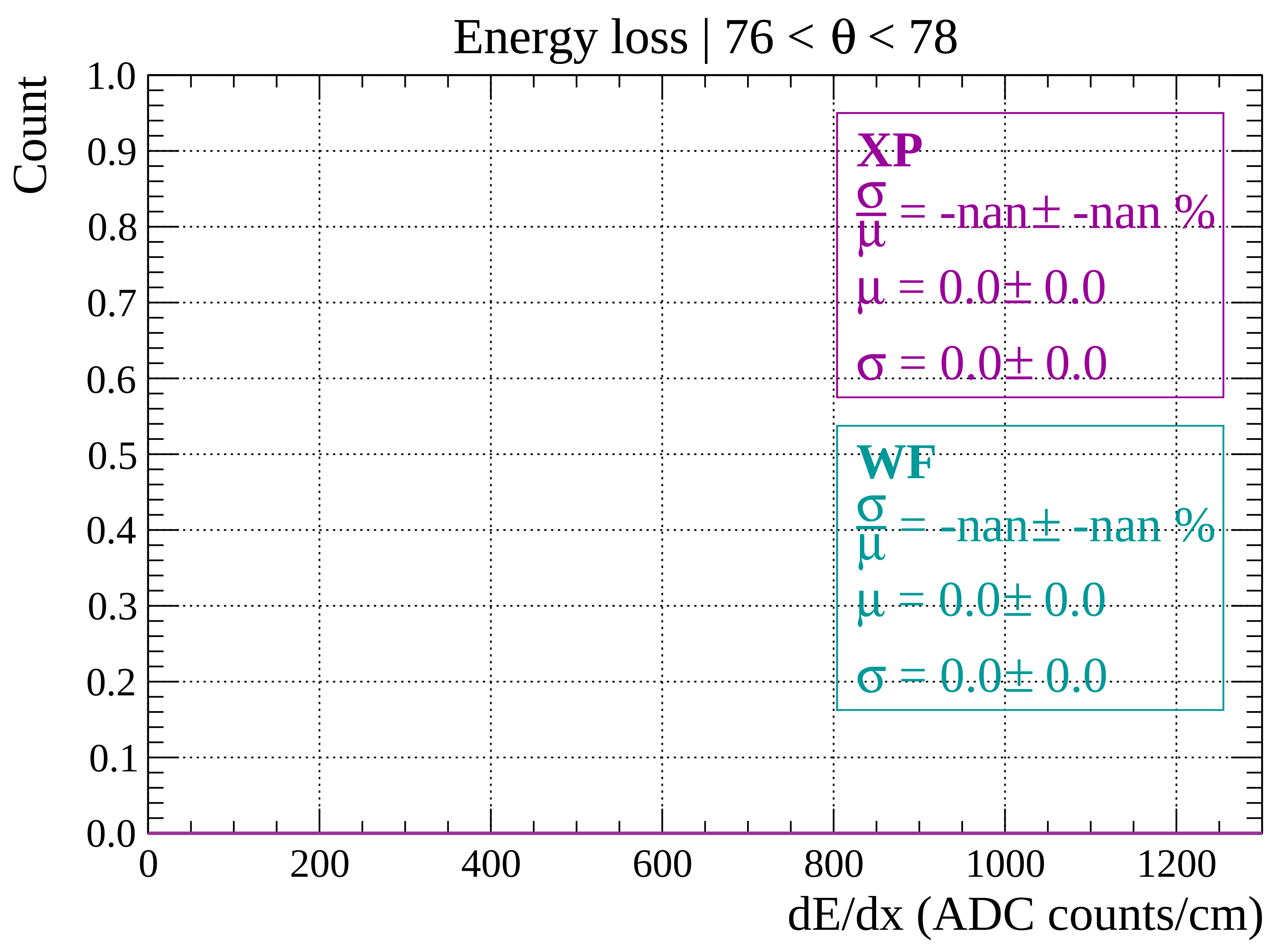
Energy loss | 68 < θ < 70
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 26.65± 31.14 %
μ = 507.4± 75.1
σ = 135.3± 138.0
WF
σ/μ = 31.99± 100.05 %
μ = 520.7± 344.4
σ = 166.6± 410.8



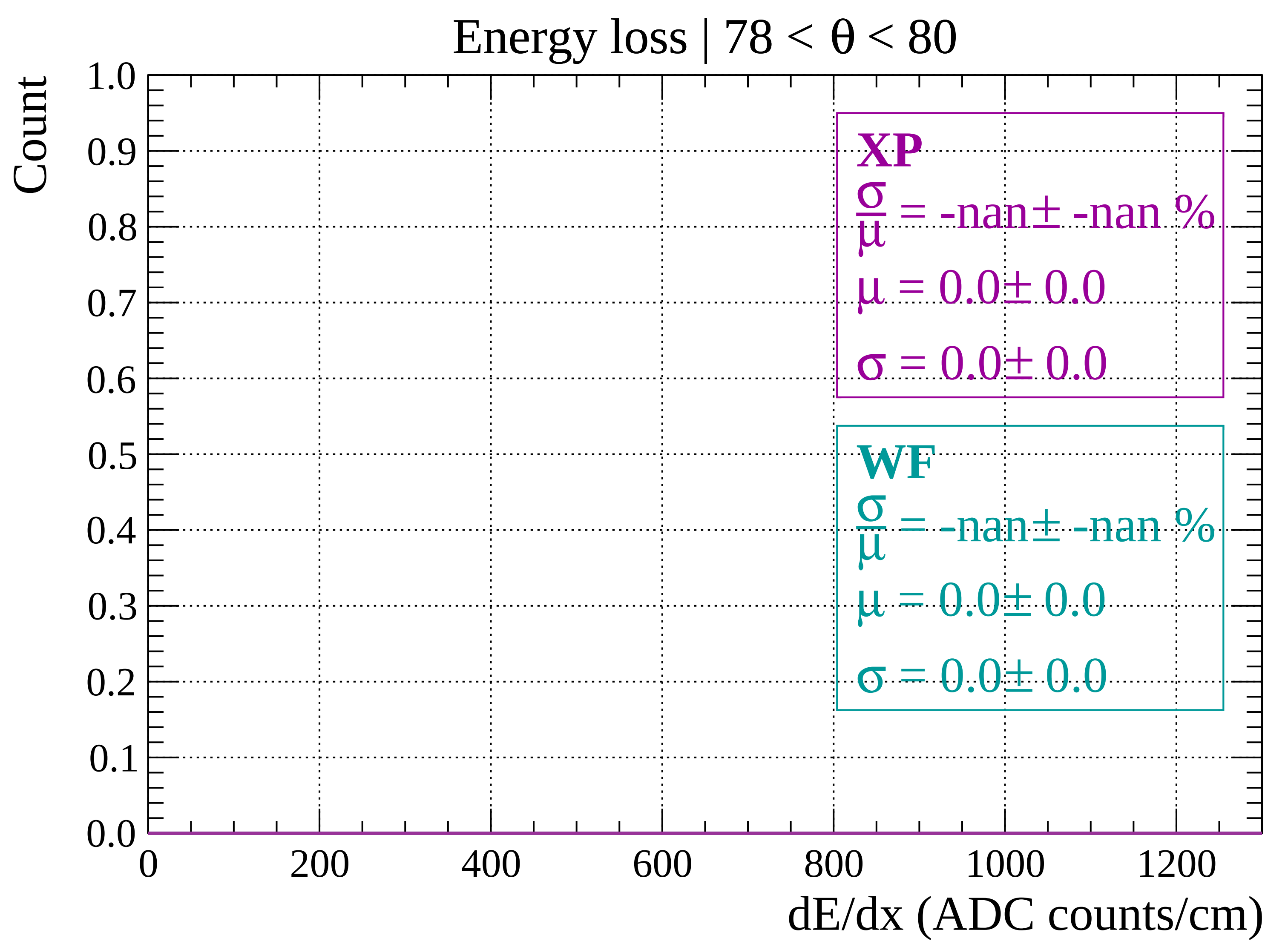
Energy loss | 70 < θ < 72
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 303.30± 3247.58 %
μ = 301.0± 2899.1
σ = 913.1± 984.0
WF
σ/μ = 12.50± 94.21 %
μ = 156.0± 64.7
σ = 19.5± 138.9

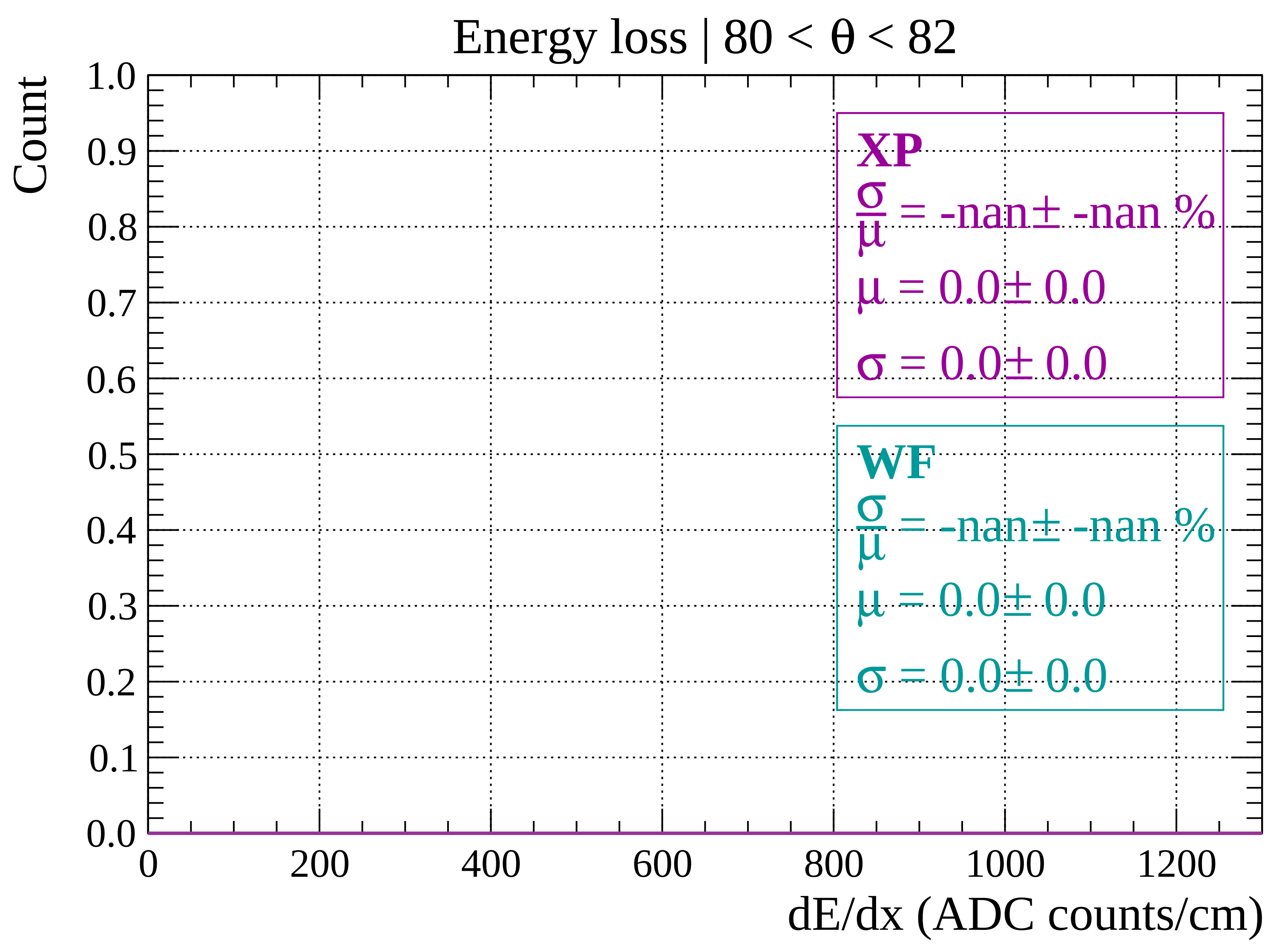

Energy loss | 72 < θ < 74
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
$\frac{\sigma}{\mu}$ = 0.05± 0.42 %
μ = 461.8± 1.8
σ = 0.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 26.71± 56.19 %
μ = 1283.2± 1655.3
σ = 342.7± 278.9


Energy loss | 74 < θ < 76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

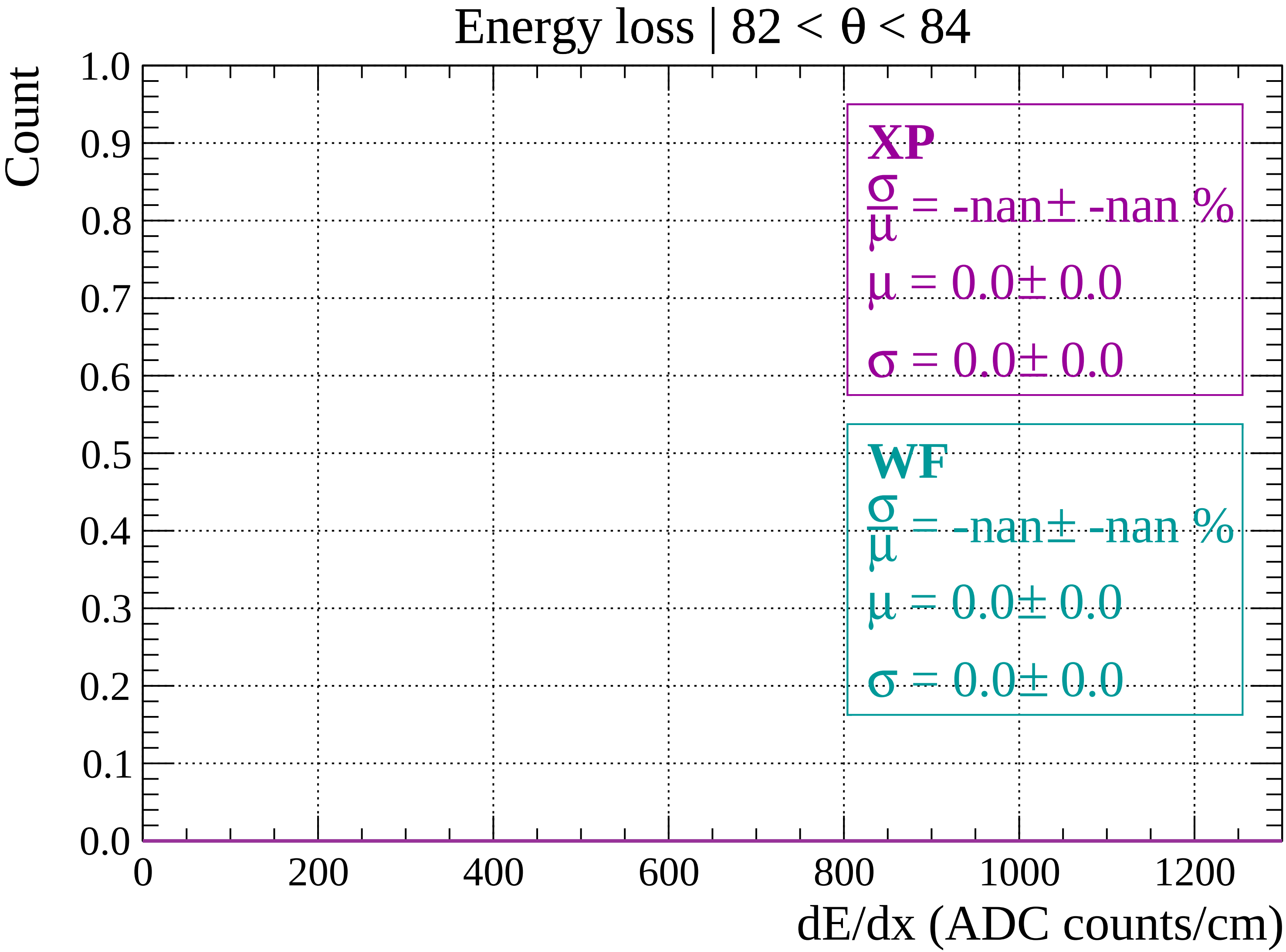


Energy loss | 76 < θ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

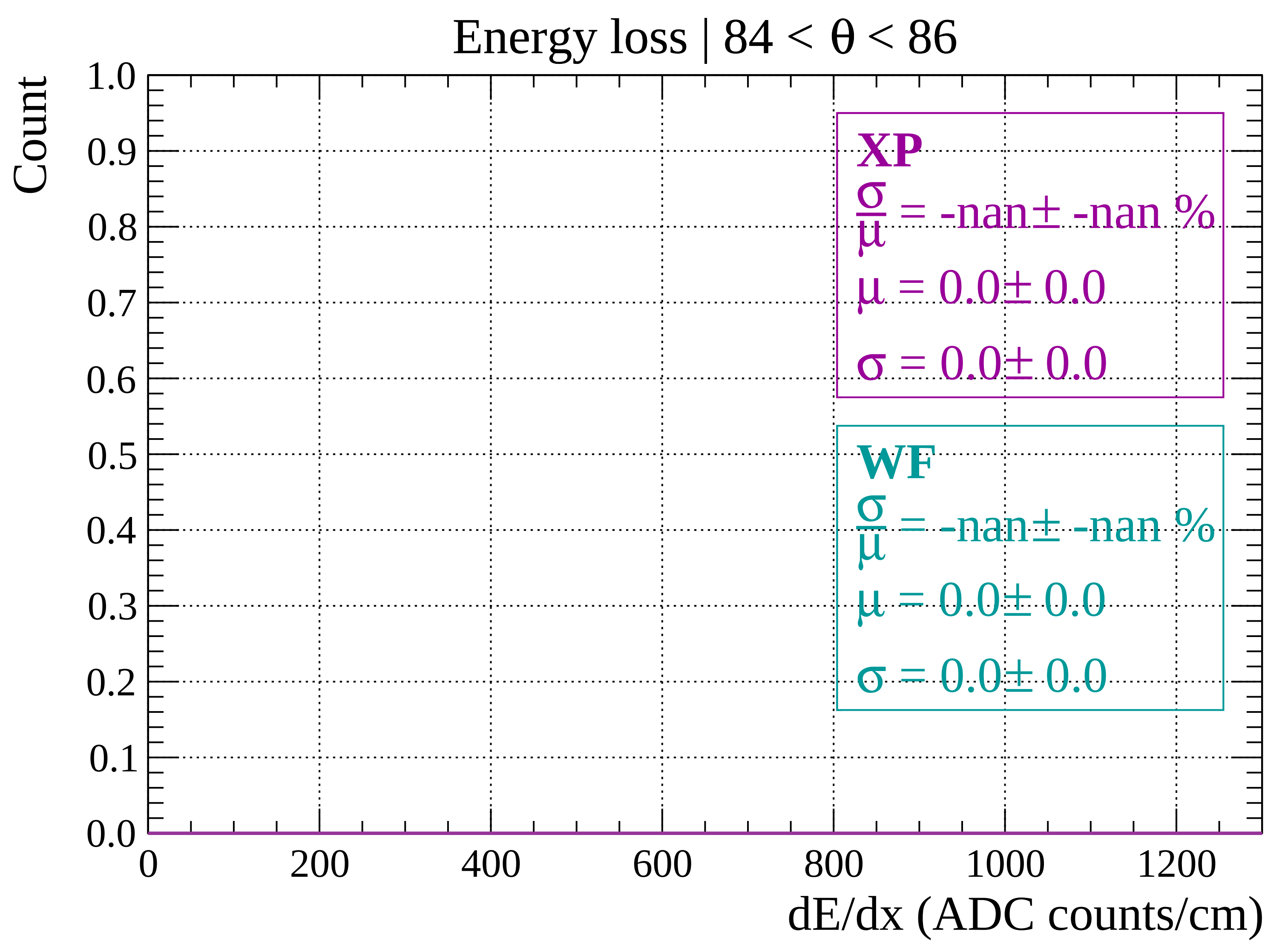

Energy loss | 78 < θ < 80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

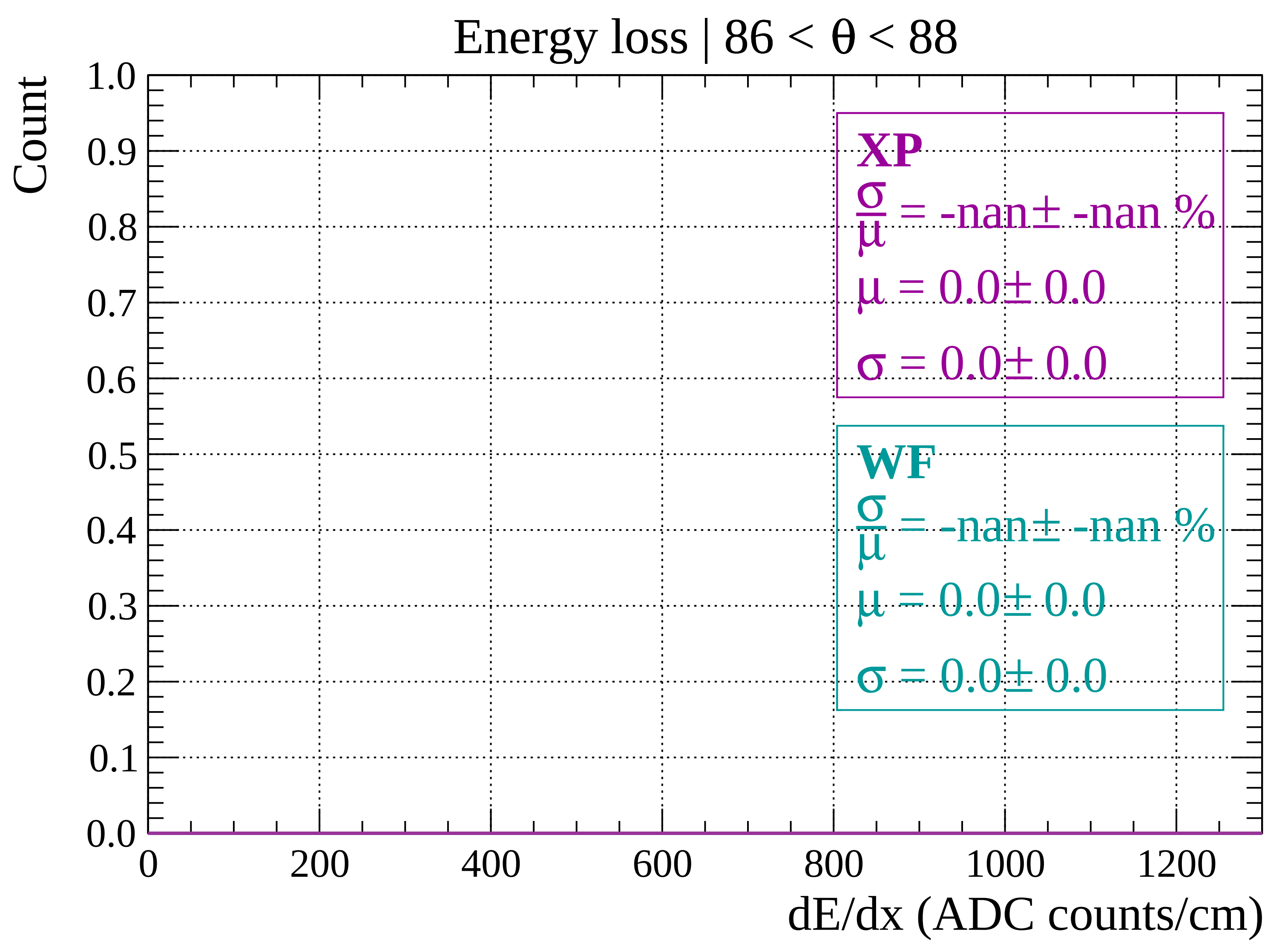


Energy loss | 80 < θ < 82
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

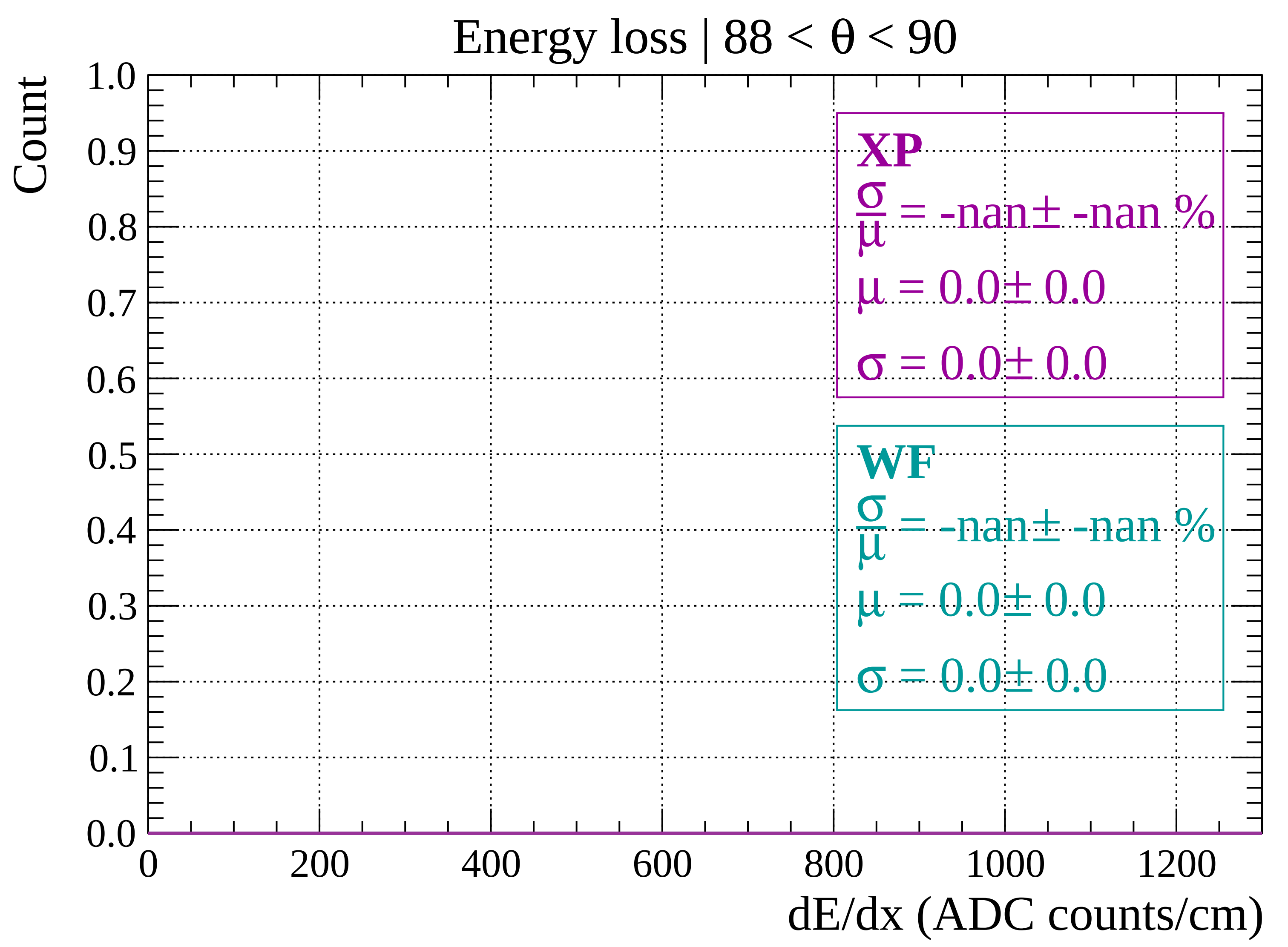


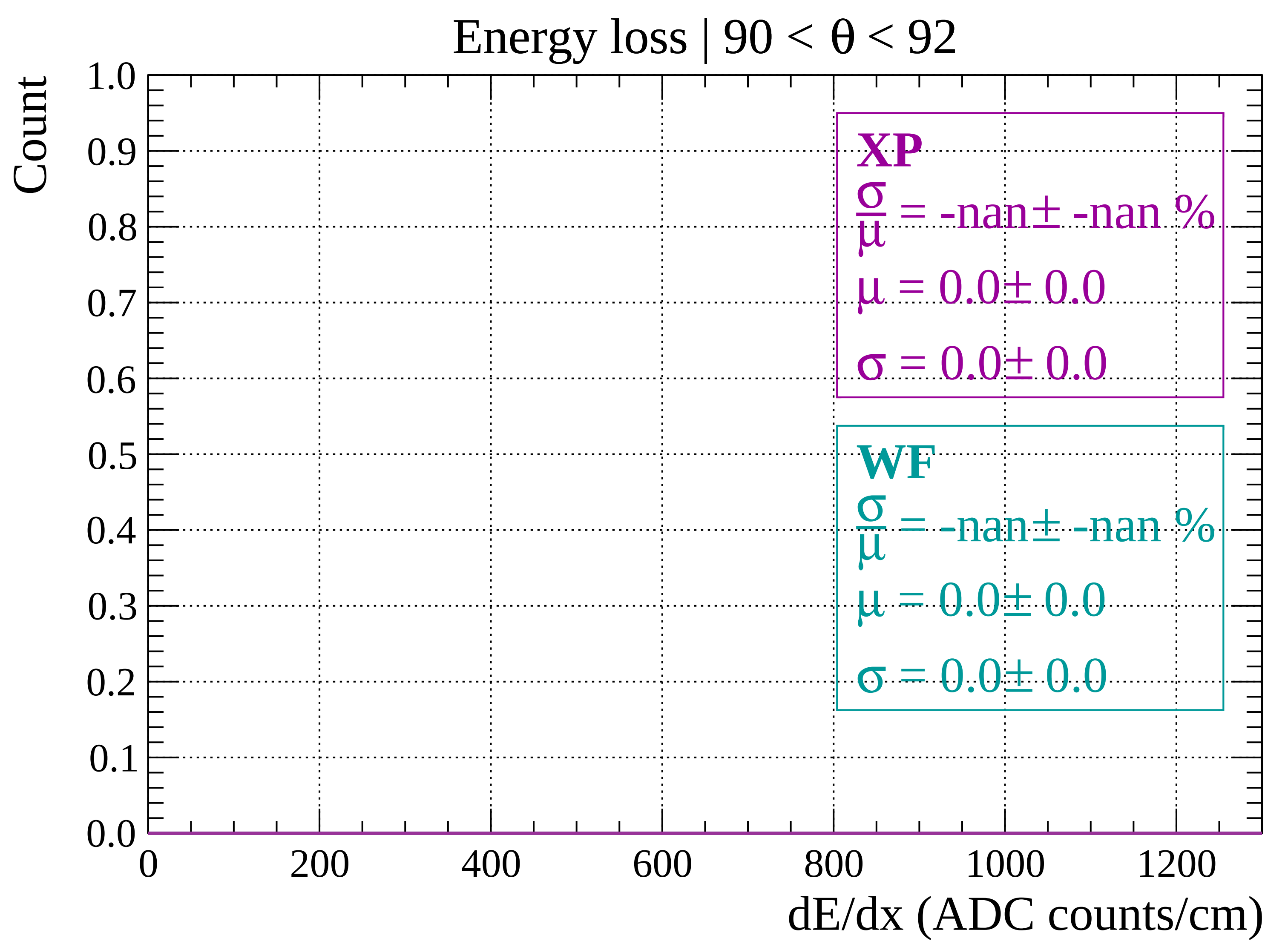
Energy loss | 82 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < θ < 86
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

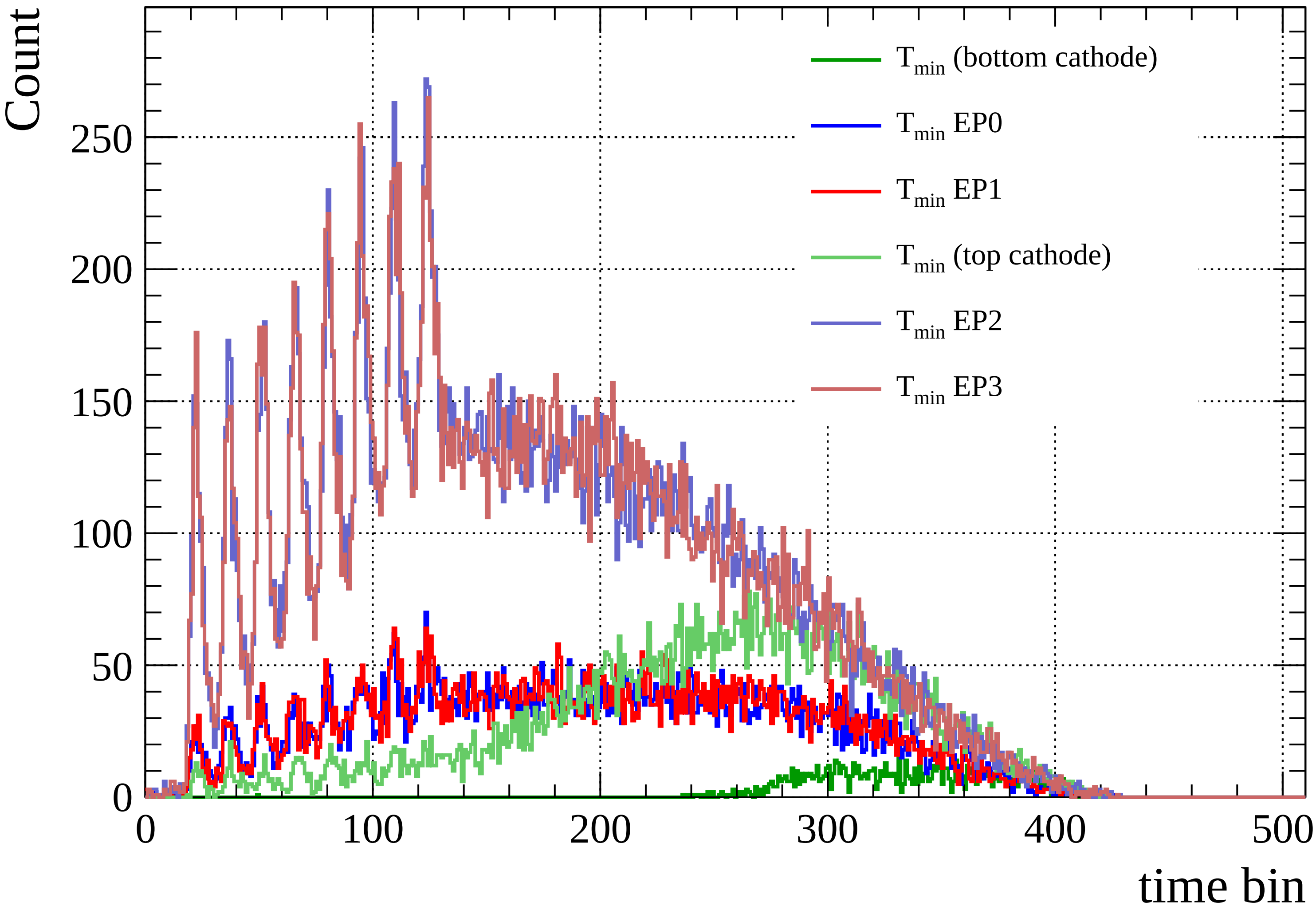


Energy loss | 86 < θ < 88
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

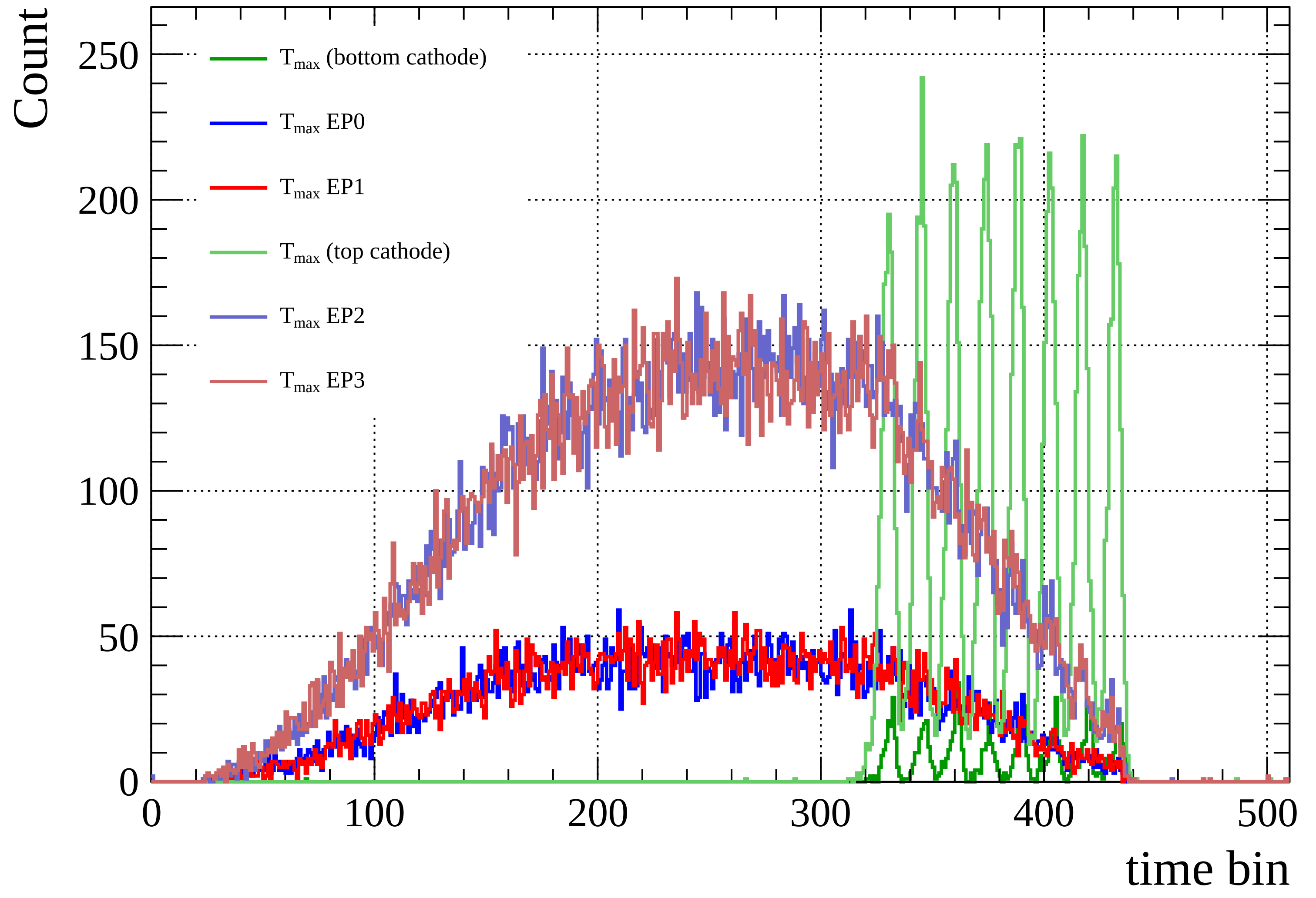

Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200



Energy loss | 90 < θ < 92
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
250
200
150
100
50
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
50
100
150
200
250
0
100
200
300
400
500